

Elementary Number Theory

Solutions Manual

Contents

1	Some Preliminary Considerations	1
1.1	Mathematical Induction	1
1.2	The Binomial Theorem	9
1.3	Early Number Theory	19
2	Divisibility Theory in the Integers	25
2.1	The Division Algorithm	25
2.2	The Greatest Common Divisor	30
2.3	The Euclidean Algorithm	46
2.4	The Diophantine Equation $ax + by = c$	53
3	Primes and Their Distribution	65
3.1	The Fundamental Theorem of Arithmetic	65
3.2	The Sieve of Eratosthenes	73
3.3	The Goldbach Conjecture	80
4	The Theory of Congruences	97
4.2	Basic Properties of Congruence	97
4.3	Special Divisibility Tests	108
4.4	Linear Congruences	123
5	Fermat's Theorem	149
5.2	Fermat's Factorization Method	149
5.3	The Little Theorem	156
5.4	Wilson's Theorem	172
6	Number-Theoretic Functions	183
6.1	The Functions τ and σ	183
6.2	The Möbius Inversion Formula	200
6.3	The Greatest Integer Function	208
6.4	An Application to the Calendar	223
7	Euler's Generalization of Fermat's Theorem	229
7.2	Euler's Phi-Function	229
7.3	Euler's Generalization of Fermat's Theorem	248

7.4	Some Properties of the Phi-Function	260
7.5	An Application to Cryptography	274
8	Primitive Roots and Indices	295
8.1	The Order of an Integer Modulo n	295
8.2	Primitive Roots for Primes	305
8.3	Composite Numbers Having Primitive Roots	313
9	The Quadratic Reciprocity Law	331
9.1	Euler's Criterion	331
9.2	The Legendre Symbol and Its Properties	339
9.3	Quadratic Reciprocity	351
9.4	Quadratic Congruence With Composite Moduli	368

Chapter 1

Some Preliminary Considerations

1.1 Mathematical Induction

1.a. Prove that $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$ for all $n \geq 1$.

Proof.

Base case: $1 = \frac{1 \cdot (1+1)}{2} = 1$.

Suppose $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$.

Then

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1)}{2} + \frac{2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

So the statement holds for $k+1$. ■

1.b. Prove that $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for all $n \geq 1$.

Proof.

Base case: $1 = 1$.

Suppose $1 + 3 + \cdots + (2k-1) = k^2$.

Then

$$1 + 3 + \cdots + (2k-1) + (2k+1) = k^2 + 2k + 1 = (k+1)^2.$$

So the statement holds for $k+1$. ■ **1.c.** Prove that $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}$, all $n \geq 1$.

Proof.

$k = 1$: $1 \cdot 2 = \frac{1 \cdot (1+1)(1+2)}{3} = \frac{1 \cdot 2 \cdot 3}{3} = 2$.

Suppose the statement is true for k . Then

$$1 \cdot 2 + \cdots + k(k+1) + (k+1)(k+2) = \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= \frac{(k+1)(k+2)[k+3]}{3} = \frac{(k+1)(k+2)(k+3)}{3}.$$

So, $k \Rightarrow k+1$.

1.d. Prove that $1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$, all $n \geq 1$.

Proof.

$$k=1: 1^2 = \frac{1 \cdot (2 \cdot 1 - 1)(2 \cdot 1 + 1)}{3} = \frac{1 \cdot 1 \cdot 3}{3} = 1.$$

$k \Rightarrow k+1$:

$$\begin{aligned} & 1^2 + 3^2 + \cdots + (2k-1)^2 + (2k+1)^2 \\ &= \frac{k(2k-1)(2k+1)}{3} + (2k+1)^2 \\ &= \frac{k(2k-1)(2k+1)}{3} + \frac{3(2k+1)^2}{3} \\ &= \frac{(2k+1)[k(2k-1) + 3(2k+1)]}{3} \\ &= \frac{(2k+1)[2k^2 - k + 6k + 3]}{3} \\ &= \frac{(2k+1)[2k^2 + 5k + 3]}{3} = \frac{(2k+1)(2k+3)(k+1)}{3} \\ &= \frac{(k+1)(2k+1)(2k+3)}{3}. \end{aligned}$$

So true for $k+1$.

1.e. Prove that $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$, all $n \geq 1$.

Proof.

$$k=1: 1^3 = \left[\frac{1 \cdot 2}{2} \right]^2 = 1^2 = 1.$$

$k \Rightarrow k+1$:

$$\begin{aligned} & 1^3 + 2^3 + \cdots + k^3 + (k+1)^3 \\ &= \left[\frac{k(k+1)}{2} \right]^2 + (k+1)^3 \\ &= \frac{k^2(k+1)^2}{4} + \frac{4(k+1)^2(k+1)}{4} \\ &= \frac{(k+1)^2[k^2 + 4k + 4]}{4} = \frac{(k+1)^2(k+2)^2}{4} \end{aligned}$$

$$= \left[\frac{(k+1)(k+2)}{2} \right]^2.$$

So, true for $k+1$.

2. If $r \neq 1$, then $a + ar + \cdots + ar^n = \frac{a(r^{n+1} - 1)}{r - 1}$, $n \geq 1$.

Proof.

$$\text{For } k = 1: a + ar^1 = a(1 + r) = \frac{a(r^2 - 1)}{r - 1} = a(r + 1).$$

$k \Rightarrow k + 1$:

$$\begin{aligned} a + ar + \cdots + ar^n + ar^{n+1} &= \frac{a(r^{n+1} - 1)}{r - 1} + ar^{n+1} \\ &= \frac{a(r^{n+1} - 1)}{r - 1} + \frac{ar^{n+1}(r - 1)}{r - 1} \\ &= \frac{ar^{n+1} - a + ar^{n+2} - ar^{n+1}}{r - 1} \\ &= \frac{ar^{n+2} - a}{r - 1} = \frac{a(r^{n+2} - 1)}{r - 1}. \end{aligned}$$

So, true for $k+1$.

3. $a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \cdots + a + 1)$, for $n \geq 1$.

Proof.

$$\text{For } k = 1: a^1 - 1 = a - 1 = (a - 1)(a^0) = a - 1.$$

$k \Rightarrow k + 1$:

$$\begin{aligned} a^{k+1} - 1 &= a^{k+1} - a^k - a + a^k + a - 1 \\ &= a^{k+1} - a + a^k - 1 - a^k + a \\ &= a(a^k - 1) + a^k - 1 - a(a^{k-1} - 1) \\ &= (a + 1)(a^k - 1) - a(a^{k-1} - 1) \end{aligned}$$

Use the second principle of finite induction for k , $k - 1$:

$$\begin{aligned} &= (a + 1) \left[(a - 1)(a^{k-1} + a^{k-2} + \cdots + a + 1) \right] \\ &\quad - a \left[(a - 1)(a^{k-2} + a^{k-3} + \cdots + a + 1) \right] \\ &= (a - 1) \left[a(a^{k-1} + a^{k-2} + \cdots + a + 1) \right. \\ &\quad \left. + (a^{k-1} + a^{k-2} + \cdots + a + 1) \right. \\ &\quad \left. - a(a^{k-2} + a^{k-3} + \cdots + a + 1) \right] \quad (\text{combine these two}) \end{aligned}$$

$$\begin{aligned}
 &= (a-1) \left[a(a^{k-1} + a^{k-2} + \cdots + a + 1) + (a^{k-1} + a^{k-2} + \cdots + a + 1) \right. \\
 &\quad \left. - (a^{k-1} + a^{k-2} + \cdots + a + 1) \right] \\
 &= (a-1) \left[(a^k + a^{k-1} + \cdots + a + 1) \right]
 \end{aligned}$$

and so works for $k+1$.

4. Cube of any integer can be written as the difference of two squares.

Proof.

$$n^3 = (1^3 + 2^3 + \cdots + n^3) - (1^3 + 2^3 + \cdots + (n-1)^3)$$

From 1(e),

$$1^3 + 2^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2.$$

So,

$$n^3 = \left[\frac{n(n+1)}{2} \right]^2 - \left[\frac{(n-1)n}{2} \right]^2.$$

If n is even, then $\frac{n}{2}$ is an integer. If n is odd, then $n+1$ and $n-1$ are even, so $\frac{n+1}{2}$ and $\frac{n-1}{2}$ are integers.

$\therefore n^3$ is the difference of two squares. **5.(a)** For $n=4$: $n! + 1 = 25 = 5^2$.

$n=5$: $n! + 1 = 121 = 11^2$.

$n=7$: $n! + 1 = 5041 = 71^2$.

(b) False.

$$(3 \cdot 2)! = 720 \neq 3! 2! = 6 \cdot 2 = 12.$$

$$(2+3)! = 120 \neq 2! + 3! = 2 + 6.$$

6.a. $n! > n^2$ for $n \geq 4$.

Proof.

$$4! = 24 > 16 = 4^2.$$

Suppose $k! > k^2$, for $k \geq 4$.

$$(k+1)! = (k+1) \cdot k! > (k+1) \cdot k^2 = k^3 + k^2.$$

Since $k \geq 4$, then $k > 2$, so $k^2 > 2k$. Since $k \geq 1$, then $k^3 > 2k$, so $k^3 \geq 2k + 1$.

So, $k^3 + k^2 \geq k^2 + 2k + 1 = (k+1)^2$.

So $(k+1)! > k^3 + k^2 \geq (k+1)^2$.

So $(k+1)! > (k+1)^2$.

So, $k \Rightarrow k+1$. **6.b.** $n! > n^3$, $n \geq 6$.

Proof.

$$6! = 720 > 216 = 6^3.$$

Suppose $k! > k^3$ for any $k \geq 6$.

$$\therefore (k+1)! = (k+1)k! > (k+1)k^3 = k^4 + k^3.$$

Since $k \geq 6$, then $k > 3$, so $k^2 > 3k$. Also, $k \geq 1$, so $k^2 > k$, and $k^2 \geq k+1$.

$$\text{So } k^4 = k^2 \cdot k^2 > 3k(k+1) = 3k^2 + 3k.$$

$$\therefore k^3 + k^4 > k^3 + 3k^2 + 3k.$$

$$\therefore k^4 + k^3 \geq k^3 + 3k^2 + 3k + 1 = (k+1)^3.$$

$$\therefore (k+1)! > k^4 + k^3 \geq (k+1)^3.$$

$$(k+1)! > (k+1)^3.$$

$$\text{So, } k \Rightarrow k+1. \quad \mathbf{7.} \quad 1(1!) + 2(2!) + \cdots + n(n!) = (n+1)! - 1, \quad n \geq 1.$$

Proof.

$$k = 1: 1(1!) = 1 = (1+1)! - 1 = 2! - 1 = 1.$$

$$k \Rightarrow k+1: \text{ Let } 1(1!) + 2(2!) + \cdots + k(k!) = (k+1)! - 1.$$

Then

$$\begin{aligned} & 1(1!) + \cdots + (k+1)(k+1)! \\ &= (k+1)! - 1 + (k+1)(k+1)! \\ &= (k+1)![1 + k + 1] - 1 \\ &= (k+1)!(k+2) - 1 \\ &= (k+2)! - 1. \end{aligned}$$

So, true for $k+1$.

$$\mathbf{8.a.} \quad 2 \cdot 6 \cdot 10 \cdots (4n-2) = \frac{(2n)!}{n!}, \quad n \geq 1.$$

Proof.

$$k = 1: 2 = \frac{2!}{1!} = 2.$$

$$k \Rightarrow k+1: \text{ Let } 2 \cdot 6 \cdot 10 \cdots (4k-2) = \frac{(2k)!}{k!}.$$

$$\text{Then } 2 \cdot 6 \cdot 10 \cdots (4k-2)(4k+2) =$$

$$\begin{aligned} & \frac{(2k)!}{k!}(4k+2) = \frac{(2k)!(2k+1) \cdot 2}{k!} \\ &= \frac{(2k)!(2k+1) \cdot 2}{k!} \cdot \frac{(k+1)}{(k+1)} \\ &= \frac{(2k)!(2k+1)(2k+2)}{k!(k+1)} = \frac{(2k+2)!}{(k+1)!}. \end{aligned}$$

So, true for $k+1$.

$$\mathbf{8.b.} \quad 2^n(n!)^2 \leq (2n)!, \quad n \geq 1.$$

Proof.

From (a), $(2n)! = 2 \cdot 6 \cdot 10 \cdots (4n - 2)(n!)$.

So the problem reduces to:

$$2^n(n!)^2 \leq 2 \cdot 6 \cdot 10 \cdots (4n - 2)(n!)$$

or

$$2^n(n!) \leq 2 \cdot 6 \cdot 10 \cdots (4n - 2).$$

For $k = 1$: $2^1(1!) = 2 \leq 2$.

$k \Rightarrow k + 1$: Let $2^k(k!) \leq 2 \cdot 6 \cdot 10 \cdots (4k - 2)$.

$$\begin{aligned} 2^{k+1}(k+1)! &= 2^k(k!) \cdot 2 \cdot (k+1) \\ &= 2^k(k!)(2k+2) \\ &< 2^k(k!)(4k+2) \\ &\leq 2 \cdot 6 \cdot 10 \cdots (4k-2)(4k+2) \\ &= 2 \cdot 6 \cdot 10 \cdots (4(k+1)-2). \end{aligned}$$

So, true for $k + 1$.

9. If $1 + a > 0$, then $(1 + a)^n \geq 1 + na$, $n \geq 1$.

Proof.

$k = 1$: $1 + a \geq 1 + a$.

$k \Rightarrow k + 1$: Let $(1 + a)^k \geq 1 + ka$.

$$\begin{aligned} (1 + a)^{k+1} &= (1 + a)^k(1 + a) \\ &\geq (1 + ka)(1 + a) \\ &= 1 + ka + a + ka^2 \\ &\geq 1 + ka + a \quad (a^2 \geq 0, \text{ so } ka^2 \geq 0) \\ &= 1 + (k + 1)a. \\ \therefore (1 + a)^{k+1} &\geq 1 + (k + 1)a. \end{aligned}$$

So, it's true for $k + 1$.

$$\mathbf{10.a.} \quad \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \leq 2 - \frac{1}{n}, \quad n \geq 1.$$

Proof.

$$k = 1: \frac{1}{1^2} = 1 \leq 2 - \frac{1}{1} = 1.$$

$$k \Rightarrow k + 1: \text{ Let } \frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{k^2} \leq 2 - \frac{1}{k}.$$

Then,

$$\frac{1}{1^2} + \cdots + \frac{1}{k^2} + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k} + \frac{1}{(k+1)^2}.$$

Since $k \geq 1$, then $k^2 + 2k < k^2 + 2k + 1$, or $\frac{k^2 + 2k}{(k+1)^2} < 1$, or $\frac{k(k+2)}{(k+1)^2} < 1$.

$$\therefore \frac{(k+1)+1}{(k+1)^2} < \frac{1}{k} \Rightarrow \frac{1}{k+1} + \frac{1}{(k+1)^2} < \frac{1}{k}.$$

$$\therefore -\frac{1}{k} + \frac{1}{(k+1)^2} < -\frac{1}{k+1}.$$

$$\therefore 2 - \frac{1}{k} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1}.$$

$$\therefore \frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1}.$$

So, $k \Rightarrow k+1$.

$$\mathbf{10.b.} \quad \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n}.$$

Proof.

$$k=1: \frac{1}{2} = 2 - \frac{1+2}{2^1} = 2 - \frac{3}{2} = \frac{1}{2}.$$

$$k \Rightarrow k+1: \text{ Let } \frac{1}{2} + \frac{2}{2^2} + \cdots + \frac{k}{2^k} = 2 - \frac{k+2}{2^k}.$$

Then,

$$\begin{aligned} \frac{1}{2} + \cdots + \frac{k}{2^k} + \frac{k+1}{2^{k+1}} &= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}} \\ &= 2 - \frac{k+2}{2^k} \cdot \frac{2}{2} + \frac{k+1}{2^{k+1}} \\ &= 2 + \frac{(k+1) - (2k+4)}{2^{k+1}} = 2 + \frac{-k-3}{2^{k+1}} \\ &= 2 - \frac{(k+1)+2}{2^{k+1}}. \end{aligned}$$

So, $k \Rightarrow k+1$. **11.** $(2n)!/2^n n!$ is an integer, $n \geq 0$.

Proof.

$$n=0: 0! = 1 \text{ by definition. So } \frac{0!}{2^0 0!} = \frac{1}{1 \cdot 1} = 1.$$

$$k \Rightarrow k+1: \text{ Suppose } \frac{(2k)!}{2^k k!} \text{ is an integer.}$$

$$\begin{aligned} \frac{[2(k+1)]!}{2^{k+1}(k+1)!} &= \frac{(2k)!(2k+1)(2k+2)}{2^k(k!) \cdot 2(k+1)} \\ &= \frac{(2k)!}{2^k k!} \cdot \frac{(2k+1)(2k+2)}{2(k+1)} \end{aligned}$$

$$= (\text{integer}) \cdot (2k + 1) = \text{integer}.$$

12. [Table of values for a recursively-defined function T ; the handwriting is small and faint in places, so the exact values below should be double-checked against the original.]

$T(21) = 32$	$T(23) = 35$
$T(T(21)) = 16$	$T(35) = 53$
$T(16) = 9$	$T(53) = 80$
$T(9) = 7$	$T(80) = 40$
$T(7) = 4$	$T(40) = 20$
$T(4) = 2$	$T(20) = 10$
$T(2) = 1$	$T(10) = 5$
	$T(5) = 8$
	$T(8) = 4$
	$T(4) = 2$
	$T(2) = 1$

13. $a_1 = 1, a_2 = 2, a_3 = 3, a_n = a_{n-1} + a_{n-2} + a_{n-3}$, for $n \geq 4$.

Prove: $a_n < 2^n, n \geq 1$.

Proof.

$$1 < 2^1, 2 < 2^2, 3 < 2^3.$$

Let $k \geq 4$, and assume $a_K < 2^K, K = 4, \dots, k$.

Then

$$\begin{aligned}
 a_{k+1} &= a_k + a_{k-1} + a_{k-2} \\
 &< 2^k + 2^{k-1} + 2^{k-2} \\
 &< 2^k + 2^{k-1} + 2^{k-1} \\
 &= 2^k + 2 \cdot 2^{k-1} = 2^k + 2^k \\
 &= 2 \cdot 2^k = 2^{k+1}. \\
 \therefore a_{k+1} &< 2^{k+1}.
 \end{aligned}$$

14. $a_1 = 11, a_2 = 21, a_n = 3a_{n-1} - 2a_{n-2}, n \geq 3$.

Prove: $a_n = 5 \cdot 2^n + 1, n \geq 1$.

Proof.

$$a_1 = 5 \cdot 2 + 1 = 11. \quad a_2 = 5 \cdot 4 + 1 = 21.$$

Suppose $a_K = 5 \cdot 2^K + 1$ for $3, 4, \dots, k$.

Then

$$\begin{aligned}
 a_{k+1} &= 3a_k - 2a_{k-1} \\
 &= 3(5 \cdot 2^k + 1) - 2(5 \cdot 2^{k-1} + 1) \\
 &= 15 \cdot 2^k + 3 - 5 \cdot 2^k - 2
 \end{aligned}$$

$$\begin{aligned}
 &= 10 \cdot 2^k + 1 \\
 &= 5 \cdot 2 \cdot 2^k + 1 \\
 &= 5 \cdot 2^{k+1} + 1.
 \end{aligned}$$

$\therefore$ if it works for $3, 4, \dots, k$, then it works for $k+1$.
 $\therefore$ works for all $k \geq 1$.

1.2 The Binomial Theorem

1.a. Newton's identity.

$$\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r}, \quad n \geq k \geq r \geq 0.$$

Proof.

$$\begin{aligned}
 \frac{n!}{k!(n-k)!} \cdot \frac{k!}{r!(k-r)!} &= \frac{n!}{r!} \cdot \frac{1}{(n-k)!(k-r)!} \\
 &= \frac{n!}{r!} \cdot \frac{(n-r)!}{(n-r)!} \cdot \frac{1}{(n-k)!(k-r)!} \\
 &= \frac{n!}{r!(n-r)!} \cdot \frac{(n-r)!}{(k-r)!(n-k)!} \\
 &= \binom{n}{r} \cdot \frac{(n-r)!}{(k-r)!(n-r-(k-r))!} = \binom{n}{r} \binom{n-r}{k-r}.
 \end{aligned}$$

1.b. $\binom{n}{k} = \frac{n-k+1}{k} \binom{n}{k-1}, \quad n \geq k \geq 1.$

Without using part (a),

$$\begin{aligned}
 \binom{n}{k} &= \frac{n!}{k!(n-k)!} = \frac{n! \cdot (n-k+1)}{k(k-1)!(n-k+1)!} \\
 &= \frac{n!}{(k-1)!(n-k+1)!} \cdot \frac{(n-k+1)}{k} = \frac{(n-k+1)}{k} \binom{n}{k-1}.
 \end{aligned}$$

To use part (a), let $r = 1$.

Then $\binom{n}{k} \binom{k}{1} = \binom{n}{1} \binom{n-1}{k-1}, \quad n \geq k \geq r \geq 0.$

So, $\binom{n}{k} k = n \binom{n-1}{k-1}$

$$= n \frac{(n-1)!}{(k-1)!(n-k)!}$$

$$= \frac{n!}{(k-1)!(n-k+1)!} \cdot (n-k+1).$$

$$\text{So, } \binom{n}{k} = \frac{n-k+1}{k} \binom{n}{k-1}.$$

2. If $2 \leq k \leq n-2$, and $n \geq 4$,

$$\binom{n}{k} = \binom{n-2}{k-2} + 2\binom{n-2}{k-1} + \binom{n-2}{k}.$$

Working from the right hand side,

$$\begin{aligned} & \frac{(n-2)!}{(k-2)!(n-k)!} + \frac{2(n-2)!}{(k-1)!(n-k-1)!} + \frac{(n-2)!}{k!(n-k-2)!} \\ &= \frac{k(k-1)(n-2)!}{k!(n-k)!} + \frac{2k(n-k)(n-2)!}{k!(n-k)!} + \frac{(n-k)(n-k-1)(n-2)!}{k!(n-k)!} \\ &= \frac{(n-2)! [k^2 - k + 2kn - 2k^2 + n^2 - nk - n - kn + k^2 + k]}{k!(n-k)!} \\ &= \frac{(n-2)! [n^2 - n]}{k!(n-k)!} = \frac{n(n-1)(n-2)!}{k!(n-k)!} = \binom{n}{k}. \end{aligned}$$

$2 \leq k$ for $(k-2)!$ in denominator to work; $n-k-2 \geq 0$, or $n-k \geq 2$, so $n \geq 4$ for $(n-k-2)!$ in denominator to work.

3.a. From the binomial theorem, letting $a = b = 1$,

$$(a+b)^n = 2^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} 1^k.$$

$$\text{So, } 2^n = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}.$$

3.b. From the binomial theorem, let $a = 1$, $b = -1$.

$$0^n = 0 = \binom{n}{0} - \binom{n}{1} + \cdots + (-1)^n \binom{n}{n}.$$

$$\textbf{3.c. } \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n2^{n-1}.$$

In the binomial theorem, let $a = 1$. Then

$$(1+b)^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} b^k.$$

So, $n(1+b)^{n-1} = n \left[\binom{n-1}{0} + \binom{n-1}{1}b + \cdots + \binom{n-1}{n-1}b^{n-1} \right]$.

Now let $b = 1$. Then

$$\begin{aligned} n2^{n-1} &= n \binom{n-1}{0} + n \binom{n-1}{1} + \cdots + n \binom{n-1}{n-1} \\ &= \sum_{k=0}^{n-1} n \binom{n-1}{k}. \end{aligned}$$

$$\begin{aligned} \text{But } n \binom{n-1}{k} &= \frac{n(n-1)!}{k!(n-1-k)!} = \frac{n!}{k!(n-(k+1))!} \cdot \frac{(k+1)}{(k+1)} \\ &= (k+1) \frac{n!}{(k+1)!(n-(k+1))!} \\ &= (k+1) \binom{n}{k+1}. \end{aligned}$$

$$\begin{aligned} \therefore n2^{n-1} &= \sum_{k=0}^{n-1} (k+1) \binom{n}{k+1} = \sum_{k=0}^{n-1} n \binom{n-1}{k} \\ &= \binom{n}{1} + 2 \binom{n}{2} + 3 \binom{n}{3} + \cdots + n \binom{n}{n}. \end{aligned}$$

3.d. $\binom{n}{0} + 2 \binom{n}{1} + 2^2 \binom{n}{2} + \cdots + 2^n \binom{n}{n} = 3^n$.

In the binomial theorem, let $a = 1$, $b = 2$.

$$\begin{aligned} (a+b)^n &= 3^n = \binom{n}{0}1^n + \binom{n}{1}1^{n-1}2 + \cdots + \binom{n}{n}2^n \\ &= \binom{n}{0} + 2 \binom{n}{1} + \cdots + 2^n \binom{n}{n}. \end{aligned}$$

3.e.

$$\begin{aligned} \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots &= 2^{n-1}, \\ \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots &= 2^{n-1}. \end{aligned}$$

Proof. Add/subtract the results of (a) and (b).

If n is even, then the last term is positive:

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$

$$+ \left[\binom{n}{0} - \binom{n}{1} + \cdots + \binom{n}{n} = 0 \right]$$

$$2 \left[\binom{n}{0} + \binom{n}{2} + \cdots + \binom{n}{n} \right] = 2^n.$$

If n is odd, the last term is $-\binom{n}{n}$:

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$

$$+ \left[\binom{n}{0} - \binom{n}{1} + \cdots - \binom{n}{n} = 0 \right]$$

$$2 \left[\binom{n}{0} + \binom{n}{2} + \cdots + \binom{n}{n-1} \right] = 2^n.$$

So, $\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = 2^{n-1}$. If n is even, then the last term is positive:

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$

$$- \left[\binom{n}{0} - \binom{n}{1} + \cdots + \binom{n}{n} = 0 \right]$$

$$2 \left[\binom{n}{1} + \binom{n}{3} + \cdots + \binom{n}{n-1} \right] = 2^n.$$

If n is odd, the last term is $-\binom{n}{n}$:

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$

$$- \left[\binom{n}{0} - \binom{n}{1} + \cdots - \binom{n}{n} = 0 \right]$$

$$2 \left[\binom{n}{1} + \binom{n}{3} + \cdots + \binom{n}{n} \right] = 2^n.$$

So, $\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots = 2^{n-1}$.

3.f.

$$\binom{n}{0} - \frac{1}{2} \binom{n}{1} + \frac{1}{3} \binom{n}{2} - \cdots + \frac{(-1)^n}{n+1} \binom{n}{n} = \frac{1}{n+1}.$$

The terms look like the terms in (b) with coefficients. So, need a relation with a coefficient in front of the binomial term. The k th term can be written as:

$$(-1)^{k-1} \frac{1}{k} \binom{n}{k-1}.$$

Note that

$$\begin{aligned} \binom{n}{k-1} &= \frac{n!}{(k-1)!(n-k+1)!} \\ &= \frac{k}{n+1} \cdot \frac{(n+1)!}{k!(n-k+1)!}. \end{aligned}$$

Thus, $\frac{1}{k} \binom{n}{k-1} = \frac{1}{n+1} \binom{n+1}{k}.$

So, the problem is equivalent to:

$$\begin{aligned} &\binom{n}{0} - \frac{1}{2} \binom{n}{1} + \cdots + \frac{(-1)^n}{n+1} \binom{n}{n} \\ &\quad (k=1) \quad (k=2) \quad (k=n+1) \\ &= \frac{1}{n+1} \binom{n+1}{1} - \frac{1}{n+1} \binom{n+1}{2} + \cdots + \frac{(-1)^n}{n+1} \binom{n+1}{n+1} \\ &= \frac{1}{n+1} \left[\binom{n+1}{1} - \binom{n+1}{2} + \binom{n+1}{3} - \cdots + (-1)^n \binom{n+1}{n+1} \right]. \end{aligned}$$

From (b), $\binom{n}{0} = \binom{n}{1} - \binom{n}{2} + \cdots - (-1)^n \binom{n}{n}.$

Substituting $n = s+1,$

$$\begin{aligned} \binom{s+1}{0} &= \binom{s+1}{1} - \binom{s+1}{2} + \cdots - (-1)^{s+1} \binom{s+1}{s+1} \\ 1 &= \binom{s+1}{1} - \binom{s+1}{2} + \cdots + (-1)^s \binom{s+1}{s+1}. \\ \therefore \binom{n}{0} - \frac{1}{2} \binom{n}{1} + \cdots + \frac{(-1)^n}{n+1} \binom{n+1}{n+1} \\ &= \frac{1}{n+1} \left[\binom{n+1}{1} - \binom{n+1}{2} + \cdots + (-1)^n \binom{n+1}{n+1} \right] \\ &= \frac{1}{n+1} [1] = \frac{1}{n+1}. \end{aligned}$$

4.a. For $n \geq 1,$ $\binom{n}{r} < \binom{n}{r+1} \iff 0 \leq r < \frac{1}{2}(n-1).$

Proof.

$$\begin{aligned}
 & \binom{n}{r} < \binom{n}{r+1} \\
 \iff & \frac{n!}{r!(n-r)!} < \frac{n!}{(r+1)!(n-r-1)!}, \quad 0 \leq r, \ 0 \leq n-r-1 \\
 \iff & \frac{(r+1)!}{r!} < \frac{(n-r)!}{(n-r-1)!}, \quad 0 \leq r \leq n-1 \\
 \iff & r+1 < n-r, \quad 0 \leq r \leq n-1 \\
 \iff & 0 \leq 2r < n-1 \\
 \iff & 0 \leq r < \frac{1}{2}(n-1).
 \end{aligned}$$

$$\mathbf{4.b.} \quad \binom{n}{r} > \binom{n}{r+1} \iff n-1 \geq r > \frac{1}{2}(n-1).$$

Proof.

$$\begin{aligned}
 & \binom{n}{r} > \binom{n}{r+1} \\
 \iff & \frac{n!}{r!(n-r)!} > \frac{n!}{(r+1)!(n-r-1)!}, \quad r \geq 0, \ n-r-1 \geq 0 \\
 \iff & \frac{(r+1)!}{r!} > \frac{(n-r)!}{(n-r-1)!}, \quad r \geq 0, \ n-r-1 \geq 0 \\
 \iff & r+1 > n-r, \quad n-1 \geq r \geq 0 \\
 \iff & 2r > n-1, \quad n-1 \geq r \geq 0 \\
 \iff & n-1 \geq r > \frac{1}{2}(n-1) \geq 0.
 \end{aligned}$$

$$\mathbf{4.c.} \quad \binom{n}{r} = \binom{n}{r+1} \iff r = \frac{1}{2}(n-1).$$

Proof. From the steps in (a) and (b),

$$\begin{aligned}
 \binom{n}{r} = \binom{n}{r+1} & \iff r+1 = n-r, \quad n-1 \geq r \geq 0 \\
 & \iff 2r = n-1, \quad n-1 \geq r \geq 0 \\
 & \iff r = \frac{1}{2}(n-1), \quad n-1 \geq r \geq 0.
 \end{aligned}$$

$$\mathbf{5.a.} \quad \text{For } n \geq 2, \quad \binom{2}{2} + \binom{3}{2} + \cdots + \binom{n}{2} = \binom{n+1}{3}.$$

Proof. For $k = 2$: $\binom{2}{2} = 1 = \binom{2+1}{3} = 1$.

$k \Rightarrow k + 1$: Assume $\binom{2}{2} + \cdots + \binom{k}{2} = \binom{k+1}{3}$.

Then,

$$\begin{aligned} & \binom{2}{2} + \cdots + \binom{k}{2} + \binom{k+1}{2} = \binom{k+1}{3} + \binom{k+1}{2} \\ & = \binom{k+2}{3} \quad (\text{from Pascal's identity } \binom{r}{s} + \binom{r}{s-1} = \binom{r+1}{s}, 1 \leq s \leq r). \end{aligned}$$

5.b. First, $m^2 = 2\binom{m}{2} + m$, $m \geq 2$.

$$2\binom{m}{2} + m \iff \frac{2m!}{2!(m-2)!} + m, \quad m \geq 2$$

$$\iff m(m-1) + m = m^2, \quad m \geq 2.$$

$$\text{Now, } 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Proof.

$$\begin{aligned} & 1^2 + 2^2 + \cdots + n^2 \\ & = 1 + 2\binom{2}{2} + 2 + 2\binom{3}{2} + 3 + \cdots + 2\binom{n}{2} + n \\ & = (1 + 2 + \cdots + n) + 2 \left[\binom{2}{2} + \binom{3}{2} + \cdots + \binom{n}{2} \right] \\ & = (1 + 2 + \cdots + n) + 2\binom{n+1}{3} \\ & = (1 + 2 + \cdots + n) + \frac{2(n+1)!}{3 \cdot 2 \cdot (n-2)!} \\ & = \frac{n(n+1)}{2} + \frac{(n+1)(n)(n-1)}{3} \\ & = \frac{3n(n+1) + 2(n+1)(n)(n-1)}{6} \\ & = \frac{n(n+1)[3 + 2n - 2]}{6} = \frac{n(n+1)(2n+1)}{6}. \end{aligned}$$

5.c. $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}$.

Proof. From (b), $m^2 = 2\binom{m}{2} + m$, or $m(m-1) = 2\binom{m}{2}$, $m \geq 2$.

$$\therefore 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n+1)$$

$$= 2\binom{2}{1} + 2\binom{3}{2} + 2\binom{4}{2} + \cdots + 2\binom{n+1}{2}$$

$$\begin{aligned}
 &= 2 \left[\binom{n+2}{3} \right], \quad \text{from (a)} \\
 &= 2 \frac{(n+2)!}{3!(n-1)!} = \frac{2(n+2)(n+1)(n)}{3 \cdot 2 \cdot 1} \\
 &= \frac{n(n+1)(n+2)}{3}.
 \end{aligned}$$

$$6. \quad \binom{2}{2} + \binom{4}{2} + \cdots + \binom{2n}{2} = \frac{n(n+1)(4n-1)}{6}, \quad n \geq 2.$$

Proof. First, $\binom{2m}{2} = \frac{(2m)!}{2!(2m-2)!} = \frac{2m(2m-1)}{2}$

$$= 2m^2 - m = m^2 + m^2 - m.$$

$$= m^2 + 2 \binom{m}{2}, \quad m \geq 2, \quad \text{from 5(c).}$$

$$\begin{aligned}
 \therefore \binom{2}{2} + \binom{4}{2} + \cdots + \binom{2n}{2} &= 1 + \left[2^2 + 2 \binom{2}{2} + \cdots + n^2 + 2 \binom{n}{2} \right] \\
 &= (1^2 + 2^2 + \cdots + n^2) + 2 \left[\binom{2}{2} + \cdots + \binom{n}{2} \right] \\
 &= \frac{n(n+1)(2n+1)}{6} + 2 \binom{n+1}{3}, \quad \text{from 5(b), 5(a), } n \geq 2 \\
 &= \frac{n(n+1)(2n+1)}{6} + \frac{2(n+1)!}{3 \cdot 2 \cdot (n-2)!} \\
 &= \frac{n(n+1)(2n+1)}{6} + \frac{2(n+1)(n)(n-1)}{6} \\
 &= \frac{n(n+1)[2n+1+2n-2]}{6} \\
 &= \frac{n(n+1)(4n-1)}{6}.
 \end{aligned}$$

$$7. \quad \text{For } n \geq 1, \quad 1^2 + 3^2 + \cdots + (2n-1)^2 = \binom{2n+1}{3}.$$

Proof. $k = 1$: $1^2 = 1 = \binom{2 \cdot 1 + 1}{3} = \binom{3}{3} = 1.$

$k \Rightarrow k+1$: Assume $1^2 + 3^2 + \cdots + (2k-1)^2 = \binom{2k+1}{3}.$

Then,

$$\begin{aligned}
& 1^2 + 3^2 + \cdots + (2k-1)^2 + (2(k+1)-1)^2 \\
&= 1^2 + \cdots + (2k-1)^2 + (2k+1)^2 \\
&= \binom{2k+1}{3} + (2k+1)^2 \\
&= \frac{(2k+1)!}{3!(2k-2)!} + (2k+1)^2 \\
&= \frac{(2k+1)!(2k)(2k-1)}{3!(2k-2)!(2k)(2k-1)} + \frac{6(2k+1)^2(2k)!}{6 \cdot (2k)!} \\
&= \frac{(2k+1)![(2k)(2k-1) + 6(2k+1)]}{3!(2k)!} \\
&= \frac{(2k+1)! [4k^2 - 2k + 12k + 6]}{3!(2k)!} \\
&= \frac{(2k+1)! [(2k+2)(2k+3)]}{3!(2k)!} \\
&= \frac{(2k+3)!}{3!(2k+3-3)!} = \binom{2k+3}{3}.
\end{aligned}$$

So, $k \Rightarrow k+1$.

$$8. \text{ For } n \geq 1, \binom{2n}{n} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} 2^{2n}.$$

Proof.

$$k = 1: \binom{2}{1} = \frac{2!}{1!1!} = 2, \quad \frac{1}{2}2^2 = \frac{4}{2} = 2.$$

$$k \Rightarrow k+1: \text{ Suppose } \binom{2k}{k} = \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2 \cdot 4 \cdot 6 \cdots 2k} 2^{2k}.$$

Then

$$\begin{aligned}
\binom{2k+2}{k+1} &= \frac{(2k+2)!}{(k+1)!(k+1)!} \\
&= \frac{(2k+2)(2k+1)(2k)!}{(k+1)(k+1)k!k!} \\
&= \frac{(2k+2)(2k+1)}{(k+1)(k+1)} \binom{2k}{k} \\
&= \frac{(2k+2)(2k+1)}{(k+1)(k+1)} \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2 \cdot 4 \cdot 6 \cdots 2k} 2^{2k} \\
&= \frac{2(k+1)}{(k+1)(k+1)} \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)(2k+1)}{2 \cdot 4 \cdot 6 \cdots 2k} 2^{2k}
\end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{(k+1)} \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2k+1)}{2 \cdot 4 \cdot 6 \cdots 2k} 2^{2k} \\
 &= \frac{4}{(2k+2)} \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2k+1)}{2 \cdot 4 \cdot 6 \cdots 2k} 2^{2k} \\
 &= \frac{1 \cdot 3 \cdot 5 \cdots (2k+1)}{2 \cdot 4 \cdot 6 \cdots (2k)(2k+2)} \cdot 2^{2k+2}.
 \end{aligned}$$

So, $k \Rightarrow k+1$.

$$\mathbf{9.} \quad 2^n < \binom{2n}{n} < 2^{2n}, \text{ for } n > 1.$$

Proof. $n! < 1 \cdot 3 \cdot 5 \cdots (2n-1)$ for $n > 1$.

Since it's true for $k=2$ ($2 < 1 \cdot 3$), and if $k! < 1 \cdot 3 \cdots (2k-1)$, then

$$\begin{aligned}
 (k+1)! &= k!(k+1) < 1 \cdot 3 \cdot 5 \cdots (2k-1)(k+1) \\
 &< 1 \cdot 3 \cdot 5 \cdots (2k-1)(2k+1).
 \end{aligned}$$

Also, $2^n n! = 2 \cdot 4 \cdot 6 \cdots 2n$, since it's true for $k=1$, and

$$\begin{aligned}
 2^{k+1}(k+1)! &= 2(k+1) \cdot 2^k k! = 2(k+1) \cdot 2 \cdot 4 \cdot 6 \cdots 2k \\
 &= 2 \cdot 4 \cdot 6 \cdots 2k \cdot 2(k+1).
 \end{aligned}$$

$$\text{So, } 2^n n! < 1 \cdot 3 \cdot 5 \cdots (2n-1) 2^n$$

$$\begin{aligned}
 &\Rightarrow 1 < \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n n!} 2^n \\
 &= \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdots 2n} 2^n. \\
 \therefore 2^n &< \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} 2^{2n} = \binom{2n}{n}, \quad \text{by (8).}
 \end{aligned}$$

Now, since $2k-1 < 2k$ for $k \geq 1$,

then $1 \cdot 3 \cdot 5 \cdots (2n-1) < 2 \cdot 4 \cdot 6 \cdots 2n$,

so $\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} < 1$, for $n \geq 1$.

$$\therefore \text{by (8), } \binom{2n}{n} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} 2^{2n} < 2^{2n}, \quad n \geq 1. \quad \mathbf{10.} \quad \text{Given } C_n =$$

$$\frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{n!(n+1)!}, \quad n \geq 0.$$

$$\text{Prove: } C_n = \frac{2(2n-1)}{n+1} C_{n-1}, \quad n \geq 1.$$

Proof.

$$k=1: C_1 = \frac{2!}{1!2!} = 1, \quad C_0 = \frac{0!}{0!1!} = 1.$$

$$\therefore \frac{2(2-1)}{1+1} = 1.$$

$$\text{So, } C_1 = \frac{2(2-1)}{1+1} C_0.$$

$$k \Rightarrow k+1: \text{ Suppose } C_k = \frac{2(2k-1)}{k+1} C_{k-1}.$$

Then

$$\begin{aligned} C_{k+1} &= \frac{(2k+2)!}{(k+1)!(k+2)!} \\ &= \frac{(2k+2)(2k+1)}{(k+1)(k+2)} \cdot \frac{(2k)!}{k!(k+1)!} \\ &= \frac{2(2k+1)}{(k+2)} C_k = \frac{2(2k+1)}{(k+2)} \cdot \frac{2(2k-1)}{(k+1)} C_{k-1} \\ &= \frac{2(2k+1) \cdot 2(2k-1)}{(k+2)(k+1)} \cdot \frac{(2k-2)!}{(k-1)!k!} \\ &= \frac{2(2k+1) \cdot 2}{(k+2)} \cdot \frac{(2k-1)!}{(k-1)!(k+1)!} \\ &= \frac{2(2k+1) \cdot 2k}{(k+2)k} \cdot \frac{(2k-1)!}{(k-1)!(k+1)!} \\ &= \frac{2(2k+1)}{(k+2)} \cdot \frac{(2k)!}{k!(k+1)!} \\ &= \frac{2[2(k+1)-1]}{[(k+1)+1]} C_{(k+1)-1}. \end{aligned}$$

So, $k \Rightarrow k+1$.

1.3 Early Number Theory

1.a. A number is triangular $\iff$ it is of the form $\frac{n(n+1)}{2}$ for some $n \geq 1$.

Proof. $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ from problem 1(a) of Problem Set 1.1.

So, if a number X is triangular then for some n , $1 + 2 + \cdots + n = X$ by definition, and so $X = \frac{n(n+1)}{2}$.

If $X = \frac{n(n+1)}{2}$ for some n , then $X = 1 + 2 + \cdots + n$.

1.b. An integer n is triangular $\iff 8n+1$ is a perfect square.

Proof. If n is triangular, then there is a k such that $n = \frac{k(k+1)}{2}$.

$$\therefore 8n = 4k(k+1),$$

$$\begin{aligned} 8n+1 &= 4k(k+1)+1 \\ &= 4k^2+4k+1 \\ &= (2k+1)^2. \end{aligned}$$

$\therefore n$ triangular $\Rightarrow 8n+1$ is a perfect square.

If $8n+1$ is a perfect square, then there is an integer k such that $k^2 = 8n+1$.

Note that $8n+1$ must be odd. $\therefore k^2$ is odd, and so k is odd. $\therefore$ There is an s such that $2s+1 = k$.

$$\therefore (2s+1)^2 = 8n+1.$$

$$\therefore 4s^2+4s+1 = 8n+1.$$

$$\therefore 4s(s+1) = 8n.$$

$$\therefore \frac{s(s+1)}{2} = n.$$

$\therefore 8n+1$ a perfect square $\Rightarrow n$ triangular. **1.c.** If a and b are consecutive triangular numbers, then $a+b$ is a perfect square.

Proof. Let $1+2+\dots+n = a$.

Then $1+2+\dots+n+n+1 = b$.

$$\begin{aligned} \therefore a+b &= \frac{n(n+1)}{2} + \frac{n(n+1)}{2} + (n+1) \\ &= n(n+1) + (n+1) \\ &= (n+1)(n+1). \end{aligned}$$

So, $a+b$ is a perfect square.

1.d. If n is triangular, then so are $9n+1$, $25n+3$, and $49n+6$.

Proof. Let $1+2+\dots+k = n$.

$$\begin{aligned} \text{Then } 9n+1 &= \frac{9k(k+1)}{2} + 1 = \frac{9k^2+9k+2}{2} \\ &= \frac{(3k+1)(3k+2)}{2} = \frac{s(s+1)}{2} \end{aligned}$$

for $s = 3k+1$, and so by 1(a), $9n+1$ is triangular.

$$\begin{aligned} 25n+3 &= \frac{25k(k+1)}{2} + 3 = \frac{25k^2+25k+6}{2} \\ &= \frac{(5k+2)(5k+3)}{2} = \frac{s(s+1)}{2} \end{aligned}$$

for $s = 5k+2$, and so by 1(a), $25n+3$ is triangular.

$$49n+6 = \frac{49k(k+1)}{2} + 6 = \frac{49k^2+49k+12}{2}$$

$$= \frac{(7k+3)(7k+4)}{2} = \frac{s(s+1)}{2}$$

for $s = 7k + 2$, and so by 1(a), $49n + 6$ is triangular. **2.** $t_n = \binom{n+1}{2}$, $n \geq 1$, t_n the n th triangular number.

Proof.

$$\binom{n+1}{2} = \frac{(n+1)!}{2!(n-1)!} = \frac{(n+1)n}{2},$$

so by 1(a), $t_n = \binom{n+1}{2}$.

$$\mathbf{3.} \quad t_1 + t_2 + \cdots + t_n = \frac{n(n+1)(n+2)}{6}, \quad n \geq 1.$$

Proof. From problem 1(c) of Problem Set 1.1,

$$1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}, \quad n \geq 1.$$

$$\therefore \frac{1 \cdot 2}{2} + \frac{2 \cdot 3}{2} + \cdots + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}.$$

Note that each term k can be written as $\frac{k(k+1)}{2} = t_k$.

$$\therefore t_1 + t_2 + \cdots + t_n = \frac{n(n+1)(n+2)}{6}. \quad \text{The hint given: } t_{k-1} + t_k = k^2.$$

$$t_{k-1} = \frac{(k-1)(k-1+1)}{2} = \frac{k(k-1)}{2}$$

$$\therefore t_{k-1} + t_k = \frac{k(k-1)}{2} + \frac{k(k+1)}{2}$$

$$= \frac{k^2 - k + k^2 + k}{2} = k^2.$$

You could prove the statement using

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

from 5(b) of Problem Set 1.2, and breaking the problem up into even and odd numbers of terms.

$$\mathbf{4.} \quad 9(2n+1)^2 = t_{9n+4} - t_{3n+1}.$$

Proof. Since $t_k = \frac{k(k+1)}{2}$,

$$t_{9n+4} = \frac{(9n+4)(9n+5)}{2}, \quad t_{3n+1} = \frac{(3n+1)(3n+2)}{2}.$$

$$\begin{aligned}
 \therefore t_{9n+4} - t_{3n+1} &= \frac{(81n^2 + 81n + 20) - (9n^2 + 9n + 2)}{2} \\
 &= \frac{72n^2 + 72n + 18}{2} = 36n^2 + 36n + 9 \\
 &= 9(4n^2 + 4n + 1) = 9(2n + 1)^2.
 \end{aligned}$$

5.a. Find two triangular numbers, t_r and t_s , such that $t_r + t_s$ and $t_r - t_s$ are triangular. The triangular numbers are:

$$\begin{aligned}
 &1, 3, 6, 10, 15, 21, 28, 36, 45, 55, \dots \\
 &15 + 21 = 36, \quad 21 - 15 = 6.
 \end{aligned}$$

6.b. Three successive triangular numbers whose product is a perfect square.

$$\begin{aligned}
 \frac{n(n+1)}{2} \cdot \frac{(n+1)(n+2)}{2} \cdot \frac{(n+2)(n+3)}{2} &= k^2 \\
 \frac{(n+1)^2(n+2)^2}{2} \cdot n(n+3) &= k^2, \quad (2k)
 \end{aligned}$$

So, if can find an n such that $\frac{n(n+1)}{2}$ is a perfect square, the problem would be solved. The perfect squares are 1, 4, 9, 16, 25, 36, 49, ...

By trial and error, if $n = 3$, then $\frac{3(3+1)}{2} = 6$.

So, $t_3, t_4, t_5 = 6, 10, 15 = 900 = 30^2$.

6.c. Three successive triangular numbers whose sum is a perfect square.

Trial and error works faster than trying to figure this out.

$$t_5 + t_6 + t_7 = 15 + 21 + 28 = 64 = 8^2.$$

6.a. If t_n is a perfect square, then $t_{4n(n+1)}$ is also a perfect square.

Proof. Assume $t_n = k^2 = \frac{n(n+1)}{2}$.

Then $2k^2 = n(n+1)$.

$$\begin{aligned}
 t_{4n(n+1)} &= \frac{4n(n+1)(4n(n+1) + 1)}{2} \\
 &= \frac{4 \cdot 2k^2 \cdot [4n^2 + 4n + 1]}{2} \\
 &= 4k^2(2n + 1)^2 \\
 &= [2k(2n + 1)]^2, \text{ and so is a square.}
 \end{aligned}$$

6. $t_1 = 1$ is a perfect square.

$t_{4 \cdot 1(1+1)} = t_8 = 36$ is a perfect square.

$$t_{4 \cdot 8(8+1)} = t_{288} = \frac{288(288+1)}{2} = 41,616 = 204^2.$$

7. $t_{n+1}^2 - t_n^2 = k^3$, for some integer k .

Proof. $t_{n+1} = \frac{(n+1)(n+2)}{2}$, $t_n = \frac{n(n+1)}{2}$.

$$\begin{aligned} \therefore t_{n+1}^2 - t_n^2 &= \frac{(n+1)^2(n+2)^2 - (n+1)^2n^2}{4} \\ &= \frac{(n+1)^2[n^2 + 4n + 4 - n^2]}{4} \\ &= \frac{(n+1)^2(4n+4)}{4} = (n+1)^3, \quad \text{for } n \geq 1. \end{aligned}$$

8. $\frac{1}{1} + \frac{1}{3} + \frac{1}{6} + \cdots + \frac{1}{t_n} < 2$.

Proof. Each term can be written as

$$\frac{1}{\frac{k(k+1)}{2}} = \frac{2}{k(k+1)} = 2 \left[\frac{1}{k} - \frac{1}{k+1} \right].$$

$$\begin{aligned} \therefore \frac{1}{1} + \frac{1}{3} + \cdots + \frac{1}{t_n} &= 2 \left[\frac{1}{1} - \frac{1}{2} \right] + 2 \left[\frac{1}{2} - \frac{1}{3} \right] + \cdots + 2 \left[\frac{1}{n} - \frac{1}{n+1} \right] \\ &= 2 \left[\frac{1}{1} - \frac{1}{n+1} \right] = 2 \left(1 - \frac{1}{n+1} \right). \end{aligned}$$

Since $n > 0$, then $n+1 > 0$, so $\frac{1}{n+1} > 0$,

and $-\frac{1}{n+1} < 0$, so $1 - \frac{1}{n+1} < 1$,

so $2 \left(1 - \frac{1}{n+1} \right) < 2$.

$$\therefore \frac{1}{1} + \frac{1}{3} + \cdots + \frac{1}{t_n} < 2. \quad \mathbf{9.a.} \quad t_x = t_y + t_z, \quad x = \frac{n(n+3)}{2} + 1, \quad y = n+1, \quad z = \frac{n(n+3)}{2}.$$

Proof.

$$\begin{aligned} t_y + t_z &= \frac{(n+1)(n+2)}{2} + \frac{n(n+3)}{2} \left[\frac{n(n+3)}{2} + 1 \right] / 2 \\ &= \frac{2 \cdot \frac{(n+1)(n+2)}{2} + \frac{n(n+3)}{2} \left[\frac{n(n+3)}{2} + 1 \right]}{2} \end{aligned}$$

$$\begin{aligned}
 &= \frac{2 \left[\frac{n^2+3n+2}{2} \right] + \frac{n(n+3)}{2} \left[\frac{n(n+3)}{2} + 1 \right]}{2} \\
 &= \frac{2 \left[\frac{n(n+3)}{2} + 1 \right] + \frac{n(n+3)}{2} \left[\frac{n(n+3)}{2} + 1 \right]}{2} \\
 &= \frac{\left[\frac{n(n+3)}{2} + 1 \right] \left[\frac{n(n+3)}{2} + 1 + 1 \right]}{2} \\
 &= t_x, \quad \text{for } n \geq 1 \quad (\text{if } n = 0 \text{ then } z = 0).
 \end{aligned}$$

6. $n = 1$: $t_3 = t_2 + t_2$, or $6 = 3 + 3$.

$n = 2$: $t_6 = t_3 + t_3$, or $21 = 6 + 15$. $n = 3$: $t_{10} = t_4 + t_6$, or $55 = 10 + 45$.

Chapter 2

Divisibility Theory in the Integers

2.1 The Division Algorithm

1. a, b integers, $b > 0$, $\exists$ unique q, r s.t. $a = qb + r$, $0 \leq r < b$.

Pf. By Division Alg., $\exists$ unique q', r' s.t.

$$a = q'b + r', \quad 0 \leq r' < b.$$

$$\therefore a = q'b + r' + 2b - 2b = (q' - 2)b + r' + 2b.$$

Let $q = q' - 2$, $r = r' + 2b$. $\therefore r, q$ unique.

Since $0 \leq r' < b$, then $2b \leq r' + 2b < 3b$, or $2b \leq r + 2b < 3b$.

2. If $a = 6k + 5$, then for some j , $a = 3j + 2$.

Pf. $a = 6k + 5 = 3 \cdot 2k + 3 + 2 = 3(2k + 1) + 2$.

Let $j = 2k + 1$. Conversely, if $a = 3j + 2$,

$8 = 6(0) + 2$, j and 2 are unique, so $8 \neq 6k + 5$.

3.a. If a is an integer, then $a^2 = 3k$ or $a^2 = 3k + 1$.

Pf. By Division Algorithm, $\exists q$ s.t.

$$a = 3q + r, \quad 0 \leq r < 3.$$

$a = 3q$ or $a = 3q + 1$ or $a = 3q + 2$.

$a = 3q$: $\therefore a^2 = 9q^2 = 3(3q^2)$. Let $k = 3q^2$.

$a = 3q + 1$: $\therefore a^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1$.

Let $k = 3q^2 + 2q$.

$a = 3q + 2$: $\therefore a^2 = 9q^2 + 12q + 4 + 1 = 3(3q^2 + 4q + 1) + 1$.

Let $k = 3q^2 + 4q + 1$.

6. If a is an integer, then $a^4 = 9k$ or $9k + 1$.

Pf. Let $a = 3q + r$, $r = 0, 1, 2$.

$$(3q)^4 = 81q^4 = 9k.$$

$$\begin{aligned} (3q+1)^4 &= \binom{4}{0}(3q)^4 + \binom{4}{1}(3q)^3 + \binom{4}{2}(3q)^2 + \binom{4}{3}(3q) + \binom{4}{4} \\ &= 81q^4 + 108q^3 + 54q^2 + 12q + 1 = 9(9q^4 + 12q^3 + 6q^2 + q) + 1 = 9k + 1. \\ (3q+2)^4 &= 27q^4 + 54q^3 + 36q^2 + 8 = 9(3q^3 + 6q^2 + 4q) + 8 = 9k + 8. \end{aligned}$$

[Note: the original gives $(3q+2)^4$ as a cubic expression in q , which is inconsistent with a 4th-power expansion — transcribed faithfully as written; flagged in issues.md.] **c.** If n an integer, then $n^4 = 5k$ or $5k+1$.

Pf. Let $n = 5q + r$, $0 \leq r < 5$.

Consider $n^4 = (5q + r)^4$.

From the binomial expansion, each term is a factor of 5 except the last term:

$$\binom{4}{0}(5q)^4 + \binom{4}{1}(5q)^3r + \binom{4}{2}(5q)^2r^2 + \binom{4}{3}(5q)r^3 + r^4.$$

$r = 0$: then $r^4 = 0$, and $n^4 = 5k$, as all other terms have 5 as a factor.

$r = 1$: then clearly $n^4 = 5k + 1$.

$r = 2$: then $r^4 = 16 = 15 + 1$, so all terms with 15 have 5 as a factor, so again, $n = 5k + 1$.

$r = 3$: then $r^4 = 81 = 80 + 1$, and $80 = 5 \cdot 16$, so, again, $n^4 = 5k + 1$.

4. Prove $3a^2 - 1$ is never a perfect square.

Pf. Suppose $3a^2 - 1 = n^2$, some integer n . By 3(a), $3a^2 - 1 \neq 3k$. or $3a^2 - 1 = 3k$.
 $\therefore 3(a^2 - k) = 2$ or $3(a^2 - k) = 1$, each impossible, since by Div. Alg., $2 = 3 \cdot 0 + 2$ and $1 = 3 \cdot 0 + 1$.

5. For $n \geq 1$, prove $\frac{n(n+1)(2n+1)}{6}$ is an integer.

Pf. $n = 6k + r$, $0 \leq r < 6$. Let $A = \frac{n(n+1)(2n+1)}{6}$.

$r = 0$: Then $A = k(6k+1)(12k+1)$, an integer.

$$r = 1: A = \frac{(6k+1)(6k+2)(12k+3)}{6}$$

$$= \frac{(6k+1)(72k^2 + 42k + 6)}{6}$$

$$= (6k+1)(12k^2 + 7k + 1), \text{ an integer.}$$

$$r = 2: A = \frac{(6k+2)(6k+3)(12k+5)}{6}$$

$$= \frac{(36k^2 + 30k + 6)(12k+5)}{6}$$

$$= (6k^2 + 5k + 1)(12k + 5), \text{ an integer.}$$

$$\begin{aligned} r = 3: A &= \frac{(6k + 3)(6k + 4)(12k + 7)}{6} \\ &= \frac{(36k^2 + 42k + 12)(12k + 7)}{6} \end{aligned}$$

$$= (6k^2 + 7k + 2)(12k + 7), \text{ an int.}$$

$$\begin{aligned} r = 4: A &= \frac{(6k + 4)(6k + 5)(12k + 9)}{6} \\ &= \frac{(72k^2 + 102k + 36)(6k + 5)}{6} \end{aligned}$$

$$= (12k^2 + 17k + 6)(6k + 5), \text{ an int.}$$

$$\begin{aligned} r = 5: A &= \frac{(6k + 5)(6k + 6)(12k + 11)}{6} \\ &= \frac{(36k^2 + 66k + 30)(12k + 11)}{6} \\ &= (6k^2 + 11k + 5)(12k + 11), \text{ an int.} \end{aligned}$$

6. If A an integer, then $A^3 = 7k$ or $7k \pm 1$, some k .

Pf. $A = 7q + r$, $0 \leq r < 7$.

$r = 0$: $A^3 = 7^3 q^3 = 7(7^2 q^3)$. Let $k = 7^2 q^3$.

$$\begin{aligned} r = 1: A^3 &= (7q + 1)^3 = \binom{3}{0}(7q)^3 + \binom{3}{1}(7q)^2 + \binom{3}{2}(7q) + \binom{3}{3} \\ &= 7^3 q^3 + 3 \cdot 7^2 q^2 + 3 \cdot 7q + 1 \\ &= 7[\dots] + 1. \end{aligned}$$

$$r = 3: A^3 = (7q + 3)^3 = 7[\dots] + 28 - 1 = 7[\dots + 9] - 1.$$

$$r = 4: A^3 = (7q + 4)^3 = 7[\dots] + \dots. \text{ Last term is } 4^3 = 64 = 7 \cdot 9 + 1.$$

$$\text{So, } A^3 = 7[\dots + 9] + 1.$$

$$r = 5: A^3 = (7q + 5)^3 = 7[\dots] + \dots. \text{ Last term} = 125 = 7 \cdot 18 - 1.$$

$$\text{So } A^3 = 7[\dots + 18] - 1.$$

$$r = 6: A^3 = (7q + 6)^3 = 7[\dots] + \dots. \text{ Last term } 6^3 = 216 = 31 \cdot 7 - 1.$$

$$\therefore A^3 = 7[\dots + 31] - 1.$$

7. For a, b s.t. $b \neq 0$, $\exists$ unique q, r s.t. $a = qb + r$ and $-\frac{|b|}{2} < r \leq \frac{|b|}{2}$.

Pf. Break up $0 \leq |b|$ into $0 \leq \frac{|b|}{2}$ and $\frac{|b|}{2} < |b|$.

$\exists$ unique q', r' s.t. $a = bq' + r'$, $0 \leq r' < |b|$.

If $0 \leq r' \leq \frac{|b|}{2}$, let $r = r'$, $q = q'$. If $\frac{|b|}{2} < r' < |b|$, then $-\frac{|b|}{2} < r' - |b| < 0$.

$\therefore a = bq' + r' - |b| + |b|$.

If $b \geq 0$, then $a = b(q' + 1) + r' - |b|$, let $r = r' - |b|$, $q = q' + 1$.

If $b < 0$, then $|b| = -b$, so $a = bq' + r' - |b| = bq' + r' + b$, so $a = b(q' + 1) + r' - |b|$, let $q = q' + 1$, $r = r' - |b|$.

9. None of the integers in the sequence below is a perfect square: 11, 111, 1111, 11111, ...

Pf. Any number in the sequence can be written as

$$A_i = 11 + 100 + 1000 + \cdots = 11 \cdot \sum_{i=2}^? 10^i.$$

Each term of $\sum 10^i$ is divisible by 4.

So, $A_i = 4(r_i + 2) + 3$, for certain r_i .

e.g. $11 = 4(2 + 0) + 3$, $111 = 4(25 + 2) + 3$. So $A_i = 4r'_i + 3$. By Div. Alg., r'_i and 3 are unique.

Suppose $A_i = s^2$. Let $s = 4q + r$.

$r \neq 0$, as $s^2 = 16q^2 + 8q \neq 4r'_i + 3$ form.

$r = 1$: $s^2 = 16q^2 + 8q + 1$, and so not of $4r'_i + 3$ form.

$r = 2$: $s^2 = 16q^2 + 16q + 4 = 4(4q^2 + 4q + 1)$, and so not of $4r'_i + 3$ form.

$r = 3$: $s^2 = 16q^2 + 24q + 9 = 4(4q^2 + 6q + 2) + 1$, and so not of $4r'_i + 3$ form.

$\therefore$ There is no s s.t. $s^2 = 4r'_i + 3$.

$\therefore$ All A_i are not perfect squares.

9. If integer $A = r^2s$ for some r, q , then $A = 7k$ or $A = 7k \pm 1$, for some k .

Pf. Let $s = 7k + r$, $0 \leq r < 7$.

[Note: this page's setup is terse/ambiguous in the original — the relation between A , r , s , q and the following $s = 7k + r$ substitution is not fully spelled out. Transcribed faithfully; flagged in issues.md.] From #6 above, $s^3 = 7k$, if $s = 0$.

$s^3 = 7k + 1$, if $s = 1, 2, 4$.

$s^3 = 7k - 1$, if $s = 3, 5, 6$.

Or, $s^3 = 7k$ if $s = 0$;

$s^3 = 7k + 1$ if $s = 1, 2, 4$;

$s^3 = 7k + 6$ if $s = 3, 5, 6$,

for some k , ($i = 0, 1, \dots, 6$).

Now look at $A = r^2$.

Let $r = 7c + d$, $0 \leq d < 7$.

$d = 0$: $r^2 = 7(7c^2) = s^3 = 7k_i$.

$d = 1$: $r^2 = 49c^2 + 14c + 1 = 7(7c^2 + 2c) + 1 = s^3 = 7k_i + 1$.

$d = 2$: $r^2 = 49c^2 + 28c + 4 = 7(7c^2 + 4c) + 4$.

So r^2 last term = $9 = 7 + 2$,

so $r^2 = 7k + 2$.

$d = 4$: r^2 last term = $16 = 14 + 2$,

so $r^2 = 7k + 2$.

$d = 5$: r^2 last term $= 25 = 21 + 4$,

so $r^2 = 7k + 4$.

$d = 6$: r^2 last term $= 36 = 35 + 1$,

so $r^2 = 7k + 1$. Thus, r^2 of form: $7k, 7k + 1, 7k + 2, 7k + 4$.

s^3 of form: $7k, 7k + 1$, or $7k + 6$.

By the uniqueness part of the Division Algorithm, A must be either of form $7k$ or $7k + 1$.

10. For $n \geq 1$, show $n(7n^2 + 5)$ is of form $6k$.

Pf. Let $n = 6k + r$, $0 \leq r < 6$. Let $A = n(7n^2 + 5)$.

$r = 0$: $A = 6k(7(6k)^2 + 5) = 6[\dots]$.

$r = 1$: $A = (6k + 1)(7(6k + 1)^2 + 5)$

$$= 7(6k + 1)^3 + 30k + 5$$

$$= 7[6(\dots) + 1] + 7 \cdot 6k + 5$$

$$= 6[7(\dots)] + 7 \cdot 6k + 12$$

$$= 6[7(\dots) + 5k + 2] = 6k'.$$

$r = 2$: $A = (6k + 2)(7(6k + 2)^2 + 5)$

$$= 7(6k + 2)^3 + 30k + 10$$

$$= 7[6(\dots) + 8] + 30k + 10$$

$$= 6[7(\dots)] + 6[5k + 10]$$

$$= 6[7(\dots) + 5k + 11] = 6k'.$$

$r = 3$: $A = (6k + 3)(7(6k + 3)^2 + 5)$

$$= 7(6k + 3)^3 + 6 \cdot 5k + 15$$

$$= 6[7(\dots)] + 6 \cdot 5k + 6 \cdot 2 + 15$$

$$= 6[7(\dots) + 5k + 34] = 6k'.$$

$r = 4$: $A = (6k + 4)(7(6k + 4)^2 + 5)$

$$= 7(6k + 4)^3 + 6 \cdot 5k + 20$$

$$= 7[6(\dots) + 64] + 6 \cdot 5k + 20$$

$$= 6[7(\dots) + 5k + 78] = 6k'.$$

$r = 5$: $A = (6k + 5)(7(6k + 5)^2 + 5)$

$$= 7(6k + 5)^3 + 6 \cdot 5k + 25$$

$$\begin{aligned}
 &= 7[6(\cdots) + 125] + 6 \cdot 5k + 25 \\
 &= 6[7(\cdots) + 5k + 153] = 6k.
 \end{aligned}$$

11. If n is odd, show $n^4 + 4n^2 + 11$ is of the form $16k$.

Pf. Let $n = 2k + 1$.

$$\begin{aligned}
 n^4 + 4n^2 + 11 &= (n^2 + 2)^2 + 7 \\
 &= [(2k + 1)^2 + 2]^2 + 7 = [4k^2 + 4k + 1 + 2]^2 + 7 \\
 &= (4k^2 + 4k + 3)^2 + 7 \\
 &= 16k^4 + 16k^3 + 12k^2 \\
 &\quad + 16k^3 + 16k^2 + 12k \\
 &\quad + 12k^2 + 12k + 9 + 7 \\
 &= 16k^4 + 32k^3 + 40k^2 + 24k + 16.
 \end{aligned}$$

$k = 2q$ or $2q + 1$.

$k = 2q$: $16(2q)^4 + 32(2q)^3 + 40(2q)^2 + 24(2q) + 16$

$$\begin{aligned}
 &= 16[(2q)^4 + 2(2q)^3 + 10q^2 + 3q + 1] \\
 &= 16x.
 \end{aligned}$$

$k = 2q + 1$: $16(2q + 1)^4 + 32(2q + 1)^3 + 40(2q + 1)^2 + 24(2q + 1) + 16$

$$\begin{aligned}
 &= 16(\quad)^4 + 32(\quad)^3 + 160q^2 + 160q + 40 + 48q + 24 + 16 \\
 &= 16[(\quad)^4 + 2(\quad)^3 + 10q^2 + 10q + 3q + 4 + 1] \\
 &= 16x.
 \end{aligned}$$

2.2 The Greatest Common Divisor

Theorem 2.2. For integers a, b, c :

(a) $a \mid 0$, since $a \cdot 0 = 0$.

$1 \mid a$, since $1 \cdot a = a$.

$a \mid a$, since $a \cdot 1 = a$.

(b) $a \mid 1 \iff a = \pm 1$.

If $a = 1$, then $a \cdot 1 = 1$. If $a = -1$, then $a \cdot (-1) = 1$.

If $a \mid 1$, then $a \cdot c = 1$ for some c .

If $|c| \neq 1$, then $|c| > 1$. By def., $|a||c| > 1$,

$\therefore |a||c| > 1$, contradicting $a \cdot c = 1$.

$\therefore |c| = 1$, $\therefore c = \pm 1$. If $c = 1$, then $ac = a \cdot 1 = a = 1$. If $c = -1$, then $ac = -a = 1$.

(c) If $a \mid b$ and $c \mid d$, then $ac \mid bd$.

$$ax = b, \quad cy = d, \quad \therefore ac(xy) = bd.$$

(d) If $a \mid b$ and $b \mid c$, then $a \mid c$.

$$ax = b, \quad by = c. \quad \therefore a(xy) = b(y) = c.$$

(e) $a \mid b$ and $b \mid a \iff a = \pm b$.

$a \mid b \Rightarrow ax = b$. $b \mid a \Rightarrow by = a$.

$\therefore axy = a$, $xy = 1$. Using (b), $x = \pm 1$.

If $x = 1$, then $bx = b \Rightarrow a = b$.

If $x = -1$, then $ax = b \Rightarrow ax = -a = b$.

$\therefore a = \pm b$.

If $a = b$, then $a \cdot 1 = a = b$, so $a \mid b$ and $b \cdot 1 = a = b$, so $b \mid a$.

If $a = -b$, then $a \cdot (-1) = (-b)(-1) = b$, so $a \mid b$ and $b \cdot (-1) = -b = a$, so $b \mid a$.

Problems

1. $a \mid 1 \Rightarrow \exists c$ s.t. $a \cdot c = b$.

(a) $a \cdot c \cdot (-a) \cdot (-c) = b \cdot (-a) \mid b$.

(d) $-(a \cdot c) = -b \Rightarrow a \cdot (-c) \Rightarrow a \mid (-b)$.

(c) $-(a \cdot c) = -b \Rightarrow (-a) \cdot c = -b \Rightarrow (-a) \mid b$.

2.(a) $a \mid b \Rightarrow \exists x$ s.t. $ax = b$.

$$axc = bc \Rightarrow a \mid bc.$$

(b) $a \mid 1$, $c \mid c \Rightarrow \exists x, y$ s.t. $ax = 1$, $cy = c$.

$$\therefore (ax)(cy) = bc = a^2 \mid bc.$$

(c) If $a \mid b$, then $\exists x$ s.t. $ax = b$.

$$\therefore acx = bc \Rightarrow ac \mid bc.$$

If $ac \mid bc$, then $\exists x$ s.t. $acx = bc$.

Since $c \neq 0$, $ax = b \Rightarrow a \mid b$.

(d) If $a \mid b$ and $c \mid d$, then $\exists x, y$ s.t.

$$ax = b, \quad cy = d.$$

$$\therefore (ax)(cy) = ac(xy) = bd \Rightarrow ac \mid bd.$$

3. Not true. Let $a = 3$, $b = 2$, $c = 7$.

Then $a \mid (b + c) \equiv 3 \mid (2 + 7)$, but $3 \nmid 2$, $3 \nmid 7$.

4.(a) $8 \mid 5^{2n} + 7$.

Pf. $n = 1$: $5^2 + 7 = 32$, and $8 \mid 32$.

Suppose $8 \mid 5^{2k} + 7$. $\therefore \exists x$ s.t. $5^{2k} + 7 = 8x$.

$$\begin{aligned} 5^{2(k+1)} + 7 &= 5^2 \cdot 5^{2k} + 7 \\ &= 5^2(8x) - 5^2 \cdot 7 + 7 \\ &= 5^2(8x) - 7(5^2 - 1) \\ &= 5^2(8x) - 7(24) \\ &= 8 \times (25) - 8(7 \cdot 3) \\ &= 8[25x - 21]. \end{aligned}$$

$\therefore 8 \mid 5^{2(k+1)} + 7$.

(b) $15 \mid 2^{4n} - 1$.

$n = 1$: $15 = 2^4 - 1 = 16 - 1$.

Assume $15 \mid (2^{4k} - 1)$. $\therefore \exists x$ s.t. $15x = 2^{4k} - 1$.

$$\begin{aligned} 2^{4(k+1)} - 1 &= 2^4 \cdot 2^{4k} - 1 + 2^4 - 2^4 \\ &= 2^4(2^{4k} - 1) + (2^4 - 1) \\ &= 2^4(15x) + 15 = 15(x2^4 + 1). \end{aligned}$$

$\therefore 15 \mid 2^{4(k+1)} - 1$.

(c) $5 \mid (3^{3n+1} + 2^{n+1})$.

$n = 1$: $3^{3+1} + 2^2 = 81 + 4 = 85$, and $5 \mid 85$.

Suppose $5 \mid (3^{3k+1} + 2^{k+1})$. $\therefore \exists x$ s.t. $5x = 3^{3k+1} + 2^{k+1}$.

$$\begin{aligned} 3^{3(k+1)+1} + 2^{k+2} &= 3^{3k+4} + 2^{k+2} \\ &= 3^3 \cdot 3^{3k+1} + 2 \cdot 2^{k+1} + 3^3 \cdot 2^{k+1} - 3^3 \cdot 2^{k+1} \\ &= 3^3(3^{3k+1} + 2^{k+1}) - 2^{k+1}(3^3 - 2) \\ &= 3^3(5x) - 2^{k+1}(25) \\ &= 5(x3^3 - 5 \cdot 2^{k+1}). \end{aligned}$$

$\therefore$ true for $k + 1$.

(d) $21 \mid 4^{n+1} + 5^{2n-1}$.

$n = 1$: $4^2 + 5^1 = 21$.

Suppose for k , $21 \mid 4^{k+1} + 5^{2k-1}$.
 $\therefore \exists x$ s.t. $21x = 4^{k+1} + 5^{2k-1}$.

$$\begin{aligned} 4^{k+2} + 5^{2(k+1)-1} &= 4^{k+2} + 5^{2k+1} \\ &= 4 \cdot 4^{k+1} + 5^2 \cdot 5^{2k-1} + 4 \cdot 5^{2k-1} - 4 \cdot 5^{2k-1} \\ &= 4(4^{k+1} + 5^{2k-1}) + 21(5^{2k-1}) \\ &= 4(21x) + 21(5^{2k-1}). \end{aligned}$$

$\therefore$ true for $k + 1$.

(e) $24 \mid 2 \cdot 7^n + 3 \cdot 5^n - 5$.

$n = 1$: $2 \cdot 7^1 + 3 \cdot 5^1 - 5 = 14 + 15 - 5 = 24$.

Suppose $24 \mid 2 \cdot 7^k + 3 \cdot 5^k - 5$.

$\therefore \exists x$ s.t. $24x = 2 \cdot 7^k + 3 \cdot 5^k - 5$.

$$\begin{aligned} 2 \cdot 7^{k+1} + 3 \cdot 5^{k+1} - 5 &= 7(2 \cdot 7^k) + 5(3 \cdot 5^k) - 5 \\ &= 2(2 \cdot 7^k) + 5(2 \cdot 7^k) + 5(3 \cdot 5^k) - 5 + 5 \cdot 5 - 5 \cdot 5 \\ &= 5(2 \cdot 7^k + 3 \cdot 5^k - 5) + 2(2 \cdot 7^k) - 5 + 5 \cdot 5 \\ &= 5(24x) + 2(2 \cdot 7^k) + 20 \quad [\text{Eq. 1}] \end{aligned}$$

But $24 \mid 4 \cdot 7^k + 20$.

pf. $k = 1$: $4 \cdot 7 + 20 = 48 = 24 \cdot 2$. Suppose $24 \mid 4 \cdot 7^s + 20$.

$\therefore \exists y$ s.t. $24y = 4 \cdot 7^s + 20$.

$$\begin{aligned} \therefore 4 \cdot 7^{s+1} + 20 &= 7(4 \cdot 7^s) + 20 \\ &= 7(4 \cdot 7^s + 20) + 20 - 140 \\ &= 7(24y) - 24 \cdot 5. \end{aligned}$$

$\therefore \exists q$ s.t. $24q = 4 \cdot 7^k + 20$.

$\therefore [\text{Eq. 1}] = 5(24x) + 24q$.

$\therefore$ true for $k + 1$.

5. For integer a , one of $a, a + 2, a + 4$ is divisible by 3.

Pf. (a) Suppose $3 \nmid a$. $\therefore a = 3q_1 + 1$ or $a = 3q_2 + 2$.

$3q_1 + 1$: Then $a + 2 = 3q_1 + 3 = 3(q_1 + 1)$,

so $3 \mid a + 2$.

$3q_2 + 2$: Then $a + 4 = 3q_2 + 6 = 3(q_2 + 2)$

so $3 \mid a + 4$.

(b) Suppose $3 \nmid a + 2$. $\therefore a + 2 = 3q_1 + 1$ or

$a + 2 = 3q_2 + 2$. $3q_1 + 1$: $\therefore a = 3q_1 - 1$, so $a + 4 = 3q_1 + 3$,

$$\therefore 3 \mid a + 4.$$

$$3q_2 + 2: \therefore a = 3q_2, \text{ so } 3 \mid a.$$

(c) Suppose $3 \nmid a + 4$. $\therefore a + 4 = 3q_1 + 1$ or
 $a + 4 = 3q_2 + 2$.

$$3q_1 + 1: \therefore a = 3q_1 - 3, \text{ so } 3 \mid a.$$

$$3q_2 + 2: \therefore a = 3q_{12} - 2, a + 2 = 3q_{12}, \\ \text{so } 3 \mid a + 2.$$

$$\mathbf{6.(a)} \quad 2 \mid a(a + 1).$$

Pf. By Div. Alg., $a = 2q$ or $a = 2q + 1$.

$$2q: \text{ Then } a(a + 1) = 2q(2q + 1)$$

$$= 2[q(2q + 1)],$$

$$\therefore 2 \mid a(a + 1).$$

$$2q + 1: \text{ Then } a(a + 1) = (2q + 1)(2q + 2)$$

$$= 2(2q + 1)(q + 1),$$

$$\therefore 2 \mid a(a + 1).$$

$$\mathbf{(b)} \quad 3 \mid a(a + 1)(a + 2).$$

$$a = 3q, a = 3q + 1, \text{ or } a = 3q + 2.$$

$$3q: a(a + 1)(a + 2) = 3q(3q + 1)(3q + 2),$$

$$\therefore 3 \mid a(a + 1)(a + 2). \quad 3q + 1: a(a + 1)(a + 2) = (3q + 1)(3q + 2)(3q + 3)$$

$$= 3(3q + 1)(3q + 2)(q + 1),$$

$$\therefore 3 \mid a(a + 1)(a + 2).$$

$$3q + 2: a(a + 1)(a + 2) = (3q + 2)(3q + 3)(3q + 4)$$

$$= 3(3q + 2)(q + 1)(3q + 4),$$

$$\therefore 3 \mid a(a + 1)(a + 2).$$

$$\mathbf{6.(b)} \quad 3 \mid a(2a^2 + 7).$$

Pf. $a = 3q, 3q + 1, 3q + 2$.

$$3q: a(2a^2 + 7) = 3q(\cdots), \therefore 3 \mid a(2a^2 + 7).$$

$$3q + 1: a(2a^2 + 7) = (3q + 1)[2(3q + 1)^2 + 7]$$

$$= (3q + 1)[2(9q^2 + 6q + 1) + 7]$$

$$= (3q + 1)(18q^2 + 12q + 9)$$

$$= 3(3q + 1)(6q^2 + 4q + 3).$$

$$\therefore 3 \mid a(2a^2 + 7).$$

$$\begin{aligned}
 3q + 2: a(2a^2 + 7) &= (3q + 2) \left[2(3q + 2)^2 + 7 \right] \\
 &= (3q + 2) \left[2(9q^2 + 12q + 4) + 7 \right] \\
 &= (3q + 2)(18q^2 + 24q + 15) \\
 &= 3(3q + 2)(6q^2 + 8q + 5).
 \end{aligned}$$

$$\therefore 3 \mid a(2a^2 + 7).$$

(c) If a is odd, then $32 \mid (a^2 + 3)(a^2 + 7)$.

Pf. $\exists q$ s.t. $a = 2q + 1$.

$$\begin{aligned}
 (a^2 + 3)(a^2 + 7) &= (4q^2 + 4q + 4)(4q^2 + 4q + 8) \\
 &= 16q^4 + 16q^3 + 32q^2 \\
 &\quad + 16q^3 + 16q^2 + 32q \\
 &\quad + 16q^2 + 16q + 32 \\
 &= 16q^4 + 32q^3 + 64q^2 + 48q + 32.
 \end{aligned}$$

If q is even, then $q = 2x$,
so $16q^4 = 16(2x)^4 = 32 \cdot 2^3x^4$,
and $48q = 96x$.

$\therefore$ all terms divisible by 32. If q is odd, $q = 2x + 1$,

$$\begin{aligned}
 \therefore 16q^4 &= 16(2x + 1)^4 = 16 \left[2^4x^4 + \binom{4}{1}2^3x^3 + \binom{4}{2}2^2x^2 + \binom{4}{3}2x + 1 \right] \\
 &\quad + 32q^3 + 64q^2 + 96x + 80 \\
 &= 16(2x + 1)^4 + 32q^3 + 64q^2 + 96x + 80 \\
 &= 32 \left[2^4x^4 + \binom{4}{1}2^2x^3 + \binom{4}{2}2x^2 + \binom{4}{3}x \right] \\
 &\quad + 32q^3 + 64q^2 + 96x + 96.
 \end{aligned}$$

So all terms divisible by 32.

7. If a, b are odd, then $16 \mid (a^4 + b^4 - 2)$.

Pf. Let $a = 2r + 1$, $b = 2s + 1$.

$$\begin{aligned}
 a^4 &= (2r + 1)^4 = 2^4r^4 + \binom{4}{1}2^3r^3 + \binom{4}{2}2^2r^2 + \binom{4}{3}2r + 1 \\
 &= 16r^4 + 32r^3 + 24r^2 + 8r + 1.
 \end{aligned}$$

$$\therefore a^4 + b^4 - 2 = 16r^4 + 32r^3 + 24r^2 + 8r + 24s^4 + 32s^3 + \dots$$

All terms divisible by 16 except perhaps $24r^2 + 8r$, $24s^2 + 8s$.

But if r is even, then $r = 2w$ for some w , and $\therefore 24r^2 + 8r = 96w^2 + 16w$, which is divisible by 16.

If r is odd, then $r = 2w + 1$, some w .

$$\begin{aligned}\therefore 24r^2 + 8r &= 24(2w + 1)^2 + 8(2w + 1) \\ &= 96w^2 + 96w + 24 + 16w + 8 \\ &= 96w^2 + 96w + 16w + 32,\end{aligned}$$

which is divisible by 16.

Similarly for $24s^2 + 8s$.

$$\therefore 16 \mid a^4 + b^4 - 2.$$

8.(a) If a, b are odd, then $a^2 + b^2 \neq c^2$ for some integer c .

Pf. Let $a = 2r + 1$, $b = 2s + 1$.

$$\begin{aligned}\therefore a^2 + b^2 &= 4r^2 + 4r + 1 + 4s^2 + 4s + 1 \\ &= 4(K) + 2 = 2(K')\end{aligned}$$

$\therefore$ if c exists, it must be even. Let $c = 2w$, some unique w .

$$\therefore c^2 = 4w^2.$$

By Div. Alg., $a^2 + b^2 = 4q + r$, where q and r are unique. From above, $a^2 + b^2 = 4K + 2$.

If $a^2 + b^2 = c^2$, then $a^2 + b^2 = 4w^2$, which means q and r are not unique.

$\therefore a^2 + b^2 \neq c^2$ if a, b are odd.

(b) Let a, b, c, d be four consecutive integers. Then $a \cdot b \cdot c \cdot d = e^2 - 1$, for some e .

Pf. A few examples show that the product of the 1st and last terms is close to the product of the middle two terms, and that the perfect square in question is the average of the two products. An average exists because the two products are even.

$$\therefore a(a + 1)(a + 2)(a + 3) = \left[\frac{a(a + 3) + (a + 1)(a + 2)}{2} \right]^2.$$

Suppose a is even. Then $a = 2n$.

$$\begin{aligned}\therefore a(a + 1)(a + 2)(a + 3) &= 2n(2n + 1)(2n + 2)(2n + 3) \\ &= (4n^2 + 2n)(4n^2 + 10n + 6) \\ &= 16n^4 + 40n^3 + 24n^2 \\ &\quad + 8n^3 + 20n^2 + 12n \\ &= 16n^4 + 48n^3 + 44n^2 + 12n.\end{aligned}$$

$$\begin{aligned}
 & \left[\frac{2a(2n+1)(2n+2) + (2n+3)}{2} \right]^2 - 1 \\
 &= \left[\frac{4n^2 + 6n + 4n^2 + 6n + 2}{2} \right]^2 - 1 \\
 &= \left[\frac{8n^2 + 12n + 2}{2} \right]^2 - 1 \\
 &= (4n^2 + 6n + 1)^2 - 1 \\
 &= (4n^2 + 6n + 1)(4n^2 + 6n + 1) - 1 \\
 &= 16n^4 + 24n^3 + 4n^2 \\
 &\quad + 24n^3 + 36n^2 + 6n \\
 &\quad + 4n^2 + 6n + 1 - 1
 \end{aligned}$$

If a is odd, then $a = 2n + 1$.

$$\begin{aligned}
 \therefore a(a+1)(a+2)(a+3) &= (2n+1)(2n+2)(2n+3)(2n+4) \\
 &= (4n^2 + 6n + 2)(4n^2 + 14n + 12) \\
 &= 16n^4 + 56n^3 + 48n^2 \\
 &\quad + 24n^3 + 84n^2 + 72n \\
 &\quad + 8n^2 + 28n + 24 \\
 &= 16n^4 + 80n^3 + 140n^2 + 100n + 24.
 \end{aligned}$$

$$\begin{aligned}
 & \left[\frac{a(a+3) + (a+1)(a+2)}{2} \right]^2 - 1 = \\
 & \left[\frac{(2n+1)(2n+4) + (2n+2)(2n+3)}{2} \right]^2 - 1 \\
 &= \left[\frac{4n^2 + 10n + 4 + 4n^2 + 10n + 6}{2} \right]^2 - 1 \\
 &= (4n^2 + 10n + 5)^2 - 1 \\
 &= (4n^2 + 10n + 5)(4n^2 + 10n + 5) - 1 \\
 &= 16n^4 + 40n^3 + 20n^2 \\
 &\quad + 40n^3 + 100n^2 + 50n \\
 &\quad + 20n^2 + 50n + 25 - 1
 \end{aligned}$$

$$= 16n^4 + 80n^3 + 140n^2 + 100n + 24. \checkmark$$

9. $(a+1)^3 - a^3$ is never divisible by 2.

Pf. Suppose a is even. $\therefore a = 2n$.

$$\begin{aligned} \therefore (a+1)^3 - a^3 &= (2n+1)^3 - (2n)^3 \\ &= 8n^3 + \binom{3}{1}4n^2 + \binom{3}{2}2n + 1 - 8n^3 \\ &= 12n^2 + 6n + 1. \end{aligned}$$

$$= 2(K) + 1, \text{ so } (a+1)^3 - a^3 \text{ is odd.}$$

Suppose a is odd. $\therefore a = 2n+1$.

$$\begin{aligned} \therefore (a+1)^3 - a^3 &= (2n+2)^3 - (2n+1)^3 \\ &= \binom{3}{0}(2n)^3 + \binom{3}{1}(2n)^2 + \binom{3}{2}(2n) + 1 - (2n+1)^3 \end{aligned}$$

[handwriting unclear on this expansion step]

$$\begin{aligned} &= (2n+2)[3(2n+1) + 3] + 1 \\ &= (2n+2)(6n+6) + 1 \\ &= 2[(2n+1)(3n+3)] + 1 \\ &= 2(K) + 1, \text{ so } (a+1)^3 - a^3 \text{ is odd.} \end{aligned}$$

10.(a) $a \neq 0$, $\gcd(a, 0) = |a|$.

Pf. From Th. 2.2 (p.21), we know that $a \mid 0$ and $a \mid a$. $\therefore |a|$ is a common divisor.

Let c be another common divisor.

$\therefore c \mid a$ and $\therefore |c| \leq |a|$ by Th. 2.2.

$\therefore |a|$ is gcd. **(b)** $a \neq 0$, $\gcd(a, a) = |a|$.

Pf. By Th. 2.2, $a \mid a$. $\therefore |a|$ is a common divisor.

Let c be another common divisor.

$\therefore c \mid a$, and $\therefore |c| \leq |a|$, by Th. 2.2.

$\therefore |a|$ is gcd.

(c) $a \neq 0$, $\gcd(a, 1) = 1$.

Pf. By Th. 2.2, $1 \mid a$, $1 \mid 1$. $\therefore 1$ is a common divisor.

Let c be another common divisor.

$\therefore c \mid 1$, and $\therefore |c| \leq 1$ (Th. 2.2).

$\therefore 1$ is gcd.

$$\mathbf{11.} \gcd(a, b) = \gcd(a - b, b) = \gcd(a, b - a).$$

Pf. Let x be the gcd of any one pair. Then

since $x \mid a$, $x \mid b \Rightarrow x \mid (a - b)$, x is a common divisor of any other pair.

Let c be another common divisor.

Since $c \mid a \iff c \mid (a + b)$ and $c \mid b \iff c \mid (b - c)$, then $c \mid x$ by Th. 2.6. $\therefore |c| \leq |x|$.

$\therefore x$ is the gcd of the other pair.

[Note: this proof's logic is compressed/hard to follow exactly from the handwriting; transcribed as legible.] $\mathbf{12.} n \neq 0$, a any integer, $\gcd(a, a + n) \mid n$.

Pf. Let $d = \gcd(a, a + n)$.

$\exists x, y$ s.t. $a = dx$, $a + n = dy$.

$$\therefore dx + n = dy, \quad n = d(y - x). \quad \therefore d \mid n.$$

And by Th. 2.2, $d \mid 1 \iff d = \pm 1$. $\therefore \gcd(a, n) = 1$.

$\mathbf{13.(a).(1)}$ Let m, n be any integers and let $d = \gcd(a, b)$.

$\exists m, n$ s.t. $a = dm$ and $b = dn$.

$c: ax + by = dmx + dny = d(mx + ny)$.

$\therefore c \mid d$. Let $d = \gcd(a, b)$.

(2) Suppose $\gcd(a, b) = d$. $\exists x_0, y_0$ s.t. $d = ax_0 + by_0$. But $d \mid c$, so that $c = d\rho$, for some ρ .

$$\therefore c = d\rho = (ax_0 + by_0)\rho = dx_0\rho + by_0\rho.$$

$\therefore$ Let $x = x_0\rho$, $y = y_0\rho$.

(b) Let x, y be s.t. $ax + by = \gcd(a, b)$. Then $\gcd(x, y) = 1$. *Pf.* Let $d = \gcd(a, b)$.

$ax + by = d$.

Since $d \mid a$ and $d \mid b$, then $\frac{a}{d}$ and $\frac{b}{d}$ are integers.

$\frac{a}{d}x + \frac{b}{d}y = 1$, and $\therefore x$ and y are relatively prime.

$\therefore \gcd(x, y) = 1$.

$\mathbf{14.(a)}$ Since $9(2a + 1) + (-2)(9a + 4) = 1$, then by Th. 2.4, $2a + 1$ and $9a + 4$ are relatively prime, so $\gcd(2a + 1, 9a + 4) = 1$.

(b) $(-7)(5a + 2) + 5(7a + 3) = 1$. $\therefore \gcd(5a + 2, 7a + 3) = 1$.

(c) $\gcd(3a, 3a + 2) \mid 2$ by problem 12.

$\therefore \gcd = 1$ or 2 . But a odd, so $3a$ is odd.

$2 \nmid 3a$, $\therefore \gcd = 1$.

$\mathbf{15.} \gcd(2a - 3b, 4a - 5b) \mid b$.

Pf. Let $d = \gcd(2a - 3b, 4a - 5b)$.

For all x, y , by the Corollary on p.23, $x(2a - 3b) + y(4a - 5b)$ is a multiple of d .

$$\therefore \exists n \text{ s.t. } dn = (2)(2a - 3b) + (1)(4a - 5b) = b.$$

$$\therefore d \mid b.$$

Now let $\delta = -1$. $\gcd(2a + 3, 4a + 5) \mid (1)$.

$$\therefore \gcd = 1.$$

16. If a is odd, $12 \mid a^2 + (a + 2)^2 + (a + 4)^2 + 1$.

Pf. Let $a = 2n + 1$.

$$\begin{aligned} \therefore (2n + 1)^2 + (2n + 3)^2 + (2n + 5)^2 + 1 \\ &= 4n^2 + 4n + 1 \\ &\quad + 4n^2 + 12n + 9 \\ &\quad + 4n^2 + 20n + 25 + 1 \\ &= 12n^2 + 36n + 36 = 12(n^2 + 3n + 3). \end{aligned}$$

17. For all $n \geq 0$, $\frac{(3n)!}{(3!)^n}$ is an integer.

Pf. $n = 1$: $\frac{3!}{3!} = 1$.

$k \Rightarrow k + 1$: Suppose $\frac{(3k)!}{(3!)^k} = R$ is an integer.

$$\begin{aligned} \therefore \frac{[3(k + 1)]!}{(3!)^{k+1}} \\ &= \frac{(3k + 1)(3k + 2)(3k + 3)(3k)!}{3 \cdot 2 \cdot 1 \cdot (3!)^k} \\ &= \frac{3(k + 1)(3k + 2)(3k + 1)}{3 \cdot 2} \cdot R \\ &= \frac{(k + 1)(3k + 2)(3k + 1)}{2} \cdot R. \end{aligned}$$

If k is odd, then $k + 1$ is even, so $\frac{k + 1}{2} = x$, some integer x .

If k is even, then $3k + 2$ is even, so $\frac{3k + 2}{2} = x$, some integer x .

$\therefore$ entire expression is an integer.

18.(a) $6 \mid a(a + 1)(a + 2)$.

Pf. $6 = 3 \cdot 2$, and $\gcd(2, 3) = 1$ (Problem 12).

Let $R = a(a + 1)(a + 2)$.

If a is even, then $2 \mid a$, $\therefore 2 \mid R$.

If a is odd, then $2 \mid (a+1)$, $\therefore 2 \mid R$.

Let $a = 3q + r$. If $r = 0$, then $3 \mid a$, $\therefore 3 \mid R$.

If $r = 1$, then $a + 2 = 3q + 3$, $3 \mid a + 2$, $3 \mid R$.

If $r = 2$, then $a + 1 = 3q + 3$, $3 \mid a + 1$, $3 \mid R$.

$\therefore 3 \mid R$ and $2 \mid R$, and by Corollary 2 on p.24, $3 \cdot 2 \mid R$.

$\therefore 6 \mid a(a+1)(a+2)$.

(b) $24 \mid a(a+1)(a+2)(a+3)$.

Pf. $n = 1$: $1 \cdot 2 \cdot 3 \cdot 4 = 24$.

$k \Rightarrow k+1$: Suppose $24 \mid k(k+1)(k+2)(k+3)$.

$\therefore 24\rho = k(k+1)(k+2)(k+3)$, some ρ .

$$\begin{aligned} \therefore (k+1)(k+2)(k+3)(k+4) &= \\ k(k+1)(k+2)(k+3) + 4(k+1)(k+2)(k+3) &= \\ = 24\rho + 4(k+1)(k+2)(k+3). \end{aligned}$$

But by (a), $(k+1)(k+2)(k+3) = 6q$ for some q .

$$\therefore (k+1)(k+2)(k+3)(k+4) = 24\rho + 24q.$$

$\therefore 24 \mid (k+1)(k+2)(k+3)(k+4)$. **(c)** $120 \mid a(a+1)(a+2)(a+3)(a+4)$.

Pf. $n = 1$: $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$.

$k \Rightarrow k+1$: Suppose $120 \mid k(k+1)(k+2)(k+3)(k+4)$.

$\therefore \exists \rho$ s.t. $120\rho = k(k+1)(k+2)(k+3)(k+4)$.

But

$$\begin{aligned} (k+1)(k+2)(k+3)(k+4)(k+5) &= \\ k(k+1)(k+2)(k+3)(k+4) + 5(k+1)(k+2)(k+3)(k+4) &= \\ = 120\rho + 5 \cdot 24q, \quad \text{for some } q \text{ by (b) above} &= \\ = 120(\rho + q). \end{aligned}$$

$\therefore 120 \mid (k+1)(k+2)(k+3)(k+4)(k+5)$.

19.(a) $6 \mid a(a^2 + 11)$.

Pf. Let $a = 6q + r$, where $0 \leq r < 6$.

Consider each case for r .

$r = 0$: $a(a^2 + 11) = 6q[(6q)^2 + 11]$. $\therefore 6 \mid a(a^2 + 11)$. $r = 1$: $a(a^2 + 11) = (6q+1)^3 + (6q+1)11$

$$\begin{aligned} &= 6^3q^3 + \binom{3}{1}(6q)^2 + \binom{3}{2}(6q) + \binom{3}{3} + (6q)11 + 11 \\ &= 6[\dots] + \binom{3}{3} + 11 \end{aligned}$$

$$= 6[\dots] + 12 = 6[\dots].$$

$$r = 2: a(a^2 + 11) = (6q + 2)^3 + (6q + 2)11$$

$$= 6[\dots] + \binom{3}{0} 2^3 + (6q)11 + 22$$

$$= 6[\dots] + 30 = 6[\dots].$$

$$r = 3: a(a^2 + 11) = (6q + 3)^3 + (6q + 3)11$$

$$= 6[\dots] + \binom{3}{0} 3^3 + 33$$

$$= 6[\dots] + 27 + 33 = 6[\dots] + 60$$

$$= 6[\dots].$$

$$r = 4: a(a^2 + 11) = (6q + 4)^3 + (6q + 4)11$$

$$= 6[\dots] + \binom{3}{0} 4^3 + 44$$

$$= 6[\dots] + 64 + 44 = 6[\dots] + 108$$

$$= 6[\dots] + 6 \cdot 18 = 6[\dots].$$

$$r = 4: a(a^2 + 11) = (6q + 5)^3 + (6q + 5) \cdot 11$$

[the original relabels this case "r = 4" again, though it follows the r = 4 case above — likely meant to be r = 5; transcribed as written]

$$= 6[\dots] + \binom{3}{0} 5^3 + 55$$

$$= 6[\dots] + 125 + 55 = 6[\dots] + 6 \cdot 30$$

$$= 6[\dots].$$

(b) a is odd, then $24 \mid a(a^2 - 1)$.

Pf. First, show a^2 is of form $8k + 1$.

Let $a = 4q + r$. $\therefore r = 1$ or 3 since a is odd.

$$\therefore a^2 = 16q^2 + 8q + 1 = 8k + 1$$

$$\text{or } a^2 = 16q^2 + 24q + 9 = 8k' + 1.$$

So, $a(a^2 - 1) = a \cdot 8k$, for some k .

$\therefore 8 \mid a(a^2 - 1)$.

By #18 above, $6 \mid (a - 1)(a)(a + 1)$, so

$3 \mid (a-1)(a)(a+1) \equiv 3 \mid a(a^2-1)$. Since $\gcd(3, 8) = 1$, $\therefore 24 \mid a(a^2-1)$ by Corollary 2 on p.24.

(c) a, b odd $\Rightarrow 8 \mid (a^2 - b^2)$.

Pf. By (b) above, a^2 is of form $8k + 1$ and b^2 is of form $8k' + 1$.

$$\begin{aligned} \therefore a^2 - b^2 &= 8k + 1 - (8k' + 1) \\ &= 8(k - k'), \text{ some } k, k'. \end{aligned}$$

$$\therefore 8 \mid (a^2 - b^2).$$

(d) $2 \nmid a, 3 \nmid a \Rightarrow 24 \mid (a^2 + 23)$.

Pf. Let $a = 12q + r, 0 \leq r < 12$.

r can only be 1, 5, 7, 11 since $2 \nmid a$, and since $3 \nmid a, r \neq 3$ or 9.

$\therefore r$ can only be 1, 5, 7, 11.

$$\begin{aligned} \therefore a^2 + 23 &= (12q + r)^2 + 23 \\ &= 144q^2 + 24qr + r^2 + 23 \\ &= 24(6q^2 + qr) + r^2 + 23. \end{aligned}$$

$$= 24[\dots] + r^2 + 23.$$

$$r = 1: r^2 + 23 = 24.$$

$$r = 5: r^2 + 23 = 48 = 24(2).$$

$$r = 7: r^2 + 23 = 72 = 24(3).$$

$$r = 11: r^2 + 23 = 144 = 24(6).$$

$$\begin{aligned} \therefore a^2 + 23 &= 24[\dots] + r^2 + 23 \\ &= 24[\dots] + 24k, \text{ some } k. \end{aligned}$$

$$\therefore 24 \mid (a^2 + 23).$$

(e) $360 \mid a^2(a^2 - 1)(a^2 - 4)$.

$$\text{Pf. } a^2(a^2 - 1)(a^2 - 4) = a^2(a + 1)(a - 1)(a + 2)(a - 2)$$

$$= (a - 2)(a - 1)(a)(a + 1)(a + 2)(a).$$

$360 = 5 \cdot 9 \cdot 8$, and 5, 9, 8 are relatively prime.

By #18, $(a - 2)(a - 1)(a)(a + 1)(a + 2)$ is divisible by 24 and 120. $\therefore$ it is divisible by 8 and 5.

Also, $(a - 2)(a - 1)(a)$ and $a(a + 1)(a + 2)$ are both divisible by 6 and so are both divisible by 3, and $\therefore$ the entire product is divisible by 9.

$\therefore$ Entire product divisible by 360 by Corollary 2, p.24.

20.(a) $\gcd(a, b) = 1$, $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$.

Pf. $1 = ax + by$, $1 = au + cv$ for some x, y, u, v .

$$\begin{aligned} 1 &= (ax + by)(au + cv) = a^2xu + acxv + abuy + bcyv \\ &= ak_1 + bck_2. \end{aligned}$$

$\therefore a, bc$ relatively prime.

(b) $\gcd(a, b) = 1$, $c \mid a$, then $\gcd(b, c) = 1$.

Pf. $\exists x, y$ s.t. $ax + by = 1$, and $\exists n$ s.t. $cn = a$.

$\therefore cnx + by = 1 \Rightarrow \gcd(c, b) = 1$. **(c)** $\gcd(a, b) = 1$, then $\gcd(ac, b) = \gcd(c, b)$.

Pf. Let $d = \gcd(c, b)$. Need to show

(1) $d \mid ac$ ($d \mid b$ by def.)

(2) if $k \mid ac$ and $k \mid b$, then $k \mid d$.

(1): Since $d \mid c$, $\exists n$ s.t. $dn = c$, so $d(na) = ca$, $\Rightarrow d \mid ca$.

(2) $\exists x, y$ s.t. $d = cx + by$.

Since $k \mid b$, then $\exists n$ s.t. $kn = b$.

$\therefore d = cx + kny$.

Since $\gcd(a, b) = 1$, $\exists p, q$ s.t. $ap + bq = 1$.

$\therefore apc + bqc = c$.

$\therefore d = (apc + bqc)x + kny$

$$= acpx + kngcx + kny.$$

But $k \mid ac \Rightarrow \exists r$ s.t. $kr = ac$.

$$\therefore d = krpx + kngcx + kny$$

$$= k(rpx + ngcx + ny).$$

$\therefore k \mid d$.

$\therefore$ By Theorem 2.6, $\gcd(c, b) = \gcd(ac, b)$.

(d) $\gcd(a, b) = 1$, $c \mid a + b \Rightarrow \gcd(a, c) = \gcd(b, c) = 1$.

Pf. $\gcd(a, b) = 1 \Rightarrow \exists x, y$ s.t. $ax + by = 1$.

$c \mid a + b \Rightarrow \exists n$ s.t. $cn = a + b$.

$\therefore cn - b = a$.

$\therefore (cn - b)x + by = 1$.

$cnx - bx + by = 1$,

so, $cnx + b(y - x) = 1 \Rightarrow \gcd(c, b) = 1$.

Similarly, $cn - a = b$, so

$ax + (cn - a)y = 1$

$ax + cny - ay = 1$,

so, $a(x - y) + cny = 1 \Rightarrow \gcd(a, c) = 1$. **(e)** $\gcd(a, b) = 1$, $d \mid ac$, $d \mid bc$, then $d \mid c$.

Pf. $\exists x, y$ s.t. $ax + by = 1$. $\therefore acx + bcy = c$.

But $ac = dn$ and $bc = dm$ for some n, m .

$\therefore dnx + dmy = c$, $d(nx + my) = c$.

$\therefore d \mid c$.

(f) $\gcd(a, b) = 1$, then $\gcd(a^2, b^2) = 1$.

Pf. From (c) above, let $c = a$.

$\therefore \gcd(a, b) = 1 \Rightarrow \gcd(a^2, b) = \gcd(a, b) = 1$.

Also, $\gcd(a, b) = \gcd(b, a) = \gcd(b^2, a) = \gcd(b, a) = 1$.

So, $\gcd(a^2, b) = \gcd(a, b^2) = 1$.

Now apply (c) again to get

$$\gcd(a \cdot a, b^2) = \gcd(a, b^2) = 1.$$

$\therefore \gcd(a^2, b^2) = 1$. **21.(a)** $d \mid n \Rightarrow 2^d - 1 \mid 2^n - 1$.

Pf. From Problems 1.1, #3,

$$a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \cdots + a + 1).$$

$\therefore 2^n - 1 = 2^{n-1} + 2^{n-2} + \cdots + 2 + 1$ (n terms).

$2^d - 1 = 2^{d-1} + 2^{d-2} + \cdots + 2 + 1$ (d terms).

Since $d \mid n$, $\exists x$ s.t. $dx = n$.

$$\begin{aligned} \therefore 2^n - 1 &= 2^{dx} - 1 = (2^d)^x - 1 \\ &= (2^d - 1)(2^{d(x-1)} + 2^{d(x-2)} + \cdots + 2^d + 1). \end{aligned}$$

$\therefore 2^d - 1 \mid 2^n - 1$.

Could also look at this explicitly:

$$\begin{aligned} \frac{2^n - 1}{2^d - 1} &= \frac{n \text{ terms}}{d \text{ terms}} = \frac{dx \text{ terms}}{d \text{ terms}} \\ &= \frac{(d \text{ terms}) + (d \text{ terms}) + \cdots + (d \text{ terms})}{(d \text{ terms})} \\ &= 2^{d(x-1)} + 2^{d(x-2)} + \cdots + 2^d + 1. \end{aligned}$$

(b) $31 = 2^5 - 1$. Since $5 \mid 35$, $2^5 - 1 \mid 2^{35} - 1$.

$127 = 2^7 - 1$, and $7 \mid 35$. $\therefore 2^7 - 1 \mid 2^{35} - 1$.

22. What values of n does $t_n \mid t_1 + t_2 + \cdots + t_n$?

From Problems 1.3, #3,

$$t_1 + t_2 + \cdots + t_n = \frac{n(n+1)(n+2)}{6},$$

and from Problems 1.3, $\#1(a)$, $t_n = \frac{n(n+1)}{2}$.

$$\therefore \frac{\frac{n(n+1)(n+2)}{6}}{\frac{n(n+1)}{2}} = \frac{n+2}{3}.$$

$\therefore t_n$ divides $t_1 + \cdots + t_n$ when $\frac{n+2}{3}$ is an integer, or $n = 1, 4, 7, 10, \dots$

23. If $a \mid bc$, show $a \mid \gcd(a, b) \gcd(a, c)$. *Pf.* Let $d_1 = \gcd(a, b)$, $d_2 = \gcd(a, c)$.

$\therefore \exists x, y, u, v$ s.t. $d_1 = ax + by$, $d_2 = au + cv$,
and $\exists n$ s.t. $an = bc$.

$$\begin{aligned} \therefore d_1 d_2 &= (ax + by)(au + cv) \\ &= a^2 xu + acxv + abuy + bcyv \\ &= a(axu + cxv + buy) + anyv \\ &= a(axu + cxv + buy + nyv). \end{aligned}$$

$\therefore a \mid d_1 d_2$.

2.3 The Euclidean Algorithm

1.(a) $\gcd(143, 227)$

$$227 = 1 \cdot 143 + 84$$

$$143 = 1 \cdot 84 + 59$$

$$84 = 1 \cdot 59 + 25$$

$$59 = 2 \cdot 25 + 9$$

$$25 = 2 \cdot 9 + 7$$

$$9 = 1 \cdot 7 + 2$$

$$7 = 3 \cdot 2 + 1$$

$$2 = 2 \cdot 1 + 0 \quad \therefore \gcd(143, 227) = 1.$$

(b) $\gcd(306, 657)$

$$657 = 2 \cdot 306 + 45$$

$$306 = 6 \cdot 45 + 36$$

$$45 = 1 \cdot 36 + 9$$

$$36 = 4 \cdot 9 + 0 \quad \therefore \gcd(306, 657) = 9.$$

(c) $\gcd(272, 1479)$

$$1479 = 5 \cdot 272 + 119$$

$$272 = 2 \cdot 119 + 34$$

$$119 = 3 \cdot 34 + 17$$

$$34 = 17 \cdot 2 + 0 \quad \therefore \gcd(272, 1479) = 17.$$

2.(a) $\gcd(56, 72) = 56x + 72y$

$$72 = 1 \cdot 56 + 16$$

$$56 = 3 \cdot 16 + 8$$

$$16 = 2 \cdot 8 + 0 \quad \gcd = 8.$$

$$\therefore 8 = 56 - 3 \cdot 16$$

$$= 56 - 3(72 - 56)$$

$$= (4)56 - (3)72.$$

(b) $\gcd(24, 138) = 24x + 138y$

$$138 = 5 \cdot 24 + 18$$

$$24 = 1 \cdot 18 + 6$$

$$18 = 3 \cdot 6 + 0 \quad \gcd = 6.$$

$$\therefore 6 = 24 - 18$$

$$= 24 - (138 - 5 \cdot 24)$$

$$= (6)24 - 138.$$

(c) $\gcd(119, 272) = 119x + 272y$

$$272 = 2 \cdot 119 + 34 \quad \therefore 17 = 119 - 3 \cdot 34$$

$$119 = 3 \cdot 34 + 17 \quad = 119 - 3(272 - 2 \cdot 119)$$

$$34 = 17 \cdot 2 + 0 \quad = (3)119 - (3)272$$

$$\gcd = 17$$

[Note: the original's final back-substitution gives $17 = (3)119 - (3)272$, but expanding $17 = 119 - 3(272 - 2 \cdot 119)$ algebraically gives $7 \cdot 119 - 3 \cdot 272$, not $3 \cdot 119 - 3 \cdot 272$ — likely a coefficient error in the original notes. Transcribed as written.] (d) $\gcd(1769, 2378) = 1769x + 2378y$

$$2378 = 1 \cdot 1769 + 609$$

$$1769 = 3 \cdot 609 + 58$$

$$609 = 10 \cdot 58 + 29$$

$$58 = 2 \cdot 29 + 0 \quad \gcd = 29.$$

$$\begin{aligned} \therefore 29 &= 609 - 10 \cdot 58 \\ &= 609 - 10(3 \cdot 609 - 1769) \\ &= (-29) \cdot 609 + (10) \cdot 1769 \\ &= (-29)(2378 - 1769) + (10)1769 \\ &= (39)1769 - (29)2378. \end{aligned}$$

$$\mathbf{3.} \quad d_1, d_2 > 0. \quad d = \gcd(d_1, d_2) \iff \gcd\left(\frac{a}{d_1}, \frac{a}{d_2}\right) = 1.$$

Pf. (a) Let m, n be s.t. $dm = d_1, dn = d_2$. Then, by Th. 2.7 (since $d = \gcd$),

$$d = \gcd(dm, dn) = d \cdot \gcd(m, n) = d \cdot \gcd\left(\frac{a}{d_1}, \frac{a}{d_2}\right).$$

$$\therefore 1 = \gcd\left(\frac{a}{d_1}, \frac{a}{d_2}\right).$$

(b) If $\gcd\left(\frac{a}{d_1}, \frac{a}{d_2}\right) = 1$, then by Th. 2.7,

$$\gcd(d_1, d_2) = \gcd\left(d \cdot \frac{a}{d_1}, d \cdot \frac{a}{d_2}\right) = |d| \gcd\left(\frac{a}{d_1}, \frac{a}{d_2}\right) = |d|.$$

$$\mathbf{4.} \quad \gcd(a, b) = 1.$$

$$\text{(a)} \quad \gcd(a + b, a - b) = 1 \text{ or } 2.$$

Pf. Let $d = \gcd(a + b, a - b)$. By Corollary p.25, d is a divisor of all linear combinations of $a + b$ and $a - b$.

$$\therefore d \mid (a + b) + (a - b) \Rightarrow d \mid 2a.$$

$$d \mid (a + b) - (a - b) \Rightarrow d \mid 2b.$$

$$\therefore d \leq \gcd(2a, 2b) = 2 \gcd(a, b) = 2.$$

$$\therefore d = 1 \text{ or } 2.$$

$$\text{(b)} \quad \gcd(2a + b, a + 2b) = 1 \text{ or } 3.$$

Pf. Let $d = \gcd(2a + b, a + 2b)$.

$$\therefore d \mid 2(2a + b) - (a + 2b) \iff d \mid 3a.$$

$$\therefore d \mid -1 \cdot (2a + b) + 2(a + 2b) \iff d \mid 3b.$$

$$\therefore d \leq \gcd(3a, 3b) = 3 \gcd(a, b) = 3.$$

$$\therefore d = 1, 2, \text{ or } 3.$$

$$\text{If } d = 2, \text{ then } d \mid 3a \Rightarrow d \mid a, d \mid 3b \Rightarrow d \mid b,$$

since $\gcd(2, 3) = 1$ and by Th. 2.5 (Euclid's lemma).

Since $\gcd(a, b) = 1$ and $d \mid a, d \mid b \Rightarrow d \mid \gcd(a, b) = 1 \Rightarrow d \mid 1 \Rightarrow d \leq 1$. But if $2 \mid a$ and $2 \mid b$, then $\gcd(a, b) \neq 1$.

$\therefore d \neq 2$.

$\therefore d = 1$ or 3 .

(c) $\gcd(a + b, a^2 + b^2) = 1$ or 2 .

Pf. Let $d = \gcd(a + b, a^2 + b^2)$.

Then $d \mid a^2 + b^2$ and $d \mid (a + b)^2 - 3ab$.

Since $d \mid (a + b)$, then $d \mid 3ab$.

As in (c) above, since $d \mid (a + b)$, then $d \mid 3ab$.

Since $d \mid a + b$ and $\gcd(a, b) = 1$, then by Problem 20(d) p.26, $\gcd(a, d) = \gcd(b, d) = 1$.

$\therefore$ By Euclid's lemma, $d \mid 3ab \Rightarrow d \mid 3$.

$\therefore d \leq 3$. Since $\gcd(2, 3) = 1$, then

if $d = 2$, then $2 \mid 3a \Rightarrow 2 \mid a$, either of which contradicts $\gcd(a, d) = 1$.

$\therefore d \neq 2$.

$\therefore d = 1$ or 3 .

[Note: this proof is dense and somewhat repetitive in the original; transcribed faithfully but the logic from " $d \mid 3ab \Rightarrow d \mid 3$ " onward is compressed.] (d) $\gcd(a + b, a^2 - ab + b^2) = 1$ or 3 .

Pf. Let $d = \gcd(a + b, a^2 - ab + b^2)$.

$\therefore d \mid a^2 - ab + b^2 \Rightarrow d \mid (a + b)^2 - 3ab$.

As in (c) above, since $d \mid (a + b)$, then $d \mid 3ab$.

Since $d \mid a + b$ and $\gcd(a, b) = 1$, then by Problem 20(d) p.26, $\gcd(a, d) = \gcd(b, d) = 1$.

$\therefore$ By Euclid's lemma, $d \mid 3ab \Rightarrow d \mid 3a \Rightarrow d \mid 3$.

$\therefore d \leq 3$. Since $\gcd(2, 3) = 1$, then

if $d = 2$, then $2 \mid 3a \Rightarrow 2 \mid a$, either of which contradicts $\gcd(a, d) = 1$.

$\therefore d \neq 2$.

$\therefore d = 1$ or 3 .

5. $a, b > 0, n \geq 1$.

(a) If $\gcd(a, b) = 1$, then $\gcd(a^n, b^n) = 1$. *Pf.* $n = 1$: $\gcd(a, b) = 1$ was assumed.

$k \Rightarrow k + 1$: Assume $\gcd(a^k, b^k) = 1$.

By Problem 20(c) p.26, $\gcd(a^k, b^k) = \gcd(a^k, b^k) = 1$.

Since $\gcd(a, b) = \gcd(b, a)$, then $\gcd(b^{k+1}, a^k) = 1$, and $\therefore$ again by 20(c) p.26, $\gcd(a^{k+1}, a^k) = \gcd(b^{k+1}, a^k) = 1$.

(b) $a^n \mid b^n \Rightarrow a \mid b$.

Pf. $n = 1$: Clearly, $a^1 \mid b^1 = a \mid b$.

$k \Rightarrow k + 1$: Assume $a^k \mid b^k \Rightarrow a \mid b$.

$\exists x$ s.t. $xa^k = b^k$, $\exists y$ s.t. $ay = b$.

$$\therefore xa^{k+1} \cdot \frac{a}{b} = (x)b^k = \frac{b^{k+1}}{y}$$

$$\therefore xy a^{k+1} = b^{k+1}.$$

$$\therefore a^{k+1} \mid b^{k+1}.$$

[Note: the algebra in this last step is hard to follow exactly from the handwriting; transcribed as legible.] Another proof, as suggested by author.

Let $d = \gcd(a, b)$ and let r, s be s.t. $a = rd$, $b = sd$.

$\gcd(r, s) = 1$ by problem 12(b), p.25.

$\therefore \gcd(r^k, s^k) = 1$ by (a) above.

But since $a^k = b^k$, then since $a^k = b^k$, $r^k d^k = s^k d^k$, so $r^k = s^k$.

$\therefore \gcd(r^k, s^k) = r^k = s^k$, so $r = 1$.

$\therefore$ from $a = rd$, $a = d$, and from $b = sd$, $b = sa$, $\therefore a \mid b$.

$$\mathbf{6.} \gcd(a, b) = 1 \Rightarrow \gcd(a + b, ab) = 1.$$

Pf. Let c be a divisor of $a + b$ and ab .

By 20(d) p.26, $\gcd(c, a) = \gcd(c, b) = 1$.

Since $c \mid ab$ and $\gcd(c, a) = 1$, then by Euclid's lemma, $c \mid b$.

Similarly, $c \mid a$, and $\gcd(c, b) = 1 \Rightarrow c \mid a$.

So, $c \mid a$, $c \mid b$. $\therefore c \leq \gcd(a, b) = 1$, $\therefore c = 1$.

$$\mathbf{7.(a)} \ a \mid b \iff \gcd(a, b) = |a|.$$

Pf. If $a \mid a$ and $a \mid b$, $\therefore a$ is a common divisor. Suppose d is another common divisor.

$\therefore \exists n$ s.t. $a = dn$, $\therefore |a| = |d||n|$.

Since $a \neq 0$, and $d \neq 0$,

$\therefore |n| \geq 1$, $\therefore |a| = |d||n| \geq |d|$.

$\therefore |a| = \gcd(a, b) = |a|$.

(2) Assume $\gcd(a, b) = |a|$. If $a > 0$, then

$\exists n$ s.t. $b = |a|n$, so that $b = an \Rightarrow a \mid b$.

If $a < 0$, then $|a| = -a \Rightarrow a \mid b = -(-an)$, $\therefore a \mid b$.

$$\mathbf{(b)} \ a \mid b \iff \text{lcm}(a, b) = |b|.$$

Pf. (1) $a \mid b \Rightarrow a \mid |b|$, and clearly $b \mid |b|$.

Let c be another common multiple.

$\therefore a \mid c$ and $b \mid c$ (and $c > 0$).

$|b| \leq |c| \Rightarrow |c| \geq |b|$, and $|c| \geq |b|$,

$\therefore |c| \geq |b| = |c|$, $\therefore |b| = |c|$, by def.

$\therefore |b| = \text{lcm}(a, b) = |b| \Rightarrow a \mid |b|$ by def.

(2) $\text{lcm}(a, b) = |b| \Rightarrow \exists n$ s.t. $an = |b|$.

If $b > 0$, then $an = b \Rightarrow a \mid b$.

If $b < 0$, then $an = -b$, $\therefore an = -b$, $a(-n) = b$,

$\therefore a \mid b$. (c) Transitivity of (a) & (b) means $\gcd(a, b) \mid a \iff \text{lcm}(a, b) = |b|$.

Or, directly,

(1) Assume $\gcd(a, b) = |a|$.

$\therefore |a| \cdot \text{lcm}(a, b) = |ab| = |a||b|$.

$\therefore \text{lcm}(a, b) = |b|$.

(2) Assume $\text{lcm}(a, b) = |b|$.

$\therefore a \mid b \Rightarrow |a| \leq |b|$.

Let c be another common divisor.

$\exists n$ s.t. $a = cn \Rightarrow |a| = |c||n|$.

But $|a| \leq |c|$, $\therefore |c||n| \leq |c|$,

$\therefore |a| = \text{gcd}(a, b)$.

8.(a) $\text{lcm}(143, 227)$

$$227 = 1 \cdot 143 + 84$$

$$143 = 2 \cdot 84 - 25$$

$$84 = 3 \cdot 25 + 9$$

$$25 = 2 \cdot 9 + 7$$

$$9 = 7 + 2$$

$$7 = 3 \cdot 2 + 1$$

$$\therefore \text{gcd}(143, 227) = 1 \quad \therefore \text{lcm} = 143 \cdot 227 = 32,461. \quad \text{(b)} \text{ lcm}(306, 657)$$

$$657 = 2 \cdot 306 + 45$$

$$306 = 7 \cdot 45 - 9$$

$$45 = 5 \cdot 9$$

$$\therefore \text{gcd} = 9, \quad \therefore \text{lcm} = \frac{306 \cdot 657}{9} = 22,338.$$

(c) $\text{lcm}(272, 1479)$

$$1479 = 5 \cdot 272 + 119$$

$$272 = 2 \cdot 119 + 34$$

$$119 = 4 \cdot 34 - 17$$

$$34 = 2 \cdot 17$$

$$\text{gcd} = 17, \quad \therefore \text{lcm} = \frac{272 \cdot 1479}{17} = 23,664.$$

9. $a, b > 0$. $\text{gcd}(a, b) \mid \text{lcm}(a, b)$.

Pf. Since $\text{gcd}(a, b) \cdot \text{lcm}(a, b) = ab$, let

$d = \text{gcd}(a, b)$. $\therefore \exists n, m$ s.t. $a = dn$, $b = dm$.

$$\therefore d \cdot \text{lcm}(a, b) = (dn)(dm).$$

$$\therefore \text{lcm}(a, b) = d(nm) \Rightarrow d \mid \text{lcm}(a, b).$$

10.(a) $\text{gcd}(a, b) = \text{lcm}(a, b) \iff a = \pm b$.

Pf. (1) Let $d = \text{gcd}(a, b)$, $l = \text{lcm}(a, b)$.

$$\therefore d \cdot l = ab.$$

Since $d = l$, $\exists x$ s.t. $dx = a$.

$$\therefore d \cdot dx = a \Rightarrow d \mid a.$$

$\therefore d \mid b$ and $b \mid d$ by Th. 2.2(c) on p.21.

$$\therefore d = \pm a.$$

Similarly, $|d| = |a| = |b| = \pm b$.

(2) If $a = \pm b$, then $a \mid b$ and $b \mid a$.

$$\therefore \gcd(a, b) = \text{lcm}(a, b) = |a| = |b|.$$

$$\text{(b) } k > 0, \text{lcm}(ka, kb) = k \cdot \text{lcm}(a, b).$$

$$\text{Pf. } \gcd(ka, kb) \cdot \text{lcm}(ka, kb) = k^2 \cdot ab.$$

$$\therefore k \cdot \gcd(a, b) \cdot \text{lcm}(ka, kb) = k^2 |ab|.$$

$$\therefore \gcd(a, b) \cdot \text{lcm}(ka, kb) = k |ab|$$

$$= k \cdot \gcd(a, b) \cdot \text{lcm}(a, b).$$

$$\therefore \text{lcm}(ka, kb) = k \cdot \text{lcm}(a, b). \quad \text{(c) If } m \text{ is a common multiple of } a, b, \text{ then } \text{lcm}(a, b) \mid m.$$

$$\text{Pf. Let } l = \text{lcm}(a, b).$$

Let q, r be s.t. $m = lq + r$, $0 \leq r < l$.

If $r = 0$, then $l \mid m$.

Assume $0 < r < l$.

$\therefore r = m - lq$. Since m, l are multiples of a and b , $\exists x, y$ s.t.

$$r = ax - ayq = a(x - yq)$$

$$= bu - bvq = b(u - vq).$$

$\therefore r$ is a multiple of a, b , and

$\therefore r \geq l$, which contradicts $r < l$.

11. Let a, b, c be s.t. no two of which are zero.

Let $d = \gcd(a, b, c)$.

$$\text{(a) } d = \gcd(\gcd(a, b), c).$$

$$\text{Pf. Let } f = \gcd(a, b) \text{ and let } g = \gcd(f, c).$$

(1) $g \mid f \Rightarrow g \mid a, g \mid b$. Since $g \mid c$, then $g \mid \gcd$ (2) Note that $d \mid f$.

$$\text{Pf. } f = ax + by, \text{ some } x, y \text{ (Th. 2.3).}$$

$$a = du, b = dv, \text{ some } u, v.$$

$$\therefore f = dux + dvy, \therefore d \mid f.$$

Since $d \mid c$, then $d \mid f$, $\therefore d \mid f$.

$$(1) + (2) \Rightarrow d = g.$$

$$\text{(b) } d = \gcd(a, \gcd(b, c)) = \gcd(\gcd(a, c), b).$$

Proofs identical to (a) above, switching letters.

$$\text{12. Find } x, y, z \text{ s.t. } \gcd(198, 288, 512) = 198x + 288y + 512z.$$

From (11) above,

$$\gcd(198, 288, 512) = \gcd(\gcd(198, 288), 512).$$

$$\begin{aligned} 288 &= 198 + 90 & 18 &= 198 - 2 \cdot 90 \\ 198 &= 2 \cdot 90 + 18 & &= 198 - 2(288 - 198) \\ 90 &= 5 \cdot 18 & &= (-2)288 + 3 \cdot 198 \end{aligned}$$

$$\therefore \gcd(198, 288) = 18.$$

Now for $\gcd(18, 512)$:

$$512 = 28 \cdot 18 + 8$$

$$\begin{aligned} 18 &= 2 \cdot 8 + 2 & \therefore 2 &= 18 - 2 \cdot 8 \\ 8 &= 4 \cdot 2 & &= 18 - 2(512 - 28 \cdot 18) \\ \therefore \gcd(18, 512) &= 2 & &= 57 \cdot 18 - 2 \cdot 512. \end{aligned}$$

$$\therefore \gcd(198, 288, 512) = 2.$$

$$\begin{aligned} \therefore 2 &= 57 \cdot 18 - 2 \cdot 512 \\ &= 57[3 \cdot 198 - 2 \cdot 288] - 2 \cdot 512 \\ &= 171 \cdot 198 - 114 \cdot 288 - 2 \cdot 512. \end{aligned}$$

2.4 The Diophantine Equation $ax + by = c$

1.(a) $6x + 8y = 22$.

Since $\gcd(6, 8) = 2$ and $2 \mid 22$, then it can be solved.

[Note: the original handwriting appears to read " $\gcd(6, 8) = 1$ ", which is incorrect ($\gcd(6, 8) = 2$); transcribed with the corrected value since the divisibility condition $2 \mid 22$ is what actually makes the equation solvable.]

(b) $33x + 14y = 115$.

$\gcd(33, 14) = 1$, $\therefore$ it can be solved.

(c) $14x + 35y = 93$.

$\gcd(14, 35) = 7$, $7 \nmid 93$. $\therefore$ can't be solved.

2. Use Euclid's Algorithm to get $d = \gcd(a, b)$, then express d in terms of a, b , then multiply d to get c and x_0, y_0 .

(a) $56x + 72y = 16$

$$\begin{aligned} 72 &= 56 + 16 & 16 &= 56 - 5 \cdot 56 - 72 \dots \\ 56 &= 3 \cdot 16 + 8 & &= 56 - 3(72 - 56) \end{aligned}$$

$$16 = 2 \cdot 8 \quad \text{gcd} = 8$$

$$\therefore 5 \cdot 56 + 20 \cdot 56 - 15 \cdot 72 \dots$$

$$\therefore (20, -15) \text{ a solution.}$$

$$\therefore x = 20 + \frac{72}{8}t, \quad y = -15 - \frac{56}{8}t$$

or $x = 20 + 9t$, $y = -15 - 7t$.

[Note: the intermediate back-substitution arithmetic on this page is hard to read precisely; the final solution form ($x = 20 + 9t$, $y = -15 - 7t$) is legible and consistent with $\text{gcd}(56, 72) = 8$, so it is transcribed with higher confidence than the intermediate steps.]

(b) $24x + 138y = 18$

$$138 = 5 \cdot 24 + 18$$

$$24 = 1 \cdot 18 + 6$$

$$18 = 3 \cdot 6 \quad \text{gcd} = 6.$$

$$\therefore 6 = 24 - 18 = 24 - (138 - 5 \cdot 24) = 6 \cdot 24 - 138.$$

$$\therefore (18, -3) \text{ a solution.}$$

$$\therefore x = 18 + \frac{138}{6}t = 18 + 23t, \quad y = -3 - \frac{24}{6}t = -3 - 4t.$$

(c) $22x + 35y = 11$

$$22 = 6 \cdot 35 + 11? \quad \therefore 11 - 5 \cdot 2$$

$$35 = 3 \cdot 11 + 2 \quad = 11 - (35 - 3 \cdot 11)$$

$$11 = 5 \cdot 2 + 1 \quad \text{gcd} = 1$$

$$2 = 2 \cdot 1$$

[Note: this page's Euclidean-algorithm steps and back-substitution are very faint/hard to read precisely; transcribed best-effort. The original gives a final solution noted as roughly $x = 176 + 35t$, $y = -111 - 22t$ (written as "(176, -1111) a solution" — the y -value's exact digits are uncertain) — needs visual double-check against the original page.]

3.(a) $18x + 45y = 48$

[continues on next page]

$$3 = 2 + 1 \quad = 2 \cdot 3 - 5$$

$$2 = 2 \cdot 1 \quad = 2(18 - 3 \cdot 5) - 5$$

$$\therefore \text{gcd} = 1 \quad = 2 \cdot 18 - 7 \cdot 5$$

$$\therefore 48 = 96 \cdot 18 - (48 \cdot 7) \cdot 5$$

$$\therefore (96, -336) \text{ a solution.}$$

$$\therefore x = 96 + 5t$$

$$y = -336 - 18t$$

Since $x, y > 0$, $96 + 5t > 0 \Rightarrow t > -19.2$

$-336 - 18t > 0 \Rightarrow t < -18.7$

$\therefore t = -19$

$$x = 96 + 5(-19) = 1$$

$$y = -336 - 18(-19) = 6$$

(b) $54x + 21y = 906$

$$54 = 2 \cdot 21 + 12 \qquad 3 = 12 - 9$$

$$21 = 1 \cdot 12 + 9 \qquad = 12 - (21 - 12) \cdot 1$$

$$12 = 1 \cdot 9 + 3 \qquad = 2(54 - 2 \cdot 21) - 21$$

$$9 = 3 \cdot 3 \qquad = 2 \cdot 54 - 5 \cdot 21$$

$$\gcd = 3 \qquad 906 = (302 \cdot 2)(54) - (302 \cdot 5)(21)$$

$\therefore (604, -1510)$ a solution.

$$\therefore x = 604 + 7t \Rightarrow t > -86.3 \Rightarrow x > -86.3$$

$$y = -1510 - 18t > 0 \Rightarrow t < -83.9$$

$$\therefore t = -84, -85, -86.$$

$$\therefore (x, y) = (16, 2), (9, 20), (2, 38).$$

(c) $123x + 360y = 99$

$$360 = 3 \cdot 123 - 9 \qquad 3 = 14 \cdot 9 - 123$$

$$123 = 14 \cdot 9 - 3 \qquad = 14(3 \cdot 123 - 360) - 123$$

$$9 = 3 \cdot 3 \qquad = 41 \cdot 123 - 14 \cdot 360$$

$$\gcd = 3 \qquad 99 = (33 \cdot 41)123 - (33 \cdot 14)360$$

$\therefore (1353, -462)$ a solution.

$$\therefore x = 1353 + 120t > 0 \Rightarrow t \geq -11.278$$

$$y = -462 - 41t > 0 \Rightarrow t < -11.3$$

$\therefore$ no t exists, so no positive solutions.

(d) $158x - 57y = 7$

$$158 = 3 \cdot 57 - 13 \qquad 1 = 3 \cdot 2 - (5 \cdot 3)$$

$$\begin{aligned}
 57 &= 4 \cdot 13 + 5 & &= 2 \cdot 3 - 5(2 \cdot 13 - 57) \cdot 5 \\
 13 &= 2 \cdot 5 + 3 & &= 2 \cdot 13 - 5 \cdot 57 - 4 \cdot 13 \\
 5 &= 3 + 2 & &= 22 \cdot 13 - 5 \cdot 57 - 5 \cdot 57 \\
 3 &= 2 + 1 & &= 22(3 \cdot 57 - 158) - 5 \cdot 57 \\
 2 &= 2 \cdot 1 & &= 61(57) - 22 \cdot 158
 \end{aligned}$$

$\gcd = 1$

$$\therefore 7 = 427 \cdot 57 - 154 \cdot 158.$$

$$\therefore (-154, -427) \text{ a solution.}$$

$$\therefore x_0 = -154 - 57t > 0 \Rightarrow t < -2.7 \Rightarrow t \leq -3$$

$$y = -427 - 158t > 0 \Rightarrow t < -2.7 \Rightarrow t \leq -3$$

4. $\gcd(a, b) = 1$, then $ax - by = c$ has infinitely many positive solutions.

Pf. Assume $a, b > 0$.

Since $1 \mid c$, a solution exists. Let x_0, y_0 be a solution. $\therefore ax_0 - by_0 = c$. By Corollary on p.26, all solutions are given by:

$$x = x_0 - bt \quad y = y_0 - at.$$

For $x, y > 0$, $x_0 - bt > 0, \Rightarrow t < \frac{x_0}{b}$

$$y_0 - at > 0 \Rightarrow t < \frac{y_0}{a}.$$

$\therefore$ if $t < \min\left(\frac{x_0}{b}, \frac{y_0}{a}\right)$, then

$$t < \frac{x_0}{b} \Rightarrow bt < x_0 \Rightarrow x_0 - bt > 0.$$

$$t < \frac{y_0}{a} \Rightarrow at < y_0 \Rightarrow y_0 - at > 0.$$

There are infinitely many t s.t. $t < \min\left(\frac{x_0}{b}, \frac{y_0}{a}\right)$. **5.(a)** $ax + by + cz = d$ is solvable in integers $\iff \gcd(a, b, c) \mid d$.

Pf. (1) Let $g = \gcd(a, b, c)$. $\therefore \exists p, q, r$ s.t.

$$gp = a, \quad gq = b, \quad gr = c.$$

$$\therefore gpx + gqy + grz = g(px + qy + rz) = d.$$

$\therefore g \mid d$.

(2) Let $g = \gcd(a, b, c)$ and suppose $g \mid d$.

By Lemma, $\exists x_0, y_0, z_0$ s.t. $g = ax_0 + by_0 + cz_0$.

Let t be s.t. $g \cdot t = d$.

$$\therefore d = gt = ax_0t + by_0t + cz_0t.$$

$\therefore$ a solution is (x_0t, y_0t, z_0t) .

Lemma. Given a, b, c , not all of which are zero. There exist integers x, y, z s.t.

$$\gcd(a, b, c) = ax + by + cz.$$

Pf. Analogous to proof of Th. 2.3. Let $S = \{au + bv + cz \mid au + bv + cz > 0, u, v, w \in \mathbb{Z}\}$.

S is non-empty: suppose $a \neq 0$.

$\therefore |a| = a \cdot d$ or $b \cdot 0 > 0$, where

$d = 1 \cdot \text{sign}(a)$.

By Well Ordering Principal, S has a minimum value, d .

By def. of S , $\exists$ integers x, y, z s.t.

$$d = ax + by + cz.$$

Let q, r be s.t. (by Division Alg.):

$$a = qd + r, \quad 0 \leq r < d.$$

$$\therefore r = a - qd = a - q(ax + by + cz)$$

$$= a(1 - qx) - by - cqz.$$

If $r > 0$, then $r \in S$ and $r < d$.

But d is smallest element. $\therefore r = 0$.

$\therefore a = qd \Rightarrow d \mid a$.

Similarly, $d \mid b, d \mid c$.

$\therefore d$ is a common divisor.

Let δ be any other common divisor of a, b, c . Let $\delta h = a, \delta j = b, \delta k = c$.

$$d = ax + by + cz = \delta hx + \delta jy + \delta kz \Rightarrow \delta \mid d.$$

By Th. 2.2, $|\delta| \leq |d|$.

$\therefore d = \gcd(a, b, c)$. **Alternate Lemma:** By #11, p.32, $\gcd(a, b, c) = \gcd(\gcd(a, b), c)$.

Let $u = \gcd(a, b)$. $\therefore \exists x, y$ s.t.

$$u = ax + by.$$

Let $\delta = \gcd(u, c)$. $\therefore \exists r, s$ s.t.

$$d = ur + cs = a(xr) + b(yr) + cs$$

$$= a(xr) + b(yr) + cs.$$

$\therefore \exists p, q, r$ s.t. $d = ap + bq + cr$.

6. Find all integer solutions of $15x + 12y + 30z = 24$.

$\gcd(15, 12, 30) = 3$, and $3 \mid 24$, so integer solutions exist by (a) above.

Now divide by gcd to simplify.

$$15x + 12y + 30z = 24 \iff 5x + 4y + 10z = 8.$$

$$\therefore 5x + 10z = 8 - 4y.$$

Since $\gcd(5, 10) = 5$,

$$5x + 10z = 5n, \text{ some } n. \quad \therefore 5n = 8 - 4y.$$

$x = n + 2t, z = -t$ is a solution.

By Th. 2.9, $x = n + 2t, z = -t$, gives all solutions for $5x + 10z = 5n$.

$\therefore (x, y, z)$ is a solution $\iff$

$$x = n + 2t \quad \text{for some } n, t$$

$$z = -t$$

$$8 - 4y = 5n, \text{ or } 4y = 8 - 5n.$$

$\therefore y = 2 - \frac{5n}{4}$, which means $4 \mid n$,

so n must be divisible by 4.

$\therefore$ Let $n = 4k$. $\therefore y = 2 - 5k$.

$\therefore$ if k and t are any integers,

$$x = 4k + 2t$$

$$y = 2 - 5k$$

$$z = -t$$

6.(a) \$455 in dimes and quarters.

(1) Determine max & min # of coins.

(2) Can # dimes = # quarters?

(1) $10d + 25q = 455, d \geq 0, q \geq 0$.

$\gcd(10, 25) = 5$.

$\therefore$ Equation $\equiv 2d + 5q = 91, (3, 17)$ a solution.

$\therefore$ All solutions of form:

$$d = 3 + 5t, \quad 3 + 5t \geq 0$$

$$q = 17 - 2t, \quad 17 - 2t \geq 0, \quad t \geq -\frac{3}{5}, t \leq 8$$

$$0 \leq t \leq 8$$

Max # coins is when $d + q$ is a max,

$d + q = 20 + 3t$, so when $t = 8$,

you will have 44 coins (43d, 1q).

Min # coins: $t = 0$, or 20 coins (3d, 17q).

(2) For $d = q$, $3 + 5t = 17 - 2t$, $7t = 14$, $t = 2$.
 $\therefore$ 13 dimes, 13 quarters is a solution.

3. $18a + 25c = 900$, $a > c$. Also, $6 \leq 20$, $c < 20$.
 $\gcd(18, 25) = 15$.
Reduce equation to $12a + 5c = 600$.
One solution is $(50, 0)$.
 $\therefore$ All solutions of form

$$a = 50 + 5t$$

$$c = -12t$$

$$a > c \Rightarrow 50 + 5t > -12t, 17t > -50, t \geq -2.$$

$$c \geq 20 \Rightarrow -12t > -20, t < \frac{20}{12}, t \leq 0.$$

$$\therefore -2 \leq t \leq 0.$$

$$t = 0, -1, -2.$$

So, 50 adults, 0 children;

45 adults, 12 children;

40 adults, 24 children. **(c)** $6x + 7y = 126$ $36x + 54y = 756$

$$6y + 7x = 114 \quad 36x + 24y = 456$$

$$\therefore 6x + 9(x) = 300, \quad y = 10$$

$$6x = 36, \quad x = 6$$

$\therefore$ 6 nines, 10 nines.

[Note: this part is hard to follow precisely from the handwriting — the final answer line ("6 nines, 10 nines") doesn't clearly match the original problem context. Transcribed best-effort; flagged for review.]

7. $c + l + p = 100$, $c, l, p \geq 1$.

$$120c + 50l + 25p = 4000$$

$$\gcd(120, 50, 25) = 5.$$

$\therefore$ Reduce to $24c + 10l + 5p = 800$.

$$24c + 10l + 5p = 800$$

$$19c + 5p = 300 \quad (0, 60) \text{ a solution.}$$

$$\therefore c = 5t$$

$$l = 60 - 19t \quad \therefore p = 100 - 5t - (60 - 19t)$$

$$p = 40 + 14t$$

$$\text{Now, } 5t \geq 1 \Rightarrow t \geq 1$$

$$60 - 19t \geq 1 \Rightarrow 19t \leq 59 \Rightarrow t \leq 3$$

$$40 + 14t \geq 1 \Rightarrow 14t \geq -39 \Rightarrow t \geq -2$$

$$\therefore 1 \leq t \leq 3$$

$t = 1$: 5 calves, 41 lambs, 54 piglets.

$t = 2$: 10 calves, 22 lambs, 68 piglets.

$t = 3$: 15 calves, 3 lambs, 82 piglets. **9.** Let original check be d dollars and c cents.

So, Mr. Smith was given $100c + d$ cents. Find smallest value of $100d + c$.

$$\therefore 100c + d - 68 = 2(100d + c), \quad d, c \geq 0.$$

$$\therefore 98c - 199d = 68.$$

$$199 = 2 \cdot 98 + 3 \quad \therefore 1 = 3 - 2 \cdot 1$$

$$98 = 32 \cdot 3 - 2 \quad = 3 - (98 - 32 \cdot 3)$$

$$3 = 2 + 1 \quad = 33 \cdot 3 - 98$$

$$2 = 2 \cdot 1 \quad = 33(199 - 2 \cdot 98) - 98$$

$$\gcd(98, 199) = 1 \quad = 33 \cdot 199 - 67 \cdot 98.$$

$$\therefore 68 = (68 \cdot 33)199 - (68 \cdot 67)98.$$

$\therefore (-4554, -2244)$ is a solution.

$\therefore$ All solutions are of form:

$$c = -4554 - 199t, \quad c = 98t + 4554(?)$$

$$d = -2244 - 198t, \quad \therefore t \geq 0 \Rightarrow 98t \geq -2244, \quad t \geq -22.9.$$

$\therefore t = -23.$

$$100d + c = -228,856 - 1999t.$$

This is smallest when t is biggest, at $t = -23$.

$$\therefore 100d + c = -228,856 - 1999(-23) = 102 \text{ cents}$$

or 10 dollars 21 cents.

Check: 21 dollars 10 cents -68 cents = 20 dollars 42 cents.

[Note: this page's arithmetic (large negative-coefficient solutions, the exact t -range derivation) is dense and partly illegible; transcribed best-effort and flagged for review.]

9.(a) $m + w + c = 100$, $m, w, c > 0$.

$$3m + 2w + \frac{1}{2}c = 100, \text{ or } 6m + 4w + c = 200.$$

$$c = 100 - m - w.$$

$$\therefore 5m + 3w = 100.$$

One solution is $(14, 10)$:

$\therefore$ All solutions of form:

$$m = 14 + 3t \quad m > 0 \Rightarrow t > -\frac{14}{3}, t \geq -4$$

$$w = 10 - 5t \quad w > 0 \Rightarrow t < 2, t \leq 1$$

$$c = 76 + 2t \quad c > 0 \Rightarrow t > -38, t \geq -37$$

$$\therefore -4 \leq t \leq 1.$$

$t = -4$: 2 men, 30 women, 68 children.

$t = -3$: 5 men, 25 women, 70 children.

$t = -2$: 8 men, 20 women, 72 children.

$t = -1$: 11 men, 15 women, 74 children.

$t = 0$: 14 men, 10 women, 76 children.

$t = 1$: 17 men, 5 women, 78 children.

(b) Let $x =$ plantain fruit in each of the 63 packs.

$$63x + 7 = \text{total \# fruit}, \quad x > 0.$$

Let $y =$ # fruit in each of the 23 travelers.

$$23y = \text{total \# fruit}, \quad y > 0.$$

$$\therefore 63x + 7 = 23y, \text{ or } 63x - 23y = -7.$$

$$63 = 3 \cdot 23 - 6 \quad \therefore 1 = 4 \cdot 6 - 23$$

$$23 = 4 \cdot 6 - 1 \quad = 4(3 \cdot 23 - 63) - 23$$

$$6 = 6 \cdot 1 \quad = 11 \cdot 23 - 4 \cdot 63$$

$$-1 = -7 \cdot 23 + 28 \cdot 63$$

$\therefore (28, 77)$ a solution.

$\therefore$ All solutions of form:

$$x = 28 - 23t \quad x > 0 \Rightarrow 23t < 28, t \leq 1$$

$$y = 77 - 13t \quad y > 0 \Rightarrow 63t < 77, t \leq 1$$

$\therefore$ Infinitely many solutions.

$t = -1$: $x < 5$, $y = 14$ (Struts/pole, 14 traveler).

$t = 0$: $x = 28$, $y = 77$ (28 fruit/pole, 77 traveler).

(c) Let $x =$ # coins on a string when make 77 strings. $\therefore 7x - 50 = \text{total \# coins}$.

Let $y =$ # coins on a string when make 78 strings. $\therefore 78y = \text{total \# coins}$.

$$\therefore 77x - 50 = 78y, \text{ or } 77x - 78y = 50, \quad x, y > 0.$$

$\gcd(77, 78) = 1$. $\therefore 50 = 50(78) - 50(77)$.

$77 = 1 \cdot 78$. $\therefore (-50, -50)$ a solution.

$\therefore$ All solutions of form:

$$x = -50 - 78t, \quad x > 0 \Rightarrow 78t < -50, \quad t = -1$$

$$y = -50 - 77t, \quad y > 0 \Rightarrow 77t < -50, \quad t = -1$$

Infinitely many solutions.

One solution is $t = -1$: $x = 28, y = 27$.

$\therefore$ total # coins is $78(27) = 2106$.

(d) $m + w + c = 20, m, w, c > 0$.

$$3m + 2w + \frac{1}{2}c = 20, \text{ or } 6m + 4w + c = 40.$$

$$c = 20 - m - w.$$

$$\therefore 5m + 3w = 20.$$

One solution is $(1, 5)$.

$\therefore$ All solutions of form:

$$m = 1 + 3t \quad m > 0 \Rightarrow t > -\frac{1}{3}, \quad t \geq 0$$

$$w = 5 - 5t \quad w > 0 \Rightarrow 5t < 5, \quad t \leq 0$$

$$c = 14 + 2t \quad c > 0 \Rightarrow t > -7, \quad t \geq -6$$

$$\therefore t = 0.$$

$\therefore$ 1 man, 5 women, 14 children.

(e) $100 = x + y, 7 \mid x, 11 \mid y, 0 \leq x, y \leq 100$.

Let $7k = x, 11n = y$.

$$\therefore 7k + 11n = 100, \quad 0 \leq 7k, 11n \leq 100.$$

$$11 = 7 + 4 \quad \therefore 1 = 4 - 3 = 4 - (7 - 4) = 2 \cdot 4 - 7$$

$$7 = 4 + 3 \quad = 2(11 - 7) - 7$$

$$4 = 3 + 1 \quad = 2 \cdot 11 - 3 \cdot 7$$

$$3 = 3 \cdot 1 \quad \therefore 100 = 200 \cdot 11 - 300 \cdot 7$$

$$\therefore (-300, 200) \text{ a solution.}$$

$$\therefore k = -300 + 11t$$

$$n = 200 - 7t$$

$$\text{Now } 0 \leq 7k \leq 100 \Rightarrow 0 \leq k \leq \frac{100}{7}, \quad 0 \leq k \leq 14.$$

$$0 \leq 11n \leq 100 \Rightarrow 0 \leq n \leq \frac{100}{11}, 0 \leq n \leq 9.$$

$$0 \leq -300 + 11t \leq 14 \Rightarrow 300 \leq 11t \leq 314, \quad 28 \leq t \leq 28.$$

$$0 \leq 200 - 7t \leq 9 \Rightarrow -200 \leq -7t \leq -191, \quad 28.6 \leq t \leq 27.3, \quad t = 28.$$

$$\begin{aligned} \therefore k &= -300 + 11(28) = 8 & 7k &= 56 \\ n &= 200 - 7(28) = 4 & 11n &= 44 \end{aligned}$$

Note on problem 5(b) – Author’s solution

The choice of $y = 3s - 5t$, $z = -s + 2t$ was ad hoc. Other choices for the coefficients of s and t are fine, as long as they are relatively prime. They were chosen so that $12y + 30z$ would eliminate one variable.

You would then reduce 2 variables (y, z) to one variable (s) . Thereby solving an equation in two variables (x, s) in terms of a parameter r using Th. 2.9. By substitution, y and z would then be expressed in terms of s and t .

Any value of y and z can be expressed in terms of s and t , as long as the coefficients are “non-parallel” lines, and are relatively prime. In other words, $y = 3s - rst$, $z = -s + 4t$ would not work, as $s = 2.5$ could give a solution in integers, but $s = 2.6$ is not “allowed.” Relatively prime coefficients precludes this.

Chapter 3

Primes and Their Distribution

3.1 The Fundamental Theorem of Arithmetic

1. $n^2 - 2$:

$$n = 2 \Rightarrow 2^2 - 2 = 2.$$

$$n = 3 \Rightarrow 9 - 2 = 7.$$

$$n = 5 \Rightarrow 25 - 2 = 23.$$

$$n = 7 \Rightarrow 49 - 2 = 47.$$

$$n = 9 \Rightarrow 81 - 2 = 79.$$

All primes.

2. $25 = \rho + a^2$.

$$a = 1, \rho = 24.$$

$$a = 2, \rho = 21.$$

$$a = 3, \rho = 16.$$

$$a = 4, \rho = 9.$$

$$a = 5, \rho = 0.$$

$\therefore$ No prime ρ for all possible values of a .

3.(a) If $3n + 1$ is prime, so is $6n + 1$.

Pf. $3n + 1$ prime $\Rightarrow 3n + 1$ is odd. Let $\rho = 3n + 1$. Then $\rho - 1 = 3n$ is even.

$\therefore n$ is even, $\therefore n = 2m$, some m .

$$\therefore \rho = 6m + 1 = 6n + 1.$$

(b) Every integer of form $3n + 2$ has a prime factor of that form.

Pf. Let ρ be any prime factor of $3n + 2$.

$\therefore \rho = 3k + 1$ or $3k + 2$, some k . By Division Alg., $\therefore 3n + 2 = (3k_1 + 1)(3k_2 + 1) \cdots (3k_r + 1)$, by
fund. thm. of arith.

But this latter product is of form $\left[3^r k_1 \cdots k_r + 1 \right]$, where every term, except 1, is a factor of 3. Product is of form $3q + 1$, a contradiction.

(c) The only prime of form $n^3 - 1$ is 7.

Pf. $n^3 - 1 = (n - 1)(n^2 + n + 1)$.

For $n = 1$, to be prime, $\rho = 1$.

For $n = 2$, $n^3 - 1 = (n + 1)(7) = 7$.

For any $n \geq 2$, $\rho = n^3 - 1$ will be a factor of two integers, neither of which is 1. $\therefore$ for $n \geq 2$, ρ can't be prime.

(d) The only prime ρ for which $3\rho + 1$ is a perfect square is $\rho = 5$.

Pf. $3(5) + 1 = 16 = 4^2$.

Suppose $3\rho + 1 = n^2$, some $n \neq 4$. $\therefore 3\rho = n^2 - 1 = (n + 1)(n - 1)$. If $n + 1 = \rho$, then $n - 1 = 3$, $n = 4$.

Assume $n + 1 \neq \rho$. $\therefore \gcd(n + 1, \rho) = 1$.

$\therefore n + 1 = 1$ or ρ^2 . $n = 2$. $\therefore 3\rho + 1 = 4$, $\rho = 1$, a contradiction.

$\therefore n + 1$ must be ρ , and n must be 4.

Similar reasoning for $n - 1$.

If $n - 1 = \rho$, then $n + 1 = 3$, $n = 2$ (leading to contradiction of $3\rho + 1 \neq \rho^2$, $\rho = 1$).

$\therefore n - 1 \neq \rho$. Then $\gcd(n - 1, \rho) = 1$, $n - 1 = 1$ or 3. $\therefore n - 1 = 1$ or 3. $\therefore n = 3$ or 4.

(e) The only prime of form $n^2 - 4$ is 5.

Pf. Let $\rho = n^2 - 4 = (n + 2)(n - 2)$, one of the factors must be 1 and the other must be ρ .

Suppose $n + 2 = \rho$, $\therefore n - 2 = 1$, $\therefore n = 3$, $\therefore \rho = 5$.

Suppose $n + 2 = 1$, $\therefore n = -1$, and $\therefore n \neq 1$.

$\therefore$ Only possibility is $n = 3$, $\therefore \rho = 5$. 4. $\rho \geq 5$, then $\rho^2 + 2$ is composite.

Pf. By Div. Alg., $\rho = 6k + r$, $0 \leq r < 6$.

$r \neq 0$ as $\rho = 6k \Rightarrow 6 \mid \rho$.

$r \neq 2$ as $\rho = 6k + 2 \Rightarrow 2 \mid \rho$.

$r \neq 3$ as $\rho = 6k + 3 \Rightarrow 3 \mid \rho$.

$r \neq 4$ as $\rho = 6k + 4 \Rightarrow 2 \mid \rho$.

$\therefore \rho = 6k + 1$ or $\rho = 6k + 5$.

$$\therefore \rho^2 + 2 = 36k^2 + 12k + 3 \quad \text{or}$$

$$\rho^2 + 2 = 36k^2 + 60k + 27.$$

In either case, $3 \mid \rho^2 + 2$, so $\rho^2 + 2$ is composite.

5.(a) If ρ prime, $\rho \mid a^n \Rightarrow \rho^n \mid a^n$.

Pf. By Corollary 1 (p.4), $\rho \mid a^n \Rightarrow a = \rho k$, some k , so $a^n = \rho^n k^n \Rightarrow \rho^n \mid a^n$.

(b) If $\gcd(a, b) = \rho$, then by (a) above, $\rho^n \mid a^n$, $k^n \mid b^n$, so $\gcd(a^n, b^n) = \rho^n$.

$$\gcd(a^2, b) = \rho$$

$$\gcd(a^3, b^2) = \rho^2$$

6.(a) For all $n > 1$, $n^4 + 4$ is composite.

Pf. $n^4 + 4 = (n^2 - 2n + 2)(n^2 + 2n + 2)$.

Since $n > 1$, $n \geq 2$, and $\therefore n^2 - 2n + 2 > 0$, and $n^2 + 2n + 2 > 0$.

$\therefore$ Both factors are positive.

Since $n^4 + 4$ has two integer positive factors, it is composite.

Find the factors by guessing the roots (or using a calculator). Note that $(1+i)(1-i) = 2i(2i)^{-1} \dots = 4$.

$\therefore 1+i$ is a root, and $\therefore$ so is $(1-i)$.

$$\begin{aligned} \therefore (n-1-i)(n-1+i) &= n^2 - 2n + 1 - i^2 \\ &= n^2 - 2n + 2, \end{aligned}$$

and so $n^2 - 2n + 2$ is a factor.

Find the other by division.

(b) If $n > 4$ is composite, then n divides $(n-1)!$. *Pf.* Since n is composite, let

$n = p_1^{k_1} \dots p_r^{k_r}$ be the unique prime factorization.

If $r > 1$, then $n > p_i^{k_i}$, so $n-1 \geq p_i^{k_i}$.

$\therefore$ since all integers $\leq n-1$ are terms of $(n-1)!$, then each $p_i^{k_i}$ is represented by one of the terms of $(n-1)!$. $\therefore p_i^{k_i} \mid (n-1)!$.

Suppose $r = 1$, so $n = \rho^k$. $k > 1$ since n is composite.

$\therefore n = \rho^{k-1} \cdot \rho$.

$\therefore n-1 \geq \rho$ and $n > \rho^{k-1}$.

$\therefore n-1 \geq \rho^{k-1}$.

If $\rho \neq \rho^{k-1}$, then each is represented in $(n-1)!$, so $\rho \cdot \rho^{k-1} = \rho^k \mid (n-1)!$.

Suppose $\rho = \rho^{k-1}$, so $k = 2$. $\therefore n = \rho^2$.

$\therefore n > 4\rho$, so $n-1 \geq \rho$.

Since $n \geq 6$, then $\rho \neq 2$.

And $2(n-1) < (n-1)!$ for $n > 4$.

$2(n-1)$ is a term of $(n-1)!$.

$\therefore$ Each of ρ and 2ρ are terms of $(n-1)!$. $\therefore 2\rho^2 \mid (n-1)!$, so

$\rho^2 \mid (n-1)!$. $\therefore \rho^k = n \mid (n-1)!$. **(c)** $8^n + 1$, $n \geq 1$, is composite.

Pf. $a^3 + 1 = (a+1)(a^2 - a + 1)$.

$\therefore (2^n)^3 + 1 = (2^n + 1)(2^{2n} - 2^n + 1)$.

$\therefore 2^{3n} + 1 \mid 2^{3n} + 1$,

and $2^{3n} + 1 = 8^n + 1$.

$\therefore 2^n + 1 \mid 8^n + 1$.

(d) $n \geq 11$, then n is the sum of two composite numbers.

Pf. Suppose n is even. Then $\exists k$ s.t. $n = 2k$.

$n = 2k = 6 + (2k - 6)$.

$\therefore n$ is the sum of $6 (= 2 \cdot 3)$ and $2(k - 3)$.

If $k \leq 5$ ($10 \geq 2k - 3 \geq 1$), $0 \leq 2(k - 3)$, so $2(k-3)$ is product of two numbers > 1 , then $2k \geq 10$, $n \geq 11$, and n is the sum of two composites.

Suppose n is odd. Then $\exists k$ s.t. $n = 2k + 1$.

If $n = 2(k - 1) + 3$: 3 is prime

$$= 2(k - 2) + 5 : \quad 5 \text{ is prime}$$

$$= 2(k - 3) + 7 : \quad 7 \text{ is prime}$$

$$= 2(k - 4) + 9$$

So, if $k \geq 6$, then $2(k - 4)$ is the product of two numbers > 1 , so $n = 2k + 1 \geq 13$, and n is the sum of two composites.

7. Find all primes that divide $50!$.

All primes < 50 will divide $50!$, since each is a term of $50!$.

By Fund. Th. of Arithmetic, each term K of $50!$ that is non-prime has a unique prime factorization, and each term of this unique factorization of K is smaller than K , and K is a prime that is < 50 . There is no prime > 50 represented in this factorization at K .

$\therefore$ All primes < 50 are all the primes that divide $50!$.

8. $\rho \geq \rho \geq 5$, ρ, q primes, $24 \mid \rho^2 - q^2$.

Pf. From #4 above, $\rho = 6r + 1$ or $6r + 5$, $q = 6s + 1$ or $6s + 5$.

Three possibilities:

$$(1) \quad \rho = 6r + 1, \quad q = 6s + 1$$

$$(2) \quad \rho = 6r + 5, \quad q = 6s + 5$$

$$(3) \quad \rho = 6r + 1, \quad q = 6s + 5 \quad \text{The situation of } \rho = 6r + 5, \quad q = 6s + 1 \text{ is equivalent to \#3.}$$

$$(1) \quad \text{Let } \rho = 6r + 1, \quad q = 6s + 1 \quad (r, s \geq 0, \rho, q \geq 7).$$

Since $\rho, q \geq 5$, then $r, s \neq 0$.

$$\therefore \rho^2 - q^2 = (\rho + q)(\rho - q)$$

$$= (6r + 6s + 2)(6r - 6s)$$

$$= (6r + 6s + 2) \cdot 6(r - s)$$

$$= 2 \cdot 6(3r + 3s + 1)(r - s).$$

If r, s are both even or both odd, then $r - s$ is even and $\neq 0$, so $r - s = 2k$.

$$\therefore \rho^2 - q^2 = 2 \cdot 6 \cdot 2(3r + 3s + 1)(k)$$

$$= 24(3r + 3s + 1)(k).$$

If one is even and one is odd, then $3r + 3s + 1$ is even, so $3r + 3s + 1 = 2k$.

$$\therefore \rho^2 - q^2 = 2 \cdot 6 \cdot 2(k)(r - s) = 24(k)(r - s).$$

$$\therefore 24 \mid \rho^2 - q^2.$$

(2) Let $\rho = 6r + 5$, $q = 6s + 5$ ($r, s \geq 0$, so $\rho, q \geq 5$).

$$\begin{aligned}\therefore \rho^2 - q^2 &= (\rho + q)(\rho - q) \\ &= (6r + 6s + 10)(6r - 6s) \\ &= (6r + 6s + 5 + 5)(6r - 6s) \\ &= 2 \cdot 6(3r + 3s + 5)(r - s).\end{aligned}$$

If r, s are both even or both odd, then $r - s$ is even and $\neq 0$, so $r - s = 2k$.

$$\begin{aligned}\therefore \rho^2 - q^2 &= 2 \cdot 6 \cdot 2(k)(3r + 3s + 5) \\ &= 24(3r + 3s + 5)(k).\end{aligned}$$

If one is even and one is odd, then $3r + 3s + 5$ is even, so $3r + 3s + 5 = 2k$.

$$\begin{aligned}\therefore \rho^2 - q^2 &= 2 \cdot 6 \cdot 2(k)(r - s) = 24(k)(r - s). \\ \therefore 24 &| \rho^2 - q^2.\end{aligned}$$

(3) $\rho = 6r + 1$, $q = 6s + 5$ ($r \geq 0, s \geq 0$, so $\rho \geq 1, q \geq 5$).

$$\begin{aligned}\therefore \rho^2 - q^2 &= (\rho + q)(\rho - q) \\ &= (6r + 6s + 5 + 1 + 10)(6r - 6s - 4) \\ &= (6r + 6s + 6)(6r - 6s - 4) \\ &= 6 \cdot 2(r + s + 1)(3r - 3s - 2).\end{aligned}$$

If one is even, one is odd, then $r + s + 1$ is even, so $r + s + 1 = 2k$.

$$\therefore \rho^2 - q^2 = 2 \cdot 6(k)(3r - 3s - 2), \text{ so } 24 | \rho^2 - q^2.$$

If both even or both odd, then $3r - 3s - 2$ is even, so $3r - 3s - 2 = 2k$.

$$\therefore \rho^2 - q^2 = 2 \cdot 6(r + s + 1)(k), \text{ so } 24 | \rho^2 - q^2.$$

$$\mathbf{9.(a)} \quad 2^4 + 1 = 17 \quad 2^8 + 1 = 257 \quad 10^2 + 1 = 101$$

$$\mathbf{(b)} \quad 2^2 + 1 = 5 \quad 6^2 + 1 = 37 \quad \mathbf{10.} \quad \rho \neq 5, \rho \neq 2, \text{ prove } 10 | \rho^4 - 1 \text{ or } 10 | \rho^4 + 1.$$

Pf. ρ is of the form: $10k + 1, 10k + 3, 10k + 7, 10k + 9$.

$10k$ even: can factor out 2, so not prime.

$$\begin{aligned}(10k + 1)^4 &= 100k^2 + 20k + 1 & \therefore 10 &| \rho^4 - 1. \\ (10k + 3)^4 &= 100k^2 + 60k + 9 & \therefore 10 &| \rho^4 - 1. \\ (10k + 7)^4 &= 100k^2 + 140k + 49 & \therefore 10 &| \rho^4 + 1. \\ (10k + 9)^4 &= 100k^2 + 180k + 81 & \therefore 10 &| \rho^4 - 1.\end{aligned}$$

$$\begin{array}{ll} \mathbf{11. (a)} & 2^3 - 1 = 7 \qquad 2^7 - 1 = 127 \\ & 2^5 - 1 = 31 \qquad 2^{11} - 1 = 807 \end{array}$$

(b) If $\rho = 2^k - 1$ is prime, show k is odd.

Pf. $a^k - b^k = (a - b)(a^{k-1} + a^{k-2}b + \cdots + ab^{k-2} + b^{k-1}).$

$$\begin{aligned} \therefore 4^n - 1 &= (4 - 1)(4^{n-1} + \cdots + 1) \\ &= 3(4^{n-1} + \cdots + 1), \quad (n \geq 1). \end{aligned}$$

$$\therefore 3 \mid 4^n - 1 \Rightarrow 3 \mid 2^{2n} - 1 \quad (n \geq 1).$$

$2n$ is even, so if $\rho = 2^k - 1$ is prime ($n \geq 1 \Rightarrow 2n \neq 2$, so $k \neq 2n$), k must be odd.

$$\mathbf{12.} \quad 1234 = 2 \cdot 617$$

$$10,146 = 10 \cdot 1014 = 2 \cdot 5 \cdot 2 \cdot 507 = 2^2 \cdot 5 \cdot 3 \cdot 13^2$$

$$\begin{aligned} 36000 &= 36 \cdot 1000 = 2^2 \cdot 3^2 \cdot 10 \cdot 25 \cdot 4 \\ &= 2^2 \cdot 3^2 \cdot 2 \cdot 5^2 \cdot 2^2 \\ &= 2^5 \cdot 3^2 \cdot 5^3. \end{aligned}$$

13. If $n > 1$ not of form $6k + 3$, then $n^k + 2^n$ is composite.

Pf. n of form $6k, 6k + 1, 6k + 2, 6k + 4, 6k + 5$.

$6k$: $n^k + 2^{6k}$. Since $k > 0$,

$2 \mid 36k^2 + \cdots$ composite.

$6k + 1$: $n^k + 2^{6k+1}$

$$= 36k^2 + 12k + 1 + 2^{6k+1}.$$

From $a^n - b^n = (a - b)(a^{n-1} + \cdots + b^{n-1})$, substitute (-1) for b and get

$$a^n - (-1)^n b^n = (a + b)(a^{n-1} - a^{n-2}b + \cdots + (-1)^{n-1}b^{n-1}) = (a + b)(\cdots).$$

$$\begin{aligned} \therefore n^k + 2^{6k+1} &= 36k^2 + 12k + (2(2)^{6k}) + 1 \\ &= 36k^2 + 12k + 3[2(2)^{6k} \cdots + (2)^{6k}] + 1. \\ \therefore 3 \mid n^k + 2^k. \end{aligned}$$

$$6k + 2: n^k + 2^{6k+2} = (6k)^2 + 24k + 4 + 2^{6k+2}$$

$$= 3[6k^2 + 24k + 4 + 2^{6k+2}].$$

[Note: this page's algebra is dense and the indices on n^k vs n^n are inconsistent in the original (e.g. " $n^k + 2^n$ " in the statement vs " $n^k + 2^{6k}$ " in the proof) — transcribed as legible; flagged for review.]

$$\therefore 2 \mid n^k + 2^n.$$

$$\begin{aligned}
 6k + 4: n^k + 2^{6k+4} &= 36k^2 + 48k + 16 + 2^{6k+4} \\
 &= 2[18k^2 + 24k + 8 + 2^{6k+3}] \\
 &\therefore 2 \mid n^k + 2^n.
 \end{aligned}$$

$$\begin{aligned}
 6k + 5: n^k + 2^{6k+5} &= 36k^2 + 60k + 25 + 2^{6k+5} \\
 &= 36k^2 + 60k + 24 + 2(2^{6k+4}) + 1 \\
 &= 36k^2 + 60k + 24 + 2(2^{6k+4} + 1)(2^{6k+\dots}) \\
 &= 3[\dots].
 \end{aligned}$$

Similar to $6k + 1$ above.

Note for $6k + 3$: $36k^2 + 36k + 9$, $9 = 8 + 1$, so can use the $a^n + b^n = (a + b)(\dots)$ trick.

14.

$$\begin{array}{lll}
 10 = 149 - 139 & 10 = 419 - 409 & 10 = 719 - 709 \\
 10 = 171 - 181 & 10 = 421 - 421 & 10 = 787 - 787 \\
 10 = 251 - 241 & 10 = 587 - 577 & 10 = 821 - 811 \\
 10 = 283 - 283 & 10 = 587 - 577 & 10 = 859 - 829 \\
 10 = 347 - 337 & 10 = 701 - 691 & 10 = 929 - 919
 \end{array}$$

15. $a > 1$ is a square $\iff a$ in canonical form has all even exponents for the primes.

Pf. Suppose a is a square. $\therefore a = n^2$.

Let $p_1^{k_1} \dots p_r^{k_r} = n$. $\therefore n^2 = p_1^{2k_1} \dots p_r^{2k_r}$. So all exponents are even.

Suppose all exponents of $p_1^{k_1} \dots p_r^{k_r} = a$ are even.

$\therefore k_i = 2m_i$, some m_i , for each k_i .

$$\begin{aligned}
 \therefore a &= p_1^{2m_1} \dots p_r^{2m_r} \\
 &= (p_1^{m_1} \dots p_r^{m_r})^2.
 \end{aligned}$$

16.(a) $n > 1$ is square free $\iff n$ can be factored into a product of distinct primes.

Suppose n is square free, and let $n = p_1^{k_1} \dots p_r^{k_r}$ be the prime factorization.

Suppose any $k_i > 1$: $k_i \geq 2$, and $\therefore p_i^2$ will divide n , a contradiction of def. of square free. $\therefore$ each $k_i = 1$.

Suppose $n = p_1 \dots p_r$, each $p_i \neq p_j$.

Suppose n is not square free, $\therefore n = x^2 a$, x an integer.

Let $a = q_1^{k_1} \dots q_s^{k_s}$.

$$\therefore p_1 \dots p_r = x^2 q_1^{2k_1} \dots q_s^{2k_s} = x^2 \cdot q_1^{k_1} \dots q_s^{k_s} \cdot p_1 \dots p_r.$$

$\therefore$ By Corollary 2 (p.41), $q_i = p_k$ for some $1 \leq k \leq r$.

After factoring out q_i and p_k , we still have

$$p_1 \cdots p_r = q_1^{2k_1} \cdots q_i^{2k_i} \cdots q_s^{2k_s}, \quad \text{so that } q_i \mid p_1 \cdots p_r.$$

But the original factorization $p_1 \cdots p_r$ was unique, and q_i was factored out.

$\therefore q_i$ can't divide the remaining factorization. $\therefore n$ must be square free.

(b) Every $n > 1$ is the product of a square free integer and a perfect square.

Pf. Let $n = p_1^{k_1} \cdots p_r^{k_r}$ be the canonical form for n . If k_i is odd and $k_i > 1$, then $k_i - 1$ is even. Let

$$a = p_1^{k_1-1} \cdots p_r^{k_r-1},$$

where $1 \leq r \leq s$ and k_r is odd and $k_r \geq 1$.

Consider $b = p_1 \cdots p_r$.

$$\therefore a = \delta p_1^{k_1-1} \cdots p_r^{k_r-1}.$$

Also, b is square free, by (a) above.

$$\rho_{r_1}^{k_1} \cdots = \rho_{r_i}^{2k_i}, \quad \text{since each } k_i - 1$$

is even. Let $c = \rho_{r_1}^{x_1} \cdots \rho_{r_m}^{x_m}$.

$\therefore a = bc^2$, let a/n , let $a/n = \rho_{r_1}^{k_1} \cdots \rho_{r_i}^{k_i}$, where all K_i are even since a/n has factored out all of the odd exponents in the canonical form of n .

By #15 above, $a/n = x^2$.

$\therefore n = bcd^2 = b(cd)^2$, when b is square free.

17. $n = 2^k m$, $m \neq 0$, $k \geq 0$, m odd.

Pf. Assume $n > 0$ (if $n < 0$, choose k, m s.t. $-n = 2^k m$, $\therefore n = 2^k(-m)$).

If n is odd, choose $k = 0$, $m = n$.

If n is even, then $n \geq 2k_1$. Note $k \leq n$.

If K_1 is odd, choose $K = 1$, $m = K_1$.

If K_1 is even, then $K_1 = 2K_2$, so $n = 2^2 K_2$. Note $K_2 < K_1$.

Continue this process till K_i is odd. Since $K_i < K_{i-1}$, this is a finite process (i.e., eventually will reach 1, as no other odd integer reached by then).

18.

3, 53	47, 97	107, 157
11, 61	53, 103	113, 163
17, 67	59, 109	131, 181
23, 73	83, 139	149, 199
29, 79	101, 151	173, 223

19. If $n > 0$ is square-full, then $n = a^2 b^3$, $b > 0$.

Pf. Let $n = p_1^{k_1} \cdots p_r^{k_r}$. Since n is square-full, $k_i \geq 2$.

Write $p_1^{k_1} \cdots p_r^{k_r} = q_{m_1}^{k_{m_1}} \cdots q_{m_s}^{k_{m_s}} \cdot q_{n_1}^{k_{n_1}} \cdots q_{n_t}^{k_{n_t}}$,

where K_{m_i} are odd (so $K_{m_i} \geq 3$), and K_{n_i} are even, such that $K_{m_i} = K_i$ and $K_{n_i} = K_i$ (i.e., writing n so that odd exponents listed first, even exponents listed last).

$\therefore K_{n_i} = 2v_i$, some v_i .

$$\therefore n = q_{m_1}^{K_{m_1}} \cdots q_{m_s}^{K_{m_s}} \left(q_{n_1}^{2v_1} \cdots q_{n_t}^{2v_t} \right)$$

$$= q_{m_1}^{K_{m_1}} \cdots q_{m_s}^{K_{m_s}} \left(q_{n_1}^{v_1} \cdots q_{n_t}^{v_t} \right)^2.$$

$$\therefore n = q_{m_1}^{K_{m_1}} \cdots q_{m_s}^{K_{m_s}} (x^2), \quad x = q_{n_1}^{v_1} \cdots q_{n_t}^{v_t}.$$

Since K_{m_i} is odd and ≥ 3 , $K_{m_i} - 3$ is even.

$$\therefore n = q_{m_1}^3 \cdots q_{m_s}^3 \left(q_{m_1}^{m_1-3} \cdots q_{m_s}^{m_s-3} \right) (x^2).$$

Let $m_i - 3 = 2w_i$, $q_{m_1} \cdots q_{m_s} = b$,

$$\therefore n = b^3 \left(q_{m_1}^{2w_1} \cdots q_{m_s}^{2w_s} \right) (x^2). \quad \text{Let } y = q_{m_1}^{w_1} \cdots q_{m_s}^{w_s}.$$

$$\therefore n = b^3 y^2 x^2. \quad \text{Let } a = yx,$$

$$\therefore n = a^2 b^3.$$

3.2 The Sieve of Eratosthenes

1. Test all primes $\rho \leq \sqrt{701}$ to see if 701 is a prime.

$\sqrt{701} = 26.5 \dots$; test 2, 3, 5, 7, 11, 13, 17, 19, 23.

All do not divide 701. $\therefore$ 701 is prime.

2. $\sqrt{1009} = 31. \dots$ 1009 is not divisible by 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31.

101	103	105	107	109	111	113	115
117	119	121	123	125	127	129	...

[Note: this page contains a hand-drawn sieve grid of numbers 101–199 with composites crossed out, used to find all primes between 100 and 200; the grid itself isn't reproduced here, only the surrounding text.]

$14 < \sqrt{200} < 15$, $\therefore$ stop at $\rho = 13$.

3. If $\rho \nmid n$ for all primes $\rho \leq \sqrt[3]{n}$, $n > 1$, then n is either prime or the product of two primes.

Pf. Assume n is composite, and let

$$n = \rho_1 \rho_2 \cdots \rho_r, \quad \text{and assume } r \geq 3.$$

Note: ρ_1 not among primes $\rho \leq \sqrt[3]{n}$; $\rho_1, \rho_2, \rho_3 \leq \sqrt[3]{n}$.

We know that $1 < \sqrt[3]{n} < \rho_1 \leq \sqrt{n}$.

$$\sqrt[3]{n} \leq \rho_1 \leq \sqrt{n}$$

$$\sqrt[3]{n} \leq \rho_2 \leq \sqrt{n}$$

$$\sqrt[3]{n} \leq \rho_3 \leq \sqrt{n}$$

$$\therefore n = (\sqrt[3]{n})(\sqrt[3]{n})(\sqrt[3]{n}) \leq \rho_1 \rho_2 \rho_3 = n,$$

or $n = n$. $\therefore r < 3$, or $r = 2$ or $r = 1$.

$\therefore n$ is either prime ($r = 1$), or is the product of two primes ($r = 2$).

4.(a) $\sqrt{\rho}$ is irrational for any prime ρ .

Pf. Assume $\sqrt{\rho} = \frac{r}{s}$, some integers r, s .

Let $d = \gcd(r, s)$. Let $\rho = \frac{r}{d}$, $s' = \frac{s}{d}$.

$\therefore \gcd(\rho, s') = 1$, by Corollary 1, p.23.

Also $\frac{r}{s} = \frac{r'}{s'}$. $\therefore \sqrt{\rho} = \frac{r'}{s'}$.

$\therefore \rho = \frac{r'^2}{s'^2}$, $\rho s'^2 = r'^2 \Rightarrow \rho \mid r'^2$.

Let $\rho = \rho x$. $\therefore r'^2 = \rho^2 x^2 = \rho s'^2$, or

$\rho x^2 = s'^2$. $\therefore \rho \mid s'^2$. $\therefore \gcd(\rho, s') \neq 1$. $\therefore$ There doesn't exist integers r, s s.t. $\sqrt{\rho} = \frac{r}{s}$.

(b) $a > 0$, $\sqrt[n]{a}$ rational, then $\sqrt[n]{a}$ is an integer.

Pf. Let $\sqrt[n]{a} = \frac{r}{s}$, r, s integers, s.t. $\gcd(r, s) = 1$.

Let $r = \rho_1 \cdots \rho_x$, $s = q_1 \cdots q_y$.

$\therefore \rho_i \neq q_j$.

$$\therefore (q_1^n \cdots q_y^n) a = \rho_1^n \cdots \rho_x^n.$$

$$\therefore \rho_1^n \cdots \rho_x^n \mid a. \quad \therefore \text{Let } a = (\rho_1^n \cdots \rho_x^n) z.$$

$$\therefore (q_1^n \cdots q_y^n) (\rho_1^n \cdots \rho_x^n) z = \rho_1^n \cdots \rho_x^n.$$

$$\therefore (q_1^n \cdots q_y^n) z = 1. \quad \therefore q_j = 1 \text{ for all } j.$$

$\therefore s = 1$, $\therefore \frac{r}{s}$ is an integer.

(c) For $n \geq 2$, $\sqrt[n]{n}$ is irrational.

Pf. Suppose $\sqrt[n]{n}$ is rational. From (b), it is an integer. Let $\sqrt[n]{n} = a$. $\therefore n = a^n$. But $n < 2^n$.

$\therefore a^n < 2^n$, so $a < 2$, or $a = 1$.

$\therefore n = 1^n = 1$, a contradiction.

5. Any composite 3-digit number must have a prime factor ≤ 31 .

Pf. 999 is largest 3-digit number. $31^2 = 961$, 31 is prime, so if a is composite, largest prime divisor is $\leq \sqrt{a}$, so 31 is largest possible prime divisor.

6. Number of primes is infinite.

Pf. Assume only finite number: $\rho_1, \rho_2, \dots, \rho_n$.

Let A be the product of any r of them, $a_i \in \{1, 2, \dots, n\}$.

So $A = \rho_{a_1} \rho_{a_2} \cdots \rho_{a_r}$.

Consider $B = \rho_1 \rho_2 \cdots \rho_n / A$

$$= \frac{\rho_1 \rho_2 \cdots \rho_n}{\rho_{a_1} \rho_{a_2} \cdots \rho_{a_r}} = \rho_{b_1} \cdots \rho_{b_s},$$

where $a_i \neq a_j$ (i.e., factoring out ρ_{a_i}), and $\{\rho_{a_i}\} \cap \{\rho_{b_j}\} = \emptyset$, and $\{\rho_{a_i}\} \cup \{\rho_{b_j}\} = \{\rho_1 \cdots \rho_n\}$. So, A and B have no common factors. Then each ρ_k of $\rho_1 \rho_2 \cdots \rho_n$ divides either A or B , but not both.

Since $A > 1$, $B > 1$, then $A + B > 1$.

$A + B$ must have a prime factor, ρ_β , and $\rho \mid (A + B)$ is an integer, and $\rho \in \{\rho_1, \rho_2, \dots, \rho_n\}$ since assuming finite primes.

Suppose $\rho \mid A$. $\therefore \rho x = A + B$, some x , and $\rho \mid A \Rightarrow \rho y = A$, some y .

$\therefore \rho x = \rho y + B$, $\therefore \rho(x - y) = B$, so $\rho \mid B$, a contradiction.

7. Prove infinitely many primes using $N = \rho_1^2 + 1$.

Pf. Assume finitely many primes, ρ_n the largest. Consider

$$N = \rho_1^2 + 1.$$

$$\therefore N = 1 \cdot 2 \cdots \rho_n + 1.$$

N must have a prime divisor ρ_k , taken since assuming finite $\#$ primes. And $\rho_k \mid 1 \cdot 2 \cdot 3 \cdots \rho_n$ since ρ_k is one of the members of ρ_n !

$\therefore \rho_k \mid (N - \rho_1 \rho_2 \cdots \rho_n)$. $\therefore \rho_k \mid 1$, $\rho_k = 1$, a contradiction.

8. Prove infinitude of primes using

$$N = \rho_2 \cdots \rho_n + \rho_1 \rho_3 \cdots \rho_n + \cdots + \rho_1 \rho_2 \cdots \rho_{n-1}.$$

Pf. Assume finite $\#$ of primes $\rho_1, \rho_2, \dots, \rho_n$.

Consider $q_k = \rho_1 \rho_2 \cdots \rho_n$, s.t. each term $\rho_i \neq q_k$.

$$\therefore q_1 = \rho_2 \rho_3 \cdots \rho_n$$

$$q_2 = \rho_1 \rho_3 \rho_4 \cdots \rho_n$$

$$\vdots$$

$$q_n = \rho_1 \rho_2 \rho_3 \cdots \rho_{n-1}.$$

$\therefore \rho_k \nmid q_k$.

Let $N = q_1 + q_2 + \cdots + q_n = \sum_{i=1}^n q_i$. N must have a prime divisor from $\rho_1 \cdots \rho_n$.
Let ρ_k ($1 \leq k \leq n$) be that prime divisor.

But since $\rho_k \mid N$ and $\rho_k \mid q_i$, $i \neq k$, then $\rho_k \mid (N - \sum_{i \neq k} q_i)$.

But then $N - \sum_{i \neq k} q_i = q_k$. $\therefore \rho_k \mid q_k$, a contradiction.

9.(a) If $n > 2$, then $\exists \rho$ s.t. $n < \rho \leq n!$.

Pf. For $n > 2$, clearly $2n < n! \cdot 1 \cdot 2 \cdots n$. From Bertrand's conjecture, $\exists$ a prime ρ s.t. $n < \rho < 2n \leq n!$.

Pf. (using author's hint):

For $n > 3$, $n < n! - 1 < n!$.

If $n! - 1$ is prime, we're done.

If $n! - 1$ is not prime, let ρ be a prime divisor. $\therefore \rho < n!$.

Assume $\rho \leq n$. Then ρ is one of the terms of $1 \cdot 2 \cdot 3 \cdots n$, so $\rho \mid n!$.

$\therefore \rho \mid n!$ and $\rho \mid (n! - 1)$.

$\therefore \rho \mid n! - (n! - 1) = 1$.

$\rho > n$. $\therefore n < \rho < n! - 1 < n!$.

(b) For $n > 1$, every prime divisor of $n! + 1$ is an odd integer $> n$.

Pf. First, $n! + 1$ is odd, since $n!$ is even, as it contains 2, and $2x$ is even for all x .

$\therefore 2$ will never divide $n! + 1$, so every prime divisor of $n! + 1$ is odd.

Now suppose every prime divisor ρ_i of $n! + 1$ is $\leq n$.

Let $\rho_i \mid n! + 1$.

Clearly, $\rho_i \mid n!$, since ρ_i is one of the members of $n!$.

Since $\rho_i \mid \rho_i$, then $\rho_i \mid ((n! + 1) - n!)$, and

$\rho_i \cdot n! = 1$. $\therefore \rho_i \mid 1$, a contradiction.

$\therefore \rho_i > n$. **10.** Let q_n be smallest prime s.t. $q_n > \rho_1 \rho_2 \cdots \rho_n + 1$. Show $q_n - (\rho_1 \rho_2 \cdots \rho_n)$ is prime for $n = 1, 2, \dots, 5$.

q_1 : $2 + 1 = 3 \therefore q_1 = 5$.

q_2 : $2 \cdot 3 + 1 = 7 \therefore q_2 = 11$.

q_3 : $2 \cdot 3 \cdot 5 + 1 = 31$ $q_3 = 37$.

q_4 : $2 \cdot 3 \cdot 5 \cdot 7 + 1 = 211$ $q_4 = 223$.

q_5 : $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 + 1 = 2311$ $q_5 = 2833$.

$$\therefore q_1 - (\rho_1) = 5 - 2 = 3$$

$$q_2 - (\rho_1 \rho_2) = 11 - 6 = 5$$

$$q_3 - (\rho_1 \rho_2 \rho_3) = 37 - 30 = 7$$

$$q_4 - (\rho_1 \rho_2 \rho_3 \rho_4) = 223 - 210 = 13$$

$$q_5 - (\rho_1 \rho_2 \rho_3 \rho_4 \rho_5) = 2833 - 2310 = 523$$

11. Let $d_n = \rho_{n+1} - \rho_n$. Find five solutions to $d_n = d_{n+1}$.

$$d_1 = \rho_2 - \rho_1 = 3 - 2 = 1$$

$$d_2 = \rho_3 - \rho_2 = 5 - 3 = 2$$

$$d_3 = \rho_4 - \rho_3 = 7 - 5 = 2 \quad \therefore d_2 = d_3$$

$$d_4 = \rho_5 - \rho_4 = 11 - 7 = 4$$

$$d_5 = \rho_6 - \rho_5 = 13 - 11 = 2$$

$$d_6 = \rho_7 - \rho_6 = 17 - 13 = 4$$

$$d_7 = \rho_8 - \rho_7 = 19 - 17 = 2$$

$$d_8 = \rho_9 - \rho_8 = 23 - 19 = 4$$

$$d_9 = 29 - 23 = 6$$

$$d_{10} = 31 - 29 = 2$$

$$d_{11} = 37 - 31 = 6$$

$$d_{12} = 41 - 37 = 4$$

$$d_{13} = 43 - 41 = 2$$

$$d_{14} = 47 - 43 = 4$$

$$d_{15} = 53 - 47 = 6$$

$$d_{16} = 59 - 53 = 6 \quad \therefore d_{15} = d_{16}$$

$$61 - 59 = 2$$

$$67 - 61 = 6$$

$$\vdots$$

$$157 - 151 = 6$$

$$163 - 157 = 6 \quad \therefore d_{36} = d_{37}$$

$$\vdots$$

$$173 - 167 = 6$$

$$179 - 173 = 6 \quad \therefore d_{40} = d_{39}$$

$$\vdots$$

$$211 - 199 = 12$$

$$223 - 211 = 12 \quad \therefore d_{47} = d_{46}$$

12. Let ρ_n be the n -th prime number. Prove:

(a) $\rho_n > 2n - 1$, for $n \geq 5$.

Pf. For $n = 5$, $\rho_n = 11 > 2(5) - 1 = 9$.

Assume true for K : $\rho_K > 2K - 1$.

$$\therefore \rho_K + 2 > (2K - 1) + 2 = 2(K + 1) - 1.$$

Since $\rho_K + 1$ is even, then next possible prime is $\rho_K + 2$.

$$\therefore \rho_{K+1} \geq \rho_K + 2.$$

$$\therefore \rho_{K+1} > \rho_K + 2 > 2(K + 1) - 1,$$

so if assertion true for K , then it's true for $K + 1$.

$\therefore$ True for all $n \geq 5$.

(b) None of $P_n = \rho_1 \rho_2 \cdots \rho_n + 1$ is a perfect square.

Pf. First note that since $\rho_1 = 2$, then $\rho_1 \rho_2 \cdots \rho_n$ is even, so $\rho_1 \rho_2 \cdots \rho_n + 1$ is odd. By Division Algorithm, $P_n = 4k + r$, $r = 0, 1, 2, 3$.

But since P_n is odd, $r = 1, 3$.

If $r = 1$, then $\rho_1 \rho_2 \cdots \rho_n + 1 = 4k + 1$, so

$$\rho_1 \rho_2 \cdots \rho_n = 4k, \text{ so } \rho_2 \rho_3 \cdots \rho_n = 2k.$$

But $\rho_2 \cdots \rho_n$ is odd since all factors are odd, and $2k$ is even.

$$\therefore r \neq 1.$$

$$\therefore P_n = 4k + 3 \text{ for all } n.$$

Suppose $P_n = s^2$, some s , and $s^2 = 4k + 3$.

Since s^2 is odd, so is s .

$$\therefore s = 2a + 1, \text{ some } a.$$

$$\therefore s^2 = (2a + 1)^2 = 4a^2 + 4a + 1 = 4k + 3.$$

$$\therefore 4a^2 + 4a = 4k + 2$$

$$2a^2 + 2a = 2k + 1.$$

But $2a^2 + 2a$ is even, and $2k + 1$ is odd.

$\therefore$ There is no s s.t. $P_n = s^2$. **(c)** $\frac{1}{\rho_1} + \frac{1}{\rho_2} + \cdots + \frac{1}{\rho_n}$ is never an integer.

Pf. Let $P = \rho_1 \rho_2 \cdots \rho_n$, and suppose

$$\frac{1}{\rho_1} + \frac{1}{\rho_2} + \cdots + \frac{1}{\rho_n} = a, \text{ some integer } a.$$

$$\therefore \frac{P}{\rho_1} + \frac{P}{\rho_2} + \cdots + \frac{P}{\rho_n} = aP.$$

For $\rho_1, \rho_1 \mid aP$ and $\rho_1 \mid \frac{P}{\rho_2}, \rho_1 \mid \frac{P}{\rho_3}, \dots, \rho_1 \mid \frac{P}{\rho_n}$.

$$\therefore \rho_1 \mid (P - \rho_2 - \rho_3 - \cdots - \rho_n).$$

[the indexing here is loose in the original — should likely read $P/\rho_2, \dots, P/\rho_n$ throughout, not $\rho_2, \dots, \rho_n$]

$$\therefore \rho_1 \mid \frac{P}{\rho_1} \Rightarrow \rho_1 \mid \rho_2 \rho_3 \cdots \rho_n, \text{ a contradiction.}$$

Similar reasoning applies for $\rho_2, \dots, \rho_n$.

$$\therefore \text{No such integer } a = \frac{1}{\rho_1} + \cdots + \frac{1}{\rho_n} \text{ exists.}$$

13.(a) If $n \mid m$, then $R_n \mid R_m$.

Pf. First prove Lemma:

If $m = Kn$, then

$$x^m - 1 = (x^n - 1)(x^{(K-1)n} + x^{(K-2)n} + \cdots + x^n + 1).$$

Pf. From problem #3, p.7, we know that

$$a^k - 1 = (a - 1)(a^{k-1} + a^{k-2} + \cdots + a + 1).$$

$\therefore$ let $a = x^n$.

$$\therefore x^{kn} - 1 = (x^n - 1)(x^{n(k-1)} + x^{n(k-2)} + \cdots + x^n + 1).$$

Since $kn = m$,

$$\therefore x^m - 1 = (x^n - 1)(x^{n(k-1)} + x^{n(k-2)} + \cdots + x^n + 1).$$

$$\text{Now } R_n = \frac{10^n - 1}{9}, R_m = \frac{10^m - 1}{9}.$$

$$\therefore \frac{R_m}{R_n} = \frac{10^m - 1}{10^n - 1} = \frac{10^{kn} - 1}{10^n - 1}.$$

$$\text{By the Lemma, } 10^{kn} - 1 = (10^n - 1)(10^{n(k-1)} + \cdots + 10^n + 1).$$

$$\begin{aligned} \therefore \frac{R_m}{R_n} &= \frac{(10^n - 1)(10^{n(k-1)} + \cdots + 10^n + 1)}{10^n - 1} \\ &= (10^{n(k-1)} + \cdots + 10^n + 1). \end{aligned}$$

$$\therefore n \mid m \Rightarrow R_n \mid R_m. \quad \text{(b) If } d \mid R_n \text{ and } d \mid R_m, \text{ then } d \mid R_{n+m}.$$

$$\text{Pf. } R_n = \frac{10^n - 1}{9}, R_m = \frac{10^m - 1}{9}.$$

$$R_{n+m} = \frac{10^{n+m} - 1}{9} = \frac{10^n 10^m - 1}{9}$$

$$\begin{aligned}
 &= \frac{10^n 10^m - 10^m + 10^m - 1}{9} \\
 &= \frac{10^m(10^n - 1) + 10^m - 1}{9} \\
 &= 10^m R_n + R_m.
 \end{aligned}$$

$\therefore d \mid R_n \Rightarrow R_n = dr$, some r .
 $d \mid R_m \Rightarrow R_m = ds$, some s .

$$\begin{aligned}
 \therefore R_{n+m} &= 10^m R_n + R_m \\
 &= 10^m dr + ds = d(10^m r + s).
 \end{aligned}$$

$\therefore d \mid R_{n+m}$. (c) If $\gcd(n, m) = 1$, then $\gcd(R_n, R_m) = 1$.

Pf. $\gcd(n, m) = 1 \Rightarrow 1 = an + bm$, some a, b .

Let $d = \gcd(R_n, R_m)$. $\therefore d \mid R_n, d \mid R_m$.

Since $n \mid an$, then $R_n \mid R_{an}$, by (a).

Since $m \mid bm$, then $R_m \mid R_{bm}$, by (a).

Since $d \mid R_n$ and $R_n \mid R_{an}$, then $d \mid R_{an}$.

Since $d \mid R_m$ and $R_m \mid R_{bm}$, then $d \mid R_{bm}$.

$\therefore$ By (b), $d \mid R_{an+bm}$.

But $R_{an+bm} = R_1 = 1$. $\therefore d \mid 1$. $\therefore d = 1$.

14. Find prime factors of R_{10} .

Since $2 \mid 10$ and $5 \mid 10$, then by 13(a), $R_2 \mid R_{10}$ and $R_5 \mid R_{10}$. $R_2 = 11$, $R_5 = 41271$.

$\therefore 11 \cdot 41 \cdot 271 \mid R_{10}$. But $R_{10}/(11 \cdot 41 \cdot 271) = 9091$, a prime.

$\therefore R_{10} = 11 \cdot 41 \cdot 271 \cdot 9091$.

3.3 The Goldbach Conjecture

1. Verify 1949 and 1951 are twin primes.

From table of primes, $\rho_{296} = 1949$, $\rho_{297} = 1951$.

Also, $\sqrt{1951} = 44.2$, and neither divisible by primes ≤ 43 .

2.(a) ρ_1, ρ_2 twin primes, show $n^2 = \rho_1 \rho_2 + 1$ for some n .

Pf. $\rho_2 = \rho_1 + 2$.

$$\begin{aligned}
 \therefore \rho_1 \rho_2 + 1 &= \rho_1(\rho_1 + 2) + 1 \\
 &= \rho_1^2 + 2\rho_1 + 1 = (\rho_1 + 1)^2.
 \end{aligned}$$

$\therefore$ Let $n = \rho_1 + 1$.

(b) The sum of twin primes $\rho, \rho + 2$ is divisible by 12, assuming $\rho \geq 3$.

Pf. Let $N = \rho + \rho + 2 = 2\rho + 2 = 2(\rho + 1)$.

Since $\rho + 1$ is even, $\rho + 1 = 2m$, some m .

$\therefore N = 4m$, $\therefore 4 \mid N$. Now let $\rho = 3q + r$, $r = 0, 1, 2$ by Div. Alg.

$r \neq 0$ since ρ is prime.

If $r = 1$, then $\rho + 2 = 3q + 3$, so $3 \mid \rho + 2$. Since $\rho + 2$ is prime, $r \neq 1$.

$\therefore r = 2$, and $\rho = 3q + 2$.

$$\therefore \rho + 2 = 3q + 4 = 3(q + 1) + 1.$$

$$\therefore N = \rho \cdot \rho + 2 = 3q + 2 + 3(q + 1) + 1$$

$$= 3(2q + 1) + 3.$$

$\therefore 3 \mid N$.

$3 \mid N$, $4 \mid N$, and since $\gcd(3, 4) = 1$, $3 \cdot 4 = 12 \mid N$ (Corollary 2, p.21).

$\therefore 12 \mid \rho(\rho + 2)$.

3. Find all pairs of primes s.t. $\rho - q = 3$.

Pf. Since $\rho = q + 3$, if q is odd, ρ is even.

And $\rho > 3$. But there is no even prime > 2 .

$\therefore q$ is even, and $\therefore q = 2$. $\therefore \rho = 5$. **4.** Every even integer $2n > 4$ is the sum of two primes, one $\leq n/2$, the other $\leq 3n/2$.

Verify for integers $6 \leq 2n \leq 38$.

Test for $3 \leq n \leq 19$.

$2n$	n	$n/2$	$3n/2$
$6 = 3 + 3$	3	1.5	4.5
$8 = 3 + 5$	4	2	6
$10 = 3 + 7$	5	2.5	7.5
$12 = 5 + 7$	6	3	9
$14 = 7 + 7$	7	3.5	10.5
$16 = 5 + 11$	8	4	12
$18 = 7 + 11$	9	4.5	13.5
$20 = 7 + 13$	10	5	15
$22 = 11 + 11$	11	5.5	16.5
$24 = 11 + 13$	12	6	18
$26 = 13 + 13$	13	6.5	19.5
$28 = 11 + 17$	14	7	21
$30 = 11 + 19$	15	7.5	22.5
$32 = 13 + 19$	16	8	24
$34 = 11 + 23$	17	8.5	25.5
$36 = 13 + 23$	18	9	27
$38 = 19 + 19$	19	9.5	28.5
$40 = 17 + 23$	20	10	30

$42 = 19 + 23$	21	10.5	31.5
$44 = 13 + 31$	22	11	33
$46 = 17 + 29$	23	11.5	34.5
$48 = 19 + 29$	24	12	36
$50 = 19 + 31$	25	12.5	37.5
$52 = 23 + 29$	26	13	39
$54 = 23 + 31$	27	13.5	40.5
$56 = 19 + 37$	28	14	42
$58 = 29 + 29$	29	14.5	43.5
$60 = 23 + 37$	30	15	45
$62 = 31 + 31$	31	15.5	46.5
$64 = 23 + 41$	32	16	48
$66 = 29 + 37$	33	16.5	49.5
$68 = 31 + 37$	34	17	51
$70 = 29 + 41$	35	17.5	52.5
$72 = 29 + 43$	36	18	54
$74 = 31 + 43$	37	18.5	55.5
$76 = 17 + 59$	38	19	57

5. Every odd integer can be written as $\rho + 2a^2$, ρ is prime or 1, $a \geq 0$. Show not the case for 5777.

$$5777 = \rho + 2a^2, \quad a = \sqrt{\frac{5777 - \rho}{2}}.$$

Minimum of ρ would be $\rho = 2$. Largest a would be $\sqrt{5775/2} = 53.7$.

Smallest a would be 0.

$\therefore$ Test $0 \leq a \leq 53$, or

test $5777 - 2a^2$ for $0 \leq a \leq 53$ and see if it is prime.

From spreadsheet, left column is a , middle column is $5777 - 2a^2$, and right column is a factor of $5777 - 2a^2$, showing that the numbers are not primes.

$\therefore$ No prime ρ exists s.t. $5777 = \rho + 2a^2$.

[Note: the original page includes a hand-drawn/spreadsheet-style table of a , $5777 - 2a^2$, and factor values for $a = 0$ to 53 verifying each is composite; the table itself is a numeric worksheet and is not reproduced here, only the surrounding explanation.]

6. Prove: (a) Every even integer > 2 is the sum of two primes $\iff$ (b) Every integer > 5 is the sum of three primes.

Pf. (a) $\implies$ (b) Let N be any integer > 5 .

If N is even, so is $N - 2$, and $N - 2 > 3$. By (a), $N - 2 = \rho_1 + \rho_2$. $\therefore N = 2 + \rho_1 + \rho_2$.

If N is odd, $N - 3$ is even and $N - 3 > 2$. $\therefore$ By (a), $N - 3 = \rho_1 + \rho_2$. $\therefore N = 3 + \rho_1 + \rho_2$.

$\therefore N$ is the sum of three primes.

(b) $\implies$ (a) Let N be any even integer > 2 .

Since $4 = 2 + 2$, let N' be N . $N' > 5$. From (b), $N' + 2 = \rho_1 + \rho_2 + \rho_3$. Since $N + 2$ is even, and $\rho_1\rho_2\rho_3$ is odd, one of ρ_1, ρ_2, ρ_3 must be even, and so one of ρ_1, ρ_2, ρ_3 must be 2, the only even prime. Let it be ρ_1 .

$\therefore N + 2 = 2 + \rho_2 + \rho_3$, $N = \rho_2 + \rho_3$. **7.** Every odd integer > 5 can be written as $\rho_1 + 2\rho_2$. Confirm for all odd integers ≤ 75 .

$$\begin{array}{ll}
7 = 3 + 2 \cdot 2 & 41 = 37 + 2 \cdot 2 \\
9 = 3 + 2 \cdot 3 & 43 = 29 + 2 \cdot 7 \\
11 = 5 + 2 \cdot 3 & 45 = 41 + 2 \cdot 2 \\
13 = 7 + 2 \cdot 3 & 47 = 37 + 2 \cdot 5 \\
15 = 11 + 2 \cdot 2 & 49 = 23 + 2 \cdot 13 \\
17 = 11 + 2 \cdot 3 & 51 = 29 + 2 \cdot 11 \\
17 = 13 + 2 \cdot 2 & 53 = 43 + 2 \cdot 5 \\
19 = 17 + 2 \cdot 1 & 55 = 25 + 2 \cdot 13 \\
21 = 17 + 2 \cdot 2 & 57 = 47 + 2 \cdot 5 \\
23 = 17 + 2 \cdot 3 & 59 = 37 + 2 \cdot 11 \\
25 = 19 + 2 \cdot 3 & 61 = 47 + 2 \cdot 7 \\
27 = 23 + 2 \cdot 2 & 63 = 59 + 2 \cdot 2 \\
29 = 23 + 2 \cdot 3 & 65 = 59 + 2 \cdot 3 \\
31 = 17 + 2 \cdot 7 & 67 = 59 + 2 \cdot 4 \\
33 = 29 + 2 \cdot 2 & 69 = 67 + 2 \cdot 1 \\
35 = 29 + 2 \cdot 3 & 71 = 67 + 2 \cdot 2 \\
37 = 31 + 2 \cdot 3 & 73 = 59 + 2 \cdot 7 \\
39 = 27 + 2 \cdot 25 (?) & 75 = 53 + 2 \cdot 11
\end{array}$$

[Note: several of these (e.g. "19 = 17 + 2 · 1", "67 = 59 + 2 · 4", "69 = 67 + 2 · 1") have a non-prime second term, which contradicts the problem statement that both terms should be prime; transcribed as legible from faint handwriting and flagged for review.]

8. $60 = \rho_1 + \rho_2$ in 6 ways.

$$\begin{array}{lll}
60 = 53 + 7 & 60 = 43 + 17 & 60 = 37 + 23 \\
60 = 47 + 13 & 60 = 41 + 19 & 60 = 31 + 29
\end{array}$$

$78 = \rho_1 + \rho_2$ in 7 ways.

$$\begin{array}{lll}
78 = 73 + 5 & 78 = 61 + 17 & 78 = 41 + 37 \\
78 = 71 + 7 & 78 = 59 + 19 & \\
78 = 67 + 11 & &
\end{array}$$

$84 = \rho_1 + \rho_2$ in 8 ways.

$$\begin{array}{lll}
84 = 79 + 5 & 84 = 53 + 31 & 84 = 67 + 17 \\
84 = 73 + 11 & 84 = 47 + 37 & 84 = 61 + 23 \\
84 = 71 + 13 & 84 = 43 + 41 &
\end{array}$$

9.(a) For $n > 3$, $n, n+2, n+4$ cannot all be prime.

Pf. By Division Alg., n can be expressed as

$$6q + r, 0 \leq r \leq 5.$$

$r \neq 0, 2, 4$, for then n would be even.

$$\therefore r = 1, 3, 5.$$

$r = 1$: $n = 6q + 1$, so $n + 2 = 6q + 3$, which is divisible by 3.

$$\therefore r \neq 1.$$

$r = 3$: $n = 6q + 3$, but then $3 \mid n$.

So $r \neq 3$.

$r = 5$: $n = 6q + 5$, then $n + 4 = 6q + 9$, so $3 \mid n + 4$. $\therefore r \neq 5$. $\therefore$ for no value of r can all three numbers be prime.

(b) Prime triplets: $\rho, \rho + 2, \rho + 6$:

$$5, 7, 11 \quad 41, 43, 47$$

$$11, 13, 17 \quad 101, 103, 107$$

$$17, 19, 23$$

10. $(n+1)! - 2, (n+1)! - 3, \dots, (n+1)! - (n+1)$ produces n consecutive composite numbers.

Pf. For each $k \leq n+1$, K is in the term $(n+1)!$, so that $K \mid [(n+1)! - K]$.

11. $f(n) = n^2 + n + 17$ Find smallest n for each
 $g(n) = n^2 + 21n + 1$ function that makes value a composite.
 $h(n) = 3n^2 + 3n + 23$

$$f(16) = 289 = 17^2$$

$$g(18) = 703 = 19 \cdot 37$$

$$h(22) = 1541 = 23 \cdot 67$$

12. Let ρ_n be n -th prime number. For $n \geq 3$, prove

$$\rho_{n+1}^2 < \rho_n \cdot \rho_{n+2}.$$

Pf. From section 3.2, $\rho_{n+1} < 2\rho_n$.

$$\therefore \rho_{n+3} < 2\rho_{n+2},$$

$$\text{so } \rho_{n+3}^2 < 4\rho_{n+2}^2 < 4\rho_{n+2}(2\rho_{n+1}) = 8\rho_{n+2}\rho_{n+1}.$$

$$\text{Since } \rho_5 = 11, 8\rho_{n+2}\rho_{n+1} < 15\rho_{n+2}\rho_{n+1}$$

$$\therefore \rho_{n+3}^2 < \rho_{n+1}\rho_{n+2} \text{ if } n \geq 5.$$

$$\text{For } n = 4, \rho_7^2 = (17)^2 = 289 < \rho_4\rho_5^2 \cdot \rho_6 = 7 \cdot 11 \cdot 13 = 1001.$$

$$n = 3: \rho_6^2 = (13)^2 = 169 < \rho_4\rho_5 \cdot \rho_4 = 5 \cdot 7 \cdot 11 = 385.$$

$$n = 2: \rho_5^2 = (11)^2 = 121 < \rho_3 \rho_4 = 5 \cdot 7 = 105.$$

[Note: the numeric checks for $n = 2, 3, 4$ on this page are hard to read precisely and the arithmetic as transcribed doesn't fully add up; transcribed best-effort and flagged for review.]

$$\therefore \text{ for } n \geq 3, \rho_{n+1}^2 < \rho_n \rho_{n+2}.$$

13. There are infinitely many primes of form $6n + 5$.

Pf. Assume only finite number of primes of form $6n + 5$. Let these be $q_1, q_2, q_3, \dots, q_s$. Consider $N = 6q_1 q_2 \cdots q_s - 1 = 6(q_1 q_2 \cdots q_s - 1) + 5$.

Let $N = r_1 r_2 \cdots r_t$ be the prime factorization. Since N is odd, $r_i \neq 2$, so each r_i can only be of form $6n + 1$, $6n + 3$, or $6n + 5$.

Since $(6n + 1)(6m + 1) = 36nm + 6n + 6m + 1 = 6(6nm + n + m) + 1$, product of two integers of $6n + 1$ form is same form.

Since $(6n + 3)(6m + 3) = 36nm + 18n + 18m + 9 = 6(6nm + 3n + 3m + 1) + 3$, product of two integers of $6n + 3$ form is same form.

Since $(6n + 1)(6m + 3) = 36nm + 18n + 6m + 3 = 6(6nm + 3n + m) + 3$, product of two integers of $6n + 1$ form and $6m + 3$ form is of $6n + 3$ form.

So, the only way for N to be of form $6n + 5$, which it is, N must contain at least one factor r_i of form $6n + 5$. But can't find such a prime among the $q_1, q_2, \dots, q_s$. If such a prime existed, then from construction of N ,

$N - 6q_1 q_2 \cdots q_s = -1$, both terms on left side would be divisible by this prime of $6n + 5$ form, so -1 , and thus 1 , would be divisible by this prime, a contradiction.

$\therefore$ Can't be finite # of primes of $6n + 5$ form.

$$\mathbf{14.} \quad 4(3 \cdot 7 \cdot 11) - 1 = 13 \cdot 71. \quad 71 \text{ is of form } 4n + 3.$$

$$4(3 \cdot 7 \cdot 11 \cdot 5) - 1 = 13,859, \text{ a prime of form } 4n + 3.$$

15. Five consecutive odd integers, 4 of which are prime.

$$5, 7, 9, 11 \quad 11, 13, 15, 17, 19 \quad 187, 193, 195, 197, 199$$

$$5, 7, 9, 11, 13 \quad 101, 103, 105, 107, 109$$

$$\mathbf{16.} \quad 23 = \rho_9 = \rho_{2 \cdot 4 + 1} = 2\rho_{2 \cdot 4} + \sum_{k_0}^{2 \cdot 4 - 1} \epsilon_k \rho_k$$

$$= 2\rho_8 + \sum_{k_0}^7 \epsilon_k \rho_k$$

$$\therefore 23 = 2 \cdot 19 + \epsilon_0 + 2\epsilon_1 + 3\epsilon_2 + 5\epsilon_3 + 7\epsilon_4 + 11\epsilon_5 + 13\epsilon_6 + 17\epsilon_7$$

$$= 38 + 1 + 2 + 3 + 5 - 7 + 11 - 13 - 17$$

$$= 38 + (2 - 17) + (1 + 3 + 5) + (-7 + 11 - 13)$$

$$= 38 - 15 + 9 - 9$$

$$\begin{aligned} 29 &= \rho_{10} = \rho_{2.5} = \rho_{2.5-1} + \sum_{k=0}^{2.5-2} \epsilon_k \rho_k = 23 + \sum_{k=0}^8 \epsilon_k \rho_k \\ &= 23 + \epsilon_0 + 2\epsilon_1 + 3\epsilon_2 + 5\epsilon_3 + 7\epsilon_4 + 11\epsilon_5 + 13\epsilon_6 + 17\epsilon_7 + 19\epsilon_8 \\ &= 23 + 6 + 12 + 6 - 6 \\ &= 23 + (1 + 2 + 3) + (5 + 7) + (-11 + 17) + (13 - 19) \\ &= 23 + 1 + 2 + 3 + 5 + 7 - 11 + 17 + 13 - 19 \end{aligned}$$

$$31 = \rho_{11} = \rho_{2.5+1} = 2\rho_{2.5} + \sum_{k=0}^{2.5-1} \epsilon_k \rho_k = 2 \cdot 29 + \sum_{k=0}^9 \epsilon_k \rho_k$$

$$= 2 \cdot 29 + \epsilon_0 + 2\epsilon_1 + 3\epsilon_2 + 5\epsilon_3 + 7\epsilon_4 + 11\epsilon_5 + 13\epsilon_6 + 17\epsilon_7 + 19\epsilon_8 + 23\epsilon_9$$

$$31 = 58 - 27, \text{ so find } -27.$$

$$-27 = 0 - 8 + (-11 - 4 - 4)$$

$$= (2 + 3 - 5) + (-1 - 7) + (-11 - 4 - 4)$$

$$\therefore 31 = 2 \cdot 29 - 1 + 2 + 3 - 5 - 7 - 11 + 13 - 17 + 19 - 23$$

$$37 = \rho_{12} = \rho_{2.6} = \rho_{2.6-1} + \sum_{k=0}^{2.6-2} \epsilon_k \rho_k = 31 + \sum_{k=0}^{10} \epsilon_k \rho_k$$

$$= 31 + \epsilon_0 + 2\epsilon_1 + 3\epsilon_2 + 5\epsilon_3 + 7\epsilon_4 + 11\epsilon_5 + 13\epsilon_6 + 17\epsilon_7 + 19\epsilon_8 + 23\epsilon_9 + 29\epsilon_{10}$$

$$37 = 31 + 6 = 31 + 6 + (-21 - 2 + -42) + 46$$

$$= 31(1 + 2 + 3) + (5 - 7 + 11 - 13 - 19 - 23) + (17 + 29)$$

$$\therefore 37 = 31 + 1 + 2 + 3 + 5 - 7 + 11 - 13 + 17 - 19 - 23 + 29$$

17. Show 509 and 877 can't be the sum of a prime and power of 2.

[Note: the signed-sum derivations on pages 172-174 (problem 16) are very dense; several intermediate arithmetic lines (e.g. "31 = 58 - 27" decomposition, "37 = 31 + 6 + (-21 - 2 - 42) + 46") are hard to verify digit-by-digit from the handwriting. Transcribed best-effort; flagged for review.] From spreadsheet,

1st column is n .

2nd column is 2^n .

3rd column is $509 - 2^n$.

4th column is $877 - 2^n$.

None of the positive entries in 3rd & 4th cols. is prime.

[Note: the spreadsheet table itself (columns $n = 0 \dots 10$, 2^n , $509 - 2^n$, $877 - 2^n$) is a numeric worksheet and is not reproduced here.]

18.(a) ρ prime, $\rho \nmid b$, show every ρ th term in $a_1, a + b, a + 2b, \dots$ is divisible by ρ .

Better restatement: there is a term within the first ρ terms that is divisible by ρ , and every ρ th term thereafter is divisible by ρ (because the ρ th term from the beginning is not always divisible by ρ).

Pf. Since $\rho \nmid b$ and ρ is prime, $\gcd(\rho, b) = 1$.

$\therefore$ There exist integers r, s s.t. $a + rb = 1$ [1]

Consider $n_k = K\rho - as$, $K = 1, 2, 3, \dots$

For $K = 1$, $n_2 = \rho - as$, and clearly $n_2 < \rho$.

Note that n_2 is the ρ th term (after n_1), n_3 is the ρ th term after n_2 , etc.

$$\begin{aligned} a + n_k b &= a + (K\rho - as)b = a + K\rho b - abs \\ &= a(1 - bs) + K\rho b \\ &= a(r\rho) + K\rho b \quad (\text{using [1]}). \end{aligned}$$

$\therefore \rho \mid a + n_k b$, so there is a term within the first ρ terms that is divisible by ρ , and every ρ th term after that is divisible by ρ .

(b) If b is odd in $a, a + b, a + 2b, \dots$,

Then since $2 \nmid b$ and 2 is prime, by (a) either a or $a + b$ is divisible by 2, and every 2nd term is also. So every other term is even.

19.

$$25 = 5 + 7 + 13 \quad 81 = 3 + 5 + 73$$

$$69 = 3 + 5 + 61 \quad 125 = 5 + 7 + 113$$

20. If ρ and $\rho^2 + 8$ are both prime, then $\rho^3 + 4$ is prime.

Pf. As in prob. #4 of Problem 3.1, if $\rho > 3$ is prime it is of form $6k + 1$ or $6k + 5$.

$$\therefore \rho^2 + 8 = 36k^2 + 12k + 9, \text{ or } 36k^2 + 60k + 33.$$

But $3 \mid (36k^2 + 12k + 9)$

and $3 \mid (36k^2 + 60k + 27)$.

So $\rho^2 + 8$ is not prime if $\rho > 3$.

$\therefore \rho \leq 3$.

$\rho = 2 \Rightarrow 17$

$\rho = 3 \Rightarrow 41$

$\rho = 4 \Rightarrow 31$ (not prime)

21.(a) Let $k > 0$ be any integer, and let a, b be integers with $\gcd(a, b) = 1$. Prove the series, $a + b, a + 2b, a + 3b, \dots$ contains k consecutive composite terms.

Pf. Let k be any integer, and let n be the integer formed by:

$$n = (a + b)(a + 2b) \cdots (a + kb).$$

Consider this series of k terms:

$$a + (n + 1)b, a + (n + 2)b, \dots, a + (n + k)b.$$

For the i -th term, ($1 \leq i \leq k$)

$$a + (n + i)b =$$

$$a + nb + ib = a + ib + nb$$

But n contains $(a + ib)$ as one of its terms by definition of n .

$\therefore (a + ib) \mid (a + (n + i)b)$ for all i .

For $k \geq 2$, $a + ib < a + (n + i)b$.

For $K = 1$, $n = a + b$, and the i -th term of our series is

$$a + (n + i)b = a + (a + b + i)b = a(1 + b) + b(b + i).$$

$\therefore a + b < a + (a + b + i)b$.

$\therefore$ for $K = 1$, $1 < a + ib$ for all i .

Also, $1 < a + ib$ for all i .

$\therefore$ all K terms of $a + (n + 1)b, \dots, a + (n + K)b$ are divisible by an integer that is > 1 and $<$ the term.

$\therefore$ all K terms are composite.

Note: proof doesn't use $\gcd(a, b) = 1$. It does assume $a \neq 0$.

(b) From our construction, let

$$\begin{aligned} n &= (6 + 5)(6 + 2 \cdot 5)(6 + 3 \cdot 5)(6 + 4 \cdot 5)(6 + 5 \cdot 5) \\ &= 2,978,976. \end{aligned}$$

$$\therefore 6 + (n + 1)5 = 14,894,891 \div (6 + 5) = 11$$

$$6 + (n + 2)5 = 14,894,896 \div (6 + 2 \cdot 5) = 16$$

$$6 + (n + 3)5 = 14,894,901 \div 21$$

$$6 + (n + 4)5 = 14,894,906 \div 26$$

$$6 + (n + 5)5 = 14,894,911 \div 31$$

$\therefore$ The above 5 consecutive terms are composite.

22. Show 13 is largest prime that can divide two successive integers of form $n^2 + 3$.

Pf. First, look at first possibilities for n .

n	$n^2 + 3$	prime fac.	n	$n^2 + 3$	prime fac.
0	3	3	9	84	$2^2 \cdot 3 \cdot 7$
1	4	2^2	10	103	103
2	7	7	11	124	$2^2 \cdot 31$
3	12	$2^2 \cdot 3$	12	147	$7^2 \cdot 3$
4	19	19	13	172	$2^2 \cdot 43$
5	28	$2^2 \cdot 7$	14	195	$3 \cdot 5 \cdot 13$
6	39	$3 \cdot 13$	15	228	$2^2 \cdot 3 \cdot 19$
7	52	$13 \cdot 2^2$	16	257	257
8	67	67			

It seems that after $n \geq 8$, there are no common factors for adjacent terms.

Adjacent terms are $n^2 + 3$, $(n + 1)^2 + 3 = n^2 + 2n + 4$.

Use Euclid's algorithm to find gcd for $n \geq 8$.

$\therefore$ Suppose 1st term is even, $n = 2s$, and $s \geq 4$.

$\therefore$ terms are $(2s)^2 + 3 = 4s^2 + 3$

$(2s + 1)^2 + 3 = 4s^2 + 4s + 4$

$$4s^2 + 4s + 4 = 1 \cdot (4s^2 + 3) + 4s + 1, \quad 4s + 1 < 4s^2 \text{ for } s \geq 4$$

$$4s^2 + 3 = (s - 3)(4s + 1) + 3s + 4 \quad (\#)$$

[the coefficient/remainder bookkeeping in this Euclidean-algorithm step is hard to follow exactly from the handwriting]

So $\gcd(\dots) = 13$ if $3s + 4 > 3$, or $s > -\frac{1}{3}$.

So, must prove $\gcd = 13$ for $4 \leq s \leq 16$ for $(\#)$.

So $(\#)$ becomes for each s :

$$3s + 4 = 1 \cdot (3s - 3) + r$$

$$s = 4: 16 = 16 \cdot 1 \quad \gcd = 1$$

$$s = 5: 19 = 9 \cdot 2 + 1, 2 \cdot 1 \quad \gcd = 1 \quad s = 6: 22 = 7 \cdot 3 + 1, 3 \cdot 3 - 1 \quad \gcd = 1$$

$$s = 7: 25 = 6 \cdot 4 + 1, 4 - 4 \cdot 1 \quad \gcd = 1$$

$$s = 8: 28 = 5 \cdot 5 + 3, 5 \div 2 = 1, 2 \cdot 2 - 1, \gcd = 1$$

$$s = 9: 31 = 5 \cdot 6 + 1, 6 - 6 \cdot 1 \quad \gcd = 1$$

$$s = 10: 34 = 4 \cdot 7 + 6, 7 - (1 \cdot 6 + 1), 2 \cdot 6 + 1 \quad \gcd = 1$$

$$s = 11: 37 = 4 \cdot 8 + 5, 8 = 7 \cdot (1 \cdot 5 + 3), 2 \cdot 3 - 1, \gcd = 1$$

$$s = 12: 40 = 2 \cdot 9 + 4, 9 = 4 \cdot 2 + 1 \quad \gcd = 1$$

$$s = 13: 43 = 4 \cdot 10 + 3, 10 = 3 \cdot 3 + 1, 1 \cdot 5 \cdot 3 + 1 \quad \gcd = 1$$

$$s = 14: 46 = 4 \cdot 11 + 2, 11 = 5 \cdot 2 + 1, 2 \cdot 5 + 1 \quad \gcd = 1$$

$$s = 15: 49 = 4 \cdot 12 + 1, 12 - 12 \cdot 1 \quad \gcd = 1$$

$$s = 16: 52 = 4 \cdot 13 \quad \gcd = 13$$

[Note: the individual Euclidean-algorithm steps for $s = 6$ through 15 are very terse/abbreviated in the original and hard to verify digit-by-digit; the final gcd value stated for each s is transcribed with more confidence than the intermediate arithmetic.]

$\therefore$ Examples show $\gcd = 1$ or $\gcd = 13$ for adjacent terms $0 \leq s \leq 16$, and above shows $\gcd = 1$ or $\gcd = 13$ for all n if 1st term is even.

Now suppose first term is odd, i.e., $n = 2s + 1$ and $s \geq 4$.

$$(2s + 1)^2 + 3 = 4s^2 + 4s + 4$$

$$(2s + 2)^2 + 3 = 4s^2 + 8s + 7$$

$$4s^2 + 8s + 7 = 1 \cdot (4s^2 + 4s + 4) + 4s + 3, \quad 4s + 3 < 4s^2 + 4s + 4 \text{ for } s \geq 4.$$

$$4s + 3 = 3(s + 4) + s - 9 \quad (\#) \quad 0 < 5 - 9 + 4 \text{ if } s \geq 9$$

$$s + 4 = 1 \cdot (3 - 9) + 13 \quad 13 < 5 - 9, \text{ if } 24 < 5$$

$$\gcd(s - 9, 13) = 1 \text{ or } 13$$

$\therefore \gcd = 1$ if $s \geq 22$. $(\#)$ becomes

for $4 \leq s \leq 22$, $4s + 3 = a(s + 4) + \dots$

$$s = 4 : \quad 19 = 2 \cdot 8 + 3, \quad 8 = 2 \cdot 3 + 2, \quad 3 = 1(2 + 1) \quad \gcd = 1$$

$$s = 5 : \quad 23 = 2 \cdot 7 + 9, \quad 9 = 1 \cdot 9 + 4, \quad s = 4 \quad \gcd = 1$$

$$s = 6 : \quad 27 = 2 \cdot 10 + 7 \quad \gcd = 1$$

$$s = 7 : \quad 31 = 2 \cdot 11 + 9 \quad \gcd = 1$$

$$s = 8 : \quad 35 = 2 \cdot 12 + 11 \quad \gcd = 1$$

$$s = 9 : \quad 39 = 3 \cdot 13 \quad \gcd = 13$$

$$s = 10 : \quad 43 = 3 \cdot 15 + 12 \quad \gcd = 1$$

$$s = 11 : \quad 47 = 3 \cdot 16 + 3 \quad \gcd = 1$$

$$s = 12 : \quad 51 = 3 \cdot 17 + 3 \quad \gcd = 1$$

$$s = 13 : \quad 55 = 3 \cdot 18 + 5 \quad \gcd = 1$$

$$s = 14 : \quad 59 = 3 \cdot 19 + 6 \quad \gcd = 1$$

$$s = 15 : \quad 63 = 3 \cdot 20 + 7 \quad \gcd = 1$$

$$s = 16 : \quad 67 = 3 \cdot 21 + 8 \quad \gcd = 1$$

$$s = 17 : \quad 71 = 3 \cdot 21 + 8 \quad \gcd = 1$$

$$s = 18 : \quad 75 = 3 \cdot 22 + 9 \quad \gcd = 1$$

$$s = 19 : \quad 79 = 3 \cdot 23 + 10 \quad \gcd = 1$$

$$s = 20 : \quad 83 = 3 \cdot 24 + 11 \quad \gcd = 1$$

$$s = 21 : \quad 87 = 3 \cdot 25 + 12, \quad 25 = 2 \cdot 12 + 1 \quad \gcd = 1$$

$$s = 22 : \quad 91 = 3 \cdot 26 + 13 \quad \gcd = 13$$

[Note: this page's Euclidean-algorithm case analysis (for the "odd first term" branch of problem 22) is dense and abbreviated; the first lines especially are hard to read precisely. Transcribed best-effort.] $\therefore$ for $4 \leq s \leq 21$, $\gcd = 1$ or $\gcd = 13$.

For $s > 22$, $\gcd = 1$ or 13 .

$\therefore$ for all $s \geq 4$, $\gcd = 1$ or 13 .

$\therefore$ for all terms of $n^2 + 3$, $(n + 1)^2 + 3$ beginning with n odd and $n \geq 7$, $\gcd = 1$ or 13 .

$\therefore$ for all $n \geq 0$, adjacent terms have $\gcd$ of 1 or 13 .

Note: it would have been difficult to start with

$$n^2 + 2n + 4 = 1 \cdot (n^2 + 3) + 2n + 1$$

$$n^2 + 3 = n(2n + 1) - 2n + 3.$$

This approach is not fruitful.

23.(a) Twin primes with a triangular mean.

Some triangular numbers: $1 + 2 = 3$, $1 + 2 + 3 = 6$, $1 + 2 + 3 + 4 = 10$, $10 + 5 = 15$, $15 + 6 = 21$, $21 + 7 = 28$.

$19, 23$ are adjacent but not twin primes.

$(5 + 7)/2 = 6$, so $5, 7$ work. Suppose $\rho > 7$.

From problem #1(a), sec. 1.3, a number is triangular $\iff$ it is of form $\frac{n(n+1)}{2}$.

$$\therefore \frac{\rho + \rho + 2}{2} = \frac{n(n+1)}{2}.$$

$$\therefore 2\rho + 2 = n^2 + n.$$

$$2\rho = n^2 + n - 2 = (n + 2)(n - 1).$$

Since 2ρ is even, one of $n + 2$ or $n - 1$ must be even.

Suppose $n - 1 = 2K$: $n + 2 = 2K + 3$.

$$\therefore 2\rho = (2K + 3)(2K).$$

$$\rho = (2K + 3)(K).$$

For ρ to be prime, $K = 1$, $\therefore \rho = 5$.

Suppose $n + 2 = 2K$: $\therefore n - 1 = 2K - 3$.

$$\therefore 2\rho = 2K(2K - 3).$$

$$\rho = K(2K - 3).$$

$2K - 3 = 1$, $K = 2$, or

$K = 1$, $2K - 3 = -1$.

$\therefore n + 2 \neq 2K$.

So, only possible twin primes are $5, 7$.

(b) Twin primes with square mean.

Suppose: $\frac{\rho + \rho + 2}{2} = n^2$.

$$\therefore \rho + 1 = n^2, \quad \rho = n^2 - 1 = (n + 1)(n - 1).$$

For ρ to be prime, $n - 1 = 1$, $\therefore n = 2$.

$$\therefore n^2 = 4.$$

$\therefore$ Only possibility is 3, 5.

24. Determine all twin primes ρ and $q = \rho + 2$ for which $\rho q - 2$ is prime.

Pf. 3, 5: $3 \cdot 5 - 2 = 13$.

Suppose $\rho > 3$. All primes > 3 are of form $6k + 1$ or $6k + 5$. But ρ must be of form $6k + 5$ since $6k + 5 = \rho$, $q + 2 = 6k + 8 = 6(k + 1) + 2$.

$\therefore$ Let $\rho = 6k + 5$, $q = 6k + 7$.

$$\begin{aligned} \therefore (6k + 5)(6k + 7) &= 36k^2 + 72k + 35 - 2 \\ &= 36k^2 + 72k + 33 \\ &= 3(12k^2 + 24k + 11). \end{aligned}$$

$\therefore$ if $\rho > 3$, there are no twin primes such that $\rho q - 2$ is prime.

3, 5 is the only pair.

25. Let ρ_n be n -th prime. For $n > 3$, show

$$\rho_n < \rho_1 + \rho_2 + \cdots + \rho_{n-1}.$$

Pf. $\rho_3 = 5 = 2 + 3 = \rho_1 + \rho_2$.

$$\rho_4 = 7 < 2 + 3 + 5 = \rho_1 + \rho_2 + \rho_3.$$

$\therefore$ Assume for $K \geq 4$,

$$\rho_K < \rho_1 + \rho_2 + \cdots + \rho_{K-1}.$$

$$\therefore 2\rho_K < \rho_1 + \cdots + \rho_{K-1} + \rho_K.$$

By Bertrand's conjecture, $\exists \rho$ s.t.

$$\rho_K < \rho < 2\rho_K.$$

But $\rho_K < \rho_{K+1} \leq \rho$.

$$\therefore \rho_{K+1} \leq \rho < 2\rho_K < \rho_1 + \cdots + \rho_{K-1} + \rho_K.$$

$\therefore$ true for $K + 1$.

$\therefore$ True for all $n > 4$.

26.(a) Infinitely many primes ending in 83.

Pf. $100 = 2^2 \cdot 5^2$ and 83 are relatively prime.

[Note: the original page appears to write " $83 = 3 \times 11$ " here, which is numerically wrong ($3 \times 11 = 33$, and 83 is itself prime); transcribed as the intended claim ($\gcd(100, 83) = 1$) rather than the erroneous factorization.] $\therefore$ By Dirichlet's Theorem, the series $33, 33 + 100, 33 + 2 \cdot 100, \dots$

$= 33, 133, 233, \dots$ contains infinitely many primes.

(b) Infinitely many primes which do not belong to any pair of twin primes.

Pf. 5 and $21 = 3 \cdot 7$ are relatively prime.

$\therefore$ By Dirichlet's Theorem, that series, $5 + 21k$, for $k = 1, 2, 3, \dots$ contains infinitely many primes.

Let ρ be one such prime.

$\therefore$ For some k , $\rho = 5 + 21k$.

$$\rho + 2 = 7 + 21k = 7(1 + 3k)$$

$$\rho - 2 = 3 + 21k = 3(1 + 7k)$$

$\therefore \rho + 2$ and $\rho - 2$ can't be prime.

$\therefore$ all the primes contained in $5 + 21k$ cannot be members of twin primes. (c) There exists a prime ending in as many consecutive 1's as desired.

Pf. $R_n = \frac{10^n - 1}{9}$ by def.

Since $1 \cdot 10^n - 9R_n = 1$, $\gcd(10^n, R_n) = 1$.

$\therefore$ Using Dirichlet's Theorem, form the series

$$10^n \cdot k + R_n, \quad k = 1, 2, 3, \dots$$

which is for $n = 1$: 11, 21, 31, $\dots$

$n = 2$: 111, 211, 311, $\dots$

Each contains infinitely many primes.

$\therefore$ each contains at least one prime ending in n 1's.

(d) There are infinitely many primes that contain but do not end in the block of digits 123456789.

Pf. Consider $10^{10} = 2^{10} \cdot 5^{10}$.

The number 123456789 is odd, so contains no factor of 2, and does not end in 0 or 5, so contains no factor of 5.

$\therefore 10^{10}$ and 123456789 are relatively prime. $\therefore$ By Dirichlet's Theorem, the series $10^{10} \cdot k + 123456789$ contains infinitely many primes, and each number in the series contains 123456789 but ends in 1.

A few numbers in the series are: 1234567891, 21234567891, 31234567891.

27. For every $n \geq 2$, there exists a prime $\rho \leq n < 2\rho$.

Pf. Suppose n is odd. $\exists k$ s.t. $n = 2k + 1$, and since $n \geq 2$, $k \geq 1$.

By Bertrand's conjecture, there is a prime ρ s.t. $k < \rho < 2k$, so $\rho < n$.

Also, $2k < 2\rho$, so $2k + 1 \leq 2\rho$. But $2k + 1 = n$ is odd, and 2ρ is even.

$\therefore n < 2\rho$. But $n \leq 2\rho$. $\therefore n < 2\rho$.

$\therefore \exists \rho$ s.t. $\rho < n < 2\rho$, so $n < 2\rho$.

$\therefore \rho \leq n < 2\rho$.

Suppose n is even. $\exists K$ s.t. $n = 2K$, $K \geq 1$.

By Bertrand's conjecture, there is a prime ρ s.t. $K < \rho < 2K = n$, so $\rho < n$.

$\therefore n = 2K < 2\rho$, so $n < 2\rho$.

$\therefore \rho < n < 2\rho$.

28.(a) If $n > 1$, show that $n!$ is never a perfect square.

Lemma 1. If $\rho_1 < \rho_2$ are adjacent primes, then if $\rho_1 < N < \rho_2^2$, then the prime factors of N are less than ρ_1 (for $\rho_1 > 3$).

Pf. Let $q_1^{k_1} \cdots q_r^{k_r} = N$, $r \geq 2$. Suppose $q_i = \rho_2$, some i .

Since $q_i \geq 2$ for all i , $N = q_1 \cdots q_r \geq \rho_2 \cdot 2^{r-2} \cdot \rho_2$.

$\therefore N = q_1 \cdots q_r \geq \rho_2 \cdot 2 \cdot 2^{r-2} \geq \rho_2^2$.

Since $2\rho_1 > \rho_2$ (top p.50, a direct consequence of Bertrand's conjecture), $N > \rho_2^2$, a contradiction.

Lemma 2. Let $q_1^{k_1} \cdots q_r^{k_r}$ be the prime canonical factorization of $n!$. Then $k_1 \geq 1$ for all $n \geq 2$. *Pf.* Clearly true for $n = 2, 3$. Let N be any integer > 3 .

Suppose true for $N!$: $N! = q_1^{k_1} \cdots q_r^{k_r}$ ($q_i < q_r$).

Since each term of $N!$ is $< N$, then the prime factors of each term (which is $<$ each term) are $< N$. $\therefore q_i < N$, and so $q_r < N$.

Consider $(N + 1)! = N!(N + 1)$.

If $N + 1$ is prime, then $q_r < N + 1$.

$\therefore (N + 1)! = q_1^{k_1} \cdots q_r^{k_r} (N + 1)$.

$\therefore$ lemma true.

Suppose $N + 1$ is not prime. Then q_r must be largest prime $\leq N + 1$. If a larger prime existed, it would be a term in $(N + 1)!$ and $\therefore$ would be represented in the prime factorization: $q_1^{k_1} \cdots q_r^{k_r}$. By Lemma 1 above, prime factors of $N + 1$ are $< q_r$.

$\therefore q_r$ remains largest prime factor of ρ it has exponent 1.

$\therefore$ Lemma true for $N + 1$ when true for N .

Back to main problem:

$\therefore$ The prime factorization of $n!$ has exponent 1 for largest factor.

$\therefore$ If $n! = a^2$, some a , all prime factors would have even exponents, as would the last factor.

$\therefore n! \neq a^2$ for any $n \geq 2$.

Note: By Lemma 2, $n!$ can't be any power of any number.

(b) Find values of $n \geq 1$ for which

$n! + (n+1)! + (n+2)!$ is a perfect square.

$$\begin{aligned} n! + (n+1)! + (n+2)! &= n![1 + (n+1) + (n+1)(n+2)] \\ &= n![1 + (n+1) + n^2 + 3n + 2] \\ &= n![n^2 + 4n + 4] \\ &= n!(n+2)^2. \end{aligned}$$

$\therefore$ Let $a^2 = n!(n+2)^2$.

From (a), all the prime factors of a have even exponents. $\therefore$ prime factors of $n! \cdot (n+2)^2$ should have even exponents.

But $n!$ has, for its largest prime factor (say $n > 2$), an exponent of 1 (from (a) above). Call this factor ρ . Even if $(n+2)^2$ had a ρ factor, its exponent would be even.

Thus, the exponent of ρ in the factorization of $n!(n+2)^2$ will be odd. This contradicts expectation of a^2 .

$\therefore n$ can't be ≥ 2 .

$\therefore$ for $n = 1$, $n! + (n+1)! + (n+2)! = 9 = 3^2$.

$\therefore$ Only $n = 1$ is statement true.

Chapter 4

The Theory of Congruences

4.2 Basic Properties of Congruence

Def. Complete set of residues modulo n .

A set $A = \{a_1, a_2, \dots, a_n\}$ is said to form a complete set of residues modulo $n \iff$ given any integer z , there is an $a_i \in A$ s.t. $a_i \equiv z - n \cdot K_n$ for some integer K_n , but for $a_j \neq a_i$ and $a_j \in A$, then there exists integers q_n, r , $0 \leq r < n$, s.t. $a_i - z = qn + r$.

Lemma. Let $A = \{a_1, a_2, \dots, a_n\}$ be a complete set of residues modulo n , and let $B = \{0, 1, 2, \dots, n-1\}$. Then there is a one-to-one correspondence between A and B .

Pf. Let $K \in B$. By def. of complete set of residues, there is an $a_i \in A$ s.t. $K \equiv a_i \pmod{n}$, and $K \not\equiv a_j \pmod{n}$ for all $a_j \neq a_i$.

Since there are n elements in B and in A , each element of B is matched with one and only one element of A . $\therefore$ Given any element of A , there is an element of B associated with it, and only one element of B . For if $a_i \in A$ is associated with two elements of B , say b_i and b_j , then $a_i \equiv b_i \pmod{n}$ and $a_i \equiv b_j \pmod{n}$, so $b_i \equiv b_j \pmod{n}$, which is impossible, since $0 \leq b_i < n$, $0 \leq b_j < n$, so $0 \leq |b_i - b_j| < n$, and so n can't divide a number less than itself.

Theorem 1. $A = \{a_1, a_2, \dots, a_n\}$ is a complete set of residues modulo $n \iff$ for $a_i \in A$, $a_i \not\equiv a_j \pmod{n}$ for $i \neq j$.

Pf. (1) Suppose A is a complete set, let $a_i, a_j \in A$ s.t. $a_i \neq a_j$, and suppose $a_i \equiv a_j \pmod{n}$.

$\therefore a_i - a_j = Kn$, some K .

Let z be s.t. $z \equiv a_i \pmod{n}$. Such a z exists since $a_i + cn \equiv a_i \pmod{n}$, where c is any integer.

$\therefore z - a_i = sn$, some s . [2] Adding [1] and [2], $z - a_j = (K + s)n$,

$\therefore z \equiv a_j \pmod{n}$, contradicting def. of complete set. $\therefore a_i \not\equiv a_j \pmod{n}$.

(2) Suppose $a_i \not\equiv a_j \pmod{n}$ for $a_i, a_j \in A$, $a_i \neq a_j$.

Consider $a_i = q_i n + r_i$, for $1 \leq i \leq n$, $0 \leq r_i < n$.

Then $r_i \neq r_j$, for if $r_i = r_j$, because it $a_i - a_j = (q_i - q_j)n$, and $\therefore a_i \equiv a_j \pmod{n}$.

Since there are n members in set A , there are n different r_i , $0 \leq r_i < n$, so there is a one-to-one correspondence between a_i and $\{0, 1, \dots, n-1\}$, i.e., given any r_i s.t. $0 \leq r_i < n$, there is an a_i s.t. $a_i \equiv r_i \pmod{n}$.

Now let z be any integer.

By Div. Algorithm, $z = qn + r$, $0 \leq r < n$.

From statement above, there is an $a_i \in A$ s.t. $a_i - r = Kn$, some K .

$$\therefore z = qn + r + qn + (a_i - Kn), \text{ so}$$

$$z = a_i + (q - K)n, \quad [3]$$

so $z \equiv a_i \pmod{n}$. Suppose $z \equiv a_j \pmod{n}$, $a_j \neq a_i$.

$\therefore z - a_j = sn$, some s . From [3],

$$a_i + (q - K)n - a_j = sn, \quad a_i - a_j = (s - q + K)n.$$

$\therefore a_i \equiv a_j \pmod{n}$, a contradiction.

$\therefore z$ is $\equiv$ to one and only one of $a_i \in A$, $\text{mod } n$.

Theorem 2. If $ab \equiv 0 \pmod{\rho}$, ρ prime, then $a \equiv 0 \pmod{\rho}$ or $b \equiv 0 \pmod{\rho}$.

Pf. Suppose $a \not\equiv 0 \pmod{\rho}$.

$\therefore a = q\rho + r$, $0 < r < \rho$. Thus, r and ρ are relatively prime.

Since $\exists K$ s.t. $ab = K\rho$, then

$$ab = q\rho b + rb, \quad K\rho = q\rho b + rb.$$

$$\rho(K - qb) = rb.$$

$\therefore$ By Euclid's lemma, $\rho \mid b$. $\therefore \exists s$ s.t. $b = \rho s$. $\therefore b \equiv 0 \pmod{\rho}$.

Theorem 3. $z \equiv a \pmod{n} \iff z + cn \equiv a + dn \pmod{n}$.

Pf. (1) Suppose $z \equiv a \pmod{n}$.

$\therefore z - a = Kn$, some K .

$$\therefore z + cn - (a + dn) = z - a + cn - dn$$

$$= Kn + (c - d)n$$

$$= (K + c - d)n.$$

$\therefore z + cn \equiv a + dn \pmod{n}$.

(2) Suppose $z + cn \equiv a + dn \pmod{n}$, some K .

$$\therefore z - a = -cn + dn + Kn$$

$$= (K - c + d)n.$$

$\therefore z \equiv a \pmod{n}$.

Problems

1.(a). If $a \equiv b \pmod{n}$ and $m \mid n$, then $a \equiv b \pmod{m}$.

Pf. $a \equiv b \pmod{n} \Rightarrow a - b = Kn$, some K .

$m \mid n \Rightarrow n = rm$, some r .

$\therefore a - b = Krm \Rightarrow a \equiv b \pmod{m}$. **(b).** If $a \equiv b \pmod{n}$, and $c > 0$, then $ca \equiv cb \pmod{cn}$.

Pf. $a - b = Kn$, some K . $\therefore ca - cb = Kcn \Rightarrow ca \equiv cb \pmod{cn}$.

(c) If $a \equiv b \pmod{n}$, and a, b, d all divisible by $d > 0$, then $a/d \equiv b/d \pmod{n/d}$.

Pf. $a - b = Kn$, some K . By assumption, $a = K_1d$, $\therefore a/d = K_1$.

$b = K_2d$, $b/d = K_2$.

$n = K_3d$, $n/d = K_3$.

$$\therefore K_1d - K_2d = K(K_3d), \quad \therefore K_1 - K_2 = K(K_3).$$

$$\therefore K_1 - K_2 = KK_3 \Rightarrow a/d - b/d = K(n/d).$$

$$\therefore a/d \equiv b/d \pmod{n/d}.$$

2. $a^2 \equiv b^2 \pmod{n} \not\Rightarrow a \equiv b \pmod{n}$.

$5^2 \equiv 4^2 \pmod{9}$, since $25 - 16 = 9 = 1 \cdot 9$.

But $5 \not\equiv 4 \pmod{9}$.

[Note: the modulus on this page is faint and could be read as 8 or 9; 9 is the value consistent with $25 - 16 = 9$, so transcribed as 9.]

3. If $a \equiv b \pmod{n}$, then $\gcd(a, n) = \gcd(b, n)$.

Pf. $a - b = Kn$, some K . Let $d = \gcd(a, n)$.

$\therefore a = dr$, $n = ds$, some r, s .

$\therefore dr - b = Kds$, $b = d(r - Ks)$. $\therefore d \mid b$.

Let $d' = \gcd(b, n)$. $\therefore$ since $d \mid n$ and $d \mid b$, $d' \leq d$.

By similar reasoning as above, $d' \mid a$.

$\therefore d \leq d'$.

$\therefore d' = d$. **4.(a)** Find remainder of $2^{50} \div 7$ and $41^{65} \div 7$.

$$2^{50} \div 7 : \quad 2^{50} = (2^5)^{10}$$

$$2^5 \equiv 4 \pmod{7}$$

$$\therefore 2^{50} \equiv 4^{10} \pmod{7}$$

But $4^{10} = 2^{20} = (2^5)^4$.

From above, $2^5 \equiv 4 \pmod{7}$?

$$\therefore 2^{20} \equiv 4^4 \pmod{7}$$

But $4^4 = 2^5 \cdot 6 + 4$

$$\therefore 4^4 \equiv 4 \pmod{7}?$$

$$4^{10} \equiv 2^5 \cdot 6 + 4 \pmod{7}$$

$$\therefore 4^{10} \cdot 4 \equiv 4^{10} \cdot 4 \equiv 2^4 \cdot 4 \equiv 4^4 \equiv 0 \pmod{7}?$$

[Note: this derivation is dense and the intermediate steps connecting 2^{50} , 4^{10} , 2^{20} , 4^4 are hard to fully reconstruct from the handwriting; transcribed best-effort. The final conclusion on the next page, " $2^{50} \equiv 4 \pmod{7}$," is the one stated with confidence.]

$$\therefore 2^{50} \equiv 4 \pmod{7},$$

so $2^{50} \div 7$ has remainder 4.

$$41^{65} \div 7: 41^{65} = (41^5)^{13}.$$

$$41 \equiv 6 \pmod{7}.$$

$$\therefore 41^5 \equiv 6^5 \pmod{7}.$$

$$\text{But } 6^5 = 7776.$$

$$\therefore 6^5 \equiv 6 \pmod{7}.$$

$$\therefore 41^{65} = (41^5)^{13} \equiv (6^5)^{13} \equiv 6^{13} \pmod{7}.$$

$$6^{13} \equiv 6 \cdot 7 + 1, \therefore 6^{12} \equiv 1 \pmod{7}, 6^{13} \equiv 6 \pmod{7}.$$

$$\therefore 41^{65} \equiv (6^5)^{13} \equiv 6^{13} \equiv 6 \pmod{7}.$$

$$\therefore 41^{65} \div 7 \text{ has remainder } 6.$$

(b) What is remainder when $1^4 + 2^4 + \cdots + 100^4 \div 4$?

Since $1^4 \equiv 1 \pmod{4}$, and since $1 \equiv 5 \equiv \cdots \equiv 100 \pmod{4}$,

$$3^2 \equiv 2^4 \equiv 0 \pmod{4} \quad 2 \equiv 6 \equiv 10 \cdots \pmod{4}$$

$$2 \equiv 3^4 \equiv 3 \pmod{4} \quad 3 \equiv 7 \equiv 11 \cdots \pmod{4}$$

$$4^4 \equiv 0 \pmod{4} \quad 4 \equiv 8 \equiv 12 \cdots \pmod{4}$$

Each block of 4 numbers will have same remainder sum.

$$\text{Since } 1^4 + 2^4 + 3^4 + 4^4 \equiv 1 + 0 + 3 + 0 \equiv 4 \equiv 0 \pmod{4}.$$

Then per 25 blocks will all have remainder 0.

$\therefore$ Entire remainder is 0.

5. Prove $53^{103} + 103^{53} \equiv 0 \pmod{39}$.

$$111^{331} + 333^{111} \equiv 0 \pmod{7}$$

Pf. $53^{103} + 103^{53} \equiv 0 \pmod{39}$

$$39 = 3 \cdot 13. \quad 53 = 3 \cdot 17 + 2 = 3 \cdot 18 - 1.$$

$$103 = 34 \cdot 3 - 1.$$

$$\therefore 53 \equiv -1 \pmod{3}, \quad 103 \equiv 1 \pmod{3}$$

$$53 \equiv -1 \pmod{13}, \quad 103 \equiv -1 \pmod{13}$$

$$53^{103} \equiv 1 \pmod{13} \quad 103^{53} \equiv -1 \pmod{13}$$

$$\therefore 53^{103} + 103^{53} \equiv -1 + 1 \equiv 0 \pmod{3}$$

$$53^{103} + 103^{53} \equiv 1 + (-1) \equiv 0 \pmod{13}$$

$\therefore$ Both 3 and 13 divide sum, and $\gcd(3, 13) = 1$, so by Corollary 2, p.24, $3 \cdot 13 = 39$ divides sum.

$$\therefore 53^{103} + 103^{53} \equiv 0 \pmod{39}.$$

$$111^{331} + 333^{111} \equiv 0 \pmod{7}.$$

$111 = 7 \cdot 15 + 6, \therefore 111 \equiv 6 \equiv -1 \pmod{7}$, or $111^{331} \equiv (-1)^{331} \pmod{7}$, or $111^{331} \equiv -1 \pmod{7}$.

$$333 = 47 \cdot 7 + 4, \therefore 333 \equiv 4 \pmod{7}, \quad 333^{111} \equiv 4^{111} \pmod{7}.$$

$$4^4 = 64 = 9 \cdot 7 + 1, \therefore 4^4 \equiv 1 \pmod{7}.$$

$$\text{And } 222 \equiv 6 \cdot 17, \therefore (4^4)^{17} \equiv 4^{68}.$$

$$\therefore 4^{68} \equiv 1^{17} \pmod{7}, \text{ or } 4^{68} \equiv 1 \pmod{7}.$$

$$\therefore 333^{111} \equiv 1 \pmod{7}.$$

$$\therefore 111^{331} + 333^{111} \equiv (-1 + 1) \pmod{7}, \text{ or}$$

$$111^{331} + 333^{111} \equiv 0 \pmod{7}.$$

$$\mathbf{6.(a)} \quad 7 \mid (5^{2n} + 3 \cdot 2^{5n-2}), \quad n \geq 1.$$

$$\text{Pf. } n = 1: 5^{2n} + 3 \cdot 2^{5n-2} = 25 + 3 \cdot 8 = 49 = 7 \cdot 7 \dots 7 = 49 \cdot 7^{-1}.$$

$$= 5^{2n}(3 \cdot 7 + 4) + 3 \cdot 2^{5n-2}(4 \cdot 7 + 4)$$

$$= 3 \cdot 7 \cdot 5^{2n} + 4 \cdot 5^{2n} + 3 \cdot 2^{5n-2} \cdot 4 \cdot 7 + 4 \cdot 7x$$

$$= 3 \cdot 7 \cdot 5^{2n} + 4(5^{2n} + 3 \cdot 2^{5n-2}) + 4 \cdot 7x$$

$$= 7 \left(3 \cdot 5^{2n} + 4 \cdot 3 \cdot 2^{5n-2} + 4x \right),$$

where x is some integer, since it was assumed that for n ,

$$5^{2n} + 3 \cdot 2^{5n-2} = 7x \text{ as in [3].}$$

$\therefore$ for $n + 1$, number is divisible by 7.

$\therefore$ true for all $n \geq 1$.

Or, $5^2 = 25 \equiv 4 \pmod{7}. \therefore 5^{2n} \equiv 4^n \pmod{7}.$

$$2^5 \equiv 4 \pmod{7}. \therefore 2^{5n} \equiv 4^n \pmod{7}.$$

$$\text{For } n \geq 1, 2^{5n} \cdot 4^{-1} \equiv 4^n \cdot 4^{-1} \pmod{7}.$$

$$\therefore 2^{5n-4} \equiv 4^{n-1} \pmod{7}.$$

$$\therefore 3 \cdot 2^{5n-2} \equiv 8 \cdot 4^{n-1} \pmod{7}.$$

$$\text{But } 4^n + 3 \cdot 4^{n-1} = 4 \cdot 4^{n-1} + 3 \cdot 4^{n-1}$$

$$= 7 \cdot 4^{n-1}.$$

$$\therefore 5^{2n} + 3 \cdot 2^{5n-2} \equiv 7 \cdot 4^{n-1} \equiv 0 \pmod{7}. \quad (\mathbf{b}) \quad 13 \mid (3^{n+2} + 4^{2n+1}).$$

$$Pf. \quad 3 \equiv 16 \pmod{13}, \quad 3 \equiv 4^2 \pmod{13}.$$

$$\therefore 3^n \equiv 4^{2n} \pmod{13}.$$

$$3^n \cdot 9 \equiv 4^{2n} \cdot 9 \pmod{13}, \quad 3^{n+2} \equiv 4^{2n} \cdot 9 \pmod{13}.$$

$$\therefore 3^{n+2} + 4^{2n+1} \equiv 4^{2n} \cdot 9 + 4^{2n+1} \pmod{13}$$

$$\equiv 4^{2n}(9 + 4) \pmod{13}$$

$$\equiv 4^{2n} \cdot 13 \pmod{13}$$

$$\equiv 0 \pmod{13}.$$

$$(\mathbf{c}) \quad 27 \mid (2^{5n+1} + 5^{n+2}).$$

$$Pf. \quad 32 \equiv 5 \pmod{27}, \quad \therefore 2^5 \equiv 5 \pmod{27}.$$

$$\therefore 2^{5n} \equiv 5^n \pmod{27}.$$

$$2^{5n} \cdot 2 \equiv 2 \cdot 5^n \pmod{27}.$$

$$\therefore 2^{5n+1} + 5^{n+2} \equiv 2 \cdot 5^n + 5^{n+2} \pmod{27}$$

$$\equiv 5^n(2 + 25) \pmod{27}$$

$$\equiv 5^n \cdot 27 \pmod{27}$$

$$\equiv 0 \pmod{27}.$$

$$(\mathbf{d}) \quad 43 \mid (6^{n+2} + 7^{2n+1}).$$

$$Pf. \quad 6 \equiv 49 \pmod{43}, \quad 6 \equiv 7^2 \pmod{43}.$$

$$\therefore 6^n \equiv 7^{2n} \pmod{43}.$$

$$6^n \cdot 36 \equiv 7^{2n} \cdot 36 \pmod{43}, \quad 6^{n+2} \equiv 7^{2n} \cdot 36 \pmod{43}.$$

$$6^{n+2} + 7^{2n+1} \equiv 7^{2n} \cdot 36 + 7^{2n+1} \pmod{43}$$

$$\equiv 7^{2n}(36 + 7) \pmod{43}$$

$$\equiv 0 \pmod{43}.$$

$$\mathbf{7.} \quad \text{For } n \geq 1, \quad (-13)^{n+1} \equiv (-13)^n + (-13)^{n-1} \pmod{181}.$$

$$Pf. \quad n = 1: \quad (-13)^2 = 169, \quad 169 + 13 = 182.$$

$$\therefore 169 \equiv -13 \pmod{181}.$$

$$K \Rightarrow K + 1: \quad \text{Suppose } (-13)^{K+1} \equiv (-13)^K + (-13)^{K-1} \pmod{181}.$$

$$\therefore (-13)^{K+2} = (-13) \cdot (-13)^{K+1} \equiv (-13)[(-13)^K + (-13)^{K-1}] \pmod{181}.$$

$$\therefore (-13)^{K+2} \equiv (-13)^{K+1} + (-13)^K \pmod{181}.$$

$\therefore$ True for all $n \geq 1$.

8.(a) If a is odd, then $a^2 \equiv 1 \pmod{8}$.

Pf. By Div. Alg., a odd means $a = 4k + 1$ or $a = 4k + 3$, some k .

$$\therefore a^2 = 16k^2 + 8k + 1 \quad \text{or} \quad a^2 = 16k^2 + 24k + 9.$$

$$\therefore a^2 - 1 = 8(2k^2 + k) \quad \text{or} \quad a^2 - 1 = 8(2k^2 + 3k + 1).$$

$$\therefore a^2 \equiv 1 \pmod{8}.$$

(b) For any a , $a^3 \equiv 0, 1$, or $6 \pmod{7}$.

Pf. By Div. Alg., $a = 7K + r$, $0 \leq r < 7$.

$$a = 7K: a^3 = (7K)^3, a^3 = 7 \cdot 7K^3 \equiv 0 \pmod{7}.$$

$$a = 7K + 1: a^3 = 7K^3 + \binom{3}{1}7K^2 + \binom{3}{2}7K + 1^3, \therefore a^3 \equiv 1 \pmod{7}.$$

$$a = 7K + 2: a^3 = 7K^3 + \binom{3}{1}7K^2 + \binom{3}{2}7K \cdot 2 + 2^3$$

$$a^3 - 6 = 7[7K^3 + \dots + 1], [3]$$

$$\therefore a^3 \equiv 6 \pmod{7}.$$

$$a = 7K + 3: a^3 = 7K^3 + \binom{3}{1}7K^2 + \binom{3}{2}7K \cdot 3 + 27$$

$$a^3 - 6 = 7[7K^3 + \dots] + 27$$

$$\therefore a^3 \equiv 6 \pmod{7}.$$

$$a = 7K + 4: a^3 = 7K^3 + \dots + 64$$

$$a^3 - 1 = 7[7K^3 + \dots] + 63 \quad \therefore a^3 \equiv 1 \pmod{7}. \quad a = 7K + 5: a^3 = 7K^3 + \dots + 125:$$

$$119 + 6$$

$$\therefore a^3 - 6 = 7[7K^3 + \dots] + 119 \quad \therefore a^3 \equiv 6 \pmod{7}.$$

$$a = 7K + 6: a^3 = 7K^3 + \dots + 216: 7K^3 + \dots + 31 \cdot 7 + 1$$

$$\therefore a^3 - 1 = 7[7K^3 + \dots + 31] \quad \therefore a^3 \equiv 1 \pmod{7}.$$

(c) For any a , $a^4 \equiv 0$ or $1 \pmod{5}$.

Pf. By Div. Alg., $a = 5k + r$, $0 \leq r < 5$.

$$a = 5k: a^4 = 5 \cdot 5^4 k^4, \therefore a^4 \equiv 0 \pmod{5}.$$

$$a = 5k + 1: a^4 = 5^4 k^4 + \binom{4}{1}5^3 k^3 + \binom{4}{2}5^2 k^2 + \binom{4}{3}5k + 1$$

$$\therefore a^4 - 1 = 5[\dots]$$

$$\therefore a^4 \equiv 1 \pmod{5}.$$

$$a = 5k + 2: a^4 = 5^4 k^4 + \dots + 16 = 5[5^3 k^4 + \dots + 3]$$

$$a^4 - 1 = 5[5^3 k^4 + \dots + 3]$$

$$a^4 \equiv 1 \pmod{5}. \quad a = 5k + 3: a^4 = 5^4 k^4 + \dots + 78$$

$$\therefore a^4 - 1 = 5[5^4 k^4 + \dots + 15 + 1]$$

$$a^4 \equiv 1 \pmod{5}.$$

$$a = 5k + 4: a^4 = 5^4 k^4 + \dots + 255 + 1$$

$$\therefore a^4 \equiv 1 \pmod{5}.$$

(d) If a is not divisible by 2 or 3, then $a^2 \equiv 1 \pmod{24}$.

Pf. By Div. Alg., $a = 24k + r$, $0 \leq r < 24$.

Since a is not divisible by 2, r must be odd.

Since a is not divisible by 3, $r = 1, 5, 7, 11, 13, 17, 19$.

$$\therefore a^2 = (24k + r)^2 = 24^2 k^2 + 48kr + r^2.$$

$r = 1$: $r^2 = 1$ $\therefore$ Let $c = 0$.

$r = 5$: $r^2 = 25 = 24 + 1$ $\therefore$ Let $c = 1$.

$r = 7$: $r^2 = 49 = 2 \cdot 24 + 1$ $\therefore$ Let $c = 2$.

$r = 11$: $r^2 = 121 = 5 \cdot 24 + 1$ $\therefore$ Let $c = 5$.

$r = 13$: $r^2 = 169 = 7 \cdot 24 + 1$ $\therefore$ Let $c = 7$.

$r = 17$: $r^2 = 289 = 12 \cdot 24 + 1$ $\therefore$ Let $c = 12$.

$r = 19$: $r^2 = 361 = 15 \cdot 24 + 1$ $\therefore$ Let $c = 15$.

$$\therefore a^2 = 24k^2 + 48kr + 24c + 1$$

$$= 24[24k^2 + 2kr + c] + 1.$$

$\therefore a^2 \equiv 1 \pmod{24}$. **9.** If ρ is prime s.t. $n < \rho < 2n$, then

$$\binom{2n}{n} \equiv 0 \pmod{\rho}.$$

Pf.

$$\binom{2n}{n} = \frac{1 \cdot 2 \cdot 3 \cdots n(n+1) \cdots (2n)}{n!n!} = \frac{(n+1) \cdots (2n)}{n!}.$$

$$\therefore n! \binom{2n}{n} = (n+1) \cdots (2n).$$

Since $n < \rho < 2n$, ρ must be one of the factors of $(n+1) \cdots (2n)$.

$$\therefore n! \binom{2n}{n} = K\rho.$$

Since $\rho > n$, it is greater than every term of $n!$, it is not a member of the prime factorization of each member.

$\therefore \gcd(n!, \rho) = 1$.

$\therefore$ By Euclid's lemma, $\rho \mid \binom{2n}{n}$.

$\therefore \binom{2n}{n} \equiv 0 \pmod{\rho}$. **10.** If $\{a_1, \dots, a_n\}$ is a complete set of residues mod n and $\gcd(a, n) = 1$, then $\{aa_1, \dots, aa_n\}$ is a complete set of residues mod n .

Pf. Consider aa_i and aa_j , $i \neq j$, $i, j \leq n$.

If aa_i and aa_j are congruent mod n , then $aa_i - aa_j = Kn$, some K . $\therefore a(a_i - a_j) = Kn$.

Since $\gcd(a, n) = 1$, then by Euclid's lemma, $n \mid (a_i - a_j)$, contradicting that $a_i \neq a_j$.

$\therefore aa_i \neq aa_j$.

By Theorem 1 at top, $\{aa_1, \dots, aa_n\}$ is a complete set.

11. Show $0, 1, 2, 2^2, \dots, 2^9$ is a complete set of residues mod 11, but that $0, 1^2, 2^2, \dots, 10^2$ is not.

Pf. Since $\gcd(11, 2^i) = 1$, for $0 \leq i \leq 9$. Then $2^i \not\equiv 0 \pmod{11}$ for $0 \leq i \leq 9$.

Consider 2^r and 2^s , $1 \leq r, s \leq 9$, $r \neq s$.

Suppose $s > r$. $\therefore 2^s - 2^r = 2^r(2^{s-r} - 1)$.

Since $\gcd(11, 2^r) = 1$, and $\gcd(2^{s-r} - 1, 11) = 1$, for $0 \leq s, r \leq 8$, then there is no $K \geq 1$ s.t. $2^s - 2^r \mid 11K$.

$\therefore 0, 1, 2^2, \dots, 2^9$ is a complete set of residues mod 11.

Another proof (more obvious).

Look at remainders from Div. Alg.

$$\begin{array}{ll} 0: r = 0 & 2^5: 5 \\ 1: r = 1 & 2^6: 3 \\ 2: r = 2 & 2^7: 7 \\ 2^3: r = 4 & 2^8: 7 \\ 2^4: r = 8 & \end{array}$$

$\therefore$ Remainders are in 1-to-1 correspondence to $\{0, 1, \dots, 10\}$, and therefore constitute a complete set of residues mod 11.

$$\begin{array}{lll} 0^2: 0 & 4^2: 5 & 8^2: 9 \\ 1^2: 1 & 5^2: 3 & 9^2: 4 \\ 2^2: 4 & 6^2: 3 & 10^2: 1 \\ 3^2: 9 & 7^2: 5 & \end{array}$$

$\therefore$ not a 1-to-1 correspondence $\therefore$ not a complete set of residues (same as to of this exercise set). **12.(a)** If $\gcd(a, n) = 1$, then

$$c, c + a, c + 2a, \dots, c + (n - 1)a$$

forms a complete set of residues mod n .

Pf. Consider $c + ra$ and $c + sa$, $r \neq s$, $0 \leq r, s \leq n - 1$. Suppose $s > r$.

$$\therefore c + sa - (c + ra) = (s - r)a.$$

$s - r < n$ since $s \leq n - 1$, $r = n - 1$.

$\therefore n \nmid (s - r)$. Since $\gcd(a, n) = 1$, then there is no integer, K , s.t. $(s - r)a = nk$.

$\therefore c + sa \neq c + ra$, so the above set is a complete set of residues.

(b) Any n consecutive integers form a complete set of residues mod n .

Pf. From (a) above, let c = first of the consecutive list, let $a = 1$.

$\therefore$ The list in (a) is

$$c, c + 1, c + 2, \dots, c + (n - 1)$$

(c) The product of any set of n consecutive integers is divisible by n .

Pf. By (b), the set of n consecutive integers forms a complete set of residues mod n . $\therefore$ One of the members is congruent to 0 mod n , which means one member is divisible by n . $\therefore$ The entire product is divisible by n .

13. If $a \equiv b \pmod{n_1}$, $a \equiv b \pmod{n_2}$, then $a \equiv b \pmod{n}$, where $n = \text{lcm}(n_1, n_2)$.

Pf. Let K_1, K_2 be the integers such that $a - b = K_1n_1$, $a - b = K_2n_2$.

Let $d = \text{gcd}(n_1, n_2)$. $\therefore n_1 = dr$, some r , $r = n_1/d$.

$$\therefore a - b = K_2n_2 = K_2n_2 \left(\frac{n_1}{dr} \right) = \frac{K_2}{r} \cdot \frac{n_1n_2}{d}.$$

But $\frac{n_1n_2}{d} = \text{lcm}(n_1, n_2)$ (Th. 2.8, p.30).

$$\therefore a - b = \frac{K_2}{r} \cdot \text{lcm}(n_1, n_2).$$

Is $\frac{K_2}{r}$ an integer?

Let s be s.t. $n_2 = ds$. $\therefore a - b = K_1n_1 = K_2n_2$.

Then $K_1dr = K_2ds$, so $K_1r = K_2s$.

Since r and s are relatively prime (see proof of Corollary 1, p.25),

by Euclid's lemma, $r \mid K_2$, so $\frac{K_2}{r}$ is an integer.

14. Show that $a^k \equiv b^k \pmod{n}$ and $k \equiv j \pmod{n}$ need not imply $a^j \equiv b^j \pmod{n}$.

Pf. $2^2 \equiv 3^2 \pmod{5}$ since $4 \equiv 9 \pmod{5}$.

$$2 \equiv 7 \pmod{5}.$$

$$2^7 = 128, 3^7 = 2187, 2187 - 128 = 2059.$$

$$70 \neq 3.$$

[Note: this counterexample's final numeric check is hard to follow exactly from the handwriting — the comparison " $70 \neq 3$ " doesn't clearly connect to the preceding line; transcribed as legible.]

15. If a is odd, then for $n \geq 1$, $a^{2^n} \equiv 1 \pmod{2^{n+2}}$.

Pf. $n = 1$: is $a^2 \equiv 1 \pmod{2^3}$?

Since a is odd, $a = 4r + 1$ or $a = 4r + 3$.

$$\therefore a^2 = 16r^2 + 8r + 1 \quad \text{or} \quad a^2 = 16r^2 + 24r + 9.$$

$$\therefore a^2 - 1 = 8(2r^2 + r) \quad \text{or} \quad a^2 - 1 = 8(2r^2 + 3r + 1).$$

$$\therefore a^2 \equiv 1 \pmod{8}.$$

$K \Rightarrow K + 1$: Suppose $a^{2^K} \equiv 1 \pmod{2^{K+2}}$.

$$\therefore a^{2^K} - 1 = (2^{K+2})r, \text{ some } r.$$

$$a^{2^{K+1}} - 1 = a^{2 \cdot 2^K} - 1 = (a^{2^K})^2 - 1$$

$$\begin{aligned}
&= (a^{2^K} - 1)(a^{2^K} + 1) \\
&= (a^{2^K} + 1)(2^{K+2})r \\
&= (2^{K+2}r + 2)(2^{K+2}r) \\
&= 2^{2K+4}r^2 + 2 \cdot 2^{K+2}r \\
&= 2^{2K+4}r^2 + 2^{K+3}r \\
&= 2^{K+3}(2^{K+1}r^2 + r) \\
&= 2^{(K+1)+2}s, \quad \text{where } s = 2^{K+1}r^2 + r.
\end{aligned}$$

$\therefore$ when true for K , true for $K + 1$.

16.(a) Show $89 \mid 2^{44} - 1$.

Idea: Look at multiple of 89 to see if close or off by 1 from powers of 2.

$$\begin{array}{lll}
2^8 = 32 & 2^5 = 256 & 2^8 \equiv -11 \cdot 8 = 267 \\
2^6 = 64 & 2^7 = 512 & 6 \cdot 8 \cdot 5 = 534 \\
2^7 = 128 & 2^{10} = 1024 & 11 \cdot 89 = 979 \\
2^8 = 2048 \dots & 12 \cdot 89 = 1068 & \\
& 23 \cdot 89 = 2047 &
\end{array}$$

$$\therefore 2^{11} \equiv -1 \pmod{89}.$$

$$\therefore 2^{22} \equiv 1 \pmod{89}.$$

Another way: $2^8 \equiv (-11) \pmod{89}$ ($8 \cdot 5 = 267$)

$$2^{11} \cdot 2^8 \equiv 2^{11} \cdot (-11) \pmod{89}$$

$$\therefore 2^{11} \equiv -1 \pmod{89}, \text{ and } 2^{22} \equiv 1 \pmod{89}.$$

[Note: this page's table of computed powers of 2 and their reduction mod 89 is hard to fully reconstruct from the handwriting; the stated conclusions ($2^{11} \equiv -1$, $2^{22} \equiv 1 \pmod{89}$) are clear and match standard results, so transcribed with higher confidence on the conclusion than the intermediate table entries.]

(b) $97 \mid 2^{48} - 1$.

97 is close to 100, so look at powers of 2 close to 100. We find that $2^{197} = 2037$.

$$2^{11} = 2048 \equiv 11 \pmod{97}.$$

$$2^{12} = 4096 \equiv 21 \pmod{97}.$$

$$2^{24} \equiv 2 \cdot 11^2 \pmod{97}.$$

$$\text{But } 2 \cdot 11^2 = (4 \cdot 12)^2 = (484), \text{ and } 5 \cdot 97 = 485.$$

$$\therefore 484 \equiv -1 \pmod{97}.$$

$$\therefore (4 \cdot 12)^2 \equiv -1 \pmod{97}.$$

$$\therefore 2^{11} \cdot (4 \cdot 12)^2 \equiv (4 \cdot 12)^2 \pmod{97}.$$

$$2^{48} \equiv 1 \pmod{97}.$$

[Note: the numeric reasoning in part (b) (powers of 2 near 100, the " $2 \cdot 11^2 = (4 \cdot 12)^2$ " step) is very dense and hard to verify digit-by-digit from the handwriting; transcribed best-effort. The final conclusion $2^{48} \equiv 1 \pmod{97}$ is stated clearly in the original.] **17.** If $ab \equiv cd \pmod{n}$, $b \equiv d \pmod{n}$, $\gcd(b, n) = 1$, then $a \equiv c \pmod{n}$.

Pf. Let $ab - cd = rn$, some r .

$$b - d = sn, \text{ some } s.$$

$$\therefore b - sn = d.$$

$$\therefore ab - cd = ab - c(b - sn) = ab - cb + csn.$$

$$rn = (a - c)b + csn.$$

$$rn - csn = (a - c)b.$$

$$(r - cs)n = (a - c)b.$$

Since $\gcd(n, b) = 1$, then by Euclid's lemma, $n \mid (a - c)$. $\therefore a \equiv c \pmod{n}$.

Alternatively, $b \equiv d \pmod{n}$, then $cb \equiv cd \pmod{n}$.

Since $ab \equiv cd \pmod{n}$, then $ab \equiv cb \pmod{n}$.

Since $\gcd(b, n) = 1$, then by Corollary 1, p.68, $a \equiv c \pmod{n}$.

18. If $a \equiv b \pmod{n_1}$ and $b \equiv c \pmod{n_2}$, then $a \equiv c \pmod{n}$, where $n = \gcd(n_1, n_2)$.

Pf. $a - b = K_1 n_1$, some K_1 . Since $n \mid n_1$, $n_1 = rn$, some r . $\therefore a - b = K_1 rn$. Similarly, since $n \mid n_2$, then $a \equiv c \pmod{n}$.

$\therefore$ By Theorem 4.2(c), $b \equiv c \pmod{n}$.

4.3 Special Divisibility Tests

1.(a) For any integer a , the units digit of a^2 is 0, 1, 4, 5, 6, or 9.

Pf. Let $a = a_n 10^n + \cdots + a_1 10 + a_0$, $0 \leq a_0 < 10$.

$$\therefore a - a_0 = 10(a_n 10^{n-1} + \cdots + a_1).$$

$$\therefore a \equiv a_0 \pmod{10}. \therefore a^2 \equiv a_0^2 \pmod{10}.$$

Note that all the other a_i , of a are associated with a factor of 10 in a^2 , and so don't contribute to units digit.

$\therefore$ only a_0^2 contributes to units digit of a^2 .

$$a_0^2 = 0, 1, 4, 9, 16, 25, 36, 49, 64, 81.$$

$$\therefore a_0^2 \equiv 0, 1, 4, 5, 6, 9 \text{ or } 4 \pmod{10}.$$

$$\therefore a^2 \equiv 0, 1, 4, 5, 6, 9 \pmod{10}.$$

(b) Any integer a_0 , $0 \leq a_0 \leq 9$, can occur in units digit of a^3 .

Pf. As in (a), let $a = a_n 10^n + \cdots + a_0$, $0 \leq a_0 < 10$.

$$\therefore a - a_0 = 10(a_n 10^{n-1} + \cdots + a_1).$$

$$\therefore a \equiv a_0 \pmod{10}. \therefore a^3 \equiv a_0^3 \pmod{10}.$$

$$a_0^3 = 0, 1, 8, 27, 64, 125, 216, 343, 512, 729.$$

$$\therefore a_0^3 \equiv 0, 1, 2, 3, 4, 5, 6, 7, 8, \text{ or } 9 \pmod{10}.$$

(c) For any a , the units digit of a^4 is 0, 1, 5, 6, or K .

Pf. As in (a), the only contributor to units digit in a^4 is a_0^4 . Look at all possibilities of a_0^4 , $0 \leq a_0 \leq 9$.

$$\text{From (a), } a_0^2 \equiv 0, 1, 16, 25, 36, 81 \pmod{10}.$$

$$\therefore a_0^4 \equiv 0, 1, 5, 7, \text{ or } 6 \pmod{10}.$$

$$\therefore a_0^4 \equiv 0, 1, 5, 7, \text{ or } 6 \pmod{10}.$$

(d) The units digit of a triangular number is 0, 1, 3, 5, 6, or K .

Pf. A number a is triangular $\iff$ there is a number n , $n \geq 1$, s.t. $a = \frac{n(n+1)}{2}$ (Problems 1.3, #1a).

Let $n = a_m 10^m + \cdots + a_0$. $\therefore n \equiv a_0 \pmod{10}$. Either n or $n+1$ is even. $\therefore \frac{n}{2}$ or $\frac{n+1}{2}$ is an integer. Similarly for $a_0/2$ or $(a_0+1)/2$.

$$\therefore \frac{n(n+1)}{2} \equiv a_0(a_0+1) \pmod{10}.$$

Consider all possibilities for a_0 .

a_0	$\frac{a_0(a_0+1)}{2}$	mod10	+5 (mod 10)
0	0	0	5
1	1	1	6
2	3	3	8
3	6	6	1
4	10	0	5
5	15	5	0
6	21	1	6
7	28	8	3
8	36	6	1
9	45	5	0

Note that for the other a_i , it is associated with a factor of 10, then $a_i 10^i \equiv 0 \pmod{10}$.

Thus, the column $1 + 5 \pmod{10}$ shows possibilities of other factors contributing to units digit if divided by 2.

$\therefore a \equiv 0, 1, 3, 5, 6, \text{ or } 8 \pmod{10}$ if a is triangular.

2. Find the last two digits of 9^{99} .

$9^9 - 9 = 5(9^2 - 1) = 5(80)$. $\therefore 9^9 \equiv 9 \pmod{10}$. $\therefore 9^9 - 9 = 10K$, some K , or $9^9 = 9 + 10K$.

$$\therefore 9^{9^9} = 9^{9+10K} = 9^9 \cdot 9^{10K}.$$

Look at the last 2 digits of 9^9 and 9^{10K} yields: $9^1 = 9$, $9^5 = 59$, $9^{10} = 01$.

$$9^4 = 81, 9^5 = 49, 9^{10} = 21.$$

$$9^2 = 27, 9^3 = 41, 9^9 = 89.$$

$$\therefore 9^9 \equiv 81 \pmod{100}, \text{ and } 9^{10K} \equiv 1 \pmod{100}.$$

$$\therefore 9^{10K} \equiv (1)^K \equiv 1 \pmod{100}.$$

$$\therefore 9^9 \cdot 9^{10K} \equiv 81 \cdot 1 \pmod{100}.$$

$$\text{But } 9^{9^9} = 9^9 \cdot 9^{10K}.$$

$$\therefore 9^{9^9} \equiv 81 \pmod{100}.$$

$\therefore$ Last two digits of 9^{9^9} are 89.

3. 176,521,221: $1 + 7 + 6 + 5 + 2 + 1 + 2 + 2 + 1 = 27$.

$\therefore$ divisible by 9.

$$1 - 2 + 1 - 2 + 5 - 6 + 7 - 1 = -3.$$

$\therefore$ not divisible by 11.

149,215,678: $1 + 4 + 9 + 2 + 1 + 5 + 6 + 7 + 8 = 36$.

$\therefore$ divisible by 9.

$$8 - 7 + 6 - 5 + 1 - 2 + 9 - 4 + 1 = 4.$$

$\therefore$ not divisible by 11. **4.(a)** Prove: If N is represented in the base b by

$$N = a_m b^m + \cdots + a_1 b + a_0, \quad 0 \leq a_K < b - 1.$$

Then $(b - 1) \mid N \iff (b - 1) \mid (a_m + a_{m-1} + \cdots + a_1 + a_0)$.

Pf. Consider $P(x) = \sum_{K=0}^m a_K x^K$, a polynomial with integer coefficients.

Note that $b \equiv 1 \pmod{b-1}$.

$\therefore P(b) \equiv P(1) \pmod{b-1}$ by Th. 4.4.

But $P(b) = N$, and

$$P(1) = a_m + \cdots + a_1 + a_0.$$

$$\therefore N \equiv a_m + \cdots + a_1 + a_0 \pmod{b-1}.$$

$$\therefore N \equiv 0 \pmod{b-1} \iff a_m + \cdots + a_1 + a_0 \equiv 0 \pmod{b-1}.$$

$$\therefore (b - 1) \mid N \iff (b - 1) \mid (a_m + \cdots + a_1 + a_0).$$

Note: $(b - 1)$ divides N (base 10) $\iff$ sum of digits (base 10) is divisible by $(b - 1)$.

(b) For N written in base 9.

(1) N is divisible by 8 $\iff$ sum of digits of N (in base 10) is divisible by 8 (in base 10).

This follows from (a). (2) N is divisible by 3 $\iff$ units digit is divisible by 3, since each term in the polynomial (other than units digit) contains a power of 9.

(c) $(997836)_9$: is divisible by 3 since $3 \mid 6$.

$4 + 9 + 7 + 8 + 3 + 6 = 32$, so is also divisible by 8.

5. Find the missing digits.

(a) $51,840 = 273,581 \times 0$.

$5 + 1 + 8 + 4 + 0 = 18$, so $9 \mid 51,840$.

$9 \mid (1 + 4 + 8 + 2 + 4 + 5 + x + 9 + 4 + x + 2 \cdot 7)$, $x = 0$ or 9 .

Since $1 - 8 + 5 - 3 + 7 - 2 = -10$, then $11 \mid 273,581$.

$0 - 9 + 0 - x + 3 - 7 + 8 - 1 + 4 + 1 = -13 - x$.

$\therefore 11 \mid (1 - 13 - x)$, $\therefore x \neq 0$, and $x = 9$.

$\therefore x = 9$.

6. $2 \times 99,561 = [3(523 + x)]^2$. Since 3^2 is on right side, $9 \mid 2 \times 99,561$.

$\therefore 2 + x + 9 + 5 + 6 + 1 = x + 32$, $\therefore x = 4$.

(c) $2784x = x \cdot 5569$.

$5 + 6 + 9 = 25$. From proof of Pr. 45,

$5569 \equiv 25 \pmod{9}$, and $25 \equiv 7 \pmod{9}$.

$\therefore 5569 \equiv 7 \pmod{9}$.

$\therefore 5569x \equiv 7x \pmod{9}$.

$2784x \equiv (2 + 7 + 8 + 4 + x) \pmod{9}$, or $6x \equiv (3 + x) \pmod{9}$.

$7 \mid (6x - 3)$, so $x = 2, 5, 8$.

$5569 \equiv (5 - 6 + 5 - 7) \equiv (K - 1) \pmod{11}$, $5569x \equiv (K - 1) \pmod{11}$.

$2784x \equiv (x - 9 + 8 - 7 + 2) \equiv (x - 1) \pmod{11}$, $2x \equiv (x - 1) \pmod{11}$.

$\therefore 3x \equiv (x - 1) \pmod{11}$, $2x \equiv -1 \equiv 10 \pmod{11}$.

$\therefore x = 5$.

(d) $572 \cdot 1 \times 5825 = 1,000,000,000$. $572 \equiv (5 + 1 + 2) \equiv P \pmod{9}$.

$(1x583) \equiv (1 + x + 5 + 8 + 3) \equiv (8 + x) \pmod{9}$.

$P \cdot (8 + x) \equiv 1,000,000,000 \equiv 1 \pmod{9}$.

$64 + 8x \equiv 6 + 4 + x \equiv (1 + x) \equiv 1 \pmod{9}$.

$\therefore x = 0$ or $x = 9$.

$572 \equiv (2 - 1 + 5) \equiv 6 \pmod{11}$.

$(1x583) \equiv (5 - 2 + 1 - 3 + 5 - x + 1) \equiv (7 - x) \pmod{11}$.

$6 \cdot (7 - x) \equiv 1 \pmod{11}$, $42 - 6x \equiv 1 \pmod{11}$, $43 \equiv 6x \pmod{11}$.

$x = 0$, and for $x = 9$, $43 \equiv 54 \pmod{11}$.

$\therefore x = 9$.

6.(a) An integer is divisible by 2^K if its units digit is 0, 2, 4, 6, or 8.

Pf. Since $10 \equiv 5 \cdot 2$, in the base 10 representation of an integer $N = a_m 10^m + \cdots + a_1 10 + a_0$, each term, except a_0 , contains a power of 10, and so is divisible by 2.

$\therefore N$ is divisible by 2 $\iff a_0$ is divisible by 2, $\therefore a_0 \equiv 0, 2, 4, 6$, or 8.

(3) An integer is divisible by 3 $\iff$ the sum of its digits is divisible by 3.

Pf. Let $N = a_m 10^m + \cdots + a_1 10 + a_0$ be the decimal expansion of N , $0 \leq a_K < 10$, and let $S = \sum a_K$.

Consider $P(x) = \sum_{K=0}^m a_K x^K$. Note $P(10) = N$, $P(1) = S$. Note also $10 \equiv 1 \pmod{3}$, so $P(10) \equiv P(1) \pmod{3}$.

$\therefore N \equiv S \pmod{3}$.

$\therefore N \equiv 0 \pmod{3} \iff S \equiv 0 \pmod{3}$.

(c) An integer is divisible by 4 $\iff$ the number formed by its tens and units digits is divisible by 4.

Pf. Let $N = a_m 10^m + \cdots + a_2 10^2 + a_1 10 + a_0$, $0 \leq a_K < 10$.

Let $K \geq 2$. Then $10^K = 10^{K-2} \cdot 10^2 = 10^{K-2} \cdot 25 \cdot 4$.

$\therefore$ Each term $a_K 10^K = a_K 10^{K-2} \cdot 25 \cdot 4$ is divisible by 4 if $K \geq 2$.

$\therefore N$ is divisible by 4 $\iff a_1 10 + a_0$ is divisible by 4.

(d) An integer is divisible by 5 $\iff$ its units digit is 0 or 5.

Pf. Let $N = a_m 10^m + \cdots + a_2 10^2 + a_1 10 + a_0$, $0 \leq a_K < 10$.

Let $C_K = a_K 10^K = a_K (5 \cdot 2)^K = a_K 5^K 2^K$. $\therefore$ Each C_K is divisible by 5, if $K \geq 1$.

$\therefore N$ is divisible by 5 $\iff a_0$ is divisible by 5, and a_0 is divisible by 5 $\iff a_0 = 0, 5$.

7. For any integer a , show that $a^2 - a + 7$ ends in one of the digits 3, 7, or 9.

Pf. If $a = a_m 10^m + \cdots + a_1 10 + a_0$, then $a \equiv a_0 \pmod{10}$.

$\therefore a^2 - a + 7 \equiv a_0^2 - a_0 + 7 \pmod{10}$.

a_0	$a_0^2 - a_0 + 7$	ends with	K
0	7	7	$K = 0$
1	7	7	$K = 0$
2	9	9	$K = 0$
3	13	3	$K = 1$
4	19	9	$K = 1$
5	27	7	$K = 2$
6	37	7	$K = 3$
7	49	9	$K = 4$
8	63	3	$K = 6$
9	79	9	$K = 7$

Since $a_0^2 - a_0 + 7 \equiv c_K - c_0 + 7 - 10K \pmod{10}$, then

$$a^2 - a + 7 \equiv 3, 7, \text{ or } 9 \pmod{10}.$$

8. Find the remainder when 4444^{4444} is divided by 9.

Note that $4444 \pmod{9} \equiv (4 + 4 + 4 + 4) = 16 \pmod{9}$.

$16 = 2^3 \cdot 2$, so $4444 \pmod{9} \equiv 2^3 \cdot 2 \pmod{9}$.

Since $2^3 \equiv -1 \pmod{9}$, then $4444 \equiv (-1) \cdot 2 \pmod{9}$.

$\therefore 4444^{4444} \equiv (-1 \cdot 2)^{4444} \equiv 2^{4444} \pmod{9}$.

But $4444 = 3 \cdot 1481 + 1$, so

$$2^{4444} = (2^3)^{1481} \cdot 2 \quad \therefore 2^{4444} \equiv (-1)^{1481} \cdot 2 \pmod{9}.$$

$$4444^{4444} \equiv 2^{4444} \equiv (-1)^{1481} \cdot 2 \equiv -2 \equiv 7 \pmod{9}.$$

$\therefore$ remainder is 7.

9. Prove that no integer whose digits add up to 15 can be a square or cube.

Pf. Let N be any integer whose digits add up to 15.

$\therefore N \equiv 15 \pmod{9}$ (sect. Th. 4.5).

But $15 \equiv 6 \pmod{9}$.

$\therefore N \equiv 6 \pmod{9}$.

Consider $a = a_m 9^m + \cdots + a_1 9 + a_0$.

$$\therefore a \equiv a_0 \pmod{9}, \text{ and } \therefore a^2 \equiv a_0^2 \pmod{9}$$

$$\text{and } a^3 \equiv a_0^3 \pmod{9}.$$

Consider all possibilities of a_0 :

for $\pmod{9}$:

a_0	a_0^2	$a_0^2 \pmod{9}$	a_0^3	$a_0^3 \pmod{9}$
0	0	0	0	0
1	1	1	1	1
2	4	4	8	8
3	9	0	27	0
4	16	7	64	1
5	25	7	125	8
6	36	0	216	0
7	49	4	343	1
8	64	1	512	8

$\therefore a^2 \equiv 0, 1, 4, \text{ or } 7 \pmod{9}$.

$\therefore$ There is no a^2 s.t. $a^2 \equiv 6 \pmod{9}$.

$a^3 \equiv 0, 1, \text{ or } 8 \pmod{9}$, and so there is no a^3 s.t. $a^3 \equiv 6 \pmod{9}$. **10.** Assuming 495 divides $273x49y5$, find x, y .

Let $495 \mid N = 273x49y5$.

$$495 \equiv 0 \pmod{9}. \therefore 495 \mid N \Rightarrow N \equiv 0 \pmod{9}.$$

$$\therefore 273x49y5 \equiv 0 \pmod{9}.$$

$$(2 + 7 + 3 + x + 4 + 9 + y + 5) = 30 + x + y \equiv 0 \pmod{9},$$

$$\text{so } 3 + x + y \equiv 0 \pmod{9}, \text{ or } x + y \equiv 6 \pmod{9}$$

$$\text{and } x - y \equiv 15 \pmod{9}.$$

Also, $5 - 9 + 4 - 0 = 0$, so $495 \equiv 0 \pmod{11}$.

$$\therefore 495 \mid N \Rightarrow N \equiv 0 \pmod{11}.$$

$$\therefore 273x49y5 \equiv 0 \pmod{11}.$$

$$5 - y + 9 - 4 + x - 3 + 7 - 2 = x - y + 12 \equiv 0 \pmod{11}$$

$$x - y \equiv -1 \pmod{11}, \text{ or } x - y \equiv 1 \pmod{11}.$$

$$\therefore x + y = 6$$

$$x - y = 1, \text{ no integer}$$

$$2y = 7, \text{ no integer; } 2x = 16, \therefore y = 8, x = 7.$$

11. Determine the last 3 digits of 7^{999} .

Need to use mod 1000 since want last 3 digits.

Also, $7^4 = 2401 \equiv (1 + 6 \cdot 400)$. Since $400^2 = 160,000$, then $400^2 \equiv 0 \pmod{1000}$ for $n \geq 2$.

$$\therefore 7^{4n} = (1 + 6 \cdot 400)^n = \sum_{k=0}^n \binom{n}{k} (6 \cdot 400)^k.$$

So for $k \geq 2$, $\binom{n}{k} (6 \cdot 400)^k \equiv 0 \pmod{1000}$.

$$\therefore 7^{4n} \equiv 1 + \binom{n}{1} 6 \cdot 400 \pmod{1000}.$$

Now $999 = 4 \cdot 249 + 3$.

$$\therefore 7^{4 \cdot 249 + 3} \equiv 1 + (249)(6 \cdot 400) \pmod{1000}.$$

$$249 \cdot 6 \cdot 400 = (249 \cdot 6)(400) = (1494)(400) = (1)(4)(400) + (1494 - 4)(400)$$

$$\therefore 249 \cdot 6 \cdot 400 \equiv 6 \cdot 4 \cdot 400 \pmod{1000}.$$

$$\therefore 7^{4 \cdot 249} \equiv 1 + 6 \cdot 4 \cdot 400 \pmod{1000}.$$

$$\equiv 1 + 6 \cdot 4 \cdot 400 \pmod{1000}.$$

$$6 \cdot 4 = 216, \therefore 1 + 6 \cdot 4 \cdot 400 = 2161.$$

$$\therefore 7^{4 \cdot 249} \equiv 161 \pmod{1000}.$$

$$7^{999} \equiv 161 \cdot 7^3 \pmod{1000}.$$

$$7^3 = 343, (601)(343) = 206,143.$$

$$\therefore 7^{999} \equiv 206,143 \equiv 143 \pmod{1000}.$$

$\therefore 143$ are the last 3 digits.

[Note: this page's arithmetic is dense and several intermediate numbers (e.g. "601" appearing where "161" was just derived) are inconsistent or hard to reconcile exactly with the handwriting; transcribed best-effort. The final answer (last three digits 143) is stated clearly in the original and is consistent with known results for $7^{999} \bmod 1000$.] **12.** If t_n is the n th triangular number, show that $t_{n+2K} \equiv t_n \pmod{K}$. $\therefore t_n, t_{n+20}$ have same last digit.

$$\begin{aligned}
 \text{Pf. } t_{n+2K} &= \frac{(n+2K)(n+2K+1)}{2} \\
 &= \frac{n^2 + 2Kn + n + 2Kn + 4K^2 + 2K}{2} \\
 &= \frac{n^2 + n + 4Kn + 4K^2 + 2K}{2}. \\
 \therefore t_{n+2K} - t_n &= \frac{n^2 + n + 4Kn + 4K^2 + 2K}{2} - \frac{n(n+1)}{2} \\
 &= \frac{4Kn + 4K^2 + 2K}{2} \\
 &= K(2n + 2K + 1).
 \end{aligned}$$

$$\therefore t_{n+2K} \equiv t_n \pmod{K}.$$

$$\therefore t_{n+20} \equiv t_n \pmod{10}, \text{ or}$$

$$t_{n+20} = t_n + 10K, \text{ some } K.$$

$\therefore$ if $t_n = (a_m a_{m-1} \cdots a_2 a_1 a_0)_{10}$, then adding $10K$ not affect $a_0, \dots$, so t_{n+20} and t_n have same units digit.

13. For any $n \geq 1$, prove there exists a prime with at least n of its digits equal to 0.

Pf. This follows from Dirichlet's Theorem (p.56). From problem #12 of section 2.2, $\gcd(a, n) = 1$. $\therefore$ consider arithmetic progressions with powers of 10.

$\therefore 9 + 10K, K = 1, 2, 3, \dots$ contains infinitely many primes.

$10^{n+1} + 9$ has n zeros, and there only a finite number less than $10^{n+1} + 9$.

By Dirichlet's Theorem, there must be a prime in the series $K \cdot 10^{n+1} + 9$, and each has n zeros.

14. Find the values of $n \geq 1$ for which $1! + 2! + \cdots + n!$ is a perfect square.

$$1! = 1 \quad 3! = 6$$

$$2! = 2 \quad 4! = 24$$

$$\therefore 1! = 1^2$$

$$1! + 2! = 3 \quad 1! + 2! + 3! = 9 = 3^2$$

$$1! + 2! + 3! + 4! = 33$$

$\therefore$ The units digit of $\sum_{K=1}^n K!$ will be 3 for $n \geq 4$.

By problem 1(a), a perfect square can't end in 3. There is no perfect square for $n \geq 4$.

$\therefore n = 1, 3$ are the only values.

15. Show that 2^n divides an integer $N \iff 2^n$ divides the last n digits of N .

Pf. Let $N = a_{n+j}10^{n+j} + \dots + a_110 + a_0$ be the decimal representation for N , $n \geq 1$, $j \geq 0$.

(a) If 2^n divides the last n digits of N , then

$$2^n \mid (a_{n+j-1}10^{n-1} + \dots + a_110 + a_0) \quad [1]$$

$$\begin{aligned} \text{But } a_{n+j}10^{n+j} + \dots + a_n10^n &= 10^n(a_{n+j}10^j + \dots + a_n) \\ &= 2^n5^n(a_{n+j}10^j + \dots + a_n). \end{aligned}$$

$$\therefore 2^n \mid (a_{n+j}10^{n+j} + \dots + a_n10^n) \quad [2]$$

From [1] and [2], $2^n \mid (a_{n+j}10^{n+j} + \dots + a_n10^n + \dots + a_0)$, so $2^n \mid N$.

(b) Suppose $2^n \mid N$. Since $2^n \mid 2^n5^n$, then

$$\begin{aligned} 2^n &\mid 10^n(a_{n+j}10^j + \dots + a_n). \\ \therefore 2^n &\mid (a_{n+j}10^{n+j} + \dots + a_n10^n). \\ \therefore 2^n &\mid N - (a_{n+j}10^{n+j} + \dots + a_n10^n) \\ &\Rightarrow 2^n \mid (a_{n-1}10^{n-1} + \dots + a_110 + a_0). \end{aligned}$$

$\therefore 2^n$ divides the last n digits of N .

16. Let $N = a_m10^m + \dots + a_110 + a_0$, $0 \leq a_K \leq 9$, be the decimal expansion of a positive integer N .

(a) Prove 7, 11 and 13 all divide $N \iff 7, 11$ and 13 divide the integer

$$M = (a_0 + 10a_1 + 100a_2) - (a_3 + 10a_4 + 100a_5) + (a_6 + 10a_7 + 100a_8) - \dots$$

Pf. Since $1001 = 1000 + 1 = 1000 - (-1)$.

$$\therefore 10^3 \equiv -1 \pmod{1001}.$$

$$\text{Also, } 10^6 - 1 = (10^3 - 1)(10^3 + 1) \geq 995(1001).$$

$$\therefore 10^6 \equiv 1 \pmod{1001}.$$

$$\text{Also, } 1001 = 7 \cdot 11 \cdot 13.$$

Consider 10^{3n} .

If n is odd, $n = 2K + 1$, for some K .

$$\therefore 10^{3n} = 10^{3(2K+1)} = 10^{6K+3} = 10^{6K} \cdot 10^3.$$

$$\text{But } 10^{6K} \equiv 1^K = 1 \pmod{1001}.$$

$$\therefore 10^{6K} \cdot 10^3 \equiv (1) \cdot (-1) \equiv -1 \pmod{1001}. \quad [1]$$

$\therefore n$ odd $\Rightarrow 10^{3n} \equiv -1 \pmod{1001}$.

If n is even, $n = 2K$, some K .

$$\therefore 10^{3n} = 10^{6K} \equiv 1^K \equiv 1 \pmod{1001}. \quad [2]$$

$\therefore n$ even $\Rightarrow 10^{3n} \equiv 1 \pmod{1001}$.

Note: $N = (a_0 + 10a_1 + 100a_2) - (a_3 + 10a_4 + 100a_5) + \dots$

$$= 10^0(a_0 + a_1 \cdot 10 + a_2 \cdot 100) - 10^3(a_3 + a_4 \cdot 10 + a_5 \cdot 100) + \dots$$

$$= 10^{3j}(a_{3j} + a_{3j+1}10 + a_{3j+2}100) \pmod{1001} \quad [3]$$

From [2], j even: $a_{3j}10^{3j} \equiv a_{3j} \pmod{1001} \quad [4]$

$$a_{3j+1}10^{3j+1} \equiv a_{3j+1}10 \pmod{1001} \quad [5]$$

$$a_{3j+2}10^{3j+2} \equiv a_{3j+2}100 \pmod{1001} \quad [6]$$

From [1], K odd: $a_{3K}10^{3K} \equiv -a_{3K} \pmod{1001} \quad [7]$

$$a_{3K+1}10^{3K+1} \equiv -a_{3K+1}10 \pmod{1001} \quad [8]$$

$$a_{3K+2}10^{3K+2} \equiv -a_{3K+2}100 \pmod{1001} \quad [9]$$

Adding [4] + [5] + [6], j even:

$$a_{3j}10^{3j} + a_{3j+1}10^{3j+1} + a_{3j+2}10^{3j+2} \equiv a_{3j} + a_{3j+1}10 + a_{3j+2}100 \pmod{1001} \quad [10]$$

Adding [7] + [8] + [9], K odd:

$$\equiv -a_{3K} - a_{3K+1}10 - a_{3K+2}100 \pmod{1001}$$

Now let $3K = 3j + 3$: $\therefore 3K$ is odd, as K is odd.

Adding [3] + [10]:

$$a_{3j}10^{3j} + a_{3j+1}10^{3j+1} + a_{3j+2}10^{3j+2} + \dots$$

$$= (a_{3j} + 10a_{3j+1} + 100a_{3j+2}) - (a_{3j+3} + 10a_{3j+4} + 100a_{3j+5}) \pmod{n}$$

$$\therefore N \equiv M \pmod{1001}, N \equiv M \pmod{7 \cdot 11 \cdot 13}. \quad \therefore N \equiv 0 \pmod{7, 11, 13} \iff M \equiv 0 \pmod{7, 11, 13}.$$

(b) Prove $6 \mid N \iff 6$ divides the integer

$$M = a_0 + 4a_1 + \dots + 4^m a_m.$$

Pf. $10 \equiv 4 \pmod{6}$, and $40 \equiv 4 \pmod{6}$.

Lemma: $10^K \equiv 4 \pmod{6}$, $n \geq 1$.

Pf. True for $n = 1$, since $10 \equiv 4 \pmod{6}$.

Suppose true for K : $10^K \equiv 4 \pmod{6}$.

$$\therefore 10^{K+1} = 10^K \cdot 10 \equiv 4 \cdot 10 = 40 \equiv 4 \pmod{6}.$$

$\therefore$ True for $K + 1$.

$\therefore a_K 10^K \equiv 4a_K, K \geq 1.$

$\therefore N = a_0 + a_1 \cdot 10 + \cdots + a_m \cdot 10^m \equiv a_0 + 4a_1 + \cdots + 4a_m \pmod{6}.$

$\therefore N \equiv M \pmod{6}.$

$\therefore N \equiv 0 \pmod{6} \iff M \equiv 0 \pmod{6}.$

17. Is 1,010,908,899 divisible by 7, 11, and 13?

$$\begin{aligned} & 9 + 9 \cdot 10 + 8 \cdot 100 - (8 + 0 \cdot 10 + 9 \cdot 100) + (0 + 1 \cdot 10 + 0 \cdot 100) \cdot (-1) \\ &= 1 + 90 - 100 - (8 + 0 + 900) + (0 + 10 + 0) \cdot (-1) \\ &= 1 + 90 - 100 - 908 + (-10) \end{aligned}$$

[Note: the intermediate arithmetic on this line is hard to reconstruct exactly from the handwriting; the stated final result is 0.]

$$= 0.$$

$\therefore N \equiv 0 \pmod{1001}, \therefore N$ divisible by 7, 11, 13. **18.(a)** Given integer N , let M be the integer formed by reversing the order of digits of N . Show $N - M$ is divisible by 9.

Pf. Let $N = a_m 10^m + \cdots + 10a_1 + a_0.$

$\therefore M = a_0 10^m + \cdots + a_{m-1} 10 + a_m.$

$$\therefore N - M = (a_m - a_0)10^m + \cdots + (a_1 - a_{m-1})10 + (a_0 - a_m).$$

The sum of the coefficients of $N - M$ is:

$$(a_m + a_{m-1} + \cdots + a_0) - (a_0 + a_{m-1} + \cdots + a_m) = 0.$$

$\therefore$ By Th. 4.5, since $9 \mid 0, 9 \mid (N - M).$

(b) Prove any palindrome with an even number of digits is divisible by 11.

Pf. Let $N = a_m 10^m + \cdots + a_0$ have even number of digits. $\therefore m$ is odd.

Also, N is a palindrome $\iff a_m = a_0, a_{m-1} = a_1$, and in general, $a_K = a_{m-K}, 0 \leq K \leq m.$ Look at $T = (a_0 - a_m) + (a_2 - a_{m-1}) + \cdots + (a_{m-1} - a_m).$

Since m is odd, there is no coefficient that is ungrouped.

Rearranging terms of T , by reversing the order of the negative coefficients,

$$\begin{aligned} T &= (a_0 - a_m) + (a_2 - a_{m-1}) + \cdots + (a_{m-1} - a_0) \\ &= 0 + 0 + \cdots + 0 \\ &= 0. \end{aligned}$$

$\therefore$ By Th. 4.6, $11 \mid N.$

19. Given repunit R_n , prove:

(a) $9 \mid R_n \iff 9 \mid n.$

Pf. Note for R_n , sum of digits, S , is n since $R_n = 11 \cdots 1$ (n digits of 1).

$\therefore$ since $R_n \equiv S \pmod{9}$ by Th. 4.5,

$$R_n \equiv n \pmod{9}.$$

$$R_n \equiv 0 \pmod{9} \iff n \equiv 0 \pmod{9}. \quad (\mathbf{b}) \quad 11 \mid R_n \iff n \text{ is even.}$$

Pf. Let $R_n = 1 \cdot 10^m + \cdots + 1 \cdot 10 + 1$.

$$\text{Look at } T = (a_0 - a_1) + (a_2 - a_3) + \cdots + (-1)^m a_m$$

$$= (1 - 1) + (1 - 1) + \cdots + (-1)^m$$

T will be 0 $\iff$ can group terms, which means m is odd.

$\therefore T = 0 \iff$ number terms is even.

$\therefore$ By Th. 4.6, $11 \mid R_n \iff T = 0$.

$\therefore 11 \mid R_n \iff n$ is even.

20. Factor $R_6 = 111,111$ into a product of primes.

Since R_6 has an even number of groups of three coefficients, and

$$M = (1 + 10 + 100) - (1 + 10 + 100)$$

Then $R_6 \equiv 0 \pmod{1001}$ by prob. 16(c).

$\therefore R_6 = 111,111 = 7 \cdot 11 \cdot 13 \cdot K = 1001 \cdot K$. Try $K = 111$ since $111 \cdot 1000 = 111,000$.

Find that $R_6 = 111 \cdot 1001$, and $111 = 3 \cdot 37$.

$\therefore R_6 = 3 \cdot 7 \cdot 11 \cdot 13 \cdot 37$.

21. Show why

$$1 \cdot 9 + 2 = 11$$

$$12 \cdot 9 + 3 = 111$$

$$\vdots$$

$$123456789 \cdot 9 + 10 = 111,111,111.$$

i.e., show that

$$(10^{n-1} + 2 \cdot 10^{n-2} + \cdots + n) \cdot 9 + (n + 1) = \frac{10^{n+1} - 1}{9}.$$

Since $9 = 10 - 1$, then

$$\begin{aligned} (10^{n-1} + 2 \cdot 10^{n-2} + \cdots + n) \cdot 9 &= (10^{n-1} + 2 \cdot 10^{n-2} + \cdots + n)(10 - 1) \\ &= (10^{n-1} + 2 \cdot 10^{n-2} + \cdots + (n - 1) \cdot 10 + n) \cdot 10 - (10^{n-1} + \cdots + n) \\ &= 10^n + 2 \cdot 10^{n-1} + 3 \cdot 10^{n-2} + \cdots + (n - 1) \cdot 10^2 + n \cdot 10 - (10^{n-1} + \cdots + n) \\ &= 10^n + 1 \cdot 10^{n-1} + 1 \cdot 10^{n-2} + \cdots + 1 \cdot 10 - n. \end{aligned}$$

$$\therefore (10^{n-1} + 2 \cdot 10^{n-2} + \cdots + n) \cdot 9 + (n + 1)$$

$$= 10^n + 1 \cdot 10^{n-1} + 1 \cdot 10^{n-2} + \cdots + 1 \cdot 10 - n + (n+1).$$

$$= 10^n + 1 \cdot 10^{n-1} + \cdots + 1 \cdot 10 + 1. \quad [1]$$

The latter basically proves the assertion. However, to go further, multiply [1] by 9 and add 1.

$$\begin{aligned} & (10^n + 1 \cdot 10^{n-1} + \cdots + 1 \cdot 10 + 1) \cdot 9 + 1 \\ &= (10^n + 10^{n-1} + \cdots + 10 + 1)(10 - 1) + 1 \\ &= 10^{n+1} + 10^n + \cdots + 10^2 + 10 \\ &\quad - 10^n - 10^{n-1} - \cdots - 10 - 1 + 1 \\ &= 10^{n+1}. \end{aligned}$$

$$\begin{aligned} \therefore 10^n + 1 \cdot 10^{n-1} + \cdots + 1 \cdot 10 + 1 & \quad [1] \\ &= \frac{10^{n+1} - 1}{9}. \end{aligned}$$

$$\therefore (10^{n-1} + 2 \cdot 10^{n-2} + \cdots + n) \cdot 9 + (n+1) = \frac{10^{n+1} - 1}{9}.$$

22. An invoice shows that 72 canned hams were purchased for \$ $x67.9y$. Find the missing digits. **Solution.** $72 \cdot N = x679y$, where N = cost in cents of one ham.

$$\text{Note } 72 = 8 \cdot 9 = 2^3 \cdot 3^2.$$

$$\text{By problem 15, } 8 \mid x679y \therefore 2^3 \mid x679y.$$

$$\text{Since } 79y = 8 \cdot 9 + 7y, \text{ and } 8 \mid 72y,$$

$$\therefore 8 \mid y79y, \text{ i.e., } 8 \mid 72y.$$

$$\therefore x = 3.$$

$$\text{Since } 9 \mid x679, 9 \mid 6792, 1 \mid x6 + 7 + 9 + 2.$$

$$\therefore x = 3. \quad x6792 = 36792 \quad (x = 3, y = 2).$$

23. If 792 divides $13xy456$, find x, y, z .

Solution. Since $792 = 8 \cdot 99 = 8 \cdot 722$, so

$$8 \mid 13xy456, \text{ and by problem 15, } 8 \mid 456.$$

$$45c = 8 \cdot 50 + 5c.$$

$$\therefore 8 \mid 5c.$$

$$\text{Since } 9 \mid 13xy456, 9 \mid 1 + 3 + x + y + 4 + 5 + 6.$$

$$\therefore 9 \mid 1 + x + y, \quad x + y + 1 \equiv 18, 17.$$

$$\text{Also, } 6 \mid 792, \text{ so, by problem 16(b), } 6 \mid 6 + 4 + 5 + 1 + 4y + 4 + 3 + 1, \text{ or}$$

$$6 \mid y + 4 + 58, \text{ or } 6 \mid 4y + 4 + 7 + 4.$$

$$6 \mid 9(1 + x + y), \text{ so } 3 \mid (1 + x + y).$$

$$\text{This doesn't help since } 9 \mid (1 + 4y). \quad \text{Note } 11 \mid 792, \text{ so } 11 \mid 13xy456.$$

$\therefore 11 \mid 6 - 5 + 4 - y + x - 3 + 1$, or $11 \mid 3 + x - y$.
 $\therefore x - y = 11$ (can't = 24, since $x, y \leq 9$).
 $\therefore x - y = 8$.
 From $9 \mid 1 + x + y$, $x + y = 8, 17$.
 $\therefore x + y = 8$
 $x - y = 8$
 $\therefore 2x = 16$, $x = 8, y = 0$.
 $2x = 25$, not a solution.
 $\therefore 13xy456 = 1,380,456$.

24. For any prime $\rho > 3$, prove $13 \mid 10^{2\rho} - 10^\rho + 1$.

Pf. Use result of problem 16(a) and look at possible coefficients of the constructed integer M , formed by alternately summing and subtracting blocks of 3 coefficients of the decimal expansion of N .

For an integer N , part of 16(a) showed

$$N \equiv M \pmod{7 \cdot 11 \cdot 13}.$$

$\therefore$ Goal is to show constructed $M \equiv 0 \pmod{13}$, and so $N \equiv 0 \pmod{13}$.

To analyze M , look at decimal expansion coefficients of $N = 10^{2\rho} - 10^\rho + 1$.

First consider 10^ρ . Since ρ is prime, ρ is odd, so there is an odd number of zeros in 10^ρ .

Example: $10^5 = 1 \cdot 10^5 + 0 \cdot 10^4 + \cdots + 0 \cdot 10 + 0 = a_5 \cdot 10^5 + a_4 \cdot 10^4 + \cdots + a_1 \cdot 10 + a_0$.

Look at M :

$$\begin{array}{cccc} 100x & 10x & 1x & +/- \\ a_{K+2} & a_{K+1} & a_K & \end{array}$$

Note $\rho \geq 3$.

For $\rho = 5, 11, 17, \dots$:	value of K :
$m = -100$	2: 1, 1 +
	3: 0, 3 -
For $\rho = 7, 13, \dots$:	8: 7, 6 +
$M = +10$	10: 9 -, 6 +
	14: 13, 12 +
	17: 16, 15 -

$\rho \neq 9, 15, \dots$ as ρ is prime.

[Note: this page's table of K -values and signs is dense and the exact layout is hard to reconstruct from the handwriting; transcribed best-effort.] From Div. Alg., $\rho = 3q + r$, and as ρ is prime, $r \neq 0$.

$\therefore$ Can restrict considerations to $\rho = 3q + 1$ and $\rho = 3q + 2$.

For $\rho = 3q + 1$, q must be even for ρ to be odd. Let $q = 2k$. $\therefore \rho = 6k + 1$ ($k = 1, 2, \dots$).

For $\rho = 3q + 2$, q must be odd, since ρ is odd. Let $q = 2k + 1$.

$$\therefore \rho = 6k + 5 \quad (k = 0, 1, \dots).$$

$$\therefore \rho = 6k + 1 \quad (k = 1, 2, \dots) \quad \text{or}$$

$$\rho = 6k + 5 \quad (k = 0, 1, \dots).$$

$$\therefore 10^\rho = 10^{6k} \cdot 10 \quad (k = 1, 2, \dots) \quad \text{or}$$

$$10^\rho = 10^{6k} \cdot 10^5 \quad (k = 0, 1, \dots).$$

From above, since $M = -100$ for 10^5 , and since $N \equiv M \pmod{13}$, letting $N \equiv 10^{2\rho}$, then $10^5 \equiv -100 \pmod{13}$.

Similarly $10^\rho \equiv 10 \pmod{13}$.

Since $10^6 \equiv 1 \pmod{13}$, $10^{6k} \equiv 1^k \pmod{13}$.

$$\therefore 10^\rho = 10^{6k} \cdot 10 \equiv 10 \pmod{13} \quad (k = 1, 2, \dots)$$

$$10^\rho = 10^{6k} \cdot 10^5 \equiv 10^5 \pmod{13} \quad (k = 0, 1, \dots)$$

and $10^5 \equiv -100 \pmod{13}$.

$$\therefore 10^\rho \equiv 10 \pmod{13} \quad (k = 1, 2, \dots) \quad [1]$$

or $10^\rho \equiv -100 \pmod{13} \quad (k' = 0, 1, 2, \dots)$.

For $10^{2\rho}$: $2\rho = 12k + 2 \quad (k = 1, 2, \dots)$ or

$2\rho = 12k + 10 \quad (k' = 0, 1, 2, \dots)$.

$$\therefore 10^{2\rho} = 10^{12k} \cdot 100 \quad (k = 1, 2, \dots) \quad \text{or}$$

$$10^{2\rho} = 10^{12k} \cdot 10^{10} \quad (k' = 0, 1, 2, \dots).$$

Using $N \equiv M \pmod{13}$, $N = 10^{2\rho}$, and using $10^6 \equiv 1 \pmod{13}$, so $10^{12k} \equiv 1 \pmod{13}$.

$$\therefore 10^{2\rho} \equiv 100 \pmod{13} \quad (k = 1, 2, 3, \dots) \quad [2]$$

or $10^{2\rho} \equiv 10^5 \cdot 10^5 \pmod{13} \quad (k' = 0, 1, 2, \dots)$.

From above, $10^5 \equiv -100 \pmod{13}$.

$$10^5 \cdot 10^5 \equiv (-100)(-100) \equiv 10000 \pmod{13}.$$

Since $\gcd(10, 13) = 1$, then $10^{-1} \equiv 1 \pmod{13}$, so

$$10^7 \cdot 10 \equiv -100/10 \pmod{13},$$

$$10000 \equiv -100/10 \pmod{13}.$$

$$\therefore \text{From [2], } 10^{2\rho} \equiv 100 \pmod{13} \quad (k = 1, 2, 3, \dots) \quad [2']$$

$$\text{or } 10^{2\rho} \equiv -10 \pmod{13} \quad (k' = 0, 1, 2, \dots).$$

$$\therefore N = 10^{2\rho} - 10^\rho + 1 = [2'] - [1] + 1 \text{ becomes}$$

$$10^{2\rho} - 10^\rho + 1 \equiv 100 - 10 + 1 = 91 \pmod{13} \quad (k = 1, 2, 3, \dots)$$

$$10^{2\rho} - 10^\rho + 1 \equiv -10 - (-100) + 1 = 91 \pmod{13} \quad (k' = 0, 1, 2, \dots)$$

But $91 = 7 \cdot 13$, so $91 \equiv 0 \pmod{13}$.

$$\begin{aligned}\therefore 10^{2\rho} - 10^\rho + 1 &\equiv 0 \pmod{13}, \text{ so} \\ 13 &\mid (10^{2\rho} - 10^\rho + 1).\end{aligned}$$

4.4 Linear Congruences

Note that since $\{0, 1, \dots, n-1\}$ is a complete set of residues mod n , then $\{0, c \cdot 2, \dots, c \cdot (n-1)\}$ is also a complete set mod n if $\gcd(c, n) = 1$, by prob. 10, p.69.

For: $cx \equiv r \pmod{n}$, to find a solution, you can just test for $x = 0, 1, \dots, n-1$, since r must be congruent to one of $0, 1, \dots, c \cdot (n-1)$ if $\gcd(c, n) = 1$.

When solving for $Nx \equiv 1 \pmod{n_K}$, you can try $x = 0, 1, \dots, n_K - 1$ to find one solution, provided $\gcd(N, n_K) = 1$.

1.(a) $25x \equiv 15 \pmod{29}$.

$\gcd(25, 29) = 1$, $\therefore$ solution exists.

$-4x \equiv -14 \pmod{29}$ (adding -29)

$2x \equiv 7 \pmod{29}$ ($\gcd(2, 29) = 1$)

$30x \equiv 105 \pmod{29}$ (mult. by 15)

$x \equiv 76 \pmod{29}$ (adding -29)

$\therefore x \equiv 18 \pmod{29}$ (adding -58 on right)

(b) $5x \equiv 2 \pmod{26}$. $\gcd(5, 26) = 1$, $\therefore$ solution exists.

$25x \equiv 10 \pmod{26}$ (mult. by 5)

$25x - 26x \equiv 10 - 26 \pmod{26}$

$-x \equiv -16 \pmod{26}$

$\therefore x \equiv 16 \pmod{26}$.

(c) $6x \equiv 15 \pmod{21}$.

$\gcd(6, 21) = 3$, $3 \mid 15$, $\therefore$ solution exists.

$2x \equiv 5 \pmod{7}$ (divide by 3)

$2x \equiv 12 \pmod{7}$ (add 7)

$x \equiv 6 \pmod{7}$ ($\gcd(2, 7) = 1$, divide by 2)

$\therefore x = 6 + 7t$.

Since $\gcd(6, 21) = 3$, then are 3 mutually incongruent solutions, by Th. 4.7, and by Th. 4.7, they are $t = 0, 1, 2$.

$\therefore x \equiv 6, 13, 20 \pmod{21}$.

(d) $36x \equiv 8 \pmod{102}$.

$\gcd(36, 102) = 6$, and $6 \nmid 8$, $\therefore$ no solution.

(e) $34x \equiv 60 \pmod{98}$.

$\gcd(34, 98) = 2$, $2 \mid 60$, $\therefore$ solution exists. $102x \equiv 180 \pmod{98}$ (mult. by 3)

$102x - 98x \equiv 180 - 2 \cdot 98 \pmod{98}$

$$4x \equiv -16 \pmod{98}$$

$$2x \equiv -7 \pmod{49}$$

$$\therefore x \equiv -4 \pmod{49} \quad (\gcd(2, 49) = 1)$$

By Th. 4.7, two incongruent solutions exist.

$$t = 0, 1 \Rightarrow x = -4, -4, 45, \text{ or}$$

$$x \equiv 45, 98 \pmod{98}.$$

$$\text{(f)} \quad 140x \equiv 133 \pmod{301}.$$

$$140 = 2^2 \cdot 5 \cdot 7, \quad 301 = 7 \cdot 43, \quad \gcd(140, 301) = 7.$$

and $7 \mid 133$. $\therefore 7$ incongruent solutions exist.

$$20x \equiv 19 \pmod{43} \quad (\text{divide by } 7)$$

$$40x \equiv 38 \pmod{43} \quad (\text{multiply by } 2)$$

$$43x - 40x \equiv 43 - 38 \pmod{43}$$

$$3x \equiv 5 \pmod{43}$$

$$42x \equiv 70 \pmod{43} \quad (\text{mult. by } 14)$$

$$43x - 42x \equiv 85 - 70 \pmod{43}$$

$$x \equiv 16 \pmod{43}.$$

$$\therefore x = 16 + 43t, \quad t: \text{ set } t = 0, 1, 2, 3, 4, 5, 6.$$

$$\therefore x \equiv 16, 59, 102, 145, 188, 231, 274 \pmod{301}. \quad \mathbf{2.(a)} \quad 4x + 51y = 9.$$

$$4x \equiv 9 \pmod{51}$$

$$52x \equiv 117 \pmod{51} \quad (\text{mult. by } 13)$$

$$x \equiv 15 \pmod{51} \quad (\text{subtract } 51x, 102)$$

$$\therefore x = 15 + 51t.$$

$$51y \equiv 9 \pmod{4}$$

$$17y \equiv 3 \pmod{4} \quad (\gcd(51, 4) = 1, \text{ divide by } 3)$$

$$17y - 16y \equiv 3 \pmod{4}$$

$$y \equiv 3 \pmod{4}$$

$$\therefore y = 3 + 4s.$$

$$\therefore 4x + 51y = 4(15 + 51t) + 51(3 + 4s)$$

$$= 60 + 204t + 153 + 204s$$

$$\therefore 9 = 213 + 204t + 204s$$

$$\therefore -204 = 204t + 204s$$

$$-1 = t + s$$

$$s = -1 - t$$

$$\therefore x = 15 + 51t$$

$$y = 3 + 4(-1 - t) = -1 - 4t.$$

(b) $12x + 25y = 331$.

$$12x \equiv 331 \pmod{25}$$

$$24x \equiv 662$$

$$25x - 24x \equiv 662 - 650 \pmod{25}$$

$$x \equiv 12 \pmod{25}$$

$$\therefore x = 12 + 25t.$$

$$25y \equiv 331 \pmod{12}$$

$$25y - 24y \equiv 331 - 324 \pmod{12}$$

$$y \equiv 7 \pmod{12}$$

$$\therefore y = 7 + 12s.$$

$$\therefore 12x + 25y = 12(12 + 25t) + 25(7 + 12s)$$

$$= 144 + 300t + 175 + 300s$$

$$\therefore 331 = 319 + 300t + 300s$$

$$12 = 300t + 300s$$

$$1 = 25t + 25s$$

$$\therefore 25t = 1 - 25s$$

$$\therefore x = 12 + 25t = 13 - 25s.$$

$$\therefore x = 13 - 25s$$

$$y = 7 + 12s.$$

(c) $5x - 53y = 17$.

$$5x \equiv 17 \pmod{53}$$

$$55x \equiv 187 \pmod{53} \quad (\text{mult. by } 11)$$

$$55x - 53x \equiv 187 - 3 \cdot 53 \pmod{53}$$

$$2x \equiv 28 \pmod{53}$$

$$x \equiv 14 \pmod{53} \quad (\gcd(2, 53) = 1, \text{ divide by } 2)$$

$$\therefore x = 14 + 53t.$$

$$\begin{aligned} -53y &\equiv 17 \pmod{5} \\ -53y + 50y &\equiv 17 \pmod{5} \\ -3y &\equiv 17 \pmod{5} \\ -y &\equiv 1 \pmod{5} \\ y &\equiv 51 + 5s \quad (\text{added } 107) \\ \therefore y &= 51 + 5s. \end{aligned}$$

$$\begin{aligned} \therefore 5x - 53y &= 5(14 + 53t) - 53(51 + 5s) = 7 + 1 \\ &\quad 70 + 265t - 2703 - 265s \\ &\quad 2650 = 265t - 265s \\ &\quad 10 = t - s, \quad s = t - 10. \end{aligned}$$

$$\begin{aligned} \therefore y &= 51 + 5(t - 10) = 5t + 1 \\ \therefore x &= 14 + 53t \\ y &= 1 + 5t. \end{aligned}$$

3. Find all solutions to $3x - 7y \equiv 11 \pmod{13}$.

$$3x \equiv 7y + 11 \pmod{13}$$

$\gcd(3, 13) = 1$, so $13 \mid (7y + 11)$. There are 13 incongruent possibilities for y ($y = 0, 1, \dots, 12$).

$$\begin{array}{ll} y = 0: 3x \equiv 11 \pmod{13} & y = 1: 3x \equiv 18 \pmod{13} \\ 12x \equiv 44 & 12x \equiv 72 \\ 12x - 13x \equiv 44 - 5 \cdot 13 & 12x - 13x \equiv 72 - 5 \cdot 13 \\ -x \equiv 5, x \equiv 8 & -x \equiv 7, x \equiv 6 \end{array}$$

$$\begin{array}{ll} y = 2: 3x \equiv 25 - 26 \pmod{13} & y = 3: 3x \equiv 32 \pmod{13} \\ 12x \equiv -4 & 12x \equiv 4 \pmod{13} \\ 12x - 13x \equiv -4 & 12x - 13x \equiv 28 - 26 \\ x \equiv 4 & x \equiv 2 \end{array}$$

$$\begin{array}{ll} y = 4: 3x \equiv 39 \pmod{13} & y = 5: 3x \equiv 46 \pmod{13} \\ 12x \equiv 4 \pmod{13} & 12x \equiv 4 \pmod{13} \\ -x \equiv 0 & -x \equiv 28 - 26 = 2 \\ x \equiv 0 & x \equiv -2 \equiv 11 \end{array}$$

$\therefore$ From pattern, all $\pmod{13}$:

$$\begin{array}{lll}
y \equiv 0, x \equiv 8 & y \equiv 4, x \equiv 0 & y \equiv 8, x \equiv 5 \\
y \equiv 1, x \equiv 6 & y \equiv 5, x \equiv 11 & y \equiv 9, x \equiv 3 \\
y \equiv 2, x \equiv 4 & y \equiv 6, x \equiv 9 & y \equiv 10, x \equiv 1 \\
y \equiv 3, x \equiv 2 & y \equiv 7, x \equiv 7 & y \equiv 11, x \equiv 12 (= -1) \\
& & y \equiv 12, x \equiv 10
\end{array}$$

4.(a)

$$x \equiv 1 \pmod{3}$$

$$x \equiv 2 \pmod{5}$$

$$x \equiv 3 \pmod{7}$$

$$N = 3 \cdot 5 \cdot 7 = 105.$$

$$N_1 = \frac{105}{3} = 35, \quad N_2 = \frac{105}{5} = 21, \quad N_3 = \frac{105}{7} = 15.$$

$$35x \equiv 1 \pmod{3} \quad 21x \equiv 1 \pmod{5} \quad 15x \equiv 1 \pmod{7}$$

$$35x - 36x \equiv 1 \quad 21x - 20x \equiv 1 \quad 15x - 14x \equiv 1$$

$$-x \equiv 1 \quad x \equiv 1 \pmod{5} \quad x \equiv 1 \pmod{7}$$

$$x \equiv -1 \pmod{3}$$

$$\therefore x_1 = -1, x_2 = 1, x_3 = 1.$$

$$\therefore a_1 N_1 x_1 + a_2 N_2 x_2 + a_3 N_3 x_3 =$$

$$1 \cdot 35 \cdot (-1) + 2 \cdot 21 \cdot 1 + 3 \cdot 15 \cdot 1 = 52.$$

$$\therefore x \equiv 52 \pmod{105}.$$

(b)

$$x \equiv 5 \pmod{11}$$

$$x \equiv 14 \pmod{29}$$

$$x \equiv 15 \pmod{31}$$

$$N = 11 \cdot 29 \cdot 31 = 9889.$$

$$N_1 = 29 \cdot 31 = 899, \quad N_2 = 11 \cdot 31 = 341, \quad N_3 = 11 \cdot 29 = 319.$$

$$899x \equiv 1 \pmod{11} \quad 341x \equiv 1 \pmod{29} \quad 319x \equiv 1 \pmod{31}$$

$$899x - 81 \cdot 11x \equiv 1 \quad 341x - 148x \equiv 1 \quad 9x \equiv 7$$

$$8x \equiv 1 \quad -7x \equiv 1 \quad x = 7$$

$$32x \equiv 4 \quad -28x \equiv 4$$

$$31x - 33x \equiv 4 \quad x \equiv 4$$

$$x \equiv -4 \pmod{11}$$

$$\therefore x_1 = -4, x_2 = 4, x_3 = 7.$$

$$\therefore a_1 N_1 x_1 + a_2 N_2 x_2 + a_3 N_3 x_3 =$$

$$5 \cdot 899 \cdot (-4) + 14 \cdot 341 \cdot 4 + 15 \cdot 319 \cdot 7 = 34,611.$$

$$\therefore x \equiv 34,611 \pmod{9889} = 34,611 \bmod 9889 = 4,944.$$

[Note: the intermediate reduction steps in this CRT computation (mod 11, 29, 31) are dense and partly illegible in the original; transcribed best-effort. The final answer line is also uncertain — "34,611 mod 9889" is shown reducing to two different values ("7885" and "4944") in the original, which is inconsistent; flagged for review.]

(c)

$$x \equiv 5 \pmod{6}$$

$$x \equiv 4 \pmod{11}$$

$$x \equiv 3 \pmod{17}$$

[Note: the second modulus is transcribed as 11 rather than the original's apparent "17" (which would duplicate the third line) — 11 is consistent with $N = 6 \cdot 11 \cdot 17$ used just below.]

$$N = 6 \cdot 11 \cdot 17 = 1122.$$

$$N_1 = 11 \cdot 17 = 187, N_2 = 6 \cdot 17 = 102, N_3 = 6 \cdot 11 = 66.$$

$$187x \equiv 1 \pmod{6} \quad 102x \equiv 1 \pmod{11} \quad 66x \equiv 1 \pmod{17}$$

$$187x - 184x \equiv 1 \quad 102x - 93x \equiv 1 \quad 66x - 68x \equiv 1$$

$$x \equiv 1 \quad 9x \equiv 7 \quad -2x \equiv 7$$

$$18x \equiv 22 \equiv -9 \quad x \equiv -9$$

$$x \equiv -7$$

$$\therefore x_1 = 1, x_2 = -7, x_3 = -9.$$

$$\therefore a_1 N_1 x_1 + a_2 N_2 x_2 + a_3 N_3 x_3 =$$

$$5 \cdot 187 \cdot 1 + 4 \cdot 102 \cdot (-7) + 3 \cdot 66 \cdot (-1) = -3703.$$

$$\therefore x \equiv -3703 + 4 \cdot 1122 = 785 \pmod{1122}.$$

(d)

$$2x \equiv 1 \pmod{5}$$

$$3x \equiv 9 \pmod{6}$$

$$4x \equiv 7 \pmod{7}$$

$$5x \equiv 7 \pmod{11}$$

$$N = 5 \cdot 2 \cdot 7 \cdot 11 = 770.$$

$$N_1 = 2 \cdot 7 \cdot 11 = 154, N_2 = 5 \cdot 7 \cdot 11 = 385,$$

$$N_3 = 5 \cdot 2 \cdot 11 = 110, N_4 = 5 \cdot 2 \cdot 7 = 70.$$

$$154x_1 \equiv 1 \pmod{5} \quad 385x_2 \equiv 1 \pmod{2}$$

$$x_1 = -1 \quad x_2 = 1$$

$$110x_3 \equiv 1 \pmod{7} \quad 70x_4 \equiv 1 \pmod{11}$$

$$15x_3 \equiv 3 \quad 70x_4 \equiv 66x_4 + 4x_4$$

$$3x_3 = 3 \quad 12x_4 = 3$$

$$x_3 = 1 \quad x_4 = 3$$

$$\therefore a_1N_1x_1 + a_2N_2x_2 + a_3N_3x_3 + a_4N_4x_4 =$$

$$(-2)(154)(-1) + (-2)(385)(1) + 2 \cdot 110 \cdot 1 + (-18)(70)(3) = -1657.$$

$$x \equiv -1657 + 3 \cdot 770 = 653 \pmod{770}.$$

[Note: the coefficients a_i used in the final combination (e.g. "-2", "-18") don't obviously match the stated congruences $2x \equiv 1$, etc., at the top of this page; the relationship between the original problem statement and these worked coefficients is unclear from the handwriting. Transcribed as legible.] 5. $17x \equiv 3 \pmod{2 \cdot 3 \cdot 5 \cdot 7}$.

$$17x \equiv 3 \pmod{2} \iff x \equiv 1 \pmod{2}$$

$$17x \equiv 3 \pmod{3} \iff 2x \equiv 0 \pmod{3} \iff x \equiv 0 \pmod{3}$$

$$17x \equiv 3 \pmod{5} \iff 2x \equiv 3 \pmod{5} \iff x \equiv -6 \pmod{5}$$

$$17x \equiv 3 \pmod{7} \iff 3x \equiv 3 \pmod{7} \iff x \equiv -6 \pmod{7}$$

$$N = 2 \cdot 3 \cdot 5 \cdot 7 = 210.$$

$$N_1 = 3 \cdot 5 \cdot 7 = 105 \quad N_3 = 2 \cdot 3 \cdot 7 = 42$$

$$N_2 = 2 \cdot 5 \cdot 7 = 70 \quad N_4 = 2 \cdot 3 \cdot 5 = 30$$

$$105x_1 \equiv 1 \pmod{2} \quad 70x_2 \equiv 1 \pmod{3}$$

$$x_1 \equiv 1$$

$$42x_3 \equiv 1 \pmod{5} \quad 30x_4 \equiv 1 \pmod{7}$$

$$84x_3 \equiv 2 \quad 90x_4 \equiv -7 \cdot 3 = -x_4 = 3$$

$$84x_3 - 85x_3 \equiv 2 \quad -x_4 \equiv 3$$

$$x_3 \equiv -2 \quad x_4 \equiv -3$$

$$\begin{aligned} \therefore a_1 N_1 x_1 + a_2 N_2 x_2 + a_3 N_3 x_3 + a_4 N_4 x_4 = \\ 1 \cdot 105 \cdot 1 + 0 \cdot 70 + (-1)(42)(2) + (-3)(30)(-3) = 1149. \end{aligned}$$

$$\therefore x \equiv 1149 - 5 \cdot 210 \equiv 99 \pmod{210}.$$

6. Find smallest integer $a > 2$ s.t. $2 \mid a$, $3 \mid a + 1$, $4 \mid a + 2$, $5 \mid a + 3$, $6 \mid a + 4$.
This is equivalent to:

$$a \equiv 0 \pmod{2} \quad \text{or} \quad a \equiv 0 \pmod{2} \quad [1]$$

$$a + 1 \equiv 0 \pmod{3} \quad a \equiv -1 \pmod{3} \quad [2]$$

$$a + 2 \equiv 0 \pmod{4} \quad a \equiv -2 \pmod{4} \quad [3]$$

$$a + 3 \equiv 0 \pmod{5} \quad a \equiv -3 \pmod{5} \quad [4]$$

$$a + 4 \equiv 0 \pmod{6} \quad a \equiv -4 \pmod{6} \quad [5]$$

Note that $\gcd(2, 4) = 2$. So eliminate #1, since #1 is true, [3] is automatically true.
Also, $\gcd(3, 6) \neq 1$. Multiply [2] by 2 and get

$$\begin{aligned} (a + 1)2 &\equiv 2 \cdot 0 \pmod{3 \cdot 2}, \text{ or} \\ 2a + 2 &\equiv 0 \pmod{6}. \end{aligned}$$

Combine this with [5] and get

$$\begin{aligned} 2a + 2 &\equiv 0 \equiv a + 4 \pmod{6} \\ \therefore a &\equiv 2 \pmod{6}. \end{aligned}$$

$\therefore$ If this is true, then [2] and [5] will be true.

So far, we have $a \equiv -2 \pmod{4}$ [3]'

$$a \equiv -3 \pmod{5} \quad [4]'$$

$$a \equiv 2 \pmod{6} \quad [5]'$$

Note $\gcd(4, 6) \neq 1$. $\therefore$ combine

[3]' becomes $3a \equiv -6 \pmod{12}$. [5]' becomes $2a \equiv 4 \pmod{12}$.

$$3a + 12 \equiv -6 + 12 = 6 \pmod{12}$$

$$\therefore 2a + 2 \equiv 4 + 2 = 6 \pmod{12}, \text{ or}$$

$$3a + 12 \equiv 2a + 2 \pmod{12}, \text{ or}$$

$$a \equiv -10 \pmod{12}.$$

$\therefore$ The system reduces to:

$$a \equiv -3 \pmod{5}$$

$$a \equiv -10 \pmod{12}$$

$$\therefore N = 5 \cdot 12 = 60. \quad N_1 = 12, N_2 = 5.$$

$$12x_1 \equiv 1 \pmod{5} \quad 5x_2 \equiv 1 \pmod{12}$$

$$24x_1 \equiv 2 \quad 25x_2 \equiv 5$$

$$24x_1 - 25x_2 \equiv -x_1 \equiv 2 \quad 25x_2 - 24x_2 \equiv x_2 \equiv 5$$

$$\therefore x_1 = -2 \quad x_2 \equiv 5$$

$$\therefore a_1 N_1 x_1 + a_2 N_2 x_2 =$$

$$(-3)(12)(-2) + (-10)(5)(5) = 72 - 250 = -178.$$

$$a \equiv -178 \pmod{60}, \text{ or } a \equiv 2 \pmod{60}.$$

$$a \equiv 62 \pmod{60}. \quad \therefore a = 62.$$

7.(a) Obtain three consecutive integers, each having a square factor.

An integer a satisfying the hint will do.

$$a \equiv 0 \pmod{2^2}$$

$$a + 1 \equiv 0 \pmod{3^2}$$

$$a + 2 \equiv 0 \pmod{5^2}$$

Note $2^2, 3^2, 5^2$ are relatively prime, so can use Chinese Remainder Theorem.

$$\therefore a \equiv 0 \pmod{4} \quad N = 4 \cdot 9 \cdot 25 = 900$$

$$a \equiv -1 \pmod{9} \quad N_1 = 9 \cdot 25 = 225$$

$$a \equiv -2 \pmod{25} \quad N_2 = 4 \cdot 25 = 100$$

$$N_3 = 4 \cdot 9 = 36$$

$$225x_1 \equiv 1 \pmod{4} \quad 100x_2 \equiv 1 \pmod{9} \quad 36x_3 \equiv 1 \pmod{25}$$

$$225x_1 - 224x_1 \equiv 1 \quad 100x_2 - 99x_2 \equiv 1 \quad 72x_3 - 75x_3 = -3x_3$$

$$x_1 \equiv 1 \quad x_2 \equiv 1 \quad 24x_3 \equiv -16$$

$$\begin{aligned} 24x_3 + 25x_3 &= -x_3 \\ x_3 &= 16 \end{aligned}$$

$$\begin{aligned} \therefore a_1N_1x_1 + a_2N_2x_2 + a_3N_3x_3 &= \\ 0 + (-1)(100)(1) + (-2)(36)(16) &= -1252. \end{aligned}$$

$$X = -1252 + 2 \cdot 900 = 548 \pmod{900}.$$

$\therefore 548, 549, 550$. **(b)** Obtain three consecutive integers, the first of which is divisible by a square, the second by a cube, and the third by a fourth power.

Consider $a \equiv 0 \pmod{5^2}$ Choose reverse order to get

$$a + 1 \equiv 0 \pmod{3^3} \quad \text{small } \# \text{ for } n^4$$

$$a + 2 \equiv 0 \pmod{2^4}$$

$2^4, 3^3, 5^2$ are relatively prime, so can use Chinese Remainder Theorem.

$$\begin{aligned} \therefore a &\equiv 0 \pmod{25} & N &= 25 \cdot 27 \cdot 16 = 10,800 \\ a &\equiv -1 \pmod{27} & N_1 &= 27 \cdot 16 = 432 \\ a &\equiv -2 \pmod{16} & N_2 &= 25 \cdot 16 = 400 \\ & & N_3 &= 25 \cdot 27 = 675 \end{aligned}$$

$$\begin{aligned} 432x_1 &\equiv 1 \pmod{25} & 400x_2 &\equiv 1 \pmod{27} \\ 432x_1 - 425x_1 &\equiv 7x_1 & 400x_2 - 15 \cdot 27x_2 &= -5x_2 \\ 7x_1 &\equiv 1, 49x_1 \equiv 7 & -5x_2 &\equiv 11, -x_2 \equiv 11 \\ x_1 &\equiv 7 & x_2 &\equiv -11 \end{aligned}$$

$$\begin{aligned} 675x_3 &\equiv 1 \pmod{16} \\ 675x_3 - 42 \cdot 16x_3 &= 3x_3 \\ 675x_3 - 672x_3 &= 3x_3 \\ 16x_3 &\equiv 5, x_3 \equiv -5 \end{aligned}$$

$$\begin{aligned} \therefore a_1N_1x_1 + a_2N_2x_2 + a_3N_3x_3 &= \\ 0 + (-1)(400)(-11) + (-2)(675)(-5) &= 11,150. \end{aligned}$$

$$\therefore 11,150 - 10,800 = 350.$$

$$\therefore 11,150 \equiv 350 \pmod{25 \cdot 27 \cdot 16}.$$

$$\therefore 350, 351, 352.$$

8. Eggs removed from a basket.

2 at a time	Remaining eggs: 1
3 at a time	2
4 "	3
5 "	4
6 "	5
7 "	0

Find smallest number of eggs in basket.

$$\begin{aligned}
 X &\equiv 1 \pmod{2} [1] && \text{Need to eliminate the} \\
 X &\equiv 2 \pmod{3} [2] && \text{non-relatively prime} \\
 X &\equiv 3 \pmod{4} [3] && \text{conditions!} \\
 X &\equiv 4 \pmod{5} [4] \\
 X &\equiv 5 \pmod{6} [5] \\
 X &\equiv 0 \pmod{7} [6]
 \end{aligned}$$

If [3] is true, then $X = 3 + 4n = 1 + 2(1 + 2n)$, so $X \equiv 1 \pmod{2}$.

$\therefore$ Eliminate [1]. Now look at [2], since $\gcd(3, 6) \neq 1$. Multiply [2] by 2 and get $2x \equiv 4 \pmod{3 \cdot 2}$.

Combine with [5]:

$$\begin{aligned}
 2x - 4 &\equiv x - 5 \pmod{6} \\
 \therefore x &\equiv 1 \pmod{6}.
 \end{aligned}$$

$\therefore$ If [5] is true, [2] and [5] will be true.

We now have

$$\begin{aligned}
 x &\equiv 3 \pmod{4} [3] \\
 x &\equiv 4 \pmod{5} [4] \\
 x &\equiv 1 \pmod{6} [5''] \\
 x &\equiv 0 \pmod{7} [6]
 \end{aligned}$$

But $\gcd(4, 6) \neq 1$. $\therefore$ multiply [3] by 3 and [5''] by 2.

$$\begin{aligned}
 3x &\equiv 9 \pmod{12} \\
 2x &\equiv 2 \pmod{12} \\
 \therefore 3x - 9 &\equiv 2x + 2 \pmod{12} \\
 x &\equiv 11 \pmod{12}.
 \end{aligned}$$

$\therefore$ Everything reduces to:

$$\begin{aligned}
 x &\equiv 4 \pmod{5} [4] \\
 x &\equiv 11 \pmod{12} [5''']
 \end{aligned}$$

$$x \equiv 0 \pmod{7} \quad [6]$$

5, 12, 7 are relatively prime, so can use Chinese Remainder Theorem.

$$N = 5 \cdot 12 \cdot 7 = 420 \quad N_1 = 12 \cdot 7 = 84$$

$$N_2 = 5 \cdot 7 = 35 \quad N_3 = 5 \cdot 12 = 60$$

$$84x_1 \equiv 1 \pmod{5} \quad 35x_2 \equiv 1 \pmod{12}$$

$$84x_1 - 85x_1 = -x_1 \quad 35x_2 - 36x_2 = -x_2$$

$$x_1 = -1 \quad x_2 = -1$$

$$60x_3 \equiv 1 \pmod{7}$$

$$60x_3 - 56x_3 = 4x_3$$

$$8x_3 \equiv 1$$

$$x_3 = 2$$

$$\therefore a_1 N_1 x_1 + a_2 N_2 x_2 + a_3 N_3 x_3 =$$

$$4 \cdot 84 \cdot (-1) + 11 \cdot 35 \cdot (-1) + 0 = -721.$$

$$-721 + 2 \cdot 420 = 119.$$

$\therefore$ 119 eggs in the basket.

9. Basket-of-eggs problem: One egg remains when the eggs are removed from the basket 2, 4, 5, or 6 at a time, but no eggs remain if removed 7 at a time. Find smallest number of eggs in the basket.

$$X \equiv 1 \pmod{2} \quad [1] \quad \text{Need to consolidate}$$

$$X \equiv 1 \pmod{3} \quad [2] \quad \text{since } \gcd(2, 4) \neq 1,$$

$$X \equiv 1 \pmod{4} \quad [3] \quad \gcd(3, 6) \neq 1,$$

$$X \equiv 1 \pmod{5} \quad [4] \quad \gcd(4, 6) \neq 1$$

$$X \equiv 1 \pmod{6} \quad [5]$$

$$X \equiv 0 \pmod{7} \quad [6]$$

If [3] is true, then $X = 1 + 4n = 1 + 2(2n)$, and so [1] must be true. $\therefore$ Eliminate [1].

If [5] is true, then $X = 1 + 6n = 1 + 3(2n)$, so [2] is true. $\therefore$ Eliminate [2].

Now multiply [3] and [5] by 2 to get:

$$3x \equiv 3 \cdot 4 = 3 \pmod{12} \quad [3']$$

$$2x \equiv 2 \pmod{2 \cdot 6} = 2 \pmod{12} \quad [5']$$

$$\therefore 3x - 3 \equiv 2x - 2 \pmod{12}$$

$$x \equiv 1 \pmod{12}. \quad [12]$$

If [12] is true, then so must [3] and [5].

Now have:

$$X \equiv 1 \pmod{5}$$

$$X \equiv 0 \pmod{7}$$

$$X \equiv 1 \pmod{12}$$

5, 7, 12 relatively prime.

$$N = 5 \cdot 7 \cdot 12 = 420 \quad N_1 = 7 \cdot 12 = 84$$

$$N_2 = 5 \cdot 12 = 60 \quad N_3 = 5 \cdot 7 = 35$$

$$84x_1 \equiv 1 \pmod{5} \quad 35x_3 \equiv 1 \pmod{12}$$

$$84x_1 - 85x_1 = -x_1 \quad 35x_3 - 36x_3 = -x_3$$

$$x_1 = -1 \quad x_3 = -1$$

$$60x_2 \equiv 1 \pmod{7}$$

irrelevant since $a_2 = 0$.

$$\therefore a_1N_1x_1 + a_2N_2x_2 + a_3N_3x_3 = 1 \cdot 84 \cdot (-1) + 0 + 1 \cdot 35 \cdot (-1)$$

$$= -84 - 35$$

$$= -119.$$

$$\therefore -119 + 420 = 301.$$

$\therefore$ 301 eggs in basket.

10. (Ancient Chinese Problem). A band of 17 pirates stole a sack of gold coins. When they tried to divide the fortune into equal portions, 3 coins remained. In the ensuing brawl over who should get the extra coins, one pirate was killed. The wealth was redistributed, but this time an equal division left 10 coins. Again an argument developed in which another pirate was killed. But now the total number of coins could finally be evenly distributed among the survivors. What was the least number of coins that could have been there?

$$X \equiv 3 \pmod{17} \quad 17, 16, 15 \text{ are relatively prime.}$$

$$X \equiv 10 \pmod{16}$$

$$X \equiv 0 \pmod{15}$$

$$N = 17 \cdot 16 \cdot 15 = 4080 \quad N_1 = 16 \cdot 15 = 240$$

$$N_2 = 17 \cdot 15 = 255 \quad N_3 = 17 \cdot 16 = 272$$

$$240x_1 \equiv 1 \pmod{17} \quad 255x_2 \equiv 1 \pmod{16}$$

$$240x_1 - 14 \cdot 17x_1 = 2x_1 \quad 255x_2 - 16 \cdot 17x_2 = -x_2$$

$$2x_1 \equiv 1, 18x_1 \equiv 9 \quad \therefore x_2 \equiv 1 \pmod{16}$$

$$\therefore x_1 \equiv 9 \pmod{17}$$

$$99x_3 \equiv 1 \pmod{15}$$

irrelevant since $a_2 = 0$.

$$\begin{aligned} \therefore a_1 N_1 x_1 + a_2 N_2 x_2 + a_3 N_3 x_3 &= 3 \cdot 240 \cdot 9 + 10 \cdot 255 \cdot (1) + 0 \\ &= 3930. \end{aligned}$$

$\therefore$ 3930 coins.

The solutions manual uses different approach.

$$X \equiv 3 \pmod{17} \Rightarrow X = 3 + 17t$$

$$X \equiv 10 \pmod{16} \Rightarrow 3 + 17t \equiv 10 \pmod{16}, \text{ or}$$

$$17t \equiv 7 \pmod{16} \Rightarrow t \equiv 7 \pmod{16}$$

$$\therefore X = 3 + 17(7 + 16K) = 122 + 272K.$$

The third condition means $122 + 272K \equiv 0 \pmod{15}$, or

$$122 + 272K - 18 \cdot 15K \equiv 0, \text{ or } 12K + 2 \equiv 0 \pmod{15}, \text{ or}$$

$$12K - 8 \cdot 15K \equiv -8, 2K \equiv -8 \equiv 13 \pmod{15}$$

$$16K \equiv 104, K \equiv 14 \pmod{16}, K = 14 + 15r$$

$$\therefore X = 122 + 272(14 + 15r) = 3930 + 4080r.$$

11. Prove $x \equiv a \pmod{n}$ and $x \equiv b \pmod{m}$ admit a simultaneous solution $\iff \gcd(n, m) \mid a - b$; if a solution exists, confirm it is unique modulo $\text{lcm}(n, m)$.

Pf. (1) Suppose a solution exists for x .

Let $d = \gcd(n, m)$. $\therefore n = dr, m = ds$.

$x \equiv a \pmod{n} \Rightarrow x = a + nt$, some integer t .

$x \equiv b \pmod{m} \Rightarrow x = b + mk$, some integer k .

$\therefore a + nt = b + mk$, or $nt - mk = b - a$.

Substituting for n and m ,

$$drt - dsk = b - a$$

$$d(rt - sk) = a - b. \quad \therefore d = \gcd(n, m) \mid a - b.$$

(2) Let $d = \gcd(n, m)$, and suppose $d \mid a - b$.

$\therefore dt = a - b$, some integer t .

By Th. 2.3, there are integers x_0 and y_0 s.t.

$$nx_0 + my_0 = d.$$

$$\therefore dt = nx_0t + my_0t = a - b.$$

$$\therefore my_0t + b = a - x_0nt.$$

Let $x = a + (-x_0t)n = b + (y_0t)m$.

$$\therefore x \equiv a \pmod{n} \quad \text{So there is a}$$

$$x \equiv b \pmod{m} \quad \text{simultaneous solution.}$$

Now let y be any other solution.

$$\therefore x \equiv a \pmod{n} \quad \text{and} \quad y \equiv a \pmod{n}$$

$$x \equiv b \pmod{m} \quad y \equiv b \pmod{m}$$

$$\therefore x \equiv y \pmod{n}$$

$$x \equiv y \pmod{m}$$

By Section 4.2, problem 13, p.69, $X \equiv y \pmod{\text{lcm}(n, m)}$.

12. $x \equiv 5 \pmod{6}$ and $x \equiv 7 \pmod{15}$.

$\gcd(6, 15) = 3$. Since $3 \nmid (7 - 5)$, there is no solution.

13. If $x \equiv a \pmod{n}$, prove either $x \equiv a \pmod{2n}$ or $x \equiv a + n \pmod{2n}$. *Pf.*

$x \equiv a \pmod{n} \Rightarrow x = a + Kn$, some K .

If K is even, then $K = 2r$, some r .

$$\therefore x = a + r(2n) \Rightarrow x \equiv a \pmod{2n}.$$

If K is odd, then $K = 2r + 1$, some r .

$$\therefore x = a + (2r + 1)n = a + n + r2n \Rightarrow$$

$$x \equiv a + n \pmod{2n}.$$

14.

$$x \equiv 1 \pmod{9}$$

$$x \equiv 2 \pmod{11}$$

$$x \equiv 6 \pmod{13}$$

and $1 < x < 1287$.

9, 11, 13 are rel. prime, so can use Chinese Remainder Theorem.

$$N = 9 \cdot 11 \cdot 13 = 1287 \quad N_1 = 11 \cdot 13 = 143 \quad N_2 = 9 \cdot 13 = 117 \quad N_3 = 9 \cdot 11 = 99$$

$$143x_1 \equiv 1 \pmod{9} \quad 117x_2 \equiv 1 \pmod{11}$$

$$143x_1 - 9 \cdot 15x_1 = 8x_1 \quad 117x_2 - 12x_2 = -4x_2$$

$$8x_1 - 7x_1 = -x_1 = 1 \quad -2x_2 \equiv 5, x_2 = -3$$

$$x_1 = -1 \quad x_2 = -3$$

$$99x_3 \equiv 1 \pmod{13}$$

$$99x_3 - 8 \cdot 13x_3 = -5x_3$$

$$-15x_3 \equiv 3, -2x_3 \equiv 3$$

$$x_3 = 18$$

$$\therefore a_1 N_1 x_1 + a_2 N_2 x_2 + a_3 N_3 x_3 =$$

$$1 \cdot 143 \cdot (-1) + 2 \cdot 117 \cdot (-3) + 6 \cdot 99 \cdot (18) = 9847.$$

$$9847 - 7 \cdot 1287 = 838.$$

$$\therefore x = 838.$$

15.(a) Find an integer having the remainders 1, 2, 5, 5 when divided by 2, 3, 6, 12 respectively.

$$X \equiv 1 \pmod{2} [2] \quad \text{divisors not relatively}$$

$$X \equiv 2 \pmod{3} [3] \quad \text{prime, so simplify.}$$

$$X \equiv 5 \pmod{6} [6]$$

$$X \equiv 5 \pmod{12} [12]$$

$\gcd(3, 6) \neq 1$, so multiply [3] by 2

$$\therefore 2x \equiv 4 \pmod{6}$$

$$X \equiv 5 \pmod{6}$$

$$\therefore 2x - 4 \equiv x - 5 \pmod{6}, \text{ or } x \equiv -1 \pmod{6} [6']$$

$\therefore$ if $[6']$ is true, then so is $[6]$ and $[3]$.

But $[6']$ is the same as $x = -1 + 6 = 5 \pmod{6}$, which is $[6]$. $\therefore$ can drop $[3]$.

$\gcd(6, 12) \neq 1$, so multiply $[6]$ by 2.

$$\therefore 2x \equiv 10 \pmod{12}$$

$$X \equiv 5 \pmod{12}$$

$$\therefore X \equiv 5 \pmod{12}, \text{ which is } [12].$$

$\therefore$ if $[12]$ is true, then so is $[6]$, and so is $[3]$.

$\therefore$ can drop $[3]$ and $[6]$.

$$\therefore X \equiv 1 \pmod{2} [2]$$

$$X \equiv 5 \pmod{12} [12]$$

But $\gcd(2, 12) \neq 1 \therefore$ multiply $[2]$ by 6.

$$6x \equiv 6 \pmod{12}$$

$$X \equiv 5 \pmod{12}$$

$$35x - 36x \equiv 7, \text{ or } -x \equiv 7$$

$$X \equiv -7 + 12 = 5$$

$$\therefore X \equiv 5 \pmod{12}$$

$$X = 5 + 12k.$$

Since want $X > 12$, choose $X = 5 + 12 = 17$. **(b)** Find an integer with remainders 2, 3, 4, 5 when divided by 3, 4, 5, 6 respectively.

$$X \equiv 2 \pmod{3} [3] \quad \text{Divisors not relatively}$$

$$X \equiv 3 \pmod{4} [4] \quad \text{prime, so simplify.}$$

$$X \equiv 4 \pmod{5} [5]$$

$$X \equiv 5 \pmod{6} [6]$$

Multiply $[3]$ by 2: $2x \equiv 4 \pmod{6}$

$$X \equiv 5 \pmod{6} [3']$$

$\therefore X \equiv -1 \pmod{6}$, or

$X \equiv 5 \pmod{6}$, which is $[6]$.

$\therefore [3]$ is superfluous since if $[6]$ is true, so is $[3]$.

Now examine $[4]$ and $[6]$. Multiply $[4]$ by 3 and $[6]$ by 2.

$$\therefore 3x \equiv 9 \pmod{12}$$

$$2x \equiv 10 \pmod{12}$$

$$\therefore x \equiv -1, \text{ or } x \equiv 11 \pmod{12} [12]$$

$\therefore$ if [12] is true, so is [4] and [6].

$$\therefore X \equiv 4 \pmod{5}$$

$$X \equiv 11 \pmod{12}$$

$\gcd(5, 12) = 1$, $\therefore$ use Chinese Remainder Th.

$$N = 5 \cdot 12 = 60 \quad N_1 = 12 \quad N_2 = 5$$

$$12x_1 \equiv 1 \pmod{5} \quad 5x_2 \equiv 1 \pmod{12}$$

$$24x_1 \equiv 2 \quad 25x_2 \equiv 5$$

$$24x_1 - 25x_2 = -x_1 = 2 \quad 25x_2 - 24x_2 = x_2 = 5$$

$$x_1 = -2 \quad x_2 = 5$$

$$\therefore a_1 N_1 x_1 + a_2 N_2 x_2 = 4 \cdot 12 \cdot 3 + 11 \cdot 5 \cdot 5 = 419.$$

$$\therefore X \equiv 419 \pmod{60}, \text{ or } X \equiv 419 - 6 \cdot 60$$

$$X \equiv 59 \pmod{60}.$$

$\therefore X = 59$.

(c) Find an integer having remainders 3, 11, 15 when divided by 10, 13, 17 respectively.

$$X \equiv 3 \pmod{10} \quad \text{All divisors or relatively}$$

$$X \equiv 11 \pmod{13} \quad \text{prime } \therefore \text{ use Chinese}$$

$$X \equiv 15 \pmod{17} \quad \text{Remainder Th.}$$

$$N = 10 \cdot 13 \cdot 17 = 2210 \quad N_1 = 13 \cdot 17 = 221$$

$$N_2 = 10 \cdot 17 = 170 \quad N_3 = 10 \cdot 13 = 130$$

$$221x_1 \equiv 1 \pmod{10} \quad 170x_2 \equiv 1 \pmod{13}$$

$$221x_1 - 220x_1 = x_1 \quad 170x_2 - 13 \cdot 15x_2 = -5x_2$$

$$x_1 \equiv 1 \quad -5x_2 \equiv 1, 18x_2 \equiv -3$$

$$x_2 \equiv -3$$

$$130x_3 \equiv 1 \pmod{17}$$

$$130x_3 - 8 \cdot 17x_3 = -6x_3$$

$$-6x_3 \equiv 1, 18x_3 \equiv -3$$

$$x_3 = -3$$

$$\therefore a_1N_1x_1 + a_2N_2x_2 + a_3N_3x_3 =$$

$$3 \cdot 221 \cdot (1) + 11 \cdot 170 \cdot (-1) + 15 \cdot 130 \cdot (-3) = -3817.$$

$$-3817 + 2(2210) = 1103.$$

$$\therefore X = 1103.$$

16. Let t_n be the n -th triangular number. For which values of n does t_n divide

$$t_1^2 + t_2^2 + \cdots + t_n^2?$$

$$t_1^2 + t_2^2 + \cdots + t_n^2 = t_n(3n^3 + 12n^2 + 13n + 2)/30$$

Pf. By induction, if $n = 1$, $t_1 = (3 + 12 + 13 + 2)/30 = 1$.

$(3 + 12 + 13 + 2)/30 = 1$. Now suppose, for $K > 1$,

$$t_1^2 + \cdots + t_K^2 = t_K(3K^3 + 12K^2 + 13K + 2)/30 \quad [1]$$

$$\begin{aligned} \therefore t_1^2 + \cdots + t_K^2 + t_{K+1}^2 &= t_K(3K^3 + 12K^2 + 13K + 2)/30 + \left[\frac{(K+1)(K+2)}{2} \right]^2 \\ &= \frac{K(K+1)}{2} \cdot \frac{3K^3 + 12K^2 + 13K + 2}{30} + \frac{(K+1)^2(K+2)^2}{4} \\ &= \frac{K+1}{2} \left[\frac{K(3K^3 + 12K^2 + 13K + 2)}{30} + \frac{(K+1)(K+2)^2}{2} \right] \\ &= \frac{K+1}{2} \left[\frac{3K^4 + 12K^3 + 13K^2 + 2K}{30} + K(K+2)^2 + 4 \right] \\ &= \frac{K+1}{2} \cdot \frac{3K^4 + 27K^3 + 88K^2 + 122K + 60}{30} \quad [2] \end{aligned}$$

Now look at right side of [1] using $K+1$:

$$\begin{aligned} &t_{K+1}(3(K+1)^3 + 12(K+1)^2 + 13(K+1) + 2)/30 \\ &= \frac{(K+1)(K+2)}{2} \cdot \frac{[3K^3 + 9K^2 + 9K + 3 + 12K^2 + 24K + 12 + 13K + 13 + 2]}{30} \\ &= \frac{(K+1)(K+2)}{2} \cdot \frac{[3K^3 + 9K^2 + 9K + 3 + 12K^2 + 24K + 12 + 13K + 13 + 2]}{30} \end{aligned}$$

$$\begin{aligned}
 &= \frac{(K+1)(K+2)}{2} \cdot \frac{3K^3 + 21K^2 + 46K + 30}{30} \\
 &= \frac{(K+1)(K+2)}{2} \left[\frac{3K^3 + 21K^2 + 46K + 30}{30} \right] \\
 &= \frac{(K+1)}{2} \left[\frac{3K^4 + 21K^3 + 46K^2 + 30K + 6K^2 + 42K + 52K + 60}{30} \right] \\
 &= \frac{(K+1)}{2} \left[\frac{3K^4 + 21K^2 + 88K^2 + 122K + 60}{30} \right] \quad [3]
 \end{aligned}$$

[Note: this line's coefficient on K^2 appears to be a transcription/arithmetic slip in the original – it should combine to match [2] on the previous page (which has $27K^3 + 88K^2$); transcribed as written.]

$\therefore [2] = [3]$, so $K \Rightarrow K+1$.

$\therefore t_n \mid (t_1^2 + \cdots + t_n^2) \Leftrightarrow (3n^3 + 12n^2 + 13n + 2)/30$ is

an integer, i.e.,

$$(3n^3 + 12n^2 + 13n + 2) \equiv 0 \pmod{30}, \text{ or}$$

$$(3n^3 + 12n^2 + 13n + 2) \equiv 0 \pmod{2 \cdot 3 \cdot 5}, \text{ or}$$

$$3n^3 + 12n^2 + 13n + 2 \equiv 0 \pmod{2} \quad [2]$$

$$3n^3 + 12n^2 + 13n + 2 \equiv 0 \pmod{3} \quad [3]$$

$$3n^3 + 12n^2 + 13n + 2 \equiv 0 \pmod{5} \quad [4]$$

since unique solutions are $\pmod{30}$ by **Chinese Remainder Theorem**.

For [2], $3n^3 - 2n^2 + 12n^2 - 3 \cdot 4 \cdot n + 13n + 2 =$

$$n^3 + n = n(n^2 + 1) \equiv 0 \pmod{2}.$$

If n is even, then $n(n^2 + 1)$ is even, and $n^2(n+1) \equiv 0 \pmod{2}$.

If n is odd, n^2 is odd, and $n^2 + 1$ is even, so $n(n^2 + 1)$ is even, so $n(n^2 + 1) \equiv 0 \pmod{2}$.

So [2] puts no restrictions on n .

For [8], $3n^3 - 3n^2 + 12n^2 - 3 \cdot 4 \cdot n + 13n + 2 \equiv$

$$n + 2 \equiv 0 \pmod{3}$$

$$\therefore n \equiv 1 \pmod{3}.$$

For [5], $3n^3 + 12n^2 - 5 \cdot 2n^2 + 13n - 5 \cdot 2n + 2 \equiv$

$$3n^3 + 2n^2 + 3n + 2 = n^2(3n + 2) + (n^2)(3n + 2)$$

[Note: this regrouping line is dense/hard to verify term-by-term from the handwriting; transcribed best-effort.]

$$\therefore (3n + 2)(n^2 + 1) \equiv 0 \pmod{5}$$

(using $ab \equiv 0 \pmod{p}$, p prime $\Rightarrow a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$ – see comments at end of section 4.2, and proof of Theorem 2 in Problems 4.2).

$\therefore$ Problem reduces to $[n \equiv 1 \pmod{3}]$

and $(n^2 + 1) \equiv 0 \pmod{5}$ or $(3n + 2) \equiv 0 \pmod{5}$

$$3n + 2 \equiv 0 \pmod{5} \Rightarrow n \equiv 1 \pmod{5} \Rightarrow n \equiv 3 \pmod{15}$$

$$n \equiv -2, 6n \equiv -4$$

$$n \equiv 4, 6n \equiv 1$$

$$\therefore n \equiv 1 \pmod{5}$$

$$n^2 + 1 \equiv 0 \pmod{5} \Rightarrow 3n^2 \equiv 6 \pmod{15}$$

$$n \equiv 1 \pmod{5} \Rightarrow 3n \equiv 6 \pmod{15} \Rightarrow 3n \equiv 5 \pmod{15}$$

$$n \equiv 2 \pmod{5} \Rightarrow 3n \equiv 6 \pmod{15}, 5n \equiv -1, 2n \equiv 14, 2n \equiv -14$$

$$\therefore n \equiv 2 \pmod{5}, n \equiv 7$$

$$n \equiv 3 \pmod{5} \Rightarrow 3n \equiv 9 \pmod{15}$$

$$n \equiv 1 \pmod{5} \Rightarrow 3n \equiv 6 \pmod{15} \Rightarrow 3n \equiv 5 \pmod{15}, 3n \equiv 9 - 4, 2n \equiv -4$$

$$\therefore n \equiv 2, n \equiv 2, n \equiv 14$$

$$\therefore n \equiv 1, \text{ or } 7, \text{ or } 13 \pmod{15}$$

[Note: pages 283-284 – the algebraic manipulation deriving $n \equiv 1, 7, 13 \pmod{15}$ is dense, with several lines hard to follow precisely from the handwriting; the final stated result is transcribed as written.]

17. Find solutions of

$$3x + 7y \equiv 5 \pmod{13} \quad [1]$$

$$2x + 5y \equiv 7 \pmod{13} \quad [2]$$

$$\text{Mult [1] by 5: } 15x + 20y \equiv 25 \pmod{13} \quad [1']$$

$$\text{Mult [2] by 4: } 8x + 20y \equiv 28 \pmod{13} \quad [2'] \quad [1'] - [2']: \quad 7x \equiv -3 \pmod{13}$$

$$\therefore 14x \equiv -6$$

$$14x - 13x \equiv -6 + 13$$

$$x \equiv 7 \pmod{13} \quad [5']$$

$$\text{Substitute [5'] into [1]: } 3x \equiv 21 \pmod{13} \quad [3']$$

$$3x \equiv 5 - 4y \pmod{13} \quad [1]$$

$$\therefore 21 \equiv 5 - 4y \pmod{13}$$

$$16 \equiv -4y$$

$$48 \equiv -12y$$

$$48 - 13 = -12y + 13y$$

$$9 \equiv y \pmod{13}$$

$$\therefore x \equiv 7 \pmod{13}$$

$$y \equiv 9 \pmod{13}$$

18. Obtain the two incongruent solutions mod 210 of the system:

$$2x \equiv 3 \pmod{5} \quad [1]$$

$$4x \equiv 2 \pmod{6} \quad [2]$$

$$3x \equiv 4 \pmod{7} \quad [3]$$

From [1]: $4x \equiv 6 \pmod{5}$

$$4x - 5x \equiv 6 - 5$$

$$-x \equiv 1$$

$$x \equiv -1 + 5$$

$$x \equiv 4 \pmod{5}$$

From [2]: $4x/2 \equiv 2/2 \pmod{6/2}$

$$2x \equiv 1 \pmod{3}$$

$$4x \equiv 2$$

$$4x - 3x = x \equiv 2 \pmod{3}, \therefore x \equiv 2 \pmod{6}$$

[Note: the modulus on the last line is written as $\pmod{6}$ but the reduction was done mod 3 in the prior steps; this likely should read $\pmod{3}$, with the conclusion $x \equiv 2$ or $5 \pmod{6}$ as the "other" solution per Theorem 4.7 below. Transcribed as written.]

Since $\gcd(4, 6) = 2$, Th. 4.7 says there are 2 incongruent solutions $x_0, x_0 + \frac{6}{2}$, where x_0 is a solution and $x_0 + 3$ is the other, so $2 + 3 = 5 \pmod{6}$ is the other.

$\therefore x \equiv 5 \pmod{6}$ is the other.

From [7]: $6x \equiv 4 \pmod{7}$

[Note: labeled [7] here but refers to equation [3] above; transcribed as written.]

$$6x - 7x = -x \equiv 4 - 7$$

$$-x \equiv -3$$

$$x \equiv 3 \pmod{7}$$

$$\therefore x \equiv 4 \pmod{5}$$

$$x \equiv 2 \pmod{6} \text{ or } x \equiv 5 \pmod{6}$$

$$x \equiv 3 \pmod{7}$$

$$N = 5 \cdot 6 \cdot 7 = 210 \quad N_1 = 6 \cdot 7 = 42$$

$$N_2 = 5 \cdot 7 = 35$$

$$N_3 = 5 \cdot 6 = 30$$

$$\therefore 42x_1 \equiv 1 \pmod{5} \quad 35x_2 \equiv 1 \pmod{6}$$

$$42x_1 - 40x_1 = 2x_1$$

$$6x_1 \equiv 3, \quad 6x_1 - 5x_1 = x_1 \equiv 3 \pmod{5}$$

$$\therefore x_1 \equiv 3 \pmod{5}$$

$$35x_2 - 36x_2 = -x_2$$

$$-x_2 \equiv -1 + 6 = 5$$

$$x_2 \equiv 5 \pmod{6}$$

$$30x_3 \equiv 1 \pmod{7}$$

$$30x_3 - 28x_3 = 2x_3$$

$$2x_3 \equiv 1, \quad 8x_3 \equiv 4$$

$$8x_3 - 7x_3 = x_3 \equiv 4 \pmod{7}$$

$$\therefore x_3 \equiv 4 \pmod{7}$$

$$\therefore a_1N_1x_1 + a_2N_2x_2 + a_3N_3x_3 =$$

$$4(42)(3) + 2(35)(5) + 3(30)(4) = 1214$$

or

$$4(42)(3) + 5(35)(5) + 3(30)(4) = 1739$$

$$\therefore X \equiv 1214 \pmod{210} \quad X \equiv 164 \pmod{210}$$

or

$$X \equiv 1739 \pmod{210} \quad X \equiv 59 \pmod{210}$$

19. Obtain the 8 incongruent solutions of $3x + 4y \equiv 5 \pmod{8}$.

Set $3x \equiv 5 - 4y \pmod{8}$. Since $\gcd(3, 8) = 1$, and $11 \mid (5 - 4y)$, Th. 4.7 says there is one solution for any value of y . Since there are 8 incongruent values of $5 - 4y$ ($y = 0, 1, \dots, 7$)

solve for each value of y .

$$3x \equiv 5 \pmod{8}, \quad 15x \equiv 25, \quad 15x - 8x = 5x \equiv 25 - 24$$

$$x \equiv -1, \quad x \equiv 7$$

$$\therefore x \equiv 7, \quad y \equiv 0 \pmod{8}$$

$$3x \equiv 1 \pmod{8}, \quad 15x \equiv 5, \quad x \equiv 5, \quad x \equiv -3$$

$$\therefore x \equiv 3, \quad y \equiv 1 \pmod{8}$$

$$3x \equiv -3 \pmod{8}, \quad 15x \equiv -15, \quad -x \equiv 1, \quad x \equiv 7$$

$$\therefore x \equiv 7, \quad y \equiv 2 \pmod{8}$$

$$3x \equiv -7 \pmod{8}, \quad 3x \equiv 1, \quad 15x \equiv 5, \quad -x \equiv 5$$

$$\therefore x \equiv 3, \quad y \equiv 3 \pmod{8}$$

$$3x \equiv -11 \pmod{8}, \quad 3x \equiv 5, \quad 15x \equiv 25, \quad -x \equiv 1$$

$$x \equiv -1, \quad x \equiv 3$$

$$\therefore x \equiv 7, \quad y \equiv 4 \pmod{8}$$

$$3x \equiv -15 \pmod{8}, \quad 3x \equiv 1, \quad 15x \equiv 5, \quad -x \equiv 5$$

$$x \equiv -5, \quad x \equiv 3$$

$$\therefore x \equiv 3, \quad y \equiv 5 \pmod{8}$$

$$3x \equiv -19 \pmod{8}, \quad 3x \equiv -3, \quad x \equiv 7 \text{ from above}$$

$$\therefore x \equiv 7, \quad y \equiv 6 \pmod{8}$$

$$3x \equiv -23 \pmod{8}, \quad 3x \equiv 1, \quad \therefore x \equiv 3 \text{ from above}$$

$$\therefore x \equiv 3, \quad y \equiv 7 \pmod{8}$$

20. Find solutions to the following systems.

(a). $5x + 3y \equiv 1 \pmod{7} \quad [1]$

$$3x + 2y \equiv 4 \pmod{7} \quad [2]$$

$$10x + 6y \equiv 2 \pmod{7} \quad [1'] : [1] \times 2$$

$$9x + 6y \equiv 12 \pmod{7} \quad [2'] : [2] \times 3$$

$$x \equiv -10 \pmod{7} \quad [1'] - [2']$$

$$x \equiv -10 + 14 \equiv 4 \pmod{7}$$

$$5x \equiv 20 \pmod{7} \quad [3] \equiv [3] \times 5$$

$$1 - 3y \equiv 20 - 14 \equiv 6 \quad [3'] \text{ in } [1]$$

$$-3y \equiv 19 - 14 = 5$$

$$-6y \equiv 10$$

$$6y \equiv -10$$

$$\therefore x \equiv 4 \pmod{7}$$

$$y \equiv 10 \pmod{7}$$

[Note: final y value should likely reduce mod 7 (e.g. $10 \equiv 3 \pmod{7}$); transcribed as written in the original.] (b). $7x + 8y \equiv 6 \pmod{11} \quad [1]$

$$4x + 5y \equiv 9 \pmod{11} \quad [2]$$

$$14x + 16y \equiv 12 \pmod{11} \quad [1'] : [1] \times 2$$

$$12x + 16y \equiv 27 \pmod{11} \quad [2'] : [2] \times 3$$

$$2x \equiv -15 \pmod{11} \quad [1'] - [2']$$

$$2x \equiv -15 + 22 = 7$$

$$10x \equiv 35$$

$$10x - 11x = -x \equiv 2$$

$$x \equiv -2 + 11 = 9 \quad [3]$$

$$4x \equiv 36 \pmod{11} \quad [3] \times 4$$

$$4x \equiv 36 - 33 = 3 \pmod{11} \quad [5'] \text{ in } [2]$$

$$3 \equiv 9 - 2y \pmod{11}$$

$$-2y \equiv -6$$

$$-8y \equiv -30$$

$$-8y + 11y = 3y \equiv -30 + 33 = 3$$

$$\therefore y \equiv 1 \pmod{11}$$

$$\therefore x \equiv 9 \pmod{11}$$

$$y \equiv 3 \pmod{11}$$

[Note: there is some inconsistency in the intermediate y value (first $y \equiv 1$, then concluding $y \equiv 3$); transcribed as written, final answer as stated in original.]

$$(c). \quad 11x + 5y \equiv 7 \pmod{20} \quad [1]$$

$$6x + 3y \equiv 8 \pmod{20} \quad [2]$$

$$33x + 15y \equiv 21 \pmod{20} \quad [1'] = [1] \times 3$$

$$30x + 15y \equiv 40 \pmod{20} \quad [2'] = [2] \times 5$$

$$3x \equiv -19 \pmod{20} \quad [3] = [1'] - [2']$$

$$3x \equiv -19 + 20 = 1 \pmod{20} \quad [3']$$

$$21x \equiv 7 \pmod{20} \quad [3'] \times 7$$

$$21x - 20x \equiv 7$$

$$x \equiv 7 \pmod{20} \quad [4]$$

$$6x \equiv 42 \pmod{20} \quad [4'] = [4] \times 6$$

$$\therefore 42 \equiv 8 - 3y \pmod{20} \quad [4'] \text{ in } [2]$$

$$-3y \equiv 34 - 20 = 14 \pmod{20} \quad [4'']$$

$$-21y \equiv 98 \pmod{20} \quad [5] = [4''] \times 7$$

$$-21y + 20y \equiv 98 - 5 \cdot 20$$

$$-y \equiv -2$$

$$y \equiv 2$$

$$\therefore x \equiv 7 \pmod{20}$$

$$y \equiv 2 \pmod{20}$$

Chapter 5

Fermat's Theorem

5.2 Fermat's Factorization Method

Problems 5.2

1. Use Fermat's method to factor each number.

(a). 2279

$$47^2 < 2279 < 48^2 \quad \frac{2279 + 1}{2} = 1140$$

$$\therefore 48^2 - 2279 = 25 = 5^2$$

$$\therefore 48 - 5 = 43, 48 + 5 = 53$$

$$\therefore 2279 = 43 \cdot 53$$

(b) 10541 $102^2 < 10541 < 103^2$, $\frac{10541+1}{2} = 5271$

$$\therefore 108^2 - 10541 = 68$$

$$104^2 - 10541 = 275$$

$$105^2 - 10541 = 484 = 22^2$$

$$\therefore 105 - 22 = 83, 105 + 22 = 127$$

$$\therefore 10541 = 83 \cdot 127$$

(c) 340663 $588^2 < 340663 < 584^2$, $\frac{340663+1}{2} = 170332$

[Note: the inequality " $588^2 < 340663 < 584^2$ " has the bounds reversed/inconsistent ($588 > 584$); likely a transcription or original slip, transcribed as written.]

$$584^2 - 340663 = 393$$

From spreadsheet,

$$592^2 - 340663 = 9801 = 99^2$$

$$\therefore 592 - 99 = 493, 592 + 99 = 691$$

691 is prime (from table of primes); 493 not

$$22^2 < 493 < 23^2 \quad \frac{493 + 1}{2} = 247$$

$$23^2 - 493 = 36 = 6^2$$

$$23 + 6 = 29, 23 - 6 = 17$$

$$\therefore 493 = 17 \cdot 29$$

$$\therefore 340663 = 17 \cdot 29 \cdot 691$$

2. Prove a perfect square must end in one of the following digits:

00 16 25 49 81 89
 01 24 36 57 76 86
 04 21 64 81
 09 25 44 64 84

[Note: this digit table is faint/cramped and the row/column grouping is hard to read precisely; transcribed best-effort as a flat list of 2-digit endings.]

Pf. First note $(x + 50)^2 = x^2 + 100x + 2500$, so $x^2 \equiv (x + 50)^2 \pmod{100}$. This means you only need to examine the last two digits of $x = 0, 1, \dots, 49$ (since $0 \equiv 50, 1 \equiv 51, \dots$). But $(x - 50)^2 = x^2 - 100x + 2500$, so $x^2 \equiv (x - 50)^2 \pmod{100}$, so for $x = 26, 27, \dots, 49$

$$26^2 \equiv 24^2, 27^2 \equiv 23^2, \dots$$

$\therefore$ Only need to look at digits $x = 0, 1, 2, \dots, 25$.

x	$x^2 \pmod{100}$	$x^2 \pmod{100}$
0	00	16 00 #
1	01	21 41
2	04	22 84
3	09	23 29
4	16	24 76
5	25	25 25 #
6	36	16
7	49	17 89
8	64	18 24 # = duplicated ending
9	81	19 61

[Note: this $x/x^2 \pmod{100}$ table is dense and hard to read precisely past $x = 9$; transcribed best-effort, several entries (marked with "#") appear to flag duplicated endings as noted on the page.]

$\therefore$ The above endings are the ones that were to be proved.

3. Factor the number $2^{11} - 1$ using Fermat's method.

$$2^{11} - 1 = 2047, \quad 45^2 < 2047 < 46^2$$

From spreadsheet, $56^2 - 2047 = 1089 = 33^2$

$$\therefore 56^2 - N^2 = 33^2 \quad N = (56 + 33)(56 - 33)$$

or $N = 89 \cdot 23$

$$\therefore 2^{11} - 1 = 89 \cdot 23$$

4. If $n = a^2 + b^2 = c^2 + d^2$, Then

$$n = \frac{(ac + bd)(ac - bd)}{(a + d)(a - d)} = \gcd(c_1, c_2) \cdot \gcd(c_1, d_1)$$

[Note: this general statement of the identity is dense/abbreviated and the exact role of $\gcd(c_1, c_2)$, $\gcd(c_1, d_1)$ is unclear from the handwriting; transcribed as written.]

(a). Factor $493 = 18^2 + 13^2 = 22^2 + 3^2$

$$\begin{aligned} 493 &= \frac{(18 \cdot 22 + 13 \cdot 3)(18 \cdot 22 - 13 \cdot 3)}{(18 + 3)(18 - 3)} = \frac{(435)(357)}{(21)(15)} \\ &= \frac{435}{15} \cdot \frac{357}{21} = 29 \cdot 17 \end{aligned}$$

(b) $38025 = 168^2 + 99^2 = 156^2 + 117^2$

$$\begin{aligned} &= \frac{(168 \cdot 156 + 99 \cdot 117)(168 \cdot 156 - 99 \cdot 117)}{(168 + 117)(168 - 117)} \\ &= \frac{(37771)(14625)}{(285)(51)} = \frac{14625}{285} \cdot \frac{37791}{51} \\ &= \frac{14625}{285} \cdot 741 \end{aligned}$$

But 741 is not prime: $741 = 3 \cdot 247 =$

$$3 \cdot 13 \cdot 19$$

$$\therefore 38025 = \frac{14625}{285} \cdot (3 \cdot 13 \cdot 19)$$

$$\frac{14625}{285} \text{ is not an integer}$$

$$\frac{14625}{285} = \frac{5 \cdot 2925}{5 \cdot 57} = \frac{25 \cdot 117}{3 \cdot 19} = \frac{5^2 \cdot 9 \cdot 13}{3 \cdot 19}$$

$$\begin{aligned}\therefore 38025 &= \frac{5^2 \cdot 9 \cdot 13}{3 \cdot 19} \cdot 3 \cdot 13 \cdot 19 \\ &= \frac{5^2 \cdot 3^2 \cdot 13}{19} \cdot 13 \cdot 19 \\ &= 3^2 \cdot 5^2 \cdot 13^2\end{aligned}$$

5. Use generalized Fermat method to factor.

(a). 2911 Use hint: $138^2 \equiv 67 \pmod{2911}$

$$\therefore \gcd(138 - 67, 2911) = \gcd(71, 2911)$$

Using Euclidean Algorithm,

$$2911 = 41 \cdot 71, \text{ and } 71, 41 \text{ both prime.}$$

(b) 4573 Use hint: $177^2 \equiv 92^2 \pmod{4573}$

$$\therefore \gcd(177 - 92, 4573) = \gcd(85, 4573)$$

$$\therefore 4573 = 53 \cdot 85 + 68$$

$$85 = 1 \cdot 68 + 17$$

$$68 = 4 \cdot 17 \quad \therefore \gcd = 17$$

$$\therefore \gcd(177 + 92, 4573) = \gcd(269, 4573)$$

$$4573 = 17 \cdot 269, \quad \therefore \gcd = 17$$

Also, 269 is prime, $\therefore 4573 = 17 \cdot 269$.

(c) 6923 From hint: $208^2 \equiv 98^2 \pmod{6923}$

$$\therefore \gcd(208 - 98, 6923) = \gcd(115, 6923)$$

$$\therefore 6923 = 60 \cdot 115 + 23$$

$$115 = 5 \cdot 23 \quad \gcd = 23$$

$$\gcd(208 + 98, 6923) = \gcd(301, 6923)$$

$$\therefore 6923 = 23 \cdot 301, \quad \therefore \gcd = 301$$

and $301 = 7 \cdot 43$

$$\therefore 6923 = 7 \cdot 23 \cdot 43$$

6. Factor 13561.

From $288^2 \equiv 8^2 \cdot 5 \pmod{13561}$

$$128^2 \equiv 2^4 \cdot 5 \pmod{13561}$$

$$(233 \cdot 128)^2 \equiv 2^4 \cdot 5^2 \cdot (2 \cdot 5)^2 \pmod{13561}$$

$$\therefore 298473^2 \equiv 60^2 \pmod{13561}$$

and $298473 - 22 \cdot 13561 = 131 \not\equiv \pm 60 \pmod{13561}$

[Note: this line's logic (asserting $131 \not\equiv \pm 60$) and the preceding modular manipulations are dense/hard to fully verify from the handwriting; transcribed best-effort.]

$$\therefore \gcd(298473 - 60, 13561) = \gcd(298413, 13561)$$

$$298473 = 22 \cdot 13561 + 71$$

$$13561 = 191 \cdot 71 \quad \therefore \gcd = 71 \text{ (a prime)}$$

and 191 is prime.

$$\therefore 13561 = 71 \cdot 191$$

7.(a). Factor 4537 by searching for x s.t. $x^2 - k \cdot 4537$ is the product of small primes.

$$\sqrt{4537} = 67.4$$

$$\therefore 67^2 - 4537 = -4 \cdot 3 \quad [1]$$

$$68^2 - 4537 = 87 = 3 \cdot 29 \quad [1']$$

$$\sqrt{2 \cdot 4537} = 95.3$$

$$95^2 - 2 \cdot 4537 = -49 = -7^2 \quad [2]$$

$$96^2 - 2 \cdot 4537 = 142 = 2 \cdot 71 \quad [2']$$

$$\sqrt{3 \cdot 4537} = 116.7$$

$$116^2 - 3 \cdot 4537 = -155 = -5 \cdot 31 \quad [3]$$

$$117^2 - 3 \cdot 4537 = 78 = 2 \cdot 3 \cdot 13 \quad [3']$$

$$\sqrt{4 \cdot 4537} = 134.7$$

$$134^2 - 4 \cdot 4537 = -192 = -2^6 \cdot 3 \quad [4]$$

$$135^2 - 4 \cdot 4537 = 77 = 7 \cdot 11 \quad [4'] \quad \text{Note: } \gcd(s) = 1 \text{ from } [1], [2], [3]$$

$$\sqrt{5 \cdot 4537} = 150.6$$

$$150^2 - 5 \cdot 4537 = -185 = -5 \cdot 37 \quad [5]$$

$$151^2 - 5 \cdot 4537 = 116 = 4 \cdot 29 \quad [5']$$

$$\sqrt{6 \cdot 4537} = 164.99$$

$$165^2 - 6 \cdot 4537 = 3 \quad [6]$$

$$\sqrt{7 \cdot 4537} = 178.2$$

$$(178)^2 - 7 \cdot 4537 = -75 = -3 \cdot 5^2 \quad [7]$$

$$179^2 - 7 \cdot 4537 = 282 = 2 \cdot 141 = 2 \cdot 3 \cdot 47 \quad [7']$$

$$\sqrt{8 \cdot 4537} = 190.6$$

$$190^2 - 8 \cdot 4537 = -196 = -2 \cdot 98 = -2^2 \cdot 7^2 \quad [8]$$

$$191^2 - 8 \cdot 4537 = 185 = 5 \cdot 37 \quad [8'] \quad \text{Note: } \gcd = 1 \text{ from } [2], [8]$$

$$\sqrt{17 \cdot 4537} = 202.1$$

$$(202)^2 - 9 \cdot 4537 = -29 \quad [9]$$

$$\therefore (\text{look at } [3], [6])$$

$$(202 \cdot 151)^2 \equiv (2 \cdot 23)^2 \pmod{4537}$$

$$(30502)^2 \equiv (58)^2 \pmod{4537}$$

$$\gcd(30502 - 58, 4537) = \gcd(30444, 4537) = \gcd(30444, 4537)$$

$$\gcd(30502 + 58, 4537) = 3 \cdot ? \quad [9']$$

$$\therefore (\text{look at } [6'], [3'], [5'])$$

$$(165 \cdot 179 \cdot 203)^2 \equiv (3 \cdot 47)^2 \pmod{4537}$$

$$(5958605)^2 \equiv (564)^2 \pmod{4537}$$

$$\gcd(5958605 - 564, 4537) = \gcd(5958041, 4537)$$

$$5958041 = 1321 \cdot 4537 + 1664$$

$$4537 = 2 \cdot 1664 + 1209$$

$$1664 = 1 \cdot 1209 + 455$$

$$1209 = 3 \cdot 455 - 156$$

$$455 = 3 \cdot 156 - 13$$

$$156 = 12 \cdot 13$$

$$\gcd = 13$$

$\therefore 4537 = 13 \cdot 349$, and 349 is prime.

(b). Factor 14429 using method in (a).

Use hint

$$120^2 - 14429 = -29$$

$$3001^2 - 625 \cdot 14429 = -116 = -2^2 \cdot 29$$

$$\therefore (120 \cdot 3001)^2 \equiv (2 \cdot 29)^2 \pmod{14429}$$

$$(360360)^2 \equiv (58)^2 \pmod{14429}$$

$$\gcd(360360 - 58, 14429) = \gcd(360302, 14429)$$

$$360302 = 25 \cdot 14429 - 423$$

$$14429 = 34 \cdot 423 + 47$$

$$423 = 9 \cdot 47$$

$$\therefore \gcd = 47 \text{ (a prime)}$$

$$\gcd(30360 + 58, 14429) = \gcd(30418, 14429)$$

$$30418 = 2 \cdot 14429 - 307$$

$$14429 = 47 \cdot 307$$

$$\therefore \gcd = 307$$

$$\therefore 14429 = 47 \cdot 307$$

8. Use Kraitchik's method to factor 20437.

$$\sqrt{20437} = 142.9$$

$$143^2 - 20437 = 12 = 2^2 \cdot 3 \quad [1]$$

$$144^2 - 20937 = 219 = 13 \cdot 23 \quad [2]$$

$$145^2 - " = 588 = 2^2 \cdot 5^2 \cdot 7^2 \quad [3]$$

$$146^2 - " = 871 = 3 \cdot 293$$

$$147^2 - " = 1172 = 3^4 \cdot 163$$

$$148^2 - " = 1462$$

[Note: several of the trial-value factorizations above (146, 147, 148) look hard to verify exactly from the handwriting; transcribed as written.]

From [1], [3], $(143 \cdot 145)^2 \equiv (2^3 \cdot 3 \cdot 7)^2 \pmod{20437}$

$$(20735)^2 \equiv (84)^2 \pmod{20437}$$

$$\gcd(20735 - 84, 20437) = \gcd(20651, 20937)$$

$$20651 = 1 \cdot 20437 + 214$$

$$20437 = 95 \cdot 214 + 107$$

$$214 = 2 \cdot 107$$

$$\therefore \gcd = 107 \text{ (a prime)}$$

$$\gcd(20735 + 84, 20437) = \gcd(20819, 20437)$$

$$\therefore 20819 = 1 \cdot 20437 + 382$$

$$20437 = 53 \cdot 382 + 191$$

$$382 = 2 \cdot 191$$

$$\therefore \gcd = 191 \text{ (a prime)}$$

$$\therefore 20437 = 107 \cdot 191$$

5.3 The Little Theorem

1. Use Fermat's Theorem to verify $17 \mid (11^{104} + 1)$.

Since $17 \nmid 11$, $11^{16} \equiv 1 \pmod{17}$ (Fermat's Thm).

$$\therefore (11^{16})^6 \equiv 11^{96} \equiv 1 \pmod{17}$$

But $121 = 11^2$ and $7 \cdot 17 = 119 = 121 - 2$.

$$\therefore 11^2 \equiv 2 \pmod{17}$$

$$\therefore 11^6 \equiv 2^3 \equiv 16 \pmod{17}$$

$$\therefore 11^{96} \cdot 11^8 \equiv 16 \pmod{17}$$

$$11^{104} \equiv 16 \pmod{17}$$

But $16 \equiv -1 \pmod{17}$

$$\therefore 11^{104} \equiv -1 \pmod{17} \Rightarrow 17 \mid 11^{104} + 1.$$

- 2.(a). If $\gcd(a, 35) = 1$, show $a^{12} \equiv 1 \pmod{35}$.

Since $35 = 5 \cdot 7$, then $\gcd(a, 7) = 1$, $\gcd(a, 5) = 1$.

$\therefore$ By Fermat's Theorem,

$$a^6 \equiv 1 \pmod{7} \text{ and } a^4 \equiv 1 \pmod{5}$$

$$\therefore a^{12} = a^6 \cdot a^6 \equiv 1 \pmod{7}, \quad (a^4)^3 = a^{12} \equiv 1^3 \equiv 1 \pmod{5}$$

Since $\gcd(5, 7) = 1$, by corollary 2, section 2.2,

$$35 \mid a^{12} - 1 \Rightarrow a^{12} \equiv 1 \pmod{35}$$

(b). If $\gcd(a, 42) = 1$, show $168 = 3 \cdot 7 \cdot 8$ divides $a^6 - 1$.

Since $42 = 7 \cdot 3 \cdot 2$, $\gcd(a, 7) = \gcd(a, 3) = \gcd(a, 2) = 1$

By Fermat's Th.,

$$a^6 \equiv 1 \pmod{7}, \quad a^2 \equiv 1 \pmod{3}, \quad a \equiv 1 \pmod{2}$$

But $a^2 \equiv 1 \pmod{3} \Rightarrow a^6 = (a^2)^3 \equiv 1 \pmod{3}$, so

$$a^6 \equiv 1 \pmod{3}$$

Also, $a^6 - 1 = (a - 1)(a^5 + a^4 + a^3 + a^2 + a + 1)$

$$= (a - 1)(a^5 + 1) + a^4 + a^2 + a$$

[Note: this factoring/regrouping line is hard to parse exactly from the handwriting; transcribed as written.]

$$= (a - 1)(a + 1)(a^4 - a^3 + a^2 - a + 1)$$

Assume $|a| > 1$. Since a is odd

if $a > 0$, Then $a \geq 3$, so $2 \mid (a + 1)$ and

$$4 \mid a + 1. \quad 8 \mid a^2 - 1$$

if $a < 0$, then $a \leq -3$ so $4 \mid a - 1$ and $2 \mid a + 1$, so

$$8 \mid a^2 - 1$$

Since $7 \mid a^6 - 1$, $3 \mid a^6 - 1$, and $8 \mid a^6 - 1$, and $3, 7, 8$ are relatively prime, Then

$$3 \cdot 7 \cdot 8 = 168 \mid a^6 - 1$$

(c). If $\gcd(a, 133) = 1$, show $133 \mid (b^{18} - 6^{18})$.

[Note: variable seems to switch between a and b here — likely should be $\gcd(a, 133) = 1$, show $133 \mid (a^{18} - 6^{18})$; transcribed as written.]

$133 = 7 \cdot 19$. $\gcd(a, 19) = \gcd(6, 19) = 1$ By Fermat's Th.,

$$a^{18} \equiv 1 \pmod{19}, \quad 6^{18} \equiv 1 \pmod{19}$$

$$\therefore a^{18} - 6^{18} \equiv 1 - 1 = 0 \pmod{19}, \quad \therefore 19 \mid a^{18} - 6^{18}$$

Also since $\gcd(a, 7) = \gcd(6, 7) = 1$, by Fermat's Th.,

$$a^6 \equiv 1 \pmod{7}, \quad 6^6 \equiv 1 \pmod{7}$$

$$\therefore a^{18} - 6^{18} \equiv 0 \pmod{7}, \therefore 7 \mid a^{18} - 6^{18}$$

Since $a^{18} - 6^{18} = (a^6)^3 - (6^6)^3$, Then $7 \mid a^{18} - 6^{18}$

$$\therefore 7 \cdot 19 = 133 \mid a^{18} - 6^{18}$$

3. From Fermat's Th, show for any integer $n \geq 0$, $13 \mid 11^{12n+6} + 1$.

Since $13 \nmid 11$, $11^{12} \equiv 1 \pmod{13}$ by Fermat's Th.

$$\therefore (11^{12})^n \equiv 1^n \equiv 1 \pmod{13}$$

But $11^2 = 121$ and $9 \cdot 13 = 117$.

$$\therefore 11^2 \equiv 4 \pmod{13}$$

$$11^6 \equiv 4^3 \equiv 64 \pmod{13} \equiv -1 \cdot 13 - 3.$$

[Note: this final reduction line is dense/abbreviated; transcribed best-effort, but the conclusion below ($11^6 \equiv -1$) is consistent with $64 \equiv -1 \pmod{13}$.]

$$11^{12n} \cdot 11^6 \equiv 1 \cdot (-1) \pmod{13}, \text{ or } 11^{12n+6} \equiv -1 \pmod{13}$$

$$\therefore 13 \mid 11^{12n+6} + 1$$

4. Derive each congruence.

(a). $a^{21} \equiv a \pmod{15}$ for all a .

$$a^5 \equiv a \pmod{5} \text{ by Fermat's Th.}$$

$$\therefore (a^5)^4 \equiv a^4 \pmod{5}, \text{ or } a^{20} \equiv a^4 \pmod{5}$$

$$\therefore a^{21} \equiv a^5 \equiv a \pmod{5}$$

Also $a^3 \equiv a \pmod{3}$, $\therefore (a^3)^7 \equiv a^7 \pmod{3}$, or $a^{21} \equiv a^7 \pmod{3}$ and $(a^3)^2 \equiv a^2 \pmod{3}$, or $a^6 \equiv a^2 \pmod{3}$

$$\therefore a^7 \equiv a^3 \equiv a \pmod{3}, \therefore a^{21} \equiv a \pmod{3}$$

$$\therefore 5 \mid a^{21} - a \text{ and } 3 \mid a^{21} - a, \therefore 15 \mid a^{21} - a,$$

$$\therefore a^{21} \equiv a \pmod{15}$$

(b). $a^9 \equiv a \pmod{42}$ for all a .

$42 = 7 \cdot 3 \cdot 2$. By Fermats Th, $a^7 \equiv a \pmod{7}$ Also $a^3 \equiv a \pmod{3}$, so $a^9 \equiv a^3 \pmod{3}$

$$\therefore a^9 \equiv a \pmod{3}$$

Also, $a^2 \equiv a \pmod{2}$, $\therefore a^9 \equiv a^2 \pmod{2}$

$$(a^2)^4 \equiv a^4 \pmod{2}, \therefore a^9 \equiv a^4 \pmod{2}$$

$$a^4 \equiv a^2 \pmod{2}, \therefore a^9 \equiv a \pmod{2}$$

$$\therefore 7 \mid a^9 - a, 3 \mid a^9 - a, 2 \mid a^9 - a, a^9 \equiv a \pmod{7 \cdot 3 \cdot 2}$$

(c). $a^{13} \equiv a \pmod{3 \cdot 7 \cdot 13}$ for all a .

By Fermats Th, $a^{13} \equiv a \pmod{13}$

Also, $a^7 \equiv a \pmod{7}$, $\therefore a^{13} \equiv a \cdot a^6 \equiv a^7 \pmod{7}$

$$\text{so } a^{13} \equiv a^7 \equiv a \pmod{7}$$

Also, $a^3 \equiv a \pmod{3}$ and; $a^{13} = a \cdot a^{12} \equiv a \cdot a^4 \pmod{3}$, or

$$a^{13} \equiv a \cdot a \cdot a^2 = a^4 \equiv a \cdot a = a^2 \equiv a \pmod{3}$$

$$\therefore 3 \mid a^{13} - a, 7 \mid a^{13} - a, \text{ and } 13 \mid a^{13} - a.$$

$$\therefore a^{13} \equiv a \pmod{3 \cdot 7 \cdot 13} \text{ by corollary 2, sec. 2.2}$$

(d). $a^9 \equiv a \pmod{30}$ for all a .

$30 = 5 \cdot 3 \cdot 2$. Using Fermats Th,

$$a^5 \equiv a \pmod{5}, \quad a^9 = (a^5)^a \equiv a \cdot a^5 \cdot a^3 = a^9 \equiv a \pmod{5}$$

$$\therefore a^9 \equiv a \pmod{3} : a^3 \equiv a \pmod{3}$$

$$a^9 = (a^3)^3 \equiv a^3 \equiv a \pmod{3}$$

$$a^2 \equiv a \pmod{2} : (a^2)^4 \equiv a^4 \pmod{2}, \quad a^8 \equiv a^4 \pmod{2}$$

$$a^9 = a \cdot a^8 \equiv a \cdot a^4 = a^5 \equiv a \pmod{2}$$

$$\therefore 5 \mid a^9 - a, 3 \mid a^9 - a, \text{ and } 2 \mid a^9 - a, \text{ so}$$

$$a^9 \equiv a \pmod{5 \cdot 3 \cdot 2}$$

5. If $\gcd(a, 30) = 1$, show that 60 divides $a^4 + 59$.

$$\gcd(a, 30) = 1 \Rightarrow \gcd(a, 2) = \gcd(a, 3) = \gcd(a, 5) = 1$$

Also, $\gcd(c, 4) = \gcd(a, 5) = 1$.

[Note: this line introduces "c" where "a" is presumably meant; transcribed as written.]

$60 = 2^2 \cdot 3 \cdot 5$. $60 \mid a^4 + 59$ is the same as

$$a^4 \equiv -59 \pmod{60}, \text{ or } a^4 \equiv 1 \pmod{60}$$

$$\gcd(a, 5) = 1 \Rightarrow a^4 \equiv 1 \pmod{5} \text{ by Fermat's Th.}$$

$$\gcd(a, 3) = 1 \Rightarrow a^2 \equiv 1 \pmod{3}, \therefore a^4 \equiv 1 \pmod{3}$$

$$\gcd(a, 2) = 1 \Rightarrow a \equiv 1 \pmod{2}, \therefore a^2 \equiv 1 \pmod{2}$$

$$a^4 = (a^2)^2 = a \cdot a - a^2 + 1 \equiv a \pmod{2}$$

[Note: this line's algebra is hard to follow exactly from the handwriting; transcribed best-effort.]

$$5 \mid a^4 - 1, 3 \mid a^4 - 1, 4 \mid a^4 - 1, \text{ and}$$

$$\gcd(5, 3) = \gcd(3, 4) = \gcd(4, 5) = 1$$

By corollary 2, sec. 2.2, $60 \mid a^4 - 1$.

$$\therefore a^4 \equiv 1 \pmod{60}, a^4 \equiv -60 \equiv -59 \pmod{60}$$

$$\therefore 60 \mid a^4 + 59$$

6.(a). Find the units digit of 3^{100} using Fermat's Th.

We need something mod 10. $10 = 5 \cdot 2$ By Fermat's Th, $3^4 \equiv 1 \pmod{5}$

$$\therefore (3^4)^{25} \equiv 1 \pmod{5}, 3^{100} \equiv 1 \pmod{5}$$

Also, $3 \equiv 1 \pmod{2}, \therefore 3^{100} \equiv 1 \pmod{2}$

$$\therefore 5 \mid 3^{100} - 1 \text{ and } 2 \mid 3^{100} - 1 \text{ by corollary 2, sec. 2.2}$$

$$\therefore 3^{100} \equiv 1 \pmod{10}$$

$\therefore$ units digit of 3^{100} is 1.

(b). For any integer a , verify that a^5 and a have same units digit.

By Fermat's Th, $a^5 \equiv a \pmod{5}$

Also, $a \equiv a \pmod{2}, \therefore a^5 \equiv a^5 \equiv a \pmod{2},$

[Note: this line is somewhat circular/abbreviated in the original; the intended argument is presumably $a^5 - a \equiv 0 \pmod{2}$ via $a(a^4 - 1) \equiv 0$, since either a even or $a^4 - 1$ even when a odd.]

so $a^5 - a \equiv a \cdot a - a \equiv a \pmod{2}$

$$\therefore 5 \mid a^5 - a \text{ and } 2 \mid a^5 - a, \therefore 10 \mid a^5 - a$$

$$\therefore a^5 \equiv a \pmod{10}, \text{ let } 0 \leq r < 10$$

$$\therefore a^5 - r \equiv a - r \pmod{10}$$

$$\therefore a^5 - r \equiv 0 \pmod{10} \Leftrightarrow a - r \equiv 0 \pmod{10}$$

$\therefore$ units digit is the same.

7. If $7 \nmid a$, prove either $7 \mid a^3 + 1$ or $7 \mid a^3 - 1$.

Pf. By Fermats Th, $a^6 \equiv 1 \pmod{7}$.

$$\therefore 7 \mid a^6 - 1. \text{ But } a^6 - 1 = (a^3 + 1)(a^3 - 1)$$

Suppose $7 \nmid a^3 + 1$. $\gcd(7, a^3 + 1) = 1$, and so $7 \mid a^3 - 1$ by Euclid's lemma.

8. Prove

$$1815^{1970} + 1986^{2061} \equiv 0 \pmod{7}$$

Pf. $1835 = 7 \cdot 262 + 1$, $\therefore 1835 \equiv 1 \pmod{7}$

[Note: the number is written as "1815" in the problem statement but "1835" in the proof; transcribed as written, likely a copy slip in the original.]

$$\therefore 1835^{1970} \equiv 1 \pmod{7}$$

$$1986 = 7 \cdot 283 + 5 \quad \therefore 1986 \equiv 5 \pmod{7}$$

Note also $5^3 \equiv (125) \equiv (126 - 1)$ and

$$126 = 7 \cdot 18 \quad \therefore 5^3 \equiv -1 \pmod{7}$$

$$2061 = 3 \cdot 687 \quad \therefore 5^{2061} = (5^3)^{687} \equiv (-1)^{687} \equiv -1 \pmod{7}$$

$$\therefore 1986^{2061} \equiv -1^{687} \equiv -1 \pmod{7}$$

$$\therefore 1835^{1970} + 1986^{2061} \equiv 1 + (-1) \equiv 0 \pmod{7}$$

9.(a) Let ρ be prime, $\gcd(a, \rho) = 1$. Use Fermats Th. to verify that $x \equiv a^{\rho-2} \pmod{\rho}$ is a solution to $ax \equiv b \pmod{\rho}$.

Pf. $ax \equiv b \pmod{\rho} \Rightarrow ax \cdot a^{\rho-2} \equiv b \cdot a^{\rho-2} \pmod{\rho}$

$$\Rightarrow x \cdot a^{\rho-1} \equiv b \cdot a^{\rho-2} \pmod{\rho}$$

But $a^{\rho-1} \equiv 1 \pmod{\rho}$ by Fermats Th.

$$\therefore x \cdot a^{\rho-1} \equiv x \pmod{\rho}$$

$$\therefore x \equiv x \cdot a^{\rho-1} \equiv b \cdot a^{\rho-2} \pmod{\rho}$$

$$\therefore ax \equiv b \Rightarrow x \equiv b \cdot a^{\rho-2} \pmod{\rho}$$

(b) $2x \equiv 1 \pmod{31} \Rightarrow x \equiv 2^{31-2} \equiv 2^{29} \pmod{31}$

But $2^5 = 32 \equiv 3 + 1$, $\therefore 2^5 \equiv 1 \pmod{31}$

$$(2^5)^5 \cdot 2^4 \equiv 1 \cdot 4 \equiv 1 \pmod{31}$$

$$2^{29} \equiv 2^4 \equiv 2^4 \equiv 1 \pmod{31}$$

$$\therefore 2x \equiv 1 \pmod{31} \Rightarrow x \equiv 16 \pmod{31}$$

$$6x \equiv 5 \pmod{11} \Rightarrow x \equiv 5 \cdot 6^{11-2} \equiv 5 \cdot 6^9 \pmod{11}$$

$$\text{But } 6^2 = 36 \equiv 3 \cdot 3, \therefore 6^2 \equiv 3 \pmod{11}$$

$$(6^2)^4 = 6^8 \equiv 3^4 \cdot 6 \equiv 4 \cdot 6 \pmod{11}$$

$$3^4 \equiv 81 \equiv 4 \pmod{11}, \therefore 3^4 \equiv 4 \pmod{11}$$

$$\therefore x \equiv 5 \cdot 6^9 = 5 \cdot (4) \cdot 6 \equiv (20)(6) \equiv 10 \pmod{11}$$

$$\therefore x \equiv 10 \pmod{11}$$

$$3x \equiv 17 \pmod{29} \Rightarrow x \equiv 17 \cdot 3^{29-2} \equiv 17 \cdot 3^{27} \pmod{29}$$

$$3^3 = 27 \equiv -2 \pmod{29}$$

$$3^{27} = (3^3)^9 \equiv (-2)^9 \pmod{29}$$

$$2^9 = 2 \cdot 2^4 = 2 \cdot 16 = 32 \equiv 3 \pmod{29}, \therefore (-2)^9 \equiv -(2^9) \equiv -3 \equiv -48 \pmod{29}$$

[Note: this line's exponent manipulation is dense and hard to fully verify from the handwriting; transcribed best-effort.]

$$\therefore (-2)^9 \equiv -48 \equiv 4 + 25 \equiv? \pmod{29}$$

$$\therefore x \equiv 17 \cdot 3^{27} \equiv 17 \cdot (-3) \equiv -51 \equiv +25 \equiv 25 \pmod{29}$$

$$\therefore x \equiv 25 \pmod{29}$$

10. Assume $\rho \nmid a$, $\rho \nmid b$, ρ prime.

(a) If $a^\rho \equiv b^\rho \pmod{\rho}$, Then $a \equiv b \pmod{\rho}$.

Pf. $a^\rho \equiv a \pmod{\rho}$, $b^\rho \equiv b \pmod{\rho}$ for any integers a, b .

$$\therefore a \equiv a^\rho \equiv b^\rho \equiv b \pmod{\rho}.$$

(b) If $a^\rho \equiv b^\rho \pmod{\rho}$, Then $a^\rho \equiv b^\rho \pmod{\rho^2}$.

By (a), $a = b + \rho k$, some K .

$$\begin{aligned} \therefore a^\rho - b^\rho &= (b + \rho k)^\rho - b^\rho \\ &= b^\rho + \sum_{i=1}^{\rho} \binom{\rho}{i} b^{\rho-i} (\rho k)^i - b^\rho \\ &= \sum_{i=1}^{\rho} \frac{\rho!}{i!(\rho-i)!} b^{\rho-i} (\rho k)^i \end{aligned}$$

Clearly, when $i \geq 2$, each term is divisible by ρ^2 , since $(\rho k)^i$ has at least ρ^2 in the term.

$$\begin{aligned} \therefore \text{Look at } i = 1 \text{ term: } & \frac{\rho!}{1!(\rho-1)!} b^{\rho-1} \rho k \\ & = \rho \cdot b^{\rho-1} \rho k = \rho^2 k \cdot b^{\rho-1} \text{ so this is} \end{aligned}$$

also divisible by ρ^2 .

$$\therefore a^\rho - b^\rho \text{ is divisible by } \rho^2.$$

11. Use Fermats Th. to prove that if ρ is an odd prime,

$$(a) \ 1^{\rho-1} + 2^{\rho-1} + \cdots + (\rho-1)^{\rho-1} \equiv -1 \pmod{\rho}$$

Pf. Since ρ is prime ≥ 3 , Then $\rho \nmid a$ if $a < \rho$.

$$\therefore \text{By Fermats Th, } a^{\rho-1} \equiv 1 \pmod{\rho}.$$

There are $\rho-1$ terms in $1^{\rho-1} + 2^{\rho-1} + \cdots + (\rho-1)^{\rho-1}$

$$\therefore 1^{\rho-1} + 2^{\rho-1} + \cdots + (\rho-1)^{\rho-1} \equiv (\rho-1) \cdot 1 \pmod{\rho},$$

$(\rho-1) \equiv -1 \pmod{\rho}$. Since $\rho \geq 3 \pmod{\rho}$,

$$1^{\rho-1} + 2^{\rho-1} + \cdots + (\rho-1)^{\rho-1} \equiv -1 \pmod{\rho}$$

Note: it's true even if $\rho = 2$

$$(b) \ 1^\rho + 2^\rho + \cdots + (\rho-1)^\rho \equiv 0 \pmod{\rho}$$

Pf. By corollary to Fermats Th., $a^\rho \equiv a \pmod{\rho}$.

$$\therefore 1^\rho + 2^\rho + \cdots + (\rho-1)^\rho \equiv 1 + 2 + \cdots + (\rho-1) \pmod{\rho}$$

Since $1 + 2 + \cdots + n = n(n+1)/2$,

$$1 + 2 + \cdots + (\rho-1) = (\rho-1)(\rho-1+1)/2 = \rho(\rho-1)/2$$

As ρ is an odd prime, $\rho-1$ is even, so $\rho-1 = 2k$, some K .

$$\therefore 1 + 2 + \cdots + (\rho-1) = \rho k, \text{ some } k.$$

$$\therefore 1^\rho + 2^\rho + \cdots + (\rho-1)^\rho \equiv \rho k \equiv 0 \pmod{\rho}$$

12. Prove that if ρ is an odd prime, K an integer s.t. $1 \leq K \leq \rho-1$, Then $\binom{\rho-1}{K} \equiv (-1)^K \pmod{\rho}$.

$$\text{Pf. } \binom{\rho-1}{K} = \frac{(\rho-1)!}{K!(\rho-K-1)!} = \frac{(\rho-1)(\rho-2)\cdots(\rho-K)}{K!}$$

$$\therefore K! \binom{\rho-1}{K} = (\rho-1)(\rho-2)\cdots(\rho-K)$$

But $\rho - j \equiv -j \pmod{\rho}$.

$$\therefore (\rho - 1)(\rho - 2) \cdots (\rho - K) \equiv (-1)(-2) \cdots (-K) \pmod{\rho}$$

$$(-1)(-2) \cdots (-K) = (-1)^K K!$$

$$\therefore K! \binom{\rho - 1}{K} \equiv (-1)^K K! \pmod{\rho}$$

Since $\rho - 1 \geq K$, $\rho > K$, $\therefore \rho \nmid K!$ ($K! = 1, 2, 3, \dots, K$)

$\gcd(\rho, K!) = 1$, $1 \leq K \leq \rho - 1$. By Corollary 1, sec. 4.2,

$$\binom{\rho - 1}{K} \equiv (-1)^K \pmod{\rho}.$$

13. If ρ, q are distinct odd primes s.t. $\rho - 1 \mid q - 1$, and if $\gcd(a, \rho q) = 1$, show $a^{q-1} \equiv 1 \pmod{\rho q}$.

Pf. $\gcd(a, \rho q) = 1 \Rightarrow \gcd(a, \rho) = \gcd(a, q) = 1$, since q, ρ are distinct primes.

$$\therefore a^{\rho-1} \equiv 1 \pmod{\rho} \text{ and } a^{q-1} \equiv 1 \pmod{q}$$

Since $\rho - 1 \mid q - 1$, then $q - 1 = K(\rho - 1)$, some k .

$$\therefore a^{\rho-1} \equiv 1 \pmod{\rho} \Rightarrow (a^{\rho-1})^K \equiv 1 \pmod{\rho}$$

$$\Rightarrow a^{K(\rho-1)} \equiv 1 \pmod{\rho} \Rightarrow a^{q-1} \equiv 1 \pmod{\rho}$$

$$\therefore \rho \mid a^{q-1} - 1 \text{ and } q \mid a^{q-1} - 1$$

$$\therefore \rho q \mid a^{q-1} - 1 \text{ by corollary 2, sec. 2.2}$$

$$\therefore a^{q-1} \equiv 1 \pmod{\rho q}$$

14. If ρ, q are distinct primes, prove

$$\rho^{q-1} + q^{\rho-1} \equiv 1 \pmod{\rho q}$$

Pf. By Fermats Th., $\rho^{q-1} \equiv 1 \pmod{q}$. Clearly, $q \mid q^{\rho-1}$, so $q^{\rho-1} \equiv 0 \pmod{q}$.

$$\therefore \rho^{q-1} + q^{\rho-1} \equiv 1 \pmod{q}$$

Similarly, $\rho \mid \rho^{q-1}$ so $\rho^{q-1} \equiv 0 \pmod{\rho}$, and $q^{\rho-1} \equiv 1 \pmod{\rho}$ by Fermats Th.

$$\therefore \rho^{q-1} + q^{\rho-1} \equiv 1 \pmod{\rho}.$$

$$\therefore q \mid (\rho^{q-1} + q^{\rho-1} - 1) \text{ and } \rho \mid (\rho^{q-1} + q^{\rho-1} - 1),$$

and $\gcd(\rho, q) = 1$. $\therefore$ By corollary 2, sec. 2.2,

$$\begin{aligned}\rho q & \mid (\rho^{q-1} + q^{\rho-1} - 1) \\ \therefore \rho^{q-1} + q^{\rho-1} & \equiv 1 \pmod{\rho q}\end{aligned}$$

15. Establish the following.

(a) If $M_\rho = 2^\rho - 1$ is composite, ρ prime, then M_ρ is pseudoprime.

Pf. Must show $2^{M_\rho} \equiv 2 \pmod{M_\rho}$.

Proof is much like proof to Th. 5.2. Since $2^\rho - 1$ is composite, $\rho \neq 2$, so $\rho \nmid 2$. By Fermats Th. (the corollary), $2^\rho \equiv 2 \pmod{\rho}$.

$$\therefore 2^\rho - 2 = K\rho, \text{ some } K.$$

$$\therefore M_\rho - 1 = 2^\rho - 1 - 1 = 2^{\rho-2} = 2^{K\rho}$$

[Note: this line is hard to follow exactly from the handwriting; the intended relation is likely $M_\rho - 1 = 2^\rho - 2 = K\rho$.]

$$\begin{aligned}2^{M_\rho-1} - 1 &= 2^{K\rho} - 1 = (2^\rho - 1)(2^{\rho(K-1)} + 2^{\rho(K-2)} + \dots + 2^\rho + 1) \\ &= M_\rho(2^{\rho(K-1)} + 2^{\rho(K-2)} + \dots + 2^\rho + 1) \\ &\equiv 0 \pmod{M_\rho} \\ \therefore 2^{M_\rho-1} &\equiv 1 \pmod{M_\rho} \\ 2 \cdot 2^{M_\rho-1} &\equiv 2 \pmod{M_\rho} \\ \therefore 2^{M_\rho} &\equiv 2 \pmod{M_\rho}\end{aligned}$$

By def, M_ρ is a pseudoprime.

(b) Every composite number $F_n = 2^{2^n} + 1$ is a pseudoprime ($n = 0, 1, 2, \dots$)

Pf. Since $n+1 \leq 2^n$ for $n \geq 0$, Then

$$2^{n+1} \leq 2^{2^n}, \text{ so } 2^{n+1} \mid 2^{2^n}$$

$\therefore$ By problem #21, sec. 2.2,

$$(2^{2^{n+1}} - 1) \mid (2^{2^{2^n}} - 1), \text{ or } (2^{2^{n+1}} - 1) \mid (2^{F_n-1} - 1) \quad [1]$$

$$\text{But } 2^{2^{n+1}} = 2^{2 \cdot 2^n} = (2^{2^n})^2 = (2^{2^n})^2$$

$$\therefore 2^{2^{n+1}} - 1 = (2^{2^n})^2 - 1 = (2^{2^n} + 1)(2^{2^n} - 1)$$

$$= (F_n)(2^{2^n} - 1) \quad [2]$$

$$\therefore \text{From [2], } F_n \mid (2^{2^{n+1}} - 1) \quad [3]$$

$$\begin{aligned}\therefore \text{From [1] and [3], } F_n &| (2^{F_n-1} - 1) \\ \therefore F_n &| 2(2^{F_n-1} - 1) = 2^{F_n} - 2 \\ \therefore F_n &\text{ is pseudoprime (whenever } F_n \text{ is composite).}\end{aligned}$$

16. Confirm the following are absolute pseudoprimes.

- (a). $1105 = 5 \cdot 13 \cdot 17$. Let a be any integer.
If $1105 \nmid a$, Then $5 \nmid a$, $13 \nmid a$, $17 \nmid a$

$\therefore$ By Fermats Th.,

$$a^4 \equiv 1 \pmod{5}, \quad a^{12} \equiv 1 \pmod{13}, \quad a^{16} \equiv 1 \pmod{17}$$

$$\therefore a^{1104} = (a^4)^{276} \equiv 1 \pmod{5}$$

$$a^{1104} = (a^{12})^{92} \equiv 1 \pmod{13}$$

$$a^{1104} = (a^{16})^{69} \equiv 1 \pmod{17}$$

[Note: this last modulus should likely read $\pmod{17}$, not $\pmod{7}$, to be consistent with the rest of the proof; transcribed as written.]

$$\therefore a^{1104} \equiv 1 \pmod{5, 13, 17} \text{ when } 1105 \nmid a$$

$$\therefore a^{1105} \equiv a \pmod{1105} \text{ when } 1105 \nmid a.$$

But when $1105 \mid a$, clearly $1105 \mid a^{1105} - a$

$$\therefore a^{1105} \equiv a \pmod{1105} \text{ for all } a.$$

- (b). $2821 = 7 \cdot 13 \cdot 31$ Let a be any integer.

If $2821 \nmid a$, Then $7 \nmid a$, $13 \nmid a$, $31 \nmid a$

$$\therefore a^6 \equiv 1 \pmod{7}, \quad a^{12} \equiv 1 \pmod{13}, \quad a^{30} \equiv 1 \pmod{31}$$

$$a^{2820} = (a^6)^{470} \equiv 1 \pmod{7}$$

$$a^{2820} = (a^{12})^{235} \equiv 1 \pmod{13}$$

$$a^{2820} = (a^{30})^{94} \equiv 1 \pmod{31}$$

$$\therefore a^{2820} \equiv 1 \pmod{7 \cdot 13 \cdot 31} \text{ when } 2821 \nmid a$$

$$\therefore a^{2821} \equiv a \pmod{2821} \text{ when } 2821 \nmid a$$

But when $2821 \mid a$, clearly $a^{2821} \equiv a \pmod{2821}$

$$\therefore \text{For all } a, \quad a^{2821} \equiv a \pmod{2821}$$

(c). $2465 = 5 \cdot 17 \cdot 29$ Let a be any integer.

If $2465 \nmid a$, Then $5 \nmid a$, $17 \nmid a$, $29 \nmid a$

$$\therefore a^4 \equiv 1 \pmod{5}, a^{16} \equiv 1 \pmod{17}, a^{28} \equiv 1 \pmod{29}$$

$$a^{2464} = (a^4)^{616} \equiv 1 \pmod{5}$$

$$a^{2464} = (a^{16})^{154} \equiv 1 \pmod{17}$$

$$a^{2464} = (a^{28})^{88} \equiv 1 \pmod{29}$$

$$\therefore a^{2464} \equiv 1 \pmod{5 \cdot 17 \cdot 29} \text{ when } 2465 \nmid a$$

But when $2465 \mid a$, clearly $a^{2465} \equiv a \pmod{2465}$

$$\therefore \text{For all } a, a^{2465} \equiv a \pmod{2465}$$

17. Show that the smallest pseudoprime 341 is not an absolute prime by showing

$$11^{340} \not\equiv 11 \pmod{341}.$$

$341 = 11 \cdot 31$. Suppose $11^{340} \equiv 11 \pmod{341}$. Then $11^{340} \equiv 11 \pmod{31}$. But $11^{12} \equiv -3 \pmod{31}$

$$\therefore 11^{2 \cdot 170} \equiv (-3)^{170} \pmod{31}$$

But $(-3)^7 = -2187$ and $-2187 - 31 \cdot (-71) = -1985$

[Note: this line's arithmetic is dense and hard to verify exactly from the handwriting; transcribed best-effort.]

$$(-3)^7 \equiv 2 \pmod{31}$$

$$(-3)^{14} = (-3)^{7 \cdot 2} \equiv 2^2 \equiv 4 \pmod{31}$$

But $2^{10} = 1024 \equiv 1 \pmod{31}$ ($31 \cdot 33 = 1023$)

$$2^8 = 256 \equiv 8 \pmod{31}, 2^8 \equiv 8 \pmod{31}$$

$$(-3)^{162} \equiv 8 \pmod{31}$$

$$(-3)^4 = 81 \equiv 2 \cdot 31 + 19 \equiv 19 \pmod{31}$$

$$(-3)^8 = 19^2 = 361 \equiv 3 \cdot 31 + 20 \equiv 20 \pmod{31}$$

$$\therefore (-3)^{162} \equiv (-3)^8 \equiv 8 \cdot 20 \equiv 160 \equiv 5 \cdot 31 + 5 \equiv 5 \pmod{31}$$

$$\therefore 11^{340} \equiv (-3)^{170} \equiv 5 \pmod{31}$$

$$\therefore 11^{340} \equiv 5 \cdot 11 = 55 \equiv 31 \cdot 24 + 24 \equiv 24 \pmod{31}$$

[Note: this line's reasoning (substituting $5 \cdot 11$) is unclear/abbreviated from the handwriting; transcribed as written.]

$$\therefore 11^{340} \equiv 24 \pmod{31}, \text{ so } 11^{340} \not\equiv 11 \pmod{31}$$

$\therefore$ Contradiction reached, so $11^{340} \not\equiv 11 \pmod{341}$

Note: assuming $11^{340} \equiv 11 \pmod{341} \Rightarrow 11^{340} \equiv 11 \pmod{11}$ But since $11^{340} \equiv 0 \pmod{11}$ and $11^{340} \not\equiv 11 \pmod{11}$, and $\therefore 11^{340} \not\equiv 11 \pmod{11}$. That's why attacked problem using mod 31.

18.(a) When $n = 2\rho$, ρ an odd prime, prove $a^{n-1} \equiv a \pmod{n}$ for any integer a .

Pf. $a^{\rho-1} \equiv 1 \pmod{\rho}$ (Fermat's Th.), and $a^\rho \equiv a \pmod{\rho}$.

$$\therefore a^\rho \cdot a^{\rho-1} = a^{2\rho-1} \equiv a \cdot 1 \equiv a \pmod{\rho}$$

$$\therefore a^{n-1} \equiv a \pmod{\rho}, \text{ so } \rho \mid a^{n-1} - a.$$

Now, if a is even, let $a = 2x$, some x .

$$a^{n-1} - a = (2x)^{n-1} - 2x = 2^{n-1}x^{n-1} - 2x = 2(2^{n-2}x^{n-1} - x)$$

Since $n \geq 2$, Then $2 \mid a^{n-1} - a$. Suppose a is odd. Since 2ρ is even, $n-1$ is odd. $\therefore a^{n-1}$ is odd, and $\therefore a^{n-1} - a$ is even, so $2 \mid a^{n-1} - a$.

$\therefore$ Both 2 and ρ divide $a^{n-1} - a$. Since $\gcd(2, \rho) = 1$, Then $2\rho \mid a^{n-1} - a$, so $a^{n-1} \equiv a \pmod{n}$.

(b) for $n = 195 = 3 \cdot 5 \cdot 13$, verify $a^{n-2} \equiv a \pmod{n}$ for any integer a .

Pf. If $195 \mid a$, then clearly $195 \mid a^{n-2} - a$. Assume $195 \nmid a$.

$$\therefore 3 \nmid a, 5 \nmid a, 13 \nmid a$$

$$\therefore \text{By Fermats Th., } a^2 \equiv 1 \pmod{3}$$

$$a^4 \equiv 1 \pmod{5}$$

$$a^{12} \equiv 1 \pmod{13}$$

$$a^{192} = a^{2 \cdot 96} \equiv 1 \pmod{3}$$

$$a^{192} = a^{4 \cdot 48} \equiv 1 \pmod{5}$$

$$a^{192} = a^{12 \cdot 16} \equiv 1 \pmod{13}$$

$$\therefore a^{192} \equiv 1 \pmod{3 \cdot 5 \cdot 13}$$

$$a^{195} \equiv a \pmod{n}$$

$$a^{n-2} \equiv a \pmod{n}$$

19. Prove any integer of the form

$$n = (6k+1)(12k+1)(18k+1)$$

is an absolute pseudoprime if all three factors are prime.

Pf. Let $\rho_1 = 6k + 1$, $\rho_2 = 12k + 1$, $\rho_3 = 18k + 1$.

Assume ρ_1, ρ_2, ρ_3 are prime.

$$\begin{aligned} n &= (72k^2 + 18k + 1)(18k + 1) \\ &= 18 \cdot 72k^3 + 72k^2 + 18 \cdot 18k^2 + 18k + 18k + 1 \end{aligned}$$

[Note: this expansion line is dense/abbreviated and hard to verify exactly from the handwriting; transcribed best-effort.]

$\therefore n - 1 = 36Kk[3k + 1]$ is divisible by $\langle (\rho_1 - 1)(\rho_1 - 1), (\rho_2 - 1)(n - 1), (\rho_3 - 1)n - 1 \rangle$

[Note: this divisibility-conditions line is hard to parse exactly from the handwriting; transcribed best-effort.]

Since ρ_1, ρ_2, ρ_3 are distinct primes, and n is square-free (each prime to power 1), Then n is absolute pseudoprime by Th. 5.3.

20. Show that $561 \mid 2^{561} - 2$ and $561 \mid 3^{561} - 3$.

(a). $2^2 \equiv 1 \pmod{3}$, $\therefore (2^2)^{280} \equiv 2^{560} \equiv 1 \pmod{3}$

$$2^{10} \equiv 1 \pmod{11} \text{ (Fermats Th.)}$$

$$\therefore (2^{10})^{56} \equiv 2^{560} \equiv 1 \pmod{11}$$

$$2^{16} \equiv 1 \pmod{17} \text{ (Fermats Th.)}$$

$$\therefore (2^{16})^{35} \equiv 2^{560} \equiv 1 \pmod{17}$$

$$\therefore 2^{560} \equiv 1 \pmod{3 \cdot 11 \cdot 17}, \text{ and } 3 \cdot 11 \cdot 17 = 561$$

$$\therefore 561 \mid 2^{560} - 2, \therefore 561 \mid 2^{561} - 2$$

(b) By Fermats Th., $3^{10} \equiv 1 \pmod{11}$, $3^{16} \equiv 1 \pmod{17}$

$$\therefore 3^{560} \equiv 1 \pmod{11}, 3^{560} \equiv 1 \pmod{17}$$

$$\therefore 11 \mid 3^{560} - 1, 17 \mid 3^{560} - 1. \text{ Clearly } 3 \mid 3^{560} - 1$$

$$\therefore 3 \cdot 11 \cdot 17 \mid 3^{560} - 1 \text{ since } \gcd(3, 11, 17) = 1$$

Alternatively, $3^7 = 2187 = 4 \cdot 561 - 57$

$$\therefore 3^7 \equiv -57 \pmod{561}$$

$$(3^7)^{80} \equiv 3^{560} \equiv (-57)^{80} \pmod{561}$$

But $57^2 = 3249 = 6 \cdot 561 - 117$

$$\therefore 57^2 \equiv (-117) \pmod{561}, \text{ and } (57)^{40} \equiv (-57)^{40}$$

$$\begin{aligned}
 \therefore 3^{560} &\equiv (-117)^{40} \pmod{561} \\
 (-117)^2 &= 13689 = 24 \cdot 561 + 225 \\
 \therefore (-117)^2 &\equiv 225 \pmod{561} \\
 \therefore 3^{560} &\equiv (-117)^{40} \equiv 225^{20} \pmod{561} \\
 225^2 &= 50625 = 90 \cdot 561 + 135 \\
 \therefore 225^2 &\equiv 135 \pmod{561} \\
 3^{560} &\equiv 225^{20} \equiv 135^{10} \pmod{561} \\
 135^2 &= 18225 = 32 \cdot 561 + 273 \\
 \therefore 135^2 &\equiv 273 \pmod{561} \\
 3^{560} &\equiv 135^{10} \equiv 273^5 \pmod{561} \\
 273^2 &= 74529 = (133 \cdot 561 - 84) \\
 273^2 &\equiv -84 \pmod{561} \\
 \therefore (273)^4 &\equiv (-84)^2 = 7056 = 12 \cdot 561 + 524 \\
 3^{560} &= 273^5 = 273^4 \cdot 273 \\
 &\equiv (-324) \cdot 273 \pmod{561}
 \end{aligned}$$

But $-324 \cdot 273 = -88452 = 767 \cdot 561 - 375$

$$\begin{aligned}
 \therefore 3^{560} &\equiv -324 \cdot 273 \equiv -375 \pmod{561} \\
 3^{561} &\equiv 3 \cdot 186 \equiv 558 \pmod{561}
 \end{aligned}$$

[Note: the intermediate steps connecting -375 to 186 are abbreviated/hard to verify exactly from the handwriting; transcribed best-effort, but the final conclusion below ($3^{561} \equiv 3 \pmod{561}$) is the expected correct result.]

$$\begin{aligned}
 3^{561} - 3 &= 558 - 3 = 0 \pmod{561}, \\
 3^{561} &\equiv 3 \pmod{561}
 \end{aligned}$$

21. Show $2222^{5555} + 5555^{2222} \equiv 0 \pmod{7}$.

$$1111 = 159 \cdot 7 + 2 \quad \therefore 1111 \equiv 2 \pmod{7}$$

$$2222 \equiv 4 \pmod{7} \quad \therefore 5555 \equiv -101 - 10 + 14 = 4 \pmod{7}$$

[Note: the arithmetic on " $5555 \equiv -101 - 10 + 14 = 4$ " is dense/hard to follow exactly; transcribed best-effort. The conclusion ($5555 \equiv 4 \pmod{7}$) is correct.]

$$\therefore 2222^{5555} \equiv (4)^{5555} \pmod{7}$$

But $(4)^3 = 16 \equiv 2 \pmod{7}$, $5555 = 3(2777) + 1$

$$\therefore (4)^{5555} = 4^{3(2777)+1} \equiv 2^{2777} \cdot (4) \pmod{7}$$

$$\therefore 2222^{5555} \equiv -2^{2779} \pmod{7}$$

[Note: exponent shown here (2779) is inconsistent with the prior page's 2777; transcribed as written.]

But $2^3 \equiv 1 \pmod{7}$, and $3 \cdot 926 = 2778$

$$(2^3)^{926} \equiv 2^{2778} \equiv 1 \pmod{7}$$

$$\therefore 2222^{5555} \equiv -2^{2778} \cdot 2 \equiv -2 \pmod{7}$$

Now $5555 \equiv 4 \pmod{7} \Rightarrow 5555^{2222} \equiv 4^{2222} \pmod{7}$

$$\therefore 5555^{2222} \equiv 4^{4944} \pmod{7}$$

[Note: exponent "4944" appears where "2222" might be expected from the substitution shown; transcribed as written.]

and $4944 = 1981 \cdot 3 + 1$, and $2^3 \equiv 1 \pmod{7}$

$$\therefore 5555^{2222} \equiv (2^3)^{1981} \cdot 2 \equiv 1 \cdot 2 \equiv 2 \pmod{7}$$

$$\therefore 2222^{5555} + 5555^{2222} \equiv -2 + 2 \equiv 0 \pmod{7}$$

Theorem 5.3 – a more explicit proof

Let n be a composite square-free integer, say, $n = \rho_1 \rho_2 \rho_3 \cdots \rho_r$, each ρ_i distinct.

If $(\rho_i - 1) \mid (n - 1)$ for $i = 1, 2, \dots, r$, Then n is absolute pseudoprime. *Pf.* Suppose initially a is some integer s.t. $\gcd(a, n) = 1$. $\therefore \gcd(a, \rho_i) = 1$ for each i .

Fermats Th. yields $\rho_i \mid a^{\rho_i-1} - 1$.

Since $(\rho_i - 1) \mid (n - 1)$, Then $n - 1 = K(\rho_i - 1)$, some K .

$$\therefore a^{K(\rho_i-1)} - 1 = (a^{\rho_i-1} - 1)(a^{(K-1)(\rho_i-1)} + \cdots + a^{\rho_i-1} + 1)$$

$$\therefore \rho_i \mid a^{n-1} - 1 \text{ for all } i.$$

$$\therefore \rho_i \mid a^n - a \text{ for all } i.$$

$\therefore$ From corollary 2, Th. 2.4 (sec. 2.2),

$$n \mid a^n - a$$

Now if $\gcd(a, n) \neq 1$, then let

$$\gcd(a, n) = \rho_{i_1} \rho_{i_2} \cdots \rho_{i_s}, \text{ where each } \rho_{i_x} \in \{\rho_1, \dots, \rho_r\}$$

Since n is square-free and composed of distinct primes, then ρ_{i_x} are distinct and have exponents of 1.

Let $a' = a/(\rho_{j_1} \cdot \rho_{j_2} \cdots \rho_{j_s})$

$$\therefore a = (\rho_{j_1} \cdot \rho_{j_2} \cdots \rho_{j_x})(a') \quad [1]$$

$\therefore \gcd(a', n) = 1$, and from above,

$$n \mid (a')^n - a$$

$$\therefore n \mid (\rho_{j_1} \cdot \rho_{j_2} \cdots \rho_{j_x})^n (a')^n - (\rho_{j_1} \rho_{j_2} \cdots \rho_{j_x})(a')$$

From [1] above, $(\rho_{j_1} \rho_{j_2} \cdots \rho_{j_x})(a') = a$

$$\therefore n \mid a^n - a \text{ if } \gcd(a, n) \neq 1$$

$$\therefore n \mid a^n - a \text{ for all } a.$$

5.4 Wilson's Theorem

1.(a). Find the remainder when $15!$ is divided by 17.

Since $(17 - 1)! \equiv -1 \pmod{17}$, Then $16! \equiv -1 \pmod{17}$ But $16 \equiv -1 \pmod{17}$.

$$\therefore 16! \equiv 16 \pmod{17}, \gcd(16, 17) = 1.$$

$$\therefore 16!/16 \equiv 16/16 \pmod{17}$$

$$\therefore 15! \equiv 1 \pmod{17}$$

(b) Find the remainder when $2(26!)$ is divided by 29

By Wilson's Th, $28! \equiv -1 \pmod{29}$

$$\therefore 28! \equiv 28 \pmod{29}, \text{ Since } \gcd(28, 29) = 1,$$

$$\therefore 27! \equiv 1 \pmod{29}, 27! \equiv 1 + 29 \pmod{29}$$

$$27! \equiv 30 \pmod{29}, 9 \cdot 26! \equiv 30 \pmod{29}$$

$$\therefore 9 \cdot 26! \equiv 10 \pmod{29}, \text{ since } \gcd(9, 29) = 1$$

$$9 \cdot 26! \equiv 39 \pmod{29}$$

$$3 \cdot 26! \equiv 13 \pmod{29}$$

$$3 \cdot 26! \equiv 13 + 29 = 42 = 3 \cdot 14$$

$$\therefore 26! \equiv 14 \pmod{29}, \therefore 2 \cdot 26! \equiv 28 \pmod{29}$$

2. Determine whether 17 is a prime by deciding whether $16! \equiv -1 \pmod{17}$.

$$4 \cdot 3 \cdot 2 \cdot 1 = 24 \equiv (17 + 7) \equiv 7 \pmod{17}$$

$$5! \equiv 5 \cdot 7 = 35 \equiv 2 \cdot 17 + 1 \equiv 1 \pmod{17}$$

$$6! \equiv 6 \pmod{17}$$

$$7! = 42 = 34 + 8 \equiv 8 \pmod{17}$$

$$8! = 64 = 68 - 4 \equiv -4 \pmod{17}$$

$$9! = -36 = -34 - 2 \equiv -2 \pmod{17}$$

$$10! = -20 = -3 \equiv -3 \pmod{17}$$

$$11! = -33 = -34 + 1 \equiv 1 \pmod{17}$$

$$12! = 12 \equiv 12 = 3 \equiv 3 \pmod{17}$$

$$13! = 5 \cdot 13 = -65 = -68 + 3 \equiv 3 \pmod{17}$$

$$14! = 3 \cdot 14 = 42 = 34 + 8 \equiv 8 \pmod{17}$$

$$15! = 8 \cdot 15 = 120 = 7 \cdot 17 + 1 \equiv 1 \pmod{17}$$

$$16! = 16 = 17 - 1 \equiv -1 \pmod{17}$$

3. Arrange $2, 3, 4, \dots, 21$ in pairs to satisfy $ab \equiv 1 \pmod{23}$

Look for $23k + 1 = 24$, next $23 \cdot 2 + 1 = 47$ (prime), $23 \cdot 5 = 70$

$$4 \cdot 27 + 1 = 73, 5 \cdot 23 + 1 = 116, 6 \cdot 23 + 1 = 138, 7 \cdot 23 + 1 = 162$$

$$8 \cdot 23 + 1 = 185, 9 \cdot 23 + 1 = 208, 10 \cdot 23 + 1 = 231$$

$$11 \cdot 23 + 1 = 254, 12 \cdot 23 + 1 = 277, 13 \cdot 23 + 1 = 300, 14 \cdot 23 + 1 = 323$$

$$2 \cdot 12 = 24 \equiv 1 \pmod{23}$$

$$3 \cdot 8 = 24 \equiv 1$$

$$4 \cdot 6 = 24 \equiv 1$$

$$5 \cdot 14 = 70 \equiv 1$$

$$7 \cdot 10 = 70 \equiv 1$$

$$9 \cdot 18 = 162 \equiv 1$$

$$11 \cdot 21 = 231 \equiv 1$$

$$13 \cdot 71 = 208 \equiv 1$$

[Note: "13 · 71" appears to be a transcription slip for "13 · 16" (matching the pattern of pairs from 2-21); transcribed as written.]

$$15 \cdot 20 = 300 \equiv 1$$

$$12 \cdot 19 = 323 \equiv 1$$

4. Show that $18! \equiv -1 \pmod{437}$

$19 \cdot 23 = 437$, since $19, 23 = 437$

[Note: this line is garbled/abbreviated in the handwriting; the intended statement is presumably that 19 and 23 are the prime factors of 437.]

By Wilson's Th., $18! \equiv -1 \pmod{19}$ Must show $23 \mid 18! + 1$ By Wilson's Th., $22! \equiv -1 \equiv 22 \pmod{23}$

$$\therefore 22!/22 \equiv 22/22 \pmod{23} \quad \gcd(21, 23) = 1$$

$$21! \equiv 1 + 23 \equiv 24 \pmod{23}$$

$$21 \cdot 20! \equiv 6 + 3 \pmod{23} \quad \gcd(3, 23) = 1$$

$$7 \cdot 20! \equiv 8 \pmod{23}$$

$$7 \cdot 20! \equiv 9 + 23 \equiv 8 \pmod{23} \quad \gcd(4, 23) = 1$$

$$1 \cdot 5 \cdot 19! \equiv 2 \pmod{23}$$

$$7 \cdot 5 \cdot 18! \equiv 2 \pmod{23}$$

$$7 \cdot 5 \cdot 19 \cdot 18! \equiv 2 + 23 = 25 \pmod{23}$$

$$\therefore 7 \cdot 19 \cdot 18! \equiv 5 \pmod{23} \quad \gcd(5, 23) = 1$$

$$7 \cdot 19 \cdot 18! \equiv 5 + 23 = 28 \pmod{23}$$

$$7 \cdot 19 \cdot 18! \equiv 5 \cdot 23 = 28 \pmod{23} \quad \gcd(7, 23) = 1$$

$$19 \cdot 18! \equiv 4 \pmod{23}$$

$$19 \cdot 18! \equiv 9 - 23 = -2 \pmod{23} \quad \gcd(19, 23) = 1$$

$$7 \cdot 19 \cdot 18! \equiv 2 \pmod{23}$$

$$\therefore 23 \mid 18! + 1, \text{ and } 19 \mid 18! + 1$$

$$\therefore 19 \cdot 23 = 437 \mid 18! + 1$$

[Note: this multi-step back-substitution (problem 4) has several dense lines whose modular reductions are hard to verify exactly digit-by-digit from the handwriting; transcribed best-effort, the final conclusion is correct and matches the problem statement.]

5.(a) Prove $n > 1$ is prime $\Leftrightarrow (n-2)! \equiv 1 \pmod{n}$

Pf. By Wilson's Th and its converse, n is prime $\Leftrightarrow (n-1)! \equiv -1 \pmod{n}$

$$\equiv -1 + n \equiv n - 1 \pmod{n}$$

Since $\gcd(n, n-1) = 1$ (prob.#12 sec.2.2)

$$\therefore (n-1)!/(n-1) \equiv (n-1)/(n-1) \pmod{n}, \text{ or}$$

$$(n-1)! \equiv -1 \pmod{n} \Leftrightarrow (n-2)! \equiv 1 \pmod{n}$$

$$\therefore n \text{ is prime} \Leftrightarrow (n-2)! \equiv 1 \pmod{n}$$

(b) If n is composite, show $(n-1)! \equiv 0 \pmod{n}$, except when $n = 4$. *Pf.* For $n = 4$, $(4-1)! = 3! = 6 \equiv 2 \pmod{4}$. $\therefore$ Assume $n > 4$.

Since n is composite, let $r \cdot s = n$. Since $\gcd(n, n-1) = 1$ (prob.#12 sec.2.2), r must be one of the factor of $(n-1)!$, $1 < r < n-1$. Similarly $1 < s < n-1$.

If $r \neq s$, Then r and s are different factors in $(n-1)!$, so $n = rs \mid (n-1)!$

$$\therefore (n-1)! \equiv 0 \pmod{n}$$

Suppose $r = s$. $n = r^2$.

Now $r < \frac{n}{2}$. For if $r \geq \frac{n}{2}$, Then

$$n = r^2 \geq \frac{n^2}{4}, \text{ or } 4n \geq n^2, \text{ or } 4 \geq n$$

But $n > 4$, $\therefore r < \frac{n}{2}$

$$\therefore 2r < n, \text{ or } 2r \leq n-1.$$

$\therefore$ Both r and $2r$ are factors of $(n-1)!$

$$\therefore r \cdot (2r) \mid (n-1)!, \text{ so } r^2 \mid (n-1)!$$

$$\therefore (n-1)! \equiv 0 \pmod{n}$$

6. Given a prime number ρ , establish

$$(\rho-1)! \equiv \rho-1 \pmod{1+2+\cdots+(\rho-1)}$$

Pf.: From Wilson's Th, $(\rho-1)! \equiv -1 \equiv -1 + \rho \pmod{\rho}$

$$\therefore \rho \mid (\rho-1)! - (\rho-1)$$

Now $1+2+\cdots+n = \frac{n(n+1)}{2}$ for all n .

$$\therefore 1+2+\cdots+(\rho-1) = \frac{(\rho-1)\rho}{2}$$

Since $\rho-1$ is even, $(\rho-1)/2$ is an integer, and clearly, $(\rho-1) \mid (\rho-1)! - (\rho-1)$.

$$\therefore \frac{\rho-1}{2} \mid (\rho-1)! - (\rho-1)$$

Also, $\gcd\left(\frac{\rho-1}{2}, \rho\right) = 1$ since ρ is prime.

$$\therefore \rho \text{ and } \frac{\rho-1}{2} \text{ divide } (\rho-1)! - (\rho-1), \text{ so}$$

$$\rho \cdot \frac{\rho-1}{2} = 1 + 2 + \cdots + (\rho-1) \text{ divides } (\rho-1)! - (\rho-1)$$

$$\therefore (\rho-1)! \equiv \rho-1 \pmod{1+2+\cdots+(\rho-1)}$$

7. If ρ is prime, prove that for any a ,

$$\rho \mid a^\rho + (\rho-1)!a \quad \text{and} \quad \rho \mid (\rho-1)!a^\rho + a$$

$$(a) \quad \rho \mid a^\rho + (\rho-1)!a$$

Pf. By corollary to Fermats Th., $a^\rho \equiv a \pmod{\rho}$ for any a . By Wilsons Th., $-1 \equiv (\rho-1)! \pmod{\rho}$

$$\therefore \text{By multiplying, } -a^\rho \equiv (\rho-1)!a \pmod{\rho},$$

$$\text{or, } a^\rho \equiv -(\rho-1)!a \pmod{\rho}$$

$$\therefore \rho \mid a^\rho + (\rho-1)!a$$

$$(b) \quad \rho \mid (\rho-1)!a^\rho + a$$

Pf. As in (a), $(\rho-1)! \equiv -1 \pmod{\rho}$

$$a^\rho \equiv a \pmod{\rho}$$

Multiplying together, $a^\rho(\rho-1)! \equiv -a \pmod{\rho}$, or

$$\rho \mid a^\rho(\rho-1)! + a$$

8. Find two odd primes $\rho \leq 13$ s.t. $(\rho-1)! \equiv -1 \pmod{\rho^2}$

5: $4! + 1 = 25$, so $\rho^2 \mid (\rho-1)! + 1$ 7: $6! + 1 = 721$, $7^2 \nmid 721$

9: $8! + 1 = 40321$, $9 \nmid 40321$

[Note: 9 is not prime, so this entry is inconsistent with the problem statement ("odd primes"); transcribed as written.]

11: $10! - 1 = 3,628,801$, $11^2 \nmid 3,628,801$

13: $(12! + 1 = 479,001,601)$, and $13^2 \nmid 479,001,601$

9. Prove for any odd prime, $1^2 \cdot 3^2 \cdot 5^2 \cdots (\rho-2)^2 \equiv (-1)^{\frac{\rho-1}{2}} \pmod{\rho}$

Pf. $K = K - \rho = -(\rho - k) \pmod{\rho}$

$$\therefore 2 \equiv -(\rho-2) \pmod{\rho}$$

$$4 \equiv -(\rho-4) \pmod{\rho}$$

$$\rho-1 \equiv -1 \pmod{\rho}$$

$$2 \cdot 4 \cdots (\rho-1) \equiv (-1)(-3) \cdots [-(\rho-2)] \pmod{\rho}, \quad \frac{\rho-1}{2} \text{ factors } (\rho > 2)$$

There are $(\rho - 1)/2$ factors, so

$$2 \cdot 4 \cdots (\rho - 1) \equiv (-1)^{\frac{\rho-1}{2}} \cdot 1 \cdot 3 \cdots (\rho - 2) \pmod{\rho}$$

Now, multiply both sides by $1 \cdot 3 \cdot 5 \cdots (\rho - 2)$

$$\therefore 1 \cdot 2 \cdot 3 \cdot 4 \cdots (\rho - 2)(\rho - 1) = (\rho - 1)! \equiv (-1)^{\frac{\rho-1}{2}} \cdot 1^2 \cdot 3^2 \cdots (\rho - 2)^2 \pmod{\rho}$$

Or, $(\rho - 1)! \equiv (-1)^{\frac{\rho-1}{2}} \cdot 1^2 \cdot 3^2 \cdots (\rho - 2)^2 \pmod{\rho}$ By Wilson's Th., $-1 \equiv (\rho - 1)! \pmod{\rho}$

$$\therefore -1 \equiv (-1)^{\frac{\rho-1}{2}} \cdot 1^2 \cdot 3^2 \cdots (\rho - 2)^2 \pmod{\rho}$$

Multiplying both sides by $(-1)^{\frac{\rho-1}{2}}$, and noting that $[(-1)^{\frac{\rho-1}{2}}]^2 = 1$ since $(-1)^{\frac{\rho-1}{2}} = \pm 1$, and noting $-1 = (-1)^{\frac{\rho-1}{2}}$,

[Note: this last claim (" $-1 = (-1)^{(\rho-1)/2}$ ") is only true for certain primes, not all odd primes; the original argument appears to be using a multiplication-by- $(-1)^{(\rho-1)/2}$ step to isolate the squares term rather than asserting this equality generally. Transcribed as written.]

$$(-1)^{\frac{\rho-1}{2}} \cdot (-1)^{\frac{\rho-1}{2}} \equiv 1^2 \cdot 3^2 \cdots (\rho - 2)^2 \pmod{\rho}$$

$$\therefore (-1)^{\frac{\rho-1}{2}} \equiv 1^2 \cdot 3^2 \cdots (\rho - 2)^2 \pmod{\rho}$$

10.(a) For a prime ρ of the form $4k + 3$, prove either

$$\left(\frac{\rho - 1}{2}\right)! \equiv 1 \pmod{\rho} \quad \text{or} \quad \left(\frac{\rho - 1}{2}\right)! \equiv -1 \pmod{\rho}$$

Pf. If ρ is any prime, $ab \equiv 0 \pmod{\rho} \Rightarrow a \equiv 0 \pmod{\rho}$ or $b \equiv 0 \pmod{\rho}$ (see end of Section 4.2, proof a beginning of solutions to problems of 4.2, (edited Theorem 2).

$$\therefore \text{If } a \text{ is any integer, } a^2 \equiv 1 \pmod{\rho} \Rightarrow$$

$$a^2 - 1 \equiv 0 \pmod{\rho} \therefore a + 1 \equiv 0 \pmod{\rho} \text{ or}$$

$$a - 1 \equiv 0 \pmod{\rho} \Rightarrow a \equiv 1 \pmod{\rho} \text{ or } a \equiv -1 \pmod{\rho}.$$

In the proof to Theorem 5.5 on $\rho 100$,

[Note: " $\rho 100$ " is unclear — likely " $p. 100$ " (page reference); transcribed as written.]

$$(-1) \equiv (-1)^{\frac{\rho+1}{2}} \left[\left(\frac{\rho - 1}{2}\right)! \right]^2 \pmod{\rho}$$

Multiplying both sides by $(-1)^{\frac{\rho+1}{2}}$ and noting $(-1) = (-1)^{\frac{\rho-1}{2}}$,

$$(-1)^{\frac{\rho+1}{2}} \equiv \left[\left(\frac{\rho - 1}{2}\right)! \right]^2 \pmod{\rho}$$

Now, if ρ is of the form $4k + 3$, Then

$$(-1)^{\frac{4k+4}{2}} = (-1)^{2k+2} = 1 \equiv \left[\left(\frac{\rho-1}{2} \right)! \right]^2 \pmod{\rho}$$

$\therefore$ From above, $\left(\frac{\rho-1}{2} \right)! \equiv 1 \pmod{\rho}$, or

$$\left(\frac{\rho-1}{2} \right)! \equiv -1 \pmod{\rho}$$

(b) If $\rho = 4k + 3$ is prime, Then the product of all the even integers $< \rho$ is congruent mod ρ to either 1 or -1 .

Pf. Let $2, 4, 6, \dots, a$ be all even integers $< \rho$

$$\therefore a = \rho - 1$$

Consider $2 \cdot 4 \cdot 6 \cdots a = 2^K (1 \cdot 2 \cdot 3 \cdots \frac{a}{2})$ where $K = \#$ of terms in $2, 4, 6, \dots, a$ Since $a/2 = (\rho - 1)/2$, Then $K = (\rho - 1)/2$

$$\begin{aligned} \therefore 2 \cdot 4 \cdots a &= 2^{\frac{\rho-1}{2}} (1 \cdot 2 \cdot 3 \cdots \frac{\rho-1}{2}) \\ &= 2^{\frac{\rho-1}{2}} \left(\frac{\rho-1}{2} \right)! \quad [1] \end{aligned}$$

By Fermats Th., $2^{\rho-1} \equiv 1 \pmod{\rho}$, since $\rho \nmid 2$ as ρ is an odd prime.

$$\therefore 2^{\rho-1} = \left(2^{\frac{\rho-1}{2}} \right)^2 \equiv 1 \pmod{\rho}, \text{ so}$$

$$2^{\frac{\rho-1}{2}} \equiv \pm 1 \pmod{\rho}$$

Multiplying both sides by $\left(\frac{\rho-1}{2} \right)!$,

$$2^{\frac{\rho-1}{2}} \left(\frac{\rho-1}{2} \right)! \equiv \pm \left(\frac{\rho-1}{2} \right)! \pmod{\rho} \text{ or } -1 \pmod{\rho}$$

$\therefore$ From [1] above,

$$2 \cdot 4 \cdots a \equiv 1 \pmod{\rho} \text{ or } 2 \cdot 4 \cdots a \equiv -1 \pmod{\rho}$$

Note: above just used ρ as an odd prime. Could be of $4k + 1$ form as well.

11. Obtain two solutions of $x^2 \equiv -1 \pmod{29}$ and $x^2 \equiv -1 \pmod{37}$

(a) $x^2 \equiv -1 \pmod{29}$

As $29 \equiv 1 \pmod{4}$, then by Th.5.5 shows that the proof of $\left[\left(\frac{\rho-1}{2} \right)! \right]^2 \equiv (-1) \pmod{\rho}$,
 $\therefore \left[\left(\frac{29-1}{2} \right)! \right]^2 \equiv -1 = 14!$

[Note: this line's notation is dense/abbreviated in the handwriting; transcribed best-effort.]

$\therefore 14!$ or $-14!$ is a solution

(b) $x^2 \equiv -1 \pmod{37}$. As $37 \equiv 1 \pmod{4}$, as in (a), $\left(\frac{37-1}{2}\right)! = 18!$. $\therefore 18!$ or $-18!$ is a solution.

12. Show that if $\rho = 4k + 3$ is prime and $a^2 + b^2 \equiv 0 \pmod{\rho}$, Then $a \equiv b \equiv 0 \pmod{\rho}$. *Pf.* Suppose $a \not\equiv 0 \pmod{\rho}$. $\therefore \rho \nmid a$.

Consider $ax \equiv 1 \pmod{\rho}$. By Th. 4.7, There is a unique solution mod ρ , and so there is a unique integer c s.t. $1 \leq c \leq \rho - 1$ and $ac \equiv 1 \pmod{\rho}$. $\therefore a^2c^2 \equiv 1 \pmod{\rho}$

From $a^2 + b^2 \equiv 0 \pmod{\rho}$, after multiplying both sides by c^2 , you get

$$a^2c^2 + b^2c^2 \equiv 0 \pmod{\rho}$$

But $a^2c^2 \equiv 1 \pmod{\rho}$.

$$\therefore 1 + b^2c^2 \equiv 0 \pmod{\rho}$$

$$\therefore x = bc \text{ is a solution to } x^2 + 1 \equiv 0 \pmod{\rho},$$

which, by Th. 5.5 means $\rho \equiv 1 \pmod{4}$. But this contradicts $\rho = 4k + 3 \Rightarrow \rho \equiv 3 \pmod{4}$.

$$\therefore a \equiv 0 \pmod{\rho}$$

The exact same reasoning applies to b , so that $b \equiv 0 \pmod{\rho}$.

13. Supply details in the proof that $\sqrt{2}$ is irrational. *Pf.* Suppose $\sqrt{2} = a/b$, $\gcd(a, b) = 1$ Then $a^2 = 2b^2$.

$$\therefore a^2 + b^2 = 3b^2, \text{ and } \therefore 3 \mid (a^2 + b^2), \text{ or}$$

$$a^2 + b^2 \equiv 0 \pmod{3}$$

$\therefore$ From problem #12 above, since 3 is a prime of form $\rho = 4k + 3$, Then

$$3 \mid a \text{ and } 3 \mid b, \text{ contradicting } \gcd(a, b) = 1.$$

14. Prove the odd prime divisors of $n^2 + 1$ are of the form $4k + 1$.

Pf. Let ρ be an odd prime divisor of $n^2 + 1$.

$$\therefore n^2 + 1 \equiv 0 \pmod{\rho}$$

n is a solution to $x^2 + 1 \equiv 0 \pmod{\rho}$, and by Th. 5.5, ρ is of form $4k + 1$.

15. Verify $4(27!) + 5!$ is divisible by 31.

By Wilson's Th., $30! \equiv -1 \pmod{31}$

$$\therefore 30 \cdot 29! \equiv 31 - 1 \equiv 30 \pmod{31}$$

$$\therefore 29! \equiv 1 \pmod{31}, \text{ as } \gcd(30, 31) = 1$$

$$\therefore 4(29!) \equiv 4 \pmod{31}$$

$$5! = 120. \therefore 4(29!) + 5! \equiv 4 + 120 \equiv 124 \pmod{31}$$

$$\text{But } 124 = 4 \cdot 31.$$

$$\therefore 4(29!) + 5! \equiv 0 \pmod{31}$$

$$\Rightarrow 31 \mid (4(29!) + 5!)$$

[Note: the problem statement says " $4(27!)+5!$ " but the worked proof uses " $29!$ " throughout; transcribed as written, likely a copy slip between the two.]

16. For a prime ρ and $0 \leq K \leq \rho - 1$, show that

$$K!(\rho - K - 1)! \equiv (-1)^{K+1} \pmod{\rho}$$

$$\text{Pf. } (\rho - 1)! = 1 \cdot 2 \cdots (\rho - K - 1)(\rho - K) \cdots (\rho - 2)(\rho - 1)$$

$$= (\rho - K - 1)!(\rho - K) \cdots (\rho - 2)(\rho - 1)$$

$$\text{But } \rho - 1 \equiv -1 \pmod{\rho}$$

$$\rho - 2 \equiv -2 \pmod{\rho}$$

$$\vdots$$

$$\rho - K \equiv -K \pmod{\rho}$$

$$\therefore (\rho - K) \cdots (\rho - 2)(\rho - 1) \equiv (-K) \cdots (-2)(-1) \pmod{\rho}$$

$$\text{But } (-K) \cdots (-2)(-1) = (-1)^K K!$$

$$\therefore (\rho - K) \cdots (\rho - 2)(\rho - 1) \equiv (-1)^K K! \pmod{\rho}$$

$$\therefore (\rho - K - 1)!(\rho - K) \cdots (\rho - 2)(\rho - 1) \equiv (-1)^K K!(\rho - K - 1)! \pmod{\rho}$$

$$\therefore (\rho - 1)! \equiv (-1)^K K!(\rho - K - 1)! \pmod{\rho}$$

$$\text{By Wilson's Th., } (\rho - 1)! \equiv -1 \pmod{\rho}$$

$$\therefore (-1) \equiv (-1)^K K!(\rho - K - 1)! \pmod{\rho} \quad [1]$$

Since $(-1)^K \cdot (-1)^K = 1$, and $(-1) \cdot (-1)^{K+1}$ after multiplying both sides of [1] by $(-1)^K$, you get

$$(-1)^{K+1} \equiv K!(\rho - K - 1)! \pmod{\rho}$$

17. If ρ, q are distinct primes, prove for any integer a ,

$$\rho q \mid a^{\rho q} - a^\rho - a^q + a$$

Pf. Consider $(a^q)^\rho - a^\rho$. By the corollary to Fermat's Th., $x^\rho \equiv x \pmod{\rho}$, so letting

$$x = a^q,$$

$$\rho \mid (a^q)^\rho - a^q. \text{ Also } a^q \equiv a \pmod{\rho}. \therefore \rho \mid a^q - a.$$

$$\rho \mid [(a^q)^\rho - a^q] - (a^q - a) = a^{\rho q} - a^q - a^q + a$$

[Note: this line's arithmetic should likely simplify to $a^{\rho q} - a^\rho - a^q + a$, not " $-a^q - a^q + a$ "; transcribed as written, possibly a copy slip from the handwriting.]

Similarly, $\rho \mid (a^q)^\rho - a^\rho$ and $\rho \mid a^\rho - a^q$

$$\therefore \rho \mid [(a^q)^\rho - a^\rho] - (a^\rho - a^q) = a^{\rho q} - a^\rho - a^q + a$$

$\therefore$ Both ρ and q divide $a^{\rho q} - a^\rho - a^q + a$, and so by corollary 2, sec. 2.2,

$$\rho q \mid a^{\rho q} - a^\rho - a^q + a$$

18. Prove if ρ and $\rho + 2$ are twin primes, then

$$4[(\rho - 1)! + 1] + \rho \equiv 0 \pmod{\rho(\rho + 2)}$$

Pf. $(\rho - 1)! \equiv -1 \pmod{\rho}$ Wilson's Th.

$$\therefore (\rho - 1)! + 1 \equiv 0 \pmod{\rho}$$

$$\therefore 4[(\rho - 1)! + 1] + \rho \equiv 0 \pmod{\rho} \quad [1]$$

Also, $(\rho + 2 - 1)! = (\rho + 1)! \equiv -1 \pmod{\rho + 2}$ Wilson's Th.

$$\therefore (\rho + 1)\rho! \equiv -1 - \rho + 2 = -\rho + 1 \pmod{\rho + 2}$$

$$\therefore \rho! \equiv 1 \pmod{\rho + 2} \text{ as } \gcd(\rho + 1, \rho + 2) = 1$$

$$\therefore 4\rho! \equiv 4 = 4 + 2\rho - 2\rho = 2(\rho + 2) - 2\rho \pmod{\rho + 2}$$

$$\therefore 4\rho(\rho - 1)! \equiv -2\rho \pmod{\rho + 2}$$

$$\therefore 4(\rho - 1)! \equiv -2 \pmod{\rho + 2} \quad \gcd(\rho, \rho + 2) = 1$$

$$\therefore 4(\rho - 1)! + \rho + 2 \equiv -2 \pmod{\rho + 2}$$

$$\therefore 4(\rho - 1)! + \rho + 4 \equiv 0 \pmod{\rho + 2}$$

$$\therefore 4[(\rho - 1)! + 1] + \rho \equiv 0 \pmod{\rho + 2} \quad [2]$$

$$\therefore \rho \text{ and } \rho + 2 \text{ divide } 4[(\rho - 1)! + 1] + \rho \text{ by } [1], [2]$$

$$\therefore \rho(\rho + 2) \text{ divides } 4[(\rho - 1)! + 1] + \rho \text{ by corollary 2,}$$

section 2.2.

$$\therefore 4[(\rho - 1)! + 1] + \rho \equiv 0 \pmod{\rho(\rho + 2)}$$

Chapter 6

Number-Theoretic Functions

6.1 The Functions τ and σ

1. Let m, n be positive integers, and $\rho_1, \dots, \rho_r$ be the distinct primes that divide at least one of m or n . Let $m = \rho_1^{k_1} \rho_2^{k_2} \cdots \rho_r^{k_r}$, $K_i \geq 0$ for $i = 1, 2, 3, \dots, r$

$$n = \rho_1^{j_1} \rho_2^{j_2} \cdots \rho_r^{j_r}, \quad j_i \geq 0 \text{ for } i = 1, 2, 3, \dots, r$$

Prove $\gcd(m, n) = \rho_1^{u_1} \rho_2^{u_2} \cdots \rho_r^{u_r}$, $\text{lcm}(m, n) = \rho_1^{v_1} \rho_2^{v_2} \cdots \rho_r^{v_r}$ where $u_i = \min\{K_i, j_i\}$, $v_i = \max\{K_i, j_i\}$

Pf. (a) Consider the integer $a = \rho_1^{u_1} \rho_2^{u_2} \cdots \rho_r^{u_r}$

(1) $u_i = \min\{K_i, j_i\} \Rightarrow 0 \leq u_i \leq K_i$, By Th. 6.1, $a \mid m$

$$u_i = \min\{K_i, j_i\} \Rightarrow 0 \leq u_i \leq j_i, \text{ By Th. 6.1, } a \mid n$$

(2) Let d be any other divisor of m and n By Th. 6.1, $d = \rho_1^{s_1} \rho_2^{s_2} \cdots \rho_r^{s_r}$ where

$$0 \leq s_i \leq K_i \text{ since } d \mid m \text{ and}$$

$$0 \leq s_i \leq j_i \text{ since } d \mid n.$$

If $K_i \leq j_i$, Then by def of u_i , $u_i = K_i \geq s_i$

$$\therefore s_i \leq u_i$$

If $K_i \leq j_i$, Then $u_i = K_i$. $s_i \leq K_i \Rightarrow s_i \leq u_i$

If $j_i < K_i$, Then $u_i = j_i$. $s_i \leq j_i \Rightarrow s_i \leq u_i$

$$\therefore s_i \leq u_i \text{ so that } d \mid a.$$

$$\therefore a = \gcd(m, n) \text{ by (1) \& (2).}$$

(b) Let $a = \rho_1^{v_1} \rho_2^{v_2} \cdots \rho_r^{v_r}$.

(1) $v_i = \max\{K_i, j_i\} \Rightarrow K_i \leq v_i$ and $j_i \leq v_i$.

$$\therefore \rho_i^{K_i} \mid \rho_i^{v_i} \text{ and } \rho_i^{j_i} \mid \rho_i^{v_i}$$

$\therefore m \mid a$ and $n \mid a$, so a is a multiple

of m and n .

(2) Suppose $m \mid b$ and $n \mid b$ (any other multiple).

$$m \mid b \Rightarrow \rho_i^{K_i} \mid b; \quad n \mid b \Rightarrow \rho_i^{j_i} \mid b$$

If $K_i \leq j_i$, Then $v_i = \max\{K_i, j_i\} \Rightarrow v_i = K_i = j_i$

[Note: this line likely should read $v_i = j_i$, not " $K_i = j_i$ "; transcribed as written.]

$\therefore \rho_i^{v_i} \mid b$, and $\therefore$ by corollary 2, sec. 2.2,

$$\rho_1^{v_1} \rho_2^{v_2} \cdots \rho_r^{v_r} \mid b, \text{ so } a \mid b$$

If $K_i < j_i$, Then $v_i = \max\{K_i, j_i\} \Rightarrow v_i = j_i$

$$\therefore \rho_i^{v_i} \mid b \Rightarrow \rho_i^{j_i} \mid b, \text{ so } \rho_1^{v_1} \rho_2^{v_2} \cdots \rho_r^{v_r} \mid b$$

by corollary 2, sec. 2.2. $\therefore a \mid b$

$\therefore$ By (1) & (2), $a = \text{lcm}(m, n)$.

2. Calculate $\text{gcd}(12378, 3054)$ and $\text{lcm}(12378, 3054)$.

$12378 = 2 \cdot 6189 = 2 \cdot 3 \cdot 2063$, and 2063 is prim.

[Note: "prim." likely abbreviates "prime"; transcribed as written.]

$3054 = 2 \cdot 1527 = 2 \cdot 3 \cdot 509$, and 509 is prime.

$$\therefore \text{gcd}(12378, 3054) = 2 \cdot 3 = 6$$

$$\text{lcm}(12378, 3054) = 2 \cdot 3 \cdot 509 \cdot 2063 = 6,304,482$$

3. Use Problem 1 to show that $\text{gcd}(m, n) \cdot \text{lcm}(m, n) = mn$.

Pf. $mn = \rho_1^{K_1+j_1} \rho_2^{K_2+j_2} \cdots \rho_r^{K_r+j_r} \therefore$ For any i , need to show $K_i + j_i = u_i + v_i$ where $u_i = \min\{K_i, j_i\}$, $v_i = \max\{K_i, j_i\}$

(1) If $K_i = j_i$, Then $u_i = K_i = j_i$, $v_i = K_i = j_i$

$$\therefore K_i + j_i = u_i + v_i$$

(2) If $K_i > j_i$, Then $u_i = j_i$, $v_i = K_i$

$$\therefore K_i + j_i = v_i + u_i$$

(3) If $K_i < j_i$, Then $u_i = K_i$, $v_i = j_i$

$$\therefore K_i + j_i = u_i + v_i$$

$$\therefore (1), (2), (3) \Rightarrow mn = \gcd(m, n) \cdot \text{lcm}(m, n)$$

4. Using notation of Problem 1, show $\gcd(m, n) = 1 \Leftrightarrow K_i j_i = 0$ for $i = 1, 2, \dots, r$

Pf. $\gcd(m, n) = \rho_1^{u_1} \rho_2^{u_2} \cdots \rho_r^{u_r}$, where $u_i = \min\{K_i, j_i\}$

(1) If $\gcd(m, n) = 1$, Then $u_1 = 0, u_2 = 0, \dots, u_r = 0$, so $u_i = 0$ for $i = 1, 2, \dots, r$

$$\therefore 0 = \min\{K_i, j_i\}, \text{ so one of } K_i \text{ or } j_i$$

must be 0, so $K_i j_i = 0$.

(2) If $K_i j_i = 0$ for $i = 1, 2, \dots, r$, Then either $K_i = 0$ or $j_i = 0$. $\therefore \min\{K_i, j_i\} = 0$

$$\therefore u_i = 0 \text{ for } i = 1, 2, \dots, r$$

$$\therefore \rho_1^{u_1} \rho_2^{u_2} \cdots \rho_r^{u_r} = 1 = \gcd(m, n)$$

5.(a) Verify $\tau(n) = \tau(n+1) = \tau(n+2) = \tau(n+3)$ for $n = 3655$ and $n = 4503$.

$$3655 = 5 \cdot 17 \cdot 43 \quad \therefore \tau(n) = 2 \cdot 2 \cdot 2 = 8$$

$$3656 = 2^3 \cdot 457 \quad \therefore \tau(n+1) = 4 \cdot 2 = 8$$

$$3657 = 3 \cdot 23 \cdot 53 \quad \therefore \tau(n+2) = 2 \cdot 2 \cdot 2 = 8$$

$$3658 = 2 \cdot 31 \cdot 59 \quad \therefore \tau(n+3) = 2 \cdot 2 \cdot 2 = 8$$

$$4503 = 3 \cdot 19 \cdot 79 \quad \therefore \tau(n) = 2 \cdot 2 \cdot 2 = 8$$

$$4504 = 2^3 \cdot 563 \quad \therefore \tau(n+1) = 4 \cdot 2 = 8$$

$$4505 = 5 \cdot 17 \cdot 53 \quad \therefore \tau(n+2) = 2 \cdot 2 \cdot 2 = 8$$

$$4506 = 2 \cdot 3 \cdot 751 \quad \therefore \tau(n+3) = 2 \cdot 2 \cdot 2 = 8$$

(b) Show $\sigma(n) = 5(n+1)$ when $n = 14, 205$ or 957 .

$$\sigma(14) = \sigma(2 \cdot 7) = \sigma(2)\sigma(7) = 3 \cdot 8 = 24$$

$$\sigma(15) = \sigma(3 \cdot 5) = \sigma(3)\sigma(5) = 4 \cdot 6 = 24$$

[Note: problem 5(b) asks to verify $\sigma(n) = 5(n+1)$, but this worked example computes $\sigma(14)$ and $\sigma(15)$ rather than directly relating to $5(n+1)$; transcribed as written.]

$$\sigma(205) = \sigma(5 \cdot 41) = \sigma(5)\sigma(41) = 6 \cdot 42 = 252$$

$$\sigma(206) = \sigma(2 \cdot 103) = \sigma(2)\sigma(103) = 3 \cdot 104 = 312$$

[Note: line reads "3 · 104 – 3 · 12" in places that are hard to parse exactly; transcribed best-effort, final value 312.]

$$\sigma(957) = \sigma(3 \cdot 11 \cdot 29) = \sigma(3)\sigma(11)\sigma(29) = 4 \cdot 12 \cdot 30 = 1440$$

$$\sigma(958) = \sigma(2 \cdot 479) = \sigma(2)\sigma(479) = 3 \cdot 480 = 1440$$

6. For any $n \geq 1$, show $\tau(n) \leq 2\sqrt{n}$

Pf. Note if $d \mid n$, Then either $d \leq \sqrt{n}$ or $n/d \leq \sqrt{n}$ For if both $d > \sqrt{n}$ and $n/d > \sqrt{n}$, Then

$$d \cdot \frac{n}{d} = n > \sqrt{n} \cdot \sqrt{n} = n, \text{ a contradiction.}$$

Let $d_1, d_2, \dots, d_K$ be the divisors of n , where $d_1 < d_2 < \dots < d_K$. Clearly, $d_1 = 1$, $d_K = n$. Since whenever d_i is a divisor of n , so is n/d_i , and so n/d_i must be one of the d_i . Pair up the divisors so that $d_i d_j = n$, where $d_i = n/d_j$. Either $d_i \leq d_j$ or $d_i \geq d_j$ for each. (1) If K is even, we have $\frac{K}{2}$ unique pairs (d_i, d_j) s.t. $d_i d_j = n$, for each pair, arrange them so that $d_i < d_j$.

Let $d_{K'}$ be the largest of the d_i . It must be that $\frac{K}{2} \leq d_{K'}$ since there are $\frac{K}{2}$ unique pairs. But $\tau(n) = K$, and from above, $d_{K'} \leq \sqrt{n}$

$$\therefore \frac{\tau(n)}{2} \leq \sqrt{n}, \text{ so } \tau(n) \leq 2\sqrt{n}.$$

(2) If K is odd, we have $\frac{K-1}{2}$ unique pairs $\{d_i, d_j\}$ s.t. $d_i d_j = n$ and one pair $\{d_r, d_r\}$ where $d_r d_r = n$.

For the unique $\frac{K-1}{2}$ pairs, arrange them so that $d_i < d_j$.

Let $d_{K'}$ be the largest of the d_i . $d_{K'} < d_r$, for if $d_r \leq d_{K'}$, let d_j be the associated pair with $d_{K'}$. By def. of $d_{K'}$, $d_{K'} < d_j$ and $d_{K'} d_j = n$

$$\therefore d_r < d_{K'} \text{ and } d_r < d_j, \text{ so } d_r d_r < d_{K'} d_j,$$

or $n < n$. $\therefore d_{K'} < d_r$

But $d_r^2 = n$, so $d_r = \sqrt{n}$ As in (1), $\frac{K-1}{2} \leq d_{K'}$ and $K = \tau(n)$

$$\therefore \frac{\tau(n) - 1}{2} \leq d_{K'} < d_r = \sqrt{n}$$

$$\therefore \frac{\tau(n) - 1}{2} < \sqrt{n}, \tau(n) - 1 < 2\sqrt{n}, \text{ so}$$

$$\tau(n) \leq 2\sqrt{n}.$$

$$\therefore \text{Whether } \tau(n) \text{ is even or odd, } \tau(n) \leq 2\sqrt{n}$$

7.(a) Prove $\tau(n)$ is odd $\Leftrightarrow n$ is a perfect square.

(1) Let $n = \rho_1^{K_1} \rho_2^{K_2} \dots \rho_r^{K_r}$ Suppose $\tau(n)$ is odd. Since, by Th. 6.2,

$$\tau(n) = (K_1 + 1)(K_2 + 1) \dots (K_r + 1), \text{ Then each}$$

$(K_i + 1)$ must be odd, so K_i is even.

$$\therefore K_i = 2j_i, \text{ so}$$

$$n = \rho_1^{2j_1} \rho_2^{2j_2} \cdots \rho_r^{2j_r} = (\rho_1^{j_1} \rho_2^{j_2} \cdots \rho_r^{j_r})^2$$

so n is a perfect square.

(2) Suppose n is a perfect square.

$$\therefore n = a^2, \text{ some } a = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$$

$$\therefore n = \rho_1^{2K_1} \rho_2^{2K_2} \cdots \rho_r^{2K_r}$$

$$\therefore \text{By Th. 6.2, } \tau(n) = (2K_1 + 1)(2K_2 + 1) \cdots (2K_r + 1)$$

Since each $2K_i + 1$ is odd, $\tau(n)$ is odd.

(b) $\sigma(n)$ is odd $\Leftrightarrow n$ is a perfect square or twice a perfect square. (1) Suppose $\sigma(n)$ is odd.

Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$. As discussed in the proof to Th. 6.2,

$$\sigma(n) = (1 + \rho_1 + \rho_1^2 + \cdots + \rho_1^{K_1}) \cdots (1 + \rho_r + \cdots + \rho_r^{K_r})$$

$\therefore$ Each term $(1 + \rho_i + \rho_i^2 + \cdots + \rho_i^{K_i})$ must be odd.

If $\rho_i = 2$, Then each ρ_i^x is even, so $1 + 2 + 2^2 + \cdots + 2^{K_i}$ is odd. $\therefore$ consider ρ_i to be odd. $\therefore \rho_i^x$ is odd.

If K_i is odd, Then you have an odd number of terms: $\rho_i^0 + \rho_i^1 + \cdots + \rho_i^{K_i}$, which must be odd. $\therefore 1 + \rho_i + \cdots + \rho_i^{K_i}$ must be even.

$\therefore K_i$ must be even for $1 + \rho_i + \cdots + \rho_i^{K_i}$ to be odd.

$$\therefore \rho_i^{K_i} = \rho_i^{2j_i} = (\rho_i^{j_i})^2$$

$\therefore$ If 2 is a factor of n ,

$$n = 2^K (\rho_2^{j_2} \rho_3^{j_3} \cdots \rho_r^{j_r})^2, \quad K = 1, 2, \dots$$

$$= 2^K a^2, \quad K = 1, 2, \dots, \quad a = \rho_2^{j_2} \cdots \rho_r^{j_r}$$

For K even, $K = 2s$, so $n = (2^s a)^2$, so n is a perfect square. For K odd, $K = 2s + 1$, so $n = 2 \cdot 2^{2s} a^2 = 2(2^s a)^2$, so n is twice a perfect square.

If 2 is not a factor of n , Then

$$n = (\rho_1^{j_1} \rho_2^{j_2} \cdots \rho_r^{j_r})^2, \text{ so } n \text{ is a}$$

perfect square.

(2) Suppose n is a perfect square or twice a perfect square.

(a) Let $n = a^2$; let $a = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$

$$\therefore n = \rho_1^{2K_1} \cdots \rho_r^{2K_r}. \text{ As in the proof to}$$

By Th. 6.2, $\sigma(n) = (1 + \rho_i + \cdots + \rho_i^{2K_i}) \cdots (1 + \rho_r + \cdots + \rho_r^{2K_r})$

If $\rho_i = 2$, Then $(1 + \rho_i + \cdots + \rho_i^{2K_i})$ is odd.

If ρ_i is odd, ρ_i^x is odd. There are an even number of terms in $\rho_i^0 + \cdots + \rho_i^{2K_i}$, which means the sum is even (odd times even is even). $\therefore 1 + \rho_i + \cdots + \rho_i^{2K_i}$ is odd

$\therefore \sigma(n)$ is odd

(b) Suppose $n = 2a^2$.

Let $a = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$. $\therefore n = 2(\rho_1^{K_1} \cdots \rho_r^{K_r})^2$

$$= 2^1 \rho_1^{2K_1} \cdots \rho_r^{2K_r}, \quad \rho_1 < \rho_2 < \cdots < \rho_r \text{ (canonical form)}$$

If $\rho_1 = 2$, Then $n = 2^{K_1+1} \rho_2^{2K_2} \cdots \rho_r^{2K_r}$, and all of $2, \rho_2$ are relatively prime.

$$\therefore \sigma(n) = \sigma(2^{K_1+1}) \cdot \sigma(\rho_2^{2K_2}) \cdots \sigma(\rho_r^{2K_r})$$

But $\sigma(\rho_1^{2K_1}) = 1 + \rho_1 + \rho_1^2 + \cdots + \rho_1^{2K_1}$. As ρ_i^s is odd for any s (ρ_i is odd).

$$\therefore \rho_i^0 + \rho_i^1 + \cdots + \rho_i^{2K_i} \text{ is even as}$$

There are an even # of terms.

$$\sigma(\rho_i^{2K_i}) = 1 + \rho_i + \cdots + \rho_i^{2K_i} \text{ is odd}$$

$2^{K_1+1} - 1$ is odd, so $\sigma(N)$ is odd.

If $\rho_1 \neq 2$, Then $n = 2\rho_1^{2K_1} \cdots \rho_r^{2K_r}$.

As all $2, \rho_i$ are relatively prime,

$$\sigma(n) = \sigma(2)\sigma(\rho_1^{2K_1}) \cdots \sigma(\rho_r^{2K_r})$$

But $\sigma(2) = 1 + 2 = 3$.

As in (a) above, $\sigma(\rho_i^{2K_i})$ is odd.

$\therefore \sigma(n)$ is odd.

8. Show that $\sum_{d|n} \frac{1}{d} = \sigma(n)/n$ for all $n > 0$

Pf. Note that d is a divisor of $n \Rightarrow n/d$ is a divisor of n , since $d \cdot \frac{n}{d} = n$. $\therefore$ The set of divisors of $n = \{d_1, \dots, d_k\}$ can also be written as $\left\{ \frac{n}{d_1}, \dots, \frac{n}{d_k} \right\}$

$$\therefore \sigma(n) = d_1 + d_2 + \cdots + d_k = \frac{n}{d_1} + \frac{n}{d_2} + \cdots + \frac{n}{d_k}$$

$$= n \left(\frac{1}{d_1} + \frac{1}{d_2} + \cdots + \frac{1}{d_k} \right)$$

$$\therefore \frac{\sigma(n)}{n} = \frac{1}{d_1} + \frac{1}{d_2} + \cdots + \frac{1}{d_k} = \sum_{d|n} \frac{1}{d}$$

9. If n is a square free integer, prove $\tau(n) = 2^r$, where r is the number of prime divisors of n .

Pf. n square-free $\Rightarrow n = \rho_1 \rho_2 \cdots \rho_r$, where each ρ_i is distinct and of exponent power 1 (see problem 16(a), section 5.1).

By Th. 6.2, $\tau(n) = (K_1 + 1) \cdots (K_r + 1)$, and

here, each $K_i = 1$.

$$\therefore \tau(n) = (1 + 1) \cdots (1 + 1) = (2) \cdots (2). \text{ As there}$$

are r terms, $\tau(n) = 2^r$ **10.** Establish the assertions below:

(a) If $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$ for $n > 1$, Then

$$1 > \frac{n}{\sigma(n)} > \left(1 - \frac{1}{\rho_1}\right) \left(1 - \frac{1}{\rho_2}\right) \cdots \left(1 - \frac{1}{\rho_r}\right)$$

Since the divisors of n include 1 and n , $\sigma(n) \geq n + 1 > n$, so $\sigma(n) > n$, $\therefore$

$$1 > \frac{n}{\sigma(n)}$$

By Th. 6.2, $\sigma(n) = \frac{\rho_1^{K_1+1}-1}{\rho_1-1} \cdots \frac{\rho_r^{K_r+1}-1}{\rho_r-1}$

$$\begin{aligned} \therefore \frac{n}{\sigma(n)} &= \frac{\rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}}{\frac{(\rho_1^{K_1+1}-1) \cdots (\rho_r^{K_r+1}-1)}{(\rho_1-1) \cdots (\rho_r-1)}} \\ &= \frac{\rho_1^{K_1} \cdots \rho_r^{K_r} (\rho_1-1) \cdots (\rho_r-1)}{(\rho_1^{K_1+1}-1) \cdots (\rho_r^{K_r+1}-1)} \\ &= \frac{(\rho_1-1) \cdots (\rho_r-1)}{\frac{(\rho_1^{K_1+1}-1) \cdots (\rho_r^{K_r+1}-1)}{\rho_1^{K_1} \cdots \rho_r^{K_r}}} \\ &= \frac{(\rho_1-1) \cdots (\rho_r-1)}{\left(\rho_1 - \frac{1}{\rho_1^{K_1}}\right) \cdots \left(\rho_r - \frac{1}{\rho_r^{K_r}}\right)} \quad [1] \end{aligned}$$

But $\rho_i > \rho_i - \frac{1}{\rho_i^{K_i}}, \therefore \frac{1}{\rho_i - \frac{1}{\rho_i^{K_i}}} > \frac{1}{\rho_i}$

$$\begin{aligned} \therefore \text{From [1], } \frac{n}{\sigma(n)} &= \frac{(\rho_1 - 1) \cdots (\rho_r - 1)}{\left(\rho_1 - \frac{1}{\rho_1^{K_1}}\right) \cdots \left(\rho_r - \frac{1}{\rho_r^{K_r}}\right)} > \frac{(\rho_1 - 1) \cdots (\rho_r - 1)}{\rho_1 \cdots \rho_r} \\ &= \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\ \therefore 1 &> \frac{n}{\sigma(n)} > \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \end{aligned}$$

(b) For any positive integer n ,

$$\frac{\sigma(n!)}{n!} \geq 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

Clearly, all $1, 2, 3, 4, \dots, n$ are divisors for $n!$. There are more divisors of $n!$ (including $n!$ itself), since $n! > n$ for $n \geq 3$, and $n! \geq n$ for $n = 2$.

$$\text{From prob. 8, } \frac{\sigma(n!)}{n!} = \sum_{d|n!} \frac{1}{d} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n!}$$

[Note: the final term should likely be $\frac{1}{n}$, not $\frac{1}{n!}$, to match the problem statement; transcribed as written.]

$$\geq 1 + \frac{1}{2} + \cdots + \frac{1}{n}$$

(c) If $n > 1$ is a composite number, then

$$\sigma(n) > n + \sqrt{n}$$

Since $\sigma(n) = 1 + d_1 + \cdots + d_K + n$, it suffices to show $1 + d_1 + \cdots + d_K > \sqrt{n}$

Since n is composite, there is a d_i s.t. $1 < d_i < n$ and $d_i \mid n$. $\therefore \frac{n}{d_i} \mid n, a = d_i$

[Note: this last fragment " $a = d_i$ " is unclear/abbreviated; transcribed as written.]

$$d_i < n \Rightarrow 1 < \frac{n}{d_i}, \text{ and } 1 < d_i \Rightarrow \frac{n}{d_i} < n$$

[Note: the second implication appears inconsistent (should likely be $\frac{n}{d_i} < n$); transcribed as written.]

$$\frac{n}{d_i} < n. \therefore 1 < \frac{n}{d_i} < n$$

(a) If $d_i > \sqrt{n}$, Then clearly $1 + d_i + \cdots + n > n + \sqrt{n}$, so

$$\sigma(n) = 1 + d_1 + \cdots + n > n + \sqrt{n}$$

(b) Suppose $d_i \leq \sqrt{n}$. $\therefore \frac{1}{\sqrt{n}} \leq \frac{1}{d_i} \Rightarrow$

$$\sqrt{n} = \frac{n}{\sqrt{n}} \leq \frac{n}{d_i}. \text{ Let } d_j = \frac{n}{d_i}, \text{ and}$$

d_j is a divisor of n and $\therefore d_j \geq \sqrt{n}$.

$$\therefore 1 + d_j + n > n + \sqrt{n}$$

$$\therefore \text{From } \sigma(n) = 1 + d_1 + d_2 + \cdots + n,$$

$$\sigma(n) > n + \sqrt{n}$$

$$\therefore \text{From (a) \& (b), } \sigma(n) > n + \sqrt{n}$$

11. Given integer $K > 1$, show there are infinitely many integers n for which $\tau(n) = K$, but at most finitely many n with $\sigma(n) = K$.

(a) Let ρ be any prime, let $n = \rho^{K-1}$

$$\therefore \tau(n) = K \text{ by Th. 6.2. Since there are}$$

infinitely many primes, there are infinitely many n s.t. $n = \rho^{K-1}$ and $\tau(n) = K$. (b) By

problem 10(a), $1 > \frac{n}{\sigma(n)}$ for any n .

$$\therefore \sigma(n) > n$$

$\therefore$ Given any K , if $\sigma(n)$ is to equal K , Then K serves as an upper bound on n .

In fact, for any $n \geq K$, $\sigma(n) > K$ by problem 10(a). $\therefore$ There are at most K positive integers s.t. $\sigma(n) \leq K$.

12.(a). Find the form of all positive integers n satisfying $\tau(n) = 10$. What is the smallest positive integer for which this is true?

Since $\tau(n) = (K_1 + 1)(K_2 + 1) \cdots (K_r + 1)$ if $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, Then $10 = (K_1 + 1) \cdots (K_r + 1)$

Possibilities include $10 = 10$ or $10 = 5 \cdot 2$

$$\therefore (K + 1) = 10 \text{ or } (K_1 + 1)(K_2 + 1) = 5 \cdot 2$$

$$\therefore n = \rho_1^9 \text{ or } n = \rho_1^4 \rho_2, \rho_1, \rho_2 \text{ distinct}$$

primes. The smallest such integer would be 2^9 or $3^4 \cdot 2$. Of the three, the smallest is $3^4 \cdot 2 = 162$.

[Note: this line says "Of the three" but only two forms (ρ_1^9 and $\rho_1^4 \rho_2$) were derived; transcribed as written.]

(b) Show there are no positive integers n satisfying $\sigma(n) = 10$. The only possible integers under considerations would be $1, 2, 3, 4, \dots, 7$. $\sigma(1) = 1$. for $2, 3, 5, 7$, $\sigma(i) = 2$

$$\sigma(4) = 7 \quad (1 + 2 + 4)$$

$$\sigma(6) = 12 \quad (1 + 2 + 3 + 6)$$

$$\sigma(8) = 15 \quad (1 + 2 + 4 + 8)$$

$$\sigma(9) = 13 \quad (1 + 3 + 9)$$

$\therefore$ No positive integers n s.t. $\sigma(n) = 10$

13. From there are infinitely many pairs of integers m, n s.t. $\sigma(m^2) = \sigma(n^2)$.

Pf. There are infinitely many K s.t. $\gcd(K, 5) = 1$.

Consider $m = 5K$, $n = 4K$. $\therefore$ There are infinitely many such m, n . Let $K = \rho$, a prime s.t. $\rho \neq 2$ or 5 .

$$m^2 = 5^2 \rho^2 \text{ and } n^2 = 4^2 \rho^2 = 2^4 \rho^2$$

$$\therefore \text{By Th. 6.2, } \sigma(m^2) = \frac{5^3 - 1}{5 - 1} \cdot \frac{\rho^3 - 1}{\rho - 1} = 31 \left(\frac{\rho^3 - 1}{\rho - 1} \right)$$

$$\sigma(n^2) = \frac{2^5 - 1}{2 - 1} \cdot \frac{\rho^3 - 1}{\rho - 1} = 31 \left(\frac{\rho^3 - 1}{\rho - 1} \right)$$

$\therefore$ There are infinitely many m, n , s.t. $\sigma(m^2) = \sigma(n^2)$

14. For $K \geq 2$, show each of the following:

(a) $n = 2^{K-1}$ satisfies $\sigma(n) = 2n - 1$

$$\text{By Th. 6.2, } \sigma(n) = \frac{2^{K-1+1} - 1}{2 - 1} = 2^K - 1$$

$$\text{But } 2n - 1 = 2(2^{K-1}) - 1 = 2^K - 1. \therefore \sigma(n) = 2n - 1$$

(b) If $2^K - 1$ is prime, Then $n = 2^{K-1}(2^K - 1)$ satisfies $\sigma(n) = 2n$

If $2^K - 1$ is prime, Then $2^{K-1} \neq 2$ since $K \geq 2$. $\therefore$ let $\rho = 2^K - 1$.

By Th. 6.2,

$$\begin{aligned} \sigma(n) &= \frac{2^{K-1+1} - 1}{2 - 1} \cdot \frac{\rho^2 - 1}{\rho - 1} = (2^K - 1)(\rho + 1) \\ &= (2^K - 1)(2^K) \end{aligned}$$

$$\text{But } 2n = 2(2^{K-1})(2^K - 1) = 2^K(2^K - 1)$$

$$\therefore \sigma(n) = 2n$$

(c) If $2^K - 3$ is prime, then $n = 2^{K-1}(2^K - 3)$ satisfies $\sigma(n) = 2n + 2$

Let $\rho = 2^K - 3$. Since ρ is prime, $K \geq 3$, so $\rho \neq 2$.

$$\therefore n = 2^{K-1} \cdot \rho, \text{ and by Th. 6.2,}$$

$$\begin{aligned}
\sigma(n) &= \frac{2^{K-1+1} - 1}{2 - 1} \cdot \frac{\rho^2 - 1}{\rho - 1} = (2^K - 1)(\rho + 1) \\
&= (2^K - 1)(2^K - 3 + 1) = (2^K - 1)(2^K - 2) \\
&= 2^{2K} - 3 \cdot 2^K + 2
\end{aligned}$$

$$\text{But } 2n + 2 = 2(2^{K-1})(2^K - 3) + 2$$

$$= 2^K(2^K - 3) + 2 = 2^{2K} - 3 \cdot 2^K + 2$$

$$\therefore \sigma(n) = 2n + 2$$

15. If n and $n + 2$ are a pair of twin primes, show that $\sigma(n + 2) = \sigma(n) + 2$.

Pf. For any prime ρ , the only divisors are 1 and ρ . $\therefore \sigma(\rho) = \rho + 1$

$$\sigma(n + 2) = (n + 2) + 1 = n + 3$$

$$\sigma(n) + 2 = (n + 1) + 2 = n + 3$$

$$\therefore n \text{ and } n + 2 \text{ prime} \Rightarrow \sigma(n + 2) = \sigma(n) + 2$$

16.(a) For any integer $n > 1$, prove there exist integers n_1 and n_2 s.t. $\tau(n_1) + \tau(n_2) = n$

Pf. If n is prime, $\tau(n) = 2$. Since $\tau(1) = 1$, Then let $n_1 = n_2 = 1$. $\therefore \tau(n_1) + \tau(n_2) = \tau(n)$.

If n is composite, let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$ be the canonical prime factorization. Since n is composite, at least one of the $K_i \geq 2$. Let $\rho_j^{K_j}$ be that factor.

$$\therefore \tau(n) = (K_1 + 1)(K_2 + 1) \cdots (K_j + 1) \cdots (K_r + 1)$$

$$= K_j(K_1 + 1)(K_2 + 1) \cdots (K_r + 1)$$

$$+ (K_1 + 1)(K_2 + 1) \cdots (K_r + 1)$$

$$= (K_j - 1 + 1)(K_1 + 1)(K_2 + 1) \cdots (K_r + 1)$$

$$+ (K_1 + 1)(K_2 + 1) \cdots (K_r + 1)$$

$$\therefore \text{Let } n_1 = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_j^{K_j-1} \cdots \rho_r^{K_r} \text{ and let } n_2 = n / \rho_j^{K_j} = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r} / (\rho_j + \rho_j)$$

[Note: this last fragment " $/(\rho_j + \rho_j)$ " is unclear/garbled in the handwriting; the intended definition is presumably $n_2 = n / \rho_j^{K_j}$. Transcribed as written.]

$$\therefore \tau(n_1) = (K_1 + 1) \cdots (K_j + 1) \cdots (K_r + 1)$$

$$= K_j(K_1 + 1) \cdots (K_r + 1)$$

$$\tau(n_2) = (K_1 + 1) \cdots (K_r + 1)$$

$$\therefore \tau(n) = \tau(n_1) + \tau(n_2)$$

(b) Prove the Goldbach conjecture implies that for each even integer $2n$, there exist integers n_1 and n_2 with $\sigma(n_1) + \sigma(n_2) = 2n$.

Pf. Since $2n$ is even, so is $2n - 2$.

Assume $2n - 2 \geq 4$. Goldbach conjecture states there exist odd primes ρ_1 and ρ_2 , s.t. $\rho_1 + \rho_2 = 2n - 2$.

$\therefore$ Let $n_1 = \rho_1$, $n_2 = \rho_2$.

$$\begin{aligned}\sigma(n_1) + \sigma(n_2) &= (\rho_1 + 1) + (\rho_2 + 1) = \rho_1 + \rho_2 + 2 \\ &= 2n - 2 + 2 = 2n\end{aligned}$$

17. For a fixed integer K , show the function f defined by $f(n) = n^K$ is multiplicative.

Pf. $f(mn) = (mn)^K = m^K n^K = f(m) \cdot f(n)$ m and n don't even have to be relatively prime.

18. Let f, g be multiplicative, not identically zero, and s.t. $f(\rho^K) = g(\rho^K)$ for each prime ρ and $K \geq 1$. Prove $f = g$. *Pf.* Let n be any positive integer > 1 .

Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$ be the prime factorization.

$$\begin{aligned}\therefore f(n) &= f(\rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}) = f(\rho_1^{K_1}) \cdots f(\rho_r^{K_r}) \\ &= g(\rho_1^{K_1}) \cdots g(\rho_r^{K_r}) = g(\rho_1^{K_1} \cdots \rho_r^{K_r}) \\ &= g(n)\end{aligned}$$

If $n = 1$, then $f(1) = g(1) = 1$ by discussion on p.107.

$$\therefore f = g$$

19. Prove if f and g are multiplicative functions, Then so is $f \cdot g$ and f/g (whenever f/g is defined).

Pf. Let m, n be integers s.t. $\gcd(m, n) = 1$.

$$\begin{aligned}f \cdot g(mn) &= f(mn) \cdot g(mn) \\ &= f(m) \cdot f(n) \cdot g(m) \cdot g(n) \\ &= f(m) \cdot g(m) \cdot f(n) \cdot g(n) \\ &= f \cdot g(m) \cdot f \cdot g(n)\end{aligned}$$

$$\begin{aligned}f/g(mn) &= f(mn)/g(mn) \\ &= f(m) \cdot f(n)/g(m) \cdot g(n) \\ &= \frac{f(m)}{g(m)} \cdot \frac{f(n)}{g(n)} \\ &= f/g(m) \cdot f/g(n)\end{aligned}$$

20. Let $\omega(n)$ denote the number of distinct prime divisors of $n > 1$, with $\omega(1) = 0$. For example, $\omega(360) = \omega(2^3 \cdot 3^2 \cdot 5) = 3$.

(a) Show $2^{\omega(n)}$ is a multiplicative function

Pf. Let m, n be integers s.t. $\gcd(m, n) = 1$, and let $f(n) = 2^{\omega(n)}$

Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$ and $m = \rho_1^{j_1} \rho_2^{j_2} \cdots \rho_s^{j_s}$

Since $\gcd(m, n) = 1$, Then $\rho_u \neq \rho_v$ for $1 \leq u \leq r, 1 \leq v \leq s$.

$\therefore n$ has r distinct primes, m has s distinct

primes.

$\therefore nm = \rho_1^{K_1} \cdots \rho_r^{K_r} \rho_1^{j_1} \cdots \rho_s^{j_s}$ has $r + s$ distinct

primes (found the Fund. Th. of Arith. says this is unique).

$\therefore \omega(n) = r, \omega(m) = s$, and $\omega(nm) = r + s$

$\therefore f(nm) = 2^{\omega(nm)} = 2^{\omega(n) + \omega(m)} = 2^{\omega(n)} \cdot 2^{\omega(m)}$

$= f(n) \cdot f(m)$

$\therefore 2^{\omega(n)}$ is multiplicative.

(b) For a positive integer n , establish

$$\tau(n^2) = \sum_{d|n} 2^{\omega(d)}$$

From (a), $F(n) = \sum_{d|n} 2^{\omega(d)}$ is multiplicative

$\therefore$ Let $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$

$\therefore F(n) = F(\rho_1^{K_1} \cdots \rho_r^{K_r})$

$= F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r})$

$= \sum_{d|\rho_1^{K_1}} 2^{\omega(d)} \cdots \sum_{d|\rho_r^{K_r}} 2^{\omega(d)}$

All the divisors of $\rho_i^{K_i}$ are $1, \rho_i, \rho_i^2, \dots, \rho_i^{K_i}$ by Th. 6.1. Also, $\omega(\rho_i^{a_i}) = 1$ for $1 \leq a_i \leq K_i$ and $\omega(\rho_i^0) = 0$.

$$\therefore \sum_{d|\rho_i^{K_i}} 2^{\omega(d)} = 2^0 + \underbrace{2^1 + \cdots + 2^1}_{K_i \text{ terms}}$$

$$= 1 + K_i 2^1 = 1 + 2K_i$$

$$\therefore F(n) = \sum_{d|\rho_1^{K_1}} 2^{\omega(d)} \cdots \sum_{d|\rho_r^{K_r}} 2^{\omega(d)}$$

$$= (1 + 2K_1) \cdots (1 + 2K_r)$$

But $n^2 = \rho_1^{2K_1} \cdots \rho_r^{2K_r}$, so by Th. 6.2

$$\tau(n^2) = (2K_1 + 1) \cdots (2K_r + 1)$$

$$\therefore \tau(n^2) = F(n) = \sum_{d|n} 2^{\omega(d)}$$

21. For any positive integer n , prove

$$\sum_{d|n} \tau(d)^3 = \left(\sum_{d|n} \tau(d) \right)^2$$

Pf. $\tau(n)$ is a multiplicative function.

$$\text{Since } [\tau(mn)]^3 = [\tau(m)\tau(n)]^3$$

$$= \tau(m)^3 \cdot \tau(n)^3,$$

Then $\tau(n)^3$ is multiplicative.

$$\therefore \text{By Th. 6.4, } F(n) = \sum_{d|n} \tau(d)^3 \text{ is}$$

multiplicative.

Also, $G(n) = \sum_{d|n} \tau(d)$ is multiplicative, since $H(n) = G^2(n)$ is multiplicative, since

$$\begin{aligned} H(mn) &= G^2(mn) = [G(m)G(n)]^2 = [G(m)]^2[G(n)]^2 \\ &= G^2(m) \cdot G^2(n) \end{aligned}$$

Let $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$.

If $F(n) = H(n)$ for $n = \rho^K$, Then since F and H are multiplicative, the statement will also be true for $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, since

$$\begin{aligned} F(\rho_1^{K_1} \cdots \rho_r^{K_r}) &= F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r}) = H(\rho_1^{K_1}) \cdots H(\rho_r^{K_r}) \\ &= H(\rho_1^{K_1} \cdots \rho_r^{K_r}) \end{aligned}$$

$\therefore$ Consider $n = \rho^K$. By Th. 6.1, all the divisors of n are $1, \rho, \rho^2, \dots, \rho^K$

$$\begin{aligned} \therefore \sum_{d|\rho^K} \tau(d)^3 &= \tau(1)^3 + \tau(\rho)^3 + \tau(\rho^2)^3 + \cdots + \tau(\rho^K)^3 \\ &= 1 + (1+1)^3 + (2+1)^3 + \cdots + (K+1)^3 \\ &= 1 + 2^3 + 3^3 + \cdots + (K+1)^3 \end{aligned}$$

$$\begin{aligned}
 &= \left[\frac{(K+1)(K+2)}{2} \right]^2 \text{ by prob. 1.e, sec. 1.1.} \\
 &\left[\sum_{d|\rho^K} \tau(d) \right]^2 = [\tau(1) + \tau(\rho) + \cdots + \tau(\rho^K)]^2 \\
 &= [1 + 2 + \cdots + (K+1)]^2 \\
 &= \left[\frac{(K+1)(K+2)}{2} \right]^2 \text{ by prob. 1.a, sec. 1.1.} \\
 \therefore \sum_{d|\rho^K} \tau(d)^3 &= \left[\sum_{d|\rho^K} \tau(d) \right]^2, \text{ so } F(n) = H(n)
 \end{aligned}$$

for $n = \rho^K$ **22.** Given $n \geq 1$, let $\sigma_s(n)$ denote the sum of the s th powers of the positive divisors of n ; that is,

$$\sigma_s(n) = \sum_{d|n} d^s$$

Verify the following:

(a) $\sigma_0 = \tau$ and $\sigma_1 = \sigma$

(1) since $d^0 = 1$ for all $d \geq 1$, $\sigma_0(n) = \sum_{d|n} 1 = \tau(n)$ by definition.

(2) since $d^1 = d$ for all $d \geq 1$, $\sigma_1(n) = \sum_{d|n} d = \sigma(n)$ by definition.

(b) σ_s is a multiplicative function.

Pf. If $f(n) = n^s$ can be shown to be multiplicative, then by Th. 6.4,

$$\sum_{d|n} f(d) = \sum_{d|n} d^s = \sigma_s(n) \text{ will be}$$

multiplicative. $\therefore$ Consider $f(n) = n^s$

$$f(mn) = (mn)^s = m^s n^s = f(m) \cdot f(n)$$

so $f(n) = n^s$ is multiplicative, and $\therefore$ so is $\sigma_s(n)$.

(c) If $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$ is the prime factorization, Then $\sigma_s(n) = \left(\frac{\rho_1^{s(K_1+1)} - 1}{\rho_1^s - 1} \right) \cdots \left(\frac{\rho_r^{s(K_r+1)} - 1}{\rho_r^s - 1} \right)$

Pf. By Th. 6.1, all positive divisors of n are of the form $\rho_1^{a_1} \cdots \rho_r^{a_r}$, $0 \leq a_i \leq K_i$
 $\therefore$ all the s th powers of the divisors of n are of the form $\rho_1^{sa_1} \rho_2^{sa_2} \cdots \rho_r^{sa_r}$

Consider the sum:

$$(1 + \rho_1^s + \rho_1^{2s} + \cdots + \rho_1^{sK_1}) \cdots (1 + \rho_r^s + \cdots + \rho_r^{sK_r})$$

Each positive divisor to the s th power occurs once and only once as a term in the expansion of the product.

$$\therefore \sigma_s(n) = (1 + \rho_1^s + \rho_1^{2s} + \cdots + \rho_1^{sK_1}) \cdots (1 + \rho_r^s + \cdots + \rho_r^{sK_r})$$

Using the formula for the sum of a finite geometric series,

$$1 + \rho_1^s + \rho_1^{2s} + \cdots + \rho_1^{sK_1} = \frac{\rho_1^{s(K_1+1)} - 1}{\rho_1^s - 1}$$

$$\therefore \sigma_s(n) = \left(\frac{\rho_1^{s(K_1+1)} - 1}{\rho_1^s - 1} \right) \cdots \left(\frac{\rho_r^{s(K_r+1)} - 1}{\rho_r^s - 1} \right)$$

23. For any positive integer n , show the following:

$$(a) \sum_{d|n} \sigma(d) = \sum_{d|n} \left(\frac{n}{d} \right) \tau(d)$$

Pf. (1) First note that since $\tau(n)$ is multiplicative, $H(n) = \sum_{d|n} \frac{n}{d} \tau(d)$ is multiplicative.

$$\begin{aligned} \text{Pf: } H(mn) &= \sum_{d|mn} \frac{mn}{d} \tau(d) = \sum_{\substack{d_1|m \\ d_2|n}} \frac{mn}{d_1 d_2} \tau(d_1 d_2) \\ &= \sum_{\substack{d_1|m \\ d_2|n}} \frac{mn}{d_1 d_2} \tau(d_1) \tau(d_2) \\ &= \sum_{\substack{d_1|m \\ d_2|n}} \frac{m}{d_1} \tau(d_1) \cdot \frac{n}{d_2} \tau(d_2) \\ &= \left(\sum_{d_1|m} \frac{m}{d_1} \tau(d_1) \right) \left(\sum_{d_2|n} \frac{n}{d_2} \tau(d_2) \right) \\ &= H(m) \cdot H(n) \end{aligned}$$

$\therefore$ The functions

$$F(n) = \sum_{d|n} \sigma(d) \text{ and } G(n) = \sum_{d|n} \frac{n}{d} \tau(d)$$

are multiplicative.

(2) Now, let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$ be the prime factorization of n . If it can be shown that $F(\rho^K) = G(\rho^K)$, Then $F(n) =$

$$\begin{aligned} F(\rho_1^{K_1} \cdots \rho_r^{K_r}) &= F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r}) = G(\rho_1^{K_1}) \cdots G(\rho_r^{K_r}) \\ &= G(\rho_1^{K_1} \cdots \rho_r^{K_r}) = G(n). \end{aligned}$$

$$(3) \therefore F(\rho^K) = \sum_{d|\rho^K} \sigma(d) = (\rho^0) +$$

$$\begin{aligned} &(\rho^0 + \rho^1) + \cdots + \\ &(\rho^0 + \rho^1 + \cdots + \rho^K) \\ &= (K+1)\rho^0 + (K)\rho^1 + \cdots + (1)\rho^K \\ &= (1) \cdot \rho^K + \cdots + (K)\rho + (K+1) \quad [1] \end{aligned}$$

$$\begin{aligned} G(\rho^K) &= \sum_{d|\rho^K} \frac{\rho^K}{d} \tau(d) \\ &= \left(\frac{\rho^K}{\rho^0} \cdot \tau(\rho^0) \right) + \left(\frac{\rho^K}{\rho^1} \cdot \tau(\rho^1) \right) + \cdots + \left(\frac{\rho^K}{\rho^K} \tau(\rho^K) \right) \\ &= (1) \cdot \rho^K + 2 \cdot \rho^{K-1} + \cdots + (K)\rho + (K+1) \quad [2] \\ &\therefore \text{Since } [1] = [2], \text{ Then } F(\rho^K) = G(\rho^K) \end{aligned}$$

$$\therefore \text{As stated in (2), because of (3), } \sum_{d|n} \sigma(d) = \sum_{d|n} \frac{n}{d} \tau(d)$$

$$(b) \sum_{d|n} \left(\frac{n}{d} \right) \sigma(d) = \sum_{d|n} d \tau(d)$$

Pf. Since $f(n) = n$ is multiplicative, so is $f \cdot \tau$

$$\therefore G(n) = \sum_{d|n} d \tau(d) \text{ is multiplicative.}$$

As in (a), the proof that $F(n) = \sum_{d|n} \left(\frac{n}{d} \right) \sigma(d)$ is multiplicative is identical to Th. 6.4.

So, as in (a) it suffices to prove $F(n) = G(n)$ for $n = \rho^K$.

$$\begin{aligned} F(\rho^K) &= \sum_{d|\rho^K} \left(\frac{\rho^K}{d} \right) \sigma(d) \\ &= \rho^K \cdot \rho^0 \\ &\quad + \rho^{K-1}(\rho^0 + \rho) \end{aligned}$$

$$\begin{aligned}
 & +\rho^{K-2}(\rho^0 + \rho + \rho^2) \\
 & + \cdots \\
 & +\rho^0(\rho^0 + \rho + \cdots + \rho^K) \\
 & = (K+1)\rho^K + K \cdot \rho^{K-1} + (K-1)\rho^{K-2} + \cdots + 1 \\
 & = 1 + \sum_{i=1}^K (i+1)\rho^i \quad [1] \\
 & G(\rho^K) = \sum_{d|\rho^K} d \tau(d) \\
 & = 1 \cdot 1 \\
 & + \rho \cdot (1+1) \\
 & + \rho^2 \cdot (1+1+1) \\
 & + \cdots \\
 & + \rho^K (K+1) \\
 & = 1 + \sum_{i=1}^K \rho^i (i+1) \quad [2] \\
 & \text{Since } [1] = [2], \quad F(\rho^K) = G(\rho^K)
 \end{aligned}$$

6.2 The Möbius Inversion Formula

1.(a). For each positive integer n , show

$$\mu(n)\mu(n+1)\mu(n+2)\mu(n+3) = 0$$

Pf. From the Division Algorithm, let $n = 4q + b$, where $0 \leq b \leq 4$.

If $b = 0$, Then $n/4 = q \Rightarrow 2^2 \mid n$, so $\mu(n) = 0$ If $b = 1$, Then $n+3 = 4q+4$, so $4 \mid n+3 \Rightarrow \mu(n+3) = 0$ so $\mu(n+3) = 0$ If $b = 2$, $n+2 = 4q+4$, so $4 \mid n+2 \Rightarrow \mu(n+2) = 0$ If $b = 3$, $n+1 = 4q+4$, so $4 \mid n+1 \Rightarrow \mu(n+1) = 0$
 $\therefore$ for any n , at least one factor will yield $\mu = 0$

(b). For any integer $n \geq 3$, show $\sum_{K=1}^n \mu(K!) = 1$

Pf. $\mu(4) = 0$ since $4 = 2^2$.

If $n \geq 4$, Then $n!$ will contain 4 as a factor. μ is multiplicative, so for $n \geq 4$,

$$\mu(n!) = \mu(n) \cdots \mu(4)\mu(3)\mu(2)\mu(1) = 0.$$

$\therefore$ Only need to consider cases of $n = 3$.

$$\mu(1) = 1, \mu(2) = -1, \mu(3) = -1.$$

$$\begin{aligned} \therefore \sum_{K=1}^3 \mu(K!) &= \mu(1!) + \mu(2!) + \mu(3!) \\ &= \mu(1) + \mu(2) + \mu(6) \\ &= 1 + (-1) + 1 = 1. \end{aligned}$$

2. The Mangoldt function Λ is defined by

$$\Lambda(n) = \begin{cases} \log(\rho), & \text{if } n = \rho^K, \rho \text{ prime}, K \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Prove } \Lambda(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) \log(d) = - \sum_{d|n} \mu(d) \log(d)$$

Pf. Let $n = \rho^K$

$$(a) \sum_{d|n} \mu\left(\frac{n}{d}\right) \log(d) = \mu(\rho^K) \log(1)$$

$$+ \mu(\rho^{K-1}) \log(\rho)$$

$$+ \cdots$$

$$+ \mu(\rho^{K-1}) \log(\rho^{K-1})$$

$$+ \mu(\rho^0) \log(\rho^K)$$

If $K = 1$, the sum is $\mu(\rho^1) \log(1) + \mu(\rho^0) \log(\rho)$

$$= \mu(1) \log(\rho) = \log(\rho)$$

If $K > 1$, $\mu(\rho^{K-i}) = 0$ except for $i = 1, 2$, and then the sum is the same as for $K = 1$.

$$\therefore \sum_{d|n} \mu\left(\frac{n}{d}\right) \log(d) = \log(\rho) = \Lambda(n)$$

$$(b) \sum_{d|n} \mu(d) \log(d) = \mu(\rho^0) \log(1)$$

$$+ \mu(\rho^1) \log(\rho)$$

$$+ \cdots$$

$$+ \mu(\rho^1) \log(\rho^1)$$

$$+ \mu(\rho^K) \log(\rho^K)$$

Because $\mu(\rho^K) = 0$ for $K > 1$, the above sum reduces to, for all K ,

$$\mu(\rho^0) \log(1) + \mu(\rho^1) \log(\rho) = -\log(\rho)$$

$$\therefore \sum_{d|n} \mu(d) \log(d) = -\Lambda(n)$$

3. Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$ for $n > 1$. If f is a multiplicative function not identically 0, prove that

$$\sum_{d|n} \mu(d) f(d) = (1 - f(\rho_1))(1 - f(\rho_2)) \cdots (1 - f(\rho_r))$$

Pf. Since μ and f are multiplicative, Then μf is multiplicative (prob.#19, sec. 6.1).

$$\therefore \text{By Th. 6.4, } F(n) = \sum_{d|n} \mu(d) f(d) \text{ is}$$

multiplicative. $\therefore$ If proven for $F(\rho^K)$, Then, since $F(\rho_1^{K_1} \cdots \rho_r^{K_r}) = F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r})$, will have proven for $F(n)$.

$$\begin{aligned} \therefore \text{Consider } F(\rho^K) &= \sum_{d|\rho^K} \mu(d) f(d) \\ &= \mu(1)f(1) + \mu(\rho)f(\rho) + \cdots + \mu(\rho^K)f(\rho^K) \\ &= \mu(1)f(1) + \mu(\rho)f(\rho), \quad \sum \mu(\rho^i) = 0 \text{ for } i \geq 2 \\ &= 1 \cdot f(1) + (-1) \cdot f(\rho) = 1 - f(\rho) \end{aligned}$$

since, for a multiplicative function not identically zero, $f(1) = 1$ (see sec. 6.1).

$$\therefore F(\rho^K) = 1 - f(\rho).$$

$$\therefore \sum_{d|n} \mu(d) f(d) = (1 - f(\rho_1)) \cdots (1 - f(\rho_r))$$

4. Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$. Establish the following:

$$(a) \sum_{d|n} \mu(d) \tau(d) = (-1)^r$$

Pf. By Prob.#3 above,

$$\sum_{d|n} \mu(d) \tau(d) = [1 - \tau(\rho_1)] \cdots [1 - \tau(\rho_r)]$$

But $\tau(\rho) = 2$, for any prime ρ . $\therefore 1 - \tau(\rho) = -1$.

$$\therefore \sum_{d|n} \mu(d)\tau(d) = (-1)^r$$

$$(b) \sum_{d|n} \mu(d)\sigma(d) = (-1)^r \rho_1 \rho_2 \cdots \rho_r$$

Pf. By Prob.#3 above,

$$\sum_{d|n} \mu(d)\sigma(d) = [1 - \sigma(\rho_1)] \cdots [1 - \sigma(\rho_r)]$$

But $\sigma(\rho) = 1 + \rho$ for any prime ρ .

$$\therefore 1 - \sigma(\rho) = -\rho$$

$$\begin{aligned} \therefore \sum_{d|n} \mu(d)\sigma(d) &= (-\rho_1)(-\rho_2) \cdots (-\rho_r) \\ &= (-1)^r \rho_1 \rho_2 \cdots \rho_r \end{aligned}$$

$$(c) \sum_{d|n} \mu(d)/d = \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right)$$

Pf. First, $f(n) = \frac{1}{n}$ is multiplicative since $f(mn) = \frac{1}{mn} = \frac{1}{m} \cdot \frac{1}{n} = f(m)f(n)$.

$\therefore$ By Prob.#3 above, where $f(n) = \frac{1}{n}$

$$\sum_{d|n} \mu(d) \frac{1}{d} = (1 - f(\rho_1)) \cdots (1 - f(\rho_r)) = \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right)$$

$$(d) \sum_{d|n} d \mu(d) = (1 - \rho_1)(1 - \rho_2) \cdots (1 - \rho_r)$$

Pf. Let $f(n) = n$. f is clearly multiplicative. $\therefore$ By Prob.#3 above,

$$\begin{aligned} \sum_{d|n} d \mu(d) &= \sum_{d|n} f(d) \mu(d) = (1 - f(\rho_1)) \cdots (1 - f(\rho_r)) \\ &= (1 - \rho_1) \cdots (1 - \rho_r) \end{aligned}$$

5. Let $S(n)$ denote the number of square-free divisors of n . Establish that

$$S(n) = \sum_{d|n} |\mu(d)| = 2^{\omega(n)}, \text{ where}$$

$\omega(n)$ = number of distinct prime divisors of n .

Pf. Note that

$$|\mu(n)| = \begin{cases} 1, & \text{if } n = 1 \\ 0, & \text{if } \rho^2 \mid n, \rho \text{ prime} \\ 1, & \text{if } n = \rho_1 \rho_2 \cdots \rho_s, \rho_i \text{ distinct} \end{cases}$$

Let $f(n) = |\mu(n)|$. Let m, n be relatively prime. Clearly, $f(1) = 1$. If $m = 1$, $f(mn) = f(n) = f(m)f(n)$. Let m, n be relatively prime. If $\rho^2 \mid m$, Then $\rho^2 \mid mn$. $\therefore f(mn) = 0$ and $f(m) = 0$. $\therefore f(mn) = f(m)f(n)$

Assume both m, n are square-free. Let $m = \rho_1 \cdots \rho_r$, $n = q_1 \cdots q_s$. $\rho_i \neq q_j$, since $\gcd(m, n) = 1$. Clearly, $f(m) = 1$, $f(n) = 1$, and $f(mn) = 1$. $\therefore f(mn) = f(m)f(n)$.

$\therefore |\mu(n)|$ is multiplicative.

$\therefore$ By Th. 6.4, $\sum_{d|n} |\mu(d)|$ is multiplicative

$\therefore S(n)$ is multiplicative.

Consider $n = \rho^K$. The divisors of n are $1, \rho, \rho^2, \dots, \rho^K$.

$$\begin{aligned} \therefore \sum_{d|n} |\mu(d)| &= |\mu(1)| + |\mu(\rho)| + |\mu(\rho^2)| + \cdots + |\mu(\rho^K)| \\ &= 1 + 1 + 0 + \cdots + 0 = 2 \end{aligned}$$

The number of square-free divisors of ρ^K is 2 and is defined by $\sum_{d|\rho^K} |\mu(d)|$ Consider

$n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$. From Th. 6.1, all the square-free divisors of n are represented by $n = \rho_1^{a_1} \rho_2^{a_2} \cdots \rho_r^{a_r}$, $0 \leq a_i \leq 1$

Since the number of square-free divisors from ρ_1 is 2 (1 and ρ_1), from ρ_2 is 2, $\dots$, from ρ_r is 2, the total number of square-free divisors is 2^r , or $2^{\omega(n)}$ where $\omega(n) = r = \#$ of distinct prime divisors of n .

$$\begin{aligned} \therefore S(n) &= S(\rho_1^{K_1} \cdots \rho_r^{K_r}) = S(\rho_1^{K_1}) \cdots S(\rho_r^{K_r}) \\ &= \left(\sum_{d|\rho_1^{K_1}} |\mu(d)| \right) \cdots \left(\sum_{d|\rho_r^{K_r}} |\mu(d)| \right) \\ &= (2) \cdots (2) = 2^r = 2^{\omega(n)} \end{aligned}$$

6. Find formulas for $\sum_{d|n} \frac{\mu^2(d)}{\tau(d)}$ and $\sum_{d|n} \frac{\mu^2(d)}{\sigma(d)}$ in terms of the prime factorization of n .

From Prob.#19, Sec. 6.1, $\frac{\mu^2(n)}{\tau(n)}$ and $\frac{\mu^2(n)}{\sigma(n)}$ are both multiplicative.

∴ First consider case for $n = \rho^K$

$$\begin{aligned} \sum_{d|n} \frac{\mu^2(d)}{\tau(d)} &= \frac{\mu^2(1)}{\tau(1)} + \frac{\mu^2(\rho)}{\tau(\rho)} + \frac{\mu^2(\rho^2)}{\tau(\rho^2)} + \cdots + \frac{\mu^2(\rho^K)}{\tau(\rho^K)} \\ &= \frac{1}{1} + \frac{1}{2} + 0 + \cdots + 0 \\ &= \frac{3}{2} \end{aligned}$$

$$\begin{aligned} \sum_{d|n} \frac{\mu^2(d)}{\sigma(d)} &= \frac{\mu^2(1)}{\sigma(1)} + \frac{\mu^2(\rho)}{\sigma(\rho)} + \frac{\mu^2(\rho^2)}{\sigma(\rho^2)} + \cdots + \frac{\mu^2(\rho^K)}{\sigma(\rho^K)} \\ &= \frac{1}{1} + \frac{1}{\rho+1} + 0 + \cdots + 0 \\ &= \frac{\rho+2}{\rho+1} \end{aligned}$$

∴ Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$

$$F(n) = \sum_{d|n} \frac{\mu^2(d)}{\tau(d)}, \quad G(n) = \sum_{d|n} \frac{\mu^2(d)}{\sigma(d)}$$

Both F and G are multiplicative.

$$\therefore F(n) = F(\rho_1^{K_1} \cdots \rho_r^{K_r}) = F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r})$$

and $G(n) = G(\rho_1^{K_1}) G(\rho_2^{K_2}) \cdots G(\rho_r^{K_r})$

$$\begin{aligned} \therefore \sum_{d|n} \frac{\mu^2(d)}{\tau(d)} &= \left(\frac{3}{2}\right) \cdots \left(\frac{3}{2}\right) \\ &= \left(\frac{3}{2}\right)^r, \quad r = \# \text{ distinct primes} \\ \sum_{d|n} \frac{\mu^2(d)}{\sigma(d)} &= G(\rho_1^{K_1}) \cdots G(\rho_r^{K_r}) \\ &= \left(\frac{\rho_1+2}{\rho_1+1}\right) \left(\frac{\rho_2+2}{\rho_2+1}\right) \cdots \left(\frac{\rho_r+2}{\rho_r+1}\right) \end{aligned}$$

7. The Liouville λ -function is defined by:

$$\lambda(1) = 1$$

$$\lambda(n) = (-1)^{K_1+K_2+\dots+K_r}, \quad n > 1, \quad n = \rho_1^{K_1} \rho_2^{K_2} \dots \rho_r^{K_r}$$

(a) Prove λ is a multiplicative function

Pf. Let m, n be relatively prime,

$$m = \rho_1^{K_1} \rho_2^{K_2} \dots \rho_r^{K_r}, \quad n = q_1^{j_1} q_2^{j_2} \dots q_s^{j_s}$$

$$mn = \rho_1^{K_1} \rho_2^{K_2} \dots \rho_r^{K_r} q_1^{j_1} \dots q_s^{j_s}, \quad \text{where } \rho_i \neq q_j$$

since $\gcd(m, n) = 1$.

$$\begin{aligned} \therefore \lambda(mn) &= (-1)^{K_1+\dots+K_r+j_1+\dots+j_s} \\ &= (-1)^{K_1+\dots+K_r} \cdot (-1)^{j_1+\dots+j_s} \\ &= \lambda(m) \cdot \lambda(n) \end{aligned}$$

(b). Given $n > 0$, verify

$$\sum_{d|n} \lambda(d) = \begin{cases} 1 & \text{if } n = m^2, \text{ some } m \\ 0 & \text{otherwise} \end{cases}$$

Pf. Let $F(n) = \sum_{d|n} \lambda(d)$. F is multiplicative by Th. 6.4. Plan is to prove for $n = \rho^K$, then extrapolate.

Let $n = \rho^K$.

$$\begin{aligned} \therefore F(n) &= \lambda(1) + \lambda(\rho) + \dots + \lambda(\rho^K) \\ &= 1 + (-1) + (-1)^2 + (-1)^3 + \dots + (-1)^{K-1} + (-1)^K \end{aligned}$$

If K is even, $n = \rho^{2w}$, where $K = 2w$.

$$\therefore \text{let } m = \rho^w. \therefore n = m^2.$$

Also, $F(n) = 1$.

If K is odd, $F(\rho^K) = 0$.

Now, let $n = \rho_1^{K_1} \rho_2^{K_2} \dots \rho_r^{K_r}$

$$\therefore F(n) = F(\rho_1^{K_1}) \dots F(\rho_r^{K_r})$$

If $n = m^2$ for some m , then all the K_i are even, so $F(\rho_i^{K_i}) = 1$ from above.

$$\therefore F(n) = 1$$

If any one of the K_i is odd, Then $F(\rho_i^{K_i}) = 0$, so $F(n) = 0$.

9. For any integer $n \geq 1$, verify formulas below:

$$(a) \sum_{d|n} \mu(d)\lambda(d) = 2^{\omega(n)}, \quad \omega(n) = \# \text{ distinct prime divisors of } n.$$

Pf. $\mu \cdot \lambda$ is multiplicative (Prob.19, sec. 6.1).

$$\therefore \text{ Consider } n = \rho^K$$

$$\begin{aligned} \therefore \sum_{d|n} \mu(d)\lambda(d) &= \mu(1)\lambda(1) \\ &+ \mu(\rho)\lambda(\rho) + \cdots + \mu(\rho^K)\lambda(\rho^K) \\ &= 1 \cdot 1 + (-1)(-1) + \cdots + 0 \cdot (-1)^K \\ &= 2 \end{aligned}$$

$$\text{That is, for } n = \rho^K, \quad F(n) = \sum_{d|n} \mu(d)\lambda(d) = 2$$

$$\therefore \text{ for } n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r},$$

$$\begin{aligned} F(n) &= F(\rho_1^{K_1})F(\rho_2^{K_2}) \cdots F(\rho_r^{K_r}) \\ &= 2 \cdot 2 \cdots 2 = 2^r = 2^{\omega(n)} \end{aligned}$$

since $\omega(n) = r$.

$$(b) \sum_{d|n} \lambda\left(\frac{n}{d}\right) 2^{\omega(d)} = 1$$

Pf: Lemma: If $f(n), g(n)$ are multiplicative, Then so is

$$F(n) = \sum_{d|n} f\left(\frac{n}{d}\right) \cdot g(d)$$

Pf: Let m, n be relatively prime positive integers.

$$\begin{aligned} \therefore F(mn) &= \sum_{d|mn} f\left(\frac{mn}{d}\right) g(d) \\ &= \sum_{\substack{d_1|m \\ d_2|n}} f\left(\frac{mn}{d_1 d_2}\right) g(d_1 d_2) = \sum_{\substack{d_1|m \\ d_2|n}} f\left(\frac{m}{d_1}\right) f\left(\frac{n}{d_2}\right) g(d_1) g(d_2) \\ &= \sum_{\substack{d_1|m \\ d_2|n}} f\left(\frac{m}{d_1}\right) g(d_1) \cdot f\left(\frac{n}{d_2}\right) g(d_2) \\ &= \left(\sum_{d_1|m} f\left(\frac{m}{d_1}\right) g(d_1) \right) \left(\sum_{d_2|n} f\left(\frac{n}{d_2}\right) g(d_2) \right) = F(m)F(n) \end{aligned}$$

$$\therefore F(n) = \sum_{d|n} \lambda\left(\frac{n}{d}\right) 2^{\omega(d)} \text{ is multiplicative}$$

by above lemma and problems 19, 20(a), Sec. 6.1

$\therefore$ Consider $n = \rho^K$

$$\begin{aligned} F(\rho^K) &= \sum_{d|\rho^K} \lambda\left(\frac{\rho^K}{d}\right) 2^{\omega(d)} \\ &= \lambda\left(\frac{\rho^K}{1}\right) 2^{\omega(1)} + \lambda\left(\frac{\rho^K}{\rho}\right) 2^{\omega(\rho)} + \cdots + \lambda\left(\frac{\rho^K}{\rho^K}\right) 2^{\omega(\rho^K)} \\ &= (-1)^K \cdot 1 + (-1)^{K-1} \cdot 2 + \cdots + (-1)^1 \cdot 2 + 1 \cdot 2 \end{aligned}$$

There are K terms of $(-1)^{K-1} \cdot 2 + \cdots + (-1)^1 \cdot 2 + 1 \cdot 2$

$\therefore$ If K is even, $(-1)^K \cdot 1 = 1$, and

$$\underbrace{[(-1)^{K-1} + (-1)^{K-2} + \cdots + (-1)^1 + 1]}_{K=\text{even} \# \text{ terms, alternating signs}} \cdot 2 = 0 \cdot 2 = 0$$

$$\therefore F(\rho^K) = 1 + 0 = 1$$

If K is odd, $(-1)^K \cdot 1 = -1$, and

$$\underbrace{[(-1)^{K-1} + (-1)^{K-2} + \cdots + (-1)^1 + 1]}_{K-1=\text{even} \# \text{ terms, alternating signs}} \cdot 2 = [0 + 1] \cdot 2 = 2$$

$$\therefore F(\rho^K) = (-1) + 2 = 1$$

$$\therefore \text{ If } n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r},$$

$$F(n) = F(\rho_1^{K_1}) F(\rho_2^{K_2}) \cdots F(\rho_r^{K_r}) = 1 \cdot 1 \cdots 1 = 1$$

6.3 The Greatest Integer Function

1. Given integers $a, b > 0$, show there exists a unique integer r with $0 \leq r < b$ satisfying $a = \lfloor a/b \rfloor b + r$. *Pf.* By def, $a/b - 1 < \lfloor a/b \rfloor \leq a/b$

$$\therefore \lfloor a/b \rfloor \cdot b \leq (a/b) \cdot b = a, \therefore 0 \leq a - \lfloor a/b \rfloor b$$

Let $r = a - \lfloor a/b \rfloor b$, $\therefore 0 \leq r$ Also, $(a/b - 1)b < \lfloor a/b \rfloor b$, so $a - \lfloor a/b \rfloor b < b$,

$$\therefore r < b. \therefore \text{ With } r = a - \lfloor a/b \rfloor b, 0 \leq r < b$$

r is unique: let $r' \geq 0$ s.t. $a = \lfloor a/b \rfloor b + r'$

$$\therefore r' = a - \lfloor a/b \rfloor b, \text{ so } r' = r$$

2. Let x, y be real numbers. Prove the greatest integer function satisfies the following properties:

(a) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$ for any integer n .

By def, $x - 1 < \lfloor x \rfloor \leq x$

$$x - 1 + n < \lfloor x \rfloor + n \leq x + n \quad [1]$$

Also, by def of $\lfloor x + n \rfloor$,

$$x + n - 1 < \lfloor x + n \rfloor \leq x + n \quad [2]$$

$$\therefore x + n < \lfloor x + n \rfloor + 1 \quad [3]$$

[1] and [3] yield $\lfloor x \rfloor + n < \lfloor x + n \rfloor + 1$ [4]

From [2], $\lfloor x + n \rfloor - 1 \leq x + n - 1$

and from [1], $x + n - 1 < \lfloor x \rfloor + n$

$$\therefore \lfloor x + n \rfloor - 1 < \lfloor x \rfloor + n \quad [5]$$

[4] and [5] yield $\lfloor x + n \rfloor - 1 < \lfloor x \rfloor + n < \lfloor x + n \rfloor + 1$

$$\therefore -1 < \lfloor x \rfloor + n - (\lfloor x + n \rfloor) < 1$$

Since all quantities are integers,

$$\lfloor x \rfloor + n - (\lfloor x + n \rfloor) = 0, \text{ or } \lfloor x \rfloor + n = \lfloor x + n \rfloor$$

(b) $\lfloor x \rfloor + \lfloor -x \rfloor = 0$ or -1 , according as x is an integer or not. (1) If x is an integer, $\lfloor x \rfloor = x$ and $\lfloor -x \rfloor = -x$.

$$\therefore \lfloor x \rfloor + \lfloor -x \rfloor = 0$$

(2) If x is not an integer, $x = \lfloor x \rfloor + \theta$, $0 < \theta < 1$

$$-x = \lfloor -x \rfloor + \theta', \quad 0 < \theta' < 1$$

Adding, $x + (-x) = \lfloor x \rfloor + \lfloor -x \rfloor + \theta + \theta'$, or

$$0 = \lfloor x \rfloor + \lfloor -x \rfloor + \theta + \theta'$$

But $0 < \theta + \theta' < 2$, and $\theta + \theta' \neq 1$

$$\therefore 0 < -(\lfloor x \rfloor + \lfloor -x \rfloor) < 2, \text{ or } -2 < \lfloor x \rfloor + \lfloor -x \rfloor < 0$$

$\lfloor x \rfloor + \lfloor -x \rfloor$ is an integer, so $\lfloor x \rfloor + \lfloor -x \rfloor = -1$.

(c) $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor$, and when $x \geq 0, y \geq 0$, $\lfloor xy \rfloor \leq \lfloor x \rfloor \lfloor y \rfloor$

(1) $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor$

If x, y are both integers, $\lfloor x \rfloor = x$, $\lfloor y \rfloor = y$,

$$x + y = x + y. \therefore \lfloor x \rfloor = \lfloor y \rfloor = \lfloor x + y \rfloor$$

If one of x, y is an integer, say x, y

[Note: "say x, y " is unclear/garbled in the handwriting (likely meant "say x "); transcribed as written.]

Then $\lfloor x \rfloor + \lfloor y \rfloor = x + y$ By (a), $\lfloor x + y \rfloor = x + y$

$$\therefore \lfloor x \rfloor + \lfloor y \rfloor = \lfloor x + y \rfloor$$

Suppose both x and y are not integers.

$$\therefore \text{By def., } x - 1 < \lfloor x \rfloor < x$$

$$y - 1 < \lfloor y \rfloor < y$$

$$\therefore \lfloor x \rfloor + \lfloor y \rfloor - 1 < x + y - 1 \quad [1]$$

From the def. of $\lfloor x + y \rfloor$,

$$x + y - 1 < \lfloor x + y \rfloor \leq x + y \quad [2]$$

From [1] and [2],

$$\lfloor x \rfloor + \lfloor y \rfloor - 1 < \lfloor x + y \rfloor \quad [3]$$

Since all quantities are integers in [3],

$$\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor$$

(2) $\lfloor x \rfloor \lfloor y \rfloor \leq \lfloor xy \rfloor$ when $x > 0, y > 0$

Let $x = \lfloor x \rfloor + \theta$, $0 \leq \theta < 1$

$$y = \lfloor y \rfloor + \theta', \quad 0 \leq \theta' < 1$$

$$\begin{aligned} \therefore \lfloor xy \rfloor &= \lfloor (\lfloor x \rfloor + \theta)(\lfloor y \rfloor + \theta') \rfloor \\ &= \lfloor \lfloor x \rfloor \lfloor y \rfloor + \theta \lfloor y \rfloor + \theta' \lfloor x \rfloor + \theta \theta' \rfloor \\ &= \lfloor x \rfloor \lfloor y \rfloor + \lfloor \theta \lfloor y \rfloor + \theta' \lfloor x \rfloor + \theta \theta' \rfloor, \text{ by} \end{aligned}$$

(a) above since $\lfloor x \rfloor \lfloor y \rfloor$ is an integer.

All quantities in $\theta \lfloor y \rfloor + \theta' \lfloor x \rfloor + \theta \theta'$ are positive.

$$\therefore \lfloor xy \rfloor \geq \lfloor x \rfloor \lfloor y \rfloor$$

(d) $\lfloor x/n \rfloor = \lfloor \lfloor x \rfloor / n \rfloor$ for any positive integer n .

Let $x/n = \lfloor x/n \rfloor + \theta$, $0 \leq \theta < 1$.

$$\therefore x = \lfloor x/n \rfloor \cdot n + \theta n$$

$$\begin{aligned}\therefore \lfloor x \rfloor &= \lfloor \lfloor x/n \rfloor \cdot n + \theta n \rfloor \\ &= \lfloor x/n \rfloor \cdot n + \lfloor \theta n \rfloor, \text{ by (a) since } \lfloor x/n \rfloor \cdot n\end{aligned}$$

is an integer

$$\therefore \lfloor x \rfloor / n = \lfloor x/n \rfloor + \lfloor \theta n \rfloor / n \quad [1']$$

But since $0 \leq \theta < 1$, Then $0 \leq \theta n < n$

$$\therefore 0 \leq \lfloor \theta n \rfloor \leq \theta n < 1, \therefore \lfloor \theta n \rfloor / n = 0$$

$$\therefore \text{From } [1'], \lfloor x \rfloor / n = \lfloor x/n \rfloor$$

(e) $\lfloor nm/K \rfloor \geq n \lfloor m/K \rfloor$ for positive integers n, m, K .

From (d), $\lfloor nm/K \rfloor = \lfloor n \rfloor \lfloor m/K \rfloor$

But $\lfloor n \rfloor = n$ for positive integer n .

$$\therefore \lfloor nm/K \rfloor \geq n \lfloor m/K \rfloor$$

$$\mathbf{3.} \quad \lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor \leq \lfloor 2x \rfloor + \lfloor 2y \rfloor$$

Pf. Let $x = \lfloor x \rfloor + \theta$, $0 \leq \theta < 1$

$$y = \lfloor y \rfloor + \theta', \quad 0 \leq \theta' < 1$$

$$\therefore x + y = \lfloor x \rfloor + \lfloor y \rfloor + \theta + \theta'$$

$$\begin{aligned}\therefore \lfloor x + y \rfloor &= \lfloor \lfloor x \rfloor + \lfloor y \rfloor + \theta + \theta' \rfloor, \text{ and using (a)} \\ &= \lfloor x \rfloor + \lfloor y \rfloor + \lfloor \theta + \theta' \rfloor \quad [1]\end{aligned}$$

(1) Suppose $0 \leq \theta < 0.5$, $0 \leq \theta' < 0.5$

$$\therefore 0 \leq \theta + \theta' < 1, \text{ so } \lfloor \theta + \theta' \rfloor = 0$$

$\therefore$ Equation [1] becomes $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$

$$\therefore \lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor = 2\lfloor x \rfloor + 2\lfloor y \rfloor$$

But $2\lfloor x \rfloor = 2x - 2\theta$, so $2x = 2\lfloor x \rfloor + 2\theta$

$$\therefore \lfloor 2x \rfloor = \lfloor 2\lfloor x \rfloor + 2\theta \rfloor$$

$$= 2\lfloor x \rfloor + \lfloor 2\theta \rfloor, \text{ by (a)}$$

Since $0 \leq \theta < 0.5$, $0 \leq 2\theta < 1$.

$$\therefore \lfloor 2\theta \rfloor = 0$$

$$\therefore \lfloor 2x \rfloor = 2\lfloor x \rfloor$$

Similarly, $\lfloor 2y \rfloor = 2\lfloor y \rfloor$

$$\therefore \lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor = 2\lfloor x \rfloor + 2\lfloor y \rfloor = \lfloor 2x \rfloor + \lfloor 2y \rfloor$$

Or, $\lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor = \lfloor 2x \rfloor + \lfloor 2y \rfloor$

(2) Suppose $0 \leq \theta < 0.5$, $0.5 \leq \theta' < 1$

$$\therefore 0 < \theta + \theta' < 1.5, \text{ so } 0 \leq \lfloor \theta + \theta' \rfloor \leq 1$$

$\therefore$ From Equation [1], $\lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1$

$$\therefore \lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor \leq 2\lfloor x \rfloor + 2\lfloor y \rfloor + 1 \quad [3]$$

But $2\lfloor x \rfloor = 2x - 2\theta$, so $2x = 2\lfloor x \rfloor + 2\theta$

$$\therefore \lfloor 2x \rfloor = \lfloor 2\lfloor x \rfloor + 2\theta \rfloor$$

$$= 2\lfloor x \rfloor + \lfloor 2\theta \rfloor \text{ by (a)}$$

And $0 \leq 2\theta < 1$, $\therefore \lfloor 2\theta \rfloor = 0$

$$\therefore \lfloor 2x \rfloor = 2\lfloor x \rfloor$$

Also $1 \leq 2\theta' < 2$, so $2y = 2\lfloor y \rfloor + 2\theta'$

$$\lfloor 2y \rfloor = \lfloor 2\lfloor y \rfloor + 2\theta' \rfloor$$

$$= 2\lfloor y \rfloor + \lfloor 2\theta' \rfloor \text{ by (a)}$$

And $1 \leq 2\theta' < 2$

$$\therefore \lfloor 2\theta' \rfloor = 1$$

$$\therefore \lfloor 2y \rfloor = 2\lfloor y \rfloor + 1$$

Or, $2\lfloor y \rfloor = \lfloor 2y \rfloor - 1 \quad [4]$

Substituting [3] and [4] into [3],

[Note: this should likely substitute into a labeled equation other than [3] itself; transcribed as written.]

$$\lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor \leq \lfloor 2x \rfloor + \lfloor 2y \rfloor - 1 + 1$$

$$= \lfloor 2x \rfloor + \lfloor 2y \rfloor$$

Similarly, if $0.5 \leq \theta < 1$, $0 \leq \theta' < 0.5$,

$$\lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor \leq \lfloor 2x \rfloor + \lfloor 2y \rfloor$$

(3) Suppose $0.5 \leq \theta < 1$, $0.5 \leq \theta' < 1$

$$\therefore 1 \leq \theta + \theta' < 2, \text{ so } \lfloor \theta + \theta' \rfloor = 1$$

$\therefore$ Equation [1] becomes $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor + 1$

$$\therefore \lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor = 2\lfloor x \rfloor + 2\lfloor y \rfloor + 1 \quad [5]$$

But $2x = 2\lfloor x \rfloor + 2\theta$

$$\therefore \lfloor 2x \rfloor = \lfloor 2\lfloor x \rfloor + 2\theta \rfloor$$

$$= 2\lfloor x \rfloor + \lfloor 2\theta \rfloor \text{ by (a)}$$

And $1 \leq 2\theta < 2$, so $\lfloor 2\theta \rfloor = 1$

$$\therefore \lfloor 2x \rfloor = 2\lfloor x \rfloor + 1$$

Similarly, $\lfloor 2y \rfloor = 2\lfloor y \rfloor + 1$

Substituting into [5],

$$\lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor = \lfloor 2x \rfloor - 1 + \lfloor 2y \rfloor - 1 + 1$$

$$= \lfloor 2x \rfloor + \lfloor 2y \rfloor - 1$$

$$< \lfloor 2x \rfloor + \lfloor 2y \rfloor$$

$$\therefore \lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor < \lfloor 2x \rfloor + \lfloor 2y \rfloor$$

$\therefore$ (1), (2), and (3) yield:

$$\lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor \leq \lfloor 2x \rfloor + \lfloor 2y \rfloor$$

3. Find the highest power of 5 dividing 1000! and the highest power of 7 dividing 2000!.

(a)

$$\begin{aligned} & \left\lfloor \frac{1000}{5} \right\rfloor + \left\lfloor \frac{1000}{5^2} \right\rfloor + \left\lfloor \frac{1000}{5^3} \right\rfloor + \left\lfloor \frac{1000}{5^4} \right\rfloor \\ &= 200 + 40 + 8 + 1 = 249 \\ &\therefore 5^{249} \text{ divides } 1000! \end{aligned}$$

(b)

$$\begin{aligned} & \left\lfloor \frac{2000}{7} \right\rfloor + \left\lfloor \frac{2000}{7^2} \right\rfloor + \left\lfloor \frac{2000}{7^3} \right\rfloor \\ &= 285 + 40 + 5 = 330 \\ &\therefore 7^{330} \text{ divides } 2000! \end{aligned}$$

4. For integer $n \geq 0$, show $\left\lceil \frac{n}{2} \right\rceil - \left\lfloor \frac{-n}{2} \right\rfloor = n$

Pf. By def., $\frac{n}{2} - 1 < \left\lfloor \frac{n}{2} \right\rfloor \leq \frac{n}{2}$ [1]

and $-\frac{n}{2} - 1 < \left\lfloor -\frac{n}{2} \right\rfloor \leq -\frac{n}{2}$ [2] From [2], $-\left\lfloor -\frac{n}{2} \right\rfloor < \frac{n}{2} + 1$

Adding to [1], $\left\lfloor \frac{n}{2} \right\rfloor - \left\lfloor -\frac{n}{2} \right\rfloor < \frac{n}{2} + \frac{n}{2} + 1 = n + 1$

$$\therefore \left\lfloor \frac{n}{2} \right\rfloor - \left\lfloor -\frac{n}{2} \right\rfloor \leq n \quad [3]$$

Also from [1], $\frac{n}{2} \leq -\left\lfloor -\frac{n}{2} \right\rfloor$

Adding to [1], $\frac{n}{2} + \frac{n}{2} - 1 < \left\lfloor \frac{n}{2} \right\rfloor - \left\lfloor -\frac{n}{2} \right\rfloor$, or, $n - 1 < \left\lfloor \frac{n}{2} \right\rfloor - \left\lfloor -\frac{n}{2} \right\rfloor$

$$\therefore n \leq \left\lfloor \frac{n}{2} \right\rfloor - \left\lfloor -\frac{n}{2} \right\rfloor \quad [4]$$

$$[3] \text{ and } [4] \Rightarrow \left\lfloor \frac{n}{2} \right\rfloor - \left\lfloor -\frac{n}{2} \right\rfloor = n$$

5.(a) Verify that $1000!$ terminates in 249 zeros.

From #3, 5^{249} divides $1000!$, but 5^{250} does not.

The greatest power of 2 dividing $1000!$ is far greater than 500 since $\left\lfloor \frac{1000}{2} \right\rfloor = 500$

$\therefore (2 \cdot 5)^{249}$ divides $1000!$, but

$(2 \cdot 5)^{250}$ does not.

$$\therefore 1000! = n \times 10^{249}, \text{ but } 1000! \neq n \times 10^{250}$$

$\therefore 1000!$ ends in 249 zeros.

(b) For what value of n does $n!$ terminate in 37 zeros?

Must find n s.t. $2^{37} \mid n!$ and $5^{37} \mid n!$ but either $2^{38} \nmid n!$ or $5^{38} \nmid n!$

$$\therefore \text{Consider } \left\lfloor \frac{n}{5} \right\rfloor = 37 \text{ since } \left\lfloor \frac{n}{2} \right\rfloor > \left\lfloor \frac{n}{5} \right\rfloor$$

$$\therefore n = 5 \cdot 37 = 185$$

2 is not the limiting factor since $\left\lfloor \frac{185}{2} \right\rfloor = 92$

Since $5^2 = 25$, $5^3 = 125$, 5^2 will contribute 6 powers of 5 for $n \geq 150$, and 5^3 will contribute at least 1 power. For $n \geq 150$, 5 contributes $\left\lfloor \frac{150}{5} \right\rfloor = 30$

For $n \geq 155$, 5 contributes 31 powers.

For $n \geq 155$, $5, 5^2, 5^3$ would contribute $31 + 6 + 1 = 38$ powers, which is too big.

$\therefore$ For $150 \leq n \leq 154$, 5 contributes exactly 37 powers of 5.

For $n = 147$, 5 contributes 29 powers, 5^2 contributes 5, 5^3 contributes 1.

$\therefore n = 147 \rightarrow 35$ powers of 5, too small.

For $150 \leq n \leq 154$, 2 easily contributes at least 37 powers as $150/2 = 75$.

$\therefore$ For $150 \leq n \leq 154$, $n!$ will terminate in 37 zeros.

6. If $n \geq 1$ and ρ is a prime, prove that (a) $\binom{2n}{n}/(n!)$ is an even integer

Pf. From Th. 6.10, letting $n = 2n$ and $r = n$ in the $\binom{n}{r}$ formula,

$$\binom{2n}{n} = \frac{(2n)!}{n!(2n-n)!} = \frac{(2n)!}{(n!)^2} \text{ is an integer}$$

$$\text{But } \frac{(2n)!}{(n!)^2} = \frac{2n \cdot (2n-1) \cdots (n+1) \cdot n!}{(n!)(n!)}$$

$$= \frac{2n(2n-1) \cdots (n+1)}{n!}$$

This is an integer containing 2 as a factor, and so it is even.

(b) The exponent of the highest power of ρ that divides $(2n)!/(n!)^2$ is $\sum_{K=1}^{\infty} \left(\left\lfloor \frac{2n}{\rho^K} \right\rfloor - 2 \left\lfloor \frac{n}{\rho^K} \right\rfloor \right)$

Pf. For any prime ρ , let S be the highest power of ρ that divides $(2n)!$ If ρ also divides $n!$, let K be the highest power of ρ dividing $n!$.

$$\therefore \frac{f}{\rho^K} = \rho^{S-K}, \text{ so } S - K \text{ is the highest}$$

power of ρ dividing $\frac{(2n)!}{n!}$.

$$\therefore \frac{f^2}{\rho^{2K}} = \rho^{S-2K}, \text{ so } S - 2K \text{ is the}$$

highest power of ρ dividing $\frac{(2n)!}{(n!)^2}$.

By Th. 6.9, the highest power of ρ dividing $(2n)!$ is $S = \sum_{K=1}^{\infty} \left\lfloor \frac{2n}{\rho^K} \right\rfloor$

and the highest power of ρ dividing $n!$ is:

$$\sum_{K=1}^{\infty} \left\lfloor \frac{n}{\rho^K} \right\rfloor$$

$\therefore$ The highest power of ρ dividing $\frac{(2n)!}{(n!)^2}$ is:

$$\begin{aligned} & \sum_{K=1}^{\infty} \left\lfloor \frac{2n}{\rho^K} \right\rfloor - 2 \sum_{K=1}^{\infty} \left\lfloor \frac{n}{\rho^K} \right\rfloor \\ &= \sum_{K=1}^{\infty} \left(\left\lfloor \frac{2n}{\rho^K} \right\rfloor - 2 \left\lfloor \frac{n}{\rho^K} \right\rfloor \right) \end{aligned}$$

(c) In the prime factorization of $\frac{(2n)!}{(n!)^2}$, the exponent of any prime ρ s.t. $n < \rho < 2n$ is 1.

Pf. Since $n < \rho$, Then $\frac{n}{\rho} < 1$, so $\left\lfloor \frac{n}{\rho} \right\rfloor = 0$. $\therefore$ for any $K > 0$, $\left\lfloor \frac{n}{\rho^K} \right\rfloor = 0$. $\therefore$ From (b), the highest power of ρ is:

$$\sum_{K=1}^{\infty} \left\lfloor \frac{2n}{\rho^K} \right\rfloor, \text{ since the contribution}$$

by $2 \lfloor n/\rho^K \rfloor = 0$.

But since $n < \rho$, $\frac{n}{\rho} < 1$, so $\frac{2n}{\rho} < 2$

As $\rho < 2n$, $1 < \frac{2n}{\rho}$. $\therefore \left\lfloor \frac{2n}{\rho} \right\rfloor = 1$ and

$$\frac{2n}{\rho^2} < \frac{2}{\rho} < 1 \text{ if } \rho \geq 2. \therefore \left\lfloor \frac{2n}{\rho^2} \right\rfloor = 0 \text{ for all } K \geq 2$$

$$\therefore \sum_{K=1}^{\infty} \left\lfloor \frac{2n}{\rho^K} \right\rfloor = 1 \text{ for any } n < \rho < 2n,$$

so the highest power of ρ is 1.

7. Let $n = a_K \rho^K + a_{K-1} \rho^{K-1} + \cdots + a_1 \rho + a_0$, where $0 \leq a_i < \rho$.

Show the exponent of highest power of ρ appearing in the prime factorization of $n!$ is:

$$\frac{n - (a_K + \cdots + a_2 + a_1 + a_0)}{\rho - 1}$$

Pf: The exponent of the highest power of ρ , by Th. 6.9, is:

$$\begin{aligned} \sum_{i=1}^{\infty} \left\lfloor \frac{n}{\rho^i} \right\rfloor &= \left\lfloor a_K \rho^{K-1} + \cdots + a_1 + \frac{a_0}{\rho} \right\rfloor \\ &+ \left\lfloor a_K \rho^{K-2} + \cdots + a_2 + \frac{a_1}{\rho} + \frac{a_0}{\rho^2} \right\rfloor \\ &\vdots \\ &+ \left\lfloor a_K + \cdots + \frac{a_1}{\rho^{K-1}} + \frac{a_0}{\rho^K} \right\rfloor \\ &+ \left\lfloor \frac{a_K}{\rho} + \cdots + \frac{a_1}{\rho^K} + \frac{a_0}{\rho^{K+1}} \right\rfloor \quad [1] \end{aligned}$$

Lemma: for $\rho > 1$, $n > 1$, $(\rho - 1) \left(\frac{1}{\rho} + \frac{1}{\rho^2} + \cdots + \frac{1}{\rho^n} \right) < 1$

By induction, let $K = 1$.

$$(\rho - 1) \left(\frac{1}{\rho} \right) = 1 - \frac{1}{\rho} < 1$$

Suppose true for K .

$$\begin{aligned} \therefore (\rho - 1) \left(\frac{1}{\rho} + \cdots + \frac{1}{\rho^K} + \frac{1}{\rho^{K+1}} \right) &= \\ (\rho - 1) \left(\frac{1}{\rho} + \cdots + \frac{1}{\rho^K} \right) + \frac{\rho - 1}{\rho^{K+1}} &= \\ \rho \left(\frac{1}{\rho} + \cdots + \frac{1}{\rho^K} \right) - \frac{1}{\rho} - \cdots - \frac{1}{\rho^K} + \frac{1}{\rho^K} - \frac{1}{\rho^{K+1}} \end{aligned}$$

But by assumption,

$$\rho \left(\frac{1}{\rho} + \cdots + \frac{1}{\rho^K} \right) - \frac{1}{\rho} - \cdots - \frac{1}{\rho^K} < 1$$

$$\therefore \rho \left(\frac{1}{\rho} + \cdots + \frac{1}{\rho^K} \right) - \frac{1}{\rho} - \cdots - \frac{1}{\rho^K} + \frac{1}{\rho^K} - \frac{1}{\rho^{K+1}} < 1 + \frac{1}{\rho^K} - \frac{1}{\rho^{K+1}}$$

or, $\rho \left(\frac{1}{\rho} + \cdots + \frac{1}{\rho^K} \right) - \frac{1}{\rho} - \cdots - \frac{1}{\rho^K} - \frac{1}{\rho^{K+1}} < 1 - \frac{1}{\rho^{K+1}} < 1 \therefore$ True for $K+1$ also. $\therefore$ True for all $n > 1$.

In [1], all $0 \leq a_i \leq \rho - 1 \therefore$ By the lemma, all terms in [1] divided by ρ^K add up to less than 1, [1] becomes:

$$\left\lfloor \frac{n}{\rho} \right\rfloor = a_K \rho^{K-1} + \cdots + a_2 \rho + a_1 \quad [a_1]$$

$$\left\lfloor \frac{n}{\rho^2} \right\rfloor = a_K \rho^{K-2} + \cdots + a_2 \quad [a_2]$$

$\vdots$

$$\left\lfloor \frac{n}{\rho^{K-1}} \right\rfloor = a_K \rho + a_{K-1} \quad [a_{K-1}]$$

$$\left\lfloor \frac{n}{\rho^K} \right\rfloor = a_K \quad [a_K]$$

Note that $\left\lfloor \frac{n}{\rho^K} \right\rfloor \rho = a_K \rho = \left\lfloor \frac{n}{\rho^{K-1}} \right\rfloor - a_{K-1}$

$$\therefore \left\lfloor \frac{n}{\rho} \right\rfloor \rho = n - a_0$$

$$\left\lfloor \frac{n}{\rho^2} \right\rfloor \rho = \left\lfloor \frac{n}{\rho} \right\rfloor - a_1$$

$\vdots$

$$\left\lfloor \frac{n}{\rho^{K-1}} \right\rfloor \rho = \left\lfloor \frac{n}{\rho^{K-2}} \right\rfloor - a_{K-2}$$

$$\left\lfloor \frac{n}{\rho^K} \right\rfloor \rho = \left\lfloor \frac{n}{\rho^{K-1}} \right\rfloor - a_{K-1}$$

$$0 = \left\lfloor \frac{n}{\rho^K} \right\rfloor - a_K$$

Adding the left column entries and right column entries, you get:

$$\left(\left\lfloor \frac{n}{\rho} \right\rfloor + \left\lfloor \frac{n}{\rho^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{\rho^K} \right\rfloor \right) \rho = \left\lfloor \frac{n}{\rho} \right\rfloor + \left\lfloor \frac{n}{\rho^2} \right\rfloor + \cdots + n - (a_0 + \cdots)$$

$$\therefore \left(\left\lfloor \frac{n}{\rho} \right\rfloor + \left\lfloor \frac{n}{\rho^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{\rho^K} \right\rfloor \right) (\rho - 1) = n - (a_0 + a_1 + \cdots + a_K)$$

or

$$\sum_{K=1}^{\infty} \left\lfloor \frac{n}{\rho^K} \right\rfloor = \frac{n - (a_0 + a_1 + \cdots + a_K)}{\rho - 1}$$

8.(a) Using #7, show that the exponent of highest power of ρ dividing $(\rho^K - 1)!$ is $\frac{\rho^K - (\rho - 1)K - 1}{\rho - 1}$

Strategy: find coefficients for the ρ -based expansion of $\rho^K - 1$ ($0 \leq a_i < \rho$).

First note $\rho^K - 1 = (\rho - 1)(\rho^{K-1} + \rho^{K-2} + \cdots + \rho + 1)$

$$= (\rho - 1)\rho^{K-1} + (\rho - 1)\rho^{K-2} + \cdots + (\rho - 1)\rho + (\rho - 1)$$

Since ρ is prime, $0 \leq \rho - 1 < \rho$, so

$$a_{K-1} = \rho - 1, a_{K-2} = \rho - 1, \dots, a_1 = \rho - 1, a_0 = \rho - 1$$

$\therefore$ Using $n = \rho^K - 1$, the formula in #7 becomes:

$$\begin{aligned} & \frac{(\rho^K - 1) - [(\rho - 1) + (\rho - 1) + \cdots + (\rho - 1)]}{\rho - 1} \\ &= \frac{(\rho^K - 1) - K(\rho - 1)}{\rho - 1} \end{aligned}$$

(b). Determine the highest power of 3 dividing $80!$ and the highest power of 7 dividing $2400!$.

(1) $80 = 81 - 1 = 3^4 - 1$, here $K = 4$, $\rho = 3$ Using the formula in (a), $\frac{(3^4 - 1) - 4(3 - 1)}{3 - 1}$

$$= \frac{80 - 8}{2} = 36$$

$\therefore 3^{36} \mid 80!$, 36 is the highest power of 3.

(2) $2400 = 7^4 - 1$, here $K = 4$, $\rho = 7$

Using the formula in (a), $\frac{(7^4 - 1) - 4(7 - 1)}{7 - 1}$

$$= \frac{2400 - 24}{6} = \frac{400 - 4}{1} = 396$$

$\therefore 7^{396} \mid 2400!$, 396 is the highest power of 7.

9. Find an integer $n \geq 1$ s.t. the highest power of 5 contained in $n!$ is 100.

Using problem #7, express n as a ρ -based number and use formula: $\frac{n-r}{\rho-1}$, where $r = \sum a_i$ (coefficients), $\rho = 5$

$$\therefore 100 = \frac{n-r}{5-1} = \frac{n-r}{4}, 400 = n-r$$

If $r = 1$, $n > 401$. $401 = 3 \cdot 5^3 + 1 \cdot 5^2 + 1$

$$r = 3 + 1 + 1 = 5$$

Because n must be > 400 , for $a_3 5^3$, a_3 r must be at least 3. $\therefore$ Try $n = 404$, $404 = 3 \cdot 5^3 + 1 \cdot 5^2 + 4$,

$$r = 3 + 1 + 4 = 8$$

$\therefore$ Try $n = 405$, $405 = 3 \cdot 5^3 + 1 \cdot 5^2 + 1 \cdot 5 + 0$

$$r = 3 + 1 + 1 = 5$$

$\therefore$ When $n = 405$, highest power of 5 dividing $405!$ is 100. **10.** Given a positive integer

N , show the following:

$$(a) \sum_{n=1}^N \mu(n) \left\lfloor \frac{N}{n} \right\rfloor = 1$$

Pf. Let $F(n) = \sum_{d|n} \mu(d)$.

By Th. 6.6, $F(n) = 1$ if $n = 1$, $F(n) = 0$, $n > 1$.

$$\text{By Th. 6.11, } \sum_{K=1}^N F(K) = \sum_{n=1}^N \mu(n) \left\lfloor \frac{N}{n} \right\rfloor$$

$$\text{But } \sum_{K=1}^N F(K) = F(1) + F(2) + \cdots + F(N)$$

$$= 1 + 0 + \cdots + 0$$

$$= 1$$

$$\therefore \sum_{n=1}^N \mu(n) \left\lfloor \frac{N}{n} \right\rfloor = 1$$

$$(b) \left| \sum_{n=1}^N \frac{\mu(n)}{n} \right| \leq 1$$

Pf: From Professor David M. Burton.

$$\text{From (a), } \sum_{n=1}^{N-1} \mu(n) \left\lfloor \frac{N}{n} \right\rfloor + \mu(N) \left\lfloor \frac{N}{N} \right\rfloor = 1$$

But $\left\lfloor \frac{N}{N} \right\rfloor = 1$, so $\sum_{n=1}^{N-1} \mu(n) \left\lfloor \frac{N}{n} \right\rfloor + \mu(N) = 1$ Dividing by N ,

$$\frac{\mu(N)}{N} = \frac{1}{N} - \frac{1}{N} \sum_{n=1}^{N-1} \mu(n) \left\lfloor \frac{N}{n} \right\rfloor \quad [1]$$

Now, $\sum_{n=1}^N \frac{\mu(n)}{n} = \sum_{n=1}^{N-1} \frac{\mu(n)}{n} + \frac{\mu(N)}{N}$

$$= \frac{1}{N} \sum_{n=1}^{N-1} \mu(n) \frac{N}{n} + \frac{\mu(N)}{N} \quad [2]$$

Substituting [1] into [2],

$$\begin{aligned} \sum_{n=1}^N \frac{\mu(n)}{n} &= \frac{1}{N} \sum_{n=1}^{N-1} \mu(n) \frac{N}{n} + \frac{1}{N} - \frac{1}{N} \sum_{n=1}^{N-1} \mu(n) \left\lfloor \frac{N}{n} \right\rfloor \\ &= \frac{1}{N} \sum_{n=1}^{N-1} \mu(n) \left(\frac{N}{n} - \left\lfloor \frac{N}{n} \right\rfloor \right) + \frac{1}{N} \end{aligned}$$

Since $|a + b| \leq |a| + |b|$, and $0 \leq \frac{N}{n} - \left\lfloor \frac{N}{n} \right\rfloor < 1$, and $|c \cdot b| = |a||b|$,

$$\begin{aligned} \left| \sum_{n=1}^N \frac{\mu(n)}{n} \right| &\leq \frac{1}{N} \sum_{n=1}^{N-1} |\mu(n)| \left| \frac{N}{n} - \left\lfloor \frac{N}{n} \right\rfloor \right| + \frac{1}{N} \\ &\leq \frac{1}{N} \sum_{n=1}^{N-1} |\mu(n)| + \frac{1}{N} \quad (\text{and } |\mu(n)| \leq 1) \\ &\leq \frac{1}{N}(N-1) + \frac{1}{N} = 1 \end{aligned}$$

11. Illustrate problem 10 when $N = 6$

$$\begin{aligned} \text{(a)} \quad \sum_{n=1}^6 \mu(n) \left\lfloor \frac{6}{n} \right\rfloor &= \mu(1) \left\lfloor \frac{6}{1} \right\rfloor + \mu(2) \left\lfloor \frac{6}{2} \right\rfloor + \\ &\quad \mu(3) \left\lfloor \frac{6}{3} \right\rfloor + \mu(4) \left\lfloor \frac{6}{4} \right\rfloor + \mu(5) \left\lfloor \frac{6}{5} \right\rfloor + \mu(6) \left\lfloor \frac{6}{6} \right\rfloor \\ &= 1 \cdot 6 + (-1) \cdot 3 + (-1) \cdot 2 + 0 \cdot 1 + (-1) \cdot 1 + 1 \cdot 1 \\ &= 6 - 3 - 2 + 0 - 1 + 1 \end{aligned}$$

$$= 1$$

$$\begin{aligned} \text{(b)} \quad \left| \sum_{n=1}^6 \frac{\mu(n)}{n} \right| &= \left| \frac{\mu(1)}{1} + \frac{\mu(2)}{2} + \frac{\mu(3)}{3} + \frac{\mu(4)}{4} + \frac{\mu(5)}{5} + \frac{\mu(6)}{6} \right| \\ &= \left| 1 + \left(-\frac{1}{2}\right) + \left(-\frac{1}{3}\right) + \frac{0}{4} + \left(-\frac{1}{5}\right) + \frac{1}{6} \right| \\ &= \left| 1 + \left(-\frac{5}{6}\right) + \left(-\frac{1}{5}\right) + \frac{1}{?} \right| \end{aligned}$$

[Note: this intermediate line has an unclear/illegible fraction (shown as "1/?" – handwriting hard to read); the next line's regrouping is transcribed as written.]

$$\begin{aligned} &= \left| 1 + \left(-\frac{2}{3}\right) + \left(-\frac{1}{5}\right) \right| \\ &= \left| 1 + \left(-\frac{13}{15}\right) \right| = \left| \frac{2}{15} \right| < 1 \end{aligned}$$

12. Verify that the formula $\sum_{n=1}^N \lambda(n) \left\lfloor \frac{N}{n} \right\rfloor = \lfloor \sqrt{N} \rfloor$ holds for any positive integer N .

Pf. Let $F(n) = \sum_{d|n} \lambda(d)$ (λ -function defined on page 116, #7, Sec. 6.2)

$$\therefore \text{By Th. 6.11, } \sum_{n=1}^N F(n) = \sum_{n=1}^N \lambda(n) \left\lfloor \frac{N}{n} \right\rfloor$$

By Prob. #7(b), Sec. 6.2,

$$F(n) = \begin{cases} 1 & \text{if } n = m^2 \text{ for some integer } m \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore \sum_{n=1}^N F(n) \text{ keeps track of the number of perfect}$$

squares $\leq N$, as $F(n)$ assigns a value of 1 to each n that can be represented as a perfect square.

$$\therefore \sum_{n=1}^N \lambda(n) \left\lfloor \frac{N}{n} \right\rfloor = \# \text{ perfect squares } \leq N.$$

Now consider $\lfloor \sqrt{N} \rfloor$ and perfect squares.

The perfect squares are $1^2, 2^2, 3^2, \dots$. $\therefore$ For any $N = m^2$, there are exactly m perfect squares (positive integers) less than or equal to N .

Suppose $\sqrt{N}$ is not an integer.

Let m be the largest integer s.t.

$$m^2 < N. \therefore m < (m+1)$$

$$\therefore m < \sqrt{N} < m+1$$

Since $m = \lfloor \sqrt{N} \rfloor$, Then $\lfloor \sqrt{N} \rfloor$ is the number of perfect squares $\leq N$.

$$\therefore \sum_{n=1}^N \lambda(n) \left\lfloor \frac{N}{n} \right\rfloor = \lfloor \sqrt{N} \rfloor$$

13. If N is a positive integer, establish the following:

$$(a) \ N = \sum_{n=1}^{2N} \tau(n) - \sum_{n=1}^N \left\lfloor \frac{2N}{n} \right\rfloor$$

$$\text{By Corollary 1 to Th. 6.11, } \sum_{n=1}^{2N} \tau(n) = \sum_{n=1}^{2N} \left\lfloor \frac{2N}{n} \right\rfloor$$

$$\begin{aligned} & \therefore \sum_{n=1}^{2N} \tau(n) - \sum_{n=1}^N \left\lfloor \frac{2N}{n} \right\rfloor \\ &= \sum_{n=1}^{2N} \left\lfloor \frac{2N}{n} \right\rfloor - \sum_{n=1}^N \left\lfloor \frac{2N}{n} \right\rfloor \\ &= \sum_{n=N+1}^{2N} \left\lfloor \frac{2N}{n} \right\rfloor \end{aligned}$$

But $1 \leq \frac{2N}{N+k} < 2$ for all $1 \leq k \leq N$

Pf: $k \leq N \Rightarrow N+k \leq N+N = 2N$

$$\therefore 1 \leq \frac{2N}{N+k}$$

Also, $1 \leq k \Rightarrow 0 < k \Rightarrow 0 < 2k$

$$\therefore 2N < 2N + 2k = 2(N+k)$$

$$\therefore \frac{2N}{N+k} < 2$$

$$\therefore \left\lfloor \frac{2N}{n} \right\rfloor = 1 \text{ for all } N+1 \leq n \leq 2N$$

$$\therefore \sum_{n=N+1}^{2N} \left\lfloor \frac{2N}{n} \right\rfloor = N \cdot 1 = N$$

$$\therefore \sum_{n=1}^{2N} \tau(n) - \sum_{n=1}^N \left\lfloor \frac{2N}{n} \right\rfloor = N$$

$$(b) \tau(N) = \sum_{n=1}^N \left(\left\lfloor \frac{N}{n} \right\rfloor - \left\lfloor \frac{N-1}{n} \right\rfloor \right)$$

By Corollary 1 to Th. 6.11,

$$\begin{aligned} \sum_{n=1}^N \left\lfloor \frac{N}{n} \right\rfloor &= \sum_{n=1}^N \tau(n) \\ \therefore \sum_{n=1}^N \left\lfloor \frac{N-1}{n} \right\rfloor &= \sum_{n=1}^{N-1} \left\lfloor \frac{N-1}{n} \right\rfloor + \left\lfloor \frac{N-1}{N} \right\rfloor \\ &= \sum_{n=1}^{N-1} \tau(n) + \left\lfloor \frac{N-1}{N} \right\rfloor \end{aligned}$$

But $\frac{N-1}{N} < 1$ for all $N > 0$. $\therefore \left\lfloor \frac{N-1}{N} \right\rfloor = 0$

$$\begin{aligned} \therefore \sum_{n=1}^N \left\lfloor \frac{N-1}{n} \right\rfloor &= \sum_{n=1}^{N-1} \tau(n) \\ \therefore \sum_{n=1}^N \left(\left\lfloor \frac{N}{n} \right\rfloor - \left\lfloor \frac{N-1}{n} \right\rfloor \right) &= \sum_{n=1}^N \tau(n) - \sum_{n=1}^{N-1} \tau(n) \\ &= \tau(N) \end{aligned}$$

6.4 An Application to the Calendar

1. Find the number n of leap years s.t. $1600 < n < Y$, for

(a) $Y = 1825$ ($c = 18, y = 25$)

$$\begin{aligned} L &= 24c + \left\lfloor \frac{c}{4} \right\rfloor + \left\lfloor \frac{y}{4} \right\rfloor - 388, \\ &= 24(18) + \left\lfloor \frac{18}{4} \right\rfloor + \left\lfloor \frac{25}{4} \right\rfloor - 388 \\ &= 432 + 4 + 6 - 388 = \underline{54} \end{aligned}$$

(b) $Y = 1950$ ($c = 19, y = 50$)

$$\begin{aligned} L &= 24c + \left\lfloor \frac{c}{4} \right\rfloor + \left\lfloor \frac{y}{4} \right\rfloor - 388 \\ &= 24(19) + \left\lfloor \frac{19}{4} \right\rfloor + \left\lfloor \frac{50}{4} \right\rfloor - 388 \\ &= 456 + 4 + 12 - 388 = \underline{84} \end{aligned}$$

(c) $Y = 2075$ ($c = 20, y = 75$)

$$\begin{aligned} L &= 24c + \left\lfloor \frac{c}{4} \right\rfloor + \left\lfloor \frac{y}{4} \right\rfloor - 388 \\ &= 24(20) + \left\lfloor \frac{20}{4} \right\rfloor + \left\lfloor \frac{75}{4} \right\rfloor - 388 \\ &= 480 + 5 + 18 - 388 = \underline{115} \end{aligned}$$

2. Determine the day of the week for which you were born.

Jan. 12, 1952. $c = 19, y = 51, m = 11, d = 12$

$$\begin{aligned} W &\equiv d + \lfloor 2.6m - 0.2 \rfloor - 2c + y + \left\lfloor \frac{c}{4} \right\rfloor + \left\lfloor \frac{y}{4} \right\rfloor \pmod{7} \\ &= 12 + \lfloor 2.6(11) - 0.2 \rfloor - 2(19) + 51 + \left\lfloor \frac{19}{4} \right\rfloor + \left\lfloor \frac{51}{4} \right\rfloor \pmod{7} \\ &= 12 + 28 - 38 + 51 + 4 + 12 \pmod{7} \\ &= 69 \equiv 9 \cdot 7 + 6 \equiv 6 \pmod{7} \Rightarrow \underline{Sat} \end{aligned}$$

3. Find the day of the week for the important dates:

(a) November 19, 1863 (Lincoln's Gettysburg Address).

$$\begin{aligned} W &\equiv d + \lfloor 2.6m - 0.2 \rfloor - 2c + y + \left\lfloor \frac{c}{4} \right\rfloor + \left\lfloor \frac{y}{4} \right\rfloor \pmod{7} \\ &= 19 + \lfloor 2.6(9) - 0.2 \rfloor - 2(18) + 63 + \left\lfloor \frac{18}{4} \right\rfloor + \left\lfloor \frac{63}{4} \right\rfloor \pmod{7} \\ &= 19 + 23 - 36 + 63 + 4 + 15 = 88 \equiv 7 \cdot 12 + 4 \\ &\equiv 4 \pmod{7} \Rightarrow \underline{Thu} \end{aligned}$$

(b) April 18, 1906 (S.F. earthquake)

$$W \equiv d + \lfloor 2.6m - 0.2 \rfloor - 2c + y + \left\lfloor \frac{c}{4} \right\rfloor + \left\lfloor \frac{y}{4} \right\rfloor \pmod{7}$$

$$\begin{aligned}
 &= 18 + \lfloor 2.6(4) - 0.2 \rfloor - 2(19) + 6 + \left\lfloor \frac{19}{4} \right\rfloor + \left\lfloor \frac{6}{4} \right\rfloor \pmod{7} \\
 &= 18 + 5 - 38 + 6 + 4 + 1 = -4 \equiv 3 \pmod{7} \Rightarrow \underline{Wed}
 \end{aligned}$$

(c) Nov. 11, 1918 (Great War ends)

$$\begin{aligned}
 W &= 11 + \lfloor 2.6(9) - 0.2 \rfloor - 2(19) + 18 + \left\lfloor \frac{19}{4} \right\rfloor + \left\lfloor \frac{18}{4} \right\rfloor \\
 &= 11 + 23 - 38 + 18 + 4 + 4 = 22 \equiv 1 \pmod{7} \Rightarrow \underline{Mon}
 \end{aligned}$$

(d) Oct. 24, 1929 (N.Y. Stock Market crash)

$$\begin{aligned}
 W &= 24 + \lfloor 2.6(8) - 0.2 \rfloor - 2(19) + 29 + \left\lfloor \frac{19}{4} \right\rfloor + \left\lfloor \frac{29}{4} \right\rfloor \\
 &= 24 + 20 - 38 + 29 + 4 + 7 = 46 \equiv 4 \pmod{7} \\
 &\Rightarrow \underline{Thu}
 \end{aligned}$$

(e) June 6, 1944 (D-Day, Allies land in Normandy)

$$\begin{aligned}
 W &= 6 + \lfloor 2.6(4) - 0.2 \rfloor - 2(19) + 44 + \left\lfloor \frac{19}{4} \right\rfloor + \left\lfloor \frac{44}{4} \right\rfloor \\
 &= 6 + 10 - 38 + 44 + 4 + 11 = 37 \equiv 2 \pmod{7} \\
 &\Rightarrow \underline{Tue}
 \end{aligned}$$

(f) Feb. 15, 1898 (Battleship Maine blown up)

$$W = 15 + \lfloor 2.6(12) - 0.2 \rfloor - 2(18) + 97 + \left\lfloor \frac{18}{4} \right\rfloor + \left\lfloor \frac{97}{4} \right\rfloor$$

[Note: y value for 1898 appears miswritten as "97" here; based on $c = 18$ the year should be $1800 + y$, so y should be 98. Transcribed as written.]

$$\begin{aligned}
 &= 15 + 31 - 36 + 97 + 4 + 24 = 135 \equiv 7 \cdot 19 + 2 \\
 &\equiv 2 \pmod{7} \Rightarrow \underline{Tue}
 \end{aligned}$$

4. Show that days with the identical calendar date in the years 1899 and 1915 fell on the same day of the week.

Pf: Let W_1 = weekday for any day in 1915

W_2 = weekday for any day in 1899

The months and days will be the same.

∴ For 1915:

$$\begin{aligned}
 W_1 &\equiv d + \lfloor 2.6m - 0.2 \rfloor - 2(19) + 15 + \left\lfloor \frac{19}{4} \right\rfloor + \left\lfloor \frac{15}{4} \right\rfloor \\
 &\equiv d + \lfloor 2.6m - 0.2 \rfloor - 38 + 15 + 4 + 3 \\
 &\equiv d + \lfloor 2.6m - 0.2 \rfloor - 16 \pmod{7} \\
 &\equiv d + \lfloor 2.6m - 0.2 \rfloor + 5 \pmod{7}
 \end{aligned}$$

For 1899:

$$\begin{aligned}
 W_2 &= d + \lfloor 2.6m - 0.2 \rfloor - 2(18) + 99 + \left\lfloor \frac{18}{4} \right\rfloor + \left\lfloor \frac{99}{4} \right\rfloor \\
 &= d + \lfloor 2.6m - 0.2 \rfloor - 36 + 99 + 4 + 24 \\
 &= d + \lfloor 2.6m - 0.2 \rfloor + 91 \pmod{7} \\
 &\equiv d + \lfloor 2.6m - 0.2 \rfloor + 5 \pmod{7} \\
 \therefore W_1 - W_2 &\equiv 0 \pmod{7}, \text{ or } W_1 \equiv W_2 \pmod{7}
 \end{aligned}$$

5. For the year 2010, determine the following:

(a) The calendar dates on which Mondays will occur in March.

Monday $\Rightarrow$ 1, so

$$\begin{aligned}
 1 &\equiv d + \lfloor 2.6(1) - 0.2 \rfloor - 2(20) + 10 + \left\lfloor \frac{20}{4} \right\rfloor + \left\lfloor \frac{10}{4} \right\rfloor \\
 &\equiv d + 2 - 40 + 10 + 5 + 2 \equiv d - 21 \pmod{7} \\
 \therefore 22 &\equiv d \pmod{7}, \text{ or } 1 \equiv d \pmod{7}
 \end{aligned}$$

∴ March 1, 8, 15, 22, and 29 will all be Mondays in 2010.

(b) The months in which the 13th will fall on Fri.

Friday $\Rightarrow$ 5, so

$$\begin{aligned}
 5 &\equiv 13 + \lfloor 2.6m - 0.2 \rfloor - 2(20) + 10 + \left\lfloor \frac{20}{4} \right\rfloor + \left\lfloor \frac{10}{4} \right\rfloor \\
 &\equiv \lfloor 2.6m - 0.2 \rfloor + 13 - 40 + 10 + 5 + 2 \\
 &\equiv \lfloor 2.6m - 0.2 \rfloor - 10 \pmod{7} \\
 \therefore 15 &\equiv \lfloor 2.6m - 0.2 \rfloor \pmod{7}
 \end{aligned}$$

for $m = 1, 2, \dots, 9, 10$, $\lfloor 2.6m - 0.2 \rfloor$ becomes

$$2, 5, 7, 10, 12, 15, 18, 20, 23, 25$$

mod 7, these values are 2, 1, 0, 3, 5, 1, 4, 6, 2, 4

$$\therefore \text{When } m = 6, \lfloor 2.6m - 0.2 \rfloor \equiv 1 \pmod{7}$$

$m = 6 \Rightarrow$ August, so August 2010 will contain a Friday 13. Now must check Jan & Feb 2010, which are in year 2009 for the formula.

$$\begin{aligned} 5 &\equiv 13 + \lfloor 2.6m - 0.2 \rfloor - 2(20) + 9 + \left\lfloor \frac{20}{4} \right\rfloor + \left\lfloor \frac{9}{4} \right\rfloor \\ &\equiv \lfloor 2.6m - 0.2 \rfloor - 40 + 9 + 5 + 2 \\ &\equiv \lfloor 2.6m - 0.2 \rfloor - 24 \pmod{7} \\ \therefore 29 &\equiv 1 \equiv \lfloor 2.6m - 0.2 \rfloor \pmod{7} \end{aligned}$$

For $m = 11, 12$, $\lfloor 2.6m - 0.2 \rfloor = 28, 31$

And $28 \not\equiv 1$ and $31 \not\equiv 1 \pmod{7}$, so Jan, Feb. contain no Fri. 13 for 2010.

$\therefore$ For 2010, August is only month with Fri. 13

6. Find the years in the decade 2000 to 2009 when Nov. 29 is on a Sunday.

Nov. $\Rightarrow m = 9$, Sunday $\Rightarrow w = 0$

$$\therefore 0 \equiv 29 + \lfloor 2.6(9) - 0.2 \rfloor - 2(20) + y + \left\lfloor \frac{20}{4} \right\rfloor + \left\lfloor \frac{y}{4} \right\rfloor$$

Chapter 7

Euler's Generalization of Fermat's Theorem

$$= 29 + 23 - 40 + y + 5 + \left\lfloor \frac{y}{4} \right\rfloor$$

$$= 17 + y + \left\lfloor \frac{y}{4} \right\rfloor$$

$$\therefore 4 \equiv y + \left\lfloor \frac{y}{4} \right\rfloor \pmod{7}, \quad 0 \leq y \leq 9$$

$$\text{For } y = 0, 1, 2, 3, 4, 5, 6, 7, 8, \quad 4 \not\equiv y + \left\lfloor \frac{y}{4} \right\rfloor \pmod{7} \quad \text{For } y = 9, \quad y + \left\lfloor \frac{y}{4} \right\rfloor = 11 \equiv 4 \pmod{7}$$

$\therefore$ Nov. 29 is a Sunday only for 2009.

7.2 Euler's Phi-Function

1. Calculate $\phi(1001)$, $\phi(5040)$, $\phi(36,000)$.

$$\phi(1001): \quad 1001 = 7 \cdot 11 \cdot 13$$

$$\therefore \phi(1001) = 1001 \left(1 - \frac{1}{7}\right) \left(1 - \frac{1}{11}\right) \left(1 - \frac{1}{13}\right)$$

$$= 1001 \left(\frac{6}{7}\right) \left(\frac{10}{11}\right) \left(\frac{12}{13}\right)$$

$$= (6)(10)(12) = \underline{720}$$

$$\phi(5040): \quad 5040 = 2^4 \cdot 3^2 \cdot 5 \cdot 7$$

$$\therefore \phi(5040) = 5040 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{7}\right)$$

$$= \frac{5040}{2 \cdot 10} (2)(4)(6) = \underline{1152}$$

[Note: the intermediate arithmetic here is condensed/hard to fully verify from the handwriting; final stated answer transcribed as written.]

$$\phi(36,000): \quad 36,000 = 2^5 \cdot 3^2 \cdot 5^3$$

$$\begin{aligned} \therefore \phi(36000) &= 36000 \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{4}{5}\right) \\ &= 36000 \left(\frac{4}{15}\right) = \underline{9600} \end{aligned}$$

2. Verify $\phi(n) = \phi(n+1) = \phi(n+2)$ is true for $n = 5186$

$$5186 = 2 \cdot 2593 \quad \phi(5186) = 5186 \left(\frac{1}{2}\right) \left(\frac{2592}{2593}\right) = 2592$$

$$5187 = 3 \cdot 7 \cdot 13 \cdot 19 \quad \phi(5187) = 5187 \left(\frac{2}{3}\right) \left(\frac{6}{7}\right) \left(\frac{12}{13}\right) \left(\frac{18}{19}\right) = 2592$$

$$5188 = 2^2 \cdot 1297 \quad \phi(5188) = 5188 \left(\frac{1}{2}\right) \left(\frac{1296}{1297}\right) = 2592$$

3. Show that $m = 3^K \cdot 568$ and $n = 3^K \cdot 638$, $K \geq 0$ satisfy simultaneously $\tau(m) = \tau(n)$, $\sigma(m) = \sigma(n)$, and $\phi(m) \neq \phi(n)$.

$$568 = 2^3 \cdot 71 \quad 638 = 2 \cdot 11 \cdot 29$$

$$\therefore \tau(m) = (K+1)(3+1)(1+1) = 6(K+1)$$

$$\tau(n) = (K+1)(1+1)(1+1)(1+1) = 6(K+1)$$

$$\sigma(m) = \frac{(3^{K+1} - 1)}{(3 - 1)} \cdot \frac{(2^4 - 1)}{(2 - 1)} \cdot \frac{(71^2 - 1)}{(71 - 1)} = \frac{(3^{K+1} - 1)}{(2)}(15)(5040)$$

[Note: this intermediate evaluation is dense/hard to verify exactly from the handwriting; transcribed best-effort.]

$$= (3^{K+1} - 1)(540)$$

$$\begin{aligned} \sigma(n) &= \frac{(3^{K+1} - 1)}{(2)} \cdot \frac{(2^2 - 1)}{(2 - 1)} \cdot \frac{(11^2 - 1)}{(11 - 1)} \cdot \frac{(29^2 - 1)}{(29 - 1)} \\ &= \frac{(3^{K+1} - 1)}{2}(3)(12)(30) = (3^{K+1} - 1)(540) \end{aligned}$$

$$\begin{aligned} \phi(m) &= 3^K \cdot 568 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{71}\right) \\ &= 3^K \cdot 2^3 \cdot 71 \left(\frac{2}{3}\right) \left(\frac{1}{2}\right) \left(\frac{70}{71}\right) \end{aligned}$$

$$\begin{aligned}
 &= 3^{K-1} \cdot 2^3 \cdot 70 = 560 \cdot 3^{K-1} \\
 \phi(n) &= 3^K \cdot 638 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{11}\right) \left(1 - \frac{1}{29}\right) \\
 &= 3^K \cdot 2 \cdot 11 \cdot 29 \left(\frac{2}{3}\right) \left(\frac{1}{2}\right) \left(\frac{10}{11}\right) \left(\frac{28}{29}\right) \\
 &= 3^{K-1} \cdot 20 \cdot 28 = 560 \cdot 3^{K-1}
 \end{aligned}$$

[Note: this computation yields $\phi(m) = \phi(n) = 560 \cdot 3^{K-1}$, which contradicts the problem statement asking to show $\phi(m) \neq \phi(n)$; transcribed as written – likely an error in either the problem statement or the computed values that the original notes did not catch.]

4. Establish each of the assertions below:

(a) If n is odd, then $\phi(2n) = \phi(n)$

Pf: Let $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$. n odd $\Rightarrow \rho_i \neq 2$

$$\begin{aligned}
 \therefore 2n &= 2\rho_1^{K_1} \cdots \rho_r^{K_r} \\
 \therefore \phi(2n) &= 2n \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\
 &= n \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\
 &= \phi(n)
 \end{aligned}$$

Another proof: n odd $\Rightarrow \gcd(2, n) = 1$ ϕ multiplicative $\Rightarrow \phi(2n) = \phi(2)\phi(n) = \phi(n)$

(b) If n is even, $\phi(2n) = 2\phi(n)$

Pf: n even $\Rightarrow n = 2^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$

$$\begin{aligned}
 \therefore 2n &= 2^{K_1+1} \rho_2^{K_2} \cdots \rho_r^{K_r} \\
 \therefore \phi(2n) &= 2n \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{\rho_2}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\
 &= n \left(1 - \frac{1}{\rho_2}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\
 2\phi(n) &= 2 \cdot n \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{\rho_2}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\
 &= n \left(1 - \frac{1}{\rho_2}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\
 \therefore \phi(2n) &= 2\phi(n)
 \end{aligned}$$

(c) $\phi(3n) = 3\phi(n) \Leftrightarrow 3 \mid n$

Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$

(1) $3 \mid n \Rightarrow$ one of $\rho_i = 3$. Let $\rho_1 = 3$. $\therefore n = 3^K q$, where $\gcd(3, q) = 1$

$$\begin{aligned} \therefore 3\phi(n) &= 3\phi(3^K q) = 3\phi(3^K)\phi(q) \\ &= 3 \cdot 3^K \left(1 - \frac{1}{3}\right) \phi(q) = 2 \cdot 3^K \phi(q) \\ \phi(3n) &= \phi(3^{K+1} q) = \phi(3^{K+1})\phi(q) \\ &= 3^{K+1} \left(1 - \frac{1}{3}\right) \phi(q) = 2 \cdot 3^K \cdot \phi(q) \\ \therefore \phi(3n) &= 3\phi(n) \end{aligned}$$

(2) Suppose $\phi(3n) = 3\phi(n)$.

If $3 \nmid n$, Then for $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, $\rho_i \neq 3$

$$\begin{aligned} \therefore \gcd(3, n) &= 1 \Rightarrow \phi(3n) = \phi(3)\phi(n) \\ &= 2\phi(n) \end{aligned}$$

This contradicts $\phi(3n) = 3\phi(n)$.

$$\therefore 3 \mid n$$

(d) $\phi(3n) = 2\phi(n) \Leftrightarrow 3 \nmid n$

(1) As in (c)(2) above, $3 \nmid n \Rightarrow \phi(3n) = 2\phi(n)$ (2) Suppose $\phi(3n) = 2\phi(n)$.

From (c) above, if $3 \mid n$, Then $\phi(3n) = 3\phi(n)$. $\therefore 3 \nmid n$

(e) $\phi(n) = n/2 \Leftrightarrow n = 2^K$ for some $K \geq 1$

$$\begin{aligned} (1) \text{ If } n = 2^K, \text{ Then } \phi(n) &= \phi(2^K) = 2^K \left(1 - \frac{1}{2}\right) \\ &= 2^K \left(\frac{1}{2}\right) = n/2 \end{aligned}$$

(2) If $\phi(n) = n/2$, Then for $n/2$ to be an integer, n must be even.

$\therefore$ Let $n = 2^K \rho_2^{K_2} \cdots \rho_r^{K_r}$, and assume $K \geq 1$

Let $q = \rho_2^{K_2} \cdots \rho_r^{K_r}$, so $q \geq 1$ and $\gcd(2^K, q) = 1$

$$\begin{aligned} \therefore \phi(n) &= \phi(2^K q) = \phi(2^K)\phi(q) \\ &= 2^K \left(1 - \frac{1}{2}\right) \phi(q) = 2^{K-1} \phi(q) \\ \therefore \phi(n) &= n/2 = 2^{K-1} \phi(q), \text{ so } n = 2 \cdot 2^{K-1} \phi(q) \end{aligned}$$

$$\therefore 2^K \rho_2^{K_2} \cdots \rho_r^{K_r} = \phi(q) = \phi(\rho_2^{K_2} \cdots \rho_r^{K_r})$$

$$= \rho_2^{K_2} \cdots \rho_r^{K_r} \left(1 - \frac{1}{\rho_2}\right) \cdots \left(1 - \frac{1}{\rho_r}\right)$$

$$\therefore \rho_2 \cdots \rho_r = (\rho_2 - 1) \cdots (\rho_r - 1)$$

$$\therefore \text{for each } \rho_i, \rho_i = (\rho_j - 1) \text{ for some } j$$

This is impossible if $K_i \geq 0$. $\therefore K_i = 0$, $\therefore$ for $n = 2^K \rho_2^{K_2} \cdots \rho_r^{K_r} = 2^K$

5. Prove $\phi(n) = \phi(n+2)$ is satisfied by $n = 2(2\rho - 1)$ whenever ρ and $2\rho - 1$ are both odd primes.

Pf: $2\rho - 1$ an odd prime $\Rightarrow \gcd(2, 2\rho - 1) = 1$.

$$\therefore \phi(n) = \phi(2)\phi(2\rho - 1) = (2\rho - 1) \left(1 - \frac{1}{2\rho - 1}\right)$$

$$= 2\rho - 2$$

$n + 2 = 2(2\rho - 1) + 2 = 4\rho$, and ρ odd prime

$$\Rightarrow \gcd(4, \rho) = 1.$$

$$\therefore \phi(n + 2) = \phi(4)\phi(\rho) = 2 \cdot \rho \left(1 - \frac{1}{\rho}\right)$$

$$= 2\rho - 2$$

$$\therefore \phi(n) = \phi(n + 2)$$

6. Show there are infinitely many integers for which $\phi(n)$ is a perfect square.

$$\text{Pf: for } K \geq 1, \phi(2^K) = 2^K \left(1 - \frac{1}{2}\right) = 2^{K-1}$$

If K is odd, Then $K - 1$ is even.

Let $K = 2m + 1$, some m .

$$\therefore \phi(2^K) = \phi(2^{2m+1}) = 2^{2m+1-1} = 2^{2m} = (2^m)^2$$

$(2^m)^2$ is a perfect square.

There are infinitely many odd integers, $\therefore$ infinitely many $n = 2^K$, K odd, and $\phi(n)$ is a perfect square.

7. Verify the following:

(a) For any positive integer n , $\frac{1}{2}\sqrt{n} \leq \phi(n) \leq n$ By def, $\phi(n) \leq n$

(1) If $n = 1$, $\frac{1}{2}\sqrt{1} = \frac{1}{2}$, $\phi(1) \geq 1$, so $\frac{1}{2}\sqrt{n} < \phi(n)$

(2) If $n = 2$, Then $\phi(2) = 1$, and $\therefore \frac{1}{2}\sqrt{2} < \phi(2)$

If $n = 2^K$, for $K \geq 1$, Then $\phi(2^K) = 2^{K-1}$

$$\frac{1}{2}\sqrt{2^K} = 2^{-1} \cdot 2^{K/2} = 2^{\frac{K}{2}-1} < 2^{K-1}, \text{ as } 3 \leq 2 < K$$

$$\therefore \frac{1}{2}\sqrt{n} < \phi(n)$$

(3) If $n = \rho^K$, $\rho \geq 2$, $K \geq 1$, Then $\phi(n) = \rho^{K-1}(\rho - 1)$ by Th. 7.2.

But for $\rho \geq 2$, $\rho^2 \geq 3\rho$, so $\rho^2 + 1 \geq 3\rho$, so

$$\rho^2 - 2\rho + 1 \geq \rho, (\rho - 1)^2 \geq \rho, \rho - 1 \geq \sqrt{\rho}$$

$$\therefore \rho^{K-1}(\rho - 1) \geq \rho^{K-1}\sqrt{\rho} \geq \rho^{K-1} \cdot \rho^{\frac{1}{2}} = \rho^K$$

[Note: this line should likely conclude $\rho^{K-1} \cdot \rho^{1/2} = \rho^{K-1/2}$, not ρ^K , and the next line uses $\sqrt{\rho^K}$; transcribed as written.]

$$\therefore \phi(\rho^K) \geq \sqrt{\rho^K} \text{ if } n = \rho^K \text{ and } \rho \geq 2$$

(4) ϕ is multiplicative. Let $n = 2^K \rho_1^{K_1} \cdots \rho_r^{K_r}$, $K \geq 0$, $K_i \geq 0$

$$\begin{aligned} \therefore \phi(n) &= \phi(2^K) \phi(\rho_1^{K_1}) \phi(\rho_2^{K_2}) \cdots \phi(\rho_r^{K_r}) \\ &> \left(\frac{1}{2}\sqrt{2^K}\right) \left(\sqrt{\rho_1^{K_1}}\right) \left(\sqrt{\rho_2^{K_2}}\right) \cdots \left(\sqrt{\rho_r^{K_r}}\right) \text{ by (2),(3)} \\ &= \frac{1}{2}\sqrt{2^K \rho_1^{K_1} \cdots \rho_r^{K_r}} \\ &= \frac{1}{2}\sqrt{n} \\ \therefore \frac{1}{2}\sqrt{n} &< \phi(n) \end{aligned}$$

(b) If $n > 1$ has r distinct prime factors, Then $\phi(n) \geq n/2^r$

Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$

$$\therefore \phi(n) = n \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right)$$

But $\rho_i \geq 2$, so $\frac{1}{2} \geq \frac{1}{\rho_i}$, $-\frac{1}{\rho_i} \geq -\frac{1}{2}$,

$$\therefore 1 - \frac{1}{\rho_i} \geq 1 - \frac{1}{2} = \frac{1}{2}$$

$$\therefore \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \geq \left(\frac{1}{2}\right) \cdots \left(\frac{1}{2}\right) = \frac{1}{2^r}$$

$$\therefore \phi(n) \geq n \cdot \frac{1}{2^r} = n/2^r$$

(c) If $n > 1$ is composite, Then $\phi(n) \leq n - \sqrt{n}$

Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$, $\rho_1 < \rho_2 < \cdots < \rho_r$, $K_i \geq 1$.

$$= \rho_1(b), \text{ and } \rho_1 \leq b \Rightarrow \rho_1^2 \leq \rho_1 b \Rightarrow \rho_1 \leq \sqrt{n}$$

$$= \rho_1(b), \text{ and } \rho_1 \leq b \Rightarrow \rho_1^2 \leq \rho_1 b \Rightarrow \rho_1^2 \leq n$$

[Note: this pair of lines is repeated/garbled in the handwriting – they appear to derive the same conclusion ($\rho_1 \leq \sqrt{n}$) via slightly different intermediate steps; transcribed as written.]

$$\therefore \frac{1}{\sqrt{n}} \leq \frac{1}{\rho_1}, \text{ or } \frac{1}{\sqrt{n}} = \frac{1}{\rho_1}, \text{ so } \sqrt{n} \leq \frac{n}{\rho_1}$$

[Note: this line's algebra is hard to parse exactly from the handwriting; transcribed as written.]

$$\therefore -\frac{n}{\rho_1} \leq -\sqrt{n}$$

$$\phi(n) = n \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right)$$

$$\leq n \left(1 - \frac{1}{\rho_1}\right), \text{ since } 1 - \frac{1}{\rho_i} < 1$$

$$= n - \frac{n}{\rho_1} \leq n - \sqrt{n}$$

8. Prove if n has r distinct odd prime factors, then $2^r \mid \phi(n)$

Pf: Let $n = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}$, $\rho_i > 2$

$$\therefore \phi(n) = \rho_1^{K_1-1}(\rho_1 - 1) \rho_2^{K_2-1}(\rho_2 - 1) \cdots \rho_r^{K_r-1}(\rho_r - 1)$$

As each ρ_i is odd, let $\rho_i = 2s_i + 1$, some s_i .

$$\therefore \phi(n) = \rho_1^{K_1-1} \rho_2^{K_2-1} \cdots \rho_r^{K_r-1} (2s_1)(2s_2) \cdots (2s_r)$$

$$= 2^r \rho_1^{K_1-1} \rho_2^{K_2-1} \cdots \rho_r^{K_r-1} s_1 s_2 \cdots s_r$$

$$\therefore 2^r \mid \phi(n)$$

9. Prove the following:

(a) If n and $n + 2$ are twin primes, Then $\phi(n + 2) = \phi(n) + 2$

Pf: For any prime ρ , $\phi(\rho) = \rho - 1$.

$$\therefore \phi(n + 2) = (n + 2) - 1 = n + 1$$

$$\phi(n) = n - 1$$

$$\therefore \phi(n) + 2 = n - 1 + 2 = n + 1 = \phi(n + 2)$$

(b) If ρ and $2\rho + 1$ are both odd primes, Then $n = 4\rho$ satisfies $\phi(n + 2) = \phi(n) + 2$

Pf: Since ρ is odd, Then $\gcd(4, \rho) = 1$, so $\phi(4\rho) = \phi(4)\phi(\rho) = \phi(4) \cdot \phi(\rho) = 2 \cdot (\rho - 1)$

$$\therefore \phi(n) + 2 = 2 \cdot (\rho - 1) + 2 = 2\rho$$

Since $2\rho + 1$ is prime, $\phi(2\rho + 1) = (2\rho + 1) - 1$

$$= 2\rho$$

$$\therefore \phi(n) + 2 = \phi(n + 2) \text{ for } n = 4\rho$$

10. If every prime that divides n also divides m , establish that $\phi(m \cdot n) = n \cdot \phi(m)$.

Pf: Let $\rho_1, \rho_2, \dots, \rho_r$ be all the primes of n that divide m .

Let $n = \rho_1^{K_1} \dots \rho_r^{K_r}$

$$m = \rho_1^{j_1} \dots \rho_r^{j_r} q_1^{m_1} \dots q_s^{m_s}, \quad q_i \text{ prime}$$

so that $q_i \neq \rho_j$

$$\therefore mn = \rho_1^{K_1+j_1} \dots \rho_r^{K_r+j_r} q_1^{m_1} \dots q_s^{m_s}$$

$$\phi(mn) = \rho_1^{K_1+j_1-1} \dots \rho_r^{K_r+j_r-1} q_1^{m_1-1} \dots q_s^{m_s-1} \left(1 - \frac{1}{\rho_1}\right) \dots \left(1 - \frac{1}{\rho_r}\right) \left(1 - \frac{1}{q_1}\right) \dots \left(1 - \frac{1}{q_s}\right)$$

$$= \rho_1^{j_1-1} \dots \rho_r^{j_r-1} \left(1 - \frac{1}{\rho_1}\right) \dots \left(1 - \frac{1}{\rho_r}\right) \cdot \rho_1^{K_1} \dots \rho_r^{K_r} \cdot q_1^{m_1-1} \dots q_s^{m_s-1} \left(1 - \frac{1}{q_1}\right) \dots \left(1 - \frac{1}{q_s}\right)$$

$$= \phi(m) \cdot \rho_1^{K_1} \dots \rho_r^{K_r} = \phi(m) \cdot n$$

11.(a) If $\phi(n) \mid n - 1$, prove n is square-free.

Pf: Let $n = \rho_1^{K_1} \dots \rho_r^{K_r}$, and assume n is not square-free, so that $K_i \geq 2$ for some i .

$$\phi(n) = (\rho_1^{K_1} - \rho_1^{K_1-1}) \dots (\rho_r^{K_r} - \rho_r^{K_r-1})$$

Since $K_i \geq 2$, $K_i - 1 \geq 1$, so $\rho_i \mid \rho_i^{K_i-1}(\rho_i - 1)$

$$\therefore \rho_i \mid (\rho_i^{K_i} - \rho_i^{K_i-1}) \Rightarrow \rho_i \mid \phi(n)$$

By assumption, $\phi(n) \mid n - 1$, so that $\rho_i \mid n - 1$. Clearly $\rho_i \mid n$,

$$\therefore \rho_i \mid n - (n - 1) \Rightarrow \rho_i \mid 1, \text{ a contradiction.}$$

$$\therefore K_i = 1 \text{ for all } i \Rightarrow n \text{ is square-free.}$$

(b) Show that if $n = 2^K$ or $2^K j^i$, K, j positive,

[Note: " $2^K j^i$ " is unclear/garbled in the handwriting (likely meant to introduce a second case with a different odd prime); transcribed as written.]

Then $\phi(n) \mid n$

Pf: If $n = 2^K$, $\phi(n) = 2^{K-1}$, and $K - 1 \geq 0$ since $K > 0$. $\therefore \phi(n) \mid n$ If $n = 2^K 3^j$, Then $\phi(n) = 2^K 3^{j-1} \left(1 + \frac{1}{2}\right) \left(1 - \frac{1}{3}\right)$

[Note: this line's expansion is hard to follow exactly; the standard expected form would be $\phi(2^K 3^j) = 2^{K-1} 3^{j-1} \cdot 2$. Transcribed as written.]

$$= 2^K 3^{j-1} \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) = 2^K 3^{j-1}$$

Since $j > 0$, $j - 1 \geq 0$. $\therefore \phi(n) \mid n$

12. If $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, derive the following inequalities:

$$(a) \sigma(n)\phi(n) \geq n^2 \left(1 - \frac{1}{\rho_1^2}\right) \cdots \left(1 - \frac{1}{\rho_r^2}\right)$$

$$\text{Pf: } \sigma(n) = \frac{\rho_1^{K_1+1} - 1}{\rho_1 - 1} \cdots \frac{\rho_r^{K_r+1} - 1}{\rho_r - 1}$$

$$\phi(n) = \rho_1^{K_1-1}(\rho_1 - 1) \cdots \rho_r^{K_r-1}(\rho_r - 1)$$

$$\begin{aligned} \therefore \sigma(n)\phi(n) &= (\rho_1^{K_1+1} - 1) \cdots (\rho_r^{K_r+1} - 1) \rho_1^{K_1-1} \cdots \rho_r^{K_r-1} \\ &= (\rho_1^{2K_1} - \rho_1^{K_1-1}) \cdots (\rho_r^{2K_r} - \rho_r^{K_r-1}) \end{aligned}$$

$$\text{But } \rho_i^{2K_i} - \rho_i^{K_i-1} = \rho_i^{2K_i} \left(1 - \frac{\rho_i^{K_i-1}}{\rho_i^{2K_i}}\right)$$

$$= (\rho_i^{K_i})^2 \left(1 - \frac{1}{\rho_i^{K_i+1}}\right)$$

$$\text{For } K_i \geq 1, \rho_i^{K_i+1} \geq \rho_i^2, \text{ so } \frac{1}{\rho_i^{K_i+1}} \leq \frac{1}{\rho_i^2}$$

$$\therefore -\frac{1}{\rho_i^{K_i+1}} \geq -\frac{1}{\rho_i^2}, \text{ so } 1 - \frac{1}{\rho_i^{K_i+1}} \geq 1 - \frac{1}{\rho_i^2}$$

$$\therefore \rho_i^{2K_i} - \rho_i^{K_i-1} \geq (\rho_i^{K_i})^2 \left(1 - \frac{1}{\rho_i^2}\right)$$

$$\begin{aligned} \sigma(n)\phi(n) &\geq (\rho_1^{K_1})^2 \left(1 - \frac{1}{\rho_1^2}\right) \cdots (\rho_r^{K_r})^2 \left(1 - \frac{1}{\rho_r^2}\right) \\ &= n^2 \left(1 - \frac{1}{\rho_1^2}\right) \cdots \left(1 - \frac{1}{\rho_r^2}\right) \end{aligned}$$

$$(b) \tau(n)\phi(n) \geq n$$

Pf: If $2 \leq \rho$, Then $\frac{1}{\rho} \leq \frac{1}{2}$, $-\frac{1}{\rho} \geq -\frac{1}{2}$, $1 - \frac{1}{\rho} \geq \frac{1}{2}$.

If $1 \leq K$, Then $2 \leq K + 1$, so $2 \cdot \frac{1}{2} \leq (K + 1) \left(\frac{1}{2}\right)$, or $1 \leq (K + 1) \left(1 - \frac{1}{\rho}\right)$

$\therefore$ Let $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, $K_i \geq 1$

$$\begin{aligned} \tau(n)\phi(n) &= (K_1 + 1) \cdots (K_r + 1) \cdot n \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\ &= n \cdot \left(1 - \frac{1}{\rho_1}\right)(K_1 + 1) \cdots \left(1 - \frac{1}{\rho_r}\right)(K_r + 1) \\ &\geq n \cdot 1 \cdots 1 = n \\ \therefore \tau(n)\phi(n) &\geq n \end{aligned}$$

13. Assuming that $d \mid n$, prove $\phi(d) \mid \phi(n)$

Pf: Let $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$. Then, by Th. 6.1,

$$d = \rho_1^{a_1} \cdots \rho_r^{a_r}, \text{ where } 0 \leq a_i \leq K_i$$

Let $d = q_1^{b_1} \cdots q_s^{b_s}$, where $q_i \in \{\rho_1, \dots, \rho_r\}$ and $1 \leq b_i$

$$\therefore \phi(d) = d \left(1 - \frac{1}{q_1}\right) \cdots \left(1 - \frac{1}{q_s}\right)$$

Since each $q_i = \rho_j$, some j s.t. $1 \leq j \leq r$, Then $1 - \frac{1}{q_i} = 1 - \frac{1}{\rho_j}$. Name this ρ_{j_i} .

$$\therefore 1 - \frac{1}{q_i} = 1 - \frac{1}{\rho_{j_i}}$$

$$\therefore \phi(d) = d \left(1 - \frac{1}{\rho_{j_1}}\right) \cdots \left(1 - \frac{1}{\rho_{j_s}}\right)$$

As each $\rho_{j_i} \in \{\rho_1, \dots, \rho_r\}$, Then

$$\left(1 - \frac{1}{\rho_{j_1}}\right) \cdots \left(1 - \frac{1}{\rho_{j_s}}\right) \mid \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right)$$

Since $d \mid n$, Then

$$\begin{aligned} d \left(1 - \frac{1}{\rho_{j_1}}\right) \cdots \left(1 - \frac{1}{\rho_{j_s}}\right) &\mid n \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\ \therefore \phi(d) &\mid \phi(n) \end{aligned}$$

14. Obtain the following two generalizations of Th. 7.2:

(a) For positive integers m and n , $d = \gcd(m, n)$,

$$\phi(m)\phi(n) = \phi(mn) \frac{\phi(d)}{d}$$

Pf: (1) If $m = 1$ or $n = 1$, Then $d = 1$, $\phi(d) = 1$ and clearly $\phi(m)\phi(n) = \phi(mn) \cdot \frac{\phi(d)}{d}$

(2) $\therefore$ Assume both $m, n > 1$.

If $m = n$, Then $m = \gcd(n, n)$ Let $m = n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, $mn = \rho_1^{2K_1} \cdots \rho_r^{2K_r}$

$$\begin{aligned} \therefore \phi(mn) \frac{\phi(d)}{d} &= mn \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \cdot \frac{n \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right)}{m} \\ &= mn \left(1 - \frac{1}{\rho_1}\right)^2 \cdots \left(1 - \frac{1}{\rho_r}\right)^2 \\ &= m \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \cdot n \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \\ &= \phi(m)\phi(n) \end{aligned}$$

(3) Now assume $m \neq n$, $m, n > 1$.

If $\gcd(m, n) = 1$, Then $d = 1$, $\phi(d) = 1$, and $\phi(m)\phi(n) = \phi(mn) \cdot \frac{\phi(d)}{d}$ by Th. 7.2.

(4) Assume $m \neq n$, $m, n > 1$, $\gcd(m, n) > 1$.

Let $d = \rho_1^{K_1} \cdots \rho_r^{K_r}$, $K_i \geq 1$.

$$m = \rho_1^{a_1} \cdots \rho_r^{a_r} q_1^{u_1} \cdots q_s^{u_s}, \quad a_i \geq K_i, \quad u_i \geq 0$$

$$n = \rho_1^{b_1} \cdots \rho_r^{b_r} w_1^{v_1} \cdots w_t^{v_t}, \quad b_i \geq K_i, \quad v_i \geq 0$$

Where ρ_i, q_i, w_i are prime, and

$$q_i \neq w_j, \quad \rho_i \neq q_j, \quad \rho_i \neq w_j, \quad \text{for any } i, j.$$

$$\begin{aligned} \therefore mn &= \rho_1^{a_1+b_1} \cdots \rho_r^{a_r+b_r} q_1^{u_1} \cdots q_s^{u_s} w_1^{v_1} \cdots w_t^{v_t} \\ \therefore \phi(mn) \frac{\phi(d)}{d} &= \rho_1^{a_1+b_1-1} \cdots \rho_r^{a_r+b_r-1} q_1^{u_1-1} \cdots q_s^{u_s-1} w_1^{v_1-1} \cdots w_t^{v_t-1} \\ &\quad \left[\left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \left(1 - \frac{1}{q_1}\right) \cdots \left(1 - \frac{1}{q_s}\right) \left(1 - \frac{1}{w_1}\right) \cdots \left(1 - \frac{1}{w_t}\right) \right] \\ &\quad \left[\frac{d}{d} \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \right] \\ &= \left[\rho_1^{a_1} \cdots \rho_r^{b_r} q_1^{u_1} \cdots q_s^{u_s} \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{q_s}\right) \right] \cdot \end{aligned}$$

$$\begin{aligned}
 & \left[\left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \left(1 - \frac{1}{q_1}\right) \cdots \left(1 - \frac{1}{q_s}\right) \right] \\
 &= \left[\rho_1^{a_1} \cdots \rho_r^{a_r} \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \left(1 - \frac{1}{q_1}\right) \cdots \left(1 - \frac{1}{q_s}\right) \right] \cdot \\
 & \left[\rho_1^{b_1} \cdots \rho_r^{b_r} \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \left(1 - \frac{1}{w_1}\right) \cdots \left(1 - \frac{1}{w_t}\right) \right] \\
 &= \phi(m)\phi(n)
 \end{aligned}$$

[Note: this derivation (pages 459-460) is dense and the intermediate regrouping of factors is hard to verify exactly term-by-term from the handwriting; transcribed best-effort. The overall structure and final conclusion are clear.]

Note: if any $u_i = 0$ or $v_i = 0$, some i, j , Then the corresponding term $(1 - \frac{1}{q_i})$ or $(1 - \frac{1}{w_j})$ is not present. (b) For positive integers m, n ,

$$\phi(m)\phi(n) = \phi(\gcd(m, n)) \cdot \phi(\text{lcm}(m, n))$$

Pf: If $m = 1$, Then $\gcd(m, n) = 1$ and $\text{lcm}(m, n) = n$

$$\therefore \text{Clearly } \phi(m)\phi(n) = \phi(\gcd(m, n)) \cdot \phi(\text{lcm}(m, n))$$

Similar reasoning applies for $n = 1$.

If $\gcd(m, n) = 1$, Then $\text{lcm}(m, n) = mn$, and $\therefore$ the relation holds.

If $m = n$, $\gcd(m, n) = m = \text{lcm}(m, n)$, so the relation holds.

If $\gcd(m, n) = m$, Then $\text{lcm}(m, n) = n$, so the relation holds, and similarly for $\gcd(m, n) = n$.

$\therefore$ Assume $m \neq n$, $\gcd(m, n) > 1$, and $\gcd(m, n) \neq m$ or n .

$\therefore$ Let $d = \gcd(m, n) = \rho_1^{K_1} \cdots \rho_r^{K_r}$, $K_i \geq 1$

$$m = \rho_1^{a_1} \cdots \rho_r^{a_r} q_1^{u_1} \cdots q_s^{u_s}, \quad a_i \geq K_i, \quad u_i \geq 0$$

$$n = \rho_1^{b_1} \cdots \rho_r^{b_r} w_1^{v_1} \cdots w_t^{v_t}, \quad b_i \geq K_i, \quad v_i \geq 0$$

with ρ_i, q_i, w_i prime,

$$\rho_i \neq q_j, \quad 1 \leq i \leq r, \quad 1 \leq j \leq s$$

$$\rho_i \neq w_j, \quad 1 \leq i \leq r, \quad 1 \leq j \leq t$$

$$q_i \neq w_j, \quad 1 \leq i \leq r, \quad 1 \leq j \leq t$$

and $u_i \neq 0$ some i , $v_j \neq 0$ some j since $\gcd(m, n) \neq m$ or n .

Since $mn = \gcd(m, n) \cdot \text{lcm}(m, n)$,

$$\text{lcm}(m, n) = mn / \gcd(m, n)$$

$$= \rho_1^{a_1+b_1-K_1} \cdots \rho_r^{a_r+b_r-K_r} q_1^{u_1} \cdots q_s^{u_s} w_1^{v_1} \cdots w_t^{v_t}$$

where $a_i + b_i - K_i \geq 0$ since $a_i \geq K_i, b_i \geq K_i$

$$\begin{aligned} \therefore \phi(\gcd(m, n)) \cdot \phi(\text{lcm}(m, n)) &= \left[\rho_1^{K_1} \cdots \rho_r^{K_r} \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \right] \cdot \\ &\quad \left[\rho_1^{a_1+b_1-K_1} \cdots \rho_r^{a_r+b_r-K_r} q_1^{u_1} \cdots q_s^{u_s} w_1^{v_1} \cdots w_t^{v_t} \right] \\ &= \left[\left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \left(1 - \frac{1}{q_1}\right) \cdots \left(1 - \frac{1}{q_s}\right) \left(1 - \frac{1}{w_1}\right) \cdots \left(1 - \frac{1}{w_t}\right) \right] \\ &\quad = \left[\rho_1^{a_1+b_1-1} \cdots \rho_r^{a_r+b_r-1} q_1^{u_1-1} \cdots q_s^{u_s-1} w_1^{v_1-1} \cdots w_t^{v_t-1} \right] \cdot \\ &\quad \left[\left(1 - \frac{1}{\rho_1}\right)^2 \cdots \left(1 - \frac{1}{\rho_r}\right)^2 \left(1 - \frac{1}{q_1}\right) \left(1 - \frac{1}{q_s}\right) \left(1 - \frac{1}{w_1}\right) \cdots \left(1 - \frac{1}{w_t}\right) \right] \\ &= \left[\rho_1^{a_1} \cdots \rho_r^{a_r} q_1^{u_1} \cdots q_s^{u_s} \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \left(1 - \frac{1}{q_1}\right) \cdots \left(1 - \frac{1}{q_s}\right) \right] \cdot \\ &\quad \left[\rho_1^{b_1} \cdots \rho_r^{b_r} w_1^{v_1} \cdots w_t^{v_t} \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) \left(1 - \frac{1}{w_1}\right) \cdots \left(1 - \frac{1}{w_t}\right) \right] \\ &= \phi(m) \cdot \phi(n) \end{aligned}$$

15. Prove the following:

(a) There are infinitely many n for which $\phi(n) = n/3$

Pf: consider $n = 2^i 3^j, i, j \geq 1$

$$\therefore \phi(n) = n \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) = n \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) = n/3$$

(b) There are no integers n for which $\phi(n) = n/4$

Pf: $\phi(1) = 1, \phi(2) = 1, \phi(3) = 2, \phi(4) = 2$.

So for $n = 1, 2, 3, 4, \phi(n) \neq n/4$. Assume $n > 4$ and $\phi(n) = n/4$.

Let $n = \rho_1^{K_1} \cdots \rho_r^{K_r}, K_i \geq 1$

$$\therefore \phi(n) = n \left(1 - \frac{1}{\rho_1}\right) \cdots \left(1 - \frac{1}{\rho_r}\right) = n/4$$

$$\therefore \frac{(\rho_1 - 1) \cdots (\rho_r - 1)}{\rho_1 \cdots \rho_r} = \frac{1}{4}, \text{ or}$$

$$4(\rho_1 - 1) \cdots (\rho_r - 1) = \rho_1 \cdots \rho_r$$

If $\rho_1 = 2$, Then $2(\rho_2 - 1) \cdots (\rho_r - 1) = \rho_2 \cdots \rho_r$

But $\rho_2 \cdots \rho_r$ is odd since $\rho_2, \dots, \rho_r$ are all odd. And $2(\rho_2 - 1) \cdots (\rho_r - 1)$ is even.

$\therefore$ Can't work for $\rho_1 = 2$.

And if all $\rho_1, \dots, \rho_r$ are odd, so is $\rho_1 \cdots \rho_r$ and $4(\rho_1 - 1) \cdots (\rho_r - 1)$ is even.

$\therefore$ No such n exists.

16. Show that the Goldbach conjecture implies that for each even integer $2n$ there exists integers n_1 and n_2 with $\phi(n_1) + \phi(n_2) = 2n$.

Pf: Goldbach conjecture says for any even integer greater than 4, there exist two odd primes n_1 and n_2 s.t. $n_1 + n_2 =$ the even integer.

Let $2n + 2$ be such an even integer, so that $n_1 + n_2 = 2n + 2$, n_1, n_2 odd primes.

If n_1 and n_2 are both prime, then $\phi(n_1) = n_1 - 1$, $\phi(n_2) = n_2 - 1$

$$\therefore \phi(n_1) + \phi(n_2) = n_1 + n_2 - 2 = 2n + 2 - 2 = 2n$$

And if $2n = 4$, choose $n_1 = n_2 = 4$, so

[Note: $n_1 = n_2 = 4$ contradicts the requirement that they be odd primes; likely meant $n_1 = n_2 = 3$ or similar. Transcribed as written.]

that $\phi(4) + \phi(4) = 2 + 2 = 2n$.

If $2n = 2$, choose $n_1 = n_2 = 1$, so that

$$\phi(1) + \phi(1) = 2$$

17. Given a positive integer K , show:

(a) There are at most a finite number of integers n for which $\phi(n) = K$. *Pf:* Need to find an integer z , s.t. whenever $n \geq z$, $\phi(n) > K$. Thus, there are at most a finite # of integers, $1, 2, \dots, z - 1$, for which $\phi(n)$ may be equal to K .

By problem 7(a) above, it was proved that $\phi(n) \geq \frac{1}{2}\sqrt{n}$, for all n .

$$\therefore \text{choose } z = 4K^2 : \phi(z) \geq \frac{1}{2}\sqrt{4K^2} = K$$

$$\therefore \text{For all integers } n > 4K^2, \phi(n) > K$$

$$\therefore \text{There are at most } n = 4K^2 \text{ integers}$$

for which $\phi(n)$ may be K .

Note: it was important in 7(c) to prove that $\phi(n) \geq \frac{1}{2}\sqrt{n}$, not just $\phi(n) \geq \sqrt{n}$, $\geq 2^2$ does not give proper bound.

(b) If the equation $\phi(n) = K$ has a unique solution, say $n = n_0$, Then $4 \mid n_0$.

Pf: Suppose $\phi(n_0) = K$, and n_0 is unique.

If n_0 is odd, Then by problem 4(a), $\phi(2n_0) = \phi(n_0) = K$, so that n_0 is not unique.

$\therefore n_0$ is even, so $n_0 = 2r$, some r .

If r is odd, Then $\gcd(2, r) = 1$, $\phi(n_0) = \phi(2r)$

$$= \phi(2)\phi(r) = \phi(r) = K,$$

so, again, uniqueness of $n_0 \Rightarrow r$ is even $\Rightarrow r = 2s$, some s .

$$\therefore n_0 = 2(2s) = 4s \Rightarrow 4 \mid n_0.$$

18. Find all solutions $\phi(n) = 16$ and $\phi(n) = 24$.

Note: if $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, Then $\phi(n) = (\rho_1^{K_1} - \rho_1^{K_1-1}) \cdots (\rho_r^{K_r} - \rho_r^{K_r-1})$

Since $(\rho_i - 1) \mid \rho_i^{K_i} - \rho_i^{K_i-1}$, Then $(\rho_i - 1) \mid \phi(n)$

(a) For $\phi(n) = 16$, $(\rho_i - 1) \mid 2^4$

$$\therefore \rho_i - 1 = \{2, 4, 8, \text{ or } 16\}$$

$$\therefore \rho_i = 2, 3, 5, 9, \text{ or } 17, \text{ and } 9 \text{ not prime,}$$

so $\rho_i = 2, 3, 5, \text{ or } 17$

$$\therefore n = 2^{K_1} 3^{K_2} 5^{K_3} 17^{K_4}$$

$$\begin{aligned} 16 &= (2^{K_1} - 2^{K_1-1})(3^{K_2} - 3^{K_2-1})(5^{K_3} - 5^{K_3-1})(17^{K_4} - 17^{K_4-1}) \\ &= 2^{K_1-1}(3^{K_2-1})(2)(5^{K_3-1})(4)(17^{K_4-1})(16) \end{aligned}$$

16 clearly has an upper bound effect.

$$\therefore K_4 \leq 1, K_3 \leq 1, K_2 \leq 3, K_1 \leq 5$$

[Note: this constraint line's second " $K_2 \leq 5$ " likely should read " $K_1 \leq 5$ "; transcribed as written.]

If $K_4 = 1$, $17^{K_4-1} \cdot 16 = 16$, so $K_1 = 0$,

$$\therefore K_2 = K_3 = 0, n = 17$$

$\therefore$ Consider cases for $K_4 = 0$.

$$\therefore 16 = 2^{K_1-1}(3^{K_2-1} \cdot 2)(5^{K_3-1} \cdot 4)$$

If $K_3 = 1$, Then $K_3 = 2$, $K_1 = 2$, $n = 60$

[Note: the case condition " $K_3 = 1$, Then $K_3 = 2$ " is confusingly self-referential; presumably meant $K_2 = 1$ in places. Transcribed as written.]

or $K_2 = 0$, $K_1 = 3$, $n = 40$

If $K_3 = 0$, Then $K_2 = 1$, $K_1 = 4$, $n = 48$ or $K_2 = 0$, $K_1 = 5$, $n = 32$

$$\therefore \text{For } \phi(n) = 16, n = 17, 34, 40, 60, 32, 48$$

(b) For $\phi(n) = 24$, $(\rho_i - 1) \mid 2^3 \cdot 3$

$$\therefore (\rho_i - 1) \mid 2^3 \text{ or } (\rho_i - 1) \mid 3$$

$$\therefore \rho_i - 1 = 1, 2, 4, 8 \text{ or } \rho_i - 1 = 3, 6, 12, 24$$

$$\therefore \rho_i = 2, 3, 5 \text{ or } \rho_i = 7, 13$$

$$\therefore n = 2^{K_1} 3^{K_2} 5^{K_3} 7^{K_4} 13^{K_5}$$

$$24 = (2^{K_1-1})(3^{K_2-1} \cdot 2)(5^{K_3-1} \cdot 4)(7^{K_4-1} \cdot 6)(13^{K_5-1} \cdot 12)$$

$$\therefore K_5 \leq 1, K_4 \leq 1, K_3 \leq 2, K_2 \leq 3, K_1 \leq 5$$

For $K_5 = 1, K_4 = 0, K_3 = 0$: $K_2 = 0, K_1 = 1, n = 39$

$$K_2 = 0, K_1 = 1 : n = 78$$

$$K_2 = 0, K_1 = 2 : n = 26$$

[Note: this case list (page 469) is dense and the K -value combinations listed are hard to fully verify against the constraints derived above; transcribed as written.]

$$\therefore \text{Now assume } K_5 = 0 \quad (n = 2^{K_1} 3^{K_2} 5^{K_3} 7^{K_4})$$

$$\therefore 24 = (2^{K_1-1})(3^{K_2-1} \cdot 2)(5^{K_3-1} \cdot 4)(7^{K_4-1} \cdot 6)$$

If $K_4 = 1$: $4 = (2^{K_1-1})(3^{K_2-1} \cdot 2)(5^{K_3-1} \cdot 4)$

$$K_3 = 1, K_2 = 0, K_1 = 0 : n = 35$$

$$K_3 = 0, K_2 = 1, K_1 = 0 : n = 70$$

$$K_3 = 0, K_2 = 0, K_1 = 3 : n = 56$$

$$K_3 = 0, K_2 = 1, K_1 = 2 : n = 84$$

Now assume $K_5 = 0, K_4 = 0$ $(n = 2^{K_1} 3^{K_2} 5^{K_3})$

$$\therefore 24 = (2^{K_1-1})(3^{K_2-1} \cdot 2)(5^{K_3-1} \cdot 4). \quad K_3 \leq 1$$

$$\therefore \text{If } K_3 = 1, 6 = (2^{K_1})(3^{K_2-1} \cdot 2)$$

$$K_2 \neq 0, K_4 \neq 1$$

[Note: this condition line is unclear/garbled; transcribed as written.]

$$K_2 = 2, K_1 = 0 : n = 45$$

$$K_2 = 2, K_1 = 2 : n = 10$$

[Note: these two cases as listed don't appear consistent with the constraint equation above (the K_1 values seem swapped or miscalculated); transcribed as written.]

$\therefore$ Assume $K_5 = K_4 = K_3 = 0$ ($n = 2^{K_1} 3^{K_2}$)

$$\therefore 24 = (2^{K_1-1})(3^{K_2-1} \cdot 2)$$

If $K_2 = 0$, no solution If $K_2 = 1$, no solution If $K_2 = 2$, $K_1 = 3$, $n = 72$ If $K_2 = 3$, no solution

$\therefore$ For $\phi(n) = 24$,

$$n = 39, 78, 52, 15, 70, 56, 84, 45, 90, 72$$

19.(a) Prove that the equation $\phi(n) = 2\rho$, where ρ is prime and $2\rho + 1$ is composite, is not solvable.

Pf: Let $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$

$$\phi(n) = \rho_1^{K_1-1} \rho_2^{K_2-1} \cdots \rho_r^{K_r-1} (\rho_1 - 1) \cdots (\rho_r - 1)$$

$$= \rho_1^{K_1-1} \cdots \rho_r^{K_r-1} (\rho_1 - 1) \cdots (\rho_r - 1)$$

Suppose $\phi(n) = 2\rho$ ($2\rho + 1$ is composite). Then $\rho \neq 2$ as $2\rho + 1 = 5$. Also, $n \neq 1$.

(a) Suppose n consists of more than one odd prime factor, $\rho_j \mid \rho_K$

[Note: " $\rho_j \mid \rho_K$ " appears garbled/abbreviated; likely meant to denote two distinct odd prime factors ρ_j, ρ_K of n .]

$$\therefore \text{In } \phi(n), (\rho_j - 1) \mid (\rho_K - 1) = 2q_j \cdot 2q_K$$

$$\therefore \phi(n) = 2q_j \cdot 2q_K \cdot Q = 2\rho, \text{ when}$$

$Q = \text{other factors in } \phi(n).$

$$\therefore q_j \cdot 2q_K \cdot Q = \rho, \text{ so } \rho \text{ is even}$$

$$\Rightarrow \rho = 2. \text{ But } 2\rho + 1 = 5 \text{ is not}$$

composite.

$\therefore n$ can consist of at most one odd prime factor.

(b) $\therefore$ Let $n = 2^K \rho_1^{K_1}$, $K \geq 0, K_1 \geq 0$

(1) Suppose $K_1 = 0$, so $n = \rho_1^{K_1}$, $K_1 > 0$

$$\therefore \phi(n) = \rho_1^{K_1-1} (\rho_1 - 1) = 2\rho$$

If $K_1 = 1$, Then $\rho_1 - 1 = 2\rho$, $\rho_1 = 2\rho + 1$, and $2\rho + 1$ composite $\Rightarrow \rho_1$ is not prime.

If $K_1 > 1$, Then let $\rho_1 - 1 = 2r$

$$\therefore \phi(n) = \rho_1^{K_1-1} \cdot 2r = 2\rho,$$

$$\rho_1^{K_1-1} \cdot r = \rho \Rightarrow r = \rho_1 = \rho$$

$$\therefore \rho_1^{K_1-1} = 1 \Rightarrow K_1 = 1$$

$$\therefore \phi(n) = (\rho - 1) = 2\rho, \quad 0 = \rho + 1.$$

$$\therefore K \neq 0, \quad n = 2^K \rho_1^{K_1}, \quad K > 0, K_1 \geq 0$$

(ii) Assume $K = 1$

(1) If $K_1 = 0$, Then $n = 2$, $\phi(n) = 1 \neq 2\rho$

(2) If $K_1 = 1$, Then $n = 2\rho_1$, $\phi(n) = \rho_1 - 1$

$$\therefore \rho_1 - 1 = 2\rho, \quad \rho_1 = 2\rho + 1 \Rightarrow \rho_1 \text{ composite.}$$

(3) If $K_1 > 1$, Then $n = 2\rho_1^{K_1}$,

$$\phi(n) = \rho_1^{K_1-1}(\rho_1 - 1) = 2\rho$$

$$\text{Let } \rho_1 - 1 = 2r, \therefore \rho_1^{K_1-1} \cdot 2r = 2\rho,$$

$$\therefore \rho_1^{K_1-1} \cdot r = \rho. \quad \rho \text{ prime} \Rightarrow r = 1$$

and $K_1 = 2 \Rightarrow \rho_1 = \rho$.

$$\therefore n = 2\rho^2, \quad \phi(n) = \rho(\rho - 1) = 2\rho$$

$$\Rightarrow \rho - 1 = 2, \quad \rho = 3 \Rightarrow 2\rho + 1 = 7,$$

which is not composite.

$$\therefore (1), (2), (3) \Rightarrow K \neq 1$$

$$\therefore (i), (ii) \Rightarrow K > 1$$

(iii) $\therefore n = 2^K \rho_1^{K_1}, \quad K > 1, K_1 \geq 0$

(1) If $K_1 = 0$, $n = 2^K$, $\phi(n) = 2^{K-1} = 2\rho$

$$\therefore \rho = 2 \Rightarrow 2\rho + 1 = 5 \text{ is composite.}$$

(2) If $K_1 = 1$, Then $n = 2^K \rho_1$,

$$\therefore \phi(n) = 2^{K-1}(\rho_1 - 1) = 2\rho$$

$$\therefore 2^{K-2}(\rho_1 - 1) = \rho$$

Only possibility is $\rho = 2$ or $\rho_1 - 1 = \rho$

$\rho = 2 \Rightarrow 2\rho + 1 = 5$ is composite $\rho_1 - 1 = \rho \Rightarrow \rho_1 = \rho + 1 \Rightarrow \rho_1$ is even and ρ_1 is supposed to be odd.

(iv) $\therefore n = 2^K \rho_1^{K_1}, \quad K > 1, K_1 > 1$

$$\therefore \phi(n) = 2^{K-1} \rho_1^{K_1-1}(\rho_1 - 1) = 2\rho$$

$$\Rightarrow \rho_1^{K_1-1}(\rho_1 - 1) = \rho$$

$\rho \neq 2$ as $2\rho + 1$ must be composite.

$\therefore \rho$ is odd but $(\rho_1 - 1)$ is even, so $\rho_1^{K_1-1}(\rho_1 - 1)$ is even.

$\therefore (i), (ii), (iii),$ and $(iv) \Rightarrow$ There is no

K, K_1 s.t. $n = 2^K \rho_1^{K_1}$, with $\phi(n) = 2\rho$ and $2\rho + 1$ composite.

(b) Prove there is no solution to the equation $\phi(n) = 14$, and that 14 is the smallest positive even integer with this property.

Pf: From (a), $\phi(n) = 2 \cdot 7$, and $2(7) + 1 = 15$ is composite,

$\therefore \phi(n) = 14$ is not solvable.

$$\phi(13) = 12, \phi(11) = 10, \phi(7) = 6,$$

$$\phi(5) = 4, \phi(3) = 2$$

for $\phi(n) = 8$, note if $n = 2^K$, $\phi(n) = 2^{K-1}$

$$\therefore 2^3 = 8 = 2^{4-1}, \text{ so } n = 16$$

$$\therefore \phi(16) = 8$$

20. If ρ is prime and $K \geq 2$, show that

$$\phi(\phi(\rho^K)) = \rho^{K-2}\phi((\rho - 1)^2)$$

Pf: $\phi(\rho^K) = \rho^{K-1}(\rho - 1)$

Since $\gcd(\rho, \rho - 1) = 1$, Then $\gcd(\rho^{K-1}, \rho - 1) = 1$ ϕ is multiplicative,

$$\therefore \phi(\phi(\rho^K)) = \phi(\rho^{K-1}(\rho - 1))$$

$$= \phi(\rho^{K-1})\phi(\rho - 1)$$

$$= \rho^{K-2}(\rho - 1)\phi(\rho - 1)$$

From problem 10, $\phi(n^K) = n^{K-1}\phi(n)$ for every positive integer n .

$$\therefore (\rho - 1)\phi(\rho - 1) = \phi((\rho - 1)^2)$$

$$\therefore \phi(\phi(\rho^K)) = \rho^{K-2}(\rho - 1)\phi(\rho - 1) = \rho^{K-2}\phi((\rho - 1)^2)$$

21. Verify that $\phi(n)\tau(n)$ is a perfect square when $n = 2345 = 23 \cdot 31 \cdot 89$.

Pf: If $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, and $K_i = 1$, Then

$$\phi(n) = (\rho_1 - 1) \cdots (\rho_r - 1)$$

$$\tau(n) = \frac{(\rho_1^{K_1+1} - 1)}{\rho_1 - 1} \cdots \frac{(\rho_r^{K_r+1} - 1)}{\rho_r - 1}$$

$$\begin{aligned}
 \therefore \phi(n)\tau(n) &= (\rho_1^2 - 1) \cdots (\rho_r^2 - 1) = (\rho_1 - 1)(\rho_1 + 1) \cdots (\rho_r - 1)(\rho_r + 1) \\
 \therefore \text{for } n = 23 \cdot 31 \cdot 89, \phi(n)\tau(n) &= (23^2 - 1)(31^2 - 1)(89^2 - 1) \\
 &= (22)(24)(30)(32)(88)(10) \\
 &= (2 \cdot 11)(2^2 \cdot 3)(2 \cdot 3 \cdot 5)(2^5)(2^3 \cdot 11)(2 \cdot 5) \\
 &= 2^{14} \cdot 3^4 \cdot 5^2 \cdot 11^2 \\
 &= (2^7 \cdot 3^2 \cdot 5 \cdot 11)^2
 \end{aligned}$$

7.3 Euler's Generalization of Fermat's Theorem

1. Use Euler's Theorem to establish the following:

(a) For any integer a , $a^{37} \equiv a \pmod{1729}$

Pf: $1729 = 7 \cdot 13 \cdot 19$. $\phi(7) = 6$, $\phi(13) = 12$, $\phi(19) = 18$

$$\therefore a^{18} \equiv 1 \pmod{19} \Rightarrow a^{36} \equiv 1 \pmod{19}$$

$$a^{12} \equiv 1 \pmod{13} \Rightarrow a^{36} \equiv 1 \pmod{13}$$

[Note: the modulus on this line should likely read $\pmod{13}$, not $\pmod{18}$; transcribed as written.]

$$a^6 \equiv 1 \pmod{7} \Rightarrow a^{36} \equiv 1 \pmod{7}$$

$$\therefore a^{36} \equiv 1 \pmod{7 \cdot 13 \cdot 19} \quad [\text{prob. \#13, sec. 4.2}]$$

$$\therefore a^{37} \equiv a \pmod{1729}$$

(b) For any integer a , $a^{13} \equiv a \pmod{2730}$

Pf: $2730 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 13$. $\phi(2) = 1$, $\phi(3) = 2$, $\phi(5) = 4$, $\phi(7) = 6$, $\phi(13) = 12$

$$\therefore a \equiv 1 \pmod{2} \Rightarrow a^{12} \equiv 1 \pmod{2}$$

$$a^2 \equiv 1 \pmod{3} \Rightarrow a^{12} \equiv 1 \pmod{3}$$

$$a^4 \equiv 1 \pmod{5} \Rightarrow a^{12} \equiv 1 \pmod{5}$$

$$a^6 \equiv 1 \pmod{7} \Rightarrow a^{12} \equiv 1 \pmod{7}$$

$$a^{12} \equiv 1 \pmod{13}$$

$$\therefore a^{12} \equiv 1 \pmod{2 \cdot 3 \cdot 5 \cdot 7 \cdot 13} \quad [\text{prob. \#13, sec. 4.2}]$$

$$\therefore a^{13} \equiv a \pmod{2730}$$

(c) For any odd integer a , $a^{33} \equiv a \pmod{4080}$

Pf: $4080 = 15 \cdot 16 \cdot 17 = 15 \cdot 2 \cdot 4 \cdot 2 \cdot 17$

[Note: this factorization line is hard to follow exactly from the handwriting; transcribed as written.]

$\gcd(a, 2 \cdot 15) = 1$ since a is odd

$$\therefore a^{\phi(30)} \equiv 1 \pmod{30}$$

$$\phi(30) = 8 \Rightarrow a^8 \equiv 1 \pmod{30}$$

$$\Rightarrow a^{32} \equiv 1 \pmod{30}$$

$\gcd(a, 2 \cdot 17) = 1$ since a is odd

$$\therefore a^{\phi(34)} \equiv 1 \pmod{34}$$

$$\phi(34) = (2-1)(17-1) = 16$$

$$\therefore a^{16} \equiv 1 \pmod{34} \Rightarrow a^{32} \equiv 1 \pmod{34}$$

$\gcd(a, 16) = 1$ since a is odd

$$\therefore a^{\phi(16)} \equiv 1 \pmod{16}.$$

$$\phi(16) = 8, \therefore a^8 \equiv 1 \pmod{16}$$

$$\therefore a^{32} \equiv 1 \pmod{16}$$

$$\text{lcm}(30, 34, 16) = 4080$$

$$\therefore a^{32} \equiv 1 \pmod{4080} \quad [\text{prob. \#13, sec. 4.2}]$$

$$\therefore a^{33} \equiv a \pmod{4080}$$

2. Use Euler's Theorem to confirm that, for any integer $n \geq 0$, $51 \mid 10^{32n+9} - 7$

Pf: $51 = 17 \cdot 3. \therefore \phi(51) = 16 \cdot 2 = 32$

$$\gcd(10, 51) = 1, \therefore 10^{\phi(51)} = 10^{32} \equiv 1 \pmod{51}$$

$$\therefore 10^{32n} \equiv 1 \pmod{51} \quad [1]$$

Is $10^9 \equiv 7 \pmod{51}$

$$10 \equiv 7 \pmod{3}$$

$$10 \equiv 1 \pmod{3} \Rightarrow 10^3 \equiv 1 \pmod{3}$$

$$\therefore 10^9 = 10^{3 \cdot 3} = 7^3 \cdot 10 \equiv 1 \pmod{3}, \text{ or}$$

$$10^9 \equiv 7 \pmod{3} \quad [2]$$

$$-10 \equiv 7 \pmod{17}$$

$$\begin{aligned}
 (-10)^3 &\equiv 7^3 = 343 \equiv -2 \pmod{17} \\
 -(10)^3 &= 10^3 = (-2)^3 = -8 \equiv 16 \equiv -1 \pmod{17} \\
 \therefore 10^9 &= 10^{3 \cdot 3} = (10^3)^3 = (-1)^3 = -1 \equiv 16 \equiv 7 \pmod{17} \quad [3] \\
 [2] + [3] &\Rightarrow 10^9 \equiv 7 \pmod{51} \quad [4] \\
 [1] + [4] &\Rightarrow 10^{32n} \cdot 10^9 \equiv 1 \cdot 7 \pmod{51}, \text{ or} \\
 10^{32n+9} &\equiv 7 \pmod{51} \\
 \therefore 51 &\mid 10^{32n+9} - 7
 \end{aligned}$$

3. Prove $2^{15} - 2^3$ divides $a^{15} - a^3$ for any integer a .

$$\begin{aligned}
 \text{Pf: } a^{15} - a^3 &= a^3(a^{12} - 1) = a^3(a^6 + 1)(a^6 - 1) \\
 &= a^3(a^6 + 1)(a^3 + 1)(a^3 - 1) \\
 &= a^3(a^6 + 1)(a^2 + 1)(a^4 - a^2 + 1)(a - 1) \\
 2^{15} - 2^3 &= 2^3(2^{12} - 1) = 2^3(2^6 + 1)(2^6 - 1) \\
 &= 2^3(2^{2 \cdot 3} + 1)(2^{3+1})(2^2 + 2 + 1)(2 - 1) \\
 &= 2^3(2^6 + 1)(2^3 + 1)(2^{2+2+1})(2 - 1) \\
 &= 8(65)(9)(7) \\
 &= 8 \cdot 5 \cdot 13 \cdot 9 \cdot 7 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13
 \end{aligned}$$

[Note: this factorization-comparison sequence is dense and somewhat hard to follow exactly from the handwriting; transcribed best-effort.]

Could go through each factor and show it divides $a^{15} - a^3$.

However, easier to use Euler's Theorem.

$$\phi(8) = 4, \phi(5) = 4, \phi(13) = 12, \phi(7) = 6, \phi(9) = 6$$

(a) If $\gcd(a, 2^{15} - 2^3) = 1$, then

Pf:

$$\begin{aligned}
 \gcd(a, 8) &= 1, \gcd(a, 4) = 1, \gcd(a, 2) = 1 \\
 \gcd(a, 5) &= 1, \gcd(a, 9) = 1 \\
 \therefore a^4 &\equiv 1 \pmod{9}, a^{12} \equiv 1 \pmod{13}, a^4 \equiv 1 \pmod{5} \\
 a^{12} &\equiv 1 \pmod{9}, a^6 \equiv 1 \pmod{9}
 \end{aligned}$$

Make all $a^{12} \equiv 1$, so that

$$a^{12} \equiv 1 \pmod{8 \cdot 5 \cdot 13 \cdot 9 \cdot 7}$$

$$\therefore a^{15} \equiv a^3 \pmod{2^{15} - 2^3}$$

(b) If $\gcd(a, 2^{15} - 2^3) \neq 1$, then

$$a = K(2^{15} - 2^3), \text{ some } K, \text{ and}$$

$$\therefore a^{15} - a^3 = (a^{12} - a)(2^{15} - 2^3)K(2^{15} - 2^3)$$

[Note: this line's algebra is hard to follow exactly from the handwriting; transcribed as written.]

$$\therefore a^{15} \equiv a^3 \pmod{2^{15} - 2^3}$$

$$\therefore (a) + (b) \Rightarrow \text{for all } a, a^{15} \equiv a^3 \pmod{2^{15} - 2^3}$$

4. Show that if $\gcd(a, n) = \gcd(a - 1, n) = 1$, then

$$1 + a + a^2 + \cdots + a^{\phi(n)-1} \equiv 0 \pmod{n}$$

Pf: By Euler, $\gcd(a, n) = 1 \Rightarrow a^{\phi(n)} \equiv 1 \pmod{n}$

$$\therefore a^{\phi(n)} - 1 \equiv 0 \pmod{n}$$

But $a^{\phi(n)} - 1 = (a - 1)(a^{\phi(n)-1} + \cdots + a^2 + a + 1)$

$$\therefore (a - 1)(a^{\phi(n)-1} + \cdots + a^2 + a + 1) \equiv 0 \pmod{n}$$

Since $\gcd(a - 1, n) = 1$, can cancel $(a - 1)$,

$$\therefore (a^{\phi(n)-1} + \cdots + a^2 + a + 1) \equiv 0 \pmod{n}$$

5. If m and n are relatively prime positive integers, prove $m^{\phi(n)} + n^{\phi(m)} \equiv 1 \pmod{mn}$

Pf: Since $\gcd(m, n) = 1$, Then

$$m^{\phi(n)} \equiv 1 \pmod{n} \text{ and } n^{\phi(m)} \equiv 1 \pmod{m}$$

But $n^{\phi(m)} \equiv 0 \pmod{n}$ and $m^{\phi(n)} \equiv 0 \pmod{m}$

$$\therefore m^{\phi(n)} + n^{\phi(m)} \equiv 1 + 0 = 1 \pmod{n}$$

$$n^{\phi(m)} + m^{\phi(n)} \equiv 1 + 0 = 1 \pmod{m}$$

By prob.#13, Sec. 4.2, $m^{\phi(n)} + n^{\phi(m)} \equiv 1 \pmod{mn}$

6. Fill in any missing details in the following proof to Euler's Theorem:

Let ρ be a prime divisor of n and $\gcd(a, \rho) = 1$.

Note: if n is prime, choose $\rho = n$, and there are $\rho - 1$ choices for a .

By Fermat's Theorem, $a^{\rho-1} \equiv 1 \pmod{\rho}$, so that

$$a^{\rho-1} = 1 + x\rho \text{ for some } x. \text{ det. } a \equiv 6 \pmod{\rho}$$

$$\therefore a^{\rho(\rho-1)} = (1 + x\rho)^\rho = 1 + (x\rho) + \cdots + (x\rho)^\rho \equiv 1 \pmod{\rho^2}$$

Expansion by Binomial Thm. Also, $\rho \mid \binom{\rho}{i}$ for $1 \leq i \leq \rho - 1$. So each term in $(1 + x\rho)^\rho$ contains ρ^2 .

By induction $a^{K\rho^{K-1}(\rho-1)} \equiv 1 \pmod{\rho^K}$, $K = 1, 2, \dots$

For $K = 1$: $a^{\rho^0(\rho-1)} = a^{\rho-1} \equiv 1 \pmod{\rho}$, by Fermat.

For $K \geq K + 1$: Assume $a^{K\rho^{K-1}(\rho-1)} \equiv 1 \pmod{\rho^K}$

$$\therefore a^{\rho^{K-1}(\rho-1)} = 1 + q\rho^K, \text{ some } q.$$

Note $\rho^{(K+1)-1}(\rho-1) = \rho^K(\rho-1) = \rho[\rho^{K-1}(\rho-1)]$

$$\begin{aligned} \therefore a^{\rho^{(K+1)-1}(\rho-1)} &= (a^{\rho^{K-1}(\rho-1)})^\rho = (1 + q\rho^K)^\rho \\ &= 1 + \binom{\rho}{1}(q\rho^K) + \cdots + (q\rho^K)^\rho \end{aligned}$$

Since $\rho \mid \binom{\rho}{i}$, Then $\rho^{K+1} \mid \binom{\rho}{i}(q\rho^K)^i + \cdots + (q\rho^K)^\rho$

$$\therefore a^{\rho^{(K+1)-1}(\rho-1)} = 1 + q'\rho^{K+1} \text{ completing induction}$$

Raise both sides of this congruence to the $\phi(n)/\rho^{K-1}(\rho-1)$ power and you get $a^{\phi(n)} \equiv 1 \pmod{\rho^K}$.

[Annotations in red ink, presumably instructor/student notes:]

Since ρ is a prime divisor of n , let K be the power of ρ in the prime factorization of n . $\therefore$ By Th. 7.3, $\phi(n) = \rho^{K-1}(\rho-1) \cdot Q$, Q contains a factor $\rho^{K-1}(\rho-1)$ divides $\phi(n)$

Thus, $a^{\phi(n)} \equiv 1 \pmod{n}$

If $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, Then by above, $a^{\phi(n)} \equiv 1 \pmod{\rho_i^{K_i}}$ Since $\gcd(\rho_i^{K_i}, \rho_j^{K_j}) = 1$ for $i \neq j$, $1 \leq i, j \leq r$, Then by prob.#13, Sec. 4.2, $a^{\phi(n)} \equiv 1 \pmod{\rho_1^{K_1} \cdots \rho_r^{K_r}}$ or, $a^{\phi(n)} \equiv 1 \pmod{n}$

Note: These are the exact same steps shown on p. 137, Prob. #6 was probably in earlier edition whereas proof on p. 137 has not, and #6 was accidentally left in. **7.** Find the

units digit of 3^{100} by Euler's Theorem.

$\gcd(10, 3) = 1$. By Euler's Th., $3^{\phi(10)} \equiv 1 \pmod{10}$

$$\phi(10) = 4. \therefore 3^4 \equiv 1 \pmod{10}; (3^4)^{25} \equiv 1 \pmod{10}$$

$$\therefore 3^{100} \equiv 1 \pmod{10}$$

$\therefore$ Units digit of 3^{100} is 1.

8.(a) If $\gcd(a, n) = 1$, show the linear congruence $ax \equiv b \pmod{n}$ has the solution

$$x \equiv b \cdot a^{\phi(n)-1} \pmod{n}.$$

Pf: If $x \equiv b \cdot a^{\phi(n)-1} \pmod{n}$, Then

$$ax \equiv a \cdot b \cdot a^{\phi(n)-1} = b \cdot a^{\phi(n)}$$

But $a^{\phi(n)} \equiv 1 \pmod{n}$, by Euler's Th., since $\gcd(a, n) = 1$.

$$\therefore ax \equiv b \cdot a^{\phi(n)} \equiv b \cdot 1 \equiv b \pmod{n}$$

(b) Use (a) to solve the linear congruences

$$3x \equiv 5 \pmod{26}, \quad 15x \equiv 2 \pmod{40}, \quad 10x \equiv 21 \pmod{49}$$

$$3x \equiv 5 \pmod{26}, \quad \gcd(3, 26) = 1, \quad \phi(26) = 12$$

$$\therefore x \equiv 5 \cdot 3^{\phi(26)-1} \pmod{26}, \text{ or}$$

$$x \equiv 5 \cdot 3^{11} \pmod{26}$$

To simplify, $3^4 = 81 \equiv 3 \cdot 26 + 3 \equiv 3 \pmod{26}$

$$3^8 \equiv 9^2 = 81 \equiv 9 \pmod{26}$$

$$\therefore 3^{11} \equiv 9 \pmod{26}$$

$$\therefore x \equiv 5 \cdot 3^{11} = 45 \equiv 19 \pmod{26}$$

$$13x \equiv 2 \pmod{40}, \quad \gcd(13, 40) = 1, \quad \phi(40) = 2^2 \cdot 2 \cdot (5 - 1) = 16$$

$$\therefore x \equiv 2 \cdot 13^{15} \pmod{40}$$

To simplify, $13^4 = 169 \equiv 9 \pmod{40}$

$$\therefore 13^8 = 81 \equiv 1 \pmod{40}, \quad 13^{12} \equiv 9 \pmod{40}$$

$$9 \cdot 13 = 117 \equiv -3 \pmod{40}$$

$$\therefore 13^{15} \equiv -3 \pmod{40}$$

$$\therefore x \equiv 2 \cdot 13^{15} \equiv -6 \equiv 34 \pmod{40}$$

$$10x \equiv 21 \pmod{49}, \quad \gcd(10, 49) = 1$$

$$\phi(49) = 7 \cdot 6 = 42$$

$$\therefore x \equiv 21 \cdot 10^{41} \pmod{49}$$

To simplify, $10^2 \equiv 2 \pmod{49}$

$$2^{10} = 1024 \equiv 44 \equiv -5 \pmod{49}$$

$$\therefore 10^{20} \equiv 2^{10} \equiv -5 \pmod{49}$$

$$\therefore 10^{40} \equiv 25 \pmod{49}, 10^{41} \equiv 250 \equiv 5 \pmod{49}$$

$$\therefore x \equiv 21 \cdot 5 = 105 \equiv 98 + 7 \equiv 7 \pmod{49}$$

$$\therefore x \equiv 7 \pmod{49}$$

9. Use Euler's Th. to evaluate $2^{100000} \pmod{77}$

$$\gcd(2, 77) = 1. \therefore 2^{\phi(77)} \equiv 1 \pmod{77}$$

$$\phi(77) = 6 \cdot 10 = 60$$

$$\therefore 2^{60000} \equiv 1 \pmod{77}, (2^{60})^{500} \equiv 1^{500} \equiv 1 \pmod{77}$$

$$\therefore 2^{60000} \equiv 1 \pmod{77}; 2^{96000} \equiv 1 \pmod{77}$$

$$\therefore 2^{99600} \equiv 1 \pmod{77}; 2^{600} \equiv 1 \pmod{77}$$

$$\therefore 2^{99940} \equiv 1 \pmod{77}$$

[Note: this chain of exponent multiples (60000, 96000, 99600, 600, 99940) is dense and hard to fully verify exactly from the handwriting; transcribed as written.]

$$\text{But } 2^{10} = 1024, 13 \cdot 77 = 1001. \therefore 2^{10} \equiv 23 \pmod{77}$$

$$\therefore 2^{40} \equiv 23^4 \pmod{77}$$

$$2^{100000} \equiv 23^4 \pmod{77}$$

$$23^2 = 529 = 6 \cdot 77 + 67. \therefore 23^2 \equiv -10 \pmod{77}$$

$$23^4 \equiv 100 \equiv 23 \pmod{77}$$

$$\therefore 2^{100000} \equiv 23 \pmod{77}$$

10. For any integer a , show that a and a^{4n+1} have the same last digit.

Pf: Need to show $a \equiv a^{4n+1} \pmod{10}$ for all a . Assume $n > 0$.

(1) If $\gcd(a, 10) = 1$, Then $a^{\phi(10)} \equiv 1 \pmod{10}$

$$\text{But } \phi(10) = 4. \therefore a^4 \equiv 1 \pmod{10},$$

$$\therefore a^{4n} \equiv 1 \pmod{10}; \therefore a^{4n+1} \equiv a \pmod{10}$$

(2) Suppose $\gcd(a, 10) = 10$, Then $\rho \mid a$, and $a = 10x$, some x .

$$a^{4n+1} = (10x)^{4n+1} = 10^{4n+1}x^{4n+1}$$

$$\begin{aligned}\therefore a^{4n+1} &\equiv 0 \pmod{10}, \text{ so} \\ a &\equiv 0 \equiv a^{4n+1} \pmod{10}\end{aligned}$$

(3) Suppose $\gcd(a, 10) = 5$

Lemma: for $n \geq 1$, $5^n \equiv 5 \pmod{10}$ By induction, clearly true for $n = 1$.

Suppose $5^K \equiv 5 \pmod{10}$.

$$5^{K+1} = 25 \equiv 20 + 5 \equiv 5 \pmod{10}$$

$\therefore K \Rightarrow K + 1$ true, $\therefore$ true for all n .

Let $a = 5\rho_1^{K_1} \cdots \rho_r^{K_r}$, $\rho_i \neq 2$, $x \geq 1$

$$\begin{aligned}\therefore \rho_1^{K_1} \cdots \rho_r^{K_r} &= 2s + 1, \text{ some } s \geq 1 \\ \therefore a &= 5^x(2s + 1) = 5^x \cdot 2s + 5^x \\ &= 10(5^{x-1} \cdot s) + 5^x \\ \therefore a &\equiv 5^x \pmod{10} \equiv 5 \pmod{10} \\ a^{4n+1} &= (5^x)^{4n+1}(\rho_1^{K_1} \cdots \rho_r^{K_r})^{4n+1} \\ &= 5^{4xn+x} \cdot \rho_1^{K'_1} \cdots \rho_r^{K'_r}\end{aligned}$$

But $\rho_1^{K'_1} \cdots \rho_r^{K'_r} = 2r + 1$, some $r \geq 1$

$$\begin{aligned}\therefore a^{4n+1} &= 5^{4xn+x} \cdot (2r + 1) \\ &= 5^x \cdot 10 \cdot 5^{4xn-1} + 5^{4xn+x} \\ \therefore a^{4n+1} &\equiv 5^{4xn+x} \equiv 5 \pmod{10} \\ \therefore a &\equiv 5 \equiv a^{4n+1} \pmod{10}\end{aligned}$$

(4) Suppose $\gcd(a, 10) = 2$

Let $a = 2^x \rho_1^{K_1} \cdots \rho_r^{K_r}$, $\rho_i \neq 5$, $x \geq 1$

Let $\rho_1^{K_1} \cdots \rho_r^{K_r} = 5q + r$, $0 < r < 5$

$$\begin{aligned}\therefore a &= 2^x(5q + r) = 2^x \cdot 10 \cdot q + 2^x \cdot r \\ \therefore a &\equiv 2^x \cdot r \pmod{10}, \quad r = 1, 2, 3, 4\end{aligned}$$

Lemma: $2^{4n+1} \equiv 2 \pmod{10}$, $n \geq 0$

Pf: Clearly true for $n = 0$ For $n = 1$, $2^5 = 32 \equiv 2 \pmod{10}$ Assume true for K

$$\therefore 2^{4K+1} \equiv 2 \pmod{10}, K \geq 0$$

$$\therefore 2^{4(K+1)+1} = 2^{4K+5}$$

But $2^5 = 32 \equiv 2 \pmod{10}$

$$\therefore 2^{4K+5} \cdot 2^5 = 2^{4K+1} \cdot 2^5 \equiv 2 \cdot 32 \pmod{10}$$

[Note: this line's algebra is hard to follow exactly from the handwriting; transcribed as written.]

$$\therefore 2^{4(K+1)+1} \equiv 2 \pmod{10}$$

$$\therefore K \Rightarrow K+1, \text{ true for all } n.$$

$r = 1: a \equiv 2^x \pmod{10}$

$$\therefore a^{4n+1} \equiv (2^x)^{4n+1} \pmod{10}$$

But $(2^x)^{4n+1} = (2^{4n+1})^x \equiv 2^x \pmod{10}$ by Lemma above.

$$\therefore a^{4n+1} \equiv 2^x \equiv a \pmod{10}$$

$r = 2: a \equiv 2^x \cdot 2 \pmod{10}$

This is same as case $r = 1$.

$r = 3: a \equiv 2^x \cdot 3 \pmod{10}$

$$\therefore a^{4n+1} \equiv (2^x \cdot 3)^{4n+1} \pmod{10}$$

$$= (2^{4n+1})^x \cdot 3^{4n+1}$$

$$= 2^x \cdot 3^{4n+1} \pmod{10}$$

by Lemma above.

Lemma: $3^{4n+1} \equiv 3 \pmod{10}, n \geq 0$

Pf: Clearly true for $n = 0$ Assume true for $K \geq 1$

$$\therefore 3^{4(K+1)+1} = 3^{4K+5} = 3^{4K+1} \cdot 3^4$$

$$3^4 = 81 = 80 + 1$$

$$\therefore 3^{4K+1} \cdot 3^4 = (3^{4K+1}) \cdot 80 +$$

$$3^{4K+1} \pmod{10}$$

$$\equiv 3^{4K+1} \pmod{10}$$

$$\equiv 3 \pmod{10}$$

$$\therefore 3^{4n+1} \equiv 3 \pmod{10} \text{ for all } n$$

$$\therefore a^{4n+1} = 2^x \cdot 3^{4n+1} \equiv 2^x \cdot 3 \pmod{10}$$

$$\therefore a^{4n+1} \equiv 2^x \cdot 3 \equiv a \pmod{10}$$

$$r = 4: a \equiv 2^x \cdot 4 \pmod{10}$$

$$2^x \cdot 4 = 2^{x+2}, \therefore a \equiv 2^{x+2} \pmod{10},$$

which is the same as the case $r = 1$.

$$\therefore \text{If } a = 2^x \rho_1^{K_1} \cdots \rho_r^{K_r}, \rho_i \neq 5, x \geq 1$$

$$\text{Then } a \equiv a^{4n+1} \pmod{10}$$

$$(1), (2), (3), \text{ and } (4) \Rightarrow a \equiv a^{4n+1} \pmod{10} \text{ for all } a.$$

11. For any prime ρ , establish each of the assertions below:

$$(a) \tau(\rho!) = 2\tau((\rho - 1)!)$$

$$\text{Pf: Let } \rho! = \rho_1^{K_1} \cdots \rho_r^{K_r} \cdot \rho = 1 \cdot 2 \cdots (\rho - 1) \cdot \rho$$

$$\therefore (\rho - 1)! = \rho_1^{K_1} \cdots \rho_r^{K_r}$$

$$\text{Since } \gcd(\rho, \rho_1^{K_1} \cdots \rho_r^{K_r}) = 1,$$

$$\tau(\rho!) = \tau(\rho \cdot \rho_1^{K_1} \cdots \rho_r^{K_r}) = \tau(\rho) \cdot \tau(\rho_1^{K_1} \cdots \rho_r^{K_r})$$

$$= \tau(\rho) \cdot \tau((\rho - 1)!)$$

$$= 2 \cdot \tau((\rho - 1)!), \text{ since } \tau(\rho) = 2$$

$$(b) \sigma(\rho!) = (\rho + 1)\sigma((\rho - 1)!)$$

$$\text{As in (a), } \rho! = \rho_1^{K_1} \cdots \rho_r^{K_r} \cdot \rho = (\rho - 1)! \cdot \rho$$

$$\therefore \sigma(\rho!) = \sigma((\rho - 1)!) \cdot \sigma(\rho)$$

$$= \sigma((\rho - 1)!) \cdot (\rho + 1), \text{ as } \sigma(\rho) = \rho + 1$$

$$(c) \phi(\rho!) = (\rho - 1)\phi((\rho - 1)!)$$

$$\text{As in (b), } \rho! = \rho_1^{K_1} \cdots \rho_r^{K_r} \cdot \rho = (\rho - 1)! \cdot \rho$$

$$\therefore \phi(\rho!) = \phi((\rho - 1)!) \cdot \phi(\rho)$$

$$= \phi((\rho - 1)!) \cdot (\rho - 1), \text{ as } \phi(\rho) = \rho - 1$$

12. Given $n \geq 1$, a set of $\phi(n)$ integers relatively prime to n is called a reduced set of residues modulo n (i.e., a subset of a complete set of residues modulo n , whose members are relatively prime to n).

Verify the following: (a) The integers $-31, -16, -8, 13, 25, 80$ form a reduced set of residues modulo 9.

The complete set of residues mod 9 =

$$\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$$

The relatively prime members are $\{1, 2, 4, 5, 7, 8\}$, and all members are incongruent mod 9 since they are a subset of the complete set.

$\therefore$ Suffices to show each integer in question is congruent to one and only one of $\{1, 2, 4, 5, 7, 8\}$.

The Division Algorithm will do this:

$$-31 = -4 \cdot 9 + 5 \quad 13 = 1 \cdot 9 + 4$$

$$-16 = -2 \cdot 9 + 2 \quad 25 = 2 \cdot 9 + 7$$

$$-8 = -1 \cdot 9 + 1 \quad 80 = 8 \cdot 9 + 8$$

$\therefore \{-31, -16, -8, 13, 25, 80\}$ forms a reduced set.

(b) $3, 3^2, 3^3, 3^4, 3^5, 3^6$ form a reduced set of residues mod 14. As in (a), the reduced set of residues mod 14 is: $\{1, 3, 5, 9, 11, 13\}$

$$3 \equiv 3 \pmod{14}$$

$$3^2 \equiv 9 \pmod{14}$$

$$3^3 = 27 \equiv 13 \pmod{14}$$

$$3^4 = 81 = 5 \cdot 14 + 11 \equiv 11 \pmod{14}$$

$$3^5 = 3^4 \cdot 3 = 11 \cdot 3 = 33 \equiv 5 \pmod{14}$$

$$3^6 = 3^4 \cdot 3^2 = 11 \cdot 9 = 99 = 7 \cdot 14 + 1 \equiv 1 \pmod{14}$$

$\therefore \{3, 3^2, 3^3, 3^4, 3^5, 3^6\}$ is congruent mod 14 to

one and only one of $\{1, 3, 5, 9, 11, 13\}$ and is $\therefore$ a reduced set of residues mod 14.

(c) The integers $2^1, 2^2, 2^3, \dots, 2^{18}$ form a reduced set of residues mod 27?

$$27 = 3^3, \therefore \phi(n) = 3^3 - 3^2 = 18.$$

Clearly, $\gcd(2^x, 3^3) = 1$, for $n \geq 1$. Since $\phi(27) = 18$, and there are 18 numbers: $2^1, 2^2, \dots, 2^{18}$, only have to show the numbers are incongruent to each other mod 27.

Since $\gcd(2, 3^3) = 1$, By Euler's Th.,

$$2^{\phi(27)} \equiv 1 \pmod{27}, \text{ or } 2^{18} \equiv 1 \pmod{27}$$

Also, $2 \not\equiv 1 \pmod{27}$.

$\therefore 2^{17} \not\equiv 1 \pmod{27}$, for if $2^{17} \equiv 1 \pmod{27}$, then $2^{18} = 2^{17} \cdot 2 \equiv 1 \cdot 2 \equiv 2 \pmod{27}$, and so $1 \equiv 2 \pmod{27}$, clearly false.

This can be done for $2^{16}, 2^{15}, \dots, 2^1$ that is, $2^x \not\equiv 1 \pmod{27}$ for $x = 1, 2, \dots, 17$

$$\therefore 2^x \not\equiv 2^m \pmod{27}, x \neq m, 0 < x, m \leq 18$$

For if $2^x \equiv 2^m \pmod{27}$, $x \neq m$, $0 < x, m \leq 18$, then (assuming for ease $x > m$)

$$2^{x-m} \equiv 1 \pmod{27}, \text{ contradicting}$$

the above: $2^x \not\equiv 1 \pmod{27}$, $0 < x < 18$.

$$\therefore \{2^1, \dots, 2^{18}\} \text{ are incongruent mod } 27,$$

There are $\phi(27) = 18$ such members, and $\therefore$ they form a reduced set of residues mod 27.

13. If ρ is an odd prime, show that the integers

$$-\frac{\rho-1}{2}, \dots, -2, -1, 1, 2, \dots, \frac{\rho-1}{2}$$

form a reduced set of residues mod ρ .

Pf: $\phi(\rho) = \rho - 1$, and there are $\rho - 1$ elements in the above set, and they are all integers since $\rho \geq 3$ so that $\rho - 1$ is even.

As all of $1, 2, \dots, \frac{\rho-1}{2} < \rho$, Then they are all incongruent to each other, mod ρ .

Similarly, $-\frac{\rho-1}{2}, \dots, -2, -1$ are all incongruent mod ρ . For if $a, b \in \left\{-\frac{\rho-1}{2}, \dots, -2, -1\right\}$

and $a \equiv b \pmod{\rho}$, Then $-a \equiv -b \pmod{\rho}$, contradicting the incongruity of $\left\{1, 2, \dots, \frac{\rho-1}{2}\right\}$!

$\therefore$ Need only show if $a \in \left\{1, 2, \dots, \frac{\rho-1}{2}\right\}$ and $b \in \left\{-\frac{\rho-1}{2}, \dots, -2, -1\right\}$, Then $a \not\equiv b \pmod{\rho}$

Suppose $a \equiv b \pmod{\rho}$. Then $a = b + \rho \pmod{\rho}$ But $b + \rho$ is of the form:

$$\rho - \frac{\rho-1}{2}, \dots, \rho - 2, \rho - 1, \text{ or}$$

$$\frac{2\rho - \rho + 1}{2}, \dots, \rho - 2, \rho - 1,$$

$$\frac{\rho + 1}{2}, \dots, \rho - 2, \rho - 1$$

$b + \rho > a$, and $b + \rho < \rho$

[Note: this line " $b + \rho < \rho$ " looks inconsistent given $b + \rho$ was just shown to be of the form $\frac{\rho+1}{2}, \dots, \rho - 1$, which is $< \rho$; transcribed as written.]

$\therefore b + \rho \not\equiv a \pmod{\rho}$, a contradiction.

$\therefore$ All $\rho - 1$ elements of $\left\{-\frac{\rho-1}{2}, \dots, -2, -1, 1, 2, \dots, \frac{\rho-1}{2}\right\}$

are incongruent mod ρ to each other and so form a reduced set of residues.

7.4 Some Properties of the Phi-Function

1. For a positive integer n , prove

$$\sum_{d|n} (-1)^{n/d} \phi(d) = \begin{cases} 0 & \text{if } n \text{ is even} \\ -n & \text{if } n \text{ is odd} \end{cases}$$

Pf:(1) If n is even, Then $n = 2^K N$, where $N = \rho_1^{K_1} \cdots \rho_r^{K_r}$, $\rho_i \neq 2$, and so N is odd.

Now must break up the summation of $d | n$ into sums of divisors that always contain 2^K and all the other divisors. For divisors that always contain 2^K as a factor, this can be expressed as

$$\sum_{d|N} (-1)^{2^K \cdot N/2^K d} \phi(2^K d) = \sum_{d|N} (-1)^{N/d} \phi(2^K d)$$

But N is odd, so is d , so N/d is odd, and so $(-1)^{N/d} = -1$ ($d = \rho_1^{a_1} \cdots \rho_r^{a_r}$, $0 \leq a_i \leq K_i$)

$$\therefore \sum_{d|N} (-1)^{N/d} \phi(2^K d) = - \sum_{d|N} \phi(2^K d)$$

All the other divisors run over $2^{K-1}N$

$$\therefore \sum_{d|n} (-1)^{n/d} \phi(d) = - \sum_{d|N} \phi(2^K d) - \sum_{d|2^{K-1}N} \phi(d)$$

But as d runs over $2^{K-1}N$, N/d will always be even as the highest power of 2 for d is 2^{K-1} , so $2^{K-1}N/d$ will always have a factor of 2. $\therefore (-1)^{2^{K-1}N/d} = 1$

$$\therefore \sum_{d|n} (-1)^{n/d} \phi(d) = \sum_{d|2^K N} \phi(d) - \sum_{d|N} \phi(2^K d)$$

By Th. 7.6 from Gauss, $n = \sum_{d|n} \phi(d)$, so

$$\sum_{d|2^K N} \phi(d) = 2^K N. \text{ Also, } \phi \text{ is}$$

multiplicative, so $\phi(2^K d) = \phi(2^K) \phi(d)$

$$\begin{aligned} \therefore \sum_{d|n} (-1)^{n/d} \phi(d) &= 2^K N - \phi(2^K) \sum_{d|N} \phi(d) \\ &= 2^K N - (2^K - 2^{K-1})(N) \\ &= 2 \cdot 2^{K-1} N - 2^{K-1} N \cdot 2 + 2^{K-1} N \end{aligned}$$

[Note: this line's algebra is hard to follow exactly from the handwriting; transcribed as written.]

$$= 2 \cdot 2^{K-1}N - 2 \cdot 2^{K-1}N = 0$$

$$\therefore n \text{ even} \Rightarrow \sum_{d|n} (-1)^{n/d} \phi(d) = 0$$

(2) If n is odd, Then $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, $\rho_i \neq 2$, $d = \rho_1^{s_1} \cdots \rho_r^{s_r}$, $0 \leq s_i \leq K_i$, by Th. 6.1 $\therefore d$ is odd

$\therefore$ as d runs over n , n/d is odd, so $(-1)^{n/d} = -1$

$$\therefore \sum_{d|n} (-1)^{n/d} \phi(d) = - \sum_{d|n} \phi(d) = -n$$

$$\therefore n \text{ odd} \Rightarrow \sum_{d|n} (-1)^{n/d} \phi(d) = -n$$

2. Confirm that $\sum_{d|36} d(d) = 36$ and $\sum_{d|36} (-1)^{36/d} \phi(d) = 0$

The divisors, d , of 36 are: 1, 2, 3, 4, 6, 9, 12, 36 $\phi(d)$: 1, 1, 2, 2, 2, 6, 4, 12

$$\therefore \sum_{d|36} \phi(d) = 1 + 1 + 2 + 2 + 2 + 6 + 4 + 12 = 36$$

$$\sum_{d|36} (-1)^{36/d} \phi(d) = 1 + 1 + 2 + 2 + 2 - 6 + 4 - 12 = 0$$

3. For a positive integer n , prove that

$$\sum_{d|n} \mu^2(d)/\phi(d) = n/\phi(n)$$

Pf: By prob.#19, Sec. 6.1, if f and g are multiplicative, so is $f \cdot g$ and f/g .

$$\therefore F(n) = \sum_{d|n} \frac{\mu^2(d)}{\phi(d)} \text{ is multiplicative.}$$

Let $n = \rho^K$

$$\begin{aligned} \therefore F(\rho^K) &= \sum_{d|\rho^K} \frac{\mu^2(d)}{\phi(d)} = \frac{\mu^2(1)}{\phi(1)} + \frac{\mu^2(\rho)}{\phi(\rho)} + \cdots + \frac{\mu^2(\rho^K)}{\phi(\rho^K)} \\ &= 1 + \frac{1}{\rho-1} + 0 + \cdots + 0 \\ &= 1 + \frac{1}{\rho-1} = \frac{\rho}{\rho-1} \end{aligned}$$

as $\mu(\rho^K) = 0$ for $K \geq 2$.

$\therefore$ if $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, Then

$$\begin{aligned} F(n) &= F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r}) \\ &= \frac{\rho_1}{\rho_1 - 1} \cdots \frac{\rho_r}{\rho_r - 1} \end{aligned}$$

From Th. 7.3, $\phi(n) = n \left(\frac{\rho_1 - 1}{\rho_1} \right) \cdots \left(\frac{\rho_r - 1}{\rho_r} \right)$

$$\therefore \left(\frac{\rho_1}{\rho_1 - 1} \right) \cdots \left(\frac{\rho_r}{\rho_r - 1} \right) = \frac{\phi(n)}{n}$$

[Note: this line should likely state the reciprocal relationship $\frac{n}{\phi(n)}$, matching the conclusion below; transcribed as written.]

$$\therefore F(n) = \sum_{d|n} \mu^2(d)/\phi(d) = n/\phi(n)$$

4. Use problem 4(c), Sec. 6.2, to give a proof of the fact that $n \sum_{d|n} \mu(d)/d = \phi(n)$

Pf: Prob. 4(c) states $\sum_{d|n} \mu(d)/d = (1 - \frac{1}{\rho_1}) \cdots (1 - \frac{1}{\rho_r})$ if $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$

$$\begin{aligned} \therefore n \sum_{d|n} \mu(d)/d &= n \left(1 - \frac{1}{\rho_1} \right) \cdots \left(1 - \frac{1}{\rho_r} \right) \\ &= \phi(n) \text{ by Th. 7.3} \end{aligned}$$

5. If $n > 1$ has the prime factorization $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, establish each of the following:

(a) $\sum_{d|n} \mu(d)\phi(d) = (2 - \rho_1)(2 - \rho_2) \cdots (2 - \rho_r)$

Pf: Since μ and ϕ are multiplicative, Then $\mu \cdot \phi$ is multiplicative (prob.19, Sec. 6.1).

$$\therefore F(n) = \sum_{d|n} \mu(d)\phi(d) \text{ is multiplicative.}$$

$$\begin{aligned} F(\rho^K) &= \sum_{d|\rho^K} \mu(d)\phi(d) \\ &= \mu(1)\phi(1) + \mu(\rho)\phi(\rho) + \cdots + \mu(\rho^K)\phi(\rho^K) \\ &= 1 + (-1)(\rho - 1) + 0 + \cdots + 0 = 2 - \rho, \end{aligned}$$

since $\mu(\rho^K) = 0$ for $K \geq 2$.

$$\begin{aligned}\therefore F(n) &= F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r}) \\ &= (2 - \rho_1) \cdots (2 - \rho_r)\end{aligned}$$

Could also have used prob.3, Sec. 6.2, which says,

$$\sum_{d|n} \mu(d)f(d) = (1 - f(\rho_1)) \cdots (1 - f(\rho_r))$$

where f is multiplicative, not identically zero. But ϕ is multiplicative and not identically zero (e.g., $f(1) = 1$, $\phi(\rho) = \rho - 1$).

$$\begin{aligned}\therefore \sum_{d|n} \mu(d)\phi(d) &= (1 - \phi(\rho_1)) \cdots (1 - \phi(\rho_r)) \\ &= (1 - (\rho_1 - 1)) \cdots (1 - (\rho_r - 1)) \\ &= (2 - \rho_1) \cdots (2 - \rho_r)\end{aligned}$$

$$(b) \sum_{d|n} d \cdot \phi(d) = \left(\frac{\rho_1^{2K_1+1} + 1}{\rho_1 + 1} \right) \cdots \left(\frac{\rho_r^{2K_r+1} + 1}{\rho_r + 1} \right)$$

Pf: $f(x) = x$ is multiplicative. $\therefore f \cdot \phi$ is also.

$$\therefore F(n) = \sum_{d|n} d \cdot \phi(d) \text{ is multiplicative.}$$

(1) Consider $F(\rho^K) = \sum_{d|\rho^K} d \cdot \phi(d)$

$$\begin{aligned}&= 1 \cdot \phi(1) + \rho \cdot \phi(\rho) + \rho^2 \cdot \phi(\rho^2) + \cdots + \rho^K \phi(\rho^K) \\ &= 1 + \rho(\rho - 1) + \rho^2(\rho^2 - \rho) + \cdots + \rho^K(\rho^K - \rho^{K-1}) \\ &= 1 + \rho^2 - \rho + \rho^4 - \rho^3 + \cdots + \rho^{2K} - \rho^{2K-1} \\ &= 1 + (-1)\rho + (-1)\rho^2 + (-1)\rho^3 + \cdots + (-1)\rho^{2K-1}\end{aligned}$$

[Note: this regrouping is hard to verify exactly term-by-term from the handwriting; transcribed best-effort.]

But $(\rho^{2K+1} + 1) = (a + 1)(a^{2K} - a^{2K-1} + a^{2K-2} - \cdots - a + 1)$
(i.e., coefficient of 1 for even exponent, -1 for odd exponent).

$$\therefore (\rho^{2K+1} + 1) = (\rho + 1)(\rho^{2K} - \rho^{2K-1} + \cdots - \rho^2 + \rho - 1)$$

$$\therefore \frac{\rho^{2K+1} + 1}{\rho + 1} = \rho^{2K} - \rho^{2K-1} + \cdots + \rho^2 - \rho + 1$$

$$\therefore F(\rho^K) = \frac{\rho^{2K+1} + 1}{\rho + 1}$$

$$\begin{aligned} (2) \therefore \sum_{d|n} d \cdot \phi(d) &= F(n) = F(\rho_1^{K_1} \cdots \rho_r^{K_r}) \\ &= F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r}) \\ &= \left(\frac{\rho_1^{2K_1+1} + 1}{\rho_1 + 1} \right) \cdots \left(\frac{\rho_r^{2K_r+1} + 1}{\rho_r + 1} \right) \end{aligned}$$

$$(c) \sum_{d|n} \frac{\phi(d)}{d} = \left(1 + \frac{K_1(\rho_1 - 1)}{\rho_1} \right) \cdots \left(1 + \frac{K_r(\rho_r - 1)}{\rho_r} \right)$$

Pf: $f(x) = 1/x$ is multiplicative.

$$\therefore F(n) = \sum_{d|n} \frac{\phi(d)}{d} \text{ is multiplicative}$$

$$(1) \text{ Consider } F(\rho^K) = \sum_{d|\rho^K} \phi(d)/d$$

$$\begin{aligned} &= \frac{\phi(1)}{1} + \frac{\phi(\rho)}{\rho} + \cdots + \frac{\phi(\rho^K)}{\rho^K} \\ &= 1 + \frac{\rho - 1}{\rho} + \frac{\rho^2 - \rho}{\rho^2} + \cdots + \frac{\rho^K - \rho^{K-1}}{\rho^K} \\ &= 1 + \frac{\rho - 1}{\rho} + \frac{\rho(\rho - 1)}{\rho^2} + \cdots + \frac{\rho^{K-1}(\rho - 1)}{\rho^K} \\ &= 1 + \frac{\rho - 1}{\rho} + \frac{\rho - 1}{\rho} + \cdots + \frac{\rho - 1}{\rho} \\ &= 1 + K \left(\frac{\rho - 1}{\rho} \right) \end{aligned}$$

$$\begin{aligned} (2) \therefore \sum_{d|n} \phi(d)/d &= F(n) = F(\rho_1^{K_1} \cdots \rho_r^{K_r}) \\ &= F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r}) \\ &= \left(1 + \frac{K_1(\rho_1 - 1)}{\rho_1} \right) \cdots \left(1 + \frac{K_r(\rho_r - 1)}{\rho_r} \right) \end{aligned}$$

6. Verify the formula $\sum_{d=1}^n \phi(d) \left\lfloor \frac{n}{d} \right\rfloor = \frac{n(n+1)}{2}, n > 0.$

Pf: From Th. 7.6, if $n > 0$, $n = \sum_{d|n} \phi(d)$

Let $F(n) = \sum_{d|n} \phi(d)$ Since $F(n) = n$, Then $\sum_{d=1}^N F(d) = \frac{N(N+1)}{2}$ But by Th. 6.11,

$$\sum_{d=1}^N F(d) = \sum_{d=1}^N \phi(d) \left\lfloor \frac{N}{d} \right\rfloor$$

$$\therefore \sum_{d=1}^N \phi(d) \left\lfloor \frac{N}{d} \right\rfloor = \frac{N(N+1)}{2}$$

7. If n is square-free, prove $\sum_{d|n} \sigma(d^K) \phi(d) = n^K$ for $K \geq 2$.

Pf: σ and ϕ are multiplicative, so

$$F(n) = \sum_{d|n} \sigma(d^K) \phi(d) = \sum_{d|n} \frac{\sigma(d^K)}{d^{K-1} \text{ times}} \cdot \phi(d)$$

is also multiplicative (since $K \geq 2$).

(1) Consider $n = \rho$ (n is square-free)

$$\begin{aligned} \therefore F(\rho) &= \sum_{d|\rho} \sigma(d^K) \phi(d) \\ &= \sigma(1) \phi(1) + \sigma(\rho^K) \phi(\rho) \\ &= 1 + \frac{\rho^{K+1} - 1}{\rho - 1} \cdot (\rho - 1) = \rho^K \end{aligned}$$

(2) $\therefore$ if $n = \rho_1 \rho_2 \cdots \rho_r$, Then

$$\begin{aligned} \sum_{d|n} \sigma(d^K) \phi(d) &= F(n) = F(\rho_1 \rho_2 \cdots \rho_r) \\ &= F(\rho_1) F(\rho_2) \cdots F(\rho_r) \\ &= \rho_1^K \rho_2^K \cdots \rho_r^K = (\rho_1 \rho_2 \cdots \rho_r)^K = n^K \end{aligned}$$

8. For a square-free integer $n > 1$, show that $\tau(n)^2 = n$ if and only if $n > 3$.

Pf: (1) If $n = 3$, Then $\tau(n)^2 = \tau(3)^2 = 2^2 = 4$, by Th. 6.2.

(2) Suppose n is square-free, $n > 1$, and $\tau(n)^2 = n$

Let $n = \rho_1 \rho_2 \cdots \rho_r$, where $\rho_i \neq \rho_j$ since n is square-free.

By Th. 6.2, $\tau(n) = \tau(\rho_1^{K_1} \cdots \rho_r^{K_r}) = 2^r$

$$\therefore \tau(n)^2 = n = \rho_1 \rho_2 \cdots \rho_r = 2^{2r}$$

By Th. 3.1 (and its corollaries), all $\rho_i = 3$, which mean $r = 1$ and $n = 3$. **9.** Prove that

$3 \mid \sigma(3n + 2)$ and $4 \mid \sigma(4n + 3)$ for any positive integer n . (a) $3 \mid \sigma(3n + 2)$

Let $3n + 2 = \rho_1^{K_1} \cdots \rho_r^{K_r}$

Since $3 \equiv 0 \pmod{3}$ and $3n + 2 \equiv 2 \pmod{3}$, Then $\rho_i^{K_i} \not\equiv 0 \pmod{3}$ for $i = 1, 2, \dots, r$

If all $\rho_i^{K_i} \equiv 1 \pmod{3}$, Then $\rho_1^{K_1} \cdots \rho_r^{K_r} \equiv 1 \pmod{3}$ Since $\rho_1^{K_1} \cdots \rho_r^{K_r} \equiv 2 \pmod{3}$, there must be one ρ_i s.t. $\rho_i^{K_i} \equiv 2 \pmod{3}$, for if $\rho_i = 0$, Then (and $\therefore \rho_i \equiv 2 \pmod{3}$), for if $\rho_i \equiv 0$, $\rho_i^{K_i} \equiv 0$, not if $\rho_i \equiv 1$, Then $\rho_i^{K_i} \equiv 1$)

Since $\rho_i \equiv 2 \pmod{3}$, Then

$$\rho_i^2 \equiv 4 \equiv 1 \pmod{3}$$

$$\therefore \rho_i^{K_i} \equiv 2 \pmod{3}, \text{ Then } r \text{ is odd}$$

$\therefore$ for $\rho_i^{K_i} \equiv 2 \pmod{3}$, K_i is odd.

$$\therefore \sigma(\rho_i^{K_i}) = \frac{\rho_i^{K_i+1} - 1}{\rho_i - 1} = (\rho_i - 1)(\rho_i^{K_i} + \rho_i^{K_i-1} + \cdots + \rho_i + 1)$$

[Note: this line's algebra is hard to follow exactly from the handwriting; transcribed as written.]

$$\rho_i^{K_i} + \rho_i^{K_i-1} + \cdots + \rho_i + 1, \text{ and } K_i \text{ is odd.}$$

Note $2 \equiv -1 \pmod{3}$, so if r is odd, $\rho_i^r \equiv -1 \pmod{3}$ if r is even, $\rho_i^r \equiv 1 \pmod{3}$

$$\therefore \sigma(\rho_i^{K_i}) = \rho_i^{K_i} + \rho_i^{K_i-1} + \cdots + \rho_i + 1$$

$$\equiv (-1) + 1 + \cdots + (-1) + 1 \pmod{3}$$

$$\equiv 0 \pmod{3}$$

$$\therefore 3 \mid \sigma(\rho_i^{K_i}) \Rightarrow 3 \mid \sigma(\rho_1^{K_1}) \cdots \sigma(\rho_r^{K_r})$$

$$\Rightarrow 3 \mid \sigma(\rho_1^{K_1} \cdots \rho_r^{K_r}) \quad [\sigma \text{ is multiplicative}]$$

$$\Rightarrow 3 \mid \sigma(3n + 2)$$

(b) $4 \mid \sigma(4n + 3)$

Let $4n + 3 = \rho_1^{K_1} \cdots \rho_r^{K_r}$

$$4n + 3 \equiv 3 \equiv -1 \pmod{4}$$

As in (a), all $\rho_i^{K_i} \not\equiv 1 \pmod{4}$, and $\rho_i^{K_i} \not\equiv 0 \pmod{4}$ since if $\rho_i^{K_i} \equiv 0 \pmod{4} \Rightarrow 4n + 3 \equiv 0 \pmod{4}$ If $\rho_i^{K_i} \equiv 2 \pmod{4}$ for any i , Then $\rho_1^{K_1} \cdots \rho_r^{K_r} \equiv 2$ or $3 \pmod{4}$, $\rho_j \neq \rho_i$, so

$$4n + 3 = \rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r} \equiv 2 \cdot 2 \text{ or } 2 \cdot 3 \pmod{4}$$

$$\equiv 0 \text{ or } 2 \pmod{4}$$

$\therefore \rho_i^{K_i} \not\equiv 2 \pmod{4}$ for any i . $\therefore \rho_i^{K_i} \equiv 3 \equiv -1 \pmod{4}$ for some i , and $\therefore \rho_i \equiv 3 \pmod{4}$
 for $\rho_i \equiv 0 \Rightarrow \rho_i^{K_i} \equiv 0 \pmod{4}$ $\rho_i \equiv 1 \Rightarrow \rho_i^{K_i} \equiv 1 \pmod{4}$ $\rho_i \equiv 2 \Rightarrow \rho_i^{K_i} \equiv 4 \pmod{4}$

$$\therefore \rho_i^2 \equiv 9 \equiv 1 \pmod{4} \Rightarrow \rho_i^{2K} \equiv 1 \pmod{4}$$

for all K . $\therefore$ if s is even, $\rho_i^s \equiv 1 \pmod{4}$ $\therefore \rho_i^{K_i} \equiv -1 \pmod{4} \Rightarrow K_i$ is odd.

As in (a), $\sigma(\rho_i^{K_i}) = \rho_i^{K_i} + \rho_i^{K_i-1} + \cdots + \rho_i + 1$

$$\equiv (-1) + 1 + \cdots + (-1) + 1$$

$$\equiv 0 \pmod{4}$$

$$\therefore 4 \mid \sigma(\rho_i^{K_i}) \Rightarrow 4 \mid \sigma(\rho_1^{K_1} \rho_2^{K_2} \cdots \rho_r^{K_r}) \Rightarrow 4 \mid \sigma(4n+3)$$

10.(a) Given $K \geq 0$, establish that there exists a sequence of K consecutive integers $n+1, n+2, \dots, n+K$ satisfying

$$\mu(n+1) = \mu(n+2) = \cdots = \mu(n+K) = 0$$

Pf: Use the author's hint. Let ρ_i be the K th prime.

$$\therefore \gcd(\rho_i^2, \rho_j^2) = 1 \text{ for } i \neq j.$$

$\therefore$ By the Chinese Remainder Th, there is a solution to: $x \equiv -1 \pmod{\rho_1^2}$

$$x \equiv -2 \pmod{\rho_2^2}$$

$$\vdots$$

$$x \equiv -K \pmod{\rho_K^2}$$

where $\rho_1 = 2, \rho_2 = 3, \dots, \rho_K = K$ th prime

By the discussion on p.138#3, if $n = \rho_1 \rho_2 \cdots \rho_K$ and $N_i = n/\rho_i^2$, Then a simultaneous solution is:

$$x = (-1)N_1^{\phi(\rho_1^2)} + (-2)N_2^{\phi(\rho_2^2)} + \cdots + (-K)N_K^{\phi(\rho_K^2)}$$

$$\therefore x = -N_1^{\phi(2^2)} - 2N_2^{\phi(3^2)} - \cdots - KN_K^{\phi(\rho_K^2)}$$

$$\therefore x + i \equiv 0 \pmod{\rho_i^2} \text{ for } i = 1, 2, \dots, K$$

$$\therefore x + i = a\rho_i^2, \text{ some } a$$

so $\mu(x+i) = 0, i = 1, 2, \dots, K$

(b) Find four consecutive integers for which $\mu(n) = 0$

By (a), consider $x \equiv -1 \pmod{2^2}$

$$x \equiv -2 \pmod{3^2}$$

$$x \equiv -3 \pmod{5^2}$$

$$x \equiv -4 \pmod{7^2}$$

Let $n = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 = 44,100$

$$N_1 = n/2^2 = 3^2 \cdot 5^2 \cdot 7^2 = 11,025$$

$$N_2 = n/3^2 = 2^2 \cdot 5^2 \cdot 7^2 = 4900$$

$$N_3 = n/5^2 = 2^2 \cdot 3^2 \cdot 7^2 = 1764$$

$$N_4 = n/7^2 = 2^2 \cdot 3^2 \cdot 5^2 = 900$$

Also, $\phi(2^2) = 2$, $\phi(3^2) = 6$, $\phi(5^2) = 20$, $\phi(7^2) = 42$

$\therefore$ a simultaneous solution is:

$$x = -(11025)^2 - 2(4900)^7 - 3(1764)^9 - 4(900)^{42}$$

Using N_1, N_2, N_3, N_4 as above, and $q_1 = -1$, $q_2 = -2$, $q_3 = -3$, $q_4 = -4$

$$11025x_1 \equiv 1 \pmod{4} \quad 4900x_2 \equiv 1 \pmod{9}$$

$$11025x_1 - 11024x_1 \equiv 1 \quad 4900x_2 - 9 \cdot 544x_2 = 28$$

$$x_1 \equiv 1 \pmod{4} \quad 4x_2 = 7 + 9 \cdot 28$$

[Note: this congruence-solving line is dense/hard to verify exactly from the handwriting; transcribed best-effort.]

$$x_2 \equiv 7 \pmod{9}$$

$$1764x_3 \equiv 1 \pmod{25} \quad 900x_4 \equiv 1 \pmod{49}$$

$$1764x_3 - 1750x_3 = 1 \quad 900x_4 - 18 \cdot 49x_4 = 3$$

$$14x_3 \equiv 1, 2 \cdot 4 = 2 + 27 \quad 18x_4 \equiv 1, 57x_4 \equiv 3$$

$$3x_3 \equiv 2 = 25 + 27 \quad 54x_4 + 3x_4 \equiv 3, x_4 = 30$$

$$x_3 \equiv 9 \pmod{25} \quad x_4 \equiv 3, 5 \equiv 30$$

[Note: this multi-line congruence solving for x_3, x_4 is dense and hard to verify exactly digit-by-digit from the handwriting; transcribed best-effort.]

$$\begin{aligned} \therefore x &= q_1 N_1 x_1 + q_2 N_2 x_2 + q_3 N_3 x_3 + q_4 N_4 x_4 \\ &= -11025(1) - 2(4900)(7) - 3(1764)(9) - 4(900)(30) \end{aligned}$$

$$= -11025 - 68600 - 47628 - 108000$$

$$= -235253$$

$\therefore x = -235,253$ is a solution.

Can make this number smaller noting that the solution is unique modulo $2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 = 44,100$.

$$\therefore x = -235,253 + 6(44,100) = 29,347$$

$$\therefore x + 1 = 29,348 = 2^2 \cdot 11 \cdot 23 \cdot 29$$

$$x + 2 = 29,349 = 3^3 \cdot 1087$$

$$x + 3 = 29,350 = 2 \cdot 5^2 \cdot 587$$

$$x + 4 = 29,351 = 7^2 \cdot 599$$

$$\therefore \mu(x + 1) = \mu(x + 2) = \mu(x + 3) = \mu(x + 4) = 0$$

11. Modify the proof of Gauss's Th. to establish

$$\sum_{K=1}^n \gcd(K, n) = \sum_{d|n} d \cdot \phi(d/d) = n \sum_{d|n} \frac{\phi(d)}{d}, \text{ for } n \geq 1$$

Pf: Let d be a divisor of n . Create set S_d s.t.

$$S_d = \{m : \gcd(m, n) = d; 1 \leq m \leq n\}$$

Let $N = \{1, 2, \dots, n\}$

If $a \in N$, then if $\gcd(a, n) = 1$, Then $a \in S_1$

If $\gcd(a, n) = c > 1$, Then $a \in S_c$

$\therefore a \in N$ is in at least one set S_i

But every $a \in N$ is in at most one set S_i . For if $a \in S_i, a \in S_j, i \neq j$, then

$$\gcd(a, n) = i = j.$$

$\therefore$ the sets $S_d, d \mid n$, partition N into a finite number of sets.

$$\gcd(m, n) = d \Leftrightarrow \gcd(m/d, n/d) = 1 \quad [1]$$

Pf:(a) By corollary 1 to Th. 2.4 in Sec. 2.2,

$$\gcd(m, n) = d \Rightarrow \gcd(m/d, n/d) = 1$$

(b) Suppose $\gcd(m/d, n/d) = 1$

$\therefore$ There are integers x, y s.t.

$$\frac{m}{d}x + \frac{n}{d}y = 1, \text{ or } mx + ny = d.$$

Suppose $\gcd(m, n) = c$. $\therefore m = ca, n = cb$

$$\therefore (ac)x + (bc)y = d \Rightarrow c \mid d$$

By Th. 2.5, Sec. 2.2, $d = \gcd(m, n)$. Also, for S_d , $1 \leq m \leq n$, so $m/d \leq n/d$.

$\therefore$ From [1], # elements in $S_d = \#$ positive

integers $\leq n/d$, that are relatively prime to n/d , and this is $\phi(n/d)$.

Thus, # elements in $S_d = \phi(n/d)$

$$\therefore \sum_{K \in S_d} \gcd(K, n) = d \cdot \phi(n/d), \text{ since } \gcd(K, n) = d$$

As mentioned above, S_d exactly partitions N .

$$\therefore \sum_{K=1}^n \gcd(K, n) = \sum_{d \mid n} d \cdot \phi(n/d)$$

Now when $d \mid n$, There is a d' s.t. $dd' = n$, so that $\{d : d \mid n\} = \{d' : d \mid n\}$

$$\begin{aligned} \therefore \sum_{d \mid n} d \cdot \phi(n/d) &= \sum_{d \mid n} d' \cdot \phi(n/d') \\ &= \sum_{d \mid n} \frac{n}{d} \cdot \phi\left(\frac{n}{n/d}\right) = n \sum_{d \mid n} \frac{\phi(d)}{d} \end{aligned}$$

12. For $n \geq 2$, establish $\phi(n^2) + \phi((n+1)^2) \leq 2n^2$

Pf: relation does work for $n = 2$

$$\phi(2^2) + \phi(2+1)^2 = \phi(4) + \phi(9) = 2 + 6 \leq 2 \cdot 2^2 = 8$$

By problem 7(c), Sec. 7.2, if K is composite,

$$\phi(K) \leq K - \sqrt{K}$$

n^2 is composite, so is $(n+1)^2$

$$\therefore \phi(n^2) \leq n^2 - \sqrt{n^2} = n^2 - n$$

$$\begin{aligned}
 \phi((n+1)^2) &\leq (n+1)^2 - \sqrt{(n+1)^2} \\
 &= n^2 + 2n + 1 - (n+1) \\
 &= n^2 + n \\
 \therefore \phi(n^2) + \phi((n+1)^2) &\leq n^2 - n + n^2 + n = 2n^2
 \end{aligned}$$

13. Given integer n , prove there exists at least one K for which $n \mid \phi(K)$.

Pf: If $K = \rho_1^{K_1} \cdots \rho_r^{K_r}$, so that

$$\phi(K) = \rho_1^{K_1-1} \cdots \rho_r^{K_r-1} (\rho_1 - 1) \cdots (\rho_r - 1)$$

We want $n = \rho_1^{K_1-1} \cdots \rho_r^{K_r-1}$ $\therefore$ Let $n = q_1^{a_1} \cdots q_s^{a_s}$

Then choose $K = q_1^{a_1+1} \cdots q_s^{a_s+1}$

$$\therefore \phi(K) = q_1^{a_1} \cdots q_s^{a_s} (q_1 - 1) \cdots (q_s - 1)$$

and clearly $n \mid \phi(K)$

14. Show that if n is the product of twin primes, say $n = \rho(\rho+2)$, Then $\phi(n)\sigma(n) = (n+1)(n-3)$

Pf: $\gcd(\rho, \rho+2) = 1$, so

$$\phi(n) = \phi(\rho) \cdot \phi(\rho+2) = (\rho-1)(\rho+2-1) = (\rho-1)(\rho+1)$$

But $\sigma(n) = \sigma(\rho)\sigma(\rho+2) = (\rho+1)(\rho+3)$

$$\therefore \phi(n)\sigma(n) = (\rho-1)(\rho+1)^2(\rho+3)$$

Now $(n+1)(n-3) = (\rho^2 + 2\rho + 1)(\rho^2 + 2\rho - 3)$

$$= (\rho+1)^2(\rho+3)(\rho-1)$$

$$\therefore \phi(n)\sigma(n) = (n+1)(n-3)$$

15. Prove (a) $\sum_{d|n} \sigma(d)\phi(n/d) = n\tau(n)$ and

(b) $\sum_{d|n} \tau(d)\phi(n/d) = \sigma(n)$

Lemma: if $d \mid n$, Then $F(d) = \phi(n/d)$ is multiplicative.

Pf: For any number-theoretic function f ,

$$\sum_{d|n} f(d) = \sum_{d|n} f(n/d)$$

$\therefore$ Let $F(n) = \sum_{d|n} \phi(d)$

Since ϕ is multiplicative, By Th. 6.4, F is multiplicative.

Let $g(d) = \phi(n/d)$

$$\therefore F(n) = \sum_{d|n} \phi(d) = \sum_{d|n} \phi(n/d) = \sum_{d|n} g(d)$$

By Th. 6.4, $g(d) = \phi(n/d)$ is multiplicative.

$\therefore$ if $\gcd(r, s) = 1$ and $rs \mid n$, Then

$$g(rs) = \phi\left(\frac{n}{rs}\right) = g(r)g(s) = \phi\left(\frac{n}{r}\right)\phi\left(\frac{n}{s}\right)$$

(a) Since $F(n) = \sum_{d|n} \sigma(d)\phi(n/d)$ is multiplicative by Lemma above and prob.19, Sec. 6.1,

it suffices to show for $n = \rho^K$, ρ prime, that

$$F(\rho^K) = \rho^K \tau(\rho^K)$$

because if $n = \rho_1^{K_1} \cdots \rho_r^{K_r}$, Then

$$\begin{aligned} F(n) &= F(\rho_1^{K_1} \cdots \rho_r^{K_r}) = F(\rho_1^{K_1}) \cdots F(\rho_r^{K_r}) \\ &= \rho_1^{K_1} \tau(\rho_1^{K_1}) \cdots \rho_r^{K_r} \tau(\rho_r^{K_r}) \\ &= n \tau(\rho_1^{K_1} \cdots \rho_r^{K_r}) = n \tau(n) \end{aligned}$$

The divisors of ρ^K are $1, \rho, \rho^2, \dots, \rho^K$ ($K+1$ divisors)

$$\begin{aligned} \therefore F(\rho^K) &= \sum_{d|\rho^K} \sigma(d)\phi\left(\frac{\rho^K}{d}\right) \\ &= \sigma(1)\phi\left(\frac{\rho^K}{1}\right) + \sigma(\rho)\phi\left(\frac{\rho^K}{\rho}\right) + \cdots + \sigma(\rho^{K-1})\phi\left(\frac{\rho^K}{\rho^{K-1}}\right) + \sigma(\rho^K)\phi\left(\frac{\rho^K}{\rho^K}\right) \\ &= 1 \cdot (\rho^K - \rho^{K-1}) + (\rho + 1)(\rho^{K-1} - \rho^{K-2}) + \cdots \\ &\quad \frac{\rho^K - 1}{\rho - 1} \cdot (\rho - 1) + \frac{\rho^{K+1} - 1}{\rho - 1} \cdot 1 \\ &= (\rho^K - \rho^{K-1}) + (\rho^{K-1} - \rho^{K-2} + \rho^K - \rho^{K-1}) + \cdots \\ &\quad (\rho^K - 1) + (\rho^K + \rho^{K-1} + \rho^{K-2} + \cdots + \rho + 1) \\ &= (\rho^K - \rho^{K-1}) + (\rho^K - \rho^{K-2}) + \cdots + (\rho^K - 1) + (\rho^K + \rho^{K-1} + \cdots + \rho + 1) \\ &= (K\rho^K - \rho^{K-1} - \rho^{K-2} - \cdots - 1) + (\rho^K + \rho^{K-1} + \cdots + \rho + 1) \end{aligned}$$

[Note: this multi-line algebraic regrouping is dense and hard to verify exactly term-by-term from the handwriting; transcribed best-effort.]

$$\begin{aligned}
 &= (K+1)\rho^K \\
 &= \rho^K \tau(\rho^K) \\
 \therefore F(\rho^K) &= \rho^K \tau(\rho^K) \\
 \therefore \sum_{d|n} \sigma(d)\phi(n/d) &= n\tau(n)
 \end{aligned}$$

(b) Since, as in (a), $F(n) = \sum_{d|n} \tau(d)\phi(n/d)$ is multiplicative, it suffices to show, for ρ prime,

$$\begin{aligned}
 F(\rho^K) &= \sigma(\rho^K), \text{ for Then } \sum_{d|n} \tau(d)\phi(n/d) = \sigma(n) \\
 \therefore F(\rho^K) &= \sum_{d|\rho^K} \tau(d)\phi\left(\frac{\rho^K}{d}\right) \\
 &= \tau(1)\phi\left(\frac{\rho^K}{1}\right) + \tau(\rho)\phi\left(\frac{\rho^K}{\rho}\right) + \cdots + \tau(\rho^{K-1})\phi\left(\frac{\rho^K}{\rho^{K-1}}\right) + \tau(\rho^K)\phi\left(\frac{\rho^K}{\rho^K}\right) \\
 &= 1 \cdot (\rho^K - \rho^{K-1}) + (2)(\rho^{K-1} - \rho^{K-2}) + \cdots + (K)(\rho - 1) \cdot 1 + (K+1) \cdot 1 \\
 &= \rho^K - \rho^{K-1} + 2\rho^{K-1} - 2\rho^{K-2} + \cdots + (K-1)\rho - (K-1) + K\rho - K + K + 1 \\
 &= \rho^K + \rho^{K-1} + \rho^{K-2} + \cdots + \rho + 1 \\
 &= \frac{\rho^{K+1} - 1}{\rho - 1} = \sigma(\rho^K) \\
 \therefore F(\rho^K) &= \sigma(\rho^K) \\
 \therefore \sum_{d|n} \tau(d)\phi(n/d) &= \sigma(n)
 \end{aligned}$$

16. If $a_1, a_2, \dots, a_{\phi(n)}$ is a reduced set of residues modulo n , show $a_1 + a_2 + \cdots + a_{\phi(n)} \equiv 0 \pmod{n}$, $n > 2$

Pf: Let $b_1, b_2, \dots, b_{\phi(n)}$ be the positive integers, less than n , that are relatively prime to n . Then by Th. 7.7, for $n \geq 2$, $b_1 + b_2 + \cdots + b_{\phi(n)} = \frac{1}{2}n\phi(n)$. But $n \equiv 0 \pmod{n}$. For $n > 2$, $\phi(n)$ is even, so $\frac{1}{2}\phi(n)$ is an integer, so

$$b_1 + \cdots + b_{\phi(n)} = \frac{1}{2}n\phi(n) \equiv 0 \pmod{n}$$

Also $b_i \not\equiv b_j \pmod{n}$ since $1 \leq b_i, b_j < n$

Now, $a_1, a_2, \dots, a_{\phi(n)}$ are congruent, not necessarily in order of appearance, to $b_1, b_2, \dots, b_{\phi(n)}$, since both are reduced sets of residues.

$$\therefore a_1 \equiv b'_1 \pmod{n}$$

$$a_2 \equiv b'_2 \pmod{n}$$

$$\vdots$$

$$a_{\phi(n)} \equiv b'_{\phi(n)} \pmod{n}$$

where $b'_1, \dots, b'_{\phi(n)}$ are the integers $b_1, \dots, b_{\phi(n)}$ in some order.

$$\therefore a_1 + \dots + a_{\phi(n)} \equiv b'_1 + \dots + b'_{\phi(n)} = b_1 + \dots + b_{\phi(n)} \equiv 0 \pmod{n}$$

$$\therefore a_1 + \dots + a_{\phi(n)} \equiv 0 \pmod{n} \quad (n > 2)$$

7.5 An Application to Cryptography

1. Encrypt the message *RETURN HOME* using the Caesar cipher.

Using $A = 00, B = 01, \dots, Z = 25$, and a space stays as a space,

$$\begin{array}{cccccccccc} R & E & T & U & R & N & H & O & M & E \\ 17 & 04 & 19 & 20 & 17 & 13 & 07 & 14 & 12 & 04 \end{array} +3$$

$$\begin{array}{cccccccccc} 20 & 07 & 22 & 23 & 20 & 16 & 10 & 17 & 15 & 07 \end{array}$$

$$\therefore UHWXUQ KRP H$$

2. If the Caesar cipher produced KDSSB ELUWKGOB, what is the plaintext message?

$$\begin{array}{ccccccc} K & D & S & S & B & E & L \\ 10 & 03 & 18 & 18 & 01 & 04 & 11 \end{array}$$

$$\begin{array}{ccccccc} U & W & K & G & O & B & \\ 20 & 22 & 10 & 06 & 14 & 01 & \end{array} -3$$

$$\begin{array}{cccccccccc} 07 & 00 & 15 & 15 & 24 & 01 & 08 & 17 & 19 & 03 & 24 & 11 \end{array}$$

$$HAPPY BIRTHDAY$$

3.(a) A linear cipher is defined by $C \equiv ap + b \pmod{26}$, where a, b are integers, $\gcd(a, 26) = 1$. Show the decrypting sequence is: $\rho \equiv a'(C - b) \pmod{26}$ where a' satisfies: $aa' \equiv 1 \pmod{26}$ Pf: Since $\gcd(a, 26) = 1$, Then by corollary to Th. 4.7 (p. 76), the linear congruence $ax \equiv 1 \pmod{26}$ has a unique solution. Let it be a' . $\therefore aa' \equiv 1 \pmod{26}$.

From $C \equiv a\rho + b \pmod{26}$

$$C - b \equiv a\rho \pmod{26}$$

$$a'(C - b) \equiv a'a\rho \pmod{26}$$

since $a'a \equiv 1 \pmod{26}$

$$\therefore \rho \equiv a'(C - b) \pmod{26}$$

(b) Using the linear cipher $C \equiv 5\rho + 11 \pmod{26}$, encrypt the message

NUMBER THEORY IS EASY

Using $A = 00, B = 01, \dots, Z = 25$,

<i>N</i>	<i>U</i>	<i>M</i>	<i>B</i>	<i>E</i>	<i>R</i>	<i>T</i>	<i>H</i>	<i>E</i>	<i>O</i>	<i>R</i>	<i>Y</i>	<i>I</i>	<i>S</i>	<i>E</i>	<i>A</i>	<i>S</i>	<i>Y</i>
13	20	12	1	4	17	19	7	4	14	17	24	8	18	4	0	18	24
76	11	71	21	31	96	64	46	31	81	96	131	51	101	31	11	101	131
24	11	19	21	5	18	12	20	5	3	18	1	25	23	5	11	23	1
<i>YAHTQFS CUFOSB XX FLXB</i>																	

[Note: this multi-row computation (page 529) is dense and the exact intermediate arithmetic for each letter is hard to fully verify from the handwriting; transcribed best-effort, matching the final ciphertext shown on the page.] (c) Decrypt REQ T6U HOZTKGH OJ

KTKMMT6, which was produced using the linear cipher $C \equiv 3\rho + 7 \pmod{26}$

From (a), $3x \equiv 1 \pmod{26}$

$$\therefore 27x \equiv 9 \pmod{26}$$

$$\therefore x \equiv 9 \pmod{26}$$

$\therefore$ let $a' = 9$ as in (a)

$$\therefore \rho \equiv 9(C - 7) = 9C - 63 \equiv 9C + 15 \pmod{26}$$

REQ T6U

$$C : 17, 25, 16, 19, 6, 20$$

$$9(C - 7) : 90, 162, 81, 108, -9, 117$$

$$9(C - 7) \pmod{26} : 12, 6, 3, 4, 17, 13$$

$$\rho : M G D E R N$$

(should have been REQ T6U $\Rightarrow$ MODERN)

HOZTKGH

$$C : 7, 14, 25, 19, 10, 6, 7$$

$$9(C - 7) : 0, 63, 162, 108, 27, -9, 0$$

$$9(C - 7) \pmod{26} : 0, 11, 6, 4, 1, 17, 0$$

$$\rho : A L G E B R A$$

OJ

$$C : 3, 9 \quad (\text{should have been}$$

$$9(C - 7) : -36, 18 \quad FG \Rightarrow IS)$$

$$9(C - 7) \pmod{26} : 16, 18$$

$$\rho : Q S$$

KTKMMT6

$$C : 10, 19, 10, 12, 12, 19, 6$$

$$9(C - 7) : 27, 108, 27, 45, 45, 108, -9$$

$$9(C - 7) \pmod{26} : 1, 4, 1, 19, 19, 4, 17$$

$$\rho : B E B T T E R$$

(should have been KTKMMT6)

4. In a lengthy ciphertext message, sent using a linear cipher $C \equiv a\rho + b \pmod{26}$, the most frequently occurring letter is Q and the second most frequent is J .

(a) Break the cipher by determining the values of a and b .

$$Q \Rightarrow 16, J \Rightarrow 9$$

$$16 \equiv a\rho_1 + b \pmod{26}$$

$$9 \equiv a\rho_2 + b \pmod{26}$$

As the message is lengthy, ρ_1 likely is E and ρ_2 likely is T .

$$\therefore \rho_1 = E \Rightarrow 4$$

$$\rho_2 = T \Rightarrow 19$$

$$16 \equiv 4a + b \pmod{26}$$

$$9 \equiv 19a + b \pmod{26}$$

$$\therefore 7 \equiv -15a \pmod{26}$$

$$50a \equiv -14, 50a - 26a = -14, 4a = -14,$$

$$a \equiv 3 \pmod{26}, 4a \equiv 12,$$

$$\therefore 4a = 12, b = 4 \pmod{26}$$

$$\therefore a = 3, b = 4$$

(b) Using (a), $C \equiv 3\rho + 4 \pmod{26}$,

$$C - 4 \equiv 3\rho \pmod{26}$$

$$9(C - 4) \equiv 27\rho \equiv \rho \pmod{26}$$

$$\therefore \rho \equiv 9(C - 4) \pmod{26}$$

WCPQ

$$C : 22, 2, 15, 16$$

$$9(C - 4) : 162, -18, 99, 108$$

$$9(C - 4) \pmod{26} : 6, 8, 21, 4$$

$$\rho : G I V E$$

JZQO

$$C : 9, 25, 16, 14$$

$$9(C - 4) : 45, 189, 108, 90$$

$$9(C - 4) \pmod{26} : 19, 7, 4, 12$$

$$\rho : T H E M$$

MX

$$C : 12, 23$$

$$9(C - 4) : 72, 171$$

$$9(C - 4) \pmod{26} : 20, 15$$

$$\rho : U P$$

5.(a) Encipher the message *HAVE A NICE TRIP* using a Vigenere cipher with the keyword *MATH*.

$$MATH \Rightarrow 12\ 00\ 19\ 07$$

$$\begin{array}{cccccccccccc} H & R & V & E & A & N & I & C & E & T & R & I & P \\ 7 & 0 & 21 & 4 & 0 & 13 & 8 & 2 & 4 & 19 & 17 & 8 & 15 \end{array}$$

$$+12\ 0\ 19\ 7\ 12\ 0\ 19\ 7\ 12\ 0\ 19\ 7\ 12$$

$$19\ 0\ 40\ 11\ 12\ 13\ 27\ 9\ 16\ 19\ 36\ 15\ 27$$

$$19\ 0\ 14\ 11\ 12\ 13\ 1\ 9\ 16\ 19\ 10\ 15\ 1 \pmod{26}$$

$$TAOL\ M\ NBJQ\ TKPB$$

(b) The ciphertext *BS FMX KFSGR JAPWL* is known to have resulted from a Vigenere cipher, whose keyword is *YES*. Obtain the deciphering congruence and read the message.

$YES = 24\ 4\ 18$. Subtract $YES \pmod{26}$ from ciphertext to get plaintext.

B	S	F	M	X	K	F	S	G	R	J	A	P	W	L
1	18	5	12	23	10	5	18	6	17	9	0	15	22	11

$$\begin{array}{cccccccccccccccc} -24 & -4 & & -18 & -24 & -4 & & -18 & -24 & -4 & -18 & -24 & & -4 & -18 & -24 & -4 & -18 \\ 3 & 14 & & 13 & 14 & 19 & & 8 & 7 & 14 & 14 & 19 & & 5 & 8 & 17 & 18 & 19 \end{array} \pmod{26}$$

DO NOT SHOOT FIRST

6.(a) Use the Hill cipher $C_1 \equiv 5\rho_1 + 2\rho_2 \pmod{26}$

$$C_2 \equiv 3\rho_1 + 4\rho_2 \pmod{26}$$

to encipher *GIVE THEM TIME*

Note $\gcd(5 \cdot 4 - 3 \cdot 2, 26) = \gcd(14, 26) = 2$, so you can encipher, but not decipher.

G	I	V	E	T	H	E	M	T	I	M	E
6	8	21	4	19	7	4	12	19	8	12	4

Break up text in blocks of 2 letters.

$$GI: C_1 \equiv 5(6) + 2(8) = 46 \equiv 14 \pmod{26} \Rightarrow O$$

$$C_2 \equiv 3(6) + 4(8) = 50 \equiv 24 \pmod{26} \Rightarrow Y$$

$$VE: C_1 \equiv 5(21) + 2(4) = 113 \equiv 9 \pmod{26} \Rightarrow J$$

$$C_2 \equiv 3(21) + 4(4) = 79 \equiv 1 \pmod{26} \Rightarrow B$$

$$TH: C_1 \equiv 5(19) + 2(7) = 109 \equiv 5 \pmod{26} \Rightarrow F$$

$$C_2 \equiv 3(19) + 4(7) = 85 \equiv 7 \pmod{26} \Rightarrow H$$

$$EM: C_1 \equiv 5(4) + 2(12) = 44 \equiv 18 \pmod{26} \Rightarrow S$$

$$C_2 \equiv 3(4) + 4(12) = 60 \equiv 8 \pmod{26} \Rightarrow I$$

$$TI: C_1 \equiv 5(19) + 2(8) = 111 \equiv 7 \pmod{26} \Rightarrow H$$

$$C_2 \equiv 3(19) + 4(8) = 89 \equiv 11 \pmod{26} \Rightarrow L$$

$$ME: C_1 \equiv 5(12) + 2(4) = 68 \equiv 16 \pmod{26} \Rightarrow Q$$

$$C_2 \equiv 3(12) + 4(4) = 52 \equiv 0 \pmod{26} \Rightarrow A$$

$\therefore OYJB FHSI HLQA$

(b) The ciphertext *ALXWU VADCOJO* has been enciphered using $C_1 \equiv 4\rho_1 + 11\rho_2 \pmod{26}$

$$C_2 \equiv 3\rho_1 + 8\rho_2 \pmod{26}$$

Derive the plaintext.

Note $\gcd(4 \cdot 8 - 3 \cdot 11, 26) = \gcd(-1, 26) = 1$

$$3C_1 \equiv 12\rho_1 + 33\rho_2 \pmod{26}$$

$$4C_2 \equiv 12\rho_1 + 32\rho_2 \pmod{26}$$

$$3C_1 - 4C_2 \equiv \rho_2 \pmod{26}$$

$$8C_1 \equiv 32\rho_1 + 88\rho_2$$

$$11C_2 \equiv 33\rho_1 + 88\rho_2$$

$$11C_2 - 8C_1 \equiv \rho_1 \pmod{26}$$

<i>A</i>	<i>L</i>	<i>X</i>	<i>W</i>	<i>U</i>	<i>V</i>	<i>A</i>	<i>D</i>	<i>C</i>	<i>O</i>	<i>J</i>	<i>O</i>
0	11	23	22	20	21	0	3	2	19	9	14

$$AL: \rho_1 \equiv -8(0) + 11(11) = 121 \equiv 17 \pmod{26} \Rightarrow R$$

$$\rho_2 \equiv 3(0) - 4(11) = -44 \equiv 8 \pmod{26} \Rightarrow I$$

$$XW: \rho_1 \equiv -8(23) + 11(22) = 58 \equiv 6 \pmod{26} \Rightarrow G$$

$$\rho_2 \equiv 3(23) - 4(22) = -19 \equiv 7 \pmod{26} \Rightarrow H$$

$$UV: \rho_1 \equiv -8(20) + 11(21) = 71 \equiv 19 \pmod{26} \Rightarrow T$$

$$\rho_2 \equiv 3(20) - 4(21) = -24 \equiv 2 \pmod{26} \Rightarrow C$$

$$AD: \rho_1 \equiv -8(0) + 11(3) = 33 \equiv 7 \pmod{26} \Rightarrow H$$

$$\rho_2 \equiv 3(0) - 4(3) = -12 \equiv 14 \pmod{26} \Rightarrow O$$

$$CO: \rho_1 \equiv -8(2) + 11(14) = 138 \equiv 8 \pmod{26} \Rightarrow I$$

$$\rho_2 \equiv 3(2) - 4(14) = -50 \equiv 2 \pmod{26} \Rightarrow C$$

$$JO: \rho_1 \equiv -8(9) + 11(14) = 82 \equiv 4 \pmod{26} \Rightarrow E$$

$$\rho_2 \equiv 3(9) - 4(14) = -29 \equiv 23 \pmod{26} \Rightarrow X$$

$\therefore$ RIGHT CHOICEX

(Last two enciphered letters should have been *GA* to make plaintext *CHOICES*).

7. A long string of ciphertext resulting from a Hill cipher

$$C_1 \equiv a\rho_1 + b\rho_2 \pmod{26}$$

$$C_2 \equiv c\rho_1 + d\rho_2 \pmod{26}$$

revealed that the most frequently occurring two-letter blocks were *HO* and *PP*, in that order.

(a) Find the values of a, b, c, d

Using the hint that the most common 2-letter blocks, in English are *TH* and then *HE* we have: $H \equiv a(T) + b(H) \pmod{26}$ [1]

$$O \equiv c(T) + d(H) \pmod{26} \quad [2]$$

and: $\rho \equiv a(H) + b(E) \pmod{26}$ [3]

$$\rho \equiv c(H) + d(E) \pmod{26} \quad [4]$$

Or, using $00 = A, \dots, 25 = Z$

$$7 \equiv 19a + 7b \pmod{26} \quad [1]$$

$$14 \equiv 17c + 7d \pmod{26} \quad [2]$$

$$15 \equiv 7a + 4b \pmod{26} \quad [3]$$

$$15 \equiv 7c + 4d \pmod{26} \quad [4]$$

From [1] and [3], $7 \equiv 19a + 7b \pmod{26}$

$$15 \equiv 7a + 4b \pmod{26}$$

Note that $\gcd(19 \cdot 4 - 7 \cdot 7, 26) = \gcd(27, 1) = 1$

[Note: this gcd expression " $\gcd(27, 1)$ " looks inconsistent (likely meant $\gcd(27, 26) = 1$); transcribed as written.]

$$28 \equiv 76c + 28b \quad \therefore 7 \equiv 19a + 7b$$

$$105 \equiv 49a + 28b \quad -12 \equiv 76$$

$$-77 \equiv 27a \pmod{26} \quad 14 \equiv 7b \pmod{26}$$

$$-77 + 26 \equiv 27a - 26a \quad 2 \equiv b \pmod{26}$$

$$1 \equiv a \pmod{26} \quad \text{or } \gcd(7, 26) = 1$$

$$\therefore a = 1, b = 2$$

From [2], [4], $14 \equiv 17c + 7d \pmod{26}$

$$15 \equiv 7c + 4d \pmod{26}$$

and $\gcd(4 \cdot 19 - 7 \cdot 7, 26) = \gcd(27, 26) = 1$

$$56 \equiv 76c + 28d \quad 14 \equiv 57 + 7d, -43 \equiv 7d$$

$$105 \equiv 49c + 28d \quad 9 \equiv 7d, 9 \equiv 7d \pmod{26}$$

$$-49 \equiv 27c \pmod{26} \quad 21 \equiv 7d, d = -21$$

$$3 \equiv c \pmod{26} \quad d = 5$$

$$\therefore c = 3, d = 5$$

$$\therefore C_1 \equiv \rho_1 + 2\rho_2 \pmod{26}$$

$$C_2 \equiv 3\rho_1 + 5\rho_2 \pmod{26}$$

(b) What is the plain text for the intercepted message: *PPIH HOG RAPVT*

From (a), $3C_1 \equiv 3\rho_1 + 6\rho_2 \pmod{26}$

$$5C_1 \equiv 5\rho_1 + 10\rho_2 \pmod{26}$$

$$C_2 \equiv 3\rho_1 + 5\rho_2 \pmod{26}$$

$$2C_2 \equiv 6\rho_1 + 10\rho_2 \pmod{26}$$

$$3C_1 - C_2 \equiv \rho_2 \quad 2C_2 - 5C_1 \equiv \rho_1$$

$$\therefore \rho_1 \equiv -5C_1 + 2C_2 \pmod{26}$$

$$\rho_2 \equiv 3C_1 - C_2 \pmod{26}$$

<i>P</i>	<i>P</i>	<i>I</i>	<i>H</i>	<i>H</i>	<i>O</i>	<i>G</i>	<i>R</i>	<i>A</i>	<i>P</i>	<i>V</i>	<i>T</i>
15	15	8	7	7	14	6	17	0	15	21	19

$$PP: HE \quad IH: 8, 7 \quad \rho_1 \equiv -5(8) + 2(7) = -26 \equiv 0 \pmod{26} \Rightarrow A$$

$$\rho_2 \equiv 3(8) - 7 = 17 \pmod{26} \Rightarrow R$$

$$HO: TH \quad GR: 6, 17 \quad \rho_1 \equiv -5(6) + 2(17) = 4 \pmod{26} \Rightarrow E$$

$$\rho_2 \equiv 3(6) - 17 = 1 \pmod{26} \Rightarrow B$$

$$AP: 0, 15 \quad \rho_1 \equiv -5(0) + 2(15) = 30 \equiv 4 \pmod{26} \Rightarrow E$$

$$\rho_2 \equiv 3(0) - 15 = -15 \equiv 11 \pmod{26} \Rightarrow L$$

$$VT: 21, 19 \quad \rho_1 \equiv -5(21) + 2(19) = -67 \equiv 11 \pmod{26} \Rightarrow L$$

$$\rho_2 \equiv 3(21) - 19 = 44 \equiv 18 \pmod{26} \Rightarrow S$$

∴ HEAR THE BELLS

8. If $n = \rho q = 278,279$, and $\phi(n) = 272,376$, find the primes ρ and q .

Use the hint. Note, since $n = \rho q$, $\phi(n) = (\rho - 1)(q - 1)$

$$\begin{aligned} n - \phi(n) &= \rho q - (\rho - 1)(q - 1) \\ &= \rho q - [\rho q - \rho - q + 1] \\ &= \rho + q - 1 \end{aligned}$$

$$\therefore \rho + q = n - \phi(n) + 1$$

$$\text{Also, } \rho - q = [(\rho - q)^2]^{1/2} = [\rho^2 - 2\rho q + q^2]^{1/2}$$

$$= [\rho^2 + 2\rho q + q^2 - 4\rho q]^{1/2}$$

$$= [(\rho + q)^2 - 4n]^{1/2}$$

$$\therefore \rho + q = n - \phi(n) + 1 = 274,279 - 272,376 + 1 = 1904$$

$$\rho - q = [(\rho + q)^2 - 4n]^{1/2}$$

$$\therefore 2\rho = n - \phi(n) + 1 + [(n - \phi(n) + 1)^2 - 4n]^{1/2}$$

$$= 1904 + [1904^2 - 4(274,279)]^{1/2}$$

$$= 1904 + [2,528,100]^{1/2} = 1904 + 1590$$

$$= 3494$$

$$\therefore \rho = 1747, \quad q = 157$$

9. When the RSA algorithm is based on the key $(n, K) = (32,633, 37)$ what is the recovery exponent for the cryptosystem?

The recovery exponent is integer j satisfying

$$Kj \equiv 1 \pmod{\phi(n)}, \text{ or}$$

$$37j \equiv 1 \pmod{\phi(32,633)}$$

The prime factorization of 32,633:

$$\sqrt{32,633} = 56.8, \text{ so } \rho \leq 56. \quad 32,633 = 53 \cdot 61$$

$$\therefore \phi(n) = 52(60) = 3120$$

$$\therefore 37j \equiv 1 \pmod{3120}$$

Now $\gcd(37, 3120) = 1$ since

$$\begin{aligned} 3120 &= 10(312) = 5 \cdot 2 \cdot 2 \cdot 156 = 5 \cdot 2^2 \cdot 156 \\ &= 2^4 \cdot 3 \cdot 5 \cdot 13 \end{aligned}$$

By prob.#8(c), sec. 7.3, the solution is $j \equiv 37^{\phi(3120)-1} \pmod{3120}$

$$\phi(3120) = (2^3 \cdot 2)(2)(4)(12) = 768$$

$$\therefore j \equiv 37^{767} \pmod{3120}$$

Note that since $3120 = 2^4 \cdot 3 \cdot 5 \cdot 13$, $\gcd$ of 37 and any of these factors is 1.

$$\therefore \phi(13) = 12, \text{ so by Euler's Th., } 37^{12} \equiv 1 \pmod{13}$$

$$\therefore 37^{12} \equiv 1 \pmod{3120}$$

$$767 = 12 \cdot 63 + 11$$

$$\therefore 37^{767} = (37^{12})^{63} \cdot 37^{11} \equiv 37^{11} \pmod{3120}$$

$$\therefore j \equiv 37^{11} \pmod{3120}$$

$$37^3 = 50653 = 3120 \cdot 16 + 733$$

$$\therefore 37^3 \equiv 733 \pmod{3120}$$

$$37^6 \equiv 733^2 = 537289 = 172 \cdot 3120 + 649$$

$$\therefore 37^6 \equiv 649 \pmod{3120}$$

$$37^9 \equiv 733 \cdot 649 = 475717 = 152 \cdot 3120 + 1477$$

$$\therefore 37^9 \equiv 1477 \pmod{3120}$$

$$37^2 \equiv 1369$$

$$\therefore 37^{11} \equiv 1477 \cdot 1369 = 2022013 = 648 \cdot 3120 + 253$$

$$\therefore 37^{11} \equiv 253 \pmod{3120}$$

$$\therefore j \equiv 253 \pmod{3120}$$

10. Encrypt the plaintext message *GOLD MEDAL* using the RSA algorithm with key

$(n, K) = (2419, 3)$.

Using 99 for the space between words,

<i>G</i>	<i>O</i>	<i>L</i>	<i>D</i>		<i>M</i>	<i>E</i>	<i>D</i>	<i>A</i>	<i>L</i>
06	14	11	03	99	12	04	03	00	11

$$\therefore M' = 06141103991204030011$$

With $n = 2419$, M' is broken into blocks of 3 digits, starting from the right.

$$\therefore 006 \ 141 \ 103 \ 991 \ 204 \ 030 \ 011$$

Now convert each block using:

$$M_i^K \equiv r \pmod{n}, \text{ or } M_i^3 \equiv r \pmod{2419}$$

$$006: 6^3 = 216 \pmod{2419}$$

$$141: 141^3 = 2,803,221 = 1158 \cdot 2419 + 2019$$

$$\therefore 141^3 \equiv 2019 \pmod{2419}$$

$$103: 103^3 = 1,092,727 = 451 \cdot 2419 + 1758$$

$$\therefore 103^3 \equiv 1758 \pmod{2419}$$

$$991: 991^3 = 973,242,271 = 402,332 \cdot 2419 + 1163$$

$$\therefore 991^3 \equiv 1163 \pmod{2419}$$

$$204: 204^3 = 8,489,664 = 3503 \cdot 2419 + 1393$$

$$\therefore 204^3 \equiv 1393 \pmod{2419}$$

$$030: 30^3 = 27,000 = 11 \cdot 2419 + 391$$

$$\therefore 30^3 \equiv 391 \pmod{2419}$$

$$011: 11^3 = 1331$$

$$\therefore 11^3 \equiv 1331 \pmod{2419}$$

$\therefore$ Cyphertext is:

$$0216 \ 2019 \ 1758 \ 1163 \ 1393 \ 0391 \ 1331$$

To check, note $2419 = 41 \cdot 59$.

$$\phi(2419) = 40 \cdot 58 = 2320$$

$$Kj \equiv 1 \pmod{\phi(n)} \Rightarrow 3j \equiv 1 \pmod{2320}$$

$$773j_1 \equiv 773, \ 2319j_1 - j_1 = 773$$

$$j \equiv -773 \equiv 2320 - 773 = 1547 \pmod{2320}$$

$$\therefore j \equiv 1547 \pmod{2320}$$

$$216^{1547} = (6^3)^{1547} = 6^{4641} = 6^{2 \cdot 2320 + 1}$$

$$\equiv 6 \pmod{2320}$$

$\therefore$ First code is 006 So can reconstitute M' , which is then broken into 2-digit numbers (starting from right) to get the letters.

11. The ciphertext message produced by the RSA algorithm with key $(n, K) = (1643, 223)$ is:

$$0833\ 0823\ 1130\ 0055\ 0329\ 1099$$

Determine the original plaintext message.

Note: to get $j = 7$ for recovery exponent, $K = 223$, not 283. Both are prime, but assume a typo in book.

On the receiving side, $1643 = 31 \cdot 53$

$$\therefore \phi(n) = 30 \cdot 52 = 1560$$

Recovery exponent: $Kj \equiv 1 \pmod{\phi(n)}$, or

$$223j \equiv 1 \pmod{1560}$$

From 8(c), sec. 7.3, solution is $j \equiv 223^{\phi(1560)-1} \pmod{1560}$

$$1560 = 2^3 \cdot 3 \cdot 5 \cdot 13$$

$$\therefore \phi(1560) = 4 \cdot 2 \cdot 4 \cdot 12 = 384$$

$$\therefore j \equiv 223^{383} \pmod{1560}$$

Since $\gcd(13, 223) = 1$, $223^{12} \equiv 1 \pmod{13}$

$$\therefore 223^{12} \equiv 1 \pmod{1560}$$

$$383 = 31 \cdot 12 + 11$$

$$\therefore 223^{383} = (223^{12})^{31} \cdot 223^{11} \equiv 223^{11} \pmod{1560}$$

$$223^2 \equiv 32 \cdot 1560 - 191. \therefore 223^2 \equiv -191 \pmod{1560}$$

$$223^4 \equiv (-191)^2 = 23 \cdot 1560 + 601$$

$$\therefore 223^4 \equiv 601 \pmod{1560}$$

$$223^8 \equiv 601^2 = 231 \cdot 1560 + 841$$

$$\therefore 223^8 \equiv 841 \pmod{1560}$$

$$223^9 \equiv 841 \cdot 223 = (841)(-191) = -16,631$$

$$\equiv 103 \cdot 1560 + 49$$

$$\therefore 223^9 \equiv 49 \pmod{1560}$$

$$223^{10} \equiv 49 \cdot 223 = 10,927 = 7 \cdot 1560 + 7$$

[Note: this last line's arithmetic should likely yield $223^{10} \equiv 223^9 \cdot 223$, and the value shown (10927) doesn't cleanly match $49 \cdot 223 = 10927$; transcribed as written.]

$$\therefore 223^{10} \equiv 7 \pmod{1560}$$

$$\therefore j \equiv 223^{11} \equiv 7 \pmod{1560}$$

$\therefore$ recovery exponent is $j = 7$

modulo 1643:

$$0833 : 833^7 : 833^2 \equiv 698,889 = 422 \cdot 1643 + 543$$

$$833^2 \equiv 543 \pmod{1643}$$

$$833^4 \equiv 543^2 = 294,849 = 179 \cdot 1643 + 752$$

$$833^4 \equiv 752 \pmod{1643}$$

$$833^6 \equiv 543 \cdot 752 = 246 \cdot 1643 + 1643 + 872$$

[Note: this line's modular reduction is hard to verify exactly from the handwriting; transcribed as written.]

$$833^6 \equiv 872 \cdot 883 = 442 \cdot 1643 + 1672$$

$$833^7 \equiv 170 \pmod{1643}$$

$$0823 : 823^7 : 823^2 \equiv 413 \pmod{1643} \text{ (using a calculator)}$$

$$823^4 \equiv 1340$$

$$823^6 \equiv 1372$$

$$823^7 \equiv \underline{415}$$

$$1130 : 1130^7 : 1130^2 \equiv 289$$

$$1130^4 \equiv 1371$$

$$1130^6 \equiv 256$$

$$1130^7 \equiv \underline{112}$$

$$0055 : 55^7 : 55^3 \equiv 432$$

$$55^6 \equiv 965$$

$$55^7 \equiv \underline{499}$$

$$0329 : 329^7 : 329^3 \equiv 807$$

$$329^6 \equiv 1149$$

$$329^7 \equiv \underline{131}$$

$$1099 : 1099^7 : 1099^3 \equiv 171$$

$$1099^6 \equiv 1310$$

$$1099^7 \equiv \underline{422}$$

$$\therefore M = 170\ 415\ 112\ 499\ 131\ 422$$

$$17\ 04\ 15\ 11\ 24\ 99\ 13\ 14\ 22$$

$$\therefore \text{REPLY} \sqcup \text{NOW}$$

12. Decrypt the ciphertext

$$1369\ 1436\ 0119\ 0385\ 0434\ 1580\ 0690$$

That was encrypted using the RSA algorithm with key $(n, K) = (2419, 211)$.

$$n = 2419 = 41 \cdot 59. \phi(n) = 40 \cdot 58 = 2320 = 2^4 \cdot 5 \cdot 29$$

$$\therefore 211j \equiv 1 \pmod{2320}; \gcd(2320, 211) = 1$$

Using prob. 8(c), sec. 7.3,

$$\phi(2320) = 8 \cdot 4 \cdot 28 = 896$$

$$\gcd(29, 211) = 1, \therefore 211^{28} \equiv 1 \pmod{29} \text{ by Fermat Theorem}$$

$$895 = 31 \cdot 29 + 7, \therefore 211^{895} = (211^{28})^{31} \cdot 211^7 \equiv 211^7 \pmod{2320}$$

$$\therefore j \equiv 211^{895} \equiv 211^7 \pmod{2320}$$

$$211^3 \equiv 251 \pmod{2320} \quad [\text{calculator}]$$

$$211^6 \equiv 251^2 \equiv 361 \pmod{2320}$$

$$211^{6+1} \equiv 361 \cdot 211 \equiv 401$$

$$211^{6+1\cdot} \equiv 401 \cdot 211 \equiv 721$$

[Note: the last two exponent labels ("6 + 1", "6 + 1\cdot") are unclear/garbled in the handwriting, presumably meant to denote successive multiplications by 211 to build up to the 7th power; transcribed as written.]

$$\therefore 211^7 \equiv 251 \cdot 721 \equiv 11 \pmod{2320}$$

$$\therefore \text{recovery exponent} = 11$$

$$\therefore \text{modulo } 2419,$$

$$1369 : 1369^2 \equiv 1855$$

$$1369^4 \equiv 1855^2 \equiv 1207$$

$$1369^8 \equiv 1207^2 \equiv 611$$

$$1369^{10} \equiv 611 \cdot 1855 \equiv 1313$$

$$1369^{11} \equiv 1313 \cdot 1369 \equiv \underline{180}$$

$$1436 : 1436^2 \equiv 1108$$

$$1436^4 \equiv 1231$$

$$1436^8 \equiv 1067$$

$$1436^{10} \equiv 1764$$

$$1436^{11} \equiv 1764 \cdot 1436 \equiv \underline{411}$$

$$0119 : 119^3 \equiv 1535$$

$$119^6 \equiv 119$$

$$119^9 \equiv 1535 \cdot 119 \equiv 1240$$

$$119^{11} \equiv 1240 \cdot 119^2 \equiv \underline{119}$$

$$0385 : 385^2 \equiv 666 \quad 385^8 \equiv 980 \quad 385^{11} \equiv \underline{918}$$

$$385^4 \equiv 879 \quad 385^{10} \equiv 1969$$

$$0434 : 434^2 \equiv 2053$$

$$434^4 \equiv 2259$$

$$434^8 \equiv 1410$$

$$434^{10} \equiv 2369$$

$$434^{11} \equiv \underline{71} \Rightarrow 071$$

$$1580 : 1580^2 \equiv 2411$$

$$1580^4 \equiv 64$$

$$1580^8 \equiv 1677$$

$$1580^{10} \equiv 1098$$

$$1580^{11} \equiv \underline{417}$$

$$0690 : 690^2 \equiv 1976$$

$$690^4 \equiv 310$$

$$690^8 \equiv 1759$$

$$690^{10} \equiv 2100$$

$$690^{11} \equiv \underline{19} \Rightarrow$$

Since $n = 2419$, plaintext should have been broken up into 3-digit blocks.

$$\therefore 180\ 411\ 119\ 918\ 071\ 417\ 19$$

$$18\ 04\ 11\ 11\ 99\ 18\ 07\ 14\ 17\ 19$$

$$S\ E\ L\ L\ \sqcup\ S\ H\ O\ R\ T$$

So, wasn't enciphered properly, since only 10 2-digits codes (20 numbers). Should have been (multiple of 3):

$$0180411199180714\ 1719$$

and $\therefore 018, 041, 111, 991, 867, 141, 719$.

i.e., for consistency, how do you know to precede 71 above so it's 071, but 19 isn't translated to 019. The only way to know is always precede a 2-digit decrypted number with 0 until get to end: if need it, add the 0, if don't, don't do it; confusing.

i.e., when decrypting, must know size of block of plaintext that was enciphered. Should then use same block size when deciphering.

13. Obtain all solutions of the Knapsack problem

$$21 = 2x_1 + 3x_2 + 5x_3 + 7x_4 + 9x_5 + 11x_6$$

Let $x_6 = 1$: $x_5 \neq 1$ since no possible "1"

$$\therefore x_5 = 0$$

$$\therefore 10 = 2x_1 + 3x_2 + 5x_3 + 7x_4$$

$$x_4 = 1 : \quad x_1 = 0, \quad x_2 = 1, \quad x_3 = 0$$

$$x_4 = 0 : \quad x_1 = 1, \quad x_2 = 1, \quad x_3 = 1$$

$$\therefore \{x_1, x_2, x_3, x_4, x_5, x_6\} = \{0, 1, 0, 1, 0, 1\} \quad \text{or} \quad \{1, 1, 1, 0, 0, 1\}$$

Let $x_6 = 0$: $21 = 2x_1 + 3x_2 + 5x_3 + 7x_4 + 9x_5$

Let $x_5 = 1$: $12 = 2x_1 + 3x_2 + 5x_3 + 7x_4$

$$\therefore x_1 = 0, \quad x_2 = 0, \quad x_3 = 1, \quad x_4 = 1$$

$$x_1 = 1, \quad x_2 = 1, \quad x_3 = 0, \quad x_4 = 1$$

no solution if $x_4 = 0$

$$\therefore \{0, 0, 1, 1, 1, 0\}, \quad \{1, 1, 0, 1, 1, 0\}$$

Let $x_5 = 0$: $21 = 2x_1 + 3x_2 + 5x_3 + 7x_4$
 No solution even if all = 1.

$\therefore$ All solutions: $x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6$

$$\begin{array}{cccccc} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \end{array}$$

14. Determine which of the sequences below is superincreasing. (a) 3, 13, 20, 37, 81
 $3 < 13$, $3 + 13 < 20$, $3 + 13 + 20 < 37$, $3 + 13 + 20 + 37 < 81$
 $\therefore$ yes, it is superincreasing.

(b) 5, 13, 25, 42, 90

No, since $5 + 13 + 25 = 43 > 42$.

(c) 7, 27, 47, 97, 197, 397

$7 + 27 < 47$, $2(47) < 97$, $2(97) < 197$, $2(197) < 397$
 $\therefore$ yes, it's superincreasing.

15. Find the unique solution of each of the following superincreasing Knapsack problems:

(a) $118 = 4x_1 + 5x_2 + 10x_3 + 20x_4 + 41x_5 + 89x_6$

Since $4 + 5 + 10 + 20 + 41 = 80 < 118$, then $x_6 \neq 0$
 $\therefore x_6 = 1$, $\therefore 19 = 4x_1 + 5x_2 + 10x_3 + 20x_4 + 41x_5$

$$\therefore x_5 = 0 \quad (41 > 19)$$

$$x_4 = 0 \quad (20 > 19)$$

$$\therefore x_1 = x_2 = x_3 = 1$$

$$\therefore \{x_1, x_2, x_3, x_4, x_5, x_6\} = \{1, 1, 1, 0, 0, 1\}$$

(b) $51 = 3x_1 + 5x_2 + 9x_3 + 18x_4 + 37x_5$

Since $3 + 5 + 9 + 18 = 35 < 51$, x_5 must be 1.
 $\therefore x_5 = 1$, $14 = 3x_1 + 5x_2 + 9x_3 + 18x_4$

$$\therefore x_4 = 0 \quad (18 > 14)$$

If $x_3 = 1$, then $x_2 = 1$, $x_1 = 0$
 $x_3 = 0$: no solution.

$$\therefore \{x_1, x_2, x_3, x_4, x_5\} = \{0, 1, 1, 0, 1\}$$

$$(c) 54 = x_1 + 2x_2 + 5x_3 + 9x_4 + 18x_5 + 40x_6$$

Since $1 + 2 + 5 + 9 + 18 = 35 < 54$, x_6 must be 1.

$$\therefore x_6 = 1: 14 = x_1 + 2x_2 + 5x_3 + 9x_4 + 18x_5$$

$$\therefore x_5 = 0 \quad (18 > 14)$$

Let $x_4 = 1$. $\therefore x_1 = 0, x_2 = 0, x_3 = 1$
 $x_4 = 0$: no solution.

$$\therefore \{x_1, x_2, x_3, x_4, x_5, x_6\} = \{0, 0, 1, 1, 0, 1\}$$

16. Consider a sequence of positive integers $a_1, \dots, a_n$, where $a_{i+1} > 2a_i$, for $i = 1, \dots, n-1$. Show that the sequence is superincreasing.

Pf: By induction, for $n = 2$: $a_2 > 2a_1$, so $a_2 > a_1$. For $n = 3$: $a_3 > 2a_2 = a_2 + a_2 > a_2 + (2a_1) > a_2 + a_1$.

$\therefore$ Assume sequence is superincreasing for $k \geq 3$ (i.e., $a_k > a_1 + a_2 + \dots + a_{k-1}$).

$\therefore a_{k+1} > 2a_k$, by definition.

$$\begin{aligned} \therefore a_{k+1} &> a_k + a_k \\ &> a_k + (a_1 + a_2 + \dots + a_{k-1}) \end{aligned}$$

$$\therefore a_{k+1} > a_1 + a_2 + \dots + a_{k-1} + a_k$$

$\therefore$ Superincreasing for all n .

17. A user of the Knapsack cryptosystem has the sequence 49, 32, 30, 43 as a listed encryption key. If the user's private key involves the modulus $m = 50$ and multiplier $a = 33$, determine the secret superincreasing sequence.

Let a_1, a_2, a_3, a_4 be the superincreasing sequence.

Note that $\gcd(33, 50) = 1$, so $b_i \equiv 33a_i \pmod{50}$ has a unique solution for a_i , given b_i (by corollary to Th. 4.7, sec. 4.4, on p. 76).

$$\therefore 33a_1 \equiv 49 \pmod{50} \quad 33a_2 \equiv 32 \pmod{50}$$

$$99a_1 \equiv 3(49) - 150 \quad 99a_2 \equiv 96 - 100$$

$$-a_1 \equiv -3 \quad -a_2 \equiv -4$$

$$a_1 = 3 \quad a_2 = 4$$

$$33a_3 \equiv 30 \pmod{50} \quad 33a_4 \equiv 43 \pmod{50}$$

$$99a_3 \equiv 90 - 100 \quad 99a_4 \equiv 3(43) - 150$$

$$\begin{array}{ll} -a_3 \equiv -10 & -a_4 \equiv -21 \\ a_3 = 10 & a_4 = 21 \end{array}$$

$$\therefore a_1, a_2, a_3, a_4 = 3, 4, 10, 21$$

18. The ciphertext message produced by the Knapsack cryptosystem employing the superincreasing sequence 1, 3, 5, 11, 35, modulus $m = 73$ and multiplier $a = 5$ is: 55, 15, 124, 109, 25, 34. Obtain the plaintext message.

(1) First find the unique solution to: (multiplier) $x \equiv 1 \pmod{\text{modulus}}$, or

$$5x \equiv 1 \pmod{73} \quad (\text{note } \gcd(5, 73) = 1).$$

$$29(5x) \equiv 29, \quad 145x - 146x \equiv 29,$$

$$x \equiv -29 + 73, \quad \underline{x \equiv 44}$$

(2) Now convert ciphertext using $S' \equiv 44S \pmod{73}$

$$55: \quad S' \equiv 44(55) \pmod{73} \quad (\text{using calculator})$$

$$S' \equiv \underline{11}$$

$$15: \quad S' \equiv 44(15) \pmod{73}$$

$$S' \equiv \underline{3}$$

$$124: \quad S' \equiv 44(124) \pmod{73}$$

$$S' \equiv \underline{54}$$

$$109: \quad S' \equiv 44(109) \pmod{73}$$

$$S' \equiv \underline{51}$$

$$25: \quad S' \equiv 44(25) \pmod{73}$$

$$S' \equiv \underline{5}$$

$$34: \quad S' \equiv 44(34) \pmod{73}$$

$$S' \equiv \underline{36}$$

(3) Now solve Knapsack problems using secret sequence, noting that $S' \equiv 44S \pmod{73}$, $S' \equiv 44(b_1x_1 + \cdots + b_5x_5)$, $b_i = aa_i$, $a = \text{multiplier}$, so $S' \equiv 44aa_1x_1 + \cdots + 44aa_5x_5$, and since $44a \equiv 1 \pmod{73}$, then $S' \equiv a_1x_1 + \cdots + a_5x_5$, where x_i is the binary code of the plaintext letter.

$$\therefore S' = 11, 3, 54, 51, 5, 36$$

$$a_1, a_2, a_3, a_4, a_5 = 1, 3, 5, 11, 35$$

$$\therefore 11: \quad 11 = x_1 + 3x_2 + 5x_3 + 11x_4 + 35x_5$$

$$\therefore x_1 = x_2 = x_3 = 0, \quad x_4 = 1, \quad x_5 = 0$$

$$\therefore 00010 \Rightarrow \underline{C}$$

$$3: \quad 3 = x_1 + 3x_2 + 5x_3 + 11x_4 + 35x_5$$

$$\therefore x_1 = 0, x_2 = 1, x_3 = x_4 = x_5 = 0$$

$$\therefore 01000 \Rightarrow \underline{I}$$

$$54: \quad 54 = x_1 + 3x_2 + 5x_3 + 11x_4 + 35x_5$$

$$\therefore x_1 = 0, x_2 = x_3 = x_4 = x_5 = 1$$

$$\therefore 01111 \Rightarrow \underline{P}$$

$$51: \quad 51 = x_1 + 3x_2 + 5x_3 + 11x_4 + 35x_5$$

$$\therefore x_1 = x_2 = 0, x_3 = x_4 = x_5 = 1$$

$$\therefore 00111 \Rightarrow \underline{H}$$

$$5: \quad 5 = x_1 + 3x_2 + 5x_3 + 11x_4 + 35x_5$$

$$\therefore x_1 = x_2 = 0, x_3 = 1, x_4 = x_5 = 0$$

$$\therefore 00100 \Rightarrow \underline{E}$$

$$36: \quad 36 = x_1 + 3x_2 + 5x_3 + 11x_4 + 35x_5$$

$$\therefore x_1 = 1, x_2 = x_3 = x_4 = 0, x_5 = 1$$

$$\therefore 10001 \Rightarrow \underline{R}$$

$$\therefore \underline{CIPHER}$$

19. A user of the Knapsack cryptosystem has a private key consisting of the superincreasing sequence 2, 3, 7, 13, 27, modulus $m = 60$, and multiplier $a = 7$.

(a) Find the user's listed public key.

$$7(2) = 14$$

$$7(3) = 21$$

$$7(7) = 49$$

$$7(13) = 91 \equiv 31 \pmod{60}$$

$$7(27) = 189 \equiv 9 \pmod{60}$$

$$\therefore 14, 21, 49, 31, 9$$

(b) With the aid of the public key, encrypt the message: SEND MONEY.

First convert to binary equivalent:

$$\text{SEND} \Rightarrow 10010 \quad 00100 \quad 01101 \quad 00011$$

$$\text{MONEY} \Rightarrow 01100 \quad 01110 \quad 01101 \quad 00100 \quad 11000$$

The public key has 5 terms, so need blocks of 5 binary digits, which in this case is each letter (represented by 5 digits).

$$10010 \Rightarrow [1, 0, 0, 1, 0] \cdot [14, 21, 49, 31, 9] = 14 + 31 = 45$$

$$00100 \Rightarrow [0, 0, 1, 0, 0] \cdot [14, 21, 49, 31, 9] = 49$$

$$01101 \Rightarrow [0, 1, 1, 0, 1] \cdot [14, 21, 49, 31, 9] = 21 + 49 + 9 = 79$$

$$00011 \Rightarrow [0, 0, 0, 1, 1] \cdot [14, 21, 49, 31, 9] = 31 + 9 = 40$$

$$01100 \Rightarrow [0, 1, 1, 0, 0] \cdot [14, 21, 49, 31, 9] = 21 + 49 = 70$$

$$01110 \Rightarrow [0, 1, 1, 1, 0] \cdot [14, 21, 49, 31, 9] = 21 + 49 + 31 = 101$$

$$01101 \Rightarrow [0, 1, 1, 0, 1] \cdot [14, 21, 49, 31, 9] = 21 + 49 + 9 = 79$$

$$00100 \Rightarrow [0, 0, 1, 0, 0] \cdot [14, 21, 49, 31, 9] = 49$$

$$11000 \Rightarrow [1, 1, 0, 0, 0] \cdot [14, 21, 49, 31, 9] = 14 + 21 = 35$$

$$\therefore 45, 49, 79, 40, 70, 101, 79, 49, 35$$

Chapter 8

Primitive Roots and Indices

8.1 The Order of an Integer Modulo n

1. Find the order of the integers 2, 3, and 5:

(a) modulo 17

$\phi(17) = 16$, $\therefore$ divisors are 1, 2, 4, 8, 16

$$2^2 \equiv 4, \quad 2^4 \equiv 16, \quad 2^8 \equiv 1 \pmod{17}$$

$$3^2 \equiv 9, \quad 3^4 \equiv 13, \quad 3^8 \equiv 16, \quad 3^{16} \equiv 1 \pmod{17}$$

$$5^2 \equiv 8, \quad 5^4 \equiv 13, \quad 5^8 \equiv 16, \quad 5^{16} \equiv 1 \pmod{17}$$

$$\therefore \text{ord}(2) = 8 \pmod{17}$$

$$\text{ord}(3) = 16 \pmod{17}$$

$$\text{ord}(5) = 16 \pmod{17}$$

(b) modulo 19

$\phi(19) = 18$, $\therefore$ divisors are 1, 2, 3, 6, 9, 18

$$2^2 \equiv 4, \quad 2^3 \equiv 8, \quad 2^6 \equiv 7, \quad 2^9 \equiv 18, \quad 2^{18} \equiv 1 \quad \therefore \text{Ord}(2) = 18$$

$$3^2 \equiv 9, \quad 3^3 \equiv 8, \quad 3^6 \equiv 7, \quad 3^9 \equiv 18, \quad 3^{18} \equiv 1 \quad \therefore \text{Ord}(3) = 18$$

$$5^2 \equiv 6, \quad 5^3 \equiv 11, \quad 5^6 \equiv 7, \quad 5^9 \equiv 1 \quad \therefore \text{Ord}(5) = 9$$

(c) modulo 23

$\phi(23) = 22$, $\therefore$ divisors are 1, 2, 11, 22

$$2^2 \equiv 4, \quad 2^{11} \equiv 1 \quad \therefore \text{Ord}(2) = 11$$

$$3^2 \equiv 9, \quad 3^{11} \equiv 1 \quad \therefore \text{Ord}(3) = 11$$

$$5^2 \equiv 2, \quad 5^{11} \equiv 22, \quad 5^{22} \equiv 1 \quad \therefore \text{Ord}(5) = 22$$

2. Establish each of the statements below:

(a) If a has order hk modulo n , then a^h has order k modulo n .

Pf: $a^{hk} \equiv 1 \pmod{n} \Rightarrow (a^h)^k \equiv 1 \pmod{n}$.

Suppose $(a^h)^r \equiv 1 \pmod{n}$, $0 < r < k$.

$\therefore 0 < hr < hk$. Then a would not have order hk since $hr < hk$ and $a^{hr} \equiv 1$.

(b) If a has order $2k$ modulo the odd prime p , then $a^k \equiv -1 \pmod{p}$.

Pf: $a^{2k} \equiv 1 \pmod{p}$. If $p = 2$, a odd, then a has order $\phi(2) = 1 \neq 2k$. $\therefore$ Assume p odd.

$$\therefore (a^k)^2 - 1 \equiv 0 \pmod{p}$$

$$\therefore (a^k - 1)(a^k + 1) \equiv 0 \pmod{p}$$

$$\therefore p \mid (a^k - 1)(a^k + 1)$$

If $p \mid (a^k - 1)$, then $a^k \equiv 1 \pmod{p}$, so a would not have order $2k$.

$\therefore p \nmid (a^k - 1)$, so $p \mid (a^k + 1)$ (by Th. 3.1).

$\therefore a^k + 1 \equiv 0 \pmod{p} \Rightarrow a^k \equiv -1 \pmod{p}$.

(c) If a has order $n - 1$ modulo n , then n is prime.

Pf: $a^{n-1} \equiv 1 \pmod{n}$ and $a^{\phi(n)} \equiv 1 \pmod{n}$.

If $\phi(n) < n - 1$, then it would contradict $n - 1$ as the order of a .

$\therefore \phi(n) = n - 1$.

If n were composite, it would have a divisor d , $1 < d < n$. n is also a divisor of n , so $\phi(n) \leq n - 2$. But $\phi(n) = n - 1$, so n is not composite; $\therefore n$ is prime.

3. Prove $\phi(2^n - 1)$ is a multiple of n , all $n > 1$.

Pf: Since $(2^n - 1) \equiv 0 \pmod{2^n - 1}$, then $2^n \equiv 1 \pmod{2^n - 1}$.

Let k be order of $2 \pmod{2^n - 1}$.

$\therefore 2^k \equiv 1 \pmod{2^n - 1}$, or $2^k - 1 = a(2^n - 1)$, $a > 0$.

But $2^k > 1$ for $k \geq 1$, and $2^k - 1 < 2^n - 1$ for $k < n$. $\therefore 2^k - 1 = a(2^n - 1)$ only if $k = n$, $a = 1$.

$\therefore$ Order of $2 \pmod{2^n - 1}$ is n .

Note that $\gcd(2, 2^n - 1) = 1$, since $2^n - 1$ is odd. $\therefore$ By Euler's Th., $2^{\phi(2^n - 1)} \equiv 1 \pmod{2^n - 1}$.

$\therefore$ By Th. 8.1, $n \mid \phi(2^n - 1)$.

4. Assume order of $a \pmod{n}$ is h , and order of $b \pmod{n}$ is k . Show the order of $ab \pmod{n}$ divides hk . In particular, if $\gcd(h, k) = 1$, then ab has order hk .

Pf: (1) We know $a^h \equiv 1 \pmod{n}$, $b^k \equiv 1 \pmod{n}$.

$$\therefore a^{hk} \equiv 1^k \equiv 1 \pmod{n}, \quad b^{kh} \equiv 1^h \equiv 1 \pmod{n}$$

$$\therefore (ab)^{hk} = a^{hk} b^{hk} \equiv 1 \pmod{n}$$

$\therefore$ By Th. 8.1, order of ab divides hk .

(2) Suppose $\gcd(h, k) = 1$. Let $h = p_1^{h_1} p_2^{h_2} \cdots p_r^{h_r}$, $k = q_1^{k_1} \cdots q_s^{k_s}$, where $q_i \neq p_i$ since $\gcd(h, k) = 1$.

Let $\omega = \text{order of } ab$. From (1), $\omega \mid hk$, so

$$\omega = p_1^{\ell_1} p_2^{\ell_2} \cdots p_r^{\ell_r} q_1^{m_1} q_2^{m_2} \cdots q_s^{m_s},$$

where $0 \leq \ell_i \leq h_i$, $0 \leq m_i \leq k_i$.

Let $\omega = h_x k_y$, where $h_x = p_1^{\ell_1} \cdots p_r^{\ell_r}$, $k_y = q_1^{m_1} \cdots q_s^{m_s}$.

$$\therefore h_x \mid h, \quad k_y \mid k$$

$\therefore$ let $h = h' h_x$, $k = k' k_y$.

$$\therefore (ab)^{h_x k_y} = a^{h_x k_y} b^{h_x k_y} \equiv 1 \pmod{n}$$

$$\therefore (a^{h_x k_y} b^{h_x k_y})^{h'} \equiv 1 \pmod{n} \quad [1]$$

But $(a^{h_x k_y} b^{h_x k_y})^{h'} = a^{h' h_x k_y} b^{h' h_x k_y} = (a^h)^{k_y} (b^h)^{k_y} \equiv (b^h)^{k_y} \pmod{n} \quad [2]$

since $a^h \equiv 1 \pmod{n}$. $\therefore$ [1] and [2] imply $(b^h)^{k_y} \equiv 1 \pmod{n}$.

Since order of b is k , then by Th. 8.1, $k \mid h k_y$. Since $\gcd(h, k) = 1$, then $k \mid k_y$.

$$\therefore k_y \mid k \text{ and } k \mid k_y \Rightarrow k_y = k. \quad \text{Similarly, } h_x = h.$$

$$\therefore \omega = hk, \text{ so } \gcd(h, k) = 1 \Rightarrow \underline{\text{order } ab = hk}$$

5. Given that a has order 3 mod p , where p is an odd prime, show $a + 1$ must have order 6 mod p .

$$\text{Pf: } a^3 \equiv 1 \pmod{p}$$

$$\therefore p \mid (a^3 - 1) \Rightarrow p \mid (a - 1)(a^2 + a + 1)$$

If $p \mid (a - 1)$, then $a \equiv 1 \pmod{p}$, which contradicts order $a = 3$. $\therefore p \nmid (a - 1)$.

$$\therefore p \mid (a^2 + a + 1) \Rightarrow a^2 + a + 1 \equiv 0 \pmod{p} \quad [1]$$

$$\therefore a^2 + 2a + 1 \equiv a \pmod{p}, \quad \text{or}$$

$$(a + 1)^2 \equiv a \pmod{p}. \quad (\therefore \text{order } a + 1 \neq 2)$$

$$(a + 1)^3 \equiv a(a + 1) = a^2 + a \equiv -1 \pmod{p} \quad \text{from [1]} \quad \therefore \text{order } a + 1 \neq 3$$

$$(a + 1)^4 = [(a + 1)^2]^2 \equiv a^2 \pmod{p} \quad \therefore \text{order } a + 1 \neq 4$$

$$(a + 1)^5 = (a + 1)^3(a + 1)^2 \equiv a(-1) = -a \pmod{p} \quad \therefore \text{order } a + 1 \neq 5$$

$$(a + 1)^6 = [(a + 1)^3]^2 \equiv (-1)^2 = 1 \pmod{p}.$$

Also, $a + 1 \neq 1$. If true, then $a \equiv 0 \Rightarrow a^3 \equiv 0$, contradicting order of a is 3.

Also, $a \neq 1$, since if true, a would have order 1.

$\therefore$ Order($a + 1$) mod p is 6.

6. Verify the following assertions:

(a) The odd prime divisors of the integer $n^2 + 1$ are of the form $4k + 1$.

Pf: When n is even, $n^2 + 1$ is odd. The prime factorization of $n^2 + 1$ will thus contain odd primes. $\therefore$ consider p as any odd prime divisor of $n^2 + 1$.

Assume $\gcd(n, p) = 1$, for if $n = kp$, $n^2 = k^2p^2$, so $p \mid n^2$. This with $p \mid n^2 + 1 \Rightarrow p \mid 1$.

$\therefore p \mid n^2 + 1$, so $n^2 + 1 \equiv 0 \pmod{p}$, or $n^2 \equiv -1 \pmod{p}$, $\therefore n^4 \equiv 1 \pmod{p}$

Let r be order of n mod p . $\therefore$ By Th. 8.1, $r \mid 4$, $\therefore r = 1, 2$, or 4 .

If order n was 1, then $n \equiv 1 \pmod{p}$, so $n^2 \equiv 1$, $n^2 + 1 \equiv 2$, but $n^2 + 1 \equiv 0$, so $2 \equiv 0 \pmod{p}$, contradicting p an odd prime.

Similarly, order of n can't be 2 since $n^2 \equiv 1$, again yielding $2 \equiv 0$.

$\therefore$ order of n mod p is 4.

$\therefore$ By Th. 8.1, $\phi(p)$ is a multiple of 4.

$\therefore 4k = \phi(p) = p - 1 \Rightarrow \underline{p = 4k + 1}$

(b) The odd prime divisors of $n^4 + 1$ are of the form $8k + 1$.

Pf: Assume $p \mid n^4 + 1$. $\therefore n^4 \equiv -1 \pmod{p}$, $\therefore n^8 \equiv 1 \pmod{p}$.

Assume $\gcd(n, p) = 1$, for if $n = kp$, then $n^4 = k^4p^4$, so $p \mid n^4$. This with $p \mid n^4 + 1 \Rightarrow p \mid 1$, so assume $\gcd(n, p) = 1$.

Let r be order of n mod p .

$\therefore r \mid 8$ by Th. 8.1. $\therefore r = 1, 2, 4$, or 8 .

Order of n can't be 1, for $n = 1 \Rightarrow n^4 = 1$.

Order of n can't be 2. If true, then $n^2 \equiv 1$, $n^4 \equiv 1$, but $n^4 \equiv -1$.

Order of n can't be 4 since $n^4 \equiv -1$.

$\therefore$ Order of n mod p must be 8.

$\therefore 8 \mid \phi(p) \Rightarrow 8 \mid p - 1 \Rightarrow 8k = p - 1$, or $\underline{p = 8k + 1}$, some k .

(c) The odd prime divisors of the integer $n^2 + n + 1$ that are different from 3 are of the form $6k + 1$.

Pf: Observe that for n odd or even, $n^2 + n + 1$ is always odd. $\therefore$ restrict divisors to primes $\neq 2$. For $n = 1$, $n^2 + n + 1 = 3$, so not of form $6k + 1$. $\therefore$ Consider $p \neq 3$.

Assume p prime $\neq 3$, and $n^2 + n + 1 \equiv 0 \pmod{p}$.

Note that $2 \mid p - 1$, since p is odd.

Also, $(n-1)(n^2+n+1) \equiv 0 \pmod{p}$,
and $(n-1)(n^2+n+1) = n^3 - 1$.

$$\therefore n^3 \equiv 1 \pmod{p}$$

Now, $n \not\equiv 1 \pmod{p}$ for that would restrict n .

Also, $n^2 \not\equiv 1 \pmod{p}$, for if $n^2 \equiv 1$, then $n^2 + n + 1 \equiv n + 2$. But $n^2 + n + 1 \equiv 0$, so $n + 2 \equiv 0$, so $n \equiv -2$, also impossible for all n .

$$\therefore \text{order of } n \bmod p \text{ is } 3 \quad [1]$$

But $\gcd(n, p) = 1$. For if $n = kp$, some k , then $n^2 = k^2p^2$. $\therefore n^2 + n = k^2p^2 + kp = (k^2p + k)p$, so $p \mid (n^2 + n)$. Since $p \mid (n^2 + n + 1)$, this implies $p \mid 1$, a contradiction.

$$\therefore n^{\phi(p)} \equiv 1 \pmod{p}, \text{ and with [1]:}$$

$$3 \mid \phi(p) \Rightarrow 3 \mid p - 1$$

Since $2 \mid p - 1$, $\therefore 2 \cdot 3 \mid p - 1$, or $6 \mid p - 1$.

$$\therefore 6k = p - 1, \quad \underline{\text{or } p = 6k + 1, \text{ some } k.}$$

7. Establish that there are infinitely many primes of the form $4k + 1$, $6k + 1$, and $8k + 1$.

Pf: (a) $4k + 1$

Assume finitely many primes of form $4k + 1$, $p_1, p_2, \dots, p_r$. Consider the integer $(2p_1p_2 \cdots p_r)^2 + 1$. This integer is odd, and so its prime factorization will contain odd primes. Let q be such a prime. By prob. 6(a), q is of the form $4k + 1$, and so q must be among $p_1, \dots, p_r$.

$$\therefore q \mid (2p_1 \cdots p_r)^2 \quad \text{and} \quad q \mid (2p_1 \cdots p_r)^2 + 1$$

$$\therefore q \mid 1, \text{ a contradiction.}$$

$\therefore$ Assumption of finitely many primes of form $4k + 1$ is false.

(b) $6k + 1$

Assume finitely many primes of form $6k + 1$, $p_1, p_2, \dots, p_r$. Note that all p_i are thus odd.

Consider the integer $(3p_1p_2 \cdots p_r)^2 + (3p_1p_2 \cdots p_r) + 1$.

This is an odd integer since $3p_1 \cdots p_r$ is odd. It must have a prime divisor other than 3, for if $3^s = (3p_1 \cdots p_r)^2 + (3p_1 \cdots p_r) + 1$, some s , then $3 \mid 1$, a contradiction.

$\therefore$ Let q be such an odd divisor of $(3p_1 \cdots p_r)^2 + (3p_1 \cdots p_r) + 1$. By prob. 6(c), q must be of form $6k + 1$, and so must be among $p_1, \dots, p_r$.

$$\therefore q \mid (3p_1 \cdots p_r)^2 + (3p_1 \cdots p_r), \quad \text{and so } q \mid 1, \text{ a contradiction.}$$

$\therefore$ Main assumption false, so infinitely many primes of form $6k + 1$.

(c) $8k + 1$

Assume finitely many primes of form $8k + 1$, $p_1, \dots, p_r$. $\therefore$ All p_i are odd.

Consider $(2p_1 \cdots p_r)^4 + 1$, an odd integer.

Let q be a prime divisor of $(2p_1 \cdots p_r)^4 + 1$.

$\therefore q$ is odd since $(2p_1 \cdots p_r)^4 + 1$ is odd, and by prob. 6(b), q is of form $8k + 1$.

$\therefore q$ must be one of $p_1, \dots, p_r$.

$$\therefore q \mid (2p_1 \cdots p_r)^4 \Rightarrow q \mid 1.$$

$\therefore$ Assumption false, so there are infinitely many primes of form $8k + 1$.

8. (a) Prove that if p and q are odd primes and $q \mid a^p - 1$, then either $q \mid a - 1$ or else $q = 2kp + 1$, some k .

Pf: First note $\gcd(a, q) = 1$. For if not, then let $d = \gcd(a, q)$, $d > 1$. $\therefore d \mid q$ and $q \mid a^p - 1 \Rightarrow d \mid a^p - 1$. Since also $d \mid a$, then $d \mid 1$, a contradiction.

$\therefore \gcd(a, q) = 1$.

Since $q \mid a^p - 1$, then $a^p \equiv 1 \pmod{q}$.

Let r be order of $a \pmod{q}$.

$\therefore$ By Th. 8.1, $r \mid p$. Since p is prime, $r = 1$ or p .

If $r = 1$, then $a \equiv 1 \pmod{q} \Rightarrow \underline{q \mid (a - 1)}$.

If $r = p$, then since $a^{\phi(q)} \equiv 1 \pmod{q}$, then by Th. 8.1, $p \mid \phi(q)$.

But $\phi(q) = q - 1$.

$$\therefore p \mid q - 1 \Rightarrow \text{there is some } k' \text{ s.t. } pk' = q - 1.$$

But q odd $\Rightarrow q - 1$ even. Since p is odd, k' must be even, so $k' = 2k$, some k .

$$\therefore p(2k) = q - 1, \quad \underline{q = 2pk + 1, \text{ some } k}.$$

(b) Use part (a) to show that if p is an odd prime, then the prime divisors of $2^p - 1$ are of the form $2kp + 1$.

Pf: 2^p is even, so $2^p - 1$ is odd, so it contains an odd prime divisor. Let it be q .

$\therefore q \mid 2^p - 1$. From (a) above, letting $a = 2$, since $q \nmid (2 - 1)$, then

$$q = 2kp + 1, \text{ some } k.$$

(c) Find the smallest prime divisors of $2^{17} - 1$ and $2^{29} - 1$.

$2^{17} - 1$: By (b), prime divisors are of form $2(17)k + 1 = 34k + 1$.

Primes of form $34k + 1$: 103, 137, 239, 307, 409, 433, 613, 647, $\dots$

However, $2^{17} - 1$ happens to be prime.

$2^{29} - 1$: By (b), prime divisors are of form $2(29)k + 1 = 58k + 1$.

$\therefore$ Primes of form $58k + 1$ are: $59, 233, \dots$

$$2^{29} \stackrel{?}{\equiv} 1 \pmod{59}$$

$$2^6 = 64 \equiv 5 \pmod{59}, \quad 2^{24} \equiv 5^4 = 625 \equiv 35 \pmod{59}, \quad 2^5 \equiv 32 \pmod{59}$$

$$\therefore 2^{29} = 2^{24} \cdot 2^5 \equiv 35 \cdot 32 \equiv 58 \pmod{59}$$

$$2^{29} \stackrel{?}{\equiv} 1 \pmod{233}$$

$$2^4 \equiv 16 \pmod{233}, \quad 2^8 \equiv 256 \equiv 23, \quad 2^{16} \equiv 23^2 = 529 \equiv 63 \pmod{233}$$

$$2^{24} \equiv 23 \cdot 63 = 1449 \equiv 51 \pmod{233}$$

$$\therefore 2^{29} = 2^{24} \cdot 2^5 \equiv 51 \cdot 32 = 1632 \equiv 1 \pmod{233}$$

$$\therefore 2^{29} \equiv 1 \pmod{233}$$

$$\therefore \underline{233 \text{ smallest prime divisor of } 2^{29} - 1}$$

9. Prove there are infinitely many primes of the form $2kp + 1$, where p is an odd prime.

Pf: Assume finitely many primes of form $2kp + 1$. Call them $q_1, q_2, \dots, q_r$.

Let $a = 2q_1q_2 \cdots q_r$, and consider the integer

$$(2q_1q_2 \cdots q_r)^p - 1 = a^p - 1.$$

Plan: Use 8(a) to show an odd prime divisor q of $a^p - 1$ must be one of q_i , and so must divide a , and so will divide 1.

$$a^p - 1 = (a - 1)(a^{p-1} + a^{p-2} + \cdots + 1) = (a - 1)(a^{p-1} + a^{p-2} + \cdots + a^{p-p})$$

$\therefore a^{p-1} + a^{p-2} + \cdots + 1$ has p terms.

If a is even, $a^{p-1} + a^{p-2} + \cdots + 1$ is odd.

If a is odd, since p is odd, $a^{p-1} + \cdots + a^2 + a$ is even ($p - 1$ terms), so $a^{p-1} + \cdots + 1$ is odd.

$\therefore a^{p-1} + a^{p-2} + \cdots + 1$ is always odd, and so must have an odd prime divisor. Call it q .

$$\therefore q \mid a^{p-1} + a^{p-2} + \cdots + 1, \quad \text{or}$$

$$a^{p-1} + a^{p-2} + \cdots + 1 \equiv 0 \pmod{q} \quad [1]$$

$$\therefore q \mid a^p - 1 \quad \text{since } a^p - 1 = (a - 1)(a^{p-1} + \cdots + 1)$$

$$\therefore \text{By 8(a), either } q \mid (a - 1) \text{ or } q = 2kp + 1.$$

Suppose $q \mid (a - 1)$. $\therefore a \equiv 1 \pmod{q}$. $\therefore a^2 \equiv 1, a^3 \equiv 1$, etc.

$$\therefore a^{p-1} + a^{p-2} + \cdots + 1 \equiv p \pmod{q} \quad [2]$$

since there are p terms in $a^{p-1} + a^{p-2} + \cdots + 1$.

$$\therefore [1] \text{ and } [2] \Rightarrow p \equiv 0 \pmod{q}$$

$\therefore p = q$ since both are prime.

$$\therefore a \equiv 1 \pmod{p} \text{ since } a \equiv 1 \pmod{q} \text{ by assumption.}$$

But $a = 2q_1q_2 \cdots q_r = 2(2k_1p + 1)(2k_2p + 1) \cdots (2k_rp + 1)$.

Since $2k_ip + 1 \equiv 1 \pmod{p}$, then $a \equiv 2 \pmod{p}$.

$$\therefore a \equiv 1 \pmod{p} \text{ and } a \equiv 2 \pmod{p}$$

$\therefore$ assumption that $q \mid (a - 1)$ is false!

$\therefore q = 2kp + 1$ by 8(a), so q must be one of $q_1, q_2, \dots, q_r$ since they are finite.

$$\therefore q \mid (2q_1q_2 \cdots q_r) \text{ and since } q \mid a^p - 1, \text{ then } q \mid (2q_1q_2 \cdots q_r)^p - 1,$$

so $q \mid 1$, an impossibility.

$\therefore$ Assumption that primes of form $2kp + 1$ is finite is false.

10. (a) Verify 2 is a primitive root of 19 but not of 17.

$$\phi(19) = 18: \quad 2^6 = 64 \equiv 7 \pmod{19}, \quad 7^2 = 49 \equiv 11 \pmod{19}, \quad 7^3 = 77 = 4 \cdot 19 + 1 \equiv 1 \pmod{19}$$

$$\therefore 2^{18} = (2^6)^3 \equiv 7^3 \equiv 1 \pmod{19}$$

$$\therefore 2^{18} = 2^{\phi(19)} \equiv 1 \pmod{19}$$

Suppose order of 2 mod 19 = r , $r < 18$. By Th. 8.1, $r \mid \phi(19) = 18$, so $r \mid 18$ and $r < 18$, meaning $r \in \{1, 2, 3, 6, 9\}$.

$$2^1 \equiv 2, \quad 2^2 \equiv 4, \quad 2^3 \equiv 8, \quad 2^6 \equiv 7, \quad 2^9 \equiv 7 \cdot 8 = 56 \equiv 18 \equiv -1 \pmod{19}$$

None of these equal 1, so no $r < 18$ with $r \mid 18$ satisfies $2^r \equiv 1 \pmod{19}$.

$\therefore$ order of 2 mod 19 = 18 = $\phi(19)$, so 2 is a primitive root of 19. $\checkmark$

For mod 17: From problem 1(a), $\text{ord}(2) = 8 \pmod{17}$.

Since $\phi(17) = 16 \neq 8$, the order of 2 does not equal $\phi(17)$.

$\therefore$ 2 is **not** a primitive root of 17. $\checkmark$ (b) Show 15 has no primitive root by calculating orders of 2, 4, 7, 8, 11, 13, and 14 mod 15.

The integers relatively prime to 15: 1, 2, 4, 7, 8, 11, 13, 14.

$\therefore \phi(15) = 8$. Divisors of 8: 1, 2, 4, 8.

$$1: \quad 1^1 = 1 \equiv 1 \pmod{15} \quad 1 < 8 \Rightarrow 1 \text{ not a primitive root.}$$

$$2: \quad 2^4 = 16 \equiv 1 \pmod{15} \quad 4 < 8 \Rightarrow 2 \text{ not a prim. root.}$$

$$4: \quad 4^2 = 16 \equiv 1 \pmod{15} \quad 2 < 8 \Rightarrow 4 \text{ not a prim. root.}$$

- 7: $7^2 = 49 \equiv 4$, $\therefore 7^4 \equiv 16 \equiv 1 \pmod{15}$ $4 < 8 \Rightarrow 7$ not a prim. root.
 8: $8^2 = 64 \equiv 4 \pmod{15}$, $8^4 \equiv 16 \equiv 1 \pmod{15}$ $4 < 8 \Rightarrow 8$ not a prim. root.
 11: $11^2 = 121 \equiv 1 \pmod{15}$ $2 < 8 \Rightarrow 11$ not a prim. root.
 13: $13^2 = 169 \equiv 4 \pmod{15}$, $13^4 \equiv 16 \equiv 1 \pmod{15}$ $4 < 8 \Rightarrow 13$ not a prim. root.
 14: $14^2 = 196 \equiv 1 \pmod{15}$ $2 < 8 \Rightarrow 14$ not a prim. root.

No element has order $8 = \phi(15)$, so 15 has no primitive root.

11. Let r be a primitive root of the integer n . Prove that r^k is a primitive root of n if and only if $\gcd(k, \phi(n)) = 1$.

Pf: Since r has order $\phi(n) \pmod{n}$, by Th. 8.3, r^k has order $\phi(n)/\gcd(k, \phi(n))$.

(a) $\therefore$ If $\gcd(k, \phi(n)) = 1$, then r^k has order $\phi(n)$.

$\therefore r^k$ is a primitive root of n .

(b) Suppose r^k is a primitive root of n .

$\therefore r^k$ has order $\phi(n)$. From above,

$$\phi(n) = \phi(n)/\gcd(k, \phi(n))$$

$$\therefore \gcd(k, \phi(n)) = 1.$$

12. (a) Find two primitive roots of 10.

$$10 = 2 \cdot 5. \therefore \phi(10) = (2^1 - 2^0)(5^1 - 5^0) = 4.$$

These relatively prime numbers are 1, 3, 7, 9.

If 10 has a primitive root, then it has exactly $\phi(\phi(10)) = \phi(4) = 2$ of them.

$$\therefore 3: 3^4 = 81 \equiv 1 \pmod{10}$$

$$\text{and } 3^1 \equiv 3 \pmod{10}, \quad 3^2 \equiv 9 \pmod{10}, \quad 3^3 \equiv 7 \pmod{10}. \quad 7: 7^2 \equiv 9 \pmod{10}.$$

$$\therefore 7^4 \equiv 81 \equiv 1 \pmod{10}.$$

$$7^1 = 7, \quad 7^2 = 49 \equiv 9, \quad 7^3 = 63 \equiv 3 \pmod{10}.$$

$\therefore$ 3, 7 are primitive roots of 10.

Note $9^2 = 81 \equiv 1 \pmod{10}$, $\therefore 9$ not a prim. root since $2 < 4$. ($9^4 \equiv 1 \pmod{10}$.)

(b) Use the information that 3 is a primitive root of 17 to obtain the eight primitive roots of 17.

$$\text{Note } \phi(\phi(17)) = \phi(16) = 2^4 - 2^3 = 8.$$

By Th. 8.3, since 3 has order $\phi(17) = 16 \pmod{17}$, then 3^r has order $16/\gcd(r, 16)$.

$\therefore$ When $\gcd(r, 16) = 1$, 3^r will have order 16, and so be a prim. root of 17.

$\therefore$ For $\gcd(r, 16) = 1$: $r = 1, 3, 5, 7, 9, 11, 13, 15$.

$$3^3 \equiv 27 \equiv \underline{10} \pmod{17}$$

$$3^5 \equiv 10 \cdot 3^2 \equiv \underline{5} \pmod{17}$$

$$3^7 \equiv 3^5 \cdot 3^2 \equiv 5 \cdot 9 = 45 \equiv \underline{11} \pmod{17}$$

$$3^9 \equiv 3^7 \cdot 3^2 \equiv 11 \cdot 9 = 85 + 14 \equiv \underline{14} \pmod{17}$$

$$3^{11} \equiv 3^9 \cdot 3^2 \equiv 14 \cdot 9 = 126 \equiv 119 + 7 \equiv \underline{7} \pmod{17}$$

$$3^{13} \equiv 3^{11} \cdot 3^2 \equiv 7 \cdot 9 = 63 = 51 + 12 \equiv \underline{12} \pmod{17}$$

$$3^{15} \equiv 3^{13} \cdot 3^2 \equiv 12 \cdot 9 = 108 \equiv 102 + 6 \equiv \underline{6} \pmod{17}$$

$\therefore$ Primitive roots of 17 are: 3, 5, 6, 7, 10, 11, 12, 14.

13. (a) Prove that if p and $q > 3$ are both odd primes and $q \mid R_p$, then $q = 2kp + 1$ for some integer k .

Pf: $R_p = \frac{10^p - 1}{9}$. $\therefore$ If $q \mid R_p$, then for some r , $qr = \frac{10^p - 1}{9}$, or $q(9r) = 10^p - 1$.

By prob. 8(a), $q \mid 10 - 1$ or $q = 2kp + 1$, some k . Since $q > 3$, then $q \nmid (10 - 1)$ since $10 - 1 = 3^2$.

$$\therefore q = 2kp + 1, \text{ some } k.$$

(b) Find the smallest prime divisors of the repunits $R_5 = 11111$ and $R_7 = 1111111$.

R_5 : First test 3: $R_5 = 3 \cdot 3700 + 11$. $\therefore 3 \nmid R_5$.

By (a), if $q > 3$, then $q = 2k(5) + 1$. $\therefore q = 10k + 1$.

$\therefore$ Test 11, 31, 41, 71, 101, ...

By trial, $11 \nmid R_5$, $31 \nmid R_5$, but $41 \mid R_5$.

$\therefore$ Smallest prime divisor of R_5 is 41.

R_7 : First test 3: $R_7 = 3 \cdot 370000 + 1111$, $1111 = 3 \cdot 370 + 1$. $\therefore 3 \nmid R_7$.

By (a), if $q > 3$, then $q = 2k(7) + 1$.

$$\therefore q = 14k + 1$$

$$\therefore q = 29, 43, 71, 113, 127, 197, 211, 239, \dots$$

By trial, $\underline{239 \mid R_7}$ is the smallest. **14.** (a) Let $p > 5$ be prime. If R_n is the smallest repunit for which $p \mid R_n$, establish that $n \mid p - 1$. For example, R_8 is the smallest repunit divisible by 73, and $8 \mid 72$.

Pf: $p \mid R_n \Rightarrow$ there is some k s.t. $pk = \frac{10^n - 1}{9}$.

$$\therefore p(9k) = 10^n - 1, \quad \text{or} \quad 10^n \equiv 1 \pmod{p}$$

Suppose $\exists m < n$ s.t. $10^m \equiv 1 \pmod{p}$. Then $10^m - 1 = kp$, some k .

But for $m \geq 1$, $9 \mid 10^m - 1$.

$\therefore 9 \mid kp$, so $9 \mid k$ since p is prime, $p > 5$.

$$\therefore \text{let } 9k' = k.$$

$$\therefore \frac{10^m - 1}{9} = \frac{kp}{9} = k'p$$

$\therefore p \mid R_m$, contradicting that R_n is the smallest repunit divisible by p .

$$\therefore \text{order of } 10 \bmod p \text{ is } n. \quad [1]$$

Since $\gcd(10, p) = 1$, $10^{\phi(p)} \equiv 1 \pmod{p}$, or $10^{p-1} \equiv 1 \pmod{p}$.

By Th. 8.1 and [1], $n \mid p-1$.

(b) Find the smallest R_n divisible by 13.

By (a), if $13 \mid R_n$, then $n \mid 12$.

$\therefore$ Consider $n = 1, 2, 3, 4, 6$. $13 \nmid R_1$ since $13 \nmid 1$.

$13 \nmid R_2$ since $13 \nmid 11$.

$13 \nmid R_3$ since $13 \nmid 111$.

$13 \nmid R_4$ since $13 \nmid 1111$.

$13 \mid R_6$ since $13 \cdot 8547 = 111,111$.

8.2 Primitive Roots for Primes

1. If p is an odd prime, prove:

(a) The only incongruent solutions of $x^2 \equiv 1 \pmod{p}$ are 1 and $p-1$.

Pf: Since p is odd prime, $2 \mid p-1$.

$\therefore$ By Corollary to Lagrange Th. 8.5, the congruence $x^2 - 1 \equiv 0 \pmod{p}$ has exactly 2 solutions.

Clearly, 1 is a solution since $1 \equiv 1 \pmod{p}$.

$p-1$ is also a solution since $(p-1)^2 = p^2 - 2p + 1 \equiv 1 \pmod{p}$.

$\therefore$ 1 and $p-1$ are solutions and they are incongruent mod p .

($1 \equiv p-1 \pmod{p} \Rightarrow 1 \equiv -1 \pmod{p}$).

(b) The congruence $x^{p-2} + \cdots + x^2 + x + 1 \equiv 0 \pmod{p}$ has exactly $p-2$ incongruent solutions, and they are $2, 3, \dots, p-1$.

Pf: By Fermat's Th., when $\gcd(x, p) = 1$, $x^{p-1} \equiv 1 \pmod{p}$. So $x^{p-1} - 1 \equiv 0 \pmod{p}$.

Factor: $x^{p-1} - 1 = (x-1)(x^{p-2} + \cdots + x + 1)$.

So $(x-1)(x^{p-2} + \cdots + x + 1) \equiv 0 \pmod{p}$.

If $x \equiv 1 \pmod{p}$, substitute into $x^{p-2} + \cdots + 1$: we get $1+1+\cdots+1 = p-1 \equiv -1 \not\equiv 0 \pmod{p}$ (since p is odd prime). So $x=1$ is not a solution of the second factor.

For $x \not\equiv 1 \pmod{p}$, $p \nmid (x-1)$, so by Th. 3.1, $p \mid (x^{p-2} + \cdots + 1)$.

The solutions of $x^{p-1} \equiv 1 \pmod{p}$ are $1, 2, \dots, p-1$ (all $p-1$ of them, by Fermat's Th.).

Removing $x=1$ (which is not a solution of $x^{p-2} + \cdots + 1 \equiv 0$), we get exactly $p-2$ incongruent solutions: $2, 3, \dots, p-1$. 2. Verify that each of the congruences:

$$x^2 \equiv 1 \pmod{15}$$

$$x^2 \equiv -1 \pmod{65}$$

$$x^2 \equiv -2 \pmod{33}$$

has four incongruent solutions; hence Lagrange's Theorem need not hold if the modulus is a composite number.

Pf: By Corollary 2 to Th. 2.4 (p. 24), if p and q are primes, $p \neq q$, and $p \mid c$ and $q \mid c$, then $pq \mid c$.

In problems above, if p, q are prime, $p \neq q$, then if $x_1^2 \equiv a \pmod{p}$ and $x_1^2 \equiv a \pmod{q}$, then $x_1^2 \equiv a \pmod{pq}$.

Pf: $p \mid x_1^2 - a$, $q \mid x_1^2 - a$, and so $pq \mid x_1^2 - a$ by above statement.

$\therefore$ Strategy is to break up the congruence into two parts, solve each part, find common congruent solutions. $x^2 \equiv 1 \pmod{15} \Leftrightarrow x^2 \equiv 1 \pmod{3}, x^2 \equiv 1 \pmod{5}$

$$(x+1)(x-1) \equiv 0 \quad (x+1)(x-1) \equiv 0$$

$$x \equiv 1, 4, 7, 10, 13 \quad x \equiv 1, 6, 11, 16$$

$$x \equiv -1, 2, 5, 8, 11, 14 \quad x \equiv -1, 4, 9, 14$$

$$\therefore x \equiv 1, 4, 11, 14 \pmod{15}$$

$$x^2 \equiv -1 \pmod{65} \Leftrightarrow$$

$$x^2 \equiv -1 \pmod{5}$$

$$x^2 \equiv 4$$

$$(x+2)(x-2) \equiv 0$$

$$x \equiv -2, 3, 8, 13, 18, 23, 28, 33, 38, 43$$

$$x \equiv 2, 7, 12, 17, 22, 27, 32, 37, 42, 47, 52, 57$$

$$x^2 \equiv -1 \pmod{13}$$

$$x^2 \equiv 12, \quad x^2 \equiv 25$$

$$(x+5)(x-5) \equiv 0$$

$$x \equiv -5, 8, 21, 34, 47, 60$$

$$x \equiv 5, 18, 31, 44, 57$$

$$\therefore x \equiv 8, 18, 47, 57 \pmod{65}$$

$$x^2 \equiv -2 \pmod{33} \Leftrightarrow$$

$$x^2 \equiv -2 \pmod{3}$$

$$x^2 \equiv 1$$

$$(x+1)(x-1) \equiv 0$$

$$x \equiv -1, 2, 5, 8, 11, 14, 17, 20, 23, 26, 29$$

$$x \equiv 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31$$

$$x^2 \equiv -2 \pmod{11}$$

$$x^2 \equiv 9$$

$$(x+3)(x-3) \equiv 0$$

$$x \equiv -3, 8, 19, 30$$

$$x \equiv 3, 14, 25$$

$\therefore x \equiv 8, 14, 19, 25 \pmod{33}$ **3.** Determine all the primitive roots of the primes $p = 11, 19$, and 23 , expressing each as a power of some one of the roots.

11: There are $\phi(10) = (5-1)(2-1) = 4$ primitive roots.

Try 2: $2^5 \equiv 10$, $\therefore 2^{10} = 100 \equiv 1 \pmod{11}$.

Divisors of 10: 1, 2, 5. $2^1 = 2$, $2^2 = 4$, $2^5 = 10$.

$\therefore 2$ is a primitive root of 11.

Any integer relatively prime to 11 is congruent mod 11 to 2^k , $1 \leq k \leq 10$, since there are $\phi(11) = 10$ such numbers, and 2^k , $1 \leq k \leq 10$, are incongruent by Th. 8.4. $\therefore$ Other

primitive roots must be among the 2^k . By Th. 8.3, these integers 2^k will have order 10 also if $\gcd(k, 10) = 1$. $\therefore k = 1, 3, 7, 9$

$\therefore$ Primitive roots of 11 are: $2^1, 2^3, 2^7, 2^9$, or $2, 6 (\equiv 2^9), 7 (\equiv 2^7), 8$

19: There are $\phi(18) = (2-1)(3^2-3^1) = 6$ primitive roots. Try 2: $2^5 \equiv 13$, $2^6 \equiv 26 \equiv 7$, $\therefore 2^{11} \equiv 81 \equiv -4$.

$\therefore 2^{17} \equiv (7)(-4) = -28 \equiv 10$, $2^{18} \equiv 20 \equiv 1 \pmod{19}$.

Divisors of 18: 1, 2, 3, 6, 9, 18.

$2^1, 2^2, 2^3 \neq 1$. $2^6 = 7 \neq 1$, $2^9 \equiv 2^6 \cdot 2^3 = 7 \cdot 8 = 56 \equiv 18$.

$\therefore 2$ is a primitive root of 19.

Other primitive roots are congruent to 2^k , $1 \leq k \leq 18$. By Th. 8.3, need to select k s.t. $\gcd(k, 18) = 1$. $\therefore k = 1, 5, 7, 11, 13, 17$

$\therefore$ Primitive roots are $2^1, 2^5, 2^7, 2^{11}, 2^{13}, 2^{17}$, or $2, 13 (\equiv 2^5), 14 (\equiv 2^7), 15 (\equiv 2^{11}), 3 (\equiv 2^{13}), 10 (\equiv 2^{17})$

23: There are $\phi(22) = (2-1)(11-1) = 10$ primitive roots.

From table on p. 166, 5 is the smallest primitive root.

All primitive roots are congruent to 5^k , $1 \leq k \leq 22$. By Th. 8.3, need to select k s.t. $\gcd(22, k) = 1$. $\therefore k = 1, 3, 5, 7, 9, 13, 15, 17, 19, 21$ $\therefore$ Primitive roots are $5^1, 5^3, 5^5, 5^7, 5^9, 5^{13}, 5^{15}, 5^{17}, 5^{19}, 5^{21}$.

Note: $5^2 = 25 \equiv 2$.

$$\begin{array}{ll} 5 (\equiv 5^1) & 21 (\equiv -2 \equiv 5^{13}) \\ 10 (\equiv 5^3) & 19 (\equiv -4 \equiv 5^{15}) \\ 20 (\equiv 5^5) & 15 (\equiv -8 \equiv 5^{17}) \\ 17 (\equiv 5^7) & 7 (\equiv 5^{19}) \\ 11 (\equiv 5^9) & 14 (\equiv 5^{21}) \end{array}$$

4. Given that 3 is a primitive root of 43, find the following:

(a) All positive integers less than 43 having order 6 mod 43.

3^k , $1 \leq k \leq 42$ are incongruent by Th. 8.2.

$\therefore$ All integers < 43 are congruent to 3^k .

3^k has order $42/\gcd(k, 42)$ mod 43 by Th. 8.3.

$$\frac{42}{\gcd(k, 42)} = 6 \Rightarrow \gcd(k, 42) = 7$$

$\therefore k = 7, 35$

$\therefore 3^7, 3^{35}$ have order 6 mod 43. $3^3 = 27$, $3^4 = 81 \equiv -5$, $\therefore 3^7 \equiv -135 \equiv -135 + 3 \cdot 43 \equiv -6 \equiv 37$

$3^7 \cdot 3^7 \equiv (-6)(-6) = 36 \equiv -7$, $3^{14} \equiv (-7)(-5) = 35 \equiv -8$

$3^{32} \equiv 3^{14} \cdot 3^{18} \equiv (-7)(-8) = 56 \equiv 13$

$3^{33} \equiv 39 \equiv -4$, $\therefore 3^{35} \equiv 9(-4) = -36 \equiv 7$

$\therefore$ Only $7 (\equiv 3^{35})$ and $37 (\equiv 3^7)$ have order 6 mod 43.

(b) All positive integers less than 43 having order 21 mod 43.

As in (a), all such integers are congruent to 3^k , $1 \leq k \leq 42$.

$$\frac{42}{\gcd(42, k)} = 21 \Rightarrow \gcd(42, k) = 2$$

$$42 = 2 \cdot 3 \cdot 7$$

$$2, 2^2, 2^3, 2^4, 2^5 \quad (= 2, 4, 8, 16, 32)$$

$$2 \cdot 5, 2 \cdot 11, 2 \cdot 13, 2 \cdot 17, 2 \cdot 19 \quad (= 10, 22, 26, 34, 38)$$

$$2^2 \cdot 5 \quad (= 20)$$

$$2^3 \cdot 5 \quad (= 40)$$

$$\therefore 3^2, 3^4, 3^8, 3^{10}, 3^{16}, 3^{20}, 3^{22}, 3^{26}, 3^{32}, 3^{34}, 3^{38}, 3^{40}$$

$$3^2 = 9 \quad 3^{16} \equiv 25^2 \equiv 23 \quad 3^{32} \equiv 400 \equiv 13$$

$$3^4 = 81 \equiv 38 \quad 3^{20} \equiv 23 \cdot 38 \equiv 14 \quad 3^{34} \equiv 31$$

$$3^8 \equiv 25 \quad 3^{22} \equiv 40 \quad 3^{38} \equiv 31 \cdot 38 \equiv 17$$

$$3^{10} \equiv 225 \equiv 10 \quad 3^{26} \equiv 3^{16} \cdot 3^{10} \equiv 230 \equiv 15 \quad 3^{40} \equiv 9 \cdot 17 \equiv 24$$

$\therefore 9, 10, 13, 14, 15, 17, 23, 24, 25, 31, 38, 40$ all have order 21 mod 43.

5. Find all positive integers less than 61 having order 4 mod 61.

Table p. 166 indicates 2 is a primitive root of 61.

$\therefore 2^k, 1 \leq k \leq 60$ are incongruent.

$$\frac{60}{\gcd(60, k)} = 4 \Rightarrow \gcd(60, k) = 15 \quad \text{by Th. 8.3}$$

$$60 = 2^2 \cdot 3 \cdot 5. \therefore 3 \cdot 5, 3^2 \cdot 5 \Rightarrow 15, 45$$

$\therefore 2^{15}, 2^{45}$ have order 4 mod 61.

$$2^6 = 64 \equiv 3, 2^{12} \equiv 9, 2^{15} \equiv 9 \cdot 8 = 72 \equiv 11$$

$2^{30} \equiv 121 \equiv -1, 2^{45} \equiv (-1)(11) = -11 \equiv 50 \therefore$ Only 11, 50 have order 4 mod 61.

6. Assuming that r is a primitive root of the odd prime p , establish the following facts:

(a) The congruence $r^{(p-1)/2} \equiv -1 \pmod{p}$ holds.

Pf: We know $r^{p-1} \equiv 1 \pmod{p}$ by Fermat's Th.

As $\frac{p-1}{2}$ is an integer, $r^{\frac{p-1}{2}}$ exists.

$$\therefore r^{p-1} - 1 \equiv 0 \Rightarrow \left(r^{\frac{p-1}{2}} - 1\right)\left(r^{\frac{p-1}{2}} + 1\right) \equiv 0$$

If $r^{\frac{p-1}{2}} - 1 \equiv 0 \pmod{p}$, then since $\frac{p-1}{2} < p-1$, r wouldn't have order $p-1$.

$$\therefore r^{\frac{p-1}{2}} + 1 \equiv 0 \Rightarrow r^{\frac{p-1}{2}} \equiv -1 \pmod{p}$$

(b) If r' is any other primitive root of p , then rr' is not a primitive root of p .

If rr' were a primitive root, its order would be $p-1$.

But by (a), $r^{\frac{p-1}{2}} \equiv -1$ and $(r')^{\frac{p-1}{2}} \equiv -1$, so $\left(r^{\frac{p-1}{2}}\right)\left((r')^{\frac{p-1}{2}}\right) \equiv (-1)(-1) \pmod{p}$, or

$$(rr')^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

Since $\frac{p-1}{2} < p-1$, this contradicts the assumption of order of rr' .
 $\therefore rr'$ not a primitive root of p .

(c) If the integer r' is such that $rr' \equiv 1 \pmod{p}$, then r' is a primitive root of p .

Pf: We can assume $1 \leq r' \leq p-1$.

For if $r' = p$, then $r' \equiv 0$, so $rr' \not\equiv 1$.

If $r' > p$, then by Div. Alg., $r' = qp + s$, $0 \leq s \leq p-1$, so $r' \equiv s \pmod{p}$ and r' and s have same order.

$\therefore \gcd(r', p) = 1$.

Consider $(r')^k$, $1 \leq k \leq p-1$.

If $k < p-1$ and $(r')^k \equiv 1 \pmod{p}$, then

$$1 = 1^k \equiv (rr')^k = r^k(r')^k \equiv r^k \pmod{p},$$

contradicting order of $r = p-1$.

$$\therefore 1 = 1^{p-1} \equiv (rr')^{p-1} = r^{p-1}(r')^{p-1} \equiv (r')^{p-1} \pmod{p}.$$

$\therefore (r')^{p-1} \equiv 1 \pmod{p}$, $(r')^k \not\equiv 1 \pmod{p}$ for $1 \leq k < p-1$, and $\gcd(r', p) = 1$.

$\therefore$ By def., r' is a primitive root of p .

7. For a prime $p > 3$, prove that the primitive roots of p occur in incongruent pairs r, r' where $rr' \equiv 1 \pmod{p}$.

Pf: By Th. 7.4, for $n > 2$, $\phi(n)$ is an even integer, so that $\phi(n) \geq 2$.

Let r be one primitive root of p . By Th. 8.4, $r, r^2, \dots, r^{p-1}$ are congruent to $1, 2, \dots, p-1$ in some order, and so $r, r^2, \dots, r^{p-1}$ are incongruent. Since $p > 3$, there are at least 3 members in this list.

Let $r' = r^{p-2}$. $\therefore r$ and r' are incongruent, and $rr' = r \cdot r^{p-2} = r^{p-1} \equiv 1 \pmod{p}$. By 6(c) above, r' is a primitive root.

$\therefore$ If r is a primitive root of p , $p > 3$, we can always find an r' incongruent to r s.t. $rr' \equiv 1 \pmod{p}$.

8. Let r be a primitive root of the odd prime p . Prove the following:

(a) If $p \equiv 1 \pmod{4}$, then $-r$ is also a primitive root of p .

Pf: Let k be s.t. $p-1 = 4k$. r a primitive root of $p \Rightarrow r^{p-1} \equiv 1 \pmod{p}$.

$\therefore r^{4k} \equiv 1 \pmod{p}$.

$\therefore (-r)^{p-1} = (-r)^{4k} = r^{4k} \equiv 1 \pmod{p}$.

Let $1 \leq s < p-1$, and consider $(-r)^s$.

s even: $(-r)^s = r^s \not\equiv 1 \pmod{p}$ as r is a primitive root, and by def., $r^s \not\equiv 1$ if for $1 \leq s < p-1$. s odd: $(-r)^s = -r^s$. For s to be odd, $s = 4k-3$ or $4k-1$ for some k .

$4k-1$: Assume $-r^{4k-1} \equiv 1 \pmod{p}$.

$\therefore (-r)(-r^{4k-1}) = r^{4k} \equiv -r$.

But $r^{4k} = r^{p-1} \equiv 1$, so $-r \equiv 1 \Rightarrow r^2 \equiv 1$, contradicting r having order $p-1$.

$4k-3$: Assume $-r^{4k-3} \equiv 1 \pmod{p}$.

$\therefore (-r^3)(-r^{4k-3}) = r^{4k} \equiv -r^3$.

But $r^{4k} = r^{p-1} \equiv 1$, so $-r^3 \equiv 1$, or $r^3 + 1 \equiv 0 \pmod{p}$.

$$\therefore (r+1)(r^2+r+1) \equiv 0.$$

$$\therefore r+1 \equiv 0 \text{ or } r^2+r+1 \equiv 0.$$

If $r+1 \equiv 0$, then $r \equiv -1 \Rightarrow r^2 \equiv 1$, contradicting order of r as $p-1$.

$$\text{If } r^2+r+1 \equiv 0, \text{ then } (r^{p-1})(r^2+r+1) \equiv 0,$$

$$r^{p+1} + r^p + r^{p-1} \equiv r^{p+1} + r^p \equiv 0.$$

$\therefore$ Since $\gcd(r, p) = 1$, $\gcd(r^p, p) = 1$, dividing by r^p ,

$$r+1 \equiv 0 \Rightarrow r \equiv -1, \quad r^2 \equiv 1,$$

contradicting $p-1$ as order of r . $\therefore$ for $1 \leq s < p-1$, $(-r)^s \not\equiv 1 \pmod{p}$. But $(-r)^{p-1} \equiv 1 \pmod{p}$.

$\therefore -r$ is a primitive root of p .

(b) If $p \equiv 3 \pmod{4}$, then $-r$ has order $\frac{p-1}{2} \pmod{p}$.

Pf: $p = 3 + 4k$, some $k \geq 0$. $\therefore p-1 = 2 + 4k$.

$$\therefore \frac{p-1}{2} = 1 + 2k, \text{ some } k \geq 0. \therefore \frac{p-1}{2} \text{ is odd.}$$

$$\therefore (-r)^{\frac{p-1}{2}} = -r^{\frac{p-1}{2}}.$$

(a) Since $r^{p-1} \equiv 1 \pmod{p}$ (r a prim. root), then

$$r^{p-1} - 1 \equiv 0 \Rightarrow \left(r^{\frac{p-1}{2}} + 1\right)\left(r^{\frac{p-1}{2}} - 1\right) \equiv 0.$$

But $r^{\frac{p-1}{2}} - 1 \not\equiv 0$. If so, $r^{\frac{p-1}{2}} \equiv 1$, contradicting order of r as $p-1$.

$$\therefore r^{\frac{p-1}{2}} + 1 \equiv 0 \Rightarrow r^{\frac{p-1}{2}} \equiv -1 \Rightarrow$$

$$(-r)^{\frac{p-1}{2}} = -r^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

(b) Now suppose $-r$ has order s , $1 \leq s < \frac{p-1}{2}$.

s can't be even. If so, then $(-r)^s = r^s \equiv 1$, so order of r would be less than $p-1$, a contradiction.

$\therefore s$ is odd, $2s < p-1$.

$$\therefore (-r)^s \equiv 1 \Rightarrow (-r)^{2s} = r^{2s} \equiv 1.$$

This contradicts order of r as $p-1$.

$\therefore$ order of $-r$ can't be $< \frac{p-1}{2}$.

(a) + (b) $\Rightarrow$ order of $-r$ is $\frac{p-1}{2}$.

9. Give a different proof of Th. 5.5 by showing that if r is a primitive root of the prime $p \equiv 1 \pmod{4}$, then $r^{(p-1)/4}$ satisfies the quadratic congruence $x^2 + 1 \equiv 0 \pmod{p}$.

Pf: r a primitive root $\Rightarrow r^{p-1} \equiv 1 \pmod{p}$. Since $p-1 = 4k$, some k , then $\frac{p-1}{4} = k$,

some integer. Consider $x^4 \equiv 1 \pmod{p}$. $r^{\frac{p-1}{4}}$ is a solution.

$$\therefore x^4 - 1 = (x^2 + 1)(x^2 - 1) \equiv 0 \pmod{p}.$$

If $r^{\frac{p-1}{4}}$ is a solution to $x^2 - 1 \equiv 0$, then $r^{\frac{p-1}{2}} \equiv 1$, contradicting order of r as $p-1$.

$\therefore r^{\frac{p-1}{4}}$ is a solution to $x^2 + 1 \equiv 0 \pmod{p}$.

Th. 5.5 says $x^2 + 1 \equiv 0 \pmod{p}$, p odd, has a solution $\Leftrightarrow p \equiv 1 \pmod{4}$.

(a) If $p \equiv 1 \pmod{4}$, p has a primitive root, since p is prime. Call it r . Above shows $r^{(p-1)/4}$ is a solution.

(b) Suppose $x^2 + 1 \equiv 0 \pmod{p}$ has a solution. Proof is same as given in text on p. 95.

10. Use the fact that each prime p has a primitive root to give a different proof of Wilson's Theorem. *Pf:* Let r be a primitive root of p .

$1, 2, 3, \dots, p-1$ are the positive integers less than p that are relatively prime to p . Also, $\phi(p) = p-1$.

$\therefore$ By Th. 8.4, $r, r^2, \dots, r^{p-1}$ are congruent mod p to $1, 2, \dots, p-1$, in some order.

$\therefore 1 \cdot 2 \cdot 3 \cdots (p-1) \equiv r \cdot r^2 \cdot r^3 \cdots r^{p-1} \pmod{p}$, or

$$(p-1)! \equiv r^{1+2+\cdots+(p-1)} \pmod{p}.$$

$$\text{But } 1 + 2 + \cdots + (p-1) = \frac{(p-1)p}{2}.$$

$$\therefore (p-1)! \equiv r^{\frac{(p-1)p}{2}} \pmod{p}.$$

$$\therefore [(p-1)!]^2 \equiv (r^{p-1})^p \pmod{p}.$$

But, since r is a primitive root of p , $\therefore r^{p-1} \equiv 1 \pmod{p}$, so $(r^{p-1})^p \equiv 1 \pmod{p}$.

$\therefore [(p-1)!]^2 \equiv 1 \pmod{p}$. $\therefore [(p-1)!]^2 - 1 \equiv 0 \pmod{p}$, so

$$[(p-1)! + 1][(p-1)! - 1] \equiv 0 \pmod{p}.$$

If $(p-1)! - 1 \equiv 0$, then $r^{\frac{(p-1)p}{2}} \equiv (p-1)! \equiv 1$. But $r^{p-1} \equiv 1$, so $r^p \equiv r$.

$$\therefore r^{\frac{(p-1)p}{2}} = (r^p)^{\frac{p-1}{2}} \equiv r^{\frac{p-1}{2}} \equiv 1 \pmod{p},$$

which contradicts order of $r = p-1$, since $\frac{p-1}{2} < p-1$.

$\therefore (p-1)! + 1 \equiv 0 \pmod{p}$, so $(p-1)! \equiv -1 \pmod{p}$.

11. If p is a prime, show that the product of the $\phi(p-1)$ primitive roots of p is congruent mod p to $(-1)^{\phi(p-1)}$.

Pf: By Th. 8.4, since r is a primitive root, then $r^1, r^2, \dots, r^{p-1}$ are congruent to $1, 2, \dots, p-1$ in some order. If s is any other primitive root, it must be congruent to one of $1, 2, \dots, p-1$, and $\therefore$ is congruent to one of $r^1, r^2, \dots, r^{p-1}$.

$\therefore$ All primitive roots of p are of the form r^k , where $1 \leq k \leq p-1$.

By Th. 8.3, the r^k will have order $p-1$ if $\gcd(k, p-1) = 1$. Clearly, $k \neq p-1$ for this to be true, so k must be of the form $1 \leq k < p-1$.

Call these $\phi(p-1)$ integers $k_1, k_2, \dots, k_{\phi(p-1)}$, where $1 \leq k_i < p-1$.

$\therefore$ The product of these primitive roots is

$$r^{k_1} \cdot r^{k_2} \cdots r^{k_{\phi(p-1)}} = r^{k_1 + k_2 + \cdots + k_{\phi(p-1)}}.$$

By Th. 7.7, $k_1 + k_2 + \cdots + k_{\phi(p-1)} = \frac{1}{2}(p-1)\phi(p-1)$.
 $\therefore r^{k_1+k_2+\cdots+k_{\phi(p-1)}} = r^{\frac{1}{2}(p-1)\phi(p-1)}$.

For $p > 2$, $\phi(p-1)$ is even by Th. 7.4, so $2 \mid \phi(p-1)$.

$$\therefore r^{\frac{1}{2}(p-1)\phi(p-1)} = (r^{p-1})^{\frac{1}{2}\phi(p-1)} \equiv (1)^{\frac{1}{2}\phi(p-1)} \equiv 1 \pmod{p},$$

since $\frac{1}{2}\phi(p-1) \geq 1$ for $p > 2$.

Since $\phi(p-1)$ is even, $(-1)^{\phi(p-1)} = 1$, so

$$r^{k_1+k_2+\cdots+k_{\phi(p-1)}} \equiv (-1)^{\phi(p-1)} \pmod{p}.$$

For $p = 2$, the only primitive root is 1, $\phi(2-1) = 1$, and so

$$r^{k_1+\cdots+r^{\phi(p-1)}} = 1 \equiv (-1)^{\phi(2-1)} = -1 \pmod{2}$$

since $1 \equiv -1 \pmod{2}$. $\therefore$ The formula holds for $p = 2$.

12. For an odd prime p , verify that the sum

$$1^n + 2^n + 3^n + \cdots + (p-1)^n \equiv \begin{cases} 0 \pmod{p} & \text{if } (p-1) \nmid n \\ -1 \pmod{p} & \text{if } (p-1) \mid n \end{cases}$$

Pf: p odd means $p \neq 2$, so sum doesn't reduce to $1^n = 1$. But for $p = 2$, $p-1 \mid n$ for all n , and so $1^n = 1 \equiv -1 \pmod{2}$, so formula works even for $p = 2$. $\therefore$ Let p be an odd prime, let r be a primitive root.

$\therefore r, r^2, \dots, r^{p-1}$ are congruent mod p to $1, 2, \dots, p-1$ in some order by Th. 8.4.

Since $r^k \equiv j \pmod{p}$, $1 \leq k \leq p-1$, $1 \leq j \leq p-1$, then $r^{kn} \equiv j^n \pmod{p}$.

Thus, $r^n, r^{2n}, \dots, r^{(p-1)n}$ are congruent mod p to $1^n, 2^n, \dots, (p-1)^n$ in some order.

$\therefore 1^n + 2^n + \cdots + (p-1)^n \equiv r^n + r^{2n} + \cdots + r^{(p-1)n} \pmod{p}$.

Since r is a primitive root of p , $r^{p-1} \equiv 1 \pmod{p}$, so $r^{(p-1)n} \equiv 1 \pmod{p}$.

$\therefore 1^n + 2^n + \cdots + (p-1)^n \equiv 1 + r^n + r^{2n} + \cdots + r^{(p-2)n} \pmod{p}$.

Note: since $p \geq 3$, $r^{(p-2)n}$ makes sense.

Suppose $(p-1) \mid n$. Then $n = (p-1)k$, some k .

$\therefore$ For $1 \leq s \leq p-2$, $r^{sn} = r^{(p-1)ks}$. $\therefore$ Since $r^{(p-1)ks} \equiv 1^{ks} = 1 \pmod{p}$, then $r^{sn} \equiv 1$.

There are $(p-2)$ terms in $r^n + r^{2n} + \cdots + r^{(p-2)n}$.

$$\therefore 1 + r^n + r^{2n} + \cdots + r^{(p-2)n} \equiv 1 + \underbrace{[1 + 1 + \cdots + 1]}_{p-2} \equiv 1 + (p-2) \equiv p-1 \equiv -1 \pmod{p}.$$

$$\therefore \text{For } (p-1) \mid n, \quad 1^n + 2^n + \cdots + (p-1)^n \equiv -1 \pmod{p}. \quad [1]$$

Suppose $(p-1) \nmid n$. $\therefore n = a(p-1) + b$, $0 < b < (p-1)$, by Div. Alg.

$$\therefore r^n = r^{a(p-1)+b} = [r^{p-1}]^a \cdot r^b \equiv 1^a \cdot r^b \equiv r^b \pmod{p},$$

since r a prim. root of $p \Rightarrow r^{p-1} \equiv 1 \pmod{p}$.

$\therefore r^b \not\equiv 1 \pmod{p}$ since $b < p-1$ and prim. root means $p-1$ is the smallest order for r .

$$\begin{aligned}
 \therefore r^n - 1 &\equiv r^b - 1 \not\equiv 0 \pmod{p}. & [2] \\
 \text{Let } S &= 1 + r^n + r^{2n} + \dots + r^{(p-2)n}. & [3] \\
 \therefore r^n S &= r^n + r^{2n} + r^{3n} + \dots + r^{(p-2)n} + r^{(p-1)n}. & [4] \\
 \text{Subtracting [3] from [4],}
 \end{aligned}$$

$$r^n S - S = (r^n - 1)S = r^{(p-1)n} - 1,$$

$$\text{or } (r^n - 1)(1 + r^n + r^{2n} + \dots + r^{(p-2)n}) = r^{(p-1)n} - 1.$$

$$\text{But } r^{p-1} \equiv 1 \pmod{p}, \text{ so } r^{(p-1)n} - 1 \equiv 0 \pmod{p}.$$

$$\therefore (r^n - 1)(1 + r^n + \dots + r^{(p-2)n}) \equiv 0 \pmod{p}.$$

$$\text{From [2], } r^n - 1 \not\equiv 0 \pmod{p}.$$

$$\therefore 1 + r^n + \dots + r^{(p-2)n} \equiv 0 \pmod{p}.$$

$$\therefore \text{when } (p-1) \nmid n,$$

$$1^n + 2^n + \dots + (p-1)^n \equiv 1 + r^n + \dots + r^{(p-2)n} \equiv 0 \pmod{p}. \quad [5]$$

$$\therefore [1] \text{ and } [5] \text{ give}$$

$$1^n + 2^n + \dots + (p-1)^n \equiv \begin{cases} 0 \pmod{p} & \text{if } (p-1) \nmid n \\ -1 \pmod{p} & \text{if } (p-1) \mid n \end{cases}$$

8.3 Composite Numbers Having Primitive Roots

1. (a) Find the four primitive roots of 26 and the eight primitive roots of 25.

Proof of Corollary to Th. 8.9 shows that if r is an odd primitive root of p^k , then it is a primitive root of $2p^k$.

$\therefore$ Since $26 = 2 \cdot 13$, find odd primitive roots of 13. There are $\phi(13-1) = \phi(12)$ incongruent primitive roots of 13. $\phi(12) = (2^2 - 2)(3 - 1) = 4$.

Check 2^{12} : By Euler's Th., $2^{\phi(13)} = 2^{12} \equiv 1 \pmod{13}$. Order of 2 mod 13 must divide 12 (Th. 8.1), and $2^1, 2^2, 2^3, 2^4, 2^5, 2^6 \not\equiv 1 \pmod{13}$.

$\therefore$ 2 is a primitive root of 13.

By Th. 8.3, other integers having order 12 mod 13 are powers of 2^k s.t. $\gcd(k, 12) = 1$.

$\therefore k = 1, 5, 7, 11$.

$$\therefore 2^1 \equiv 2, 2^5 \equiv 6, 2^7 \equiv 11, 2^{11} \equiv 7 \pmod{13}.$$

So four incongruent prim. roots of 13 are 2, 6, 7, 11. $\therefore$ Four odd primitive roots of 13 are $(2+13), (6+13), 7, 11$, or 7, 11, 15, 19.

$\therefore$ 7, 11, 15, 19 will be prim. roots for $2 \cdot 13 = 26$.

For 25, $25 = 5^2$. Proof of Lemma 2 and Th. 8.9 show if r is a primitive root of p s.t. $r^{p-1} \not\equiv 1 \pmod{p^2}$, then r is a prim. root of p^k .

$$2^{5-1} = 16 \not\equiv 1 \pmod{5^2} \Rightarrow 2 \text{ a prim. root of } 25 \text{ since } 2 \text{ is a prim. root of } 5.$$

$$\phi(25) = 5^2 - 5 = 20. \therefore \text{Look at } 2^k \text{ s.t. } \gcd(k, 20) = 1.$$

$\therefore k = 1, 3, 7, 9, 11, 13, 17, 19$ (and $2^k \not\equiv 1 \pmod{25}$).

$$\begin{aligned}
 2^3 &\equiv 8, & 2^7 &= 128 \equiv 3, & 2^9 &= 2^7 \cdot 2^2 \equiv 12, \\
 2^{11} &= 2^9 \cdot 2^2 \equiv 12 \cdot 4 = 48 \equiv 23, & 2^{13} &= 2^{11} \cdot 2^2 \equiv 23 \cdot 4 = 92 \equiv 17, \\
 2^{17} &= 2^9 \cdot 2^7 \cdot 2 \equiv 12 \cdot 3 \cdot 2 = 72 \equiv 22, & 2^{19} &= 2^{17} \cdot 2^2 \equiv 22 \cdot 4 = 88 \equiv 13.
 \end{aligned}$$

$\therefore$ Primitive roots of 25: 2, 3, 8, 12, 13, 17, 22, 23. (b) Determine all the primitive roots of 3, 3^2 , 3^3 , and 3^4 .

Note 3^k has primitive roots by Th. 8.10.

2 is a primitive root of 3. $\therefore$ either 2 or $2 + 3$ will be primitive roots of 3^k , $k \geq 2$.

Since, if r is a prim. root of p , then order of $r \pmod{p^2}$ is $(p - 1)$ or $p(p - 1)$.

$\therefore$ Order of 2 $\pmod{3^2}$ is $(3 - 1)$ or $\phi(3^2)$. But $2^2 = 4 \not\equiv 1 \pmod{3^2}$, so $2^{\phi(3^2)} \equiv 1 \pmod{3^2}$, so 2 is a primitive root of 3^2 , 3^3 , and 3^4 .

3^2 : There are $\phi(\phi(3^2)) = \phi(6) = 2$ prim. roots.

Since $\phi(3^2) = 6$, by Th. 8.3, 2^h will have order 6 $\Leftrightarrow \gcd(h, 6) = 1$, or $h = 1, 5$.

$\therefore 2^5 = 32 \equiv 5 \pmod{3^2}$.

$\therefore$ primitive roots are 2, 5.

3^3 : $\phi(3^3) = 3^3 - 3^2 = 18$, $\phi(18) = 6$.

$\therefore$ 6 prim. roots, and all are of form 2^k , s.t. $\gcd(k, 18) = 1$, or $k = 1, 5, 7, 11, 13, 17$.

$2^5 = 32 \equiv 5 \pmod{27}$.

$$2^7 \equiv 5 \cdot 2^2 = 20 \pmod{27}$$

$$2^{11} \equiv 5 \cdot 5 \cdot 2 = 50 \equiv 23 \pmod{27}$$

$$2^{13} \equiv 23 \cdot 2^2 = 92 \equiv 11 \pmod{27}$$

$$2^{17} = 2^{11} \cdot 2^5 \cdot 2 \equiv 23 \cdot 5 \cdot 2 = 230 \equiv -40 \equiv 14 \pmod{27}$$

$\therefore$ prim. roots are 2, 5, 11, 14, 20, 23.

3^4 : $\phi(3^4) = 3^4 - 3^3 = 81 - 27 = 54 = 2 \cdot 3^3$, $\phi(54) = \phi(3^3) = 18$.

$\therefore$ 18 primitive roots, all of form 2^k s.t. $\gcd(k, 54) = 1$.

$\therefore k = 1, 5, 5^2, 7, 7^2, 5 \cdot 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53$.

$$2^1 \equiv 2, \quad 2^5 \equiv 32, \quad 2^7 = 128 \equiv 47 \pmod{81}$$

$$2^{11} \equiv 47 \cdot 2^4 \equiv 23, \quad 2^{13} \equiv 23 \cdot 2^2 \equiv 11$$

$$2^{17} \equiv 11 \cdot 2^4 \equiv 14, \quad 2^{19} \equiv 14 \cdot 2^2 \equiv 56$$

$$2^{23} \equiv 56 \cdot 2^4 \equiv 5, \quad 2^{29} \equiv 5 \cdot 2^5 \cdot 2 = 320 \equiv 77$$

$$2^{31} \equiv 77 \cdot 2^2 \equiv 65, \quad 2^{37} \equiv 65 \cdot 2^5 \cdot 2 \equiv 29$$

$$2^{41} \equiv 29 \cdot 2^4 \equiv 59, \quad 2^{43} \equiv 59 \cdot 2^2 \equiv 74$$

$$2^{47} \equiv 74 \cdot 2^4 \equiv 50, \quad 2^{53} \equiv 50 \cdot 2^5 \cdot 2 \equiv 41$$

$$2^{52} = 2^{25} \equiv 5 \cdot 2^2 \equiv 20, \quad 2^{72} = 2^{49} \equiv 50 \cdot 4 \equiv 38$$

$$2^{5 \cdot 7} = 2^{35} \equiv 65 \cdot 2^4 \equiv 68$$

$\therefore 2, 5, 11, 14, 20, 23, 29, 32, 38, 41, 47, 50, 56, 59, 65, 68, 74, 77$ are prim. roots. **2.** For an odd prime p , establish:

(a) There are as many incongruent primitive roots of $2p^n$ as p^n .

Pf: By Th. 8.9 and its corollary, p^n and $2p^n$ have prim. roots, where p is an odd prime and $n \geq 1$.

By corollary to Th. 8.4 (p. 161), there are exactly $\phi(\phi(2p^n))$ prim. roots for $2p^n$, and exactly $\phi(\phi(p^n))$ prim. roots for p^n (i.e., incongruent prim. roots).

For m, n s.t. $\gcd(m, n) = 1$, $\phi(mn) = \phi(m)\phi(n)$, and $\gcd(2, p^n) = 1$ since p is odd.

$\therefore \phi(2p^n) = \phi(2)\phi(p^n) = \phi(p^n)$, as $\phi(2) = 1$.

$$\therefore \phi(2p^n) = \phi(p^n) \Rightarrow \phi(\phi(2p^n)) = \phi(\phi(p^n)).$$

Note also that proof of corollary to Th. 8.9 shows if r is an odd prim. root of p^k , it is also a prim. root of $2p^k$. Similarly, if r is a primitive root of $2p^k$, then $\gcd(r, 2p^k) = 1$, r is odd. If n is order of $r \pmod{p^k}$, then $n = \phi(p^k)$.

Also, $r^n \equiv 1 \pmod{p^k} \Rightarrow r^n - 1 = xp^k$, some x .

But $r^n - 1$ is even, so $x = 2y$, some y , so $r^n \equiv 1 \pmod{2p^k}$.

$\therefore n \geq \phi(2p^k) = \phi(p^k)$. $\therefore n \geq \phi(p^k)$ and $n \leq \phi(p^k) \Rightarrow n = \phi(p^k)$.

$\therefore r$ is a primitive root of p^k .

So, if r is an even prim. root of p^k , then choose $r' = r + p^k$, so r' is odd and $\therefore$ a prim. root of $2p^k$.

(b) Any primitive root r of p^n is a primitive root of p .

Pf: $\gcd(r, p^n) = 1 \Rightarrow \gcd(r, p) = 1$.

Let k be order of $r \pmod{p}$. $\therefore r^k \equiv 1 \pmod{p}$.

$\therefore k \mid \phi(p) \Rightarrow k \mid (p - 1)$.

[1]

Also, $r^k = 1 + sp$, some s . So, for $n > 1$,

$$\begin{aligned} r^{kp^{n-1}} &= (1 + sp)^{p^{n-1}} \\ &= 1 + \binom{p^{n-1}}{1} sp + \binom{p^{n-1}}{2} (sp)^2 + \cdots + (sp)^{p^{n-1}}. \end{aligned}$$

But $p^{n-1} \mid \binom{p^{n-1}}{l}$, for $1 \leq l < p^{n-1}$, and $p \mid (sp)^k$, for $1 \leq k \leq p^{n-1}$.

$\therefore p^n \mid \left[\binom{p^{n-1}}{1} sp + \binom{p^{n-1}}{2} (sp)^2 + \cdots + (sp)^{p^{n-1}} \right]$.

$\therefore r^{kp^{n-1}} \equiv 1 \pmod{p^n}$.

$\therefore$ Since r is a prim. root of p^n , then $\phi(p^n) \mid kp^{n-1}$, by Th. 8.1.

$\phi(p^n) = p^{n-1}(p - 1)$.

$\therefore (p - 1) \mid k$.

[2]

[1] and [2] $\Rightarrow k = p - 1$, so order of $r \pmod{p}$ is $p - 1$, so r is a prim. root of p . (c) A primitive root of p^2 is also a prim. root of p^n for $n \geq 2$.

Pf: Let r be a primitive root of p^2 . By (b), r is also a prim. root of p .

Note $r^{p-1} \not\equiv 1 \pmod{p^2}$ since $\phi(p^2) = p(p - 1)$ and r is a prim. root of p^2 .

$\therefore r$ is a prim. root of p s.t. $r^{p-1} \not\equiv 1 \pmod{p^2}$, and proof to Th. 8.9 shows this r is a prim. root of p^n for $n \geq 1$.

$\therefore$ This r is certainly a prim. root for p^n , $n \geq 2$.

3. If r is a primitive root of p^2 , p being an odd prime, show that the solutions of the congruence $x^{p-1} \equiv 1 \pmod{p^2}$ are precisely the integers $r^p, r^{2p}, \dots, r^{(p-1)p}$.

Pf: Note if x is a solution to $x^{p-1} \equiv 1 \pmod{p^2}$, then it is a solution to $x^{p-1} \equiv 1 \pmod{p}$, since $a \equiv b \pmod{p^2} \Rightarrow a \equiv b \pmod{p}$. The corollary to Th. 8.5 says $x^{p-1} \equiv 1 \pmod{p}$ has exactly $p-1$ solutions.

$\therefore x^{p-1} \equiv 1 \pmod{p^2}$ has *at most* $(p-1)$ solutions.

Note that for $k \geq 1$,

$$(r^{kp})^{p-1} = (r^{p(p-1)})^k = (r^{\phi(p^2)})^k \equiv 1^k = 1 \pmod{p^2},$$

as r is a prim. root of p^2 .

$\therefore$ The $p-1$ integers $r^p, r^{2p}, \dots, r^{(p-1)p}$ satisfy $x^{p-1} \equiv 1 \pmod{p^2}$.

Note that $r^p, r^{2p}, \dots, r^{(p-1)p}$ are incongruent. For let $r^{ap} \equiv r^{bp} \pmod{p^2}$ for $1 \leq a, b \leq p-1$, $a \neq b$. Assume $a > b$ (same proof for $b > a$).

$\therefore r^{(a-b)p} \equiv 1 \pmod{p^2}$, contradicting order of r as $p(p-1)$ since $a-b < p-1$.

$\therefore$ The $p-1$ integers $r^p, r^{2p}, \dots, r^{(p-1)p}$ are incongruent solutions to $x^{p-1} \equiv 1 \pmod{p^2}$ and so are a complete set of solutions.

4. (a) Prove that 3 is a primitive root of all integers of the form 7^k and $2 \cdot 7^k$.

Pf: Computing powers of 3 mod 7: $3^1 \equiv 3$, $3^2 \equiv 2$, $3^3 \equiv 6$, $3^4 \equiv 4$, $3^5 \equiv 5$, $3^6 \equiv 1 \pmod{7}$.

$\therefore 3^{\phi(7)} = 3^6 \equiv 1 \pmod{7}$ and 3 has order $6 = \phi(7)$, so 3 is a prim. root of 7.

Order of 3 (mod 7^2) is then either $(7-1)$ or $7(7-1)$.

Computing $3^6 \pmod{49}$: $3^4 = 81 \equiv 32$, $3^5 \equiv 96 \equiv 47$, $3^6 \equiv 141 \equiv 43 \pmod{49}$.

$\therefore 3^6 \equiv 43 \not\equiv 1 \pmod{49 = 7^2}$.

$\therefore$ Order of 3 (mod 7^2) is $7(7-1) = \phi(7^2)$, so 3 is a prim. root of 7^2 .

Lemma 2 (p. 170) then shows 3 is a prim. root of 7^k for all $k \geq 1$.

Since 3 is an odd prim. root of 7^k , the Corollary (p. 171) shows 3 is a prim. root of $2 \cdot 7^k$. (b) Find a primitive root for any integer of the form 17^k .

We need to find a prim. root r of 17 s.t. $r^{16} \not\equiv 1 \pmod{17^2}$.

Try $r = 2$: $2^4 = 16 \equiv -1 \pmod{17}$, $2^8 \equiv 1 \pmod{17}$. $\therefore$ order of 2 mod 17 $\neq \phi(17)$, so 2 is not a prim. root of 17.

Try $r = 3$: Computing mod 17:

$$3^1 = 3, 3^2 = 9, 3^3 \equiv 10, 3^4 \equiv -4, 3^5 \equiv 5, 3^6 \equiv 15, 3^7 \equiv -6, 3^8 \equiv -1,$$

$$3^9 \equiv -3, 3^{10} \equiv -9, 3^{11} \equiv 7, 3^{12} \equiv 4, 3^{13} \equiv 12, 3^{14} \equiv 2, 3^{15} \equiv 6, 3^{16} \equiv 1 \pmod{17}.$$

$\therefore$ 3 is a prim. root of 17.

Computing mod $17^2 = 289$: $3^4 = 81$, $3^8 = 81^2 = 6561$. $6561 \div 289 = 22$ remainder 203, so $3^8 \equiv 203 \pmod{289}$. $3^{16} \equiv 203^2 = 41209 \pmod{289}$. $41209 \div 289 = 142$ remainder 171, so $3^{16} \equiv 171 \not\equiv 1 \pmod{17^2}$.

$\therefore 3$ is a prim. root of 17^k for all $k \geq 1$.

5. Obtain all the primitive roots of 41 and 82.

Table on p. 166 states 6 is a prim. root of 41.

The proof of Th. 8.4 shows all other primitive roots are congruent to one of $6^1, \dots, 6^{40}$.

41 has $\phi(\phi(41)) = \phi(40) = (2^3 - 2^2)(5 - 1) = 16$ incongruent prim. roots.

By Th. 8.3, since 6 has order 40, then 6^h has order $40/\gcd(h, 40)$.

$\therefore$ If $\gcd(h, 40) = 1$, then 6^h has order 40 and is a prim. root.

$\gcd(h, 40) = 1$ for $h = 1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39$. Computing powers of 6 mod 41 for each valid h :

$6^1 \equiv 6, 6^3 \equiv 11, 6^7 \equiv 29, 6^9 \equiv 19, 6^{11} \equiv 28, 6^{13} \equiv 24, 6^{17} \equiv 26, 6^{19} \equiv 34, 6^{21} \equiv 35, 6^{23} \equiv 30, 6^{27} \equiv 12, 6^{29} \equiv 22, 6^{31} \equiv 13, 6^{33} \equiv 17, 6^{37} \equiv 15, 6^{39} \equiv 7$.

$\therefore$ Prim. roots of 41: $\{6, 7, 11, 12, 13, 15, 17, 19, 22, 24, 26, 28, 29, 30, 34, 35\}$.

For 82: $\phi(82) = \phi(2 \cdot 41) = \phi(41) = 40$, so 82 also has 16 prim. roots.

If r is a prim. root of 41, then r or $r + 41$ (whichever is odd) is a prim. root of 82.

Checking parity of each prim. root of 41 and adjusting:

$6 \rightarrow 47, 7 \rightarrow 7, 11 \rightarrow 11, 12 \rightarrow 53, 13 \rightarrow 13, 15 \rightarrow 15, 17 \rightarrow 17, 19 \rightarrow 19, 22 \rightarrow 63, 24 \rightarrow 65, 26 \rightarrow 67, 28 \rightarrow 69, 29 \rightarrow 29, 30 \rightarrow 71, 34 \rightarrow 75, 35 \rightarrow 35$.

$\therefore$ Prim. roots of 82: $\{7, 11, 13, 15, 17, 19, 29, 35, 47, 53, 63, 65, 67, 69, 71, 75\}$. **6.** (a)

Prove that a prim. root r of p^k , p an odd prime, is a prim. root of $2p^k$ if and only if r is an odd integer.

Pf:

$(\Rightarrow)$ Suppose r is odd. Then $\gcd(r, 2p^k) = 1$. [1]

Let n be order of $r \pmod{2p^k}$. By Th. 8.1, $n \mid \phi(2p^k)$.

But $\phi(2p^k) = \phi(2)\phi(p^k) = \phi(p^k)$ by [1] and ϕ multiplicative.

Now $r^n \equiv 1 \pmod{2p^k} \Rightarrow r^n - 1 = a \cdot 2p^k$ for some $a \Rightarrow r^n \equiv 1 \pmod{p^k}$.

But r a prim. root of $p^k \Rightarrow \phi(p^k) \mid n$ by Th. 8.1.

$\therefore n \mid \phi(p^k)$ and $\phi(p^k) \mid n \Rightarrow n = \phi(p^k) = \phi(2p^k)$.

$\therefore$ odd r prim. root of $p^k \Rightarrow r$ a prim. root of $2p^k$.

$(\Leftarrow)$ Let r be a prim. root of p^k , p odd, and suppose r is also a prim. root of $2p^k$. Then $\gcd(r, 2p^k) = 1$, and since $2p^k$ is even, r must be odd. (b) Confirm that $3, 3^3, 3^5$, and 3^9 are prim. roots of $578 = 2 \cdot 17^2$, but that 3^4 and 3^{17} are not.

By 4(b), 3 is a prim. root of 17^2 , and so by 6(a), 3 is a prim. root of $2 \cdot 17^2$.

By Th. 8.3, 3^h has order $\phi(17^2)/\gcd(h, \phi(17^2))$; it is a prim. root iff $\gcd(h, \phi(17^2)) = 1$. $\phi(17^2) = 17^2 - 17 = 17 \cdot 16 = 2^4 \cdot 17$.

For $h = 1, 3, 5, 9$: $\gcd(h, 2^4 \cdot 17) = 1$, so $3, 3^3, 3^5, 3^9$ are prim. roots of 17^2 .

All of $3, 3^3, 3^5, 3^9$ are odd, so by 6(a), they are also prim. roots of $2 \cdot 17^2 = 578$.

For 3^4 : $\gcd(4, 2^4 \cdot 17) = 4$, so order of $3^4 \pmod{17^2}$ is $\phi(17^2)/4 = 2^2 \cdot 17 \neq \phi(17^2)$.

For 3^{17} : $\gcd(17, 2^4 \cdot 17) = 17$, so order of $3^{17} \pmod{17^2}$ is $\phi(17^2)/17 = 2^4 \neq \phi(17^2)$.

$\therefore 3^4$ and 3^{17} are not prim. roots of 17^2 , and hence not prim. roots of $2 \cdot 17^2$ either.

7. Assume r is a primitive root of the odd prime p and $(r + tp)^{p-1} \not\equiv 1 \pmod{p^2}$. Show $r + tp$ is a primitive root of p^k for each $k \geq 1$.

Pf: Since $r \equiv r + tp \pmod{p}$, then r and $r + tp$ have the same order mod p . $\therefore r + tp$ is also a prim. root of p .

Since any prim. root of p has order mod p^2 equal to either $(p - 1)$ or $p(p - 1)$, then $r + tp$ has order mod p^2 equal to $(p - 1)$ or $p(p - 1)$.

Since $(r + tp)^{p-1} \not\equiv 1 \pmod{p^2}$, the order of $r + tp$ is not $(p - 1)$, so it must be $p(p - 1) = \phi(p^2)$.

$\therefore r + tp$ is a prim. root of p^2 . The proof of Lemma 2 and Th. 8.9 then show that $r + tp$ is a prim. root of p^k for each $k \geq 1$.

8. If $n = 2^{k_0} p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$ is the prime factorisation of $n > 1$, define the *universal exponent* $\lambda(n)$ of n by

$$\lambda(n) = \text{lcm}(\lambda(2^{k_0}), \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})),$$

where $\lambda(2) = 1$, $\lambda(2^2) = 2$, $\lambda(2^k) = 2^{k-2}$ for $k \geq 3$.

Prove the following statements concerning the universal exponent:

(a) For $n = 2, 4, p^k, 2p^k$ (where p is an odd prime), $\lambda(n) = \phi(n)$.

Pf:

$\lambda(2) = 1$ by definition; $\phi(2) = 2^1 - 2^0 = 1$ by Th. 7.3.

$\lambda(4) = \lambda(2^2) = 2$ by definition; $\phi(2^2) = 2^2 - 2^1 = 2$. $\therefore \lambda(4) = \phi(4)$.

For $n = 2p^k$: note $\text{lcm}(1, x) = x$.

$$\lambda(2p^k) = \text{lcm}(\lambda(2), \phi(p^k)) = \text{lcm}(1, \phi(p^k)) = \phi(p^k) = \phi(2p^k).$$

For $n = p^k$:

$$\lambda(p^k) = \text{lcm}(\phi(p^k)) = \phi(p^k) = \phi(n).$$

(b) If $\gcd(a, 2^k) = 1$, then $a^{\lambda(2^k)} \equiv 1 \pmod{2^k}$.

Pf: By Euler's Theorem, $a^{\phi(2^k)} \equiv 1 \pmod{2^k}$.

$$\phi(2^k) = 2^k - 2^{k-1} = 2^{k-1}.$$

For $k = 1$: $\phi(2^1) = 1 = \lambda(2)$, so $a^{\lambda(2)} \equiv 1 \pmod{2}$.

For $k = 2$: $\phi(2^2) = 2 = \lambda(2^2)$, so $a^{\lambda(4)} \equiv 1 \pmod{4}$.

$\therefore$ For $k = 1, 2$, $a^{\lambda(2^k)} \equiv 1 \pmod{2^k}$.

For $k \geq 3$: $\lambda(2^k) = 2^{k-2}$.

The proof of Th. 8.3 showed that for $k \geq 3$, $a^{2^{k-2}} \equiv 1 \pmod{2^k}$.

$\therefore a^{\lambda(2^k)} \equiv 1 \pmod{2^k}$.

Alternatively, note $\lambda(2^{k+1}) = 2^{(k+1)-2} = 2^{k-1} = 2 \cdot 2^{k-2} = 2\lambda(2^k)$ for $k \geq 3$.

Using induction: proved true for $k = 1, 2$. Assume true for some $k \geq 3$: Induction step: assume $a^{\lambda(2^k)} \equiv 1 \pmod{2^k}$ for some $k \geq 3$.

Then $a^{\lambda(2^k)} = 1 + m \cdot 2^k$ for some integer m .

$$a^{\lambda(2^{k+1})} = a^{2\lambda(2^k)} = \left(a^{\lambda(2^k)}\right)^2 = (1 + m \cdot 2^k)^2 = 1 + m \cdot 2^{k+1} + m^2 \cdot 2^{2k}.$$

Since $2k \geq k + 1$ for $k \geq 1$, we have $2^{k+1} \mid m \cdot 2^{k+1}$ and $2^{k+1} \mid m^2 \cdot 2^{2k}$.

$\therefore a^{\lambda(2^{k+1})} \equiv 1 \pmod{2^{k+1}}$.

$\therefore$ By induction, $a^{\lambda(2^k)} \equiv 1 \pmod{2^k}$ for all $k \geq 1$. Induction hypothesis: $a^{\lambda(2^k)} \equiv 1 \pmod{2^k}$, so $a^{\lambda(2^k)} = 1 + r \cdot 2^k$ for some integer r .

$$\therefore a^{2\lambda(2^k)} = (1 + r \cdot 2^k)^2$$

$$a^{\lambda(2^{k+1})} = 1 + 2r \cdot 2^k + r^2 \cdot 2^{2k} = 1 + r \cdot 2^{k+1} + r^2 \cdot 2^{k-1} \cdot 2^{k+1} = 1 + (r + r^2 \cdot 2^{k-1}) 2^{k+1}.$$

$$\therefore a^{\lambda(2^{k+1})} \equiv 1 \pmod{2^{k+1}}.$$

$\therefore$ True for all k .

(c) If $\gcd(a, n) = 1$, then $a^{\lambda(n)} \equiv 1 \pmod{n}$.

Pf: Let $n = p^k$, p odd. $\therefore \lambda(n) = \phi(n)$ by (a).

$\therefore a^{\lambda(n)} = a^{\phi(n)} \equiv 1 \pmod{n}$, by Euler's Theorem.

Note that by Corollary 2, p. 24: if $c \equiv 0 \pmod{p_1}$ and $c \equiv 0 \pmod{p_2}$, then $c \equiv 0 \pmod{p_1 p_2}$, where $\gcd(p_1, p_2) = 1$. $\therefore$ If $n = 2^{k_0} p_1^{k_1} \cdots p_r^{k_r}$, then by (b) and (c):

$$a^{\lambda(2^{k_0})} \equiv 1 \pmod{2^{k_0}}.$$

$$\therefore a^{\text{lcm}[\lambda(2^{k_0}), \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})]} \equiv 1 \pmod{2^{k_0}},$$

$$\text{since } \lambda(2^{k_0}) \mid \text{lcm}[\lambda(2^{k_0}), \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})].$$

$$\therefore a^{\lambda(n)} \equiv 1 \pmod{2^{k_0}}. \quad [1]$$

$$\text{Also, } a^{\lambda(p_i^{k_i})} \equiv 1 \pmod{p_i^{k_i}}.$$

$$\therefore a^{\text{lcm}[\lambda(2^{k_0}), \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})]} \equiv 1 \pmod{p_i^{k_i}},$$

$$\text{since } \lambda(p_i^{k_i}) \mid \text{lcm}[\lambda(2^{k_0}), \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})], \text{ as } \lambda(p_i^{k_i}) = \phi(p_i^{k_i}) \text{ by (a).}$$

$$\therefore a^{\lambda(n)} \equiv 1 \pmod{p_i^{k_i}}. \quad [2]$$

$\therefore$ By [1] and [2], $a^{\lambda(n)} \equiv 1 \pmod{n}$. **9.** Verify that, for $5040 = 2^4 \cdot 3^2 \cdot 5 \cdot 7$, $\lambda(5040) = 12$ and $\phi(5040) = 1152$.

By def. in #8:

$$\lambda(5040) = \text{lcm}(\lambda(2^4), \phi(3^2), \phi(5), \phi(7))$$

$$= \text{lcm}(2^2, 3^2 - 3, 4, 6)$$

$$= \text{lcm}(4, 6, 4, 6) = 2^2 \cdot 3 = \underline{12}.$$

$$\phi(5040) = (2^4 - 2^3)(3^2 - 3)(4)(6) = (8)(6)(4)(6) = \underline{1152}.$$

10. Use Problem 8 to show that if $n \neq 2, 4, p^k, 2p^k$ (where p is an odd prime), then n has no primitive root.

Pf: Let $n = 2^{k_0} p_1^{k_1} \cdots p_r^{k_r}$ be the prime factorization.

Note from Th. 2.8, $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$.

$$\therefore \text{lcm}(a, b) \mid ab \text{ and } \text{lcm}(a, b) \leq ab.$$

Case $k_0 \neq 0$: Since $n \neq 2, 4$, or $2p^k$, then $k_0 \geq 3$, so $\lambda(2^{k_0}) = 2^{k_0-2} < 2^{k_0-1} = \phi(2^{k_0})$.

$$\therefore \lambda(2^{k_0}) = \frac{1}{2} \phi(2^{k_0}).$$

$$\begin{aligned}
 \therefore \lambda(n) &= \text{lcm}(\lambda(2^{k_0}), \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})) \\
 &= \text{lcm}(2^{k_0-2}, \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})) \\
 &< \text{lcm}(2^{k_0-1}, \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})), \quad \text{by prob. \#1, Section 6.1} \\
 &= \text{lcm}(\phi(2^{k_0}), \phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})) \\
 &\leq \phi(n).
 \end{aligned}$$

$\therefore \lambda(n) < \phi(n)$, and from 8(c), $a^{\lambda(n)} \equiv 1 \pmod{n}$.

$\therefore n$ has no primitive root for $k_0 \neq 0$.

Case $k_0 = 0$: Then $\lambda(n) = \text{lcm}(\phi(p_1^{k_1}), \dots, \phi(p_r^{k_r}))$.

Each $\phi(p_i^{k_i})$ is even, and $r > 1$ (more than one prime factor).

$\therefore$ Let $\phi(p_i^{k_i}) = 2s_i$.

$\therefore \gcd(\phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})) = \gcd(2s_1, \dots, 2s_r) \geq 2$.

$\therefore \text{lcm}(\phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})) \cdot \gcd(\phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})) \geq 2 \text{lcm}(\phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})). \therefore$

Since $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$, then

$$\begin{aligned}
 \phi(n) &= \phi(p_1^{k_1}) \cdots \phi(p_r^{k_r}) \geq 2 \text{lcm}(\phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})) \\
 &> \text{lcm}(\phi(p_1^{k_1}), \dots, \phi(p_r^{k_r})) = \lambda(n).
 \end{aligned}$$

$\therefore \lambda(n) < \phi(n)$ for $k_0 = 0$, and from 8(c), $a^{\lambda(n)} \equiv 1 \pmod{n}$.

$\therefore$ For $n \neq 2, 4, p^k, 2p^k$: $\lambda(n) < \phi(n)$ and $a^{\lambda(n)} \equiv 1 \pmod{n}$ for all a with $\gcd(a, n) = 1$.

$\therefore n$ has no primitive root.

11. (a) Prove that if $\gcd(a, n) = 1$, then the linear congruence $ax \equiv b \pmod{n}$ has the solution $x \equiv ba^{\lambda(n)-1} \pmod{n}$.

Pf: By 8(c), $a^{\lambda(n)} \equiv 1 \pmod{n}$, so

$$b \cdot a^{\lambda(n)} \equiv b \pmod{n}.$$

$\therefore b \cdot a \cdot a^{\lambda(n)-1} \equiv b \pmod{n}$

$\therefore a(ba^{\lambda(n)-1}) \equiv b \pmod{n}$

$\therefore x \equiv ba^{\lambda(n)-1} \pmod{n}$ is a solution.

(b) Use part (a) to solve $13x \equiv 2 \pmod{40}$ and $3x \equiv 13 \pmod{77}$.

$13x \equiv 2 \pmod{40}$. Note $\gcd(13, 40) = 1$.

$\therefore$ By (a), $x \equiv 2 \cdot 13^{\lambda(40)-1}$.

Since $40 = 2^3 \cdot 5$:

$$\lambda(40) = \text{lcm}(\lambda(2^3), \phi(5)) = \text{lcm}(2, 4) = 4.$$

$\therefore x \equiv 2 \cdot 13^3 \pmod{40}$. $2 \cdot 13^3 = 4394 \equiv 34 \pmod{40}$.

$\therefore x \equiv 2 \cdot 13^3 \equiv \underline{34} \pmod{40}$.

$3x \equiv 13 \pmod{77}$. Note $\gcd(3, 77) = 1$.

$\therefore$ By (a), $x \equiv 13 \cdot 3^{\lambda(77)-1}$.

$77 = 7 \cdot 11$, so $\lambda(77) = \text{lcm}(\phi(7), \phi(11)) = \text{lcm}(6, 10) = 30$.
 $\therefore x \equiv 13 \cdot 3^{29} \pmod{77}$.
 Computing powers of 3 mod 77: $3^4 \equiv 4$, $3^8 \equiv 16$, $3^{12} \equiv 64$, $3^{24} \equiv 4096 \equiv 15$, $3^{28} \equiv 60$,
 $3^{29} \equiv 180 \equiv 26 \pmod{77}$.
 $13 \cdot 3^{29} \equiv 13 \cdot 26 = 338 \equiv 30 \pmod{77}$.
 $\therefore x \equiv 13 \cdot 3^{29} \equiv \underline{30} \pmod{77}$.

8.4 The Theory of Indices

1. Find the index of 5 relative to each of the primitive roots of 13.

There are $\phi(\phi(13)) = \phi(12) = 4$ prim. roots of 13.
 2 is a prim. root since $2^{12} \equiv 1 \pmod{13}$ and $2^1, 2^2, \dots, 2^6 \not\equiv 1 \pmod{13}$.
 Other prim. roots are 2^k for $1 \leq k \leq 12$ with $\text{gcd}(k, 12) = 1$, by Th. 8.3 and 8.4.
 $\text{gcd}(k, 12) = 1 \Rightarrow k = 1, 5, 7, 11$.
 $2^1 \equiv 2$, $2^5 \equiv 6$, $2^7 \equiv 11$, $2^{11} \equiv 7 \pmod{13}$.
 $\therefore$ Prim. roots of 13: 2, 6, 7, 11.

Compute powers to locate 5:
 2: $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 \equiv 3$, $2^5 \equiv 6$, $2^6 \equiv 12$, $2^7 \equiv 11$, $2^8 \equiv 9$, $2^9 \equiv 5$.
 6: $6^1 \equiv 6$, $6^2 \equiv 10$, $6^3 \equiv 8$, $6^4 \equiv 9$, $6^5 \equiv 2$, $6^6 \equiv -1$, $6^7 \equiv 7$, $6^8 \equiv 3$, $6^9 \equiv 5$.
 7: $7^1 \equiv 7$, $7^2 \equiv -3 \equiv 10$, $7^3 \equiv -21 \equiv 5 \pmod{13}$. 11: $11^1 \equiv -2$, $11^2 \equiv -22 \equiv$
 4, $11^3 \equiv 44 \equiv 5 \pmod{13}$.
 $\therefore \text{ind}_2 5 = 9$, $\text{ind}_6 5 = 9$, $\text{ind}_7 5 = 3$, $\text{ind}_{11} 5 = 3$.

2. Using a table of indices for the primitive root 2 of 11, solve:

(a) $7x^3 \equiv 3 \pmod{11}$ (b) $3x^4 \equiv 5 \pmod{11}$ (c) $x^8 \equiv 10 \pmod{11}$

By table on p. 166, 2 is a prim. root of 11.

Powers of 2 mod 11: $2^1 \equiv 2$, $2^2 \equiv 4$, $2^3 \equiv 8$, $2^4 \equiv 5$, $2^5 \equiv 10$, $2^6 \equiv 9$, $2^7 \equiv 7$, $2^8 \equiv$
 3, $2^9 \equiv 6$, $2^{10} \equiv 1$.

a	1	2	3	4	5	6	7	8	9	10
$\text{ind}_2 a$	10	1	8	2	4	9	7	3	6	5

(a) $7x^3 \equiv 3 \pmod{11}$. Note $\text{gcd}(3, 11) = 1$ and 11 has a prim. root.

$$7x^3 \equiv 3 \pmod{11} \Leftrightarrow \text{ind}_2 7 + 3\text{ind}_2 x \equiv \text{ind}_2 3 \pmod{10} \Leftrightarrow 7 + 3\text{ind}_2 x \equiv 8 \pmod{10} \Leftrightarrow 3\text{ind}_2 x \equiv 1 \pmod{10}$$

Since $\text{gcd}(3, 10) = 1$ and $1 \mid 1$, by Th. 4.7 there is one incongruent solution.

$\text{ind}_2 x \equiv 7 \pmod{10}$. From the table, $x = 7$.

$\therefore x \equiv \underline{7} \pmod{11}$.

(b) $3x^4 \equiv 5 \pmod{11}$. Note $\text{gcd}(5, 11) = 1$ and 11 has a prim. root.

$$3x^4 \equiv 5 \pmod{11} \Leftrightarrow \text{ind}_2 3 + 4\text{ind}_2 x \equiv \text{ind}_2 5 \pmod{10} \Leftrightarrow 8 + 4\text{ind}_2 x \equiv 4 \pmod{10} \Leftrightarrow 4\text{ind}_2 x \equiv -4 \pmod{10}$$

$\gcd(4, 10) = 2$ and $2 \mid 6$, so 2 incongruent solutions.

$$4 \operatorname{ind}_2 x \equiv 6 \pmod{10} \Rightarrow 2 \operatorname{ind}_2 x \equiv 3 \pmod{5} \Rightarrow \operatorname{ind}_2 x = 4, 9.$$

From the table: $\operatorname{ind}_2 x = 4 \Rightarrow x = 5$; $\operatorname{ind}_2 x = 9 \Rightarrow x = 6$.

$\therefore x \equiv \underline{5}$ or $\underline{6} \pmod{11}$. (c) $x^8 \equiv 10 \pmod{11}$. Note $\gcd(10, 11) = 1$ and 11 has a prim. root.

$$x^8 \equiv 10 \pmod{11} \Leftrightarrow 8 \operatorname{ind}_2 x \equiv \operatorname{ind}_2 10 \pmod{10} \Leftrightarrow 8 \operatorname{ind}_2 x \equiv 5 \pmod{10}.$$

But $\gcd(8, 10) = 2$ and $2 \nmid 5$.

$\therefore$ By Th. 4.7, no solution to $8 \operatorname{ind}_2 x \equiv 5 \pmod{10}$.

$\therefore$ no solution to $x^8 \equiv 10 \pmod{11}$.

3. The following is a table of indices for the prime 17 relative to the primitive root 3:

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$\operatorname{ind}_3 a$	16	14	1	12	5	15	11	10	2	3	7	13	4	9	6	8

With the aid of this table, solve: (a) $x^{12} \equiv 13 \pmod{17}$. (b) $8x^5 \equiv 10 \pmod{17}$.
(c) $9x^8 \equiv 8 \pmod{17}$. (d) $7^x \equiv 7 \pmod{17}$.

(a) $x^{12} \equiv 13 \pmod{17}$. $\gcd(13, 17) = 1$. $\therefore 12 \operatorname{ind}_3 x \equiv \operatorname{ind}_3 13 \pmod{16}$, $\operatorname{ind}_3 13 = 4$.

$\therefore 12 \operatorname{ind}_3 x \equiv 4 \pmod{16}$, $\gcd(12, 16) = 4 \Rightarrow 4$ incongruent solutions.

Dividing by 4: $3 \operatorname{ind}_3 x \equiv 1 \pmod{4}$.

$\therefore \operatorname{ind}_3 x = 3, 7, 11, 15$.

$\therefore x = 10, 11, 7, 6$ from table.

$\therefore x \equiv \underline{6, 7, 10, 11} \pmod{17}$.

(b) $8x^5 \equiv 10 \pmod{17}$. $\gcd(10, 17) = 1$.

$$\operatorname{ind}_3 8 + 5 \operatorname{ind}_3 x \equiv \operatorname{ind}_3 10 \pmod{16} \Rightarrow 10 + 5 \operatorname{ind}_3 x \equiv 3 \pmod{16}.$$

$5 \operatorname{ind}_3 x \equiv -7 \pmod{16}$. $\gcd(5, 16) = 1 \Rightarrow 1$ solution.

Multiplying by -3 (since $5 \cdot (-3) = -15 \equiv 1 \pmod{16}$): $-\operatorname{ind}_3 x \equiv -21$, $\operatorname{ind}_3 x \equiv 21 \equiv 5 \pmod{16}$.

$\therefore x = 5$ from table.

$\therefore x \equiv \underline{5} \pmod{17}$. (c) $9x^8 \equiv 8 \pmod{17}$. $\gcd(8, 17) = 1$.

$$\operatorname{ind}_3 9 + 8 \operatorname{ind}_3 x \equiv \operatorname{ind}_3 8 \pmod{16} \Rightarrow 2 + 8 \operatorname{ind}_3 x \equiv 10 \pmod{16}.$$

$8 \operatorname{ind}_3 x \equiv 8 \pmod{16}$. $\gcd(8, 16) = 8 \Rightarrow 8$ incongruent solutions.

$\therefore \operatorname{ind}_3 x \equiv 1 \pmod{2}$, so $\operatorname{ind}_3 x = 1, 3, 5, 7, 9, 11, 13, 15$.

$\therefore x \equiv \underline{3, 10, 5, 11, 14, 7, 12, 6} \pmod{17}$.

(d) $7^x \equiv 7 \pmod{17}$. $\gcd(7, 17) = 1$.

$\therefore x \operatorname{ind}_3 7 \equiv \operatorname{ind}_3 7 \pmod{16}$.

$11x \equiv 11 \pmod{16}$. $\gcd(11, 16) = 1$, so just one solution.

$\therefore x \equiv \underline{1} \pmod{16}$ [don't need table at this point].

4. Find the remainder when $3^{24} \cdot 5^{13}$ is divided by 17.

$3^{24} \cdot 5^{13} \equiv x \pmod{17}$. Use 3 as prim. root of 17 with table in #3.

$\therefore 24\text{ind}_3 3 + 13\text{ind}_3 5 \equiv \text{ind}_3 x \pmod{16}$.

$\therefore 24(1) + 13(5) = 24 + 65 = 89 \equiv 9 \equiv \text{ind}_3 x \pmod{16}$.

From table, $x = 14$.

$\therefore x \equiv 14 \pmod{17}$, so the remainder is 14.

5. If r and r' are both prim. roots of the odd prime p , show that for $\gcd(a, p) = 1$:

$$\text{ind}_{r'} a \equiv (\text{ind}_r a)(\text{ind}_{r'} r) \pmod{p-1}.$$

Pf: Let $x = \text{ind}_{r'} a$, $y = \text{ind}_r a$, $z = \text{ind}_{r'} r$. By def., $(r')^x \equiv a \pmod{p}$, $r^y \equiv a \pmod{p}$, and $(r')^z \equiv r \pmod{p}$.

$(r')^z \equiv r \pmod{p} \Rightarrow (r')^{zy} \equiv r^y \pmod{p}$.

$\therefore (r')^x \equiv r^y \equiv (r')^{zy} \pmod{p}$.

By Th. 8.2, $x \equiv zy \pmod{p-1}$.

$\therefore \text{ind}_{r'} a \equiv (\text{ind}_r a)(\text{ind}_{r'} r) \pmod{p-1}$.

6. (a) Construct a table of indices for the prime 17 with respect to the primitive root 5.

[*Hint:* By the previous problem, $\text{ind}_5 a \equiv 13\text{ind}_3 a \pmod{16}$.]

(b) Solve the congruences in Problem 3, using the table in part (a).

(a) By #5, $\text{ind}_5 a \equiv (\text{ind}_5 3)(\text{ind}_3 a) \pmod{16}$.

Let $\text{ind}_5 3 = x$. $\therefore 5^x \equiv 3 \pmod{17}$.

$\therefore x\text{ind}_3 5 \equiv \text{ind}_3 3 \equiv 1 \pmod{16}$.

From table in #3, $\text{ind}_3 5 = 5$.

$\therefore 5x \equiv 1 \pmod{16}$. $\gcd(5, 16) = 1$, so just one solution $\pmod{16}$.

$\therefore 15x \equiv 3$, $-x \equiv 3$, $x \equiv -3 \equiv 13 \pmod{16}$.

$\therefore \text{ind}_5 3 = 13$, and hence $\text{ind}_5 a \equiv 13\text{ind}_3 a \pmod{16}$. Computing $\text{ind}_5 a \equiv 13\text{ind}_3 a \pmod{16}$ for each a :

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$\text{ind}_3 a$	16	14	1	12	5	15	11	10	2	3	7	13	4	9	6	8
$13\text{ind}_3 a$	208	182	13	156	65	195	143	130	26	39	91	169	52	117	78	104
$\text{ind}_5 a$	16	6	13	12	1	3	15	2	10	7	11	9	4	5	14	8

(b) Solving the congruences of Problem 3 using the ind_5 table:

$x^{12} \equiv 13 \pmod{17}$: $12\text{ind}_5 x \equiv \text{ind}_5 13 \equiv 4 \pmod{16}$. $\gcd(12, 16) = 4$ solutions.
 $3\text{ind}_5 x \equiv 1 \pmod{4}$, $\text{ind}_5 x = 3, 7, 11, 15$. $\therefore x \equiv 6, 10, 11, 7 \pmod{17}$ — same as in #3.

$8x^5 \equiv 10 \pmod{17}$: $\text{ind}_5 8 + 5\text{ind}_5 x \equiv \text{ind}_5 10 \pmod{16}$, $2 + 5\text{ind}_5 x \equiv 7 \pmod{16}$.
 $5\text{ind}_5 x \equiv 5 \pmod{16}$. $\gcd(5, 16) = 1$ solution. $\therefore \text{ind}_5 x \equiv 1 \pmod{16}$, $x \equiv 5 \pmod{17}$ — same as in #3.

$9x^8 \equiv 8 \pmod{17}$: $\text{ind}_5 9 + 8\text{ind}_5 x \equiv \text{ind}_5 8 \pmod{16}$. $10 + 8\text{ind}_5 x \equiv 2 \pmod{16}$,
 $8\text{ind}_5 x \equiv -8 \equiv 8 \pmod{16}$. $\gcd(8, 16) = 8$ solutions.

$\text{ind}_5 x \equiv 1 \pmod{2}$, $\text{ind}_5 x = 1, 3, 5, 7, 9, 11, 13, 15$.

$\therefore x \equiv 5, 6, 14, 10, 12, 11, 3, 7 \pmod{17}$ — same as in #3.

$7x \equiv 7 \pmod{17}$: $x\text{ind}_5 7 \equiv \text{ind}_5 7 \pmod{16}$, $15x \equiv 15 \pmod{16}$. $\gcd(15, 16) = 1$
 solution. $\therefore x \equiv 1 \pmod{16}$ — same as in #3.

7. If r is a primitive root of the odd prime p , verify that

$$\text{ind}_r(-1) = \text{ind}_r(p-1) = \frac{1}{2}(p-1).$$

Pf: (a) Since $-1 \equiv p-1 \pmod{p}$, then $\text{ind}_r(-1) = \text{ind}_r(p-1)$.

(b) Let $x = \text{ind}_r(p-1)$. Then $r^x \equiv p-1 \pmod{p}$.

As p is odd, $p-1$ is even and $\therefore \frac{p-1}{2}$ exists.

$\therefore r^{p-1} \equiv 1 \equiv p^2 - 2p + 1 = (p-1)^2 \pmod{p}$.

$\therefore r^{p-1} \equiv (p-1)^2 \pmod{p}$. $\therefore r^{\frac{p-1}{2}} \equiv p-1$ or $-(p-1) = -p+1$.

If $r^{\frac{p-1}{2}} \equiv -p+1 \equiv 1 \pmod{p}$, then since $\frac{p-1}{2} < p-1$, r would not have order $p-1$.

$\therefore r^{\frac{p-1}{2}} \not\equiv -p+1$.

$\therefore r^{\frac{p-1}{2}} \equiv p-1 \pmod{p}$.

$\therefore$ By def., $\text{ind}_r(p-1) = \frac{p-1}{2}$.

8. (a) Determine the integers a ($1 \leq a \leq 12$) s.t. the congruence $ax^4 \equiv b \pmod{13}$
 has a solution for $b = 2, 5, 6$.

Note $\gcd(b, 13) = 1$ for each of these b values.

$\therefore \text{ind } a + 4\text{ind } x \equiv \text{ind } b \pmod{12}$.

$\therefore 4\text{ind } x \equiv \text{ind } b - \text{ind } a \pmod{12}$.

$\gcd(4, 12) = 4$, so for a solution to exist, $4 \mid (\text{ind } b - \text{ind } a)$. $\therefore \text{ind } b - \text{ind } a =$
 $0, 4 \text{ (or } -4), 8 \text{ (or } -8)$.

Using table of indices for prim. root 2 of 13 (p. 175): $\text{ind}_2 2 = 1$, $\text{ind}_2 5 = 9$, $\text{ind}_2 6 =$
 5.

(1) $\text{ind } b - \text{ind } a = 0 \Rightarrow \text{ind } b = \text{ind } a \Rightarrow b = a$.

$\therefore a = 2, 5, 6$ when $b = 2, 5, 6$ respectively.

(2) $\text{ind } b - \text{ind } a = 4 \text{ (or } -4)$:

$1 - \text{ind}_2 a = -4 \Rightarrow \text{ind}_2 a = 5 \Rightarrow a = 6$. ($b = 2$)

$9 - \text{ind}_2 a = 4 \Rightarrow \text{ind}_2 a = 5 \Rightarrow a = 6$. ($b = 5$)

$5 - \text{ind}_2 a = 4, -4 \Rightarrow \text{ind}_2 a = 1, 9 \Rightarrow a = 2, 5$. ($b = 6$)

$\therefore b = 2 : a = 6$; $b = 5 : a = 6$; $b = 6 : a = 2$ or 5 .

(3) $\text{ind } b - \text{ind } a = 8 \text{ (or } -8)$:

$1 - \text{ind}_2 a = -8 \Rightarrow \text{ind}_2 a = 9 \Rightarrow a = 5$. ($b = 2$)

$9 - \text{ind}_2 a = 8 \Rightarrow \text{ind}_2 a = 1 \Rightarrow a = 2$. ($b = 5$)

$5 - \text{ind}_2 a = -8 \Rightarrow \text{ind}_2 a = 13 \Rightarrow$ no solution. ($b = 6$) Combining all three cases:

When $b = 2$: $a = 2, 6$, or 5 .

When $b = 5$: $a = 5, 6$, or 2 .

When $b = 6$: $a = 6, 2$, or 5 .

(b) Determine the integers a ($1 \leq a \leq p - 1$) s.t. the congruence $x^4 \equiv a \pmod{p}$ has a solution, for $p = 7, 11, 13$.

Construct tables of indices. 3 is a prim. root of 7; 2 is a prim. root of 11.

Powers of 3 mod 7: $3^1 \equiv 3, 3^2 \equiv 2, 3^3 \equiv 6, 3^4 \equiv 4, 3^5 \equiv 5, 3^6 \equiv 1$.

Powers of 2 mod 11: $2^1 \equiv 2, 2^2 \equiv 4, 2^3 \equiv 8, 2^4 \equiv 5, 2^5 \equiv 10, 2^6 \equiv 9, 2^7 \equiv 7, 2^8 \equiv 3, 2^9 \equiv 6, 2^{10} \equiv 1$.

a	1	2	3	4	5	6
$\text{ind}_3 a \pmod{7}$	6	2	1	4	5	3

a	1	2	3	4	5	6	7	8	9	10
$\text{ind}_2 a \pmod{11}$	10	1	8	2	4	9	7	3	6	5

$p = 7$: $x^4 \equiv a \pmod{7}$.

$4 \text{ind}_3 x \equiv \text{ind}_3 a \pmod{6}$. $\gcd(4, 6) = 2$.

$\therefore 2 \mid \text{ind}_3 a \Rightarrow \text{ind}_3 a = 2, 4, 6$. From table: $\text{ind}_3 a = 2 \Rightarrow a = 2$; $\text{ind}_3 a = 4 \Rightarrow a = 4$; $\text{ind}_3 a = 6 \Rightarrow a = 1$.

$\therefore$ For $p = 7$, the solutions are $a \in \{1, 2, 4\}$.

$p = 11$: $x^4 \equiv a \pmod{11}$.

$4 \text{ind}_2 x \equiv \text{ind}_2 a \pmod{10}$. $\gcd(4, 10) = 2$.

$\therefore 2 \mid \text{ind}_2 a \Rightarrow \text{ind}_2 a = 2, 4, 6, 8, 10$.

From table: $a = 4, 5, 9, 3, 1$.

$\therefore$ For $p = 11$, the solutions are $a \in \{1, 3, 4, 5, 9\}$.

$p = 13$: $x^4 \equiv a \pmod{13}$.

$4 \text{ind}_2 x \equiv \text{ind}_2 a \pmod{12}$. $\gcd(4, 12) = 4$.

$\therefore 4 \mid \text{ind}_2 a \Rightarrow \text{ind}_2 a = 4, 8, 12$.

From table (p. 175): $\text{ind}_2 a = 4 \Rightarrow a = 3$; $\text{ind}_2 a = 8 \Rightarrow a = 9$; $\text{ind}_2 a = 12 \Rightarrow a = 1$.

$\therefore$ For $p = 13$, the solutions are $a \in \{1, 3, 9\}$. **9.** Employ the corollary to Th. 8.12 to establish that if p is an odd prime, then:

(a) $x^2 \equiv -1 \pmod{p}$ is solvable $\Leftrightarrow p \equiv 1 \pmod{4}$.

Since $-1 \equiv p - 1 \pmod{p}$ and $\gcd(p - 1, p) = 1$, then $\gcd(-1, p) = 1$.

Using the corollary to Th. 8.12, $x^2 \equiv -1 \pmod{p}$ is solvable $\Leftrightarrow (-1)^{\frac{p-1}{2}} \equiv 1 \pmod{p}$, as $2 = \gcd(2, p - 1)$ since p is odd.

$$(-1)^{\frac{p-1}{2}} = \begin{cases} 1 & \text{if } \frac{p-1}{2} \text{ is even} \\ -1 & \text{if } \frac{p-1}{2} \text{ is odd} \end{cases}$$

So, $(-1)^{\frac{p-1}{2}} \equiv 1 \Leftrightarrow (-1)^{\frac{p-1}{2}} = 1 \Leftrightarrow \frac{p-1}{2}$ is even.

$$\therefore \frac{p-1}{2} = 2k, \text{ some } k, \text{ so } p = 1 + 4k, \text{ or } p \equiv 1 \pmod{4}.$$

$$\therefore x^2 \equiv -1 \pmod{p} \text{ solvable} \Leftrightarrow p \equiv 1 \pmod{4}.$$

$$(b) x^4 \equiv -1 \pmod{p} \text{ is solvable} \Leftrightarrow p \equiv 1 \pmod{8}$$

As in (a), $\gcd(-1, p) = 1$. Using corollary to Th. 8.12,

$$x^4 \equiv -1 \pmod{p} \text{ solvable} \Leftrightarrow (-1)^{\frac{p-1}{d}} \equiv 1 \pmod{p},$$

where $d = \gcd(4, p-1)$.

If $d = 2$, then as in (a), $p = 1 + 4k$, some k , so $p \equiv 1 \pmod{4}$ and $\therefore p \equiv 1 \pmod{8}$.
[Note: the implication $p \equiv 1 \pmod{4} \Rightarrow p \equiv 1 \pmod{8}$ stated here appears to be a gap in the argument; likely the author intends to show $d = 4$ is required.]

If $d = 4$, then $\frac{p-1}{4} = 2k$, $p = 1 + 8k$, some k , so $p \equiv 1 \pmod{8}$. $\therefore$, if $(-1)^{\frac{p-1}{d}} \equiv 1 \pmod{p}$ then $p \equiv 1 \pmod{8}$.

But if $p \equiv 1 \pmod{8}$, then $p-1 = 8k$, some k , so

$$(-1)^{\frac{p-1}{d}} = (-1)^{\frac{8k}{d}} = 1 \quad \text{whether } d = 2 \text{ or } 4,$$

so $(-1)^{\frac{p-1}{d}} \equiv 1 \pmod{p}$, so $x^4 \equiv -1 \pmod{p}$ is solvable.

$$\therefore x^4 \equiv -1 \pmod{p} \text{ solvable} \Leftrightarrow p \equiv 1 \pmod{8}.$$

10. Given the congruence $x^3 \equiv a \pmod{p}$, where $p \geq 5$ is a prime and $\gcd(a, p) = 1$, prove the following:

(a) If $p \equiv 1 \pmod{6}$, then the congruence has either no solutions or 3 incongruent solutions mod p .

Pf. If $p \equiv 1 \pmod{6}$, then $p-1 = 6k$, some k . $\therefore \gcd(3, 6k) = 3$.

Since p is prime, it has a primitive root, so

$$x^3 \equiv a \pmod{p} \Leftrightarrow 3 \operatorname{ind} x \equiv \operatorname{ind} a \pmod{6k}.$$

By Th. 4.7, if $3 \nmid \operatorname{ind} a$, there is no solution. If $3 \mid \operatorname{ind} a$, then there are 3 incongruent solutions.

(b) If $p \equiv 5 \pmod{6}$, then the congruence has a unique solution mod p .

Pf. If $p \equiv 5 \pmod{6}$, then $p-5 = 6k$, some k . $\therefore p-1 = 6k+4 = 2(3k+2)$.

$\therefore \gcd(3, p-1) = \gcd(3, 2(3k+2)) = 1$, since $\gcd(3, 3k+2) = 1$ for all k .

$\therefore$

$$x^3 \equiv a \pmod{p} \Leftrightarrow 3 \operatorname{ind} x \equiv \operatorname{ind} a \pmod{p-1} \Leftrightarrow 3 \operatorname{ind} x \equiv \operatorname{ind} a \pmod{2(3k+2)}.$$

Since $1 \mid \text{ind } a$, by Th. 4.7, the latter congruence has a unique solution mod $p-1$, and so $x^3 \equiv a \pmod{p}$ has a unique solution mod p .

11. Show that the congruence $x^3 \equiv 3 \pmod{19}$ has no solutions, whereas $x^3 \equiv 11 \pmod{19}$ has three incongruent solutions. (1) $x^3 \equiv 3 \pmod{19}$, $\gcd(3, 19) = 1$

Since $\gcd(3, \phi(19)) = \gcd(3, 18) = 3$, by Th. 8.12, since $3^{\frac{18}{3}} = 3^6 = 3^3 \cdot 3^3 \equiv 8 \cdot 8 = 64 \equiv 7 \not\equiv 1 \pmod{19}$, then $x^3 \equiv 3 \pmod{19}$ has no solutions.

(2) $x^3 \equiv 11 \pmod{19}$, $\gcd(11, 19) = 1$

Since $\gcd(3, \phi(19)) = 3$, by Th. 8.12, since

$$11^{\frac{18}{3}} = 11^6 \equiv (-8)^6 = (64)^3 \equiv 7^3 = 49 \cdot 7 \equiv 11 \cdot 7 \equiv 1 \pmod{19},$$

then there are $3 = \gcd(3, \phi(19))$ incongruent solutions.

12. Determine whether the two congruences $x^5 \equiv 13 \pmod{23}$ and $x^7 \equiv 15 \pmod{29}$ are solvable.

(1) $x^5 \equiv 13 \pmod{23}$

Using Th. 8.12, $\gcd(13, 23) = 1$, and $\gcd(5, 22) = 1$. $\therefore$ solvable $\Leftrightarrow 13^{22} \equiv 1 \pmod{23}$.

$$13^2 = 169 \equiv 8, \quad 13^4 \equiv 64 \equiv -5, \quad 13^8 \equiv 25 \equiv 2,$$

$$13^{16} \equiv 4, \quad 13^{20} \equiv -20 \equiv 3, \quad 13^{22} \equiv 8 \cdot 3 = 24 \equiv 1 \pmod{23}.$$

$\therefore 13^{22} \equiv 1 \pmod{23}$, and so $x^5 \equiv 13 \pmod{23}$ is solvable.

(2) $x^7 \equiv 15 \pmod{29}$

$\gcd(15, 29) = 1$, $\gcd(7, 28) = 7$. $\therefore$ By Th. 8.12, solvable $\Leftrightarrow 15^{\frac{28}{7}} = 15^4 \equiv 1 \pmod{29}$.

$$15^2 = 225 \equiv 22, \quad 15^4 \equiv 22^2 = 484 \equiv 20 \not\equiv 1 \pmod{29}.$$

$\therefore$ not solvable.

13. If p is a prime and $\gcd(k, p-1) = 1$, prove that the integers $1^k, 2^k, 3^k, \dots, (p-1)^k$ form a reduced set of residues mod p .

Pf. $1, 2, \dots, p-1$ form a reduced set of residues mod p . Thus, each of $1^k, 2^k, \dots, (p-1)^k$ must be congruent to one of $1, 2, \dots, p-1$. Let $1 \leq a \leq p-1$, $1 \leq b \leq p-1$, and $a \neq b$. Suppose $a^k \equiv b^k \pmod{p}$. $\therefore \text{ind } a^k = \text{ind } b^k$, so

$$k(\text{ind } a) \equiv k(\text{ind } b) \pmod{p-1}.$$

Since $\gcd(k, p-1) = 1$, then

$$\text{ind } a \equiv \text{ind } b \pmod{p-1}.$$

By definition, $1 \leq \text{ind } a \leq p-1$, $1 \leq \text{ind } b \leq p-1$. $\therefore \text{ind } a = \text{ind } b$.

If r is a primitive root of p , then $r^{\text{ind } a} = r^{\text{ind } b}$. But by definition, $a \equiv r^{\text{ind } a} \pmod{p}$, $b \equiv r^{\text{ind } b} \pmod{p}$, $\therefore a \equiv b \pmod{p} \Rightarrow a = b$, a contradiction.

$\therefore a \not\equiv b \pmod{p}$, so each of the $p - 1$ integers $1^k, 2^k, \dots, (p - 1)^k$ is incongruent to the other mod p .

$\therefore 1^k, 2^k, \dots, (p - 1)^k$ form a complete set of residues mod p .

Now need to prove $\gcd(a^k, p) = 1$ for $1 \leq a \leq p - 1$. [Note: handwriting reads " $\gcd(a^k, p - 1) = 1$ "; context requires $\gcd(a^k, p) = 1$.]

Consider $x^k \equiv a \pmod{p}$. Clearly $\gcd(a, p) = 1$. Since $\gcd(k, p - 1) = 1$, then by Th. 8.12, since $a^{p-1} \equiv 1 \pmod{p}$ by Fermat's Th., then $x^k \equiv a \pmod{p}$ has exactly $\gcd(k, p - 1) = 1$ solution mod p . $\therefore$ The solution x must be among $1, 2, \dots, p - 1$ since the solution is mod p . $\therefore$ since $\gcd(a, p) = 1$, then $\gcd(x^k, p) = 1$ as $x^k \equiv a$ (and using prob. #3, sec. 4.2).

$\therefore$ each of $1^k, 2^k, \dots, (p - 1)^k$ is relatively prime to p .

$\therefore 1^k, 2^k, \dots, (p - 1)^k$ forms a reduced set of residues mod p .

14. Let r be a primitive root of the odd prime p , and let $d = \gcd(k, p - 1)$. Prove that the values of a for which the congruence $x^k \equiv a \pmod{p}$ is solvable are $r^d, r^{2d}, \dots, r^{[(p-1)/d]d}$.

Pf. (1) Let $s = 1, 2, \dots, \frac{p-1}{d}$, let $a = r^{sd}$.

Since $(r^{sd})^{\frac{\phi(p)}{d}} = (r^{\phi(p)})^s \equiv 1^s = 1 \pmod{p}$, as r is a primitive root, then by Th. 8.12, $x^k \equiv a \pmod{p}$ has a solution when $a = r^d, r^{2d}, \dots, r^{\frac{p-1}{d} \cdot d}$, or $a = r^d, r^{2d}, \dots, r^{p-1}$. (2) If $x^k \equiv a \pmod{p}$ has a solution, then if r is a primitive root of p ,

$$\text{ind}_r x^k = \text{ind}_r a, \quad \text{so} \quad k \text{ind}_r x \equiv \text{ind}_r a \pmod{p - 1}.$$

Let $d = \gcd(k, p - 1)$. Note $1 \leq d \leq p - 1$.

$\therefore$ By Th. 4.7, $d \mid \text{ind}_r a$. By definition, $1 \leq \text{ind}_r a \leq p - 1$.

Let m be s.t. $dm = \text{ind}_r a$. By definition, $r^{dm} \equiv a \pmod{p}$.

Since $1 \leq d \leq p - 1$ and $1 \leq \text{ind}_r a \leq p - 1$, then it must be true that $1 \leq m \leq p - 1$.

$\therefore$ (1) shows that when $a = r^d, r^{2d}, \dots, r^{p-1}$, then $x^k \equiv a \pmod{p}$ is solvable, and (2) shows that if $x^k \equiv a \pmod{p}$ is solvable, a must be congruent mod p to r^{dm} , where $m = 1, 2, \dots, p - 1$. $\therefore a = r^d, r^{2d}, \dots, r^{p-1}$ are all the values, mod p , for which $x^k \equiv a \pmod{p}$ is solvable. **15.** If r is a primitive root of the odd prime p , show that

$$\text{ind}_r(p - a) \equiv \text{ind}_r a + \frac{p - 1}{2} \pmod{p - 1}$$

and consequently, that only half of an index table need be calculated to complete the table.

Pf. By definition, $r^{\text{ind}_r(p-a)} \equiv p - a \equiv (-a) \pmod{p}$. $\therefore \text{ind}_r r^{\text{ind}_r(p-a)} = \text{ind}_r(-a)$. Since $\text{ind}_r r = 1$,

$$\text{ind}_r(p - a) \equiv \text{ind}_r(-a) \equiv \text{ind}_r(-1) + \text{ind}_r(a) \pmod{p - 1}.$$

By prob. #7, $\text{ind}_r(-1) = \frac{1}{2}(p-1)$.

$$\therefore \text{ind}_r(p-a) \equiv \frac{1}{2}(p-1) + \text{ind}_r a \pmod{p-1}.$$

16. (a) Let r be a primitive root of the odd prime p . Establish that the exponential congruence

$$a^x \equiv b \pmod{p}$$

has a solution $\Leftrightarrow d \mid \text{ind}_r b$, where $d = \gcd(\text{ind}_r a, p-1)$; in this case, there are d incongruent solutions mod $p-1$. *Pf.* $a^x \equiv b \pmod{p}$

$$\Leftrightarrow x \text{ind}_r a \equiv \text{ind}_r b \pmod{p-1}. \quad [1]$$

By Th. 4.7, [1] has a solution $\Leftrightarrow \gcd(\text{ind}_r a, p-1) = d \mid \text{ind}_r b$, in which case there are d incongruent solutions, mod $p-1$.

(b) Solve the exponential congruences $4^x \equiv 13 \pmod{17}$ and $5^x \equiv 4 \pmod{19}$.

(1) $4^x \equiv 13 \pmod{17}$. 3 is a primitive root of 17.

$$x \text{ind}_3 4 \equiv \text{ind}_3 13 \pmod{16}$$

From table in prob. #3, $\text{ind}_3 4 = 12$, $\text{ind}_3 13 = 4$.

$$\therefore 12x \equiv 4 \pmod{16}, \quad \gcd(12, 16) = 4.$$

$4 \mid 4$, so 4 incongruent solutions mod 16.

$$\therefore 3x \equiv 1 \pmod{4}, \quad 9x \equiv 3, \quad x \equiv 3 \pmod{4}.$$

$$\therefore x \equiv 3, 7, 11, 15 \pmod{16}.$$

(2) $5^x \equiv 4 \pmod{19}$. 2 is a primitive root of 19.

$$\therefore x \text{ind}_2 5 \equiv \text{ind}_2 4 \pmod{18}$$

Develop table of indices for 19 relative to 2:

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\text{ind}_2 a$	18	1	13	2	16	14	6	3	8	17	12	15	5	7	11	4	10	9

$$\therefore \text{ind}_2 5 = 16, \quad \text{ind}_2 4 = 2.$$

$$\therefore 16x \equiv 2 \pmod{18}, \quad \gcd(16, 18) = 2.$$

$\therefore 2$ incongruent solutions mod 18.

$$\therefore 8x \equiv 1 \pmod{9}, \quad -8x \equiv -1, \quad x \equiv -1 \equiv 8.$$

$$\therefore x \equiv 8, 17 \pmod{18}.$$

17. For which values of b is the exponential congruence $9^x \equiv b \pmod{13}$ solvable?

2 is a primitive root of 13. Use table on p. 175. $\therefore x \text{ind}_2 9 \equiv \text{ind}_2 b \pmod{12}$

$$\text{ind}_2 9 = 8.$$

$$\therefore 8x \equiv \text{ind}_2 b \pmod{12}, \quad \gcd(8, 12) = 4.$$

$\therefore 4 \mid \text{ind}_2 b$, so $\text{ind}_2 b = 4, 8, 12$.

$\therefore b$ (using table) = 3, 9, 1.

$$\therefore b \equiv 1, 3, 9 \pmod{13}.$$

Chapter 9

The Quadratic Reciprocity Law

9.1 Euler's Criterion

1. Solve the following quadratic congruences.

(a) $x^2 + 7x + 10 \equiv 0 \pmod{11}$

Let $y = 2ax + b = 2x + 7$, $d = b^2 - 4ac = 9$.

$\therefore y^2 \equiv 9 \pmod{11}$, $\therefore y \equiv 3, 8 (= 11 - 3)$.

$$2x + 7 \equiv 3 \pmod{11} \quad 2x + 7 \equiv 8 \pmod{11}$$

$$2x \equiv -4 \quad 2x \equiv 1, \quad 10x \equiv 5$$

$$x \equiv -2 \equiv 9 \quad -x \equiv 5, \quad x \equiv -5 \equiv 6$$

$$\therefore x \equiv 6, 9 \pmod{11}.$$

(b) $3x^2 + 9x + 7 \equiv 0 \pmod{13}$

$y = 2ax + b = 6x + 9$, $d = b^2 - 4ac = -3$.

$\therefore y^2 \equiv -3 \equiv 10 \equiv 10 + 2 \cdot 13 = 36 \pmod{13}$, $\therefore y \equiv 6, 7 (= 13 - 6)$.

$$6x + 9 \equiv 6 \pmod{13} \quad 6x + 9 \equiv 7 \pmod{13}$$

$$6x \equiv -3 \equiv 10, \quad x \equiv 6 \quad 6x \equiv -2, \quad 12x \equiv -4, \quad -x \equiv -4, \quad x \equiv 4$$

$$\therefore x \equiv 4, 6 \pmod{13}.$$

(c) $5x^2 + 6x + 1 \equiv 0 \pmod{23}$

$y = 2ax + b = 10x + 6$, $d = b^2 - 4ac = 16$.

$\therefore y^2 \equiv 16 \pmod{23}$, $y \equiv 4, 19 (= 23 - 4)$.

$$10x + 6 \equiv 4 \pmod{23} \quad 10x + 6 \equiv 19 \pmod{23}$$

$$\begin{aligned} 10x &\equiv -2, & 20x &\equiv -4 & 10x &\equiv 13, & 20x &\equiv 26 \\ -3x &\equiv -4, & -24x &\equiv -32 & -3x &\equiv 3, & x &\equiv -1 \\ -x &\equiv -32, & x &\equiv 9 & x &\equiv 22 \end{aligned}$$

$$\therefore x \equiv 9, 22 \pmod{23}.$$

2. Prove that the quadratic congruence $6x^2 + 5x + 1 \equiv 0 \pmod{p}$ has a solution for every prime p , even though $6x^2 + 5x + 1 = 0$ has no solution in integers.

$$\text{Pf. } 6x^2 + 5x + 1 = 0, \quad x = \frac{-5 \pm \sqrt{25 - 24}}{12} = \frac{-5 \pm 1}{12}.$$

$$\therefore x = -\frac{1}{2}, -\frac{1}{3}.$$

$$6x^2 + 5x + 1 = (3x + 1)(2x + 1) \equiv 0 \pmod{p}$$

$$\therefore 3x + 1 \equiv 0 \pmod{p} \quad \text{or} \quad (2x + 1) \equiv 0 \pmod{p}.$$

(1) If p is odd, then choose x s.t. $2x + 1 = p$: $\therefore 2x + 1 \equiv 0 \pmod{p} \Rightarrow 6x^2 + 5x + 1 \equiv 0 \pmod{p}$.

(2) If $p = 2$, then $3x + 1 \equiv 0 \pmod{2}$:

$$3x \equiv -1 \equiv 1, \quad x \equiv 1.$$

$$\therefore x \equiv 1 \pmod{2} \Rightarrow 3x \equiv 3, \quad 3x + 1 \equiv 4 \equiv 0 \pmod{2} \Rightarrow 6x^2 + 5x + 1 \equiv 0 \pmod{2}.$$

$\therefore$ There is a solution to $6x^2 + 5x + 1 \equiv 0 \pmod{p}$ for all prime p .

3. (a) For an odd prime p , prove that the quadratic residues of p are congruent mod p to the integers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$.

$$\text{Pf. (1) For } a = 1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2,$$

$$a^{\frac{p-1}{2}} = 1^{p-1}, 2^{p-1}, \dots, \left(\frac{p-1}{2}\right)^{p-1}.$$

But for $b = 1, 2, \dots, \frac{p-1}{2}$, $\gcd(b, p) = 1$ as $1 \leq b < p - 1$ and p is prime. By Fermat's Th., $b^{p-1} \equiv 1 \pmod{p}$.

$$\therefore a^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

$\therefore$ By Euler's Criterion, $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are quadratic residues of p .

(2) $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are incongruent mod p .

For if $a^2 \equiv b^2 \pmod{p}$, $1 \leq a, b \leq \frac{p-1}{2}$, $a \neq b$, then $a^2 - b^2 \equiv 0 \Leftrightarrow (a - b)(a + b) \equiv 0 \pmod{p}$. But $a + b \leq \frac{p-1}{2} + \frac{p-1}{2} = p - 1$. $\therefore \gcd(a + b, p) = 1$, so can divide by $a + b$. $\therefore a - b \equiv 0 \pmod{p} \Rightarrow a = b \Rightarrow a \neq b$, a contradiction.

(3) Let a be any quadratic residue of p . $\therefore x^2 \equiv a \pmod{p}$ has a solution. Let it be x_0 s.t. $1 \leq x_0 \leq p - 1$. $\therefore p - x_0$ is also a solution. One of $x_0, p - x_0$ must be $\leq \frac{p-1}{2}$. For if $x_0 > \frac{p-1}{2}$, then $-x_0 < -\frac{p-1}{2}$,

$$\text{so } p - x_0 < p - \frac{p-1}{2} = \frac{p+1}{2}. \quad [\text{Note: original writes } \frac{p-1}{2}; \text{ should be } \frac{p+1}{2}.]$$

$\therefore$ One of x_0^2 or $(p - x_0)^2$ is equal to $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$.

Since $x_0^2 \equiv (p - x_0)^2 \equiv a$, then a must be congruent to one of $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$.

(b) Verify that the quadratic residues of 17 are 1, 2, 4, 8, 9, 13, 15, 16.

By (a):

$$\begin{aligned} 1^2 &\equiv 1, & 5^2 &\equiv 25 \equiv 8, & 2^2 &\equiv 4, & 6^2 &\equiv 36 \equiv 2, \\ 3^2 &\equiv 9, & 7^2 &\equiv 49 \equiv 15, & 4^2 &\equiv 16, & 8^2 &\equiv 64 \equiv 13 \pmod{17}. \end{aligned}$$

4. Show that 3 is a quadratic residue of 23, but a nonresidue of 31.

$$3^{\frac{23-1}{2}} = 3^{11} = 3^2 \cdot (3^3)^3 = 9 \cdot (27)^3 \equiv 9 \cdot (4)^3 \pmod{23} \equiv 9 \cdot 64 \equiv 9 \cdot (-5) = -45 + 46 \equiv 1.$$

$$\therefore 3^{\frac{23-1}{2}} \equiv 1 \pmod{23} \Rightarrow 3 \text{ a quadratic residue of } 23.$$

$$3^{\frac{31-1}{2}} = 3^{15} = (3^3)^5 = 27^5 \equiv (-4)^5 \pmod{31} \equiv -4^3 \cdot 4^2 \equiv (-64)(16) \equiv (-64 + 62)(16) \equiv -2(16) \equiv -32 \equiv -1.$$

$$\therefore 3^{\frac{31-1}{2}} \equiv -1 \pmod{31} \Rightarrow 3 \text{ a quadratic nonresidue of } 31.$$

5. Given that a is a quadratic residue of odd prime p , prove the following.

(a) a is not a primitive root of p .

Pf. If a were a primitive root of p , then $a^n \not\equiv 1 \pmod{p}$ for $1 \leq n < p - 1$, by definition.

But by Euler's criterion, $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. p is odd so $\frac{p-1}{2}$ is an integer, and $1 \leq \frac{p-1}{2} < p - 1$, which contradicts the above.

(b) The integer $p - a$ is a quadratic residue or nonresidue of p according as $p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{4}$.

Pf. Consider $x^2 \equiv p - a \pmod{p}$. This is equivalent to $x^2 \equiv -a \pmod{p}$.

$\therefore -a$ (or $p - a$) is a quadratic residue or nonresidue according to whether

$$(-a)^{\frac{p-1}{2}} \equiv 1 \pmod{p} \quad \text{or} \quad (-a)^{\frac{p-1}{2}} \equiv -1 \pmod{p}.$$

Since a is a quadratic residue, then $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. But $(-a)^{\frac{p-1}{2}} = (-1)^{\frac{p-1}{2}} \cdot a^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{2}} \pmod{p}$.

$\therefore p - a$ is a quadratic residue or nonresidue according to whether $(-1)^{\frac{p-1}{2}}$ is 1 or -1 , and this according to whether $\frac{p-1}{2}$ is even or odd.

$\frac{p-1}{2}$ is even $\Leftrightarrow \frac{p-1}{2} = 2k$, some k , $\Leftrightarrow p - 1 = 4k \Leftrightarrow p \equiv 1 \pmod{4}$.

$\frac{p-1}{2}$ is odd $\Leftrightarrow \frac{p-1}{2} = 2k + 1$, some k $\Leftrightarrow p - 1 = 4k + 2 \Leftrightarrow p = 3 + 4k \Leftrightarrow p \equiv 3 \pmod{4}$.

$\therefore p - a$ is a quadratic residue or nonresidue according to whether $p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{4}$.

(c) If $p \equiv 3 \pmod{4}$, then $x \equiv \pm a^{\frac{p+1}{4}} \pmod{p}$ are the solutions of the congruence $x^2 \equiv a \pmod{p}$.

Pf. $x \equiv \pm a^{\frac{p+1}{4}} \Rightarrow x^2 \equiv a^{\frac{p+1}{2}} = a^{\frac{p-1+2}{2}} = a^{\frac{p-1}{2}} \cdot a$ Since a is a quadratic residue, $a^{\frac{p-1}{2}} = 1$.

$$\therefore a^{\frac{p-1}{2}} \cdot a \equiv a.$$

$\therefore x^2 \equiv a \pmod{p}$ when $x \equiv \pm a^{\frac{p+1}{4}}$.

$\frac{p+1}{4}$ is an integer when $\frac{p+1}{4} = k$, some k , $\therefore p+1 = 4k$, $p = -1 + 4k$, $p \equiv -1 \pmod{4}$, or $p \equiv 3 \pmod{4}$.

Also, by Lagrange's Th. (Th. 8.5), there are at most 2 solutions, so $x \equiv \pm a^{\frac{p+1}{4}}$ are the exact solutions.

6. Let p be an odd prime and $\gcd(a, p) = 1$. Establish that the quadratic congruence $ax^2 + bx + c \equiv 0 \pmod{p}$ is solvable $\Leftrightarrow b^2 - 4ac$ is zero or a quadratic residue of p .

Pf. Since $\gcd(a, p) = 1$ and p is an odd prime, $\gcd(4a, p) = 1$.

$\therefore$ Solutions to

$$4a(ax^2 + bx + c) \equiv 0 \pmod{p} \quad [1]$$

are equivalent to $ax^2 + bx + c \equiv 0 \pmod{p} \quad [2]$

since you can divide [1] by $4a$ to get [2].

But $4a(ax^2 + bx + c) = 4a^2x^2 + 4abx + 4ac$

$$= (2ax + b)^2 - b^2 + 4ac = (2ax + b)^2 - (b^2 - 4ac).$$

$\therefore$ Solutions to $ax^2 + bx + c \equiv 0 \pmod{p} \quad [2]$ are equivalent to

$$(2ax + b)^2 \equiv b^2 - 4ac \pmod{p}. \quad [3]$$

(a) Suppose $b^2 - 4ac \equiv 0 \pmod{p}$.

$\therefore$ Solutions to [2] are equivalent to $(2ax + b)^2 \equiv 0 \pmod{p}$, which is equivalent to

$$2ax \equiv -b \pmod{p}.$$

Since $\gcd(2a, p) = 1$, this has a unique solution mod p (Th. 4.7).

$$\therefore ax^2 + bx + c \equiv 0 \pmod{p} \text{ is solvable } \Leftrightarrow b^2 - 4ac \equiv 0 \pmod{p}.$$

(b) $b^2 - 4ac \not\equiv 0 \pmod{p}$ and is a quadratic residue of p .

$\therefore y^2 \equiv b^2 - 4ac \pmod{p}$ has a solution by definition. Let y_1 be s.t. $y_1^2 \equiv b^2 - 4ac \pmod{p}$, $1 \leq y_1 \leq p-1$. $\therefore p - y_1$ is also a solution. By Lagrange's Th., these are the only solutions. They are also incongruent. For if $y_1 \equiv p - y_1 \pmod{p}$, then $2y_1 \equiv p \equiv 0 \pmod{p} \Leftrightarrow y_1 \equiv 0 \pmod{p}$ as $\gcd(p, 2) = 1$. $\therefore b^2 - 4ac \equiv 0$, a contradiction.

Letting $y_1 = 2ax + b$, then $y^2 \equiv b^2 - 4ac \pmod{p}$ is equivalent to

$$2ax + b \equiv y_1 \pmod{p} \quad \text{and} \quad p - (2ax + b) \equiv y_1 \pmod{p}, \quad \text{or}$$

$$2ax \equiv b^2 - 4ac - b \pmod{p} \quad [4]$$

$$\text{and } 2ax \equiv 4ac - b^2 - b \pmod{p}. \quad [5]$$

Since $\gcd(2a, p) = 1$, by Th. 4.7, [4] and [5] have unique solutions.

$\therefore$ Assuming $b^2 - 4ac \not\equiv 0 \pmod{p}$,

$b^2 - 4ac$ a quadratic residue of $p \Leftrightarrow (2ax + b)^2 \equiv b^2 - 4ac \pmod{p}$ is solvable

$$\Leftrightarrow ax^2 + bx + c \equiv 0 \pmod{p} \text{ is solvable.}$$

7. If $p = 2^k + 1$ is prime, verify that every quadratic nonresidue of p is a primitive root of p .

Pf. Let a be a quadratic nonresidue of p . $\therefore$ By Euler's criterion, $a^{\frac{p-1}{2}} \equiv -1 \pmod{p}$. Since $2^k + 1$ is prime, $k \geq 1$. $p - 1 = 2^k$. $\therefore \frac{p-1}{2} = 2^{k-1}$.

$$\therefore a^{2^{k-1}} \equiv -1 \pmod{p}. \quad [1]$$

$$\therefore \left(a^{2^{k-1}}\right)^2 = a^{2^k} \equiv 1 \pmod{p}, \text{ and } \phi(p) = p - 1 = 2^k.$$

Let n be order of $a \pmod{p}$. $\therefore n \mid 2^k$ by Th. 8.1. $\therefore$ if $n \neq 2^k$, then $n = 2^r$, $r < k$.

$$\therefore a^{2^r} \equiv 1 \pmod{p}. \quad [2]$$

If $r = k - 1$, then a contradiction is reached by [1]. If $r < k - 1$, then square [2] $k - 1 - r$ times:

$$\begin{aligned} (a^{2^r})^2 &= a^{2 \cdot 2^r} = a^{2^{r+1}} \equiv 1 \pmod{p} \\ (a^{2^{r+1}})^2 &= a^{2 \cdot 2^{r+1}} = a^{2^{r+2}} \equiv 1 \pmod{p} \\ &\vdots \\ (a^{2^{k-2}})^2 &= a^{2 \cdot 2^{k-2}} = a^{2^{k-1}} \equiv 1 \pmod{p}. \end{aligned}$$

$\therefore$ Again, a contradiction is reached by [1].

$$\therefore n = 2^k, \text{ so order of } a \text{ is } p - 1 = \phi(p).$$

8. Assume r is a primitive root of prime p , where $p \equiv 1 \pmod{8}$.

(a) Show that the solutions of the quadratic congruence $x^2 \equiv 2 \pmod{p}$ are given by

$$x \equiv \pm(r^{7(p-1)/8} + r^{(p-1)/8}) \pmod{p}.$$

Pf. Since r is a primitive root of p , $r^{p-1} \equiv 1 \pmod{p}$. But $p - 1 = 8k$, some k , or $(p - 1)/8 = k$, an integer.

Let $x \equiv \pm(r^{7(p-1)/8} + r^{(p-1)/8}) \pmod{p}$.

$$\begin{aligned} \therefore x^2 &\equiv (r^{7(p-1)/8} + r^{(p-1)/8})^2 \pmod{p} = r^{14(p-1)/8} + r^{2(p-1)/8} + 2r^{p-1} \\ &\equiv r^{14(p-1)/8} + r^{2(p-1)/8} + 2 \pmod{p}. \end{aligned}$$

$\therefore$ Need to show $r^{14(p-1)/8} + r^{2(p-1)/8} \equiv 0 \pmod{p}$ to show $x^2 \equiv 2 \pmod{p}$.

$$r^{14(p-1)/8} + r^{2(p-1)/8} = r^{2(p-1)/8}(r^{12(p-1)/8} + 1).$$

But $\gcd(r, p) = 1$, so $\gcd(r^{2(p-1)/8}, p) = 1$. $\therefore$ If can show $r^{12(p-1)/8} \equiv -1 \pmod{p}$, then $x^2 \equiv 2 \pmod{p}$.

$$\begin{aligned} r^{12(p-1)/8} &= r^{8(p-1)/8} \cdot r^{4(p-1)/8} = r^{p-1} \cdot r^{4(p-1)/8} \\ &\equiv r^{4(p-1)/8} = r^{(p-1)/2}. \end{aligned}$$

Since $(r^{\frac{p-1}{2}} + 1)(r^{\frac{p-1}{2}} - 1) = r^{p-1} - 1 \equiv 0 \pmod{p}$, then $r^{\frac{p-1}{2}} \equiv -1$ or $r^{\frac{p-1}{2}} \equiv 1 \pmod{p}$, but not both, since otherwise $-1 \equiv 1 \pmod{p}$, so $p \mid 2$, a contradiction to $p \equiv 1 \pmod{8}$.

But $r^{\frac{p-1}{2}} \not\equiv 1$ since r is a primitive root (order of $r = p - 1$).

$$\therefore r^{\frac{p-1}{2}} \equiv -1 \pmod{p}.$$

$$\therefore r^{12(p-1)/8} \equiv -1 \pmod{p}.$$

$$\therefore \text{if } x \equiv \pm(r^{7(p-1)/8} + r^{(p-1)/8}) \pmod{p},$$

then $x^2 \equiv 2 \pmod{p}$, and Lagrange's Th. shows there are no more solutions.

(b) Use part (a) to find all solutions to the two congruences $x^2 \equiv 2 \pmod{17}$ and $x^2 \equiv 2 \pmod{41}$.

$$(1) x^2 \equiv 2 \pmod{17}$$

3 is a primitive root of 17.

$$\begin{aligned} \therefore x &\equiv \pm(3^{7(17-1)/8} + 3^{(17-1)/8}) \pmod{17} \\ &= \pm(3^{14} + 3^2) \pmod{17} \\ &= \pm 3^2(3^{12} + 1) = \pm 9(3^{12} + 1). \end{aligned}$$

$$3^2 \equiv 9, \quad 3^4 \equiv -4, \quad 3^8 \equiv 16 \equiv -1, \quad 3^{12} \equiv 4 \pmod{17}.$$

$$\therefore x \equiv \pm 9(4 + 1) = \pm 45 \equiv 6, 11 \pmod{17}.$$

$$(2) x^2 \equiv 2 \pmod{41}$$

6 is a primitive root of 41 (table p. 166).

$$\begin{aligned} \therefore x &\equiv \pm(6^{7(41-1)/8} + 6^{(41-1)/8}) \pmod{41} \\ &= \pm(6^{35} + 6^5) = \pm 6^5(6^{30} + 1). \end{aligned}$$

$$\begin{aligned}
6^2 &\equiv 36 \equiv -5, & 6^3 &\equiv -30 \equiv 11, & 6^4 &\equiv 66 \equiv 25 \equiv -16, \\
6^5 &\equiv -96 \equiv -14, & 6^6 &\equiv -84 \equiv -2 \pmod{41}. \\
\therefore 6^{30} &\equiv (-2)^5 \equiv -32 \equiv 9 \pmod{41}.
\end{aligned}$$

$$\therefore x \equiv \pm 14(9 + 1) = \pm 140 \equiv \pm 17 = 17, 24 \pmod{41}.$$

9. (a) If $ab \equiv r \pmod{p}$, where r is a quadratic residue of the odd prime p , prove that a and b are both quadratic residues of p or both nonresidues of p .

Pf. Since $\gcd(r, p) = 1$, then $\gcd(ab, p) = 1$, $\therefore \gcd(a, p) = 1$ and $\gcd(b, p) = 1$. Suppose a is a quadratic residue and b a nonresidue.

$$\therefore a^{\frac{p-1}{2}} \equiv 1, \quad b^{\frac{p-1}{2}} \equiv -1 \pmod{p}.$$

$$\therefore r^{\frac{p-1}{2}} \equiv a^{\frac{p-1}{2}} \cdot b^{\frac{p-1}{2}} \equiv -1 \pmod{p},$$

which makes r a nonresidue by corollary to Euler's criterion.

$$\therefore a^{\frac{p-1}{2}} \equiv b^{\frac{p-1}{2}} \equiv 1 \quad \text{or} \quad a^{\frac{p-1}{2}} \equiv b^{\frac{p-1}{2}} \equiv -1 \pmod{p}.$$

(b) If a and b are both quadratic residues of the odd prime p or both nonresidues of p , show that the congruence $ax^2 \equiv b \pmod{p}$ has a solution.

Pf. Assume $\gcd(a, p) = \gcd(b, p) = 1$.

$\therefore ax^2 \equiv b \pmod{p}$ is equivalent to $a^2x^2 \equiv ab \pmod{p}$, or

$$(ax)^2 \equiv ab \pmod{p}.$$

$\therefore ax^2 \equiv b \pmod{p}$ has a solution $\Leftrightarrow ab$ is a quadratic residue. By (a) ab is a quadratic residue $\Leftrightarrow a, b$ are both quadratic residues or both nonresidues.

$\therefore ax^2 \equiv b \pmod{p}$ has a solution $\Leftrightarrow a, b$ are both quadratic residues or both nonresidues.

10. Let p be an odd prime and $\gcd(a, p) = \gcd(b, p) = 1$. Prove that either all three of the quadratic congruences $x^2 \equiv a \pmod{p}$, $x^2 \equiv b \pmod{p}$, $x^2 \equiv ab \pmod{p}$ are solvable or exactly one of them admits a solution.

Pf. Suppose more than one congruence admits a solution.

(1) Assume $x^2 \equiv a \pmod{p}$, $x^2 \equiv b \pmod{p}$ are solvable. By Euler's criterion, $a^{\frac{p-1}{2}} \equiv b^{\frac{p-1}{2}} \equiv 1 \pmod{p}$.

$$\therefore a^{\frac{p-1}{2}} \cdot b^{\frac{p-1}{2}} \equiv (ab)^{\frac{p-1}{2}} \equiv 1 \pmod{p},$$

and so $x^2 \equiv ab \pmod{p}$ is solvable.

(2) Assume $x^2 \equiv a \pmod{p}$, $x^2 \equiv ab \pmod{p}$ are solvable (case of $x^2 \equiv b$ and $x^2 \equiv ab$ is analogous). $\therefore$ By Euler's criterion, $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$ and $(ab)^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. [1]

Since $\gcd(a, p) = 1$, then $\gcd\left(a^{\frac{p-1}{2}}, p\right) = 1$. $\therefore$ Dividing [1] by $a^{\frac{p-1}{2}}$, we get $b^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. $\therefore$ By Euler's criterion, $x^2 \equiv b \pmod{p}$ is solvable.

11. (a) Knowing 2 is a primitive root of 19, find all the quadratic residues of 19.

The proof of Th. 9.1 shows that if a is a quadratic residue, then if r is a primitive root of 19, then $r^k \equiv a \pmod{p}$, $1 \leq k \leq p-1$, and k is even. $\therefore$ Look at all 2^k for k even and $1 \leq k \leq 18$:

$$\begin{aligned} 2^2 &\equiv 4, & 2^8 &\equiv 4 \cdot 7 \equiv 9, & 2^{14} &\equiv -32 \equiv 6 \pmod{19} \\ 2^4 &\equiv 16, & 2^{10} &\equiv 4 \cdot 9 \equiv -2 \equiv 17, & 2^{16} &\equiv 24 \equiv 5 \\ 2^6 &\equiv 64 \equiv 7, & 2^{12} &\equiv -8 \equiv 11, & 2^{18} &\equiv 20 \equiv 1 \end{aligned}$$

$\therefore$ 1, 4, 5, 6, 7, 9, 11, 16, 17 are the quadratic residues of 19.

(b) Find the quadratic residues of 29 and 31.

Can use method in (a), or easier, method in Example 9.1.

29:

$$\begin{aligned} 1^2 &\equiv 28^2 \equiv 1, & 8^2 &\equiv 21^2 \equiv 64 \equiv 6, \\ 2^2 &\equiv 27^2 \equiv 4, & 9^2 &\equiv 20^2 \equiv 81 \equiv 23, \\ 3^2 &\equiv 26^2 \equiv 9, & 10^2 &\equiv 19^2 \equiv 13, \\ 4^2 &\equiv 25^2 \equiv 16, & 11^2 &\equiv 18^2 \equiv 121 \equiv 5, \\ 5^2 &\equiv 24^2 \equiv 25, & 12^2 &\equiv 17^2 \equiv 144 \equiv -1 \equiv 28, \\ 6^2 &\equiv 23^2 \equiv 7, & 13^2 &\equiv 16^2 \equiv 169 \equiv 24, \\ 7^2 &\equiv 22^2 \equiv 20, & 14^2 &\equiv 15^2 \equiv 196 \equiv -7 \equiv 22 \pmod{29}. \end{aligned}$$

$\therefore$ 1, 4, 5, 6, 7, 9, 13, 16, 20, 22, 23, 24, 25, 28.

31:

$$\begin{aligned} 1^2 &\equiv 30^2 \equiv 1, & 9^2 &\equiv 22^2 \equiv 81 \equiv 19, \\ 2^2 &\equiv 29^2 \equiv 4, & 10^2 &\equiv 21^2 \equiv 7, \\ 3^2 &\equiv 28^2 \equiv 9, & 11^2 &\equiv 20^2 \equiv -3 \equiv 28, \\ 4^2 &\equiv 27^2 \equiv 16, & 12^2 &\equiv 19^2 \equiv 20, \\ 5^2 &\equiv 26^2 \equiv 25, & 13^2 &\equiv 18^2 \equiv 165 \equiv 14, & [Note: 13^2 = 169, not 165.] \\ 6^2 &\equiv 25^2 \equiv 5, & 14^2 &\equiv 17^2 \equiv 10, \\ 7^2 &\equiv 24^2 \equiv 18, & 15^2 &\equiv 16^2 \equiv 225 \equiv 8, \\ 8^2 &\equiv 23^2 \equiv 2 \pmod{31}. \end{aligned}$$

$\therefore$ 1, 2, 4, 5, 7, 8, 9, 10, 14, 16, 18, 19, 20, 25, 28.

12. If $n > 2$ and $\gcd(a, n) = 1$, then a is called a quadratic residue of n whenever there exists an integer x s.t. $x^2 \equiv a \pmod{n}$. Prove that if a is a quadratic residue of $n > 2$, then $a^{\phi(n)/2} \equiv 1 \pmod{n}$.

Pf. Since $\gcd(a, n) = 1$ and $x^2 \equiv a \pmod{n}$, then $\gcd(x^2, n) = 1$, and so $\gcd(x, n) = 1$ (if x and n had a common divisor $d > 1$, then $d \mid x \Rightarrow d \mid x^2$).

By Euler's Th., $x^{\phi(n)} \equiv 1 \pmod{n}$.

$$\therefore a^{\phi(n)/2} \equiv (x^2)^{\phi(n)/2} = x^{\phi(n)} \equiv 1 \pmod{n}.$$

13. Show that the result of the previous problem does not provide a sufficient condition for the existence of a quadratic residue of n ; i.e., find relatively prime integers a and n , with $a^{\phi(n)/2} \equiv 1 \pmod{n}$, for which the congruence $x^2 \equiv a \pmod{n}$ is not solvable.

Intuition suggests, from section 8.3, that if n is composite and doesn't have a primitive root, then finding such an a will be easier.

$\therefore$ Let $n = 6$. $\pmod{6}$: $1^2 \equiv 1, 4^2 \equiv 4, 2^2 \equiv 4, 5^2 \equiv 1, 3^2 \equiv 3$.

$\therefore x^2 \equiv 5 \pmod{6}$ is not solvable. $\phi(6) = 2$, however $5^1 \not\equiv 1 \pmod{6}$.

Try $n = 8$. $\pmod{8}$: $1^2 \equiv 1, 4^2 \equiv 0, 7^2 \equiv 1, 2^2 \equiv 4, 5^2 \equiv 1, 3^2 \equiv 1, 6^2 \equiv 4$.

$\therefore x^2 \equiv a \pmod{8}$ not solvable if $a = 3, 5, 7$.

$\phi(8) = 4 \therefore \phi(8)/2 = 2$. And $3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$.

$$\therefore \underline{n = 8, \quad a = 3, 5, \text{ or } 7.}$$

9.2 The Legendre Symbol and Its Properties

[Note: Section is labelled "9.1" in the manuscript, but 9.1 was used earlier for Euler's Criterion; likely should be 9.2.]

1. Find the value of the following Legendre symbols:

(a) $(19/23)$

$$19 \equiv -4 \pmod{23}.$$

$$\therefore (19/23) = (-4/23) = (-1 \cdot 2^2/23) = (-1/23) = (-1)^{(23-1)/2} = -1.$$

(b) $(-23/59)$

$$-23 \equiv -23 + 59 \pmod{59} = 36 = 6^2.$$

$$\therefore (-23/59) = (6^2/59) = 1.$$

(c) $(20/31) = (2^2 \cdot 5/31) = (5/31)$.

$$5 \equiv 36 \pmod{31}.$$

$$\therefore (20/31) = (36/31) = (6^2/31) = 1.$$

(d) $(18/43) = (2 \cdot 3^2/43) = (2/43)$.

$$43 = 5 \cdot 8 + 3, \text{ so by Th. 9.6, } (2/43) = -1.$$

(e) $(-72/131) = (-3^2 \cdot 2^2 \cdot 2/131) = (-1/131)(2/131)$.
 $(-1/131) = (-1)^{(131-1)/2} = -1$. $\therefore (-72/131) = -(2/131)$. $131 = (16) \cdot 8 + 3$, so by
 Th. 9.6, $(2/131) = -1$.

$$\therefore (-72/131) = (-1)(-1) = 1.$$

2. Use Gauss's lemma to compute each of the Legendre symbols below (that is, in each case obtain the integer n for which $(a/p) = (-1)^n$):

(a) $(8/11)$

$$(p-1)/2 = 5, \quad p/2 = 5.5.$$

$$S = \{8, 16, 24, 32, 40\} = \{1 \cdot 8, 2 \cdot 8, \dots, 5 \cdot 8\} \equiv \{8, 5, 2, 9, 7\} \pmod{11}.$$

$\therefore 8, 9, 7 > p/2$, so $n = 3$.

$$\therefore (8/11) = (-1)^3 = -1.$$

(b) $(7/13)$

$$(p-1)/2 = 6, \quad p/2 = 6.5.$$

$$S = \{7, 14, 21, 28, 35, 42\} \equiv \{7, 1, 8, 2, 9, 3\} \pmod{13}.$$

$\therefore 7, 8, 9 > p/2$, so $n = 3$.

$\therefore (7/13) = (-1)^3 = -1$. [Note: manuscript writes " $(8/13)$ " here, likely a slip for " $(7/13)$ ".]

(c) $(5/19)$

$$(p-1)/2 = 9, \quad p/2 = 9.5.$$

$$S = \{5, 10, 15, 20, 25, 30, 35, 40, 45\} \equiv \{5, 10, 15, 1, 6, 11, 16, 2, 7\} \pmod{19}.$$

$\therefore 10, 15, 11, 16 > 9.5$, so $n = 4$.

$$\therefore (-1)^4 = 1.$$

(d) $(11/23)$

$$(p-1)/2 = 11, \quad p/2 = 11.5.$$

$$S = \{11, 22, 33, 44, 55, 66, 77, 88, 99, 110, 121\} \equiv \{11, 22, 10, 21, 9, 20, 8, 19, 7, 18, 6\} \pmod{23}.$$

$\therefore 22, 21, 20, 19, 18 > 11.5$, so $n = 5$.

$$\therefore (-1)^5 = -1.$$

(e) $(6/31)$

$$(p-1)/2 = 15, \quad p/2 = 15.5.$$

$$S = \{6, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, 78, 84, 90\}$$

$$\equiv \{6, 12, 18, 24, 30, 5, 11, 17, 23, 29, 4, 10, 16, 22, 28\} \pmod{31}.$$

$\therefore 18, 24, 30, 17, 23, 29, 16, 22, 28 > 15.5$, so $n = 9$.

$$\therefore (-1)^9 = -1.$$

3. For an odd prime p , prove there are $\frac{p-1}{2} - \phi(p-1)$ quadratic nonresidues of p that are not primitive roots of p .

Pf. By Th. 9.4, there are $\frac{p-1}{2}$ quadratic residues of p and $\frac{p-1}{2}$ quadratic nonresidues of p .

If a is a quadratic residue of p , it cannot be a primitive root, because $a^{(p-1)/2} \equiv 1 \pmod{p}$ by Th. 9.1, and $\frac{p-1}{2} < p-1 = \phi(p)$.

$\therefore$ if r is a primitive root of p , it must be congruent to a quadratic nonresidue of p .

Let S be the set of quadratic nonresidues of p . There are thus $\frac{p-1}{2}$ elements of S .

$\therefore$ All the primitive roots of p are congruent to some of the $\frac{p-1}{2}$ elements of S .

By the corollary to Th. 8.6 (p. 165), there are $\phi(p-1)$ primitive roots of p , and so $\phi(p-1)$ elements of S are primitive roots.

$$\therefore \frac{p-1}{2} - \phi(p-1) \text{ elements of } S \text{ are not primitive roots of } p.$$

4. (a) Let p be an odd prime. Show that the Diophantine equation $x^2 + py + a = 0$, $\gcd(a, p) = 1$, has an integral solution if and only if $(-a/p) = 1$.

(1) If $x^2 + py + a = 0$ has a solution, then $x^2 + a = -yp \Rightarrow x^2 + a \equiv 0 \pmod{p}$

$$\Rightarrow x^2 \equiv -a \pmod{p} \Rightarrow (-a) \text{ is a quadratic residue of } p \Rightarrow (-a/p) = 1.$$

(2) If $(-a/p) = 1$, then $(-a)$ is a quadratic residue of $p \Rightarrow x^2 \equiv -a \pmod{p}$ has a solution $x = x_0$, so $x_0^2 \equiv -a \pmod{p}$, i.e. $x_0^2 + a \equiv 0 \pmod{p}$.

$$\therefore p \mid (x_0^2 + a) \Rightarrow \exists y_0 \in \mathbb{Z} \text{ s.t. } x_0^2 + a = -y_0p,$$

i.e. $x_0^2 + py_0 + a = 0$ is an integral solution. $\square$ solution in $x \Rightarrow$ there is an integer k s.t. $x^2 = -a + kp$, or $x^2 + a - kp = 0$, or letting $y = -k$, $x^2 + py + a = 0$. $\square$

(b) Determine whether $x^2 + 7y - 2 = 0$ has a solution in integers.

$\gcd(-2, 7) = 1$. By (a), since $(2/7) = 1$ by Th. 9.6, then there is a solution.

5. Prove that 2 is not a primitive root of any prime of the form $p = 3 \cdot 2^n + 1$, except when $p = 13$.

Pf. Strategy: show 2 is a quadratic residue of p , and so cannot be a primitive root.

For $p-1 = 3 \cdot 2^n$, $p-1 \equiv 0 \pmod{8}$ when $n \geq 3$.

$$\therefore \text{For } n \geq 3, p \equiv 1 \pmod{8}, \text{ so by Th. 9.6, } (2/p) = 1.$$

$$\therefore \text{For } n \geq 3, 2 \text{ is a quadratic residue}$$

and $\therefore$ not a primitive root.

$\therefore$ Consider $n = 1, 2$:

$n = 1$: $p = 3 \cdot 2 + 1 = 7$. $\therefore p \equiv 7 \pmod{8}$.

$\therefore$ By Th. 9.6, $(2/p) = 1$, so 2 is a quadratic residue, and so 2 is not a primitive root.

$n = 2$: $p = 3 \cdot 2^2 + 1 = 13$. $\therefore p \equiv 5 \pmod{8}$.

$\therefore (2/p) = -1$, so 2 is a quadratic nonresidue, and 2 is a known primitive root of 13.

$\therefore$ For $n \geq 1$, $n \neq 2$, if $p = 3 \cdot 2^n + 1$ is prime, then 2 is not a primitive root of p . $\square$

6. (a) If p is an odd prime, and $\gcd(ab, p) = 1$, prove that at least one of a , b , or ab is a quadratic residue of p .

Pf. $(ab/p) = (a/p)(b/p)$. (ab/p) , (a/p) , and (b/p) are each equal to 1 or -1 .

$\therefore$ if $(ab/p) = 1$, by def., ab is a quadratic residue of p .

$\therefore$ Suppose $(ab/p) = -1$.

$\therefore (a/p)(b/p) = -1$.

(a/p) and (b/p) cannot both be -1 , since $(-1)(-1) \neq -1$.

$\therefore$ Either $(a/p) = 1$ or $(b/p) = 1$.

$\therefore$ either a or b is a quadratic residue of p . $\square$

(b) Given a prime p , show that for some choice of $n > 0$, p divides $(n^2 - 2)(n^2 - 3)(n^2 - 6)$.

Pf. $p = 2$: Let $n = 3$. Then $(n^2 - 2)(n^2 - 3)(n^2 - 6) = (9 - 2)(9 - 3)(9 - 6) = 7 \cdot 6 \cdot 3$.

$$\therefore p \mid 7 \cdot 6 \cdot 3.$$

$p = 3$: Let $n = 3$, as above, $p \mid 7 \cdot 6 \cdot 3$.

$p > 3$: $\therefore \gcd(2 \cdot 3, p) = 1$.

$\therefore$ By (a), one of 2, 3, or $2 \cdot 3$ is a quadratic residue of p .

$\therefore$ By def., there must be an n s.t. $n^2 \equiv 2 \pmod{p}$, or $n^2 \equiv 3 \pmod{p}$, or $n^2 \equiv 6 \pmod{p}$.

$$\therefore n^2 - 2 \equiv 0 \pmod{p}, \text{ or } n^2 - 3 \equiv 0 \pmod{p}, \text{ or } n^2 - 6 \equiv 0 \pmod{p}.$$

$$\therefore (n^2 - 2)(n^2 - 3)(n^2 - 6) \equiv 0 \pmod{p}.$$

$$\therefore \text{There is an } n \text{ s.t. } p \mid (n^2 - 2)(n^2 - 3)(n^2 - 6). \quad \square$$

7. If p is an odd prime, show that

$$\sum_{a=1}^{p-2} (a(a+1)/p) = -1 \quad [(/) \text{ is Legendre symbol}].$$

Pf. Use the hint. Let a' be defined by $aa' \equiv 1 \pmod{p}$. Note that since $\gcd(a, p) = 1$, then a' exists, by Th. 4.7, for each $1 \leq a \leq p-2$.

Note that as a runs from 1 to $p-2$, a' also runs from 1 to $p-2$ (not $p-1$, for if $a' = p-1$, then $a(p-1) \equiv 1$, $ap-a \equiv 1$, $-a \equiv 1$, $0 \equiv 1+a$, $p \equiv 1+a$, $p-1 \equiv a \Rightarrow p-1 = a$, a contradiction). Also, if $a_1 a' \equiv 1$ and $a_2 a' \equiv 1$, then $a_1 a' \equiv a_2 a' \Rightarrow a_1 \equiv a_2 \Rightarrow a_1 = a_2$.

$\therefore$ As a runs through 1 to $p-2$, each a' from 1 to $p-2$ is represented only once.

$\therefore$ as a runs through 1 to $p-2$, $1+a'$ runs through 2 to $p-1$.

Since $aa' \equiv 1 \pmod{p} \Rightarrow a + aa' \equiv a + 1 \Rightarrow a(1+a') \equiv a + 1$.

$$\therefore a(a+1) \equiv a^2(1+a') \pmod{p}.$$

$$\therefore (a(a+1)/p) = (a^2(1+a')/p) = ((1+a')/p).$$

$$\begin{aligned} \therefore \sum_{a=1}^{p-2} (a(a+1)/p) &= \sum_{1+a'=2}^{p-1} ((1+a')/p) = \sum_{a'=2}^{p-1} (a'/p) = \sum_{a=2}^{p-1} (a/p). \\ &= \sum_{a=1}^{p-1} (a/p) - (1/p). \end{aligned}$$

But by Th. 9.4, $\sum_{a=1}^{p-1} (a/p) = 0$ and $(1/p) = 1$.

$$\therefore \sum_{a=1}^{p-2} (a(a+1)/p) = -1. \quad \square$$

8. Prove the statements below:

(a) If p and $q = 2p + 1$ are both odd primes, then -4 is a primitive root of q .

Pf. Since $-4 = -2^2$ and $\gcd(2, q) = 1$, then $\gcd(-4, q) = 1$.

Since $\phi(q) = q - 1 = 2p$, then the order of -4 must be 1, 2, p , or $2p$ by Th. 8.1.

(1) If $-4 \equiv 1 \pmod{q}$, then $5 \equiv 0 \pmod{q} \Rightarrow q \mid 5$. But $q > 5$ since p is odd.

$\therefore$ order of $-4 \pmod{q}$ is not 1.

(2) If $(-4)^2 \equiv 1 \pmod{q}$, then $15 \equiv 0 \pmod{q}$, or $q \mid 15 \Rightarrow q = 3$ or 5 , again contradicting $q > 5$.

$\therefore$ order of $-4 \pmod{q}$ is not 2. (3) Suppose $(-4)^p \equiv 1 \pmod{q}$.

Since $(-4)^p = (-4)^{\frac{q-1}{2}}$, then $(-4)^{\frac{q-1}{2}} \equiv 1 \pmod{q}$.

But $(-4/q) \equiv (-4)^{\frac{q-1}{2}} \pmod{q}$.

$\therefore (-4/q) \equiv 1 \pmod{q} \Rightarrow (-4/q) = 1. \quad [I]$

However, $(-4/q) = (-1/q)(2/q)(2/q)$.

$(-1/q) = (-1)^{\frac{q-1}{2}} = (-1)^{\frac{2p}{2}} = (-1)^p = -1$ as p is odd.

$(2/q) \equiv 2^{\frac{q-1}{2}} = 2^p \pmod{q}$.

If $p \equiv 1 \pmod{4}$, then $p = 1 + 4k$, some k , so $q = 2p + 1 = 3 + 8k$, so $q \equiv 3 \pmod{8}$, so by Th. 9.6, $(2/q) = 1$.

$\therefore (-4/q) = (-1)(1)(1) = -1$. This contradicts [I].

If $p \equiv 3 \pmod{4}$, then, as above, $q \equiv 7 \pmod{8}$, so by Th. 9.6, $(2/q) = 1$.

$\therefore (-4/q) = (-1)(1)(1) = -1$, contradicting [I].

$\therefore (-4)^p \not\equiv 1 \pmod{q}$, so order of $-4 \pmod{q}$ is not p .

(1), (2), (3) $\Rightarrow$ order of $(-4) \pmod{q}$ is $2p = q - 1 = \phi(q)$.

$\therefore -4$ is a primitive root of $q = 2p + 1$. $\square$

(b) If $p \equiv 1 \pmod{4}$ is a prime, then -4 and $\frac{p-1}{4}$ are quadratic residues of p .

Pf. (1) $(-4/p) = (-1/p)(2/p)(2/p)$.

By corollary to Th. 9.2 (p. 187), $(-1/p) = 1$.

$(2/p) = \pm 1$, so $(2/p)(2/p) = 1$.

$\therefore (-4/p) = 1 \Rightarrow -4$ is a quadratic residue of p .

(2) Since $\gcd(4, p) = 1$, then

$$x^2 \equiv \frac{p-1}{4} \pmod{p} \iff 4x^2 \equiv p-1 \pmod{p} \iff 4x^2 \equiv -1 \pmod{p} \iff (2x)^2 \equiv -1 \pmod{p}.$$

Let $y = 2x$.

$\therefore y^2 \equiv -1 \pmod{p}$ has a solution by corollary to Th. 9.2 (p. 187) since $p \equiv 1 \pmod{4}$.

$\therefore x^2 \equiv \frac{p-1}{4} \pmod{p}$ has a solution, and so $\frac{p-1}{4}$ is a quadratic residue of p . $\square$

9. For a prime $p \equiv 7 \pmod{8}$, show that $p \mid 2^{\frac{p-1}{2}} - 1$.

Pf. By Th. 9.2, $(2/p) \equiv 2^{\frac{p-1}{2}} \pmod{p}$.

By Th. 9.6, $(2/p) = 1$ since $p \equiv 7 \pmod{8}$.

$$\therefore 1 \equiv 2^{\frac{p-1}{2}} \pmod{p} \Rightarrow 2^{\frac{p-1}{2}} - 1 \equiv 0 \pmod{p} \Rightarrow p \mid 2^{\frac{p-1}{2}} - 1. \quad \square$$

10. Use Problem 9 to confirm that the numbers $2^n - 1$ are composite for $n = 11, 23, 83, 131, 179, 183, 239, 251$. Need to show $n = \frac{p-1}{2}$ for p of form $7 + 8k$. Then, by #9, $p \mid 2^n - 1$, so $2^n - 1$ is composite.

$$11 : 11 = \frac{p-1}{2}, \quad p = 23 = 7 + (2)8$$

$$23 : 23 = \frac{p-1}{2}, \quad p = 47 = 7 + (5)8$$

$$83 : 83 = \frac{p-1}{2}, \quad p = 167 = 7 + (20)8$$

$$131 : 131 = \frac{p-1}{2}, \quad p = 263 = 7 + (32)8$$

$$179 : 179 = \frac{p-1}{2}, \quad p = 359 = 7 + (44)8$$

$$183 : 183 = \frac{p-1}{2}, \quad p = 367 = 7 + (45)8$$

$$239 : 239 = \frac{p-1}{2}, \quad p = 479 = 7 + (57)8$$

$$251 : 251 = \frac{p-1}{2}, \quad p = 503 = 7 + (62)8$$

All the p numbers above are prime.

11. Given that p and $q = 4p + 1$ are both prime, prove the following: (a) Any quadratic nonresidue of q is either a primitive root of q or has order 4 mod q .

Pf. Let a be a quadratic nonresidue of q . It is assumed that $\gcd(a, q) = 1$.

$$\therefore -1 = (a/q) \equiv a^{(q-1)/2} = a^{4p/2} = a^{2p} \pmod{q}.$$

$$\therefore a^{4p} \equiv 1 \pmod{q}.$$

$\therefore$ order of a mod q is 1, 2, 4, p , $2p$, or $4p$.

(1) If order of a mod q is $4p = \phi(q)$, then a is a primitive root of q .

(2) $\therefore$ Assume order of a mod q is not $4p$.

1: Order of $a \neq 1$, for if $a \equiv 1 \pmod{q}$, then $x^2 \equiv a \equiv 1 \pmod{q}$ would have a solution ($x = 1$), which contradicts a as a quadratic nonresidue.

2: If $a^2 \equiv 1 \pmod{q}$, then $a^{2p} \equiv 1 \pmod{q}$. But $-1 \equiv a^{2p} \pmod{q} \Rightarrow -1 \equiv 1$, or $q \mid 2$, an impossibility. $\therefore$ order of a mod q is not 2.

p : Suppose $a^p \equiv 1 \pmod{q}$. $\therefore a^{2p} \equiv 1 \pmod{q}$, and since $a^{2p} \equiv -1 \pmod{q}$, then $1 \equiv -1 \pmod{q} \Rightarrow q \mid 2$, an impossibility. $\therefore$ order of a is not p .

$2p$: As above $a^{2p} \equiv 1 \pmod{q}$ contradicts $a^{2p} \equiv -1 \pmod{q}$, so order of a can't be $2p$.

$\therefore$ Since order can't be 1, 2, p , $2p$, $4p$, order of a mod q must be 4.

$\therefore$ (1) + (2) $\Rightarrow$ order of a mod q is either $4p$ or 4. If $4p = \phi(q)$, then a is a primitive root of q . $\square$

(b) The integer 2 is a primitive root of q ; in particular, 2 is a primitive root of the primes 13, 29, 53, and 173.

Pf. $\phi(q) = q - 1 = 4p + 1 - 1 = 4p$.

$\therefore$ Need to show $2^{4p} \equiv 1 \pmod{q}$, and order of 2 mod q can't be 1, 2, 4, p , $2p$.

Note $p \equiv 1 \pmod{4} \Rightarrow p = 1 + 4k$, so $q = 4p + 1 = 4 + 16k + 1 = 5 + 16k = 5 + 8(2k)$.

$\therefore p \equiv 1 \pmod{4} \Rightarrow q \equiv 5 \pmod{8}$.

$p \equiv 3 \pmod{4} \Rightarrow p = 3 + 4k$, so $q = 12 + 16k + 1 = 13 + 16k = 5 + 8(1 + 2k)$.

$\therefore p \equiv 3 \pmod{4} \Rightarrow q \equiv 5 \pmod{8}$.

$\therefore p$ prime and $q = 4p + 1$ prime $\Rightarrow q \equiv 5 \pmod{8}$.

$\therefore (2/q) = -1$ by Th. 9.6.

$\therefore -1 = (2/q) \equiv 2^{(q-1)/2} \pmod{q} = 2^{2p} \pmod{q}$.

$\therefore 2^{2p} \equiv -1 \pmod{q}$. [I]

(a) Squaring [I]: $2^{4p} \equiv 1 \pmod{q}$.

(b) $2p$: Order of 2 can't be $2p$ by [I], for $2^{2p} \equiv 1 \Rightarrow 1 \equiv -1 \pmod{q} \Rightarrow q \mid 2$.

p : Order of 2 can't be p , for $2^p \equiv 1 \Rightarrow 2^{2p} \equiv 1 \pmod{q}$ by squaring [I].

1: $2^1 \equiv 1 \pmod{q} \Rightarrow 1 \equiv 0 \Rightarrow q \mid 1$, so order can't be 1.

2: $2^2 \equiv 1 \pmod{q} \Rightarrow 3 \equiv 0 \Rightarrow q \mid 3$, and impossibility since $q = 4p + 1$.

4: $2^4 \equiv 1 \pmod{q} \Rightarrow 15 \equiv 0 \Rightarrow q \mid 15 \Rightarrow q = 3, 5$, or 15, also an impossibility since $q = 4p + 1$ and p is prime.

$\therefore$ Order of 2 mod q is not 1, 2, 4, p , $2p$, and so must be $4p$.

$\therefore (a) + (b) \Rightarrow 2$ is a primitive root of q .

$$\begin{aligned} 13 &= 1 + 4(3), \\ 29 &= 1 + 4(7), \\ 53 &= 1 + 4(13), \\ 173 &= 1 + 4(43). \end{aligned}$$

$\therefore 13, 29, 53$, and 173 satisfy the condition and so 2 is a primitive root for each. $\square$

12. If r is a primitive root of the odd prime p , prove that the product of the quadratic residues of p is congruent mod p to $r^{(p^2-1)/4}$ and the product of the nonresidues of p is congruent mod p to $r^{(p-1)^2/4}$.

Pf. (a) Product of quadratic residues is congruent to $r^{(p^2-1)/4}$.

The quadratic residues are congruent to the even powers of r (corollary to Th. 9.4).

Let $a_1, a_2, \dots, a_{\frac{p-1}{2}}$ be the $\frac{p-1}{2}$ quadratic residues of p .

$$\begin{aligned} \therefore a_1 a_2 \cdots a_{\frac{p-1}{2}} &\equiv r^2 \cdot r^4 \cdots r^{p-1} \pmod{p} \\ &= (r^1)^2 \cdot (r^2)^2 \cdots \left(r^{\frac{p-1}{2}}\right)^2 \pmod{p} \\ &= \left[r^1 \cdot r^2 \cdots r^{\frac{p-1}{2}}\right]^2 \pmod{p} \\ &= \left(r^{1+2+\cdots+\frac{p-1}{2}}\right)^2 \pmod{p}. \end{aligned}$$

$$\begin{aligned} \text{But } 1 + 2 + \cdots + \frac{p-1}{2} &= \frac{p-1}{2} \cdot \frac{\frac{p-1}{2} + 1}{2} \\ &= \frac{p-1}{2} \cdot \frac{p+1}{2} = \frac{p^2-1}{4} / 2 = \frac{p^2-1}{8}. \end{aligned}$$

Wait — more simply: $1 + 2 + \cdots + \frac{p-1}{2} = \frac{p-1}{2} \cdot \frac{p+1}{2} / 2 = \frac{(p^2-1)/4}{2}$. So:

$$\begin{aligned} \therefore a_1 a_2 \cdots a_{\frac{p-1}{2}} &\equiv \left(r^{1+2+\cdots+\frac{p-1}{2}}\right)^2 \pmod{p} \\ &= \left[r^{(p^2-1)/4 / 2}\right]^2 \pmod{p} \\ &= r^{\frac{p^2-1}{4}} \pmod{p}. \end{aligned}$$

$\therefore$ product of quadratic residues is congruent mod p to $r^{(p^2-1)/4}$.

(b) Product of quadratic nonresidues is congruent to $r^{(p-1)^2/4}$.

The quadratic nonresidues are congruent to the odd powers of r (corollary p. 188).

$\therefore$ Let $q_1, q_2, \dots, q_{\frac{p-1}{2}}$ be the quadratic nonresidues.

$$\begin{aligned} \therefore q_1 \cdot q_2 \cdots q_{\frac{p-1}{2}} &\equiv r^1 \cdot r^3 \cdots r^{p-2} \pmod{p} \\ &= r^{1+3+\cdots+(p-2)} \pmod{p}. \end{aligned}$$

But $1 + 3 + \cdots + p - 2 = \frac{p-1}{2}(p-1)/2$, as there are $\frac{p-1}{2}$ terms.

$$\therefore 1 + 3 + \cdots + (p-2) = \left(\frac{p-1}{2}\right)^2 \cdot \frac{1}{1} = \frac{(p-1)^2}{4}.$$

$$\therefore q_1 \cdot q_2 \cdots q_{\frac{p-1}{2}} \equiv r^{(p-1)^2/4} \pmod{p}. \quad \square$$

13. Establish that the product of the quadratic residues of the odd prime p is congruent mod p to 1 or -1 according as $p \equiv 3 \pmod{4}$ or $p \equiv 1 \pmod{4}$.

Pf. Let r be a primitive root of p .

Since $r^{p-1} \equiv 1 \pmod{p}$, then

$$r^{p-1} - 1 = \left(r^{\frac{p-1}{2}} + 1\right)\left(r^{\frac{p-1}{2}} - 1\right) \equiv 0 \pmod{p}.$$

As r is a primitive root of p , $r^{\frac{p-1}{2}} - 1 \not\equiv 0 \pmod{p}$.

$$\therefore r^{\frac{p-1}{2}} + 1 \equiv 0 \pmod{p} \Rightarrow r^{\frac{p-1}{2}} \equiv -1 \pmod{p}. \quad [\text{I}]$$

Let $a_1, a_2, \dots, a_{\frac{p-1}{2}}$ be the quadratic residues of p . $\therefore$ By prob. #12 above,

$$\begin{aligned} a_1 \cdot a_2 \cdots a_{\frac{p-1}{2}} &\equiv r^{(p^2-1)/4} \pmod{p} \\ &= \left(r^{\frac{p-1}{2}}\right)^{\frac{p+1}{2}} \pmod{p} \\ &\equiv (-1)^{\frac{p+1}{2}} \pmod{p} \quad \text{by [I]}. \end{aligned}$$

If $p \equiv 1 \pmod{4}$, then $p = 1 + 4K$, some K , so $\frac{p+1}{2} = \frac{2+4K}{2} = 1 + 2K$, an odd integer.

$$\therefore (-1)^{\frac{p+1}{2}} = -1.$$

$$\therefore p \equiv 1 \pmod{4} \Rightarrow a_1 a_2 \cdots a_{\frac{p-1}{2}} \equiv -1 \pmod{p}.$$

If $p \equiv 3 \pmod{4}$, then $p = 3 + 4K$, some K , so $\frac{p+1}{2} = \frac{4+4K}{2} = 2 + 2K$, an even integer.

$$\therefore (-1)^{\frac{p+1}{2}} = 1.$$

$$\therefore p \equiv 3 \pmod{4} \Rightarrow a_1 a_2 \cdots a_{\frac{p-1}{2}} \equiv 1 \pmod{p}. \quad \square$$

14. (a) If the prime $p > 3$, show that p divides the sum of its quadratic residues. *Pf.* Let r be a primitive root of p . Let $a_1, a_2, \dots, a_{\frac{p-1}{2}}$ be the quadratic residues of p .

By corollary to Th. 9.4 (p. 188), $r^2, r^4, \dots, r^{p-1}$ are congruent to the quadratic residues of p .

$$\therefore a_1 + a_2 + \cdots + a_{\frac{p-1}{2}} \equiv r^2 + r^4 + \cdots + r^{p-1} \pmod{p}.$$

But $r^2 + r^4 + \cdots + r^{p-1} = r^2(1 + r^2 + \cdots + r^{p-3})$, for $p > 3$.

$$\therefore a_1 + a_2 + \cdots + a_{\frac{p-1}{2}} \equiv r^2(1 + r^2 + \cdots + r^{p-3}) \pmod{p}, \quad [\text{I}] \quad \text{for } p > 3.$$

But $r^{p-1} \equiv 1 \pmod{p}$, as r is a primitive root.

$$\therefore r^2 + r^4 + \cdots + r^{p-1} \equiv 1 + r^2 + r^4 + \cdots + r^{p-3} \pmod{p}, \quad \text{for } p > 3.$$

$$\therefore a_1 + a_2 + \cdots + a_{\frac{p-1}{2}} \equiv 1 + r^2 + r^4 + \cdots + r^{p-3} \pmod{p}, \quad [\text{II}] \quad \text{for } p > 3.$$

Equating [I] and [II], for $p > 3$,

$$r^2(1 + r^2 + \cdots + r^{p-3}) \equiv (1 + r^2 + \cdots + r^{p-3}) \pmod{p}. \quad [\text{III}]$$

But p must divide $(1 + r^2 + \cdots + r^{p-3})$, for if not, then we can cancel it on both sides of [III], and obtain $r^2 \equiv 1 \pmod{p}$. This is a contradiction since r is a primitive root of $p > 3$, so that $p - 1 > 2$ and $r^{p-1} \equiv 1 \pmod{p}$.

Since $p \mid (1 + r^2 + \cdots + r^{p-3})$, then from [I] or [II], $p \mid (a_1 + a_2 + \cdots + a_{\frac{p-1}{2}})$.

Note: Since $1 + 2 + \cdots + (p-1) = \frac{p-1}{2} \cdot p$, then $p \mid (1 + 2 + \cdots + (p-1))$.

$\therefore p$ also divides the sum of the quadratic nonresidues of p , for $p > 3$. $\square$

(b) If the prime $p > 5$, show that p divides the sum of the squares of its quadratic nonresidues.

Pf. Let r be a primitive root of p , and let $a_1, a_2, \dots, a_{\frac{p-1}{2}}$ be the quadratic nonresidues of p .

By corollary to Th. 9.4 (p. 188), $r^1, r^3, \dots, r^{p-2}$ are congruent to the quadratic nonresidues of p .

$$\therefore a_1^2 + a_2^2 + \cdots + \left(a_{\frac{p-1}{2}}\right)^2 \equiv r^2 + r^6 + \cdots + r^{2(p-2)} \pmod{p}. \quad [\text{I}]$$

Each term on the right side of [I] is described by r^{4k-2} , $1 \leq k \leq \frac{p-1}{2}$.

And all terms on the right of [I] are incongruent mod p . For if $r^{4k_1-2} \equiv r^{4k_2-2} \pmod{p}$, then by Th. 8.2, $4k_1 - 2 \equiv 4k_2 - 2 \pmod{p-1} \iff 4k_1 \equiv 4k_2 \pmod{p-1} \iff k_1 \equiv k_2 \pmod{p-1}$.

Since $p > 5$, $p-1 > 4$, $\therefore \gcd(4, p-1) = 1$. But $1 \leq k_1, k_2 \leq \frac{p-1}{2}$, so $k_1, k_2 < p-1$.

$\therefore k_1 = k_2$.

$\therefore r^{4k_1-2} \equiv r^{4k_2-2} \pmod{p} \iff k_1 = k_2$, so all terms on right side of [I] are incongruent mod p .

All the terms on the right side of [I] are even, and there are $\frac{p-1}{2}$ such terms. Since they are incongruent, then the terms are congruent to $r^2, r^4, \dots, r^{p-1}$, in some order.

$$\therefore r^2 + r^6 + \cdots + r^{2(p-2)} \equiv r^2 + r^4 + \cdots + r^{p-1} \pmod{p}. \quad [\text{II}]$$

By (a), the right side of [II] is divisible by p , and so is the left side of [II], and so is the left side of [I]. $\square$

15. Prove that for any prime $p > 5$ there exist integers $1 \leq a, b \leq p-1$ for which $(a/p) = (a+1/p) = 1$ and $(b/p) = (b+1/p) = -1$. That is, there are consecutive quadratic residues of p and consecutive nonresidues.

Pf. Since $x^2 \equiv 1$, $x^2 \equiv 4$, and $x^2 \equiv 9$ have solutions for all $p > 5$, then consider $x^2 \equiv 2$, $x^2 \equiv 5$, $x^2 \equiv 10$.

Now use 6(a). For $p > 5$: $\gcd(2, p) = 1$, $\gcd(5, p) = 1$, so $\gcd(10, p) = 1$. By 6(a), one of 2, 5, or 10 must be a quadratic residue of p .

If $(2/p) = 1$, then 1, 2 are consecutive residues.

If $(5/p) = 1$, then 4 and 5 are consecutive residues.

If $(10/p) = 1$, then 9 and 10 are consecutive residues.

Since the above showed at least one pair of consecutive residues for $p > 5$, then consider the remaining $p-3$ terms (there are $p-1$ residues + nonresidues, so subtract out the 2 consecutive residues). Let a_k, a_{k+1} be the consecutive residues.

$\therefore$ Consider the terms: $1, 2, \dots, a_k, a_{k+1}, \dots, a_{p-1}$.

If 2, 3 are consecutive nonresidues, then we're done.

So suppose 2, 3 are not consecutive nonresidues.

$\therefore$ Since 1 and 4 are always residues, then in the list, there are at least 3 residues among 1, 2, 3, 4.

Suppose the remaining even number of terms $5, 6, \dots, p-1$ alternate with respect to residue/nonresidue. Then in the remaining terms $5, 6, \dots, p-1$, the number of residues = number of nonresidues, and so overall, the number of residues is at least 2 greater than the number of nonresidues.

$\therefore$ There must be consecutive nonresidues in the list $1, 2, \dots, p-1$. $\square$

16. (a) Let p be an odd prime and $\gcd(a, p) = \gcd(k, p) = 1$. Show that if the equation $x^2 - ay^2 = kp$ admits a solution, then $(a/p) = 1$; for example, $(2/7) = 1$ because $6^2 - 2 \cdot 2^2 = 4 \cdot 7$.

Pf. Suppose x, y solve $x^2 - ay^2 = kp$.

(1) $\gcd(p, y) = 1$.

For if $\gcd(p, y) = n > 1$, then since p is prime, then $y = np$.

$\therefore x^2 = a(np)^2 + kp = an^2p^2 + kp$.

$\therefore p \mid x$, so $x = mp$, some $m \geq 1$.

$$\therefore m^2p^2 = an^2p^2 + kp, \text{ or } m^2p - an^2p = \rho(m^2 - an^2) = k \Rightarrow \rho \mid k.$$

But $\gcd(k, \rho) = 1$, $\therefore \gcd(\rho, y) = 1$.

(2) By (1), $y^{p-1} \equiv 1 \pmod{p}$ by Euler's Th.

$$\therefore x^2 - ay^2 = kp \iff x^2 \equiv ay^2 \pmod{p}.$$

$$\therefore x^2y^{p-1} \equiv ay^2 \pmod{p}.$$

p is odd, so $p-1 \geq 2$, so $y^{p-1} \geq y^2$. Using (1) again to divide by y^2 :

$$x^2y^{p-3} \equiv a \pmod{p}.$$

$$\therefore x^2 y^{p-3} \cdot y^{p-1} \equiv a \pmod{p}.$$

$$x^2 y^{2p-4} = (xy^{p-2})^2 \equiv a \pmod{p}.$$

$\therefore$ Let $z = xy^{p-2}$, so $z^2 \equiv a \pmod{p}$.

$$\therefore (a/p) = 1. \quad \square$$

Incidentally, $\gcd(x, p) = 1$.

For if $x = np$, then $(np)^2 - ay^2 = kp$, or $n^2 p^2 - kp = ay^2$, $p(n^2 p - k) = ay^2$.

$\therefore p \mid a$ or $p \mid y$. This contradicts (1) and $\gcd(a, p) = 1$.

(b) By considering the equation $x^2 + 5y^2 = 7$, demonstrate that the converse of the result in part (a) need not hold.

$(-5/7) = 1$, for $(-5)^{(7-1)/2} = -5^3 = -125 = -7 \cdot 18 + 1$, so $(-5)^{(7-1)/2} \equiv 1 \pmod{7}$.

But there is no integer solution to $x^2 + 5y^2 = 7$.

Only possibilities are $x = 0, 1, 2$, $y = 0, 1$ and these don't work.

(c) Show that for any prime $p \equiv \pm 3 \pmod{8}$, the equation $x^2 - 2y^2 = p$ has no solution. *Pf.* Suppose $x^2 - 2y^2 = p$ has a solution.

Since p is odd, $\gcd(2, p) = 1$ and clearly $\gcd(1, p) = 1$.

$\therefore$ By (a), $(2/p) = 1$.

But $p \equiv \pm 3 \pmod{8} \iff p \equiv 3 \pmod{8}$ or $p \equiv 5 \pmod{8}$, and by Th. 9.6, $(2/p) = -1$.

$\therefore x^2 - 2y^2 = p$ can't have a solution. $\square$

17. Prove that the odd prime divisors p of the integers $9^n + 1$ are of the form $p \equiv 1 \pmod{4}$.

Pf. Let p be an odd prime divisor of $9^n + 1$.

$\therefore 9^n + 1 \equiv 0 \pmod{p}$, or $(3^2)^n + 1 \equiv 0 \pmod{p}$, or $(3^n)^2 \equiv -1 \pmod{p}$.

Since p is odd, either $p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{4}$.

By corollary to Th. 9.2 (p. 187), if $p \equiv 3 \pmod{4}$, there can be no solution to $x^2 \equiv -1 \pmod{p}$, so that there is no n s.t. $(3^n)^2 \equiv -1 \pmod{p}$.

$\therefore$ If p is to be a divisor of $9^n + 1$, it must be of the form $p \equiv 1 \pmod{4}$. $\square$ **18.**

For a prime $p \equiv 1 \pmod{4}$, verify that the sum of the quadratic residues of p is equal to $p(p-1)/4$.

Pf. (1) If $(a/p) = 1$, then $(p-a/p) = 1$.

For since $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$, and

$$(p-a)^n = p^n + \cdots + (-a)^n = p^n + \cdots + (-1)^n a^n,$$

then $(p-a)^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{2}} a^{\frac{p-1}{2}} \pmod{p}$.

For $p \equiv 1 \pmod{4}$, $p-1 = 4k$, some k , so $(-1)^{\frac{p-1}{2}} = (-1)^{2k} = 1$.

$$\therefore (p-a)^{\frac{p-1}{2}} \equiv a^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

$\therefore$ for $p \equiv 1 \pmod{4}$, $p-a$ is also a quadratic residue.

(2) Let $a_1, \dots, a_r$ be the quadratic residues of p less than $p/2$.

$\therefore p - a_1, \dots, p - a_r$ are quadratic residues of p , all less than p , and $p - a_i > p/2$ (since $a_i < p/2$, then $-a_i > -p/2$, $p - a_i > p - p/2 = p/2$).

Since all these residues are less than p , they are all incongruent.

(3) The a_i and $p - a_i$ are all the quadratic residues of p .

For if a_x is a quadratic residue of p s.t. $p/2 < a_x < p$, then $p - a_x$ is a residue by (2), and $0 < p - a_x < p/2$, so $p - a_x$ must be one of the a_i , since that set consisted of all residues less than $p/2$.

$\therefore p - (p - a_x)$ must be one of the $p - a_i$. But $p - (p - a_x) = a_x$, so a_x is one of $p - a_i$.

(4) Since there are a total of $\frac{p-1}{2}$ quadratic residues (Th. 9.4), (3) $\Rightarrow$ there are $\frac{p-1}{4}$ residues $< p/2$ and $\frac{p-1}{4}$ residues $> p/2$.

$\therefore r = \frac{p-1}{4}$ in the sequence $a_1, \dots, a_r$.

(5) $\therefore \text{Sum} = (a_1 + \dots + a_r) + [(p - a_1) + \dots + (p - a_r)]$

$$= p + \dots + p = rp = \left(\frac{p-1}{4}\right)p. \quad \square$$

Note that from (1), for $p \equiv 3 \pmod{4}$,

$$(p - a)^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{2}} a^{\frac{p-1}{2}} \equiv -a^{\frac{p-1}{2}} \equiv -1 \pmod{p},$$

so if a is a quadratic residue of p , $p - a$ is a quadratic nonresidue.

9.3 Quadratic Reciprocity

1. Evaluate the following Legendre symbols:

(a) $(71/73)$

$71 \equiv -1 \pmod{4}$, $73 \equiv 1 \pmod{4}$. $\therefore (71/73) = (73/71)$.

$(73/71) = (71 + 2/71) = (2/71)$.

$71 = 7 + 8(8) \Rightarrow 71 \equiv 7 \pmod{8}$, so $(2/71) = 1$.

$$\therefore (71/73) = 1.$$

(b) $(-219/383)$

$219 = 3 \cdot 73$, $383 \equiv 3 \pmod{4}$.

$$\therefore (-219/383) = (-1/383)(3/383)(73/383)$$

$$= (-1)[-(383/3)](383/73)$$

$$= (2/3)(18/73) \quad [\text{Note: } 383 \bmod 3 = 2, 383 \bmod 73 = 18; \text{ manuscript may read 14.}]$$

$$= (-1)(2 \cdot 3^2/73) = -(2/73)$$

$$= -1 \quad \text{as } 73 \equiv 1 \pmod{8}.$$

$$\therefore (-219/383) = -1.$$

(c) $(461/773)$

$$461 \equiv 1 \pmod{4}.$$

$$\begin{aligned} \therefore (461/773) &= (773/461) = (312/461) = (2^3 \cdot 3 \cdot 13/461) \\ &= (2/461)(3/461)(13/461) \\ &= (-1)(461/3)(461/13) \quad [461 \equiv 5 \pmod{8}, 3 \equiv 3 \pmod{4}, 13 \equiv 1 \pmod{4}] \\ &= (-1)(2/3)(6/13) \\ &= (-1)(-1)(2/13)(3/13) \\ &= (-1)(3/13) = (-1)(13/3) \quad \text{as } 13 \equiv 1 \pmod{4} \\ &= (-1)(1/3) = -1. \end{aligned}$$

$$\therefore (461/773) = -1.$$

(d) $(1234/4567)$, $1234 = 2 \cdot 617$.

$$4567 \equiv 3 \pmod{4}, 617 \equiv 1 \pmod{4}.$$

$$\begin{aligned} \therefore (1234/4567) &= (2/4567)(617/4567) \\ &= (1)(4567/617) \quad \text{as } 4567 \equiv 7 \pmod{8} \\ &= (248/617). \end{aligned}$$

$$\begin{aligned} (248/617) &= (2^3 \cdot 31/617) = (2/617)(31/617) \\ &= (1)(31/617) \quad \text{as } 617 \equiv 1 \pmod{8} \\ &= (617/31) = (28/31) \\ &= (4 \cdot 7/31) = (7/31) \\ &= -(31/7) \quad \text{as } 7 \equiv 3 \pmod{4}, 31 \equiv 3 \pmod{4} \\ &= -(3/7) = (7/3) = (1/3) = 1. \end{aligned}$$

$$\therefore (1234/4567) = 1.$$

(e) $(3658/12703)$, $3658 = 2 \cdot 31 \cdot 59$.

$$12703 \equiv 7 \pmod{8}, \therefore (2/12703) = 1.$$

$$\begin{aligned} \therefore (3658/12703) &= (2/12703)(31/12703)(59/12703) \\ &= (31/12703)(59/12703). \end{aligned}$$

$$12703 \equiv 3 \pmod{4}, 31 \equiv 3 \pmod{4}, 59 \equiv 3 \pmod{4}:$$

$$\begin{aligned} &= (12703/31)(12703/59) \\ &= (24/31)(18/59) \\ &= (6/31)(2/59) = -(6/31) \quad \text{as } 59 \equiv 3 \pmod{8} \\ &= -(2/31)(3/31) = -(3/31) \quad \text{as } (2/31) = 1 \\ &= (31/3) = (1/3) = 1. \end{aligned}$$

$$\therefore (3658/12703) = 1.$$

2. Prove that 3 is a quadratic nonresidue of all primes of the form $2^{2n} + 1$, and all primes of the form $2^p - 1$, where p is an odd prime.

Pf. (1) For all n , $4^n \equiv 4 \pmod{12}$.

Clearly true for $n = 1$. Assume true for n . Then $4^{n+1} = 4^n \cdot 4 \equiv 4 \cdot 4 = 16 \equiv 4 \pmod{12}$.

(2) $\therefore 2^{2n} = 4^n \equiv 4 \pmod{12}$.

$\therefore 2^{2n} + 1 \equiv 5 \pmod{12}$.

(3) Let p be a prime of form $2^{2n} + 1$.

$\therefore$ By Th. 9.10 and (2) above, $(3/p) = -1$, and so 3 is a quadratic nonresidue of prime $2^{2n} + 1$.

(4) If p is an odd prime, $p = 2n + 1$, some n .

$$\therefore 2^p - 1 = 2^{2n+1} - 1 = 4^n \cdot 2 - 1.$$

By (1), $4^n \cdot 2 - 1 \equiv 4 \cdot 2 - 1 \pmod{12} \equiv 7 \pmod{12} \equiv -5 \pmod{12}$.

$\therefore 2^p - 1 \equiv -5 \pmod{12}$.

$\therefore$ By Th. 9.10, $(3/(2^p - 1)) = -1$, so 3 is a quadratic nonresidue of prime $2^p - 1$.

□ **3.** Determine whether the following quadratic congruences are solvable:

(a) $x^2 \equiv 219 \pmod{419}$

419 is prime. $\therefore$ Consider $(219/419)$.

$219 = 3 \cdot 73$, $419 \equiv 3 \pmod{4}$, $73 \equiv 1 \pmod{4}$.

$(219/419) = (3/419) \cdot (73/419)$.

$(3/419) = -(419/3) = -(2/3) = -(-1) = 1$.

$(73/419) = (419/73) = (5 \cdot 73 + 54/73) = (54/73) = (2 \cdot 3^3/73) = (2 \cdot 3/73) = (2/73) \cdot (3/73) = 1 \cdot (3/73) = (73/3) = (1/3) = 1$.

$\therefore (219/419) = 1$, so solvable.

(b) $3x^2 + 6x + 5 \equiv 0 \pmod{89}$ [I]

$\gcd(4, 89) = 1$, $\gcd(3, 89) = 1$.

$\therefore$ [I] $\iff 12(3x^2 + 6x + 5) \equiv 0 \pmod{89}$

$\iff 36x^2 + 72x + 60 \equiv 0 \pmod{89}$

$$\iff (6x + 6)^2 + 24 \equiv 0 \pmod{89} \iff (6x + 6)^2 \equiv 65 \pmod{89}.$$

Let $y = 6x + 6$. $\therefore$ [I] $\iff y^2 \equiv 65 \pmod{89}$.

$\therefore$ Consider $(65/89) = (13/89)(5/89)$.

$13 \equiv 1 \pmod{4}$, $5 \equiv 1 \pmod{4}$, $89 \equiv 1 \pmod{89}$ (i.e. $89 \equiv 1 \pmod{4}$).

$(13/89) = (89/13) = (11/13) = (13/11)$, [since $13 \equiv 1 \pmod{4}$] $= (2/11) = -1$ as $11 \equiv 3 \pmod{8}$.

$(5/89) = (89/5) = (4/5) = (2^2/5) = 1$.

$\therefore (65/89) = -1$.

$\therefore$ [I] is not solvable.

(c) $2x^2 + 5x - 9 \equiv 0 \pmod{101}$ [I]

As in (a), let $y = 2ax + b$, $d = b^2 - 4ac$, so $[I] \iff y^2 \equiv d = 97 \pmod{101}$.
 $\therefore$ Consider $(97/101)$. 97 is prime, $97 \equiv 1 \pmod{4}$, $101 \equiv 1 \pmod{4}$.

$$\therefore (97/101) = (101/97) = (4/97) = (2^2/97) = 1.$$

$\therefore [I]$ is solvable.

4. Verify that if p is an odd prime, then

$$(-2/p) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{8} \text{ or } p \equiv 3 \pmod{8} \\ -1 & \text{if } p \equiv 5 \pmod{8} \text{ or } p \equiv 7 \pmod{8} \end{cases}$$

$$\text{Pf. } (-2/p) = (-1/p)(2/p).$$

$$(-1/p) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases} \quad (\text{Corollary to Th. 9.2, p. 187})$$

$$(2/p) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{8} \text{ or } p \equiv 7 \pmod{8} \\ -1 & \text{if } p \equiv 3 \pmod{8} \text{ or } p \equiv 5 \pmod{8} \end{cases} \quad (\text{Th. 9.6})$$

If $p \equiv 1 \pmod{8}$, then $p \equiv 1 \pmod{4}$, so $(-2/p) = (-1/p)(2/p) = 1 \cdot 1 = 1$. [I]

If $p \equiv 3 \pmod{8}$, then $p \equiv 3 \pmod{4}$, so $(-2/p) = (-1/p)(2/p) = -1 \cdot -1 = 1$. [II]

If $p \equiv 5 \pmod{8}$, then $p \equiv 5 \pmod{4} \equiv 1 \pmod{4}$. $(-2/p) = (-1/p)(2/p) = 1 \cdot -1 = -1$. [III] If $p \equiv 7 \pmod{8}$, then $p \equiv 7 \pmod{4} \equiv 3 \pmod{4}$. $(-2/p) = (-1/p)(2/p) = -1 \cdot 1 = -1$. [IV]

$$\therefore (-2/p) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{8} \text{ or } p \equiv 3 \pmod{8} & [[I], [II]] \\ -1 & \text{if } p \equiv 5 \pmod{8} \text{ or } p \equiv 7 \pmod{8} & [[III], [IV]] \end{cases} \quad \square$$

5. (a) Prove that if $p > 3$ is an odd prime, then

$$(-3/p) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{6} \\ -1 & \text{if } p \equiv 5 \pmod{6} \end{cases}$$

$$\text{Pf. } (-3/p) = (-1/p) \cdot (3/p).$$

(1) Suppose $p \equiv 1 \pmod{6}$. Then $p - 1 = 6k$, some k .

(a) If k is even, $k = 2k'$, so $p - 1 = 12k'$, or $p \equiv 1 \pmod{12}$.

$$\therefore (-1/p) = (-1)^{\frac{p-1}{2}} = (-1)^{6k'} = 1. \quad (3/p) = 1 \text{ by Th. 9.10.}$$

$$\therefore (-3/p) = 1 \cdot 1 = 1.$$

(b) If k is odd, $k = 2k' + 1$, so $p - 1 = 6(2k' + 1) = 6 + 12k'$, or $p \equiv 7 \pmod{12} \iff p \equiv -5 \pmod{12}$.

$$\therefore (-1/p) = (-1)^{\frac{p-1}{2}} = (-1)^{3+6k'} = -1. \quad (3/p) = -1 \text{ by Th. 9.10.}$$

$$\therefore (-3/p) = -1 \cdot -1 = 1.$$

$\therefore p \equiv 1 \pmod{6} \Rightarrow (-3/p) = 1$. (2) Suppose $p \equiv 5 \pmod{6}$. Then $p - 5 = 6k$, some k .

(a) If k is even, $k = 2k'$, some k' , so $p - 5 = 12k'$, or $p \equiv 5 \pmod{12}$.

$\therefore (-1/p) = (-1)^{\frac{p-1}{2}} = (-1)^{4+12k'/2} \dots$ wait, $\frac{p-1}{2} = \frac{4+12k'}{2} = 2 + 6k'$.

$(-1/p) = (-1)^{2+6k'} = 1$. $(3/p) = -1$ by Th. 9.10.

$\therefore (-3/p) = 1 \cdot -1 = -1$.

(b) If k is odd, $k = 2k' + 1$, so $p - 5 = 12k' + 6$, or $p \equiv 11 \pmod{12} \iff p \equiv -1 \pmod{12}$.

$(-1/p) = (-1)^{\frac{p-1}{2}} = (-1)^{5+6k'} = -1$. $(3/p) = 1$ by Th. 9.10.

$\therefore (-3/p) = -1 \cdot 1 = -1$.

$\therefore p \equiv 5 \pmod{6} \Rightarrow (-3/p) = -1$. $\square$

Note: $p \not\equiv 3 \pmod{6}$, for if so, then $p - 3 = k \cdot 3 \cdot 2$, so $3 \mid p - 3 \Rightarrow 3 \mid p$. Similarly for Th. 9.10.

(b) Using part (a), show that there are infinitely many primes of the form $6K + 1$.

Pf. Assume there are a finite number of primes of form $6k + 1$, say $p_1, \dots, p_r$.

Consider $N = (2p_1p_2 \cdots p_r)^2 + 3$.

$3 \nmid N$, for if $3 \mid N$, then $3 \mid (2p_1 \cdots p_r)^2$, so 3 must be one of p_i , but 3 is not of form $6k + 1$.

$\therefore$ There must be some odd prime divisor, $p > 3$, of N . And $p \neq p_i$, for if $p = p_i$, some i , then $p \mid N \Rightarrow p \mid 3$, a contradiction. $\therefore N \equiv 0 \pmod{p}$, or equivalently, $(2p_1p_2 \cdots p_r)^2 \equiv -3 \pmod{p}$.

$\therefore (-3/p) = 1$

$\therefore$ by (a), $p \equiv 1 \pmod{6}$, for if $p \equiv 5 \pmod{6}$, then $(-3/p) = -1$, and $p \not\equiv 3 \pmod{6}$ as shown above.

$\therefore p$ is of form $6k + 1$, contradicting $p \neq p_i$.

$\therefore$ infinitely many primes of form $6k + 1$. $\square$

6. Use Theorem 9.2 and problems 4 and 5 to determine which primes can divide integers of the forms $n^2 + 1$, $n^2 + 2$, or $n^2 + 3$ for some value of n .

(a) $p \mid n^2 + 1 \iff n^2 \equiv -1 \pmod{p}$

For p odd, $(-1/p) = (-1)^{\frac{p-1}{2}}$, so $(-1/p) = 1 \iff \frac{p-1}{2}$ is even.

$\therefore p > 3$, $p = 2k + 1$, so $\frac{(2k+1)-1}{2} = k$ is even, so $k = 2k'$, $\therefore p = 2(2k') + 1 = 4k' + 1$. $\therefore$ for p odd, $p \equiv 1 \pmod{4}$.

If $p = 2$, $n^2 + 1$ must be even.

$$\therefore p \mid n^2 + 1 \begin{cases} \text{if } n \text{ is odd, } & p = 2 \text{ or } p \equiv 1 \pmod{4} \\ \text{if } n \text{ is even, } & p \equiv 1 \pmod{4} \end{cases}$$

(b) $p \mid n^2 + 2 \iff n^2 \equiv -2 \pmod{p}$

If n is even, $2 \mid n^2 + 2$. If n is odd, $n^2 + 2$ is odd, $2 \nmid n^2 + 2$.

For p odd, $n^2 \equiv -2 \pmod{p}$ is solvable $\iff (-2/p) = 1$.

By prob. (4) above, $(-2/p) = 1$ if $p \equiv 1 \pmod{8}$ or $p \equiv 3 \pmod{8}$.

$$\therefore p \mid n^2 + 2 \begin{cases} \text{if } n \text{ is even, } p = 2 \text{ or } p \equiv 1 \pmod{8} \text{ or } p \equiv 3 \pmod{8} \\ \text{if } n \text{ is odd, } p \equiv 1 \pmod{8} \text{ or } p \equiv 3 \pmod{8} \end{cases}$$

$$(c) p \mid n^2 + 3 \iff n^2 \equiv -3 \pmod{p}$$

Problem (5) above addresses $p \geq 3$, in which case $(-3/p) = 1$ if $p \equiv 1 \pmod{6}$.

For $p = 2$: if n is even, $n^2 + 3$ is odd, $2 \nmid n^2 + 3$. If n is odd, $n^2 + 3$ is even, so $2 \mid n^2 + 3$.

For $p = 3$: $n^2 + 3 \equiv 0 \pmod{3} \iff n^2 \equiv 0 \pmod{3} \iff 3 \mid n$.

$$\therefore p \mid n^2 + 3 \begin{cases} \text{if } n \text{ is even and } p \equiv 1 \pmod{6}; \\ \quad \text{if also } 3 \mid n, p = 3 \\ \text{if } n \text{ is odd, } p = 2 \text{ or } p \equiv 1 \pmod{6}; \\ \quad \text{if also } 3 \mid n, p = 3 \end{cases}$$

7. Prove there exist infinitely many primes of form $8k + 3$.

Pf. Use prob. (4) above since it has conditions for $(-2/p)$ using $\pmod{8}$.

Assume finitely many primes of form $8k + 3$, which is odd, say $p_1, p_2, \dots, p_r$.

Consider $N = (p_1 p_2 \cdots p_r)^2 + 2$ (N is odd). N must contain an odd prime divisor p s.t. $p \neq p_i$, for if $p = p_i$, some i , then $p \mid N$ and $p \mid (p_i \cdots p_r)^2 \Rightarrow p \mid 2$.

$$\therefore N \equiv 0 \pmod{p} \text{ or } (p_1 \cdots p_r)^2 \equiv -2 \pmod{p}.$$

$$\therefore \text{By prob. (4) above, } p \equiv 1 \pmod{8} \text{ or } p \equiv 3 \pmod{8}.$$

Suppose $N = q_1^{k_1} \cdots q_s^{k_s}$ and all q_i are s.t. $q_i \equiv 1 \pmod{8}$.

$$\therefore N = q_1^{k_1} q_2^{k_2} \cdots q_s^{k_s} \equiv 1 \pmod{8} \quad [1]$$

But $p_i \equiv 3 \pmod{8}$, so $p_i^2 \equiv 9 \equiv 1 \pmod{8}$.

$$\therefore p_1^2 \cdots p_r^2 \equiv 1 \pmod{8}$$

$$\therefore (p_1 \cdots p_r)^2 + 2 \equiv 3 \pmod{8}$$

$$\therefore N \equiv 3 \pmod{8}, \text{ a contradiction to [1].}$$

$\therefore$ All q_i can't be s.t. $q_i \equiv 1 \pmod{8}$, so there must be some odd prime divisor $q_i = p$ of N s.t. $p \equiv 3 \pmod{8}$. And this contradicts p_i above being finite. $\square$ **8.** Find a prime number p that is simultaneously expressible in the forms $x^2 + y^2$, $u^2 + 2v^2$, and $r^2 + 3s^2$.

If $x^2 + y^2 = p$, then $x^2 \equiv -y^2 \pmod{p}$, or $\frac{x^2}{y^2} \equiv -1 \pmod{p}$.

Similarly, $\frac{u^2}{v^2} \equiv -2 \pmod{p}$, $\frac{r^2}{s^2} \equiv -3 \pmod{p}$, where $\left(\frac{x}{y}\right)^2$, $\left(\frac{u}{v}\right)^2$, and $\left(\frac{r}{s}\right)^2$ are integers.

$\therefore$ Look at $(-1/p) = (-2/p) = (-3/p) = 1$ as a minimum condition.

$$(-1/p) = 1, \quad \text{if } p \equiv 1 \pmod{4}$$

$$(-2/p) = 1, \quad \text{if } p \equiv 1 \pmod{8} \text{ or } p \equiv 3 \pmod{8} \quad [\text{prob. 4}]$$

$$(-3/p) = 1, \quad \text{if } p \equiv 1 \pmod{6} \quad [\text{prob. 5}]$$

$\therefore$ if $p \equiv 1 \pmod{8}$, then $p \equiv 1 \pmod{4}$.

If $p \equiv 1 \pmod{24}$, then $p \equiv 1 \pmod{4}$, $p \equiv 1 \pmod{8}$, and $p \equiv 1 \pmod{6}$. $\therefore$

Consider $p = 1 + 24k$, for $k = 1, 2, 3, \dots$

$\therefore$ Look at 25, 49, 73, $\dots$

73 is prime and $73 \equiv 1 \pmod{4}$, $73 \equiv 1 \pmod{8}$, and $73 \equiv 1 \pmod{6}$.

To clean up:

$$8^2 + 3^2 = 73, \quad 1^2 + 2(6)^2 = 73, \quad 5^2 + 3(4)^2 = 73.$$

These were obtained by trial and error, but could have chosen a different prime s.t. $p = 1 + 24k$ and then chosen x, y s.t. $\left(\frac{x}{y}\right)^2$ is an integer, etc.

9. If p and q are odd primes satisfying $p = q + 4a$ for some a , establish that $(a/p) = (a/q)$, and in particular, that $(6/37) = (6/13)$.

Pf. Since $p = q + 4a$, $(p/q) = (q + 4a/q)$.

But $q + 4a \equiv 4a \pmod{q}$, so $(q + 4a/q) = (4a/q) = (2^2 a/q) = (a/q)$.

$\therefore (p/q) = (a/q)$ [1] Similarly, $q = p - 4a$, so $(q/p) = (p - 4a/p) = (-a/p)$.

$\therefore (q/p) = (-a/p) = (-1/p)(a/p)$ [2]

If $p \equiv 1 \pmod{4}$, then $(-1/p) = 1$, [2] becomes $(q/p) = (a/p)$ [2'].

But by Corollary 2 to Th. 9.9 (p. 198), $(p/q) = (q/p)$.

$\therefore$ By [1] and [2'], $(a/q) = (a/p)$.

If $p \equiv 3 \pmod{4}$, then $(-1/p) = -1$, [2] becomes $(q/p) = -(a/p)$ [2''].

Note that $p \equiv 4a + q \pmod{4} \Rightarrow p \equiv q \pmod{4}$.

$\therefore q \equiv 3 \pmod{4}$.

$\therefore$ By Corollary 2 to Th. 9.9, $(p/q) = -(q/p)$.

$\therefore$ By [1] and [2''], $(a/q) = (p/q) = -(q/p) = -(-(a/p))$.

$\therefore (a/q) = (a/p)$. $\square$ **10.** Establish each of the following assertions:

(a) $(5/p) = 1 \iff p \equiv 1, 9, 11, \text{ or } 19 \pmod{20}$

Pf. By definition of $(5/p)$, p is an odd prime.

(1) If $p \equiv 1, 9, 11, \text{ or } 19 \pmod{20}$, then $p \equiv 1, 9, 11, \text{ or } 19 \pmod{5} \Rightarrow p \equiv 1 \text{ or } 4 \pmod{5}$.

Since $5 \equiv 1 \pmod{4}$, then $(5/p) = (p/5)$.

$\therefore (5/p) = (p/5) = (1/5) \text{ or } (4/5) = (1/5)$, and $(1/5) = 1$.

$\therefore (5/p) = 1$.

(2) Suppose $(5/p) = 1$. Since $5 \equiv 1 \pmod{4}$, then $(5/p) = (p/5) = 1$.

$\therefore 1 \equiv p^{\frac{5-1}{2}} \pmod{5}$, or $1 \equiv p^2 \pmod{5}$.

$\therefore p \equiv 1 \text{ or } 4 \pmod{5}$.

Also, general for any odd number, $p \equiv 1 \text{ or } 3 \pmod{4}$.

$\therefore p \equiv 1 \pmod{5} \text{ or } p \equiv 4 \pmod{5}$ and $p \equiv 1 \pmod{4} \text{ or } p \equiv 3 \pmod{4}$. $\therefore 4p \equiv 4 \pmod{20} \text{ or } 4p \equiv 16 \pmod{20}$, and $5p \equiv 5 \pmod{20} \text{ or } 5p \equiv 15 \pmod{20}$.

Subtracting: $p \equiv 1 \text{ or } -11 \pmod{20}$, or $p \equiv 11 \text{ or } -1 \pmod{20}$.

$\Rightarrow p \equiv 1, 9, 11, \text{ or } 19 \pmod{20}$. $\square$

(b) $(6/p) = 1 \iff p \equiv 1, 5, 19, \text{ or } 23 \pmod{24}$

Pf. (1) Suppose $p \equiv 1, 5, 19, \text{ or } 23 \pmod{24}$.

Let $p \equiv 1 \pmod{24}$:

$\therefore p \equiv 1 \pmod{12} \Rightarrow (3/p) = 1$ (Th. 9.10), and $p \equiv 1 \pmod{8} \Rightarrow (2/p) = 1$ (Th. 9.6).

$\therefore (6/p) = (3/p)(2/p) = 1$.

Let $p \equiv 5 \pmod{24}$:

$\therefore p \equiv 5 \pmod{12} \Rightarrow (3/p) = -1$ (Th. 9.10), and $p \equiv 5 \pmod{8} \Rightarrow (2/p) = -1$ (Th. 9.6).

$\therefore (6/p) = (3/p)(2/p) = (-1)(-1) = 1$. Let $p \equiv 19 \pmod{24}$:

$\therefore p \equiv 19 \equiv 7 \equiv -5 \pmod{12} \Rightarrow (3/p) = -1$ (Th. 9.10), and $p \equiv 19 \equiv 3 \pmod{8} \Rightarrow (2/p) = -1$ (Th. 9.6).

$\therefore (6/p) = (3/p)(2/p) = (-1)(-1) = 1$.

Let $p \equiv 23 \pmod{24}$:

$\therefore p \equiv 23 \equiv 11 \equiv -1 \pmod{12} \Rightarrow (3/p) = 1$ (Th. 9.10), and $p \equiv 23 \equiv 7 \pmod{8} \Rightarrow (2/p) = 1$ (Th. 9.6).

$\therefore (6/p) = (3/p)(2/p) = 1$.

(2) Suppose $(6/p) = 1$.

$\therefore (3/p)(2/p) = 1 \Rightarrow$

$$(3/p) = 1 \text{ and } (2/p) = 1, \quad \text{or} \quad (3/p) = -1 \text{ and } (2/p) = -1.$$

(a) If $(3/p) = 1$, then $p \equiv \pm 1 \pmod{12}$, by Th. 9.10, for $p \not\equiv 3 \text{ or } 9 \pmod{12}$, since then $3 \mid p$.

Also, $(2/p) = 1 \Rightarrow p \equiv \pm 1 \pmod{8}$, by Th. 9.6.

(i) $p \equiv 1 \pmod{12}$ and $p \equiv 1 \pmod{8} \Rightarrow$

$$(1) \quad 2p \equiv 2 \pmod{24}, \quad 3p \equiv 3 \pmod{24} \} \Rightarrow p \equiv 1 \pmod{24} \quad \text{by subtracting.}$$

$$(2) \quad p \equiv 1 \pmod{12} \text{ and } p \equiv -1 \pmod{8} \Rightarrow 2p \equiv 2 \pmod{24}, \quad 3p \equiv -3 \pmod{24} \Rightarrow p \equiv -5 \Rightarrow p \equiv 19 \pmod{24}$$

$$(3) \quad p \equiv -1 \pmod{12} \text{ and } p \equiv 1 \pmod{8} \Rightarrow 2p \equiv -2 \pmod{24}, \quad 3p \equiv 3 \pmod{24} \Rightarrow p \equiv 5 \pmod{24}$$

$$(4) \quad p \equiv -1 \pmod{12} \text{ and } p \equiv -1 \pmod{8} \Rightarrow 2p \equiv -2 \pmod{24}, \quad 3p \equiv -3 \pmod{24} \Rightarrow p \equiv -1 \Rightarrow p \equiv 23 \pmod{24}$$

$\therefore (6/p) = 1 \Rightarrow p \equiv 1, 5, 19, \text{ or } 23 \pmod{24}$. $\square$

(c) $(7/p) = 1 \iff p \equiv 1, 3, 9, 19, 25, \text{ or } 27 \pmod{28}$

Pf. (1) Suppose $(7/p) = 1$.

(a) If $p \equiv 1 \pmod{4}$, then $(7/p) = (p/7)$.

$\therefore 1 \equiv p^{\frac{7-1}{2}} = p^3 \pmod{7}$.

$\therefore$ running through $p \equiv 1, 2, 3, 4, 5, 6$ and looking at p^3 , we get $p \equiv 1, 2$, or $4 \pmod{7}$.

$\therefore p \equiv 1 \pmod{4}$ and $p \equiv 1, 2$, or $4 \pmod{7}$.

(i) $p \equiv 1 \pmod{4}$, $p \equiv 1 \pmod{7}$: $7p \equiv 7 \pmod{28}$, $4p \equiv 4 \pmod{28} \Rightarrow 8p \equiv 8 \Rightarrow p \equiv 1$

(ii) $p \equiv 1 \pmod{4}$, $p \equiv 2 \pmod{7}$: $7p \equiv 7 \pmod{28}$, $4p \equiv 8 \pmod{28} \Rightarrow 8p \equiv 16 \Rightarrow p \equiv 9$

(iii) $p \equiv 1 \pmod{4}$, $p \equiv 4 \pmod{7}$: $7p \equiv 7 \pmod{28}$, $4p \equiv 16 \pmod{28} \Rightarrow 8p \equiv 32 \Rightarrow p \equiv$

(b) If $p \equiv 3 \pmod{4}$, then $(7/p) = -(p/7)$ since $7 \equiv 3 \pmod{4}$.

$\therefore -1 \equiv p^{\frac{7-1}{2}} = p^3 \pmod{7}$.

$\therefore p \equiv 3, 5$, or $6 \pmod{7} \Rightarrow 4p \equiv 12, 20$, or $24 \pmod{28} \Rightarrow 8p \equiv 24, 40$, or $48 \pmod{28}$.

$p \equiv 3 \pmod{4} \Rightarrow 7p \equiv 21 \pmod{28}$.

(1) $7p \equiv 21 \pmod{28}$, $8p \equiv 24 \pmod{28} \Rightarrow p \equiv 3 \pmod{28}$.

(2) $7p \equiv 21 \pmod{28}$, $8p \equiv 40 \pmod{28} \Rightarrow p \equiv 19 \pmod{28}$.

(3) $7p \equiv 21 \pmod{28}$, $8p \equiv 48 \pmod{28} \Rightarrow p \equiv 27 \pmod{28}$.

$\therefore (7/p) = 1 \Rightarrow p \equiv 1, 3, 9, 19, 25$, or $27 \pmod{28}$.

(2) Suppose $p \equiv 1, 3, 9, 19, 25$, or $27 \pmod{28}$.

By definition of $(7/p)$, p is an odd prime. Also, note $7 \equiv 3 \pmod{4}$.

(a) If $p \equiv 3, 19$, or $27 \pmod{28}$, then $p \equiv 3, 19$, or $27 \pmod{4}$, so $p \equiv 3 \pmod{4}$.

$\therefore (7/p) = -(p/7)$ (corollary 1, p. 198).

Also, $p \equiv 3, 19$, or $27 \pmod{7} \Rightarrow p \equiv 3, 5, 6 \pmod{7}$.

$\therefore (p/7) = (3/7)$, $(5/7)$, or $(6/7)$. $(3/7) = -(7/3) = -(1/3) = -1$

$(5/7) = (7/5) = (2/5) = -1$

$(6/7) = (3/7)(2/7) = (-1)(1) = -1$

$\therefore (p/7) = -1$.

$\therefore (7/p) = -(p/7) = 1$.

(b) If $p \equiv 1, 9$, or $25 \pmod{28}$, then $p \equiv 1, 9$, or $25 \pmod{4}$, so $p \equiv 1 \pmod{4}$.

$\therefore (7/p) = (p/7)$ (corollary 1, p. 198).

Also, $p \equiv 1, 9, 25 \pmod{7}$.

$\therefore (p/7) = (1/7)$, $(9/7)$, $(25/7) = 1$, $(2/7)$, $(4/7) = 1$, 1 , 1 .

$\therefore (7/p) = (p/7) = 1$.

$\therefore p \equiv 1, 3, 9, 19, 25, 27 \pmod{28} \Rightarrow (7/p) = 1$. $\square$ **11.** Prove that there are infinitely many primes of the form $5K - 1$.

Pf. Assume finitely many primes of form $5k - 1$. Call them $p_1, p_2, \dots, p_r$, where $p_r > p_i$, $1 \leq i \leq r$.

Consider the integer $M = 5(n!)^2 - 1$. For $n \geq 1$, M is odd, since $n!$ is even as it contains 2. $\therefore M$ has an odd prime divisor. Note that any odd prime divisor p of M must be s.t. $p > n$; for if $p \leq n$, then $p \mid n!$, so $p \mid M$ and $p \mid 5(n!)^2 \Rightarrow p \mid 1$, a contradiction.

$\therefore$ For $N = 5(p_r!)^2 - 1$, let p be any odd prime divisor. $\therefore p > p_r$ and p cannot be of form $5k - 1$.

$$5(p_r!)^2 \equiv 1 \pmod{p} \iff 25(p_r!)^2 \equiv 5 \pmod{p} \quad [1]$$

Since $p > p_r \geq 19 = 5k - 1$ for $k = 4$. $\therefore \gcd(5, p) = 1$.

$\therefore [1] \Rightarrow (5/p) = 1$. But from prob. 10(a) above, $p \equiv 1, 9, 11$, or $19 \pmod{20} \Rightarrow p \equiv 1, 9, 11, 19 \pmod{5} \Rightarrow p \equiv 1$ or $4 \pmod{5}$.

If $p \equiv 4 \pmod{5}$, then $p \equiv -1 \pmod{5} \Rightarrow p = 5k - 1$, some k . This can't be since $p > p_r$ and p_r is the largest prime of form $5k - 1$.

$\therefore p \equiv 1 \pmod{5}$, or $p = 5k + 1$.

Since p is any odd prime divisor of N , then

$$N = (5k_1 + 1)^{n_1} (5k_2 + 1)^{n_2} \cdots (5k_s + 1)^{n_s}.$$

But $(5k_i + 1)^{n_i}$ is of form $5k + 1$, so N is of form $5k'' + 1$.

But this contradicts $N = 5(p_r!)^2 - 1$ of form $5k - 1$. (If $5k - 1 = 5k'' + 1$, then $5(k - k'') = 2 \Rightarrow 5 \mid 2$.)

$\therefore$ Assumption of finite number of primes of form $5k - 1$ is false. $\square$ **12.** Verify the following:

(a) The prime divisors $p \neq 3$ of the integer $n^2 - n + 1$ are of form $6k + 1$.

Pf. First note $n^2 - n + 1$ is odd for all $n \geq 1$. For if n is odd, n^2 is odd, so $n^2 - n$ is even, so $n^2 - n + 1$ is odd. If n is even, n^2 is even, $n^2 - n$ is even, so $n^2 - n + 1$ is odd.

$\therefore$ prime divisors p of $n^2 - n + 1 \neq 2$. With assumption $p \neq 3$, then $p > 3$.

If $p \mid n^2 - n + 1$, then $p \mid 4n^2 - 4n + 4$. Since $(2n - 1)^2 = 4n^2 - 4n + 1$, $\therefore p \mid [(2n - 1)^2 + 3]$.

$\therefore (2n - 1)^2 \equiv -3 \pmod{p} \Rightarrow (-3/p) = 1$.

$\therefore p \equiv 0, 1, 2, 3, 4, 5 \pmod{6}$. If $p \equiv 0, 2, 4 \pmod{6}$, then $p \equiv 0, 2, 4 \pmod{2} \Rightarrow 2 \mid p$, which can't be since $p > 3$. If $p \equiv 3 \pmod{6}$, then $p \equiv 3 \pmod{3} \Rightarrow 3 \mid p$, which can't be since $p > 3$. $\therefore p \equiv 1$ or $5 \pmod{6}$. By prob. 5(a) above, $p \equiv 1 \pmod{6}$; for $p \equiv 5 \Rightarrow (-3/p) = -1$.

$\therefore p \equiv 1 \pmod{6} \Rightarrow p = 1 + 6k$, some k . $\square$

(b) The prime divisors $p \neq 5$ of the integer $n^2 + n - 1$ are of the form $10k + 1$ or $10k + 9$.

Pf. If $p \mid n^2 + n - 1$, then $p \mid 4n^2 + 4n - 4$. Since $(2n + 1)^2 - 5 = 4n^2 + 4n - 4$,

$\therefore p \mid n^2 + n - 1 \Rightarrow p \mid (2n + 1)^2 - 5 \Rightarrow (2n + 1)^2 \equiv 5 \pmod{p}$.

If $p \neq 5$, then $\gcd(p, 5) = 1$, so $(5/p)$ is defined.

$\therefore$ If $p \neq 5$, $p \mid n^2 + n - 1 \Rightarrow (5/p) = 1$.

By Prob. 10(a), $p \equiv 1, 9, 11, 19 \pmod{20} \Rightarrow p \equiv 1, 9, 11, 19 \pmod{10} \Rightarrow p \equiv 1$ or $9 \pmod{10}$

$\Rightarrow p = 1 + 10k$ or $p = 9 + 10k$, some k . $\square$ (c) The prime divisors p of the integer $2n(n+1)+1$ are of the form $p \equiv 1 \pmod{4}$.

Pf. $2n(n+1)+1 = 2n^2 + 2n + 1$.

If $p \mid 2n(n+1)+1$, then $p \mid 4n^2 + 4n + 2 \Rightarrow p \mid (2n+1)^2 + 1 \Rightarrow (2n+1)^2 \equiv -1 \pmod{p}$.

$\therefore p \mid 2n(n+1)+1 \Rightarrow (-1/p) = 1 \Rightarrow p \equiv 1 \pmod{4}$, by corollary on p. 187. $\square$

(d) The prime divisors p of the integer $3n(n+1)+1$ are of the form $p \equiv 1 \pmod{6}$.

Pf. $3n(n+1)+1 = 3n^2 + 3n + 1$.

$\therefore p \mid 3n(n+1)+1 \Rightarrow p \mid 36n^2 + 36n + 12 \Rightarrow p \mid (6n+3)^2 + 3$

$\Rightarrow (6n+3)^2 \equiv -3 \pmod{p}$ [1] Note that if n is even or odd, $n(n+1)$ is even.

$\therefore 3n(n+1)+1$ is odd, so $p \neq 2$. If $p = 3$, then $p \mid 3n(n+1)+1 \Rightarrow p \mid 1$. $\therefore p \neq 3$.

$\therefore p > 3$, so $\gcd(-3, p) = 1$, so [1] $\Rightarrow (-3/p) = 1$.

By prob. 5(a), $p \equiv 1 \pmod{6}$. $\square$

13.(a) Show that if p is a prime divisor of $839 = 38^2 - 5 \cdot 11^2$, then $(5/p) = 1$. Use this fact to conclude that 839 is a prime number.

Pf. (1) If $p \mid 38^2 - 5 \cdot 11^2$, then

$$38^2 - 5 \cdot 11^2 \equiv 0 \pmod{p} \iff 38^2 \equiv 5 \cdot 11^2 \pmod{p}.$$

$\therefore 38$ is a solution to $x^2 \equiv 5 \cdot 11^2 \pmod{p}$, and so $(5 \cdot 11^2/p) = 1 \Rightarrow (5/p)(11^2/p) = 1 \Rightarrow (5/p) = 1$.

(2) Since $29^2 = 841 > 839$, only need to consider primes < 29 (discussion p. 45). By prob. 10(a), $(5/p) = 1 \Rightarrow p \equiv 1, 9, 11, 19 \pmod{20} \Rightarrow p \equiv 1, 9, 11, 19 \pmod{10} \Rightarrow p \equiv 1$ or $9 \pmod{10} \Rightarrow p = 11$ or 19 .

If $19 \mid 38^2 - 5 \cdot 11^2$, then $19 \mid 5 \cdot 11^2$. If $11 \mid 38^2 - 5 \cdot 11^2$, then $11 \mid 38^2$. Both of these are false, so there is no prime p s.t. $(5/p) = 1$.

$\therefore (5/p) \neq 1$, assumption that 839 is divisible by a prime is false $\Rightarrow$ 839 is prime.

$\square$

(b) Prove that both $397 = 20^2 - 3$ and $733 = 29^2 - 3 \cdot 6^2$ are primes.

(1) $397 = 20^2 - 3$.

Assume there is a prime divisor p s.t. $p \mid 397$. Since $20^2 = 400$, only consider $p < 20$.

$\therefore 20^2 \equiv 3 \pmod{p} \Rightarrow (3/p) = 1$. By Th. 9.10, $p \equiv \pm 1 \pmod{12}$. $\therefore p = 11$ or 13 .

But $11 \nmid 397$ and $13 \nmid 397$, so there is no p s.t. $p \mid 397 \Rightarrow 397$ is prime.

(2) $733 = 29^2 - 3 \cdot 6^2$.

Assume there is a prime p s.t. $p \mid 733$. $28^2 = 784$, so just consider $p < 28$.

$29^2 \equiv 3 \cdot 6^2 \pmod{p} \Rightarrow (3 \cdot 6^2/p) = 1 \Rightarrow (3/p) = 1$.

$\therefore$ By Th. 9.10, $p \equiv \pm 1 \pmod{12}$. $\therefore p = 11, 13$, or 23 . But $11 \nmid 733$, $13 \nmid 733$, and $23 \nmid 733$. $\therefore$ no prime p s.t. $p \mid 733 \Rightarrow 733$ is prime.

14. Solve the quadratic congruence $x^2 \equiv 11 \pmod{35}$.

Since $35 = 7 \cdot 5$, x is a solution $\iff x$ is a solution to: $x^2 \equiv 11 \pmod{5}$ and $x^2 \equiv 11 \pmod{7}$.

$$\therefore x^2 \equiv 11 \pmod{5} \iff x^2 \equiv 1 \pmod{5} \iff x \equiv \pm 1 \pmod{5}.$$

$$x^2 \equiv 11 \pmod{7} \iff x^2 \equiv 4 \pmod{7} \iff x \equiv \pm 2 \pmod{7}.$$

$$\therefore \text{(a) } x \equiv 1 \pmod{5} \text{ and } x \equiv 2 \pmod{7}. \quad \text{(b) } x \equiv 1 \pmod{5} \text{ and } x \equiv -2 \pmod{7}.$$

$$\text{(c) } x \equiv -1 \pmod{5} \text{ and } x \equiv 2 \pmod{7}.$$

$$\text{(d) } x \equiv -1 \pmod{5} \text{ and } x \equiv -2 \pmod{7}.$$

For all of these, $n = 5 \cdot 7$, $N_1 = 7$, $N_2 = 5$.

$$\therefore 7x_1 \equiv 1 \pmod{5} : \quad 2x_1 = 1, \quad 6x_1 = 3 \Rightarrow x_1 = 3.$$

$$5x_2 \equiv 1 \pmod{7} : \quad 15x_2 \equiv 3 \Rightarrow x_2 = 3.$$

$$\text{(a) } x \equiv 1 \pmod{5} \text{ and } x \equiv 2 \pmod{7}:$$

$$\therefore x = 1 \cdot 7 \cdot 3 + 2 \cdot 5 \cdot 3 = 21 + 30 = 51 \equiv \underline{16} \pmod{35}.$$

$$\text{(b) } x \equiv 1 \pmod{5} \text{ and } x \equiv -2 \pmod{7}:$$

$$\therefore x = 1 \cdot 7 \cdot 3 + (-2) \cdot 5 \cdot 3 = 21 - 30 = -9 \equiv \underline{26} \pmod{35}.$$

$$\text{(c) } x \equiv -1 \pmod{5} \text{ and } x \equiv 2 \pmod{7}:$$

$$\therefore x = (-1) \cdot 7 \cdot 3 + 2 \cdot 5 \cdot 3 = -21 + 30 = 9 \equiv \underline{9} \pmod{35}.$$

$$\text{(d) } x \equiv -1 \pmod{5} \text{ and } x \equiv -2 \pmod{7} : \therefore x = (-1) \cdot 7 \cdot 3 + (-2) \cdot 5 \cdot 3 = -21 - 30 = -51 \equiv \underline{19} \pmod{35}.$$

$$\therefore X \equiv 9, 16, 19, \text{ or } 26 \pmod{35}.$$

15. Establish that 7 is a primitive root of any prime of the form $2^{4n} + 1$.

Pf. For $n = 0$, $2^{4n} + 1 = 2$, $\phi(2) = 1$, $7^1 = 7$, and $7 \equiv 1 \pmod{2}$, so you could say 7 (and every odd integer) is a primitive root of 2.

So assume $n > 0$. $\therefore 2^{4n} + 1$ is odd.

By prob. 7 of section 9.1 (p. 184), every quadratic non-residue of prime $p = 2^k + 1$ is a primitive root of p .

$2^{4n} + 1$ is of form $2^k + 1$, so just need to show 7 is a quadratic nonresidue of prime $p = 2^{4n} + 1$, i.e., $(7/p) = -1$.

Note that for $n = 3k$, $2^{4n} + 1$ is not prime, for $2^{4n} + 1 = 2^{12k} + 1 = (2^{4k})^3 + 1$ which can be factored to $(2^{4k} + 1)[(2^{4k})^2 - 2^{4k} + 1]$.

$$\therefore 2^{4n} + 1 \text{ can only be prime if } n = 3k + 1 \text{ or } n = 3k + 2, \text{ for } k = 0, 1, 2, \dots$$

Note: $2^{4n} + 1 = 2^{2 \cdot 2n} + 1 = 4^{2n} + 1 \equiv 1 \pmod{4}$. $\therefore (7/p) = (p/7)$, by corollary 2, p. 198.

$\therefore$ need to show $(p/7) = -1$ for $n = 3k + 1$ or $n = 3k + 2$, $k = 0, 1, 2, \dots$, assuming p prime.

$$n = 3k + 1: \text{ Look at } (2^{4(3k+1)} + 1) \pmod{7}, \text{ and note } 8 \equiv 1 \pmod{7}.$$

$$2^{4(3k+1)} + 1 = 2^{3(3k+1)} \cdot 2^{(3k+1)} + 1 = (8^{3k+1})(8^k \cdot 2) + 1 \equiv (1)(1 \cdot 2) + 1 = 3 \pmod{7}.$$

$$\therefore (2^{4(3k+1)} + 1/7) = (3/7) = -(7/3) = -(1/3) = -1.$$

$$n = 3k + 2: 2^{4(3k+2)} + 1 = 2^{3(3k+2)} \cdot 2^{3k+2} + 1 = (8^{3k+2})(8^k \cdot 4) + 1 \equiv (1)(1 \cdot 4) + 1 = 5 \pmod{7}.$$

$\therefore (2^{4(3k+2)} + 1/7) = (5/7) = (7/5) = (2/5) = -1$. $\therefore$ When $2^{4n} + 1$ is prime, $(p/7) = -1$, so $(7/p) = -1$, so 7 is a quadratic nonresidue, and $\therefore$ by prob. 7, section 9.1, 7 is a primitive root of $2^{4n} + 1$. $\square$

16. Let a and $b > 1$ be relatively prime integers, with b odd. If $b = p_1 p_2 \cdots p_r$ is the decomposition of b into odd primes (not necessarily distinct), then the Jacobi symbol (a/b) is defined by

$$(a/b) = (a/p_1)(a/p_2) \cdots (a/p_r),$$

where (a/p_i) is the Legendre symbol.

Evaluate $(21/221)$, $(215/253)$, $(631/1099)$.

(a) $(21/221)$: $221 = 13 \cdot 17$.

$$\begin{aligned} \therefore (21/221) &= (21/13)(21/17) \\ &= (3/13)(7/13)(3/17)(7/17) \\ &= (13/3)(13/7)(17/3)(17/7) \\ &= (1/3)(6/7)(2/3)(3/7) \end{aligned}$$

$$\begin{aligned} &= (1)(3/7)(2/7)(-1)(3/7) \\ &= (-1)(3^2/7)(2/7) = (-1)(1)(1) = -1 \end{aligned}$$

$$\therefore (21/221) = \underline{-1}.$$

(b) $(215/253)$: $253 = 11 \cdot 23$, $215 = 5 \cdot 43$.

$$\begin{aligned} \therefore (215/253) &= (215/11)(215/23) \\ &= (5/11)(43/11)(5/23)(43/23) \\ &= (11/5)(10/11)(23/5)(20/23) \\ &= (1/5)(2/11)(5/11)(3/5)(4/23)(5/23) \\ &= (1)(-1)(1)(5/3)(1)(23/5) \\ &= (-1)(2/3)(3/5) \\ &= (-1)(-1)(5/3) = (2/3) = -1 \end{aligned}$$

$$\therefore (215/253) = \underline{-1}.$$

(c) $(631/1099)$: $1099 = 7 \cdot 157$, 631 is prime.

$$\begin{aligned} \therefore (631/1099) &= (631/7)(631/157) \\ &= (1/7)(3/157) \\ &= (1)(157/3) = (1/3) = 1 \end{aligned}$$

$$\therefore (631/1099) = \underline{1}.$$

17. Under the hypothesis of the previous problem, show that if a is a quadratic residue of b , then $(a/b) = 1$; but the converse is false. *Pf.* (a) Assume $x^2 \equiv a \pmod{b}$ has a solution. Let $b = p_1 p_2 \cdots p_r$, and note $\gcd(a, b) = 1$.

$$\begin{aligned} \therefore x^2 &\equiv a \pmod{p_i}, x^2 \equiv a \pmod{p_2}, \dots, x^2 \equiv a \pmod{p_r}, \text{ and note } \gcd(a, p_i) = 1. \\ \therefore (a/p_i) &= 1, \therefore (a/b) = (a/p_1)(a/p_2) \cdots (a/p_r) = (1)(1) \cdots (1) = 1. \end{aligned}$$

(b) Now assume $(a/b) = 1$ and let $b = p_1 p_2$.

$$\therefore (a/b) = (a/p_1)(a/p_2).$$

$$\therefore \text{it may be that } (a/p_1) = (a/p_2) = -1.$$

$$\therefore \text{There would not be a solution to } x^2 \equiv a \pmod{b}.$$

As a concrete example, $(2/3) = -1$ and $(2/5) = -1$. $\therefore (2/15) = 1$, but $x^2 \equiv 2 \pmod{15}$ can't have a solution. If it did, then so would $x^2 \equiv 2 \pmod{3}$. **18.** Prove that the following properties of the Jacobi symbol hold: If b and b' are positive odd integers and $\gcd(aa', bb') = 1$, then

$$(a) \ a \equiv a' \pmod{b} \text{ implies that } (a/b) = (a'/b).$$

Pf. Let $b = p_1 p_2 \cdots p_r$ be the decomposition of b into odd primes (not necessarily distinct). Then by definition of (a/b) , where (a/p_i) is the Legendre symbol,

$$(a/b) = (a/p_1)(a/p_2) \cdots (a/p_r).$$

From $\gcd(aa', bb') = 1$, we get $\gcd(a, p_i) = 1$ and $\gcd(a', p_i) = 1$.

From $a \equiv a' \pmod{b}$, we get $a \equiv a' \pmod{p_i}$.

$$\therefore \text{from Th. 9.2, } (a/p_i) = (a'/p_i).$$

$$\therefore (a/p_1) \cdots (a/p_r) = (a'/p_1) \cdots (a'/p_r).$$

$$\therefore (a/b) = (a'/b). \quad \square$$

(b) $(aa'/b) = (a/b)(a'/b)$ As in (a) above, let $b = p_1 p_2 \cdots p_r$.

Note that $\gcd(aa', bb') = 1 \Rightarrow \gcd(a, b) = 1$ and $\gcd(a', b) = 1$.

Using Th. 9.2,

$$\begin{aligned} (aa'/b) &= (aa'/p_1)(aa'/p_2) \cdots (aa'/p_r) \\ &= (a/p_1)(a'/p_1)(a/p_2)(a'/p_2) \cdots (a/p_r)(a'/p_r) \\ &= (a/p_1)(a/p_2) \cdots (a/p_r) \cdot (a'/p_1)(a'/p_2) \cdots (a'/p_r) \\ &= (a/b)(a'/b). \quad \square \end{aligned}$$

(c) $(a/bb') = (a/b)(a/b')$

First note $\gcd(aa', bb') = 1 \Rightarrow \gcd(a, b) = 1$ and $\gcd(a, b') = 1$.

As in (a), let $b = p_1 p_2 \cdots p_r$ and let $b' = p'_1 p'_2 \cdots p'_{r'}$. $\therefore bb' = p_1 p_2 \cdots p_r p'_1 p'_2 \cdots p'_{r'}$.

$$\therefore (a/bb') = (a/p_1)(a/p_2) \cdots (a/p_r)(a/p'_1)(a/p'_2) \cdots (a/p'_{r'}) = (a/b)(a/b'). \quad \square$$

(d) $(a^2/b) = (a/b^2) = 1$

From (b), letting $a' = a$:

$$\begin{aligned}(a^2/b) &= (a \cdot a/b) = (a/b)(a/b) \\ &= (a/p_1) \cdots (a/p_r)(a/p_1) \cdots (a/p_r) \\ &= (a/p_1)^2 \cdots (a/p_r)^2 = 1 \cdots 1 = 1.\end{aligned}$$

Similarly, letting $b = b'$ in (c): $(a/b^2) = (a/b)(a/b) = 1$ as above.

(e) $(1/b) = 1$

This follows from (d) letting $a = 1$, so that $1 = (a^2/b) = (1^2/b) = (1/b)$.

(f) $(-1/b) = (-1)^{(b-1)/2}$

If $b = p_1 p_2 \cdots p_r$, then by definition, and using Th. 9.2:

$$\begin{aligned}(-1/b) &= (-1/p_1)(-1/p_2) \cdots (-1/p_r) \\ &= (-1)^{(p_1-1)/2} (-1)^{(p_2-1)/2} \cdots (-1)^{(p_r-1)/2} \quad [1]\end{aligned}$$

Now use the hint: if u and v are odd integers, then $u = 2r + 1$, $v = 2s + 1$, some r, s .

$$\therefore \frac{u-1}{2} = r, \quad \frac{v-1}{2} = s.$$

$$\frac{uv-1}{2} = \frac{(2r+1)(2s+1)-1}{2} = \frac{4rs+2r+2s}{2} = 2rs+r+s.$$

$\therefore r+s \equiv r+s \pmod{2}$, and $r+s \equiv 2rs+r+s \pmod{2}$, so

$$\frac{u-1}{2} + \frac{v-1}{2} \equiv \frac{uv-1}{2} \pmod{2}.$$

$\therefore \lfloor \frac{u-1}{2} + \frac{v-1}{2} \rfloor$ and $\lfloor \frac{uv-1}{2} \rfloor$ must both be odd, or both must be even.

$$\therefore (-1)^{\frac{u-1}{2} + \frac{v-1}{2}} = (-1)^{\frac{uv-1}{2}}.$$

$$\begin{aligned}\therefore [1] \text{ becomes } & (-1)^{\lfloor \frac{p_1-1}{2} \rfloor} (-1)^{\lfloor \frac{p_2-1}{2} \rfloor} \cdots (-1)^{\lfloor \frac{p_r-1}{2} \rfloor} = (-1)^{\lfloor \frac{p_1 p_2 \cdots p_r - 1}{2} \rfloor} \cdots (-1)^{\lfloor \frac{p_r-1}{2} \rfloor} = (-1)^{\lfloor \frac{p_1 p_2 \cdots p_r - 1}{2} \rfloor} = \\ & (-1)^{\frac{b-1}{2}}. \quad \therefore (-1/b) = (-1)^{(b-1)/2}. \quad \square\end{aligned}$$

$$(g) \quad (2/b) = (-1)^{(b^2-1)/8}$$

Let $b = p_1 \cdots p_r$.

By def., $(2/b) = (2/p_1) \cdots (2/p_r)$.

Using corollary to Th. 9.6, p. 191, $(2/p_i) = (-1)^{(p_i^2-1)/8}$.

$$\therefore (2/b) = (-1)^{(p_1^2-1)/8} (-1)^{(p_2^2-1)/8} \cdots (-1)^{(p_r^2-1)/8}.$$

Now use the hint: if u, v are odd integers, then $u = 4r + 1$ or $u = 4r + 3$, some r , and $v = 4s + 1$ or $v = 4s + 3$, some s .

(1) $u = 4r + 1, v = 4s + 1$:

$$u^2 - 1 = 16r^2 + 8r, \quad v^2 - 1 = 16s^2 + 8s.$$

$$\therefore \frac{u^2-1}{8} + \frac{v^2-1}{8} = 2r^2 + r + 2s^2 + s.$$

$$\therefore \frac{u^2-1}{8} + \frac{v^2-1}{8} \equiv r + s \pmod{2} \quad [1]$$

$$uv = 16rs + 4r + 4s + 1.$$

$$\begin{aligned}(uv)^2 &= 16^2r^2s^2 + 64r^2s + 64rs^2 + 16rs \\ &\quad + 64r^2s + 16r^2 + 16rs + 4r \\ &\quad + 64rs^2 + 16rs + 16s^2 + 4s \\ &\quad + 16rs + 4r + 4s + 1\end{aligned}$$

$$\therefore (uv)^2 - 1 = 16^2r^2s^2 + 128r^2s + 128rs^2 + 64rs + 16r^2 + 16s^2 + 8r + 8s.$$

$$\therefore \frac{(uv)^2 - 1}{8} \equiv r + s \pmod{2} \quad [2]$$

$$\therefore [1], [2] \Rightarrow \frac{u^2 - 1}{8} + \frac{v^2 - 1}{8} \equiv \frac{(uv)^2 - 1}{8} \pmod{2}.$$

$$(2) \ u = 4r + 1, v = 4s + 3:$$

$$u^2 - 1 = 16r^2 + 8r, v^2 - 1 = 16s^2 + 24s + 8.$$

$$\therefore \frac{u^2 - 1}{8} + \frac{v^2 - 1}{8} = 2r^2 + r + 2s^2 + 3s + 1.$$

$$\therefore \frac{u^2 - 1}{8} + \frac{v^2 - 1}{8} \equiv r + s + 1 \pmod{2} \quad [1]$$

$$uv = 16rs + 12r + 4s + 3.$$

$$\begin{aligned}(uv)^2 &= 16^2r^2s^2 + (16)(12)r^2s + 64rs^2 + 48rs + (12)(16)r^2s + 144r^2 + 48rs + 36r \\ &\quad + 64rs^2 + 48rs + 16s^2 + 12s + 48rs + 36r + 12s + 9. \quad \therefore (uv)^2 - 1 = 16^2r^2s^2 + (24)(16)r^2s + \\ &\quad 128rs^2 + (4)(48)rs + 144r^2 + 16s^2 + 72r + 24s + 8.\end{aligned}$$

$$\therefore \frac{(uv)^2 - 1}{8} \equiv 9r + 3s + 1 \equiv r + s + 1 \pmod{2} \quad [2]$$

$$\therefore [1], [2] \Rightarrow \frac{u^2 - 1}{8} + \frac{v^2 - 1}{8} \equiv \frac{(uv)^2 - 1}{8} \pmod{2}.$$

$$(3) \ u = 4r + 3, v = 4s + 1: \text{ same as } \#(2) \text{ above, by symmetry.}$$

$$(4) \ u = 4r + 3, v = 4s + 3:$$

$$u^2 - 1 = 16r^2 + 24r + 8, v^2 - 1 = 16s^2 + 24s + 8.$$

$$\therefore \frac{u^2 - 1}{8} + \frac{v^2 - 1}{8} = 2r^2 + 3r + 1 + 2s^2 + 3s + 1.$$

$$\therefore \frac{u^2 - 1}{8} + \frac{v^2 - 1}{8} \equiv 3r + 3s \equiv r + s \pmod{2} \quad [1]$$

$$uv = 16rs + 12r + 12s + 9.$$

$$\begin{aligned}(uv)^2 - 1 &= 16^2r^2s^2 + (16)(12)r^2s + (16)(12)rs^2 + 144rs + (12)(16)r^2s + 144r^2 + 144rs + \\ &\quad (9)(12)r + (12)(16)rs^2 + 144rs + 144s^2 + (9)(12)s + 144rs + (9)(12)r + (9)(12)s + 80.\end{aligned}$$

$$= 16^2r^2s^2 + (24)(16)r^2s + (24)(16)rs^2 + (4)(144)rs + 144r^2 + 144s^2 + (9)(24)r + (9)(24)s + 80.$$

$$\therefore \frac{(uv)^2 - 1}{8} \equiv 27r + 27s \equiv r + s \pmod{2} \quad [2]$$

$$\therefore [1], [2] \Rightarrow \frac{u^2 - 1}{8} + \frac{v^2 - 1}{8} \equiv \frac{(uv)^2 - 1}{8} \pmod{2}.$$

$$\therefore (1), (2), (3), (4) \Rightarrow \frac{u^2 - 1}{8} + \frac{v^2 - 1}{8} \equiv \frac{(uv)^2 - 1}{8} \pmod{2}.$$

$$\begin{aligned}\therefore (2/b) &= (-1)^{(p_1^2 - 1)/8} (-1)^{(p_2^2 - 1)/8} \dots (-1)^{(p_r^2 - 1)/8} = (-1)^{[(p_1 p_2)^2 - 1]/8} \dots (-1)^{(p_r^2 - 1)/8} \\ &= (-1)^{[(p_1 p_2 \dots p_r)^2 - 1]/8} = (-1)^{[b^2 - 1]/8}. \quad \square\end{aligned}$$

19. Derive the Generalized Quadratic Reciprocity Law: If a and b are relatively prime positive odd integers, each greater than 1, then

$$(a/b)(b/a) = (-1)^{\frac{a-1}{2} \cdot \frac{b-1}{2}}.$$

Pf. Let $a = p_1 p_2 \cdots p_r$, $b = q_1 q_2 \cdots q_s$ be the prime decompositions of a and b , where, since a and b are odd, p_i and q_j are odd primes, not necessarily distinct. Since $\gcd(a, b) = 1$, then $p_i \neq q_j$ for any i, j .

By definition of (a/b) , and using Th. 9.2(d),

$$\begin{aligned}
 (a/b) &= (a/q_1)(a/q_2) \cdots (a/q_s) \\
 &= (p_1 \cdots p_r/q_1)(p_1 \cdots p_r/q_2) \cdots (p_1 \cdots p_r/q_s) \\
 &= (p_1/q_1) \cdots (p_r/q_1) \cdot (p_1/q_2) \cdots (p_r/q_2) \cdots (p_1/q_s) \cdots (p_r/q_s) \\
 &= (p_1/q_1)(p_1/q_2) \cdots (p_1/q_s) \cdot \quad [\text{rearranging rows \& cols}] \\
 &\quad (p_2/q_1)(p_2/q_2) \cdots (p_2/q_s) \cdot \quad \text{to get } r \text{ rows, } s \text{ cols}] \\
 &\quad \vdots \\
 &\quad (p_r/q_1)(p_r/q_2) \cdots (p_r/q_s).
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 (b/a) &= (q_1/p_1) \cdots (q_s/p_1) \cdot \quad [r \text{ rows, } s \text{ cols}] \\
 &\quad (q_1/p_2) \cdots (q_s/p_2) \cdot \\
 &\quad \vdots \\
 &\quad (q_1/p_r) \cdots (q_s/p_r).
 \end{aligned}$$

$$\therefore (a/b)(b/a) = \quad [\text{aligning } (a/b) \text{ rows with } (b/a) \text{ rows}]$$

$$\begin{aligned}
 &(p_1/q_1)(p_1/q_2) \cdots (p_1/q_s) \cdot (q_1/p_1) \cdots (q_s/p_1) \cdot \\
 &(p_2/q_1)(p_2/q_2) \cdots (p_2/q_s) \cdot (q_1/p_2) \cdots (q_s/p_2) \cdot \\
 &\quad \vdots \\
 &(p_r/q_1)(p_r/q_2) \cdots (p_r/q_s) \cdot (q_1/p_r) \cdots (q_s/p_r) \\
 &= (p_1/q_1)(q_1/p_1) \cdots (p_1/q_s)(q_s/p_1) \cdot \\
 &\quad (p_2/q_1)(q_1/p_2) \cdots (p_2/q_s)(q_s/p_2) \cdot \\
 &\quad \vdots \\
 &\quad (p_r/q_1)(q_1/p_r) \cdots (p_r/q_s)(q_s/p_r).
 \end{aligned}$$

Now using quadratic reciprocity on (p_i/q_j) :

$$\begin{aligned}
 &= (-1)^{\frac{p_1-1}{2} \cdot \frac{q_1-1}{2}} \cdots (-1)^{\frac{p_1-1}{2} \cdot \frac{q_s-1}{2}} \cdot \\
 &\quad (-1)^{\frac{p_2-1}{2} \cdot \frac{q_1-1}{2}} \cdots (-1)^{\frac{p_2-1}{2} \cdot \frac{q_s-1}{2}} \cdot \\
 &\quad \vdots \\
 &\quad (-1)^{\frac{p_r-1}{2} \cdot \frac{q_1-1}{2}} \cdots (-1)^{\frac{p_r-1}{2} \cdot \frac{q_s-1}{2}} \quad [1] \\
 &= (-1)^{\frac{p_1-1}{2} \left[\frac{q_1-1}{2} + \cdots + \frac{q_s-1}{2} \right]} \cdot (-1)^{\frac{p_2-1}{2} \left[\frac{q_1-1}{2} + \cdots + \frac{q_s-1}{2} \right]} \cdots (-1)^{\frac{p_r-1}{2} \left[\frac{q_1-1}{2} + \cdots + \frac{q_s-1}{2} \right]}.
 \end{aligned}$$

By prob. 18(f) above, when u, v are odd integers, $(-1)^{\frac{u-1}{2} + \frac{v-1}{2}} = (-1)^{\frac{uv-1}{2}}$.
 $\therefore [1]$ becomes

$$\begin{aligned} (a/b)(b/a) &= (-1)^{\frac{p_1-1}{2} \cdot \frac{q_1 \cdots q_s-1}{2}} \cdot (-1)^{\frac{p_2-1}{2} \cdot \frac{q_1 \cdots q_s-1}{2}} \cdots (-1)^{\frac{p_r-1}{2} \cdot \frac{q_1 \cdots q_s-1}{2}} \\ &= (-1)^{\frac{p_1-1}{2} \cdot \frac{b-1}{2}} \cdot (-1)^{\frac{p_2-1}{2} \cdot \frac{b-1}{2}} \cdots (-1)^{\frac{p_r-1}{2} \cdot \frac{b-1}{2}} \\ &= (-1)^{\left[\frac{p_1-1}{2} + \frac{p_2-1}{2} + \cdots + \frac{p_r-1}{2}\right] \cdot \frac{b-1}{2}} \\ &= (-1)^{\left[\frac{p_1 p_2 \cdots p_r - 1}{2}\right] \cdot \frac{b-1}{2}} \\ &= (-1)^{\frac{a-1}{2} \cdot \frac{b-1}{2}}. \end{aligned}$$

$$\therefore (a/b)(b/a) = (-1)^{\frac{a-1}{2} \cdot \frac{b-1}{2}}. \quad \square$$

20. Using the Generalized Quadratic Reciprocity Law, determine whether the congruence $x^2 \equiv 231 \pmod{1105}$ is solvable.

First, note $231 = 3 \cdot 7 \cdot 11$, $1105 = 5 \cdot 13 \cdot 17$. $\therefore \gcd(231, 1105) = 1$.

By prob. 17 above, if 231 is a quadratic residue of 1105 (i.e., $x^2 \equiv 231 \pmod{1105}$ has a solution), then $(231/1105) = 1$. Can't use converse, but if can show $(231/1105) = -1$, then can state $x^2 \equiv 231 \pmod{1105}$ is not solvable.

$$\therefore (231/1105)(1105/231) = (-1)^{\frac{231-1}{2} \cdot \frac{1105-1}{2}} = (-1)^{(115)(552)} = 1.$$

$$\therefore (231/1105) = (1105/231).$$

Using prob. 18(d),

$$\begin{aligned} (231/1105) &= (1105/231) \\ &= (181 + 4 \cdot 231/231) \quad [181 \text{ is prime}] \\ &= (181/231) \quad [\text{prob. 18(a)}] \\ &= (181/3)(181/7)(181/11) \\ &= (1/3)(6/7)(5/11) \\ &= (2/7)(3/7)(11/5) \\ &= (1)(-1)(1) = -1. \end{aligned}$$

$$\therefore (231/1105) = -1, \text{ so } x^2 \equiv 231 \pmod{1105} \text{ is } \underline{\text{not}} \text{ solvable.} \quad \square$$

9.4 Quadratic Congruence With Composite Moduli

1.

(a) Show that 7 and 18 are the only incongruent solutions of $x^2 \equiv -1 \pmod{5^2}$.

By Th. 8.12, $x^k \equiv -1 \pmod{25}$ has a solution $\iff (-1)^{\varphi(25)/d} \equiv 1 \pmod{25}$, where $\varphi(25) = 5^2 - 5 = 20$ and $d = \gcd(k, \varphi(25))$. Here $k = 2$, $d = \gcd(2, 20) = 2$, and so

$(-1)^{20/2} = (-1)^{10} = 1 \equiv 1 \pmod{25}$. $\therefore$ it has a solution, and Th. 8.12 gives exactly $d = 2$ incongruent solutions.

To derive 7, 18, first solve $x^2 \equiv -1 \pmod{5}$, or $x^2 \equiv 4 \pmod{5}$. $\therefore x_0 = 2$.

$x_0^2 = 4 = -1 + (1) \cdot 5$, so $b = 1$.

Now solve $2x_0y \equiv -b \pmod{5}$, or $4y \equiv -1 \pmod{5}$. $\therefore y_0 = 1$.

$\therefore x_1 = x_0 + y_0p^{k-1} = 2 + 1 \cdot 5 = 7$ is a solution to $x^2 \equiv -1 \pmod{5^2}$, and so is $-7 \equiv 18$.

$\therefore x \equiv 7, 18 \pmod{5^2}$. $\square$

(b) Use part (a) to find the solutions of $x^2 \equiv -1 \pmod{5^3}$.

(1) Solve $x^2 \equiv -1 \pmod{5^2}$. From (a), $x_0 = 7$.

$\therefore x_0^2 = a + bp^2$: $7^2 = (-1) + b(5^2)$, i.e., $49 = -1 + 2 \cdot 25$, so $b = 2$.

(2) Solve $2x_0y \equiv -b \pmod{p}$: $14y \equiv -2 \pmod{5}$, i.e., $-y \equiv -2$, so $y \equiv 2 \pmod{5}$.

(3) $\therefore x_1 = x_0 + y_0p^2 = 7 + 2(25) = 57$. Also $-57 \equiv 68 \pmod{125}$.

$\therefore x \equiv 57, 68 \pmod{5^3}$.

By Th. 8.12, $d = \gcd(k, \varphi(5^3)) = \gcd(2, 100) = 2$, so there are exactly 2 solutions.

$\square$ **2.** Solve each of the following quadratic congruences:

(a) $x^2 \equiv 7 \pmod{3^3}$

(1) First solve $x^2 \equiv 7 \pmod{3}$, i.e., $x^2 \equiv 1 \pmod{3}$. Clearly $x = \pm 1$. Choose $x = 1$.

$\therefore 1^2 = 7 + (-2) \cdot 3$, so $x_0 = 1$, $b = -2$.

(2) Solve $2x_0y \equiv -b \pmod{3}$: $2y \equiv 2 \pmod{3}$, so $y = 1$.

(3) $\therefore$ a solution to $x^2 \equiv 7 \pmod{3^2}$ is $x_0 + y_0p = 1 + (1)(3) = 4 = x'_0$.

(4) $4^2 = 7 + b' \cdot 3^2$: $16 = 7 + b' \cdot 9$, so $b' = 1$.

(5) Solve $2x'_0y' \equiv -b' \pmod{p}$: $8y' \equiv -1 \pmod{3}$, i.e., $2y' \equiv 2 \pmod{3}$ (since $8 \equiv 2$), so $y'_0 = 1$.

(6) $\therefore$ a solution to $x^2 \equiv 7 \pmod{3^3}$ is $x'_0 + y'_0 \cdot 3^2 = 4 + (1)(9) = 13$. Also $-13 \equiv 14 \pmod{27}$.

$\therefore x \equiv 13, 14 \pmod{3^3}$.

(b) $x^2 \equiv 14 \pmod{5^3}$

(1) Solve $x^2 \equiv 14 \pmod{5}$, i.e., $x^2 \equiv 4 \pmod{5}$. $\therefore x_0 = 2$. $2^2 = 14 + b \cdot 5$: $b = -2$.

(2) Solve $2x_0y \equiv -b \pmod{5}$: $4y \equiv 2 \pmod{5}$, $2y \equiv 1 \pmod{5}$, $y = 3$.

(3) $\therefore$ a solution to $x^2 \equiv 14 \pmod{5^2}$ is $x_0 + y_0p = 2 + 3(5) = 17$.

(4) $17^2 = 14 + b(5^2)$: $289 = 14 + 11 \cdot 25$, so $b = 11$.

(5) Solve $2x_0y \equiv -b \pmod{p}$: $34y \equiv -11 \pmod{5}$, i.e., $4y \equiv 4 \pmod{5}$, $y = 1$.

(6) $\therefore$ a solution to $x^2 \equiv 14 \pmod{5^3}$ is $17 + (1)(5^2) = 42$. Also $125 - 42 = 83$.
 $\therefore x \equiv 42, 83 \pmod{5^3}$.

(c) $x^2 \equiv 2 \pmod{7^3}$

(1) Solve $x^2 \equiv 2 \pmod{7}$. $x_0 = 3$. $3^2 = 2 + (1) \cdot 7$, so $b = 1$.

(2) Solve $2x_0y \equiv -b \pmod{7}$: $6y \equiv 6 \pmod{7}$, $y = 1$.

(3) $\therefore$ a solution to $x^2 \equiv 2 \pmod{7^2}$ is $x_0 + y_0p = 3 + (1)(7) = 10$.
 $10^2 = 2 + b(7^2)$: $100 = 2 + 2 \cdot 49$, so $b = 2$.

(4) Solve $2x_0y \equiv -b \pmod{7}$: $20y \equiv -2 \pmod{7}$, i.e., $-y \equiv -2 \pmod{7}$, $y = 2$.

(5) $\therefore$ a solution to $x^2 \equiv 2 \pmod{7^3}$ is $10 + (2)(7^2) = 10 + 98 = 108$. Also $7^3 - 108 = 343 - 108 = 235$.
 $\therefore x \equiv 108, 235 \pmod{7^3}$.

3. Solve the congruence $x^2 \equiv 31 \pmod{11^4}$.

(1) Solve $x^2 \equiv 31 \pmod{11}$, i.e., $x^2 \equiv 9 \pmod{11}$. $\therefore x = 3$. $3^2 = 31 + b \cdot 11$:
 $9 = 31 + (-2)(11)$, so $b = -2$.

(2) Solve $2x_0y \equiv -b \pmod{11}$: $6y \equiv 2 \pmod{11}$, $y = 4$.

(3) $\therefore$ a solution to $x^2 \equiv 31 \pmod{11^2}$ is $x_0 + y_0p = 3 + (4)(11) = 47$.
 $47^2 = 31 + b(11^2)$: $2209 = 31 + 18 \cdot 121$, so $b = 18$.

(4) Solve $2x_0y \equiv -b \pmod{11}$: $2(47)y \equiv -18 \pmod{11}$, i.e., $94y \equiv 4 \pmod{11}$, or
 $6y \equiv 4 \pmod{11}$. $12y \equiv 8 \pmod{11}$. $y \equiv 8 \pmod{11}$.

(5) $\therefore$ a solution to $x^2 \equiv 31 \pmod{11^3}$ is $47 + 8(11^2) = 47 + 968 = 1015$.
 $1015^2 = 31 + b(11^3)$: $1030225 = 31 + 774 \cdot 1331$, so $b = 774$.

(6) Solve $2x_0y \equiv -b \pmod{11}$: $2030y \equiv -774 \pmod{11}$, i.e., $6y \equiv 7 \pmod{11}$. $12y \equiv 14 \pmod{11}$. $y \equiv 3 \pmod{11}$.

(7) $\therefore$ a solution to $x^2 \equiv 31 \pmod{11^4}$ is $1015 + 3(11^3) = 1015 + 3993 = 5008$. Also
 $11^4 - 5008 = 14641 - 5008 = 9633$.
 $\therefore x \equiv 5008, 9633 \pmod{11^4}$.

4. Find the solutions of $x^2 + 5x + 6 \equiv 0 \pmod{5^3}$ and $x^2 + x + 3 \equiv 0 \pmod{3^3}$.

$$(a) \quad x^2 + 5x + 6 \equiv 0 \pmod{5^3}$$

$$(x+3)(x+2) \equiv 0 \pmod{5^3}. \quad \therefore x \equiv -3, -2, \text{ i.e., } x \equiv 122, 123 \pmod{5^3}.$$

$$(b) \quad x^2 + x + 3 \equiv 0 \pmod{3^3}$$

$$\gcd(4, 3^3) = 1, \text{ so } x^2 + x + 3 \equiv 0 \iff 4x^2 + 4x + 12 \equiv 0 \iff (2x+1)^2 + 11 \equiv 0 \iff (2x+1)^2 \equiv -11 \equiv 16 \pmod{3^3}.$$

$$\therefore 2x+1 \equiv \pm 4.$$

$$2x \equiv 3,$$

$$2x \equiv -5.$$

$$28x \equiv 42,$$

$$28x \equiv -70.$$

$$(\gcd(14, 3^3) = 1.) \quad \therefore x \equiv 15 \pmod{27} \text{ or } x \equiv 11 \pmod{27}.$$

$$\therefore x \equiv 11, 15 \pmod{3^3}.$$

5. Prove that if the congruence $x^2 \equiv a \pmod{2^n}$, where a is odd and $n \geq 3$, has a solution, then it has exactly four incongruent solutions. *Proof.* Note: we cannot invoke Th. 8.12 since 2^n has no primitive roots for $n \geq 3$.

Since a is odd, if x is a solution then x must be odd. Also, $-x$ is a solution.

Suppose y is any other solution. Then $y^2 \equiv a \pmod{2^n}$, so $x^2 \equiv y^2 \pmod{2^n}$, $n \geq 3$.

$$\therefore (x-y)(x+y) \equiv 0 \pmod{2^n}$$

$$\iff \frac{x-y}{2} \cdot \frac{x+y}{2} \equiv 0 \pmod{2^{n-2}}, \quad n \geq 3 \quad [\text{by Th. 4.3}].$$

But note $\frac{x-y}{2} + \frac{x+y}{2} = x$, which is odd. $\therefore$ only one of $\frac{x-y}{2}, \frac{x+y}{2}$ is even.

(1) Suppose $\frac{x-y}{2}$ is the even factor; $\frac{x+y}{2}$ is odd.

$$\therefore (x-y) \cdot \frac{x+y}{2} \equiv 0 \pmod{2^{n-1}} \implies x-y \equiv 0 \pmod{2^{n-1}} \implies x \equiv y \pmod{2^{n-1}}.$$

(2) Suppose $\frac{x+y}{2}$ is the even factor, so $\frac{x-y}{2}$ is the odd factor.

$$\therefore (x+y) \cdot \frac{x-y}{2} \equiv 0 \pmod{2^{n-1}} \implies$$

$$x+y \equiv 0 \pmod{2^{n-1}} \implies x \equiv -y \pmod{2^{n-1}}.$$

(1), (2) $\implies$ if y is any other solution, $y = \pm x + k \cdot 2^{n-1}$.

For k odd, $k = 2r + 1$, some r . $\therefore y = \pm x + 2^{n-1} + r \cdot 2^n$.

$$\therefore y \equiv \pm x + 2^{n-1} \pmod{2^n} \quad [1]$$

For k even, $k = 2r$, some r . $\therefore y = \pm x + r \cdot 2^n$.

$$\therefore y \equiv \pm x \pmod{2^n} \quad [2]$$

$\therefore [1], [2] \implies$ only incongruent solutions $\pmod{2^n}$ are $x, -x, x + 2^{n-1}, -x + 2^{n-1}$.

□

6. From $23^2 \equiv 17 \pmod{2^7}$, find three other solutions of the quadratic congruence $x^2 \equiv 17 \pmod{2^7}$.

From #5 above, solutions are $23, -23, 23 + 2^6$, and $-23 + 2^6 \pmod{2^7}$.

$2^6 = 64, 2^7 = 128. \therefore -23 + 2^7 = 105.$

$$\therefore 23, 105, 87, 41 \pmod{2^7}.$$

7. First determine the values of a for which the congruences below are solvable, and then find the solutions of these congruences.

(a) $x^2 \equiv a \pmod{2^4}$

By Th. 9.12, solvable $\iff a \equiv 1 \pmod{8}$. $2^4 = 16. \therefore a = 1$ or 9 .

Now use #5 above.

$a = 1: x = 1, -1, 1 + 2^3, -1 + 2^3 \pmod{2^4} \Rightarrow x \equiv 1, -1 + 16, 9, 7$

$$\therefore x \equiv 1, 7, 9, 15 \pmod{2^4}.$$

$a = 9: x = 3, -3, 3 + 2^3, -3 + 2^3$

$$\therefore x \equiv 3, 13, 11, 5.$$

(b) $x^2 \equiv a \pmod{2^5}$

$2^5 = 32$. Solvable $\iff a \equiv 1 \pmod{8}$. $\therefore a = 1, 9, 17$, or 25 .

$a = 1: x \equiv \pm 1, \pm 1 + 2^4. \therefore x \equiv 1, 31, 17, 15.$

$a = 9: x \equiv \pm 3, \pm 3 + 2^4. \therefore x \equiv 3, 29, 19, 13.$

$a = 17: \therefore x^2 \equiv 17 + 32 = 49. \therefore x \equiv \pm 7, \pm 7 + 2^4. \therefore x \equiv 7, 25, 23, 9.$

$a = 25: x \equiv \pm 5, \pm 5 + 2^4. \therefore x \equiv 5, 27, 21, 11.$

(c) $x^2 \equiv a \pmod{2^6}$

$2^6 = 64$. $\therefore$ solvable $\iff a \equiv 1 \pmod{8}$. $\therefore a = 1, 9, 17, 25, 33, 41, 49, 57.$

$a = 1: \pm 1, \pm 1 + 2^5. \therefore x \equiv 1, 63, 33, 31.$

$a = 9: \pm 3, \pm 3 + 2^5. \therefore x \equiv 3, 61, 35, 29.$

$a = 17: 17 + 64 = 81. \therefore \pm 9, \pm 9 + 2^5. \therefore x \equiv 9, 55, 41, 23.$

$a = 25: \pm 5, \pm 5 + 2^5. \therefore x \equiv 5, 59, 37, 27.$

$a = 33: 33 + 64 = 97, 33 + 128 = 161, 33 + 192 = 225. \therefore \pm 15, \pm 15 + 2^5. \therefore x \equiv 15, 49, 47, 17.$

$a = 41: 41 + 64 = 105, 41 + 128 = 169. \therefore \pm 13, \pm 13 + 2^5. \therefore x \equiv 13, 51, 45, 19.$

$a = 49: \pm 7, \pm 7 + 2^5. \therefore x \equiv 7, 57, 39, 25.$

$a = 57: 57 + 64 = 121. \therefore \pm 11, \pm 11 + 2^5. \therefore x \equiv 11, 53, 43, 21.$

8. For fixed $n > 1$, show that all the solvable congruences $x^2 \equiv a \pmod{n}$ with $\gcd(a, n) = 1$ have the same number of solutions.

Proof. Let $n = 2^{k_0} p_1^{k_1} \cdots p_r^{k_r}, k_i \geq 0$.

For $p_i^{k_i}, k_i \geq 1$, we can use Th. 8.12 to state that $x^2 \equiv a \pmod{p_i^{k_i}}$ has exactly 2 solutions. The actual value of a doesn't matter, as long as $\gcd(a, p_i^{k_i}) = 1$.

For $x^2 \equiv a \pmod{2^{k_0}}$, $k_0 \geq 1$, Th. 9.12 states there are just 2 solutions for $x^2 \equiv a \pmod{2}$ (i.e., 1, -1), 2 solutions for $x^2 \equiv a \pmod{4}$ when $a \equiv 1 \pmod{4}$ (i.e., $x = 1, 3$), and for $x^2 \equiv a \pmod{2^n}$, $n \geq 3$ when $a \equiv 1 \pmod{8}$, problem #5 shows there are exactly 4 solutions.

$\therefore$ if $\gcd(a, n) = 1$, then there are $2 \cdot 2^r$ possible solutions if $a \equiv 1 \pmod{4}$ and $a \not\equiv 1 \pmod{8}$, and $4 \cdot 2^r$ possible solutions if $a \equiv 1 \pmod{8}$. Label these solutions x_{ik_i} , so that, for example, x_{1k_1} and x_{2k_1} are the solutions to $x^2 \equiv a \pmod{p_1^{k_1}}$.

$\therefore$ Consider the $4 \cdot 2^r$ ($a \equiv 1 \pmod{8}$) or $2 \cdot 2^r$ ($a \equiv 1 \pmod{4}$, $a \not\equiv 1 \pmod{8}$) linear equation systems:

$$\begin{array}{ccc} \left\{ \begin{array}{l} x \equiv x_{1k_0} \pmod{2^{k_0}} \\ \vdots \\ x \equiv x_{1k_r} \pmod{p_r^{k_r}} \end{array} \right. & \cdots & \left\{ \begin{array}{l} x \equiv x_{2k_0} \pmod{2^{k_0}} \\ \vdots \\ x \equiv x_{1k_r} \pmod{p_r^{k_r}} \end{array} \right. \\ & & \vdots \\ & & \left\{ \begin{array}{l} x \equiv x_{2k_0} \pmod{2^{k_0}} \\ \vdots \\ x \equiv x_{2k_r} \pmod{p_r^{k_r}} \end{array} \right. \end{array}$$

By the Chinese Remainder Theorem, there is a simultaneous solution unique to $n = 2^{k_0} p_1^{k_1} \cdots p_r^{k_r}$ for each system.

Thus, the number of solutions is, for any a with $\gcd(a, n) = 1$:

- 2^r , if $n = p_1^{k_1} \cdots p_r^{k_r}$
- $2 \cdot 2^r$, if $n = 2p_1^{k_1} \cdots p_r^{k_r}$
- $2 \cdot 2^r$, if $n = 2^2 p_1^{k_1} \cdots p_r^{k_r}$ and $a \equiv 1 \pmod{4}$
- $4 \cdot 2^r$, if $n = 2^{k_0} p_1^{k_1} \cdots p_r^{k_r}$, $k_0 \geq 3$ and $a \equiv 1 \pmod{8}$

□

9. (a) Without actually finding them, determine the number of solutions of the congruences $x^2 \equiv 3 \pmod{11^2 \cdot 23^2}$ and $x^2 \equiv 9 \pmod{2^3 \cdot 3 \cdot 5^2}$.

(1) $x^2 \equiv 3 \pmod{11^2 \cdot 23^2}$:

$x^2 \equiv 3 \pmod{11^2}$ and $x^2 \equiv 3 \pmod{23^2}$ each have 2 solutions (by Th. 8.12), so $2 \cdot 2 = 4$ solutions. (2) $x^2 \equiv 9 \pmod{2^3 \cdot 3 \cdot 5^2}$:

$x^2 \equiv 9 \pmod{2^3}$ has 4 solutions (by prob. #5).

$x^2 \equiv 9 \pmod{3} \iff x^2 \equiv 0 \pmod{3}$, so just 1 solution ($x \equiv 0$).

$x^2 \equiv 9 \pmod{5^2}$ has 2 solutions.

$\therefore 4 \cdot 1 \cdot 2 = 8$ solutions.

(b) Solve $x^2 \equiv 9 \pmod{2^3 \cdot 3 \cdot 5^2}$.

$x^2 \equiv 9 \pmod{2^3}$: $x = \pm 3, \pm 3 + 2^2$ by prob. #5. $\therefore x \equiv 3, 5, 7, 1 \pmod{2^3}$.

$$x^2 \equiv 9 \pmod{3} \iff x \equiv 0 \pmod{3}.$$

$$x^2 \equiv 9 \pmod{5^2}: x = \pm 3, \text{ i.e., } x \equiv 3, 22 \pmod{5^2}.$$

$$(1) \quad x \equiv 1 \pmod{8}, \quad n = 600 = 2^3 \cdot 3 \cdot 5^2$$

$$x \equiv 0 \pmod{3}, \quad N_1 = 75 = 3 \cdot 5^2, \quad N_2 = 200 = 2^3 \cdot 5^2, \quad N_3 = 24 = 2^3 \cdot 3$$

$$x \equiv 3 \pmod{25}$$

$$75x_1 \equiv 1 \pmod{8}, \quad 3x_1 \equiv 1, \quad 9x_1 \equiv 3, \quad x_1 \equiv 5 \text{ [Note: } 3 \cdot 3 = 9 \equiv 1 \pmod{8} \text{ gives } x_1 = 3; \text{ the manuscript writes } x_1 \equiv 5.]$$

$$200x_2 \equiv 1 \pmod{3}, \quad 2x_2 \equiv 1, \quad 4x_2 \equiv 2, \quad x_2 \equiv 2$$

$$24x_3 \equiv 1 \pmod{25}, \quad -x_3 \equiv 1, \quad x_3 \equiv -1 \equiv 24 \quad 24x_3 \equiv 1 \pmod{25}: \quad -x_3 \equiv 1, \\ x_3 \equiv -1 \equiv 24.$$

$$\therefore (1)(75)(3) + 0 \cdot (200)(2) + (3)(24)(-1) = 153.$$

$$\therefore x \equiv \underline{153} \pmod{2^3 \cdot 3 \cdot 5^2}.$$

$$(2) \quad x \equiv 3 \pmod{8}, \quad \text{as above, } N = 2^3 \cdot 3 \cdot 5^2$$

$$x \equiv 0 \pmod{3}, \quad N_1 = 75, N_2 = 200, N_3 = 24$$

$$x \equiv 3 \pmod{25}, \quad x_1 = 3, x_2 = 2, x_3 = -1$$

$$\therefore x = (3)(75)(3) + 0 + (3)(24)(-1) = 603.$$

$$\therefore x \equiv \underline{3} \pmod{2^3 \cdot 3 \cdot 5^2}.$$

$$(3) \quad x \equiv 5 \pmod{8}, \quad \text{as in (1), } N = 2^3 \cdot 3 \cdot 5^2$$

$$x \equiv 0 \pmod{3}, \quad N_1 = 75, N_2 = 200, N_3 = 24$$

$$x \equiv 3 \pmod{25}, \quad x_1 = 3, x_2 = 2, x_3 = -1$$

$$\therefore x = (5)(75)(3) + 0 + (3)(24)(-1) = 1053.$$

$$\therefore x \equiv \underline{453} \pmod{2^5 \cdot 3 \cdot 5^2}.$$

[Note: exponent should be 2^3 ; manuscript writes 2^5 .]

$$(4) \quad x \equiv 7 \pmod{8}, \quad \text{as in (1), } N = 2^3 \cdot 3 \cdot 5^2$$

$$x \equiv 0 \pmod{3}, \quad N_1 = 75, N_2 = 200, N_3 = 24$$

$$x \equiv 3 \pmod{25}, \quad x_1 = 3, x_2 = 2, x_3 = -1 \quad \therefore x = (7)(75)(3) + 0 + (3)(24)(-1) = 1503.$$

$$\therefore x \equiv \underline{303} \pmod{2^3 \cdot 3 \cdot 5^2}.$$

$$(5) \quad x \equiv 1 \pmod{8}, \quad \text{as in (1), } N = 2^3 \cdot 3 \cdot 5^2 = 600$$

$$x \equiv 0 \pmod{3}, \quad N_1 = 75, N_2 = 200, N_3 = 24$$

$$x \equiv 22 \pmod{25}, \quad x_1 = 3, x_2 = 2, x_3 = -1$$

$$x = (1)(75)(3) + 0 + (22)(24)(-1) = -303.$$

$$\therefore x \equiv \underline{297} \pmod{2^3 \cdot 3 \cdot 5^2}.$$

$$(6) \quad x \equiv 3 \pmod{8}, \quad \text{as in (1), } N = 2^3 \cdot 3 \cdot 5^2 = 600$$

$$\begin{aligned} x &\equiv 0 \pmod{3}, \quad N_1 = 75, N_2 = 200, N_3 = 24 \\ x &\equiv 22 \pmod{25}, \quad x_1 = 3, x_2 = 2, x_3 = -1 \\ x &= (3)(75)(3) + 0 + (22)(24)(-1) = 147. \end{aligned}$$

$$\therefore x \equiv \underline{147} \pmod{2^3 \cdot 3 \cdot 5^2}.$$

$$\begin{aligned} (7) \quad x &\equiv 5 \pmod{8}, \quad \text{as in (1), } N = 2^3 \cdot 3 \cdot 5^2 = 600 \\ x &\equiv 0 \pmod{3}, \quad N_1 = 75, N_2 = 200, N_3 = 24 \\ x &\equiv 22 \pmod{25}, \quad x_1 = 3, x_2 = 2, x_3 = -1 \\ x &= (5)(75)(3) + 0 + (22)(24)(-1) = 597 \end{aligned}$$

$$\therefore x \equiv \underline{597} \pmod{2^3 \cdot 3 \cdot 5^2}.$$

$$\begin{aligned} (8) \quad x &\equiv 7 \pmod{8}, \quad \text{as in (1), } N = 2^3 \cdot 3 \cdot 5^2 = 600 \\ x &\equiv 0 \pmod{3}, \quad N_1 = 75, N_2 = 200, N_3 = 24 \\ x &\equiv 22 \pmod{25}, \quad x_1 = 3, x_2 = 2, x_3 = -1 \\ x &= (7)(75)(3) + 0 + (22)(24)(-1) = 1047. \end{aligned}$$

$$\therefore x \equiv \underline{447} \pmod{2^3 \cdot 3 \cdot 5^2}.$$

$$\therefore x \equiv 3, 147, 153, 297, 303, 447, 453, 597 \pmod{2^3 \cdot 3 \cdot 5^2}.$$

10. (a) For an odd prime p , prove that the congruence $2x^2 + 1 \equiv 0 \pmod{p}$ has a solution if and only if $p \equiv 1$ or $3 \pmod{8}$.

Proof. As $\gcd(8, p) = 1$, $2x^2 + 1 \equiv 0 \pmod{p}$ has a solution $\iff 8(2x^2 + 1) \equiv 0 \pmod{p}$ has a solution. $\therefore$ look at $16x^2 = (4x)^2 \equiv -8 \pmod{p}$.

Let $y = 4x$. Then we can solve $4x \equiv y \pmod{p}$: since $y^2 \equiv -8$, $\gcd(y^2, p) = \gcd(y, p) = 1$, so $4x \equiv y \pmod{p}$ has a unique solution.

$$\therefore y^2 \equiv -8 \pmod{p} \text{ has a solution } \iff (-8/p) = 1.$$

$$(-8/p) = (-1/p)(2^3/p) = (-1/p)(2/p).$$

$$\therefore \text{need } (-1/p) = (2/p).$$

$$(a) \quad (2/p) = -1 \iff p \equiv 3 \pmod{8} \text{ or } p \equiv 5 \pmod{8}.$$

$$(-1/p) = (-1)^{(p-1)/2} = -1 \iff \frac{p-1}{2} \text{ is odd } \iff \frac{p-1}{2} = 2k+1, \text{ some } k$$

$$\iff p-1 = 4k+2 \iff p = 3 + 4k.$$

$$\therefore (2/p) = (-1/p) = -1 \iff p \equiv 3 \pmod{8}.$$

$$(b) \quad (2/p) = 1 \iff p \equiv 1 \pmod{8} \text{ or } p \equiv 7 \pmod{8}.$$

$$(-1/p) = (-1)^{(p-1)/2} = 1 \iff \frac{p-1}{2} \text{ is even } \iff \frac{p-1}{2} = 2k, \text{ some } k$$

$$\iff p-1 = 4k \iff p = 1 + 4k.$$

$$\therefore (2/p) = (-1/p) = 1 \iff p \equiv 1 \pmod{8}.$$

$$\therefore (a) \text{ and } (b) \text{ show } 2x^2 + 1 \equiv 0 \pmod{p} \iff p \equiv 1 \text{ or } 3 \pmod{8}. \quad \square$$

(b) Solve the congruence $2x^2 + 1 \equiv 0 \pmod{11^2}$.

Let $z = 4x$, $\therefore z^2 \equiv -8 \pmod{11^2}$.

Use method described in proof to Th. 9.11.

First solve $z^2 \equiv -8 \pmod{11}$:

$$z^2 \equiv 3 \pmod{11}, \quad z \equiv 5 \pmod{11}, \quad \therefore x_0 = 5.$$

$$\therefore x_0^2 = -8 + b \cdot 11: 25 = -8 + 3 \cdot 11, \text{ so } b = 3.$$

Now solve $2x_0y \equiv -b \pmod{p}$: $2(5)y \equiv -3 \pmod{11}$. $\therefore 2(5)y \equiv -3 \pmod{11}$:
 $10y \equiv -3 \pmod{11}$, i.e., $-y \equiv -3 \pmod{11}$, so $y \equiv 3 \pmod{11}$, $y_0 = 3$.

$$\therefore x_0 + y_0p = 5 + 3(11) = 38.$$

$$\therefore z = 38 \text{ solves } z^2 \equiv -8 \pmod{11^2}.$$

Now convert back using $z \equiv 4x \pmod{11^2}$:

$$\therefore 4x \equiv 38 \pmod{11^2}$$

$$2x \equiv 19 \pmod{121}$$

$$122x \equiv 19 \cdot 61 = 1159 \pmod{121}$$

$$x \equiv 1159 \equiv 70 \pmod{11^2}.$$

$$\therefore x \equiv \pm 70 \pmod{11^2}.$$

$$\therefore x \equiv \underline{51, 70} \pmod{11^2}.$$